CBSE
CLASS 10

EduSpark

MATHEMATICS

ASTRA SERIES

CBSE CLASS 10 • BOARD EXAM 2027

60 DIN

MEIN PURA SYLLABUS
KHATAM!

INSIDE THIS BOOK

- **Complete Notes**
  Concepts, Formulas, Solved Examples
- **Clean Diagrams**
  Every figure redrawn, easy to read
- **MCQs, Assertion-Reason**
  Case-Based and Board-Pattern Questions
- **All 14 Chapters**
  NCERT 2026-27 rationalised syllabus
- **English + Hinglish Tips**
  Tricks, Common Mistakes, Board Tips
- **60-Day Study Plan**
  With Formula Day and 4 Full Mocks



By Shivam Bhaiya
SHIVAM KUMAR

CBSE CLASS 10 MATHEMATICS

Astra Series

by **Shivam Kumar** (Shivam Bhaiya)

BOOK DETAILS

Book	**Astra Series: Mathematics**
Subject	**CBSE Class 10 Mathematics (Standard and Basic)**
Author	**Shivam Kumar (Shivam Bhaiya)**
Publisher	**Eduspark**
Edition	**First Edition, 2026**
Copyright	**© 2026 Shivam Kumar & Eduspark**
Address	**Bawan Bigha, Road No-1, Mithanpura, PO: Ramna, DIST: Muzaffarpur, Bihar - 842002**

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Printed in India First published 2026



BEFORE DAY 1

A NOTE FROM SHIVAM BHAIYA

> *"Maths is not learnt by reading. It is learnt with a pen in your hand."*

Dear student,

Class 10 Maths has fourteen chapters, and every one of them has formulas, methods and the kind of question that looks new even when it is not. Open all of it at once and it feels like a mountain. That is why this book is written **Day by Day**. Every Day tells you exactly what to learn, which examples to follow and which questions to solve. You do not have to plan anything. You only have to show up with a notebook and a pen.

The notes are short on purpose. Every Day keeps only what the board actually asks: the concept, the formula, solved examples written step by step, and the traps where marks are usually lost. Wherever students usually go wrong, you will find me in a bubble, explaining it the way an elder brother would, in our own language.

The practice after every Day is board pattern: MCQs, Assertion-Reason, case-based questions, and short and long answers, with full worked solutions. **Solve first, check later.** When you get one wrong, write it in your **Mistake Book** with the correct step. That notebook will be worth more than any guide in the last week.

By Day 42 the whole syllabus is done, with a revision day after every pair of chapters and a full Formula Day. Then come board-practice days, four full mock papers in board timing and a calm last week. Follow the plan honestly, and on exam day you will walk in knowing that you have already written this paper four times.

Ab bas ek kaam: page palto aur Day 1 shuru karo. Main har page pe saath hoon.

Shivam Bhaiya

Shivam Kumar, Astra Series

MY 60-DAY PROMISE

I, _____, will finish one Day of this book every day.

Start date: _____ Day 42 (syllabus done) on: _____

Board exam on: _____ Signature: _____

ASTRA SERIES | CLASS 10 MATHEMATICS

CONTENTS

60 Days | 14 chapters
Formula Day + 4 full mocks

BEFORE YOU START — FRONT PAGES

--	Start	Publisher information	2
--	Start	A note from Shivam Bhaiya	3
--	Start	How to use this book	6
--	Start	The chapter plan (read first)	7
--	Start	60-day roadmap	8
--	Start	Syllabus, marks and paper pattern	11
--	Start	My trackers	12

PHASE 1 | LEARN — DAYS 1 TO 42

01	Ch1	Real Numbers: Prime Factorisation and the Fundamental Theorem of ...	13
02	Ch1	Real Numbers: HCF, LCM and Proving Irrationality	17
03	Ch2	Polynomials: Zeroes of a Polynomial and their Geometrical Meaning	21
04	Ch2	Polynomials: Relationship between Zeroes and Coefficients	25
05	Revision	Revision 1: Real Numbers + Polynomials	29
06	Ch3	Linear Equations: Graphical Method and Consistency	33
07	Ch3	Linear Equations: Substitution and Elimination	37
08	Ch3	Linear Equations: Word Problems	40
09	Ch4	Quadratic Equations: Standard Form and Factorisation	43
10	Ch4	Quadratic Equations: Quadratic Formula, Discriminant and Word Probl...	47
11	Revision	Revision 2: Linear Equations + Quadratic Equations	51
12	Ch5	Arithmetic Progressions: What is an AP, and its nth Term	55
13	Ch5	Arithmetic Progressions: Sum of First n Terms	59
14	Ch5	Arithmetic Progressions: AP Word Problems and Board Mix	62
15	Ch7	Coordinate Geometry: Distance Formula	66
16	Ch7	Coordinate Geometry: Section Formula and Midpoint	70
17	Revision	Revision 3: Arithmetic Progressions + Coordinate Geometry	74
18	Ch6	Triangles: Similar Figures and the Basic Proportionality Theorem	78
19	Ch6	Triangles: Criteria for Similarity of Triangles	82
20	Ch6	Triangles: Similarity Proofs and Board Problems	86
21	Ch10	Circles: Tangent to a Circle	90

22	Ch10	Circles: Tangents from an External Point	95
23	Revision	Revision 4: Triangles + Circles	99
24	Ch8	Trigonometry: Trigonometric Ratios	103
25	Ch8	Trigonometry: Ratios of Specific Angles	107
26	Ch8	Trigonometry: Trigonometric Identities	111
27	Ch9	Applications of Trig: Heights and Distances: Angle of Elevation	115
28	Ch9	Applications of Trig: Angle of Depression and Two-Triangle Problems	120
29	Revision	Revision 5: Trigonometry (Chapters 8 and 9)	125
30	Ch11	Areas Related to Circles: Arc Length and Area of a Sector	129
31	Ch11	Areas Related to Circles: Segments and Combination Figures	133
32	Ch12	Surface Areas and Volumes: Surface Area of Combined Solids	137
33	Ch12	Surface Areas and Volumes: Volume of Combined Solids	142
34	Ch12	Surface Areas and Volumes: Mixed Board Problems	146
35	Revision	Revision 6: Mensuration (Chapters 11 and 12)	150
36	Ch13	Statistics: Mean of Grouped Data	154
37	Ch13	Statistics: Mode of Grouped Data	159
38	Ch13	Statistics: Median of Grouped Data	162
39	Ch14	Probability: Theoretical Probability	166
40	Ch14	Probability: Coins, Dice, Cards and Mixed Problems	170
41	Revision	Revision 7: Statistics + Probability	174
42	Formula	Formula Day: All 14 chapters on one sheet	178

PHASE 2 AND 3 | PRACTICE, MOCKS, LAST WEEK — DAYS 43 TO 60

43	Practice	Board Practice: Number Systems + Algebra I (Ch 1 to 3)	183
44	Practice	Board Practice: Algebra II (Ch 4 and 5)	189
45	Practice	Board Practice: Coordinate Geometry + Triangles + Circles	194
46	Practice	Board Practice: Trigonometry (Ch 8 and 9)	200
47	Practice	Board Practice: Mensuration + Statistics + Probability	205
48	Practice	Board Practice: Case-Based and Assertion-Reason Marathon	211
49	Mock	Mock Test 1: Full Paper, 80 marks, 3 hours	218
50	Practice	Mock 1 Review: Solutions and Mistake Fixing	222
51	Mock	Mock Test 2: Full Paper, 80 marks, 3 hours	228
52	Practice	Mock 2 Review: Solutions and Mistake Fixing	232
53	Mock	Mock Test 3: Full Paper, 80 marks, 3 hours	237
54	Practice	Mock 3 Review: Solutions and Mistake Fixing	241
55	Practice	Weak-Spot Sprint: Algebra Traps (Ch 1 to 5)	248
56	Practice	Weak-Spot Sprint: Geometry, Coordinate and Trigonometry Traps	253
57	Practice	Weak-Spot Sprint: Mensuration, Statistics and Probability Traps	258
58	Mock	Mock Test 4: Final Full Paper, timed	264
59	Practice	Mock 4 Review: Solutions and Last Corrections	268
60	Exam Eve	Exam Eve: Formula Flash, Paper Strategy and Final Checklist	274

HOW TO USE THIS BOOK

THE 60-DAY MISSION

14 chapters
7 revision days
4 full mocks

This book is not a book you read once from cover to cover. It is a **60-day plan**. Every chapter is broken into **Days**. Each Day tells you what to learn, which examples to follow and which questions to solve. Follow one Day a day, tick it off, and by **Day 42** the full syllabus is done. Days 43 to 60 are for board practice, four full mock papers and a calm last week.

PHASE 1 | DAYS 1 TO 42

Learn all 14 chapters

Two chapters at a time, then a revision day on both. Day 42 is the Formula Day: every formula of the book on a few pages.

PHASE 2 | DAYS 43 TO 54

Board practice + 3 mocks

Unit-wise board questions, a case-based and Assertion-Reason marathon, then three full papers, each with a review day.

PHASE 3 | DAYS 55 TO 60

Last 6 days

Weak-spot sprints, the final mock, and an exam-eve formula flash.

Your daily routine on a learn day

about 110 minutes, one sitting or two



1. Warm-up
10 min Without opening the book, write the last Day's formulas from memory.

2. Learn
35 min Read the notes. Copy every solved example in your notebook, step by step.

3. Solve
45 min Solve the Day's practice questions without looking at the notes.

4. Check
10 min Check with the worked solutions. Write every mistake, with the right step, in a Mistake Book.

5. Recall
10 min Do Tonight's 2-minute recall and tick the Day complete line.

FIVE KINDS OF DAYS

Learn day
Notes, formula box, solved examples, 12 to 18 board-pattern questions with worked solutions.

Revision day
No new topic. Formula recap of two chapters and 15 to 25 mixed questions.

Formula and practice days
Day 42 and Day 60: every formula on a few pages. Days 43 to 48: unit-wise board practice.

Mock day
One full 80-mark paper in 3 hours at board timing; the next day is its review.

READ THIS FIRST

THE CHAPTER PLAN

2 chapters, then 1 revision day
Days weighted by board marks

The 14 chapters are learnt **two at a time**, and a **revision day** comes right after each pair, while both chapters are still fresh. Chapters that carry more marks or need more practice get **three days** (Linear Equations, AP, Triangles, Trigonometry, Surface Areas and Volumes, Statistics); lighter ones get two. Algebra carries 20 of the 80 marks, so it gets the most days.

SHIVAM BHAIYA KA RULE

Chapter khatam hote hi revision day skip mat karna. Wahi din decide karta hai ki formula 60 din tak yaad rahega ya 6 din tak.

Ch	Chapter	Unit	Days	No.	Revised on
1	**Real Numbers**	Number Systems	Days 1-2	2	Day 5
2	**Polynomials**	Algebra	Days 3-4	2	Day 5
3	**Pair of Linear Equations in Two Variables**	Algebra	Days 6-8	3	Day 11
4	**Quadratic Equations**	Algebra	Days 9-10	2	Day 11
5	**Arithmetic Progressions**	Algebra	Days 12-14	3	Day 17
7	**Coordinate Geometry**	Coordinate Geometry	Days 15-16	2	Day 17
6	**Triangles**	Geometry	Days 18-20	3	Day 23
10	**Circles**	Geometry	Days 21-22	2	Day 23
8	**Introduction to Trigonometry**	Trigonometry	Days 24-26	3	Day 29
9	**Some Applications of Trigonometry**	Trigonometry	Days 27-28	2	Day 29
11	**Areas Related to Circles**	Mensuration	Days 30-31	2	Day 35
12	**Surface Areas and Volumes**	Mensuration	Days 32-34	3	Day 35
13	**Statistics**	Statistics and Probability	Days 36-38	3	Day 41
14	**Probability**	Statistics and Probability	Days 39-40	2	Day 41

After Day 41	What happens
Day 42	Formula Day: every formula of all 14 chapters, memory tricks, rapid quiz
Days 43 to 48	Board practice, unit by unit, then a case-based and Assertion-Reason marathon
Days 49 to 54	Mock 1, 2 and 3, each followed by a review day
Days 55 to 60	Three weak-spot sprints, Mock 4, its review and the exam-eve formula flash

60-DAY ROADMAP | PART 1

DAYS 1 TO 20

Tick the box the day you finish.
Qs = practice questions that Day.

☐ Learn day ☐ Revision / formula ☐ Full mock ☐ Last 6 days

Pages = pages of this book that Day

Day	Type	Chapter and topic	Pages	Qs	Min	Done
1	Ch1	Real Numbers: Prime Factorisation and the Fundamental Theorem of Arithmetic	4	16	110	☐
2	Ch1	Real Numbers: HCF, LCM and Proving Irrationality	4	16	110	☐
3	Ch2	Polynomials: Zeroes of a Polynomial and their Geometrical Meaning	4	16	110	☐
4	Ch2	Polynomials: Relationship between Zeroes and Coefficients	4	17	110	☐
5	Revision	Revision 1: Real Numbers + Polynomials	4	18	120	☐
6	Ch3	Linear Equations: Graphical Method and Consistency	4	15	110	☐
7	Ch3	Linear Equations: Substitution and Elimination	3	14	110	☐
8	Ch3	Linear Equations: Word Problems	3	13	110	☐
9	Ch4	Quadratic Equations: Standard Form and Factorisation	4	15	110	☐
10	Ch4	Quadratic Equations: Quadratic Formula, Discriminant and Word Problems	4	16	110	☐
11	Revision	Revision 2: Linear Equations + Quadratic Equations	4	20	120	☐
12	Ch5	Arithmetic Progressions: What is an AP, and its nth Term	4	16	110	☐
13	Ch5	Arithmetic Progressions: Sum of First n Terms	3	15	110	☐
14	Ch5	Arithmetic Progressions: AP Word Problems and Board Mix	4	15	110	☐
15	Ch7	Coordinate Geometry: Distance Formula	4	16	110	☐
16	Ch7	Coordinate Geometry: Section Formula and Midpoint	4	16	110	☐
17	Revision	Revision 3: Arithmetic Progressions + Coordinate Geometry	4	21	120	☐
18	Ch6	Triangles: Similar Figures and the Basic Proportionality Theorem	4	15	110	☐
19	Ch6	Triangles: Criteria for Similarity of Triangles	4	15	110	☐
20	Ch6	Triangles: Similarity Proofs and Board Problems	4	15	110	☐

60-DAY ROADMAP | PART 2

DAYS 21 TO 40

Tick the box the day you finish.
Qs = practice questions that Day.

☐ Learn day ☐ Revision / formula ☐ Full mock ☐ Last 6 days

Pages = pages of this book that Day

Day	Type	Chapter and topic	Pages	Qs	Min	Done
21	Ch10	Circles: Tangent to a Circle	5	16	110	☐
22	Ch10	Circles: Tangents from an External Point	4	16	110	☐
23	Revision	Revision 4: Triangles + Circles	4	22	120	☐
24	Ch8	Trigonometry: Trigonometric Ratios	4	16	110	☐
25	Ch8	Trigonometry: Ratios of Specific Angles	4	17	110	☐
26	Ch8	Trigonometry: Trigonometric Identities	4	17	110	☐
27	Ch9	Applications of Trig: Heights and Distances: Angle of Elevation	5	16	110	☐
28	Ch9	Applications of Trig: Angle of Depression and Two-Triangle Problems	5	16	110	☐
29	Revision	Revision 5: Trigonometry (Chapters 8 and 9)	4	20	120	☐
30	Ch11	Areas Related to Circles: Arc Length and Area of a Sector	4	16	110	☐
31	Ch11	Areas Related to Circles: Segments and Combination Figures	4	16	110	☐
32	Ch12	Surface Areas and Volumes: Surface Area of Combined Solids	5	16	110	☐
33	Ch12	Surface Areas and Volumes: Volume of Combined Solids	4	16	110	☐
34	Ch12	Surface Areas and Volumes: Mixed Board Problems	4	15	110	☐
35	Revision	Revision 6: Mensuration (Chapters 11 and 12)	4	18	120	☐
36	Ch13	Statistics: Mean of Grouped Data	5	15	110	☐
37	Ch13	Statistics: Mode of Grouped Data	3	15	110	☐
38	Ch13	Statistics: Median of Grouped Data	4	14	110	☐
39	Ch14	Probability: Theoretical Probability	4	16	110	☐
40	Ch14	Probability: Coins, Dice, Cards and Mixed Problems	4	16	110	☐

60-DAY ROADMAP | PART 3

DAYS 41 TO 60

Tick the box the day you finish.
Qs = practice questions that Day.

☐ Learn day ☐ Revision / formula ☐ Full mock ☐ Last 6 days

Pages = pages of this book that Day

Day	Type	Chapter and topic	Pages	Qs	Min	Done
41	Revision	Revision 7: Statistics + Probability	4	18	120	☐
42	Formula	Formula Day: All 14 chapters on one sheet	5	5	90	☐
43	Practice	Board Practice: Number Systems + Algebra I (Ch 1 to 3)	6	29	120	☐
44	Practice	Board Practice: Algebra II (Ch 4 and 5)	5	29	120	☐
45	Practice	Board Practice: Coordinate Geometry + Triangles + Circles	6	30	120	☐
46	Practice	Board Practice: Trigonometry (Ch 8 and 9)	5	30	120	☐
47	Practice	Board Practice: Mensuration + Statistics + Probability	6	30	120	☐
48	Practice	Board Practice: Case-Based and Assertion-Reason Marathon	7	23	120	☐
49	Mock	Mock Test 1: Full Paper, 80 marks, 3 hours	4	38	180	☐
50	Practice	Mock 1 Review: Solutions and Mistake Fixing	6	10	120	☐
51	Mock	Mock Test 2: Full Paper, 80 marks, 3 hours	4	38	180	☐
52	Practice	Mock 2 Review: Solutions and Mistake Fixing	5	10	120	☐
53	Mock	Mock Test 3: Full Paper, 80 marks, 3 hours	4	38	180	☐
54	Practice	Mock 3 Review: Solutions and Mistake Fixing	7	10	120	☐
55	Practice	Weak-Spot Sprint: Algebra Traps (Ch 1 to 5)	5	18	120	☐
56	Practice	Weak-Spot Sprint: Geometry, Coordinate and Trigonometry Traps	5	18	120	☐
57	Practice	Weak-Spot Sprint: Mensuration, Statistics and Probability Traps	6	18	120	☐
58	Mock	Mock Test 4: Final Full Paper, timed	4	38	180	☐
59	Practice	Mock 4 Review: Solutions and Last Corrections	6	10	120	☐
60	Exam Eve	Exam Eve: Formula Flash, Paper Strategy and Final Checklist	3	15	90	☐

CBSE CLASS 10 MATHEMATICS | 2026-27

SYLLABUS, MARKS, PAPER PATTERN

Theory 80 + Internal 20
3 hours | 38 questions

Unit	Chapters in this book (NCERT, rationalised)	Marks
Number Systems	Ch1 Real Numbers	**6**
Algebra	Ch2 Polynomials; Ch3 Pair of Linear Equations in Two Variables; Ch4 Quadratic Equations; Ch5 Arithmetic Progressions	**20**
Coordinate Geometry	Ch7 Coordinate Geometry (distance and section formula)	**6**
Geometry	Ch6 Triangles; Ch10 Circles	**15**
Trigonometry	Ch8 Introduction to Trigonometry; Ch9 Some Applications of Trigonometry	**12**
Mensuration	Ch11 Areas Related to Circles; Ch12 Surface Areas and Volumes	**10**
Statistics and Probability	Ch13 Statistics; Ch14 Probability	**11**
Theory paper (board exam)		**80**
Internal assessment (periodic tests, portfolio, lab activities)		20

BOARD PAPER PATTERN: FIVE SECTIONS

A | Objective — 20
Q1-18 MCQ, 1 each
Q19-20 Assertion-Reason, 1 each

B | VSA — 10
Q21-25 2 marks each
Internal choice in 2

C | SA — 18
Q26-31 3 marks each
Internal choice in 2

D | LA — 20
Q32-35 5 marks each
Internal choice in 2

E | Case-based — 12
Q36-38 4 marks each
1 + 1 + 2 sub-parts

STANDARD AND BASIC

Both papers have the same syllabus and pattern; **Basic** has easier questions. This book teaches the full syllabus; questions marked **Standard** are a level harder. Deleted topics (Euclid's division algorithm, decimal expansions, cross-multiplication, completing the square, area of a triangle by coordinates, frustum, conversion of solids and others) are not in this book. The pattern follows the latest CBSE sample paper; when the 2026-27 sample paper is out, check it once for any change.

FILL THESE IN AS YOU GO

MY TRACKERS

Mock scores
Chapter confidence

MY MOCK MARKS TRACKER

Paper	A /20	B /10	C /18	D /20	E /12	Total /80
Mock 1 (Day 49)						
Mock 2 (Day 51)						
Mock 3 (Day 53)						
Mock 4 (Day 58)						

CHAPTER CONFIDENCE TRACKER

Ch	Chapter	Learnt	Revised	Strong	Weak topics to redo
1	Real Numbers	☐	☐	☐	
2	Polynomials	☐	☐	☐	
3	Pair of Linear Equations in Two Variables	☐	☐	☐	
4	Quadratic Equations	☐	☐	☐	
5	Arithmetic Progressions	☐	☐	☐	
6	Triangles	☐	☐	☐	
7	Coordinate Geometry	☐	☐	☐	
8	Introduction to Trigonometry	☐	☐	☐	
9	Some Applications of Trigonometry	☐	☐	☐	
10	Circles	☐	☐	☐	
11	Areas Related to Circles	☐	☐	☐	
12	Surface Areas and Volumes	☐	☐	☐	
13	Statistics	☐	☐	☐	
14	Probability	☐	☐	☐	

DAY 01

REAL NUMBERS | CHAPTER 1 | PART 1 OF 2

Prime Factorisation and the Fundamental Theorem of Arithmetic

Real Numbers

Aaj ka target

- [ ] Factor tree aur division ladder dono haath se banana seekho
- [ ] FTA ka statement word-to-word yaad karo
- [ ] HCF = chhoti power, LCM = badi power: rule pakka
- [ ] 15 sawal khud se, phir answers milao (lagbhag 90 min)

1.1 Primes, Composites and the Fundamental Theorem of Arithmetic

A **prime number** has exactly two factors, 1 and itself (2, 3, 5, 7, 11, ...). A **composite number** has more than two factors (4, 6, 8, 9, ...). The number **1 is neither prime nor composite**, and **2 is the only even prime**.

KEY TERM: FUNDAMENTAL THEOREM OF ARITHMETIC (FTA)

Every composite number can be expressed (factorised) as a product of primes, and this factorisation is **unique**, apart from the order in which the prime factors occur.

- So every natural number greater than 1 is either a prime or a **unique** product of primes.
- Writing the primes in ascending order with powers is the standard form: $1260 = 2^2 \times 3^2 \times 5 \times 7$.
- Uniqueness is the power of the theorem: no other set of primes multiplies to 1260.

MEME SAMJHO

Astra meme, maths inside



REALISING PRIME FACTORISATION IS UNIQUE

Factor tree kaise bhi banao, end mein wahi primes aayenge? ACHHAA!!

Asli baat: By the FTA, $3825 = 3^2 \times 5^2 \times 17$ whichever split you start with (3 × 1275, 5 × 765 or 25 × 153). Only the order of the primes can change.

1.2 Two Ways to Prime-Factorise

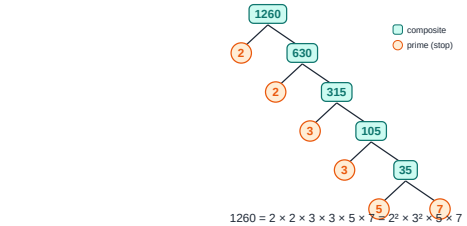


Fig. 1.1 Factor tree of 1260: keep splitting the composite branch until every leaf is a prime.

1. **Factor tree:** write the number, split it into any two factors, and keep splitting each composite branch. Stop when every leaf is a prime.
2. **Division ladder:** divide repeatedly by the smallest prime that divides the number (2, then 3, then 5, ...) until the quotient is 1.
3. Collect all the primes and write them with powers in ascending order.

SOLVED EXAMPLE 1

Express 5040 as a product of its prime factors.

SOLUTION

1. $5040 \div 2 = 2520$, $2520 \div 2 = 1260$, $1260 \div 2 = 630$, $630 \div 2 = 315$
2. $315 \div 3 = 105$, $105 \div 3 = 35$, $35 \div 5 = 7$, $7 \div 7 = 1$
3. So $5040 = 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 5 \times 7$

Answer: $5040 = 2^4 \times 3^2 \times 5 \times 7$



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Board copy mein sirf $2^4 \times 3^2 \times 5 \times 7$ likh ke mat chhodo. Division ladder ya factor tree bhi dikhao, warna working ke marks kat sakte hain. Aur check karo ki last leaf sach mein prime hai: 51 = 3 × 17 hai, prime nahi!

1.3 HCF and LCM by Prime Factorisation

FORMULA BOX: HCF AND LCM

HCF = product of the **smallest** power of each **common** prime factor

LCM = product of the **greatest** power of **every** prime factor that appears

For any two positive integers a and b:

$$\mathrm{HCF}(a, b) \times \mathrm{LCM}(a, b) = a \times b$$

This product rule is **not** true for three numbers.

SOLVED EXAMPLE 2

Find the HCF and LCM of 96 and 120 by prime factorisation and verify that HCF x LCM = product of the numbers.

SOLVED EXAMPLE 2

SOLUTION

1. $96 = 2^5 \times 3$ and $120 = 2^3 \times 3 \times 5$
2. Common primes 2 and 3; smallest powers 2^3 and 3^1, so HCF $= 2^3 \times 3 = 24$
3. All primes 2, 3, 5 with greatest powers: LCM $= 2^5 \times 3 \times 5 = 480$
4. Check: $24 \times 480 = 11520$ and $96 \times 120 = 11520$. Verified.

Answer: HCF = 24, LCM = 480

SOLVED EXAMPLE 3

Find the HCF and LCM of 12, 18 and 30.

SOLUTION

1. $12 = 2^2 \times 3$, $18 = 2 \times 3^2$, $30 = 2 \times 3 \times 5$
2. HCF: common primes 2 and 3 with smallest powers, $2 \times 3 = 6$
3. LCM: $2^2 \times 3^2 \times 5 = 180$
4. Note: $6 \times 180 = 1080$ but $12 \times 18 \times 30 = 6480$, so the product rule fails for three numbers.

Answer: HCF = 6, LCM = 180



SHIVAM BHAIYA SAYS

Yaad rakhne ka shortcut: **HCF = Hamesha Chhoti power** (sirf common primes ki), **LCM = Lambi yaani badi power** (saare primes ki). Aur HCF hamesha LCM ko divide karta hai, ye answer check karne ka sabse fast tarika hai.

1.4 Using the FTA: Last Digits and Composite Expressions

A number ends in the digit **0** only if it is divisible by 10, i.e. its prime factorisation contains **both 2 and 5**. Since $6^n = 2^n \times 3^n$ has no factor 5, 6^n **can never end with 0** for any natural number n. The same argument works for 4^n, 8^n, 12^n, 9^n and so on.

SOLVED EXAMPLE 4

Show that $3 \times 7 \times 13 + 13$ is a composite number.

SOLUTION

1. Take 13 common: $3 \times 7 \times 13 + 13 = 13\,(3 \times 7 + 1) = 13 \times 22$
2. So the number has factors other than 1 and itself (13 and 22).
3. Indeed $286 = 2 \times 11 \times 13$.

Answer: It is composite.



EXAM ALERT!

Real Numbers ka board mein **6 marks** ke aas paas aata hai. "k^n zero se end ho sakta hai kya?" aur "composite dikhao" wale 2-markers bar-bar repeat hote hain. Free marks, chhodna mat!



SHIVAM BHAIYA SAYS: BOARD TIP

"Can 15^n end with 0?" jaise sawal mein trap hai: $15 = 3 \times 5$, isme 5 hai par **2 nahi**, isliye 0 nahi aayega (5 se end hoga). Dono primes, 2 aur 5, chahiye.

Day 1 Practice

16 board-pattern questions | solve first, then check

MCQ

1 How many factors does a prime number have? (1)

(a) Exactly 1 (b) Exactly 3
(c) Exactly 2 (d) It depends on the prime

MCQ

2 The LCM of the smallest composite number and the smallest two-digit prime number is (1)

(a) 22 (b) 11 (c) 88 (d) 44

MCQ

3 If $a = p^2q^3$ and $b = p^4q$, where p and q are primes, then HCF(a, b) is (1)

(a) p^2q (b) p^4q^3 (c) p^2q^3 (d) pq

MCQ

4 The number $(5 \times 7 \times 11 \times 13) + 13$ is (1)

(a) a prime number
(b) a composite number
(c) neither prime nor composite
(d) an even prime

MCQ

5 The LCM of two numbers is 1800. Which of these can **not** be their HCF? (1)

(a) 300 (b) 360 (c) 400 (d) 450

MCQ

6 For a natural number n, the digit with which 8^n can never end is (1)

(a) 2 (b) 4 (c) 0 (d) 6

ASSERTION-REASON

7 **Assertion (A):** 12^n can never end with the digit 0 for any natural number n.
Reason (R): A number ends with 0 only if its prime factorisation contains both 2 and 5. (1)
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

8 **Assertion (A):** The HCF of any two distinct prime numbers is 1.
Reason (R): The LCM of two distinct prime numbers is equal to their sum. (1)
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

CASE-BASED

9 **Workshop rooms. A school is running a science workshop with 48 participants in Physics, 72 in Chemistry and 120 in Biology. Every room must hold the same number of participants, and each room may hold students of only one subject.** (4)
(i) What is the maximum number of participants that can sit in one room? (1)
(ii) Write the prime factorisation of 120. (1)
(iii) How many rooms are needed in all if each room holds the maximum number of participants? (2)
(iii) OR: Find the least number of badges that can be shared equally among the Physics, the Chemistry or the Biology group, with none left over. (2)

VERY SHORT ANSWER

10 **Express 4620 as a product of its prime factors.** (2)

SHORT ANSWER

11 **Check whether 15^n can end with the digit 0 for any natural number n.** (2)

SHORT ANSWER

12 **Find the HCF and LCM of 18, 24 and 40 by the prime factorisation method.** (2)

SHORT ANSWER

13 **Given that HCF(84, 270) = 6, find LCM(84, 270). Also verify your answer by prime factorisation.** (3)

SHORT ANSWER

14 **Explain why $3 \times 5 \times 11 + 11$ and $6 \times 5 \times 4 \times 3 \times 2 \times 1 + 3$ are composite numbers.** (3)

LONG ANSWER

15 **Three school bells ring at intervals of 15, 20 and 36 minutes. They ring together at 8:00 a.m. (a) At what time will they next ring together? (b) How many more times will they ring together up to and including 5:00 p.m.?** (4)

LONG ANSWER

16 **(a) Find the HCF and LCM of 378 and 132 by prime factorisation and verify that HCF x LCM = product of the two numbers. (b) Find the greatest number that divides 245 and 1029 leaving remainder 5 in each case.** (5)

Day 1 Answers with worked solutions

1 **(c)**
A prime has only the factors 1 and itself, so exactly 2 factors.

2 **(d)**
Smallest composite = 4, smallest two-digit prime = 11.
LCM(4, 11) = 44 since they share no prime factor.

3 **(a)**
HCF takes the smallest power of each common prime: $p^{\min(2,4)}q^{\min(3,1)} = p^2q$.

4 **(b)**
$= 13\,(5 \times 7 \times 11 + 1) = 13 \times 386$, so it has factors other than 1 and itself: composite.

5 **(c)**
HCF always divides the LCM.
$1800 \div 300 = 6$, $1800 \div 360 = 5$, $1800 \div 450 = 4$, but $1800 \div 400 = 4.5$, so 400 cannot be the HCF.

6 **(c)**
$8^n = 2^{3n}$ has no prime factor 5, so it is never divisible by 10 and can never end with 0. (Its last digits cycle 8, 4, 2, 6.)

7 **(a)**
$12^n = 2^{2n} \times 3^n$ contains no 5, so A is true, and R is exactly the reason. Option (a).

8 **(c)**
Two distinct primes have no common factor except 1, so A is true.
Their LCM is their **product** (e.g. LCM(3, 5) = 15, not 8), so R is false. Option (c).

9 **Ans:** 24
$48 = 2^4 \times 3$, $72 = 2^3 \times 3^2$, $120 = 2^3 \times 3 \times 5$
HCF $= 2^3 \times 3 = 24$
Ans: $2^3 \times 3 \times 5$
$120 = 2 \times 2 \times 2 \times 3 \times 5$
Ans: 10 rooms
Rooms = $\frac{48}{24} + \frac{72}{24} + \frac{120}{24} = 2 + 3 + 5 = 10$
OR: Ans: 720
We need LCM(48, 72, 120) $= 2^4 \times 3^2 \times 5 = 720$

10 **Ans:** $4620 = 2^2 \times 3 \times 5 \times 7 \times 11$
$4620 \div 2 = 2310$, $2310 \div 2 = 1155$, $1155 \div 3 = 385$, $385 \div 5 = 77$, $77 \div 7 = 11$, $11 \div 11 = 1$
So $4620 = 2^2 \times 3 \times 5 \times 7 \times 11$

11 **Ans:** No
$15^n = 3^n \times 5^n$.
To end in 0 a number must have both 2 and 5 as prime factors. 2 is missing.
By the uniqueness in the FTA, no other factorisation exists, so 15^n never ends with 0 (it always ends with 5).

12 **Ans:** HCF = 2, LCM = 360
$18 = 2 \times 3^2$, $24 = 2^3 \times 3$, $40 = 2^3 \times 5$
Only 2 is common, smallest power 2^1: HCF = 2
LCM $= 2^3 \times 3^2 \times 5 = 360$

13 **Ans:** 3780
$$\text{LCM} = \frac{84 \times 270}{6} = \frac{22680}{6} = 3780$$
Check: $84 = 2^2 \times 3 \times 7$, $270 = 2 \times 3^3 \times 5$
LCM $= 2^2 \times 3^3 \times 5 \times 7 = 3780$. Verified.

14 **Ans:** Both have factors other than 1 and themselves.
$3 \times 5 \times 11 + 11 = 11(15 + 1) = 11 \times 16 = 176$: composite.
$6 \times 5 \times 4 \times 3 \times 2 \times 1 + 3 = 3(6 \times 5 \times 4 \times 2 \times 1 + 1) = 3 \times 241 = 723$: composite.
Each number has a factor (11 and 3 respectively) other than 1 and itself.

15 **Ans:** (a) 11:00 a.m. (b) 3 times
$15 = 3 \times 5$, $20 = 2^2 \times 5$, $36 = 2^2 \times 3^2$
LCM $= 2^2 \times 3^2 \times 5 = 180$ minutes = 3 hours
(a) Next together at 8:00 + 3 h = 11:00 a.m.
(b) Together at 11:00 a.m., 2:00 p.m. and 5:00 p.m.: 3 more times.

16 **Ans:** (a) HCF = 6, LCM = 8316 (b) 16
(a) $378 = 2 \times 3^3 \times 7$, $132 = 2^2 \times 3 \times 11$
HCF $= 2 \times 3 = 6$; LCM $= 2^2 \times 3^3 \times 7 \times 11 = 8316$
$6 \times 8316 = 49896$ and $378 \times 132 = 49896$. Verified.
(b) The number must divide $245 - 5 = 240$ and $1029 - 5 = 1024$ exactly.
$240 = 2^4 \times 3 \times 5$, $1024 = 2^{10}$, so HCF $= 2^4 = 16$
Required number = 16 (and 16 > 5, so remainder 5 is possible).

TONIGHT'S 2-MINUTE RECALL
Answer from memory, then check the notes

1. Is 1 prime? : Neither prime nor composite
2. Only even prime : 2
3. FTA in one line : every composite = unique product of primes (order aside)
4. HCF rule : smallest power of common primes
5. LCM rule : greatest power of all primes
6. HCF x LCM = product holds for : two numbers only
7. Can 6^n end in 0? : No, no factor 5
8. 1260 in primes : $2^2 \times 3^2 \times 5 \times 7$

DAY 1 COMPLETE? ☐ Notes read
☐ 16 questions solved ☐ Answers checked
☐ Mistakes noted Next: Day 2, Real Numbers

MY MISTAKES AND NOTES
Write every mistake of this Day with the correct step

DAY 02

REAL NUMBERS | CHAPTER 1 | PART 2 OF 2

HCF, LCM and Proving Irrationality

Real Numbers

Aaj ka target

- [ ] Word problem padhte hi pehchano: HCF wala ya LCM wala
- [ ] √2, √3, √5 ka proof bina dekhe likho
- [ ] Remainder wale sawal: pehle remainder ghatao ya jodo
- [ ] 3 + 2√5 type proof + 15 sawal (lagbhag 100 min)

2.1 HCF or LCM? Reading a Word Problem

The question says...	Use	Why
greatest / largest / maximum size, length, number per group	HCF	the biggest thing that fits into all of them exactly
least / smallest / minimum, together again, at the same time	LCM	the first number all of them reach
divides ... leaving remainders r_1, r_2	HCF of (number − remainder)	subtract first, then the division is exact
when divided by ... leaves remainder r (smallest number)	LCM $+r$	add the remainder after the LCM

Keyword guide for Real Numbers word problems

SOLVED EXAMPLE 1 (HCF)

A hall floor measures 10.8 m by 6.3 m. Find the side of the largest square tile that can pave it exactly, and the number of tiles needed.

SOLUTION

1. Convert to cm: 1080 cm and 630 cm.
2. $1080 = 2^3 \times 3^3 \times 5$, $630 = 2 \times 3^2 \times 5 \times 7$
3. HCF $= 2 \times 3^2 \times 5 = 90$ cm
4. Tiles $= \frac{1080}{90} \times \frac{630}{90} = 12 \times 7 = 84$

Answer: Side 90 cm, 84 tiles

SOLVED EXAMPLE 2 (LCM)

On a circular track, Riya takes 16 minutes and Kabir takes 24 minutes for one round. They start together from the same point in the same direction. After how many minutes will they meet again at the starting point?

SOLUTION

1. $16 = 2^4$, $24 = 2^3 \times 3$
2. LCM $= 2^4 \times 3 = 48$

Answer: After 48 minutes

SOLVED EXAMPLE 3 (REMAINDER)

Find the smallest number which, when divided by 12, 16 and 24, leaves remainder 5 in each case.

SOLUTION

1. LCM(12, 16, 24): $12 = 2^2 \times 3$, $16 = 2^4$, $24 = 2^3 \times 3$, so LCM $= 2^4 \times 3 = 48$
2. 48 is divisible by all three; add the remainder: $48 + 5 = 53$
3. Check: $53 = 4 \times 12 + 5 = 3 \times 16 + 5 = 2 \times 24 + 5$

Answer: 53



SHIVAM BHAIYA SAYS

Remainder ka funda: **"greatest number that divides ... leaving remainder"** = pehle remainder **ghatao**, phir HCF. **"smallest number which when divided by ... leaves remainder"** = pehle LCM, phir remainder **jodo**. Ulta kiya to answer gaya!

MEME SAMJHO

Astra meme, maths inside



STUDENT AFTER USING HCF X LCM = PRODUCT ON THREE NUMBERS

Teen numbers pe HCF x LCM = product laga diya, answer match hi nahi hua... ab baith ke soch raha hoon.

Asli baat: HCF(12, 18, 30) = 6 and LCM = 180, but $6 \times 180 = 1080$ while $12 \times 18 \times 30 = 6480$. The product rule works only for two numbers.

2.2 Irrational Numbers and the Key Theorem

KEY TERM: IRRATIONAL NUMBER

A number s is **irrational** if it cannot be written as $\frac{p}{q}$ where p, q are integers and $q \neq 0$. Examples: $\sqrt{2}, \sqrt{3}, \sqrt{5}, \pi, 2 + \sqrt{3}$.

FORMULA BOX: FACTS USED IN PROOFS

Theorem: if a prime p divides a^2

FORMULA BOX: FACTS USED IN PROOFS

(a a positive integer), then p divides a.

rational $\pm$ irrational = irrational

(non-zero rational) $\times$ irrational = irrational; irrational $\div$ (non-zero rational) = irrational

irrational + irrational can be rational: $\sqrt{2} + (-\sqrt{2}) = 0$

SOLVED EXAMPLE 4 (PROOF BY CONTRADICTION)

Prove that $\sqrt{3}$ is irrational.

SOLUTION

1. Assume, to the contrary, that $\sqrt{3}$ is rational. Then $\sqrt{3} = \frac{a}{b}$, where a, b are integers, $b \neq 0$, and a, b are **COPRIME**.
2. Squaring, $3b^2 = a^2$, so 3 divides a^2, hence 3 divides a (theorem above).
3. Write $a = 3c$. Then $3b^2 = 9c^2$, i.e. $b^2 = 3c^2$, so 3 divides b^2 and hence 3 divides b.
4. So 3 divides both a and b, which contradicts that a and b are coprime.
5. The contradiction arose from assuming $\sqrt{3}$ rational.

Answer: Hence $\sqrt{3}$ is irrational. (The same steps with 2 or 5 prove $\sqrt{2}$ and $\sqrt{5}$ irrational.)

SOLVED EXAMPLE 5

Given that $\sqrt{5}$ is irrational, prove that $3 + 2\sqrt{5}$ is irrational.

SOLUTION

1. Assume $3 + 2\sqrt{5}$ is rational: $3 + 2\sqrt{5} = \frac{a}{b}$, a, b integers, $b \neq 0$.
2. Then $2\sqrt{5} = \frac{a}{b} - 3$, so $\sqrt{5} = \frac{a - 3b}{2b}$.
3. The right side is a ratio of integers with $2b \neq 0$, so it is rational, which would make $\sqrt{5}$ rational.
4. This contradicts the fact that $\sqrt{5}$ is irrational.

Answer: Hence $3 + 2\sqrt{5}$ is irrational.



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Proof mein ye 3 line likhna **compulsory** hai: (1) "assume to the contrary", (2) "a and b are coprime, $b \neq 0$", (3) "this contradicts ... hence irrational". Coprime likhna bhool gaye to contradiction kis baat ka? Seedha marks katenge.



EXAM ALERT!

"Prove that $\sqrt{5}$ is irrational" ya "$\sqrt{3}$" lagbhag har saal **3 marks** mein aata hai, aksar "hence show $2 + \sqrt{3}$ is irrational" ke saath. Isko **ratta nahi, samajh ke** 3 baar likh ke practice karo.



SHIVAM BHAIYA SAYS: BOARD TIP

$5 - \sqrt{3}$ ya $3 + 2\sqrt{5}$ jaise sawal mein agar likha ho "given that $\sqrt{3}$ is irrational", to $\sqrt{3}$ ka poora proof dobara mat likho. Sirf rearrange karke dikhao ki $\sqrt{3}$ rational ban jaayega: time bachao.

Day 2 Practice

16 board-pattern questions | solve first, then check

MCQ

1 The product of a non-zero rational number and an irrational number is (1)

(a) always rational (b) rational or irrational
(c) always irrational (d) always an integer

MCQ

2 The greatest number that divides 70 and 125 leaving remainders 5 and 8 respectively is (1)

(a) 65 (b) 13 (c) 5 (d) 117

MCQ

3 If HCF(96, 156) is written as $7m - 2$, then the value of m is (1)

(a) 1 (b) 3 (c) 4 (d) 2

MCQ

4 Which of the following is irrational? (1)

(a) $\sqrt{49}$ (b) $(2 - \sqrt{3})^2$
(c) $\frac{\sqrt{12}}{\sqrt{3}}$ (d) $(\sqrt{5} + 1)(\sqrt{5} - 1)$

MCQ

5 Two runners complete one round of a track in 18 minutes and p minutes. They meet again at the start first after 72 minutes, and HCF$(18, p) = 6$. The value of p is (1)

(a) 12 (b) 24 (c) 36 (d) 48

MCQ

6 The LCM of two prime numbers p and q $(p > q)$ is 323. The value of $3p - q$ is (1)

(a) 34 **(b)** 57 **(c)** 40 **(d)** 23

ASSERTION-REASON

7 **Assertion (A):** $\sqrt{2} + (-\sqrt{2})$ **is a rational number.** **Reason (R):** **The sum of two irrational numbers is always irrational.** (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

8 **Assertion (A):** $5 - \sqrt{3}$ **is an irrational number.** **Reason (R):** **The difference of a rational number and an irrational number is irrational.** (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

CASE-BASED

9 **Sports Day parade. On Sports Day, a contingent of 612 students has to march behind the school band of 48 members. Both groups must march in the same number of columns, and the organisers want as many columns as possible.** (4)

(i) What is the maximum number of columns in which they can march? (1)

(ii) Write 612 as a product of primes. (1)

(iii) How many students march in each column of the contingent, and how many band members in each column? (2)

(iii) OR: Find LCM(612, 48). (2)

SHORT ANSWER

10 **Find the smallest number which leaves remainder 4 when divided by 15, 20 and 25.** (2)

SHORT ANSWER

11 **Show that $\frac{3}{\sqrt{7}}$ is irrational, given that $\sqrt{7}$ is irrational.** (2)

SHORT ANSWER

12 **Prove that $\sqrt{5}$ is irrational.** (3)

SHORT ANSWER

13 **Given that $\sqrt{3}$ is irrational, prove that $7 - 2\sqrt{3}$ is irrational.** (3)

SHORT ANSWER

14 **Three buses leave a depot together at 7:00 a.m. They return to the depot every 16, 24 and 40 minutes respectively. When will all three next be at the depot together?** (3)

LONG ANSWER

15 **Prove that $\sqrt{2}$ is irrational. Hence show that $4 + \sqrt{2}$ is irrational.** (4)

LONG ANSWER

16 **A fruit seller has 336 apples, 240 oranges and 192 mangoes. He packs them in boxes so that every box holds the same number of fruits and only one kind of fruit. (a) Find the greatest number of fruits per box and the number of boxes used. (b) Find the least number of fruits that can be packed exactly into bags of 12, of 15 or of 18 fruits.** (5)

Day 2 Answers with worked solutions

1 **(c)**
If $r \neq 0$ is rational and rs were rational, then $s = \frac{rs}{r}$ would be rational. So rs is always irrational.

2 **(b)**
The number must divide $70 - 5 = 65$ and $125 - 8 = 117$ exactly.
$65 = 5 \times 13$, $117 = 3^2 \times 13$, HCF = 13 (and 13 > 8).

3 **(d)**
$96 = 2^5 \times 3$, $156 = 2^2 \times 3 \times 13$, HCF $= 2^2 \times 3 = 12$
$7m - 2 = 12 \Rightarrow m = 2$

4 **(b)**
$\sqrt{49} = 7$; $\frac{\sqrt{12}}{\sqrt{3}} = \sqrt{4} = 2$; $(\sqrt{5}+1)(\sqrt{5}-1) = 5 - 1 = 4$: all rational.
$(2 - \sqrt{3})^2 = 4 - 4\sqrt{3} + 3 = 7 - 4\sqrt{3}$: irrational.

5 **(b)**
HCF x LCM = product: $6 \times 72 = 18 \times p$, so $p = 24$.
Check: HCF(18, 24) = 6, LCM(18, 24) = 72.

6 **(c)**
The LCM of two distinct primes is their product: $pq = 323 = 17 \times 19$.
So $p = 19$, $q = 17$ and $3p - q = 57 - 17 = 40$.

7 **(c)**
$\sqrt{2} + (-\sqrt{2}) = 0$, which is rational: A true.
The example itself shows the sum of two irrationals can be rational: R false. Option (c).

8 **(a)**
5 is rational and $\sqrt{3}$ is irrational, so by R, $5 - \sqrt{3}$ is irrational. Both true, R explains A: (a).

9 **Ans:** 12

$612 = 2^2 \times 3^2 \times 17$, $48 = 2^4 \times 3$, HCF $= 2^2 \times 3 = 12$

Ans: $2^2 \times 3^2 \times 17$

$612 = 2 \times 2 \times 3 \times 3 \times 17$

Ans: 51 students, 4 band members

$612 \div 12 = 51$ and $48 \div 12 = 4$

OR: Ans: 2448

LCM $= 2^4 \times 3^2 \times 17 = 2448$

Check: $12 \times 2448 = 29376 = 612 \times 48$

10 **Ans:** 304

$15 = 3 \times 5$, $20 = 2^2 \times 5$, $25 = 5^2$, LCM $= 2^2 \times 3 \times 5^2 = 300$

Required number $= 300 + 4 = 304$

11 **Ans:** Irrational

Assume $\frac{3}{\sqrt{7}} = \frac{a}{b}$ with integers a, b, $a \neq 0$, $b \neq 0$.

Then $\sqrt{7} = \frac{3b}{a}$, a rational number.

This contradicts that $\sqrt{7}$ is irrational. Hence $\frac{3}{\sqrt{7}}$ is irrational.

12 **Ans:** Proof below

Assume $\sqrt{5} = \frac{a}{b}$ with a, b coprime integers, $b \neq 0$.

Then $5b^2 = a^2$, so 5 divides a^2, hence 5 divides a. Let $a = 5c$.

$5b^2 = 25c^2 \Rightarrow b^2 = 5c^2$, so 5 divides b^2 and hence b.

5 is a common factor of a and b, contradicting that they are coprime. So $\sqrt{5}$ is irrational.

13 **Ans:** Proof below

Assume $7 - 2\sqrt{3} = \frac{a}{b}$, a, b integers, $b \neq 0$.

$2\sqrt{3} = 7 - \frac{a}{b} = \frac{7b-a}{b}$, so $\sqrt{3} = \frac{7b-a}{2b}$.

The right side is rational, so $\sqrt{3}$ would be rational: a contradiction.

Hence $7 - 2\sqrt{3}$ is irrational.

14 **Ans:** 11:00 a.m.

$16 = 2^4$, $24 = 2^3 \times 3$, $40 = 2^3 \times 5$

LCM $= 2^4 \times 3 \times 5 = 240$ minutes = 4 hours

Together again at 7:00 a.m. + 4 h = 11:00 a.m.

15 **Ans:** Proof below

Assume $\sqrt{2} = \frac{a}{b}$, a, b coprime integers, $b \neq 0$.

$2b^2 = a^2$, so 2 divides a^2 and hence a. Let $a = 2c$: $2b^2 = 4c^2 \Rightarrow b^2 = 2c^2$.

So 2 divides b^2 and hence b: 2 divides both a and b, contradicting coprimeness. So $\sqrt{2}$ is irrational.

Now assume $4 + \sqrt{2} = \frac{p}{q}$ (integers, $q \neq 0$).

Then $\sqrt{2} = \frac{p-4q}{q}$, rational: a contradiction.

Hence $4 + \sqrt{2}$ is irrational.

16 **Ans:** (a) 48 per box, 16 boxes (b) 180

(a) $336 = 2^4 \times 3 \times 7$, $240 = 2^4 \times 3 \times 5$, $192 = 2^6 \times 3$

HCF $= 2^4 \times 3 = 48$ fruits per box

Boxes $= 7 + 5 + 4 = 16$

(b) $12 = 2^2 \times 3$, $15 = 3 \times 5$, $18 = 2 \times 3^2$

LCM $= 2^2 \times 3^2 \times 5 = 180$

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. 'Greatest / maximum' in a word problem : HCF
2. 'Least / together again' : LCM
3. Divides leaving remainders : subtract remainders, then HCF
4. Smallest number leaving remainder r : LCM + r
5. Irrational number : cannot be written as $\frac{p}{q}$, $q \neq 0$
6. Key theorem : prime $p \mid a^2 \Rightarrow p \mid a$
7. Proof method for $\sqrt{3}$: contradiction with coprime a, b
8. Is $\sqrt{2} + (-\sqrt{2})$ irrational? : No, it is 0

DAY 2 COMPLETE? ☐ Notes read ☐ 16 questions solved ☐ Answers checked ☐ Mistakes noted

Next: Day 3, Polynomials

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 03

POLYNOMIALS | CHAPTER 2 | PART 1 OF 2

Zeroes of a Polynomial and their Geometrical Meaning

Polynomials

Aaj ka target

- ☐ Degree dekh ke polynomial ka type turant batao
- ☐ Zero ka matlab: jahan $p(x) = 0$, graph pe jahan curve x-axis ko mile
- ☐ Graph dekh ke zeroes ginna: cut + touch points
- ☐ 15 sawal khud se, phir answers milao (lagbhag 90 min)

3.1 Polynomials and their Degree

A **polynomial** in x is an expression like $3x^2 - 5x + 2$ in which every power of x is a whole number. The highest power of x is its **degree**. A non-zero constant such as 7 has degree 0; the degree of the **zero polynomial** is not defined.

Type	Degree	General form	Example
Linear	1	$ax + b,\ a \neq 0$	$2x - 3$
Quadratic	2	$ax^2 + bx + c,\ a \neq 0$	$x^2 - 2x - 3$
Cubic	3	$ax^3 + bx^2 + cx + d,\ a \neq 0$	$x^3 - 4x$

Types of polynomials by degree

- $\sqrt{x} + 1$ and $\frac{1}{x} + 2$ are **not** polynomials: the powers $\frac{1}{2}$ and -1 are not whole numbers.
- $\sqrt{3}\,x^2 - x$ **is** a polynomial: only the powers of x must be whole numbers, the coefficients can be irrational.

3.2 Zero of a Polynomial

KEY TERM: ZERO OF A POLYNOMIAL

A real number k is a **zero** of $p(x)$ if $p(k) = 0$. The zeroes of $p(x)$ are the solutions (roots) of the equation $p(x) = 0$.

- A linear polynomial $ax + b$ has exactly one zero, $x = -\frac{b}{a}$.
- A polynomial of degree n has **at most** n zeroes. So a quadratic has at most 2, a cubic at most 3.

SOLVED EXAMPLE 1

Find the zeroes of $p(x) = 6x^2 - 7x - 3$.

SOLUTION

1. Split the middle term: product $6 \times (-3) = -18$, sum -7, so use -9 and $+2$.
2. $6x^2 - 9x + 2x - 3 = 3x(2x - 3) + 1(2x - 3) = (2x - 3)(3x + 1)$.
3. $p(x) = 0 \Rightarrow 2x - 3 = 0$ or $3x + 1 = 0$.

Answer: Zeroes are $\frac{3}{2}$ and $-\frac{1}{3}$.

3.3 Geometrical Meaning of the Zeroes

The zeroes of $p(x)$ are exactly the **x-coordinates of the points where the graph of $y = p(x)$ meets the x-axis** (cuts it or touches it). So we can read the **number of zeroes** straight from a graph.

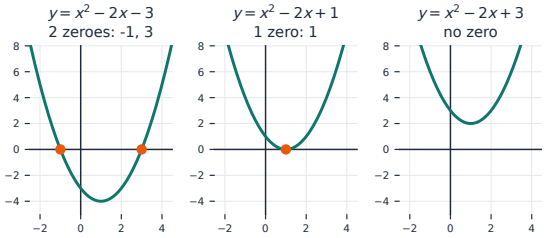


Fig. 3.1 The graph of a quadratic meets the x-axis at two points, one point, or no point.

- The graph of $y = ax^2 + bx + c$ is a **parabola**: it opens **upward** if $a > 0$ and **downward** if $a < 0$.
- Two meeting points: 2 zeroes. Touches at one point: 1 zero (two equal zeroes). Does not meet: no real zero.
- The graph of a linear polynomial is a straight line that cuts the x-axis exactly once.

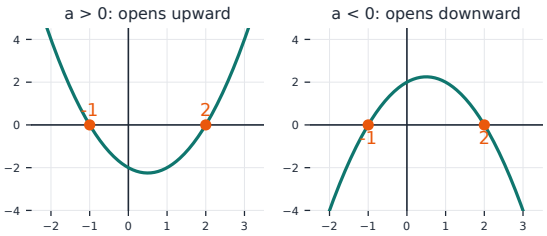


Fig. 3.2 Same zeroes -1 and 2; the sign of a only flips the opening.



SHIVAM BHAIYA SAYS

Graph wala sawal aaye to equation dhoondhne mat baitho. Bas gino curve x-axis ko **kitni jagah cut ya touch** karta hai, utne hi zeroes. Y-axis wale point ka zeroes se koi lena-dena nahi.

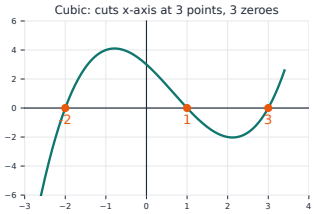


Fig. 3.3 A cubic can meet the x-axis at up to three points.

SOLVED EXAMPLE 2

The graph of $y = p(x)$ cuts the x-axis at $(-2, 0)$ and $(4, 0)$ and passes through $(0, -8)$. Find $p(x)$.

SOLUTION

1. Zeroes are -2 and 4, so $p(x) = k(x + 2)(x - 4)$.
2. At $x = 0$: $p(0) = k(2)(-4) = -8k$. Given $p(0) = -8$, so $k = 1$.
3. $p(x) = (x + 2)(x - 4) = x^2 - 2x - 8$.

Answer: $p(x) = x^2 - 2x - 8$

MEME SAMJHO

Astra meme, maths inside



STUDENT WHO JUST LEARNT THE GEOMETRIC MEANING OF ZEROES

Graph x-axis ko jitni jagah mile, utne zeroes. Equation solve karne ki zaroorat hi nahi.

Asli baat: Asli baat: The number of zeroes of $p(x)$ equals the number of distinct points where the graph of $y = p(x)$ cuts or touches the x-axis.

SOLVED EXAMPLE 3

If $x = 2$ is a zero of $p(x) = 3x^2 - x + k$, find k and the other zero.

SOLUTION

1. $p(2) = 0 \Rightarrow 3(4) - 2 + k = 0 \Rightarrow k = -10$.
2. $p(x) = 3x^2 - x - 10 = 3x^2 - 6x + 5x - 10 = (x - 2)(3x + 5)$.
3. Other zero: $3x + 5 = 0$.

Answer: $k = -10$, other zero $-\frac{5}{3}$



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Sabse common galti: zero ko y-intercept samajh lena. $y = x^2 - 2x - 8$ y-axis ko $(0, -8)$ pe kaatta hai, par -8 zero **nahi** hai. Zeroes hamesha x-axis wale points ke x-coordinates hote hain.



SHIVAM BHAIYA SAYS: BOARD TIP

Zeroes nikaalne ke baad **ek line mein verify** kar do, jaise $p(3/2) = 6 \cdot \frac{9}{4} - \frac{21}{2} - 3 = 0$. Board checker ko yeh line bahut pasand aati hai aur galti bhi pakdi jaati hai.



EXAM ALERT!

Graph dekh ke 'number of zeroes' wala 1 mark MCQ har saal jaisa aata hai. Cubic graph (3 points) aur 'touch' wala case dhyan se dekho.

Day 3 Practice

16 board-pattern questions | solve first, then check

MCQ

1 The graph of $y = p(x)$ meets the x-axis at exactly three distinct points. Which of these could $p(x)$ be? (1)

(a) A linear polynomial (b) A cubic polynomial
(c) A quadratic polynomial (d) A non-zero constant polynomial

MCQ

2 The straight-line graph of a linear polynomial $p(x)$ passes through $(0, 6)$ and $(2, 0)$. The zero of $p(x)$ is (1)

(a) 6 (b) -2 (c) 0 (d) 2

MCQ

3 The zeroes of $x^2 - 49$ are (1)

(a) 7 only (b) 49 and -49
(c) 7 and -7 (d) 0 and 7

MCQ

4 The zeroes of $p(x) = 2x^2 - 50$ are (1)

(a) both positive
(b) both negative
(c) one positive and one negative
(d) both equal

MCQ

5 The zeroes of $x^2 + 45x + 324$ are (1)

(a) both negative
(b) both positive
(c) one positive and one negative
(d) not real

MCQ

6 If $x = -1$ is a zero of $x^2 - 5x + k$, then k equals (1)

(a) 6 (b) -6 (c) 4 (d) -4

ASSERTION-REASON

7 **Assertion (A):** The polynomial $x^2 + 5x + 6$ has two real zeroes.
Reason (R): A quadratic polynomial can have at most two zeroes. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

8 **Assertion (A):** If the graph of $y = p(x)$ touches the x-axis at one point and cuts it at another point, then $p(x)$ has exactly two distinct zeroes.
Reason (R): The number of distinct zeroes of $p(x)$ equals the number of distinct points where its graph cuts or touches the x-axis. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

CASE-BASED

9 Read the passage and answer the questions that follow. A drone climbs and dips over a field. Its height (in metres) above a reference line after t minutes is modelled by $p(t) = 6t^2 - 13t + 6$. (4)

(i) What shape is the graph of $p(t)$? (1)

(ii) How many zeroes can $p(t)$ have at most? (1)

(iii) Find the zeroes of $p(t)$. (2)

SHORT ANSWER

10 Find the zeroes of $4x^2 + 8x - 5$. (2)

SHORT ANSWER

11 Find the zeroes of $\sqrt{2}\,x^2 + 7x + 5\sqrt{2}$. (2)

SHORT ANSWER

12 Find the zeroes of $5x^2 - 3$. (2)

SHORT ANSWER

13 Given that $x = 2$ is a zero of $3x^2 - x + k$, find k and the other zero. (3)

SHORT ANSWER

14 A parabola $y = p(x)$ cuts the x-axis at $(-3, 0)$ and $(1, 0)$ and the y-axis at $(0, 6)$. Find $p(x)$ and say whether it opens upward or downward. (3)

LONG ANSWER

15 For the polynomial $(k - 2)x^2 + kx - 3$, one zero is 3. Find k and the other zero. (4)

LONG ANSWER

16 Draw the graph of $y = x^2 - x - 6$ for $-3 \le x \le 4$ and read off its zeroes. Check them by factorising. (5)

Day 3 Answers with worked solutions

1 (b)
A linear polynomial has at most 1 zero and a quadratic at most 2, so three meeting points need degree at least 3: a cubic.

2 (d)
The zero is the x-coordinate where the line meets the x-axis, that is at $(2, 0)$. (Here $p(x) = 6 - 3x$ and $p(2) = 0$.)

3 (c)
$x^2 - 49 = (x - 7)(x + 7) = 0 \Rightarrow x = 7$ or $x = -7$.

4 (c)
$2x^2 - 50 = 0 \Rightarrow x^2 = 25 \Rightarrow x = \pm 5$: one positive, one negative.

5 (a)
Product $= 324 > 0$ so both have the same sign; sum $= -45 < 0$ so both are negative.
Indeed $x^2 + 45x + 324 = (x + 9)(x + 36)$, zeroes $-9, -36$.

6 (b)
$(-1)^2 - 5(-1) + k = 0 \Rightarrow 1 + 5 + k = 0 \Rightarrow k = -6$.

7 (b)
$x^2 + 5x + 6 = (x + 2)(x + 3)$, zeroes $-2, -3$: A is true.
R is true, but R does not explain why this particular quadratic has two zeroes (some quadratics have none).

8 (a)
Both statements are true and R is exactly the reason for A (for example $p(x) = (x - 1)^2(x + 2)$ has distinct zeroes 1 and -2).

9 **Ans:** A parabola (opening upward)

$p(t)$ is quadratic with $a = 6 > 0$, so its graph is an upward parabola.

Ans: 2

A quadratic polynomial has at most 2 zeroes.

Ans: $\frac{3}{2}$ and $\frac{2}{3}$

Product $6 \times 6 = 36$, sum -13: use $-9, -4$.

$6t^2 - 9t - 4t + 6 = 3t(2t - 3) - 2(2t - 3) = (2t - 3)(3t - 2)$.

Zeroes: $t = \frac{3}{2}$ and $t = \frac{2}{3}$.

10 **Ans:** $\frac{1}{2}$ and $-\frac{5}{2}$

Product $4 \times (-5) = -20$, sum 8: use 10 and -2.

$4x^2 + 10x - 2x - 5 = 2x(2x + 5) - 1(2x + 5) = (2x + 5)(2x - 1)$.

Zeroes: $x = \frac{1}{2}$, $x = -\frac{5}{2}$.

11 **Ans:** $-\sqrt{2}$ and $-\frac{5}{\sqrt{2}} = -\frac{5\sqrt{2}}{2}$

Product of end terms $\sqrt{2} \times 5\sqrt{2} = 10$, sum 7: split $7x = 2x + 5x$.

$\sqrt{2}x^2 + 2x + 5x + 5\sqrt{2} = \sqrt{2}x(x + \sqrt{2}) + 5(x + \sqrt{2}) = (x + \sqrt{2})(\sqrt{2}x + 5)$.

Zeroes: $x = -\sqrt{2}$ and $x = -\frac{5}{\sqrt{2}}$.

12 **Ans:** $\pm\sqrt{\frac{3}{5}} = \pm\frac{\sqrt{15}}{5}$

$5x^2 - 3 = 0 \Rightarrow x^2 = \frac{3}{5}$.

$x = \pm\sqrt{\frac{3}{5}} = \pm\frac{\sqrt{15}}{5}$.

13 **Ans:** $k = -10$; other zero $-\frac{5}{3}$

$p(2) = 0$: $12 - 2 + k = 0$, so $k = -10$.

$3x^2 - x - 10 = 3x^2 - 6x + 5x - 10 = (x - 2)(3x + 5)$.

Other zero: $3x + 5 = 0 \Rightarrow x = -\frac{5}{3}$.

14 **Ans:** $p(x) = -2x^2 - 4x + 6$; opens downward

Zeroes -3 and 1: $p(x) = k(x + 3)(x - 1)$.

$p(0) = k(3)(-1) = -3k = 6 \Rightarrow k = -2$.

$p(x) = -2(x^2 + 2x - 3) = -2x^2 - 4x + 6$; $a = -2 < 0$, so it opens downward.

15 **Ans:** $k = \frac{7}{4}$; other zero 4

$p(3) = 0$: $9(k - 2) + 3k - 3 = 0 \Rightarrow 12k - 21 = 0 \Rightarrow k = \frac{7}{4}$.

Then $p(x) = -\frac{1}{4}x^2 + \frac{7}{4}x - 3 = -\frac{1}{4}(x^2 - 7x + 12)$.

$x^2 - 7x + 12 = (x - 3)(x - 4)$.

Other zero is 4.

16 **Ans:** Zeroes -2 and 3

Table: $x = -3, -2, -1, 0, 1, 2, 3, 4$ gives $y = 6, 0, -4, -6, -6, -4, 0, 6$.

Plot the points and join them with a smooth upward parabola (lowest point near $x = \frac{1}{2}$, $y = -6.25$).

The curve meets the x-axis at $(-2, 0)$ and $(3, 0)$.

Check: $x^2 - x - 6 = (x - 3)(x + 2)$, zeroes 3 and -2.

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Degree of 5 (a non-zero constant)? : 0
2. Degree of the zero polynomial? : not defined
3. Maximum zeroes of a cubic? : 3
4. Zero of $ax + b$? : $-\frac{b}{a}$
5. Graph of a quadratic is called? : a parabola
6. Parabola opens downward when? : $a < 0$
7. Graph touches the x-axis at one point only: how many distinct zeroes? : 1
8. Is $(0, -8)$ on the graph a zero? : No, zeroes lie on the x-axis

DAY 3 COMPLETE? ☐ Notes read

☐ 16 questions solved ☐ Answers checked

☐ Mistakes noted Next: Day 4, Polynomials

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 04

POLYNOMIALS | CHAPTER 2 | PART 2 OF 2

Relationship between Zeroes and Coefficients

Polynomials

Aaj ka target

- [ ] $\alpha+\beta=-b/a$ aur $\alpha\beta=c/a$: sign ke saath yaad
- [ ] $\alpha^2+\beta^2$, $1/\alpha+1/\beta$ jaise expressions bina zeroes nikale
- [ ] Sum aur product se quadratic banana: x^2-Sx+P
- [ ] 16 sawal khud se, phir answers milao (lagbhag 100 min)

4.1 Sum and Product of Zeroes of a Quadratic

If α and β are the zeroes of $p(x)=ax^2+bx+c$ ($a\neq 0$), then $p(x)=a(x-\alpha)(x-\beta)=ax^2-a(\alpha+\beta)x+a\alpha\beta$. Comparing coefficients gives the two most-used results of this chapter.

ZEROES AND COEFFICIENTS (QUADRATIC)

$$\alpha+\beta=-\frac{b}{a}=-\frac{\text{coefficient of } x}{\text{coefficient of } x^2}$$

$$\alpha\beta=\frac{c}{a}=\frac{\text{constant term}}{\text{coefficient of } x^2}$$

SOLVED EXAMPLE 1

Find the zeroes of $6x^2-7x-3$ and verify the relationship between the zeroes and the coefficients.

SOLUTION

1. $6x^2-7x-3=(2x-3)(3x+1)$, so $\alpha=\frac{3}{2}$, $\beta=-\frac{1}{3}$.
2. Sum $=\frac{3}{2}-\frac{1}{3}=\frac{7}{6}$ and $-\frac{b}{a}=-\frac{-7}{6}=\frac{7}{6}$. Verified.
3. Product $=\frac{3}{2}\times\left(-\frac{1}{3}\right)=-\frac{1}{2}$ and $\frac{c}{a}=\frac{-3}{6}=-\frac{1}{2}$. Verified.

Answer: Zeroes $\frac{3}{2}$, $-\frac{1}{3}$; both relations hold.

4.2 Forming a Quadratic from its Zeroes

QUADRATIC WITH GIVEN SUM S AND PRODUCT P

$$p(x)=k\left[x^2-Sx+P\right],\quad k\neq 0$$

With zeroes α,β: $k(x-\alpha)(x-\beta)$

- Infinitely many quadratics share the same zeroes: they differ only by the non-zero constant k.
- Choose k to clear fractions: sum $-\frac{1}{4}$, product $\frac{1}{4}$ gives $x^2+\frac{1}{4}x+\frac{1}{4}$, so take $k=4$: $4x^2+x+1$.

SOLVED EXAMPLE 2

Find a quadratic polynomial whose zeroes are $\frac{2+\sqrt{3}}{4}$ and $\frac{2-\sqrt{3}}{4}$.

SOLUTION

1. $S=\frac{2+\sqrt{3}+2-\sqrt{3}}{4}=1$.
2. $P=\frac{(2+\sqrt{3})(2-\sqrt{3})}{16}=\frac{4-3}{16}=\frac{1}{16}$.
3. $p(x)=x^2-x+\frac{1}{16}$; multiply by 16.

Answer: $16x^2-16x+1$

MEME SAMJHO

Astra meme, maths inside



STUDENT HEARING THAT $AB=C/A$ FOR EVERY QUADRATIC

Har quadratic pe $\alpha+\beta=-b/a$ chalega? Guarantee deta hai??

Asli baat: Asli baat: Yes, for every quadratic ax^2+bx+c with $a\neq 0$, because $a(x-\alpha)(x-\beta)$ must equal it term by term.

4.3 Expressions in alpha and beta (without finding the zeroes)

Expression	Write it as
$\alpha^2+\beta^2$	$(\alpha+\beta)^2-2\alpha\beta$
$(\alpha-\beta)^2$	$(\alpha+\beta)^2-4\alpha\beta$
$\frac{1}{\alpha}+\frac{1}{\beta}$	$\frac{\alpha+\beta}{\alpha\beta}$
$\alpha^3+\beta^3$	$(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$
$\frac{\alpha}{\beta}+\frac{\beta}{\alpha}$	$\frac{\alpha^2+\beta^2}{\alpha\beta}$

Symmetric expressions: rewrite, then substitute S and P

SOLVED EXAMPLE 3

If α,β are the zeroes of $2x^2-7x+3$, find $\alpha^2+\beta^2$ and a quadratic whose zeroes are $\frac{1}{\alpha}$ and $\frac{1}{\beta}$.

SOLUTION

1. $\alpha+\beta=\frac{7}{2}$, $\alpha\beta=\frac{3}{2}$.
2. $\alpha^2+\beta^2=\frac{49}{4}-2\cdot\frac{3}{2}=\frac{37}{4}$.

SOLVED EXAMPLE 3

1. New sum $= \dfrac{\alpha+\beta}{\alpha\beta} = \dfrac{7}{3}$, new product $= \dfrac{1}{\alpha\beta} = \dfrac{2}{3}$.
2. $x^2 - \dfrac{7}{3}x + \dfrac{2}{3}$, multiply by 3.

Answer: $\alpha^2+\beta^2 = \dfrac{37}{4}$; polynomial $3x^2 - 7x + 2$



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Sum mein **minus** lagta hai, product mein nahi: $\alpha+\beta = -\dfrac{b}{a}$, $\alpha\beta = +\dfrac{c}{a}$. Aur pehle polynomial ko standard form ax^2+bx+c mein likho: $6x^2 - 3 - 7x$ mein $b=-7$, $c=-3$ hai.

4.4 Special Conditions on the Zeroes

Condition on zeroes	What it means
Reciprocal ($\alpha\beta = 1$)	$c = a$
Negatives of each other ($\alpha+\beta=0$)	$b = 0$
One zero is 0	$c = 0$
Equal zeroes	$b^2 = 4ac$, so a and c have the same sign
Both negative	$\alpha+\beta<0$ and $\alpha\beta>0$



SHIVAM BHAIYA SAYS

Signs ka shortcut: product c/a positive matlab dono zeroes same sign; phir sum $-b/a$ ka sign bata dega dono positive hain ya negative. x^2+kx+k mein sum $-k$ aur product k ulte sign ke hain, isliye dono zeroes kabhi positive nahi ho sakte.

4.5 Cubic Polynomials (for reference)

ZEROES OF $AX^3 + BX^2 + CX + D$

$$\alpha+\beta+\gamma = -\frac{b}{a}, \quad \alpha\beta+\beta\gamma+\gamma\alpha = \frac{c}{a}, \quad \alpha\beta\gamma$$



SHIVAM BHAIYA SAYS: BOARD TIP

Board mein 'k find karo' wale sawal mein condition ko **pehle formula mein likho** ($\alpha+\beta = \frac{1}{2}\alpha\beta$ etc.), phir substitute karo. Step marks yahin se milte hain, final answer galat ho tab bhi.



EXAM ALERT!

Zeroes and coefficients se 2-3 marks ka sawal lagbhag har board paper mein aata hai: 'form the polynomial' ya '$\alpha^2+\beta^2$ / k find karo'. Division algorithm syllabus se bahar hai, uspe time mat lagao.

Day 4 Practice

17 board-pattern questions | solve first, then check

MCQ

1 A quadratic polynomial whose zeroes are -2 and 5 is (1)

(a) $x^2+3x-10$ (b) $2x^2-6x-20$
(c) $x^2-3x+10$ (d) $x^2+3x+10$

MCQ

2 If 3 and -4 are the zeroes of $x^2+(a-1)x+b$, then (1)

(a) $a=0,\ b=12$ (b) $a=2,\ b=12$
(c) $a=0,\ b=-12$ (d) $a=2,\ b=-12$

MCQ

3 If the sum of the zeroes of $4x^2-kx+5$ is 2, then k is (1)

(a) 8 (b) -8 (c) 2 (d) $\dfrac{5}{4}$

MCQ

4 If the zeroes of $7x^2-11x+m$ are reciprocals of each other, then m equals (1)

(a) -7 (b) 11 (c) 7 (d) $\dfrac{1}{7}$

MCQ

5 For $p(x) = x^2+kx+k$ with $k \neq 0$, the zeroes (1)

(a) are always equal (b) cannot both be positive
(c) cannot both be negative (d) are always unequal

ASSERTION-REASON

6 **Assertion (A):** A quadratic polynomial whose zeroes have sum $-\frac{1}{4}$ and product $\frac{1}{4}$ is $4x^2 + x + 1$.
Reason (R): A quadratic with sum of zeroes S and product P is $k(x^2 - Sx + P)$, $k \neq 0$. (1)
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

7 **Assertion (A):** If the zeroes of $x^2 - (k-3)x + 5$ are negatives of each other, then $k = -3$.
Reason (R): If the zeroes of a quadratic are negatives of each other, their sum is 0. (1)
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

CASE-BASED

8 Read the passage and answer the questions that follow. An arch over a park gate follows the curve of $p(x) = x^2 - 2x - 15$, with its feet at the zeroes of $p(x)$ on the ground line (the x-axis). (4)
(i) Find the sum of the zeroes of $p(x)$. (1)
(ii) Find the product of the zeroes. (1)
(iii) Find a quadratic polynomial whose zeroes are twice the zeroes of $p(x)$. (2)

SHORT ANSWER

9 One zero of $x^2 - 6x + 7$ is $3 + \sqrt{2}$. Find the other zero without solving. (2)

SHORT ANSWER

10 Find the zeroes of $8x^2 - 2x - 3$ and verify the relationship between the zeroes and the coefficients. (2)

SHORT ANSWER

11 Find a quadratic polynomial whose zeroes are $\frac{3+\sqrt{5}}{2}$ and $\frac{3-\sqrt{5}}{2}$. (2)

SHORT ANSWER

12 If α, β are the zeroes of $3x^2 - 6x + k$ and $\alpha^2 + \beta^2 + \alpha\beta = \frac{10}{3}$, find k. (3)

SHORT ANSWER

13 For $x^2 - (k+2)x + 3(k+1)$, the sum of the zeroes is half their product. Find k. (3)

SHORT ANSWER

14 Find p and q if the zeroes of $x^2 + px + q$ are three times the zeroes of $3x^2 - x - 2$. (3)

LONG ANSWER

15 If α, β are the zeroes of $x^2 + px + q$, show that $\left(\frac{\alpha}{\beta} + 1\right)\left(\frac{\beta}{\alpha} + 1\right) = \frac{p^2}{q}$. (4)

LONG ANSWER

16 The zeroes of $3x^2 + (2k-6)x - 5$ are negatives of each other. Find k, and then find the zeroes of $x^2 + kx + 2$. (4)

LONG ANSWER

17 α and β ($\alpha > \beta$) are the zeroes of $2x^2 - 7x + 3$. Without finding them first, find (i) $\frac{1}{\alpha} + \frac{1}{\beta}$ (ii) $(\alpha - \beta)^2$, then (iii) find $\alpha - \beta$ and check by finding the zeroes. (5)

Day 4 Answers with worked solutions

1 **(b)**
$S = 3$, $P = -10$: $k(x^2 - 3x - 10)$. With $k = 2$: $2x^2 - 6x - 20$.

2 **(d)**
Sum $= -1 = -(a-1) \Rightarrow a = 2$.
Product $= -12 = b$.

3 **(a)**
$\alpha + \beta = \frac{k}{4} = 2 \Rightarrow k = 8$.

4 **(c)**
Reciprocal zeroes: $\alpha\beta = 1 \Rightarrow \frac{m}{7} = 1 \Rightarrow m = 7$.

5 **(b)**
Sum $= -k$, product $= k$. Both positive would need $-k > 0$ and $k > 0$ together, which is impossible.

6 **(a)**
$x^2 + \frac{1}{4}x + \frac{1}{4}$, times 4, is $4x^2 + x + 1$: A true, and R is the rule used, so R explains A.

7 **(d)**
Sum $= k - 3 = 0 \Rightarrow k = 3$, not -3: A is false.
R is true.

8 **Ans:** 2

$\alpha + \beta = -\dfrac{-2}{1} = 2.$

Ans: -15

$\alpha\beta = \dfrac{-15}{1} = -15.$

Ans: $x^2 - 4x - 60$

Zeroes of p: $(x-5)(x+3)$, so 5 and -3.

Doubled: 10 and -6; $S = 4,\ P = -60$.

Polynomial: $x^2 - 4x - 60$.

9 **Ans:** $3 - \sqrt{2}$

$\alpha + \beta = 6.$

Other zero $= 6 - (3 + \sqrt{2}) = 3 - \sqrt{2}.$

Check: $(3+\sqrt{2})(3-\sqrt{2}) = 7 = \dfrac{c}{a}.$

10 **Ans:** $\dfrac{3}{4}$ and $-\dfrac{1}{2}$

$8x^2 - 6x + 4x - 3 = 2x(4x-3) + 1(4x-3) =$ $(4x-3)(2x+1)$; zeroes $\dfrac{3}{4}, -\dfrac{1}{2}.$

Sum $= \dfrac{1}{4} = -\dfrac{-2}{8}$. Product $= -\dfrac{3}{8} = \dfrac{-3}{8}.$

Verified.

11 **Ans:** $x^2 - 3x + 1$

$S = \dfrac{6}{2} = 3.$

$P = \dfrac{9-5}{4} = 1.$

$x^2 - 3x + 1.$

12 **Ans:** $k = 2$

$\alpha + \beta = 2,\ \alpha\beta = \dfrac{k}{3}.$

$\alpha^2 + \beta^2 + \alpha\beta = (\alpha+\beta)^2 - \alpha\beta = 4 - \dfrac{k}{3}.$

$4 - \dfrac{k}{3} = \dfrac{10}{3} \Rightarrow \dfrac{k}{3} = \dfrac{2}{3} \Rightarrow k = 2.$

13 **Ans:** $k = 1$

Sum $= k + 2$, product $= 3(k+1)$.

$k + 2 = \dfrac{3(k+1)}{2} \Rightarrow 2k + 4 = 3k + 3 \Rightarrow k = 1.$

14 **Ans:** $p = -1,\ q = -6$

$3x^2 - x - 2 = (3x+2)(x-1)$: zeroes $1,\ -\dfrac{2}{3}.$

Tripled: 3 and -2; sum 1, product -6.

$x^2 - x - 6$, so $p = -1,\ q = -6$.

15 **Ans:** Shown

$\alpha + \beta = -p,\ \alpha\beta = q.$

Expand: $1 + \dfrac{\alpha}{\beta} + \dfrac{\beta}{\alpha} + 1 = 2 + \dfrac{\alpha^2 + \beta^2}{\alpha\beta}.$

$= \dfrac{2\alpha\beta + \alpha^2 + \beta^2}{\alpha\beta} = \dfrac{(\alpha+\beta)^2}{\alpha\beta} = \dfrac{p^2}{q}.$

16 **Ans:** $k = 3$; zeroes $-1, -2$

Zeroes negatives of each other: sum $= 0$.

$-\dfrac{2k-6}{3} = 0 \Rightarrow k = 3.$

$x^2 + 3x + 2 = (x+1)(x+2)$: zeroes $-1, -2$.

17 **Ans:** (i) $\dfrac{7}{3}$ (ii) $\dfrac{25}{4}$ (iii) $\dfrac{5}{2}$

$\alpha + \beta = \dfrac{7}{2},\ \alpha\beta = \dfrac{3}{2}.$

(i) $\dfrac{\alpha+\beta}{\alpha\beta} = \dfrac{7/2}{3/2} = \dfrac{7}{3}.$

(ii) $(\alpha+\beta)^2 - 4\alpha\beta = \dfrac{49}{4} - 6 = \dfrac{25}{4}.$

(iii) $\alpha > \beta$ so $\alpha - \beta = \dfrac{5}{2}.$

Check: $2x^2 - 7x + 3 = (2x-1)(x-3)$, zeroes $3, \dfrac{1}{2}$; $3 - \dfrac{1}{2} = \dfrac{5}{2}.$

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Sum of zeroes of $ax^2 + bx + c$? : $-\dfrac{b}{a}$
2. Product of zeroes? : $\dfrac{c}{a}$
3. Quadratic with sum S, product P? : $k(x^2 - Sx + P)$
4. $\alpha^2 + \beta^2$ in S and P? : $S^2 - 2P$
5. Reciprocal zeroes condition? : $c = a$
6. Zeroes negatives of each other? : $b = 0$
7. Sum of zeroes of a cubic $ax^3 + bx^2 + cx + d$? : $-\dfrac{b}{a}$
8. How many quadratics have zeroes 2 and 3? : infinitely many

DAY 4 COMPLETE? ☐ Notes read ☐ 17 questions solved ☐ Answers checked ☐ Mistakes noted

Next: Day 5, Revision 1: Real Numbers + Polynomials

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 05

REVISION 1 | REAL NUMBERS + POLYNOMIALS

Real Numbers + Polynomials

Revision 1

Aaj ka target

- [ ] FTA, HCF-LCM aur irrationality proofs ek baar bol ke sunao
- [ ] Zeroes ka graph meaning + sum/product formulas bina notes dekhe likho
- [ ] 18 mixed sawal bina chapter dekhe solve karo
- [ ] Galtiyon ka chhota list banao (lagbhag 100 min)

5.1 Real Numbers: Six things to remember

PRIME FACTORISATION AND HCF-LCM

Every composite has a unique prime factorisation, ignoring order (FTA).

$$\text{HCF}: \text{ common primes, smallest powers; } \text{I}$$

$$\text{HCF}(a,b) \times \text{LCM}(a,b) = a \times b \qquad (\text{for TW}$$

- Use **HCF** for maximum equal groups or largest equal pieces; use **LCM** for 'first time together again' or smallest common multiple.
- Proof $\sqrt{p}$ irrational (prime p): assume $\sqrt{p} = a/b$ in lowest terms. Then $a^2 = pb^2$; $p \mid a$, so $p \mid b$, contradiction.
- A non-zero rational plus an irrational is irrational. But a product of two irrationals need **not** be irrational: $\sqrt{2} \times \sqrt{2} = 2$.

5.2 Polynomials: Graph and coefficient checklist

ZEROES AT A GLANCE

$p(k) = 0$: k is a zero; graph $y = p(x)$ meets the x-axis at $(k, 0)$.

$$\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a} \quad \text{for } ax^2 + bx + c,\ a$$

$$p(x) = k[x^2 - (\alpha+\beta)x + \alpha\beta], \quad k \neq 0$$

Graph meets x-axis at	Distinct zeroes
No point	0
One point (touches)	1
Two points (cuts)	2
Three points (cubic)	3

REVISION EXAMPLE 1

Find HCF and LCM of 180 and 252 by prime factorisation.

SOLUTION

1. $180 = 2^2 \times 3^2 \times 5$ and $252 = 2^2 \times 3^2 \times 7$.
2. HCF: shared primes at the smaller powers $= 2^2 \times 3^2 = 36$.
3. LCM: all primes at the larger powers $= 2^2 \times 3^2 \times 5 \times 7 = 1260$.
4. Check $36 \times 1260 = 45360 = 180 \times 252$.

Answer: HCF 36; LCM 1260

REVISION EXAMPLE 2

If α, β are zeroes of $2x^2 - 5x - 3$, find $\alpha^2 + \beta^2$ without finding the zeroes.

SOLUTION

1. $S = \alpha + \beta = \frac{5}{2}$, $P = \alpha\beta = -\frac{3}{2}$.
2. $\alpha^2 + \beta^2 = S^2 - 2P = \frac{25}{4} + 3 = \frac{37}{4}$.

Answer: $\frac{37}{4}$

MEME SAMJHO

Astra meme, maths inside



STUDENT ON REVISION DAY, BOTH CHAPTERS IN ONE GO

HCF x LCM = a x b bhi yaad, $\alpha + \beta = -b/a$ bhi yaad. Revision day ki smile.

Asli baat: Asli baat: The HCF-LCM product rule (for two numbers) and the sum-product relations of zeroes are the two most reused tools from Chapters 1 and 2.



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

HCF $\times$ LCM $= a \times b$ ko **teen numbers** pe laga diya? Galat. Yeh identity **sirf do positive integers** ke liye guaranteed hai.



SHIVAM BHAIYA SAYS: BOARD TIP

Irrationality proof mein 'assume a/b in lowest terms' aur end mein 'contradiction to coprime' **dono** likhna. Beech ke algebra ke saath yeh dono lines marks dilati hain.



EXAM ALERT!
Mixed revision: answer mat dekho jab tak 18 questions attempt na ho jayein. Jo galat ho, us formula ko apne mistakes page pe likho.

Day 5 Practice

18 board-pattern questions | solve first, then check

MCQ

1 The exponent of 2 in the prime factorisation of 1440 is (1)

(a) 3 **(b)** 4 **(c)** 6 **(d)** 5

MCQ

2 Three lights blink every 36 s, 48 s and 60 s. They blink together at 9:00:00. They next blink together after (1)

(a) 6 minutes **(b)** 12 minutes
(c) 18 minutes **(d)** 24 minutes

MCQ

3 The largest number that divides 70 and 125 leaving remainders 5 and 8 respectively is (1)

(a) 13 **(b)** 65 **(c)** 875 **(d)** 1750

MCQ

4 The product of two different irrational numbers is (1)

(a) always irrational **(b)** always rational
(c) always an integer **(d)** rational or irrational

MCQ

5 If 3 and $\frac{1}{3}$ are the zeroes of $px^2 + 10x + r$, then (1)

(a) $p = 3,\ r = 3$ **(b)** $p = -3,\ r = -3$
(c) $p = 3,\ r = -3$ **(d)** $p = -3,\ r = 3$

MCQ

6 If the sum of the zeroes of $kx^2 - 6x + 2k$ equals their product, then k is (1)

(a) 2 **(b)** $\frac{1}{3}$ **(c)** 3 **(d)** 6

MCQ

7 The zeroes of $(k+2)x^2 - 7x + 5$ are reciprocals of each other. Then k is (1)

(a) 3 **(b)** 5 **(c)** -2 **(d)** 7

ASSERTION-REASON

8 **Assertion (A):** If the product of two numbers is 4500 and their HCF is 15, then their LCM is 300.
Reason (R): The HCF of two numbers is always a factor of their LCM. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

9 **Assertion (A):** $(2+\sqrt{3})(2-\sqrt{3})$ is a rational number.
Reason (R): The product of two irrational numbers is always irrational. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

10 **Assertion (A):** The zeroes of $x^2 - 2x - 8$ are 4 and -2.
Reason (R): The zeroes of $x^2 - 6x + 7$ are $3 \pm \sqrt{3}$. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

CASE-BASED

11 Read the passage and answer the questions that follow. A school is seating Class 9 (96 students), Class 10 (72 students) and Class 11 (120 students) in a hall. Every row must have the same number of students, and a row may contain students of only one class. (4)

(i) What is the largest possible number of students in a row? (1)

(ii) Write 120 as a product of primes. (1)

(iii) Find the minimum number of rows needed. (2)

SHORT ANSWER

12 Find a quadratic polynomial whose zeroes are $\frac{2}{5}$ and $-\frac{1}{2}$. (2)

SHORT ANSWER

13 If α, β are the zeroes of $x^2 - 3x - 5$, find $\frac{1}{\alpha} + \frac{1}{\beta} - \alpha\beta$. (2)

SHORT ANSWER

14 Find a quadratic polynomial whose sum of zeroes is 2 and product is -15, and find its zeroes. (2)

SHORT ANSWER

15 Find the HCF and LCM of 336 and 54 by prime factorisation and verify that HCF $\times$ LCM $=$ product of the numbers. (3)

SHORT ANSWER

16 Given that $\sqrt{5}$ is irrational, prove that $\frac{3-\sqrt{5}}{4}$ is irrational. (3)

SHORT ANSWER

17 The graph of $y = p(x)$, a quadratic, cuts the x-axis at $(-3, 0)$ and $(3, 0)$ and the y-axis at $(0, -18)$. Find $p(x)$ and $p(4)$. (3)

LONG ANSWER

18 Prove that $\sqrt{3}$ is irrational. Hence show that $2\sqrt{3} - 1$ is irrational. (4)

Day 5 Answers with worked solutions

1 **(d)**
$1440 = 2^5 \times 3^2 \times 5$, so the exponent of 2 is 5.

2 **(b)**
$36 = 2^2 3^2,\ 48 = 2^4 3,\ 60 = 2^2 \cdot 3 \cdot 5.$
LCM $= 2^4 \times 3^2 \times 5 = 720$ s $= 12$ minutes.

3 **(a)**
Required number divides $70 - 5 = 65$ and $125 - 8 = 117$.
HCF$(65, 117)$: $65 = 5 \times 13$, $117 = 3^2 \times 13$, so HCF $= 13$.

4 **(d)**
$\sqrt{2} \times \sqrt{8} = 4$ (rational) but $\sqrt{2} \times \sqrt{3} = \sqrt{6}$ (irrational).

5 **(b)**
Product $= 1 = \frac{r}{p} \Rightarrow r = p.$
Sum $= \frac{10}{3} = -\frac{10}{p} \Rightarrow p = -3$, so $r = -3$.

6 **(c)**
Sum $= \frac{6}{k}$, product $= \frac{2k}{k} = 2.$
$\frac{6}{k} = 2 \Rightarrow k = 3.$

7 **(a)**
$\alpha\beta = \frac{5}{k+2} = 1 \Rightarrow k = 3.$

8 **(b)**
A: LCM $= \frac{4500}{15} = 300$, true.
R is true too, but A follows from HCF $\times$ LCM $=$ product, not from R.

9 **(c)**
$(2+\sqrt{3})(2-\sqrt{3}) = 4 - 3 = 1$: A true.
R is false, and A itself is a counter-example.

10 **(c)**
$x^2 - 2x - 8 = (x-4)(x+2)$: A true.
$x^2 - 6x + 7$: $x = 3 \pm \sqrt{2}$, not $3 \pm \sqrt{3}$: R false.

11 **Ans:** 24
HCF$(96, 72, 120)$: $96 = 2^5 3,\ 72 = 2^3 3^2,\ 120 = 2^3 \cdot 3 \cdot 5$; HCF $= 2^3 \times 3 = 24.$
Ans: $2^3 \times 3 \times 5$
$120 = 8 \times 15.$
Ans: 12 rows
Rows $= \frac{96}{24} + \frac{72}{24} + \frac{120}{24} = 4 + 3 + 5 = 12.$

12 **Ans:** $10x^2 + x - 2$
$S = \frac{2}{5} - \frac{1}{2} = -\frac{1}{10},\ P = -\frac{1}{5}.$
$x^2 + \frac{1}{10}x - \frac{1}{5}$; multiply by 10.

13 **Ans:** $\frac{22}{5}$
$\alpha + \beta = 3,\ \alpha\beta = -5.$
$\frac{\alpha+\beta}{\alpha\beta} - \alpha\beta = -\frac{3}{5} + 5 = \frac{22}{5}.$

14 **Ans:** $x^2 - 2x - 15$; zeroes $5, -3$
$x^2 - 2x - 15.$
$= (x-5)(x+3)$, zeroes 5 and -3.

15 **Ans:** HCF 6, LCM 3024
$336 = 2^4 \times 3 \times 7,\ 54 = 2 \times 3^3.$
HCF $= 2 \times 3 = 6$; LCM $= 2^4 \times 3^3 \times 7 = 3024.$
$6 \times 3024 = 18144$ and $336 \times 54 = 18144.$
Verified.

16 **Ans:** Proved
Suppose $\frac{3-\sqrt{5}}{4} = r$, a rational number.
Then $\sqrt{5} = 3 - 4r$, which is rational (rationals are closed under subtraction and multiplication).
This contradicts $\sqrt{5}$ being irrational, so $\frac{3-\sqrt{5}}{4}$ is irrational.

17 **Ans:** $p(x) = 2x^2 - 18$; $p(4) = 14$

$p(x) = k(x+3)(x-3) = k(x^2-9)$.

$p(0) = -9k = -18 \Rightarrow k = 2$.

$p(x) = 2x^2 - 18$, $p(4) = 32 - 18 = 14$.

18 **Ans:** Proved

Suppose $\sqrt{3} = \frac{a}{b}$ with a, b coprime integers, $b \neq 0$. Then $a^2 = 3b^2$, so 3 divides a^2 and hence 3 divides a.

Write $a = 3c$: $9c^2 = 3b^2 \Rightarrow b^2 = 3c^2$, so 3 divides b.

3 divides both a and b, contradicting coprimality. So $\sqrt{3}$ is irrational.

If $2\sqrt{3} - 1 = r$ (rational), then $\sqrt{3} = \frac{r+1}{2}$ is rational: contradiction. So $2\sqrt{3} - 1$ is irrational.

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. HCF uses which powers? : the smallest powers of common primes
2. LCM uses which powers? : the largest powers of all primes
3. HCF x LCM = product holds for? : two positive integers
4. Is $\sqrt{2} \times \sqrt{2}$ irrational? : No, it is 2
5. Sum of zeroes of $ax^2 + bx + c$? : $-b/a$
6. Product of zeroes? : c/a
7. A parabola touching the x-axis has how many distinct zeroes? : 1
8. Max zeroes of a cubic? : 3

DAY 5 COMPLETE?

- [ ] Both chapters revised
- [ ] 18 questions solved
- [ ] Answers checked
- [ ] Mistakes page updated

Next: Day 6, Pair of Linear Equations in Two Variables

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 06

PAIR OF LINEAR EQUATIONS | CHAPTER 3 | PART 1 OF 3

Graphical Method and Consistency

Pair of Linear Equations in Two Variables

Aaj ka target

- ☐ Do lines: intersect, parallel ya coincident, ratios se turant batao
- ☐ a_1/a_2, b_1/b_2, c_1/c_2 wali table ratt lo
- ☐ Ek pair ko graph paper pe plot karke solution padhna
- ☐ 15 sawal khud se, phir answers milao (lagbhag 90 min)

6.1 Linear Equations in Two Variables

A **linear equation in two variables** has the form $ax + by + c = 0$, where a, b, c are real numbers and a, b are not both zero. Its graph is a **straight line**, and every point on the line is a solution. Two such equations in the same variables form a **pair of linear equations**:

GENERAL PAIR

$$a_1x + b_1y + c_1 = 0, \qquad a_2x + b_2y + c_2 = 0$$

KEY TERM: SOLUTION OF A PAIR

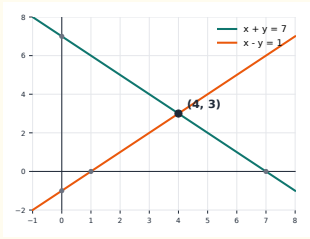
A pair of values (x, y) that satisfies **both** equations. On the graph it is a point lying on **both** lines.

6.2 Graphical Method

1. For each equation, find at least two (better three) points: put $x = 0$ to get y, put $y = 0$ to get x.
2. Plot the points on the same axes and draw both lines.
3. Read the result: the intersection point is the solution; parallel lines mean no solution; one line on top of the other means infinitely many.

SOLVED EXAMPLE 1

Solve graphically: $x + y = 7$ and $x - y = 1$.



SOLUTION

1. $x + y = 7$: points $(0, 7), (7, 0), (3, 4)$.
2. $x - y = 1$: points $(1, 0), (0, -1), (3, 2)$.
3. Draw both lines; they meet at $(4, 3)$.
4. Check: $4 + 3 = 7$ and $4 - 3 = 1$.

Answer: $x = 4,\ y = 3$

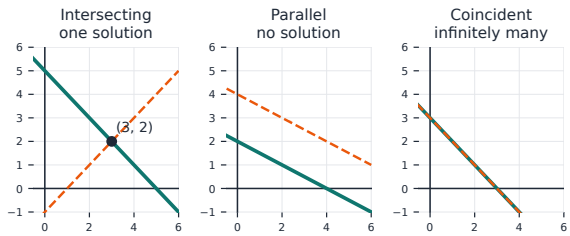


Fig. 6.1 Two lines in a plane: intersecting, parallel or coincident.

6.3 Consistency from the Coefficients

Compare ratios	Lines	Solutions	System
$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$	Intersecting	Exactly one	Consistent
$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$	Coincident	Infinitely many	Consistent (dependent)
$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$	Parallel	None	Inconsistent

The ratio test (memorise this table)

- **Consistent** = at least one solution (intersecting or coincident). **Inconsistent** = no solution (parallel).
- $x = a$ is a line parallel to the y-axis; $y = b$ is parallel to the x-axis; $y = mx$ passes through the origin.
- The lines $x = a$ and $y = b$ always meet at the single point (a, b).

SOLVED EXAMPLE 2

Without drawing, decide whether the pair $7x - y = 5$ and $21x - 3y = 10$ is consistent.

SOLVED EXAMPLE 2

SOLUTION

1. $\frac{a_1}{a_2} = \frac{7}{21} = \frac{1}{3}$, $\frac{b_1}{b_2} = \frac{-1}{-3} = \frac{1}{3}$, $\frac{c_1}{c_2} = \frac{5}{10} = \frac{1}{2}$.
2. $\frac{1}{3} = \frac{1}{3} \neq \frac{1}{2}$: the lines are parallel.

Answer: No solution, so the pair is **inconsistent**.

MEME SAMJHO

Astra meme, maths inside



REPORTER INTERVIEWING TWO PARALLEL LINES

Reporter: Aap dono kabhi miloge? Parallel lines: $a_1/a_2 = b_1/b_2 \neq c_1/c_2$, kabhi nahi.

Asli baat: Asli baat: Parallel lines have equal slopes but different intercepts, so the pair has no common solution: it is inconsistent.

SOLVED EXAMPLE 3

Find k so that $x + (k+1)y = 5$ and $(k+1)x + 9y = 8k - 1$ have infinitely many solutions.

SOLUTION

1. Need $\frac{1}{k+1} = \frac{k+1}{9} = \frac{5}{8k-1}$.
2. First two: $(k+1)^2 = 9 \Rightarrow k = 2$ or $k = -4$.
3. First and third: $8k - 1 = 5k + 5 \Rightarrow k = 2$.
4. Only $k = 2$ satisfies all three ($k = -4$ gives $-\frac{1}{3}$ vs $-\frac{5}{33}$).

Answer: $k = 2$



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Ratio lene se pehle dono equations ko **same form** mein likho: ya to $ax + by + c = 0$ ya $ax + by = c$. Ek ko $= 0$ aur doosre ko $= c$ form mein rakh ke c_1/c_2 liya to sign ulta ho jaata hai aur answer galat.



SHIVAM BHAIYA SAYS

'Infinitely many' ya 'no solution' wale k ke sawal mein **teeno ratios** check karo. Pehle do ratios se do values aati hain, teesra ratio decide karta hai kaunsi value sahi hai.



SHIVAM BHAIYA SAYS: BOARD TIP

Graph wale 3-4 mark sawal mein har line ke liye **3 points ki table** banao, axes label karo, scale likho aur intersection point ko naam do. Inhi cheezon ke marks kat-te hain.



EXAM ALERT!

Ratio table se 'unique / no / infinitely many solutions ke liye k' wala 1-2 mark sawal almost har saal aata hai. Cross-multiplication method syllabus se hata diya gaya hai.

Day 6 Practice

15 board-pattern questions | solve first, then check

MCQ

1 The pair $4x - 2y = 7$ and $-6x + 3y = 5$ has (1)

(a) a unique solution (b) no solution (c) infinitely many solutions (d) exactly two solutions

MCQ

2 The lines $x = 3$ and $x = -2$ have (1)

(a) one common point
(b) two common points
(c) infinitely many common points
(d) no common point

MCQ

3 If $3x + ky = 2$ is parallel to $5x - 4y = 9$, then k is (1)

(a) $\frac{12}{5}$ (b) $-\frac{12}{5}$ (c) $-\frac{5}{12}$ (d) $\frac{15}{4}$

MCQ

4 For what value of k does $4x + 7y = 0,\ kx + 14y = 0$ have a non-zero solution? (1)

(a) 2 (b) 4 (c) 7 (d) 8

MCQ

5 A pair of linear equations is consistent when the lines are (1)

(a) parallel
(b) intersecting or coincident
(c) parallel or coincident
(d) only coincident

ASSERTION-REASON

6 **Assertion (A):** The pair $kx - y - 2 = 0$ and $6x - 2y - 3 = 0$ is inconsistent when $k = 3$.

Reason (R): A pair $a_1x + b_1y + c_1 = 0,\ a_2x + b_2y + c_2 = 0$ is inconsistent if $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

7 **Assertion (A):** The pair $x - 2y - 3 = 0$ and $3x + 4y - 20 = 0$ has exactly one solution.
Reason (R): The pair $2x + 3y - 9 = 0$ and $4x + 6y - 18 = 0$ has exactly one solution. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

CASE-BASED

8 Read the passage and answer the questions that follow. On a graph sheet, Riya draws line ℓ_1: $x + 2y = 4$ and line ℓ_2: $x + 2y = 8$. She then draws a third line ℓ_3 through $(3, 0)$ and $(5, 2)$. (4)

(i) How are ℓ_1 and ℓ_2 placed? (1)

(ii) Where does ℓ_1 cut the y-axis? (1)

(iii) Find the equation of ℓ_3 and the point where it meets ℓ_1. (2)

VERY SHORT ANSWER

9 For what value of k does $kx + 3y = 5, 2x + 6y = 7$ have a unique solution? (1)

SHORT ANSWER

10 Find k for which $x + 3y = 4$ and $2x + ky + 12 = 0$ has no solution. (2)

SHORT ANSWER

11 Find c so that $cx + 2y + (2 - c) = 0$ and $8x + cy - c = 0$ have infinitely many solutions. (2)

SHORT ANSWER

12 Find k so that $2x + 3y = 7$ and $(k + 1)x + (2k - 1)y = 4k + 1$ has infinitely many solutions. (3)

SHORT ANSWER

13 Show graphically that $2x - 3y + 6 = 0$ and $4x - 6y + 18 = 0$ have no solution. Where does each line cut the y-axis? (3)

LONG ANSWER

14 Draw the graphs of $2x - y = 2$ and $x + 2y = 16$. Find their point of intersection and the vertices of the triangle they form with the y-axis, and its area. (4)

LONG ANSWER

15 For which value of p does $(2p - 1)x + (p - 1)y = 2p + 1, 3x + y = 1$ have no solution? Also show that for this p the lines are parallel. (4)

Day 6 Answers with worked solutions

1 **(b)**
$\frac{4}{-6} = -\frac{2}{3}$, $\frac{-2}{3} = -\frac{2}{3}$, $\frac{7}{5}$: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$, so no solution.

2 **(d)**
Both lines are parallel to the y-axis and distinct, so they never meet.

3 **(b)**
Parallel: $\frac{3}{5} = \frac{k}{-4} \Rightarrow k = -\frac{12}{5}$ (and $\frac{2}{9} \neq \frac{3}{5}$).

4 **(d)**
A non-zero solution needs coincident lines: $\frac{4}{k} = \frac{7}{14} \Rightarrow k = 8$.

5 **(b)**
Consistent means at least one solution: intersecting (one) or coincident (infinitely many).

6 **(a)**
$k = 3$: $\frac{3}{6} = \frac{1}{2}$, $\frac{-1}{-2} = \frac{1}{2}$, $\frac{-2}{-3} = \frac{2}{3}$. Ratios $\frac{1}{2} = \frac{1}{2} \neq \frac{2}{3}$: A true, and R is the reason.

7 **(c)**
A: $\frac{1}{3} \neq \frac{-2}{4}$, so unique solution: true.
R: $\frac{2}{4} = \frac{3}{6} = \frac{-9}{-18}$, coincident lines, infinitely many solutions: false.

8 **Ans:** Parallel
$\frac{1}{1} = \frac{2}{2} \neq \frac{4}{8}$: parallel lines.
Ans: $(0, 2)$
Put $x = 0$: $2y = 4$, $y = 2$.
Ans: $x - y = 3$; $\left(\frac{10}{3}, \frac{1}{3}\right)$
Slope $= \frac{2 - 0}{5 - 3} = 1$, through $(3, 0)$: $y = x - 3$, i.e. $x - y = 3$.
With $x + 2y = 4$: subtract, $3y = 1$, $y = \frac{1}{3}$, $x = \frac{10}{3}$.

9 **Ans:** $k \neq 1$
Unique: $\frac{k}{2} \neq \frac{3}{6} \Rightarrow k \neq 1$.

10 **Ans:** $k = 6$
Write as $x + 3y - 4 = 0$, $2x + ky + 12 = 0$.
No solution: $\frac{1}{2} = \frac{3}{k} \neq \frac{-4}{12}$.
$k = 6$, and $\frac{1}{2} \neq -\frac{1}{3}$ holds.

11 **Ans:** $c = 4$

$\frac{c}{8} = \frac{2}{c} \Rightarrow c^2 = 16 \Rightarrow c = \pm 4.$

$\frac{2}{c} = \frac{2-c}{-c} \Rightarrow -2c = c(2-c) \Rightarrow c^2 - 4c = 0 \Rightarrow$ $c = 0$ or 4.

Common value: $c = 4$.

12 **Ans:** $k = 5$

$\frac{2}{k+1} = \frac{3}{2k-1} \Rightarrow 4k - 2 = 3k + 3 \Rightarrow k = 5.$

Check third ratio: $\frac{3}{9} = \frac{7}{21} = \frac{1}{3}$. True.

13 **Ans:** Parallel lines; $(0, 2)$ and $(0, 3)$

Line 1: $(0, 2), (-3, 0), (3, 4)$. Line 2: $(0, 3), (-4.5, 0), (3, 5)$.

Ratios $\frac{2}{4} = \frac{-3}{-6} \neq \frac{6}{18}$, and the drawn lines are parallel, so no solution.

y-axis points: $(0, 2)$ and $(0, 3)$.

14 **Ans:** $(4, 6)$; vertices $(0, -2), (0, 8), (4, 6)$; area 20 sq units

$2x - y = 2$: $(0, -2), (1, 0), (4, 6)$. $x + 2y = 16$: $(0, 8), (16, 0), (4, 6)$.

They meet at $(4, 6)$.

Triangle with the y-axis: $(0, -2), (0, 8), (4, 6)$.

Base on the y-axis $= 8 - (-2) = 10$, height $= 4$;

area $= \frac{1}{2} \times 10 \times 4 = 20.$

15 **Ans:** $p = 2$

$\frac{2p-1}{3} = \frac{p-1}{1} \Rightarrow 2p - 1 = 3p - 3 \Rightarrow p = 2.$

For $p = 2$: $3x + y = 5$ and $3x + y = 1$; $\frac{3}{3} = \frac{1}{1} \neq \frac{5}{1}$

.

Same coefficients, different constants: parallel lines, no solution.

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Graph of $ax + by + c = 0$? : a straight line
2. Unique solution condition? : $a_1/a_2 \neq b_1/b_2$
3. No solution condition? : $a_1/a_2 = b_1/b_2 \neq c_1/c_2$
4. Infinitely many condition? : all three ratios equal
5. Consistent means? : at least one solution
6. $x = -2$ is parallel to? : the y-axis
7. Lines $x = a,\ y = b$ meet at? : (a, b)
8. Coincident lines are called? : dependent (consistent)

DAY 6 COMPLETE?

- [ ] Notes read
- [ ] 15 questions solved
- [ ] Answers checked
- [ ] Mistakes noted

Next: Day 7, Pair of Linear Equations in Two Variables

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 07

PAIR OF LINEAR EQUATIONS | CHAPTER 3 | PART 2 OF 3

Substitution and Elimination

Pair of Linear Equations in Two Variables

Aaj ka target

- [ ] Substitution: ek variable ko doosre ke terms mein likho
- [ ] Elimination: coefficients barabar karo, add ya subtract
- [ ] Har answer ko dono equations mein daal ke check karo
- [ ] 15 sawal khud se, phir answers milao (lagbhag 90 min)

7.1 Substitution Method

1. From one equation, express one variable in terms of the other (pick the variable with coefficient 1 or -1 if possible).
2. Substitute this into the other equation: you get an equation in one variable. Solve it.
3. Put the value back to find the second variable.
4. Check both values in both original equations.

SOLVED EXAMPLE 1

Solve by substitution: $y - 4x = 1$ and $6x - 5y = 9$.

SOLUTION

1. From the first: $y = 4x + 1$.
2. Substitute: $6x - 5(4x + 1) = 9 \Rightarrow 6x - 20x - 5 = 9 \Rightarrow -14x = 14 \Rightarrow x = -1$.
3. $y = 4(-1) + 1 = -3$.
4. Check: $-3 - 4(-1) = 1$ and $6(-1) - 5(-3) = 9$.

Answer: $x = -1,\ y = -3$

- If substitution gives a **false** statement like $0 = 5$, the pair has **no solution** (parallel lines).
- If it gives a **true** statement like $0 = 0$, the pair has **infinitely many** solutions (coincident lines).

7.2 Elimination Method

1. Multiply one or both equations by suitable non-zero numbers so that the coefficients of one variable become equal (or opposite).
2. Subtract (or add) the equations to eliminate that variable.
3. Solve for the remaining variable, then substitute back.
4. Check in both equations.

SOLVED EXAMPLE 2

Solve by elimination: $3x + 4y = 10$ and $2x - 2y = 2$.

SOLUTION

1. Multiply the second by 2: $4x - 4y = 4$.
2. Add to the first: $7x = 14 \Rightarrow x = 2$.
3. Then $2(2) - 2y = 2 \Rightarrow y = 1$.

Answer: $x = 2,\ y = 1$

MEME SAMJHO

Astra meme, maths inside



STUDENT AFTER FINDING X AND MOVING ON TO THE NEXT QUESTION

x nikal ke khush ho gaya, y nikalna hi bhool gaya. Ek mark gaya.

Asli baat: Asli baat: A solution of a pair is an ordered pair (x, y). Always find both values and write them together.

SOLVED EXAMPLE 3

Solve: $ax + by = a - b$ and $bx - ay = a + b$.

SOLUTION

1. Multiply the first by a and the second by b: $a^2x + aby = a^2 - ab$ and $b^2x - aby = ab + b^2$.
2. Add: $(a^2 + b^2)x = a^2 + b^2 \Rightarrow x = 1$.
3. Put in the first: $a + by = a - b \Rightarrow y = -1$.

Answer: $x = 1,\ y = -1$

7.3 Symmetric Pairs (a quick trick)

When the coefficients are swapped, like $37x + 43y = 123$ and $43x + 37y = 117$, **add** and **subtract** the equations: $80x + 80y = 240$ gives $x + y = 3$, and $-6x + 6y = 6$ gives $y - x = 1$. So $x = 1,\ y = 2$.



SHIVAM BHAIYA SAYS

Coefficients ulte-palte (swap) hon to seedha add aur subtract karo: $x + y$ aur $x - y$ do chhoti equations mil jaati hain. Bade numbers se multiply karne ki zaroorat hi nahi.



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Subtract karte waqt **poori** doosri equation ka sign badlo, constant term ka bhi. $(3x + 4y = 10) - (3x - 6y = -5)$ mein $10 - (-5) = 15$ hota hai, 5 nahi.



SHIVAM BHAIYA SAYS: BOARD TIP

Question mein method diya ho ('by substitution', 'by elimination') to **wahi method** use karo, warna poore marks nahi milte. Last mein ek line 'Check:' zaroor likho.



EXAM ALERT!

Algebraic solving 2-3 marks mein aata hai, aksar word problem ke andar. Equations reducible to linear form (jaise $\frac{2}{x}+\frac{3}{y}=13$) aur cross-multiplication ab syllabus mein nahi hain.

Day 7 Practice

14 board-pattern questions | solve first, then check

MCQ

1 If $x+2y=10$ and $y=6$, then x equals (1)

(a) 2 **(b)** -2 **(c)** 22 **(d)** -22

MCQ

2 If $x=a,\ y=b$ is the solution of $x+y=7$ and $2x-3y=4$, then (a,b) is (1)

(a) $(5,2)$ **(b)** $(2,5)$

(c) $\left(\frac{18}{5},\frac{17}{5}\right)$ **(d)** $(4,3)$

MCQ

3 Solving $2x+3y=12$ and $x-y=1$ gives (1)

(a) $x=2,\ y=3$ **(b)** $x=3,\ y=3$

(c) $x=3,\ y=2$ **(d)** $x=4,\ y=3$

MCQ

4 For the pair $23x+29y=98$ and $29x+23y=110$, the value of $x+y$ is (1)

(a) 4 **(b)** 2 **(c)** 6 **(d)** 5

ASSERTION-REASON

5 **Assertion (A):** If $(3,1)$ is a solution of $2x+y-q^2-3=0$, then $q=\pm 2$.
Reason (R): Any solution of an equation satisfies it. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

6 **Assertion (A):** If substitution in a pair of linear equations leads to $0=7$, the pair has no solution.
Reason (R): Coincident lines have no common point. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

CASE-BASED

7 Read the passage and answer the questions that follow. A school canteen sells sandwiches and juice. Aman paid Rs 110 for 2 sandwiches and 3 juices; Neha paid Rs 70 for 1 sandwich and 2 juices. (4)

(i) Write the pair of equations (sandwich Rs x, juice Rs y). (1)

(ii) Find the price of one juice. (1)

(iii) Find the price of a sandwich and the cost of 3 sandwiches and 1 juice. (2)

SHORT ANSWER

8 Solve by substitution: $y-3x=2$ and $4x-3y=4$. (2)

SHORT ANSWER

9 Solve by elimination: $5x-3y=11$ and $2x+3y=17$. (2)

SHORT ANSWER

10 Solve: $6(ax+by)=3a+2b$ and $6(bx-ay)=3b-2a$. (3)

SHORT ANSWER

11 Solve: $\frac{ax}{b}-\frac{by}{a}=a+b$ and $ax-by=2ab$. (3)

SHORT ANSWER

12 Solve $47x+31y=63$ and $31x+47y=15$. (3)

LONG ANSWER

13 Find the integral values of m for which the x-coordinate of the point of intersection of $y=mx+1$ and $3x+4y=9$ is an integer. (4)

LONG ANSWER

14 Solve $3x+2y=11$ and $2x+3y=4$ by elimination, and hence find p if $y=3x+p$ passes through the solution. (4)

Day 7 Answers with worked solutions

1 **(b)**

$x+12=10\Rightarrow x=-2.$

2 **(a)**
$x = 7 - y$: $2(7-y) - 3y = 4 \Rightarrow 14 - 5y = 4 \Rightarrow y = 2,\ x = 5.$

3 **(c)**
$x = y + 1$: $2y + 2 + 3y = 12 \Rightarrow y = 2,\ x = 3.$

4 **(a)**
Add: $52x + 52y = 208 \Rightarrow x + y = 4.$

5 **(a)**
$6 + 1 - q^2 - 3 = 0 \Rightarrow q^2 = 4 \Rightarrow q = \pm 2$. Both true, and substituting a solution is exactly why A holds.

6 **(c)**
A false statement means no (x, y) satisfies both: A true.
Coincident lines share every point: R false.

7 **Ans:** $2x + 3y = 110,\ x + 2y = 70$
From the two bills.
Ans: Rs 30
$x = 70 - 2y$: $140 - 4y + 3y = 110 \Rightarrow y = 30.$
Ans: Rs 10; Rs 60
$x = 70 - 60 = 10.$
$3(10) + 30 = 60.$

8 **Ans:** $x = -2,\ y = -4$
$y = 3x + 2.$
$4x - 3(3x + 2) = 4 \Rightarrow -5x = 10 \Rightarrow x = -2.$
$y = -6 + 2 = -4.$

9 **Ans:** $x = 4,\ y = 3$
Add: $7x = 28 \Rightarrow x = 4.$
$2(4) + 3y = 17 \Rightarrow y = 3.$

10 **Ans:** $x = \frac{1}{2},\ y = \frac{1}{3}$
$ax + by = \frac{3a + 2b}{6}$ and $bx - ay = \frac{3b - 2a}{6}$.
Multiply the first by a, the second by b and add: $(a^2 + b^2)x = \frac{3a^2 + 3b^2}{6}$, so $x = \frac{1}{2}$.
Back in the first: $\frac{a}{2} + by = \frac{a}{2} + \frac{b}{3} \Rightarrow y = \frac{1}{3}$.

11 **Ans:** $x = b,\ y = -a$
From the second: $by = ax - 2ab$.
First $\times ab$: $a^2x - b^2y = a^2b + ab^2$; substitute $b^2y = b(ax - 2ab)$: $a^2x - abx + 2ab^2 = a^2b + ab^2$.
$ax(a - b) = ab(a - b) \Rightarrow x = b$ (for $a \neq b$), then $by = ab - 2ab \Rightarrow y = -a.$

12 **Ans:** $x = 2,\ y = -1$
Add: $78x + 78y = 78 \Rightarrow x + y = 1.$
Subtract: $16x - 16y = 48 \Rightarrow x - y = 3.$
$x = 2,\ y = -1.$

13 **Ans:** $m = -1$ and $m = -2$
Substitute: $3x + 4(mx + 1) = 9 \Rightarrow x(3 + 4m) = 5 \Rightarrow x = \frac{5}{3 + 4m}$.
x is an integer when $3 + 4m \in \{\pm 1, \pm 5\}$: $m = -\frac{1}{2}, -1, \frac{1}{2}, -2$.
Integral m: -1 (gives $x = -5$) and -2 (gives $x = -1$).

14 **Ans:** $x = 5,\ y = -2;\ p = -17$
Multiply first by 3, second by 2: $9x + 6y = 33$, $4x + 6y = 8$.
Subtract: $5x = 25 \Rightarrow x = 5$; then $15 + 2y = 11 \Rightarrow y = -2$.
$-2 = 15 + p \Rightarrow p = -17.$

TONIGHT'S 2-MINUTE RECALL
Answer from memory, then check the notes

1. Substitution first step? : write one variable in terms of the other
2. Elimination first step? : make one coefficient equal
3. Substitution gives $0 = 5$? : no solution
4. Substitution gives $0 = 0$? : infinitely many solutions
5. Swapped coefficients trick? : add and subtract the equations
6. Solution of a pair is written as? : an ordered pair (x, y)
7. Is cross-multiplication in the 2026-27 syllabus? : No

DAY 7 COMPLETE?
☐ Notes read
☐ 14 questions solved ☐ Answers checked
☐ Mistakes noted
Next: Day 8, Pair of Linear Equations in Two Variables

MY MISTAKES AND NOTES
Write every mistake of this Day with the correct step

DAY 08

PAIR OF LINEAR EQUATIONS | CHAPTER 3 | PART 3 OF 3

Word Problems

Pair of Linear Equations in Two Variables

Aaj ka target

- [ ] Word problem padhte hi 'Let … = x, … = y' likho
- [ ] Ages, fractions, digits, speed, money: har type ka ek sawal
- [ ] Answer ko story mein wapas daal ke check karo
- [ ] 15 sawal khud se, phir answers milao (lagbhag 100 min)

8.1 Turning Words into Two Equations

Type	How to set up
Ages	Present ages x, y; after n years: $x+n,\ y+n$; n years ago: $x-n,\ y-n$
Fraction	Let it be $\frac{x}{y}$; change numerator/denominator as told
Two-digit number	Tens x, units y: number $= 10x+y$, reversed $= 10y+x$
Speed (same / opposite)	Relative speed $u-v$ (same direction), $u+v$ (opposite)
Boat in a stream	Downstream $u+v$, upstream $u-v$
Fixed + variable charge	Total $=$ fixed x + rate $y\times$ quantity
Geometry	Use angle sum $180°$, rectangle area lb, perimeter $2(l+b)$

Setting up the common types

SOLVED EXAMPLE 1 (AGES)

A father is three times as old as his son. After 12 years he will be twice as old as the son. Find their present ages.

SOLUTION

1. Let the father be x years and the son y years: $x = 3y$.
2. After 12 years: $x+12 = 2(y+12) \Rightarrow x - 2y = 12$.
3. Substitute: $3y - 2y = 12 \Rightarrow y = 12,\ x = 36$.
4. Check: $48 = 2 \times 24$.

Answer: Father 36 years, son 12 years

SOLVED EXAMPLE 2 (DIGITS)

The tens digit of a two-digit number is three times its units digit. Subtracting 54 from the number reverses its digits. Find the number.

SOLUTION

1. Let tens $= x$, units $= y$: $x = 3y$.
2. $(10x+y) - 54 = 10y + x \Rightarrow 9x - 9y = 54 \Rightarrow x - y = 6$.
3. $3y - y = 6 \Rightarrow y = 3,\ x = 9$.

Answer: The number is 93 (check: $93 - 54 = 39$).

MEME SAMJHO

Astra meme, maths inside



TOPPER OPENING ANY WORD PROBLEM

Sawal kitna bhi lamba ho, pehli line hamesha: 'Let the first be x and the second be y.'

Asli baat: Asli baat: Naming the unknowns clearly is the first marking step in every board word problem, and it stops mix-ups between x and y later.

SOLVED EXAMPLE 3 (SPEED)

Two cars start from places A and B, 80 km apart. Moving in the same direction they meet in 8 hours; moving towards each other they meet in 1 hour 20 minutes. Find their speeds.

SOLUTION

1. Let speeds be x and y km/h ($x > y$).
2. Same direction: $8(x-y) = 80 \Rightarrow x - y = 10$.
3. Opposite: $\frac{4}{3}(x+y) = 80 \Rightarrow x + y = 60$.
4. Add: $2x = 70 \Rightarrow x = 35,\ y = 25$.

Answer: 35 km/h and 25 km/h



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Time ko hamesha **hours** mein badlo: 1 hour 20 min $= \frac{4}{3}$ h, 40 min $= \frac{2}{3}$ h. Minutes aur hours mila diye to poora sawal galat.



SHIVAM BHAIYA SAYS

Digits wale sawal mein number aur reverse ka difference hamesha $9(x-y)$ hota hai aur sum $11(x+y)$. Yeh yaad ho to equation ek line mein ban jaati hai.



SHIVAM BHAIYA SAYS: BOARD TIP

Last mein answer **story ki language** mein likho: 'Cost of a pen = Rs 12', sirf '$x = 12$' nahi. Aur 'Let ...' line skip mat karna, uske marks alag se milte hain.



EXAM ALERT!

Linear equations ka 3 ya 5 marks ka word problem board paper ka pakka sawal hai, aur case-based (tickets, fares, canteen) mein bhi yahi aata hai. Har type ka kam se kam ek sawal khud solve karke jao.

Day 8 Practice

13 board-pattern questions | solve first, then check

MCQ

1 The length of a rectangular garden is 4 m more than its width and its perimeter is 48 m. Its dimensions are (1)

(a) 16 m by 8 m (b) 14 m by 10 m
(c) 12 m by 12 m (d) 15 m by 9 m

MCQ

2 If the tens digit of a two-digit number is x and the units digit is y, the number formed by reversing its digits is (1)

(a) $10x + y$ (b) xy (c) $10y + x$ (d) $x + y$

MCQ

3 A boat goes 24 km downstream in 2 hours and 16 km upstream in 2 hours. The speed of the stream is (1)

(a) 2 km/h (b) 4 km/h (c) 8 km/h (d) 10 km/h

MCQ

4 In $\triangle ABC$, $\angle C = 3\angle B = 2(\angle A + \angle B)$. Then $\angle B$ is (1)

(a) 20° (b) 60° (c) 40° (d) 120°

ASSERTION-REASON

5 **Assertion (A):** If the present ages of A and B are x and y years, then their ages 5 years ago were $x - 5$ and $y - 5$.
Reason (R): Every person's age changes by the same amount in the same time. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

CASE-BASED

6 Read the passage and answer the questions that follow. At a science fair, 600 tickets were sold. A child ticket costs Rs 100 and an adult ticket Rs 250. The total collection was Rs 99000. (4)

(i) Write the two equations (children x, adults y). (1)

(ii) How many adults attended? (1)

(iii) How many children attended, and how much more was collected from adults than from children? (2)

SHORT ANSWER

7 A taxi charges a fixed amount plus a rate per km. A 10 km ride costs Rs 105 and a 25 km ride costs Rs 255. What will a 30 km ride cost? (2)

SHORT ANSWER

8 A fraction becomes $\frac{1}{3}$ when 1 is subtracted from its numerator, and $\frac{1}{2}$ when 2 is subtracted from its denominator. Find the fraction. (3)

SHORT ANSWER

9 A farm has deer and ducks. There are 50 heads and 140 legs in all. How many of each? (3)

SHORT ANSWER

10 A mother's age is twice the sum of the ages of her two children. After 20 years, her age will be equal to the sum of their ages then. Find her present age. (3)

SHORT ANSWER

11 The sum of the digits of a two-digit number is 9. Nine times the number equals twice the number obtained by reversing its digits. Find the number. (3)

LONG ANSWER

12 If 8 students move from hall A to hall B, the two halls have equal numbers. If 12 students move from B to A, A has twice as many as B. Find the number in each hall. (4)

LONG ANSWER

13 If the length of a rectangle is increased by 5 m and its breadth decreased by 3 m, the area decreases by 15 m^2. If the length is decreased by 4 m and the breadth increased by 4 m, the area increases by 16 m^2. Find the dimensions. (5)

Day 8 Answers with worked solutions

1 **(b)**
$l = w + 4$ and $2(l + w) = 48 \Rightarrow 2w + 4 = 24 \Rightarrow w = 10,\ l = 14.$

2 **(c)**
Reversing puts y in the tens place: $10y + x$.

3 **(a)**
$u + v = 12,\ u - v = 8$: subtract, $2v = 4 \Rightarrow v = 2$ km/h.

4 **(c)**
$3B = 2A + 2B \Rightarrow B = 2A$, and $C = 3B = 6A$.
$A + 2A + 6A = 180^\circ \Rightarrow A = 20^\circ$, so $B = 40^\circ$.

5 **(a)**
Both true, and R is exactly why we subtract the same 5 from both.

6 **Ans:** $x + y = 600,\ 100x + 250y = 99000$
Count and money conditions.
Ans: 260
$100(600 - y) + 250y = 99000 \Rightarrow 150y = 39000 \Rightarrow y = 260.$
Ans: 340 children; Rs 31000 more
$x = 600 - 260 = 340.$
Adults: $260 \times 250 = 65000$; children: $340 \times 100 = 34000$; difference Rs 31000.

7 **Ans:** Rs 305
Let fixed charge Rs x, rate Rs y/km: $x + 10y = 105,\ x + 25y = 255.$
Subtract: $15y = 150 \Rightarrow y = 10,\ x = 5.$
30 km: $5 + 300 =$ Rs 305.

8 **Ans:** $\frac{5}{12}$
Let it be $\frac{x}{y}$: $3(x - 1) = y \Rightarrow 3x - y = 3.$
$2x = y - 2 \Rightarrow 2x - y = -2.$
Subtract: $x = 5,\ y = 12$. Check: $\frac{4}{12} = \frac{1}{3},\ \frac{5}{10} = \frac{1}{2}$.

9 **Ans:** 20 deer, 30 ducks
Let deer x, ducks y: $x + y = 50,\ 4x + 2y = 140.$
$4x + 2(50 - x) = 140 \Rightarrow 2x = 40 \Rightarrow x = 20,\ y = 30.$

10 **Ans:** 40 years
Let mother x, sum of children y: $x = 2y$.
After 20 years each child gains 20: $x + 20 = y + 40.$
$2y + 20 = y + 40 \Rightarrow y = 20,\ x = 40.$

11 **Ans:** 18
Tens x, units y: $x + y = 9.$
$9(10x + y) = 2(10y + x) \Rightarrow 88x = 11y \Rightarrow y = 8x.$
$9x = 9 \Rightarrow x = 1,\ y = 8$: number 18 (check $162 = 2 \times 81$).

12 **Ans:** A: 68, B: 52
Let A have x, B have y: $x - 8 = y + 8 \Rightarrow x - y = 16.$
$x + 12 = 2(y - 12) \Rightarrow x - 2y = -36.$
Subtract: $y = 52,\ x = 68$. Check: $60 = 60;\ 80 = 2 \times 40.$

13 **Ans:** Length 20 m, breadth 12 m
Let length l, breadth b.
$(l + 5)(b - 3) = lb - 15 \Rightarrow -3l + 5b - 15 = -15 \Rightarrow 3l = 5b.$
$(l - 4)(b + 4) = lb + 16 \Rightarrow 4l - 4b - 16 = 16 \Rightarrow l - b = 8.$
$l = b + 8$: $3b + 24 = 5b \Rightarrow b = 12,\ l = 20.$

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Two-digit number with tens x, units y? : $10x + y$
2. Ages after n years? : $x + n,\ y + n$
3. Relative speed, same direction? : $u - v$
4. Downstream speed of a boat? : $u + v$
5. Fixed charge plus rate: equation shape? : $x + ny =$ total
6. 1 h 20 min in hours? : $\frac{4}{3}$
7. Last line of a word problem? : answer in words with units

DAY 8 COMPLETE? ☐ Notes read
☐ 13 questions solved ☐ Answers checked
☐ Mistakes noted

Next: Day 9, Quadratic Equations

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 09

QUADRATIC EQUATIONS | CHAPTER 4 | PART 1 OF 2

Standard Form and Factorisation

Quadratic Equations

Aaj ka target

- [ ] Quadratic equation pehchaan lo: expand karke check karo
- [ ] Word problem se equation banana seekho
- [ ] Splitting the middle term se roots nikaalo
- [ ] Root check: value daalo, LHS = 0 aana chahiye

9.1 What is a Quadratic Equation?

KEY TERM: QUADRATIC EQUATION

An equation that can be written as $ax^2 + bx + c = 0$, where a, b, c are real numbers and $a \neq 0$. This is its **standard form**. The highest power of the variable is 2.

- If $a = 0$, the equation becomes $bx + c = 0$, which is **linear**, not quadratic.
- A quadratic equation has **at most two** roots.
- Always simplify both sides first: an x^2 term may cancel (then it is not quadratic), or an x^3 term may cancel (then it may become quadratic).

SOLVED EXAMPLE 1

Check whether each is a quadratic equation: (i) $(x-2)^2 + 1 = 2x - 3$ (ii) $x^3 - 4x^2 - x + 1 = (x-2)^3$

SOLUTION

1. (i) LHS $= x^2 - 4x + 4 + 1 = x^2 - 4x + 5$. So $x^2 - 4x + 5 = 2x - 3$.
2. $\Rightarrow x^2 - 6x + 8 = 0$: degree 2, so it **IS** quadratic.
3. (ii) RHS $= x^3 - 6x^2 + 12x - 8$. Subtract: $x^3 - 4x^2 - x + 1 - x^3 + 6x^2 - 12x + 8 = 0$.
4. $\Rightarrow 2x^2 - 13x + 9 = 0$: the x^3 terms cancel and it **IS** quadratic.

Answer: Both are quadratic equations.



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Bachcho, sirf dekh ke decide mat karo! $(x+2)^2 - 1 = (x-1)^2$ quadratic dikhta hai, par expand karo to x^2 cut jaata hai aur bachta hai $6x + 2 = 0$, jo linear hai. Pehle expand, phir judge.

9.2 Forming a Quadratic Equation from a Situation

1. Let the unknown be x (write what x stands for).
2. Translate each sentence into an expression in x.
3. Form the equation and bring it to $ax^2 + bx + c = 0$.

SOLVED EXAMPLE 2

Rs 3600 is shared equally among some children. If there were 6 fewer children, each would get Rs 20 more. Form the quadratic equation and find the number of children.

SOLUTION

1. Let the number of children be x. Share of each $= \frac{3600}{x}$.
2. With $(x-6)$ children: $\frac{3600}{x-6} - \frac{3600}{x} = 20$.
3. $\Rightarrow 3600 \times 6 = 20x(x-6) \Rightarrow x^2 - 6x - 1080 = 0$.
4. $x^2 - 36x + 30x - 1080 = 0 \Rightarrow (x-36)(x+30) = 0$.
5. $x = 36$ (reject $x = -30$, a count cannot be negative).

Answer: $x^2 - 6x - 1080 = 0$; there are **36 children**.

9.3 Roots of a Quadratic Equation

A real number α is a **root** of $ax^2 + bx + c = 0$ if $a\alpha^2 + b\alpha + c = 0$. The roots of the equation are the same as the zeroes of the polynomial $ax^2 + bx + c$.

SOLVED EXAMPLE 3

If $x = \frac{2}{3}$ is a root of $3x^2 + kx - 2 = 0$, find k and the other root.

SOLUTION

1. Put $x = \frac{2}{3}$: $3 \times \frac{4}{9} + \frac{2k}{3} - 2 = 0$.
2. $\frac{4}{3} + \frac{2k}{3} = 2 \Rightarrow 2k = 2 \Rightarrow k = 1$.
3. Equation: $3x^2 + x - 2 = 0 \Rightarrow (3x-2)(x+1) = 0$.

Answer: $k = 1$; the other root is -1.

9.4 Solving by Factorisation (Splitting the Middle Term)

SPLITTING RULE FOR $AX^2 + BX + C = 0$

Find two numbers p and q such that

$$p + q = b \quad \text{and} \quad p \times q = a \times c$$

Write $bx = px + qx$, group in pairs, take the common factor, and put each factor $= 0$.

SOLVED EXAMPLE 4

Solve: (i) $6x^2 - x - 2 = 0$ (ii) $\sqrt{3}x^2 + 10x + 7\sqrt{3} = 0$

SOLUTION

1. (i) $ac = -12$, $b = -1$: the numbers are -4 and 3.
2. $6x^2 - 4x + 3x - 2 = 2x(3x - 2) + 1(3x - 2) = (3x - 2)(2x + 1) = 0$, so $x = \frac{2}{3}, -\frac{1}{2}$.
3. (ii) $ac = \sqrt{3} \times 7\sqrt{3} = 21$, $b = 10$: the numbers are 7 and 3.
4. $\sqrt{3}x^2 + 7x + 3x + 7\sqrt{3} = x(\sqrt{3}x + 7) + \sqrt{3}(\sqrt{3}x + 7) = (\sqrt{3}x + 7)(x + \sqrt{3}) = 0$.
5. $x = -\sqrt{3}$ or $x = -\frac{7}{\sqrt{3}} = -\frac{7\sqrt{3}}{3}$.

Answer: (i) $\frac{2}{3}, -\frac{1}{2}$ (ii) $-\sqrt{3}, -\frac{7\sqrt{3}}{3}$



SHIVAM BHAIYA SAYS

Surd wale question mein $a \times c$ poora multiply karo, surd ke saath: $\sqrt{3} \times 7\sqrt{3} = 21$. Phir 21 ko aise todo ki sum 10 aaye: 7 aur 3. Ho gaya kaam!



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

$3x^2 = 6x$ mein dono side x se divide mat karna! Root $x = 0$ gayab ho jaayega. Sahi tarika: $3x^2 - 6x = 0 \Rightarrow 3x(x - 2) = 0 \Rightarrow x = 0, 2$.

9.5 Equations with Fractions

SOLVED EXAMPLE 5

Solve $\frac{1}{x+3} - \frac{1}{x-6} = \frac{9}{20}$, $x \neq -3, 6$.

SOLUTION

1. LHS $= \frac{(x-6)-(x+3)}{(x+3)(x-6)} = \frac{-9}{x^2 - 3x - 18}$.
2. So $\frac{-9}{x^2 - 3x - 18} = \frac{9}{20} \Rightarrow x^2 - 3x - 18 = -20$.
3. $x^2 - 3x + 2 = 0 \Rightarrow (x-1)(x-2) = 0$.

Answer: $x = 1$ or $x = 2$



SHIVAM BHAIYA SAYS: BOARD TIP

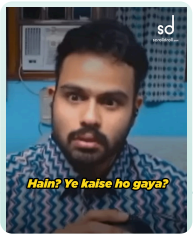
Board mein splitting wali line zaroor likho ($6x^2 - 4x + 3x - 2$). Seedha answer likha to step marks kat sakte hain. Aur negative ya fraction answer ko word problem mein reject karte waqt reason bhi likho.



EXAM ALERT!

Har saal 'form the quadratic equation and solve' wala 3-4 marks ka word problem aata hai. Equation banana hi aadha kaam hai!

MEME SAMJHO — Astra meme, maths inside



STUDENT EXPANDING $(X+2)^2 - 1 = (X-1)^2$ TO CALL IT QUADRATIC

Expand kiya aur x^2 hi cut gaya... Hain? Ye kaise ho gaya?

Asli baat: when the x^2 terms cancel, the equation is linear ($6x + 2 = 0$), not quadratic. Always simplify before you decide.

Day 9 Practice

15 board-pattern questions | solve first, then check

MCQ

1 Which of the following is a quadratic equation? (1)

(a) $(x+2)^2 = x^2 + 5$
(b) $x^3 + 2x = (x+1)^3$
(c) $x(x+1) + 3 = (x-2)(x+4)$
(d) $x^2 + \frac{1}{x^2} = 3$

MCQ

2 The roots of $4x^2 = 12x$ are (1)

(a) 3 only
(b) 0 and -3
(c) 0 and 3
(d) 4 and 12

MCQ

3 If $x = -\frac{1}{2}$ is a root of $2x^2 + kx - 3 = 0$, then $k =$ (1)

(a) 5 (b) -5 (c) $\frac{5}{2}$ (d) -1

MCQ

4 The roots of $x^2 - 2x - k(k+2) = 0$ are (1)

(a) k and $k+2$ (b) $-k$ and $-(k+2)$

(c) k and $-(k+2)$ (d) $-k$ and $k+2$

MCQ

5 Rs 4800 is divided equally among x students. If 6 more students join, each gets Rs 40 less. The equation is (1)

(a) $x^2 + 6x - 720 = 0$ (b) $x^2 - 6x - 720 = 0$

(c) $x^2 + 6x + 720 = 0$ (d) $x^2 + 40x - 4800 = 0$

ASSERTION-REASON

6 **Assertion (A):** $(x+2)^2 - 1 = (x-1)^2$ is a quadratic equation.
Reason (R): An equation of the form $ax^2 + bx + c = 0$ with $a \neq 0$ is a quadratic equation. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

7 **Assertion (A):** $x = 3$ is a root of $x^2 - 5x + 6 = 0$.
Reason (R): α is a root of $ax^2 + bx + c = 0$ if $a\alpha^2 + b\alpha + c = 0$. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

VERY SHORT ANSWER

8 Is $x = 0.3$ a root of $x^2 - 0.9 = 0$? Justify. (1)

SHORT ANSWER

9 Solve by factorisation: $10x^2 - x - 3 = 0$. (2)

SHORT ANSWER

10 Find the roots of $\sqrt{2}x^2 + 7x + 5\sqrt{2} = 0$ by factorisation. (2)

SHORT ANSWER

11 Solve: $\frac{1}{x+4} - \frac{1}{x-7} = \frac{11}{30}$, $x \neq -4, 7$. (3)

SHORT ANSWER

12 Solve for x: $8a^2x^2 - 2abx - 3b^2 = 0$, $a \neq 0$. (3)

LONG ANSWER

13 Meera's present age is 4 more than the square of her daughter Tara's age. When Tara reaches Meera's present age, Meera's age will be 2 less than 11 times Tara's present age. Find their present ages. (4)

LONG ANSWER

14 Riya's marks in Maths and Science add up to 40. Had she got 3 more marks in Maths and 4 fewer in Science, the product of her marks would have been 360. Find her marks in the two subjects. (5)

CASE-BASED

15 Case study: Lighthouse triangle A lighthouse L, a rock R and a buoy B form a right triangle with the right angle at the buoy. The shortest side is x m, the hypotenuse is $(2x+3)$ m and the third side is $(2x+2)$ m. (4)

(i) Write the quadratic equation in x. (1)

(ii) Find the length of the shortest side. (1)

(iii) Find the perimeter and the area of the triangle. (2)

Day 9 Answers with worked solutions

1 (b)
(a) and (c) reduce to linear equations after expanding; (d) is not a polynomial equation.
(b): $x^3 + 2x = x^3 + 3x^2 + 3x + 1 \Rightarrow 3x^2 + x + 1 = 0$, which is quadratic.

2 (c)
$4x^2 - 12x = 0 \Rightarrow 4x(x-3) = 0 \Rightarrow x = 0, 3$.

3 (b)
$2 \times \frac{1}{4} - \frac{k}{2} - 3 = 0 \Rightarrow \frac{1}{2} - \frac{k}{2} = 3 \Rightarrow k = -5$.

4 (d)
Sum of $-k$ and $k+2$ is 2 and product is $-k(k+2)$, matching the equation. So $(x+k)(x-k-2) = 0$.

5 (a)
$\frac{4800}{x} - \frac{4800}{x+6} = 40 \Rightarrow 4800 \times 6 = 40x(x+6)$.
$\Rightarrow x^2 + 6x - 720 = 0$ (which gives $x = 24$).

6 (d)
Expanding: $x^2 + 4x + 3 = x^2 - 2x + 1 \Rightarrow 6x + 2 = 0$, which is linear. So A is false; R is the correct definition, so R is true.

7 (a)
$9 - 15 + 6 = 0$, so A is true. R is the definition used to check it, so R explains A.

8 **Ans:** No

$(0.3)^2 - 0.9 = 0.09 - 0.9 = -0.81 \neq 0$, so 0.3 is not a root.

9 **Ans:** $x = \frac{3}{5}, -\frac{1}{2}$

$ac = -30$, $b = -1$: numbers -6 and 5.

$10x^2 - 6x + 5x - 3 = 2x(5x - 3) + 1(5x - 3) = (5x - 3)(2x + 1) = 0$.

$x = \frac{3}{5}$ or $x = -\frac{1}{2}$.

10 **Ans:** $x = -\sqrt{2}, -\frac{5\sqrt{2}}{2}$

$ac = \sqrt{2} \times 5\sqrt{2} = 10$, $b = 7$: numbers 5 and 2.

$\sqrt{2}x^2 + 5x + 2x + 5\sqrt{2} = x(\sqrt{2}x + 5) + \sqrt{2}(\sqrt{2}x + 5) = (\sqrt{2}x + 5)(x + \sqrt{2}) = 0$.

$x = -\sqrt{2}$ or $x = -\frac{5}{\sqrt{2}} = -\frac{5\sqrt{2}}{2}$.

11 **Ans:** $x = 1, 2$

LHS $= \frac{(x-7)-(x+4)}{(x+4)(x-7)} = \frac{-11}{x^2 - 3x - 28}$.

$\frac{-11}{x^2 - 3x - 28} = \frac{11}{30} \Rightarrow x^2 - 3x - 28 = -30$.

$x^2 - 3x + 2 = 0 \Rightarrow (x-1)(x-2) = 0 \Rightarrow x = 1, 2$.

12 **Ans:** $x = \frac{3b}{4a}, -\frac{b}{2a}$

Product $= 8a^2 \times (-3b^2) = -24a^2b^2$, sum $= -2ab$: the parts are $-6ab$ and $4ab$.

$8a^2x^2 - 6abx + 4abx - 3b^2 = 2ax(4ax - 3b) + b(4ax - 3b) = (4ax - 3b)(2ax + b) = 0$.

$x = \frac{3b}{4a}$ or $x = -\frac{b}{2a}$.

13 **Ans:** Tara 5 years, Meera 29 years

Let Tara's age be s. Then Meera's age $= s^2 + 4$.

Tara reaches Meera's age after $(s^2 + 4 - s)$ years; Meera will then be $(s^2 + 4) + (s^2 + 4 - s) = 2s^2 - s + 8$.

$2s^2 - s + 8 = 11s - 2 \Rightarrow 2s^2 - 12s + 10 = 0 \Rightarrow s^2 - 6s + 5 = 0$.

$(s-1)(s-5) = 0 \Rightarrow s = 1$ or 5.

$s = 1$ gives Meera 5 years, impossible for a mother, so $s = 5$, Meera $= 29$.

14 **Ans:** Maths 12, Science 28; or Maths 21, Science 19

Let Maths marks $= x$, Science $= 40 - x$.

$(x + 3)(40 - x - 4) = 360 \Rightarrow (x + 3)(36 - x) = 360$.

$36x - x^2 + 108 - 3x = 360 \Rightarrow x^2 - 33x + 252 = 0$.

$(x - 12)(x - 21) = 0 \Rightarrow x = 12$ or 21.

Maths 12, Science 28; or Maths 21, Science 19.

Check: $15 \times 24 = 360$ and $24 \times 15 = 360$.

15 **Ans:** $x^2 - 4x - 5 = 0$

$(2x + 3)^2 = (2x + 2)^2 + x^2 \Rightarrow 4x + 5 = x^2 \Rightarrow x^2 - 4x - 5 = 0$.

Ans: 5 m

$(x - 5)(x + 1) = 0 \Rightarrow x = 5$ (length cannot be -1).

Ans: Perimeter 30 m, area 30 m^2

Sides: 5 m, 12 m, 13 m.

Perimeter $= 5 + 12 + 13 = 30$ m; area $= \frac{1}{2} \times 5 \times 12 = 30$ m^2.

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Standard form : $ax^2 + bx + c = 0,\ a \neq 0$
2. If $a = 0$ the equation is : linear
3. Maximum number of roots : two
4. Splitting rule : $p + q = b$, $pq = ac$
5. Roots of $3x^2 = 6x$: 0 and 2
6. Is $(x + 2)^2 - 1 = (x - 1)^2$ quadratic? : No, it is linear
7. Check a root : put it in, LHS must be 0

DAY 9 COMPLETE? ☐ Notes read ☐ 15 questions solved ☐ Answers checked ☐ Mistakes noted

Next: Day 10, Quadratic Equations

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 10

QUADRATIC EQUATIONS | CHAPTER 4 | PART 2 OF 2

Quadratic Formula, Discriminant and Word Problems

Quadratic Equations

Aaj ka target

- [ ] Quadratic formula yaad aur fast lagana seekho
- [ ] Equal roots wale 'find k' questions pakke karo
- [ ] $D = b^2 - 4ac$ se nature of roots batao
- [ ] Speed, time, work wale word problems solve karo

10.1 The Quadratic Formula

Every quadratic equation $ax^2 + bx + c = 0$ $(a \neq 0)$ with $b^2 - 4ac \geq 0$ can be solved by the **quadratic formula**, even when factorisation is hard to spot.

QUADRATIC FORMULA

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Discriminant:

$$D = b^2 - 4ac$$

SOLVED EXAMPLE 1

Solve using the quadratic formula: (i) $2x^2 - 7x + 3 = 0$ (ii) $x^2 + 4x - 1 = 0$

SOLUTION

1. (i) $a = 2, b = -7, c = 3$; $D = 49 - 24 = 25$.
2. $x = \frac{7 \pm 5}{4}$, so $x = 3$ or $x = \frac{1}{2}$.
3. (ii) $a = 1, b = 4, c = -1$; $D = 16 + 4 = 20$, $\sqrt{20} = 2\sqrt{5}$.
4. $x = \frac{-4 \pm 2\sqrt{5}}{2} = -2 \pm \sqrt{5}$.

Answer: (i) $3, \frac{1}{2}$ (ii) $-2 + \sqrt{5}, -2 - \sqrt{5}$



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

b negative ho to $-b$ positive ho jaata hai: $b = -7$ to $-b = +7$. Aur b^2 hamesha positive: $(-7)^2 = 49$, -49 nahi! Sabse zyada marks yahin katte hain.

10.2 Nature of Roots (Discriminant)

Value of $D = b^2 - 4ac$	Nature of roots
$D > 0$	Two distinct real roots
$D = 0$	Two equal real roots, each $= -\frac{b}{2a}$
$D < 0$	No real roots

Discriminant test

- 'Real roots' means $D \geq 0$ (equal roots are also real).
- 'Distinct real roots' means $D > 0$.
- 'Equal roots' or 'coincident roots' means $D = 0$.

SOLVED EXAMPLE 2

Find the nature of the roots of $3x^2 - 4\sqrt{3}x + 4 = 0$. If real, find them.

SOLUTION

1. $a = 3, b = -4\sqrt{3}, c = 4$.
2. $D = (-4\sqrt{3})^2 - 4(3)(4) = 48 - 48 = 0$, so the roots are real and equal.
3. Each root $= -\frac{b}{2a} = \frac{4\sqrt{3}}{6} = \frac{2\sqrt{3}}{3}$.

Answer: Real and equal; $x = \frac{2\sqrt{3}}{3}, \frac{2\sqrt{3}}{3}$

SOLVED EXAMPLE 3

Find k so that $3x^2 - kx + 12 = 0$ has equal roots.

SOLUTION

1. Equal roots $\Rightarrow D = 0$.
2. $(-k)^2 - 4(3)(12) = 0 \Rightarrow k^2 = 144$.
3. $k = \pm 12$.

Answer: $k = 12$ or $k = -12$



SHIVAM BHAIYA SAYS

Pehle D nikaal lo. Agar D perfect square aaya (1, 4, 25, 49...) to roots rational hain aur factorisation bhi ho jaayega. Agar perfect square nahi hai to seedha formula lagao.



SHIVAM BHAIYA SAYS: BOARD TIP

Completing the square method 2026-27 syllabus se hata diya gaya hai. Formula se hi solve karo. Aur question mein 'by quadratic formula' likha ho to formula wali line zaroor dikhao.

10.3 Word Problems on Speed, Time and Work

1. Take the unknown (speed, time or number) as x.

1. Write time $= \frac{\text{distance}}{\text{speed}}$ or work per hour $= \frac{1}{\text{time}}$ for each case.
2. Form the equation from the difference or the total given.
3. Solve and reject values that make no sense (negative speed, negative time).

SOLVED EXAMPLE 4

A plane covers 1200 km. If its speed were 100 km/h more, it would take 1 hour less. Find its usual speed.

SOLUTION

1. Let the speed be x km/h. $\frac{1200}{x} - \frac{1200}{x+100} = 1$.
2. $1200 \times 100 = x(x+100) \Rightarrow x^2 + 100x - 120000 = 0$.
3. $D = 10000 + 480000 = 490000$, $\sqrt{D} = 700$.
4. $x = \frac{-100 \pm 700}{2} = 300$ or -400 (rejected).

Answer: Usual speed $= 300$ km/h

SOLVED EXAMPLE 5

A boat goes 36 km upstream and 36 km back. Its speed in still water is 15 km/h, and the upstream trip takes 1 hour more. Find the speed of the stream.

SOLUTION

1. Let the stream speed be x km/h. Upstream speed $= 15 - x$, downstream $= 15 + x$.
2. $\frac{36}{15-x} - \frac{36}{15+x} = 1 \Rightarrow 72x = 225 - x^2$.
3. $x^2 + 72x - 225 = 0$; $D = 5184 + 900 = 6084 = 78^2$.
4. $x = \frac{-72 + 78}{2} = 3$ (reject -75).

Answer: Speed of stream $= 3$ km/h



EXAM ALERT!

'Find k for equal roots' aur 'nature of roots' lagbhag har paper mein 1-2 marks ke aate hain. Speed-time wala word problem 3-5 marks ka favourite hai!

MEME SAMJHO — Astra meme, maths inside



STUDENT STUCK TRYING TO SPLIT THE MIDDLE TERM

Middle term split nahi ho raha? Quadratic formula laga do, woh kabhi mana nahi karta

Asli baat: the formula $x = \frac{-b \pm \sqrt{D}}{2a}$ works for every quadratic with $D \geq 0$; factorisation is only quicker when D is a perfect square.

Day 10 Practice

16 board-pattern questions | solve first, then check

MCQ

1 The discriminant of $2x^2 - 5x + 3 = 0$ is (1)

(a) 49 (b) 1 (c) -1 (d) 25

MCQ

2 Which equation has no real roots? (1)

(a) $x^2 - 4x + 4 = 0$ (b) $3x^2 + 2x - 1 = 0$
(c) $x^2 - 7 = 0$ (d) $2x^2 + x + 5 = 0$

MCQ

3 If $x^2 - kx + 16 = 0$ has equal roots, then $k =$ (1)

(a) ± 4 (b) ± 8 (c) 8 only (d) ± 16

MCQ

4 For $x^2 - 6x + 3 = 0$, the sum of the roots is ___ their product. (1)

(a) half (b) equal to
(c) thrice (d) twice

MCQ

5 In $px^2 + qx + r = 0$, p, q, r are positive and $q^2 - 4pr > 0$. Then the roots are (1)

(a) real, distinct and both negative
(b) real, distinct and both positive
(c) equal
(d) not real

ASSERTION-REASON

6 **Assertion (A):** $3x^2 - 4x + 2 = 0$ has no real roots.
Reason (R): If $b^2 - 4ac < 0$, a quadratic equation has no real roots. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

7 **Assertion (A):** $x^2 + 5x + 7 = 0$ has two distinct real roots.
Reason (R): A quadratic equation has at most two roots. (1)
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

VERY SHORT ANSWER

8 Find the values of k for which $x^2 + 6x + k = 0$ has real roots. (1)

VERY SHORT ANSWER

9 Find the discriminant of $\sqrt{3}x^2 + 2\sqrt{2}x - 2\sqrt{3} = 0$. (1)

SHORT ANSWER

10 Solve $3x^2 - 5x - 2 = 0$ using the quadratic formula. (2)

SHORT ANSWER

11 Solve $x^2 - 2x - 4 = 0$. (2)

SHORT ANSWER

12 Find k for which $(k+1)x^2 - 2(k-1)x + 1 = 0$ has equal roots. (3)

SHORT ANSWER

13 Using the quadratic formula, a student got $x = \dfrac{-3 \pm \sqrt{41}}{4}$ as the roots of $2x^2 + bx + c = 0$. Find b and c. (3)

LONG ANSWER

14 Because of bad weather, an aircraft flying 1200 km had to slow down by 200 km/h, and the trip took 1 hour longer. Find the scheduled duration of the flight. (4)

LONG ANSWER

15 Two taps together fill a tank in 6 hours. The larger tap alone takes 5 hours less than the smaller tap alone. Find the time each tap takes alone. (5)

CASE-BASED

16 Case study: Train journey A train covers 480 km at a uniform speed. If the speed had been 8 km/h more, the journey would have taken 2 hours less. Let the speed be x km/h. (4)

(i) Form the quadratic equation in x. (1)

(ii) Find the discriminant and state the nature of the roots. (1)

(iii) Find the speed of the train and the time taken. (2)

Day 10 Answers with worked solutions

1 **(b)**
$D = (-5)^2 - 4(2)(3) = 25 - 24 = 1.$

2 **(d)**
D values: 0, 16, 28 and $1 - 40 = -39$. Only (d) has $D < 0$.

3 **(b)**
$D = k^2 - 64 = 0 \Rightarrow k = \pm 8.$

4 **(d)**
Sum $= 6$, product $= 3$; $6 = 2 \times 3$.

5 **(a)**
$D > 0$: real and distinct.
Sum $= -\dfrac{q}{p} < 0$ and product $= \dfrac{r}{p} > 0$, so both roots are negative.

6 **(a)**
$D = 16 - 24 = -8 < 0$, so A is true, and R is exactly the reason.

7 **(d)**
$D = 25 - 28 = -3 < 0$, so there are no real roots: A is false. R is true.

8 **Ans:** $k \le 9$
Real roots $\Rightarrow D \ge 0$: $36 - 4k \ge 0 \Rightarrow k \le 9$.

9 **Ans:** 32
$D = (2\sqrt{2})^2 - 4(\sqrt{3})(-2\sqrt{3}) = 8 + 24 = 32.$

10 **Ans:** $x = 2, -\dfrac{1}{3}$
$D = 25 + 24 = 49.$
$x = \dfrac{5 \pm 7}{6}$, so $x = 2$ or $x = -\dfrac{1}{3}$.

11 **Ans:** $x = 1 \pm \sqrt{5}$
$D = 4 + 16 = 20$, $\sqrt{20} = 2\sqrt{5}$.
$x = \dfrac{2 \pm 2\sqrt{5}}{2} = 1 \pm \sqrt{5}.$

12 **Ans:** $k = 0$ or $k = 3$
$D = 4(k-1)^2 - 4(k+1)(1) = 0.$
$k^2 - 2k + 1 - k - 1 = 0 \Rightarrow k^2 - 3k = 0.$
$k(k-3) = 0 \Rightarrow k = 0$ or 3 (both keep $k + 1 \neq 0$).

13 **Ans:** $b = 3, c = -4$
$2a = 4$ so $a = 2$; $-b = -3$ so $b = 3$.
$b^2 - 4ac = 41 \Rightarrow 9 - 8c = 41 \Rightarrow c = -4.$
Equation: $2x^2 + 3x - 4 = 0.$

14 **Ans:** 2 hours

Let the usual speed be x km/h. $\frac{1200}{x-200} - \frac{1200}{x} = 1$.

$1200 \times 200 = x(x-200) \Rightarrow x^2 - 200x - 240000 = 0$.

$D = 40000 + 960000 = 1000000$, $\sqrt{D} = 1000$.

$x = \frac{200 + 1000}{2} = 600$ (reject -400).

Scheduled duration $= \frac{1200}{600} = 2$ hours.

15 **Ans:** Smaller 15 h, larger 10 h

Let the smaller tap take x h; the larger takes $(x-5)$ h.

$\frac{1}{x} + \frac{1}{x-5} = \frac{1}{6} \Rightarrow 6(2x-5) = x(x-5)$.

$x^2 - 17x + 30 = 0 \Rightarrow (x-15)(x-2) = 0$.

$x = 2$ gives $x - 5 < 0$, so reject; $x = 15$.

Smaller tap 15 h, larger tap 10 h.

16 **Ans:** $x^2 + 8x - 1920 = 0$

$\frac{480}{x} - \frac{480}{x+8} = 2 \Rightarrow 3840 = 2x(x+8) \Rightarrow x^2 + 8x - 1920 = 0$.

Ans: $D = 7744$; two distinct real roots

$D = 64 + 7680 = 7744 = 88^2 > 0$.

Ans: 40 km/h; 12 hours

$x = \frac{-8 + 88}{2} = 40$ (reject -48).

Time $= \frac{480}{40} = 12$ hours.

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Quadratic formula : $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
2. $D > 0$: two distinct real roots
3. $D = 0$: two equal roots, each $-\frac{b}{2a}$
4. $D < 0$: no real roots
5. 'Real roots' condition : $D \geq 0$
6. k for equal roots of $x^2 - kx + 16 = 0$: ± 8
7. Upstream speed : boat speed minus stream speed
8. Completing the square in 2026-27? : deleted, use the formula

DAY 10 COMPLETE?

- [ ] Notes read
- [ ] 16 questions solved
- [ ] Answers checked
- [ ] Mistakes noted

Next: Day 11, Revision 2: Linear Equations + Quadratic Equations

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 11

REVISION 2 | LINEAR EQUATIONS + QUADRATIC EQUATIONS

Linear Equations + Quadratic Equations

Revision 2

Aaj ka target

- [ ] Consistency ka ratio table zubaani yaad karo
- [ ] Substitution aur elimination dono fast karo
- [ ] Quadratic formula, D aur 'find k' ek baar phir
- [ ] 20 mixed questions timer laga ke solve karo

11.1 Pair of Linear Equations: One-Page Recap

Ratio condition	Graph	Solutions	Type
$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$	Intersecting lines	Exactly one	Consistent
$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$	Coincident lines	Infinitely many	Consistent (dependent)
$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$	Parallel lines	None	Inconsistent

For $a_1x + b_1y + c_1 = 0$ *and* $a_2x + b_2y + c_2 = 0$

- **Substitution:** make x or y the subject in one equation and put it in the other.
- **Elimination:** make the coefficients of one variable equal, then add or subtract.
- Graph: plot two points of each line; the meeting point is the solution.

11.2 Quadratic Equations: One-Page Recap

MUST-KNOW RESULTS

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}, \qquad D = b^2 - 4ac$$

$D > 0$: two distinct real roots; $D = 0$: equal roots $-\frac{b}{2a}$; $D < 0$: no real roots

Sum of roots $= -\frac{b}{a}$, product $= \frac{c}{a}$



SHIVAM BHAIYA SAYS

Linear pair ka type batana ho to teen ratios ek line mein likho: $\frac{a_1}{a_2}, \frac{b_1}{b_2}, \frac{c_1}{c_2}$. Pehle do equal nahi to unique solution, wahin ruk jao. Equal hain tabhi teesra dekho.



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

'Infinitely many solutions' ke liye teeno ratios equal chahiye, sirf pehle do nahi. Aur 'find k' mein jo value baaki condition tod de (jaise $k = -6$ yahan), use reject karo aur reason likho.



SHIVAM BHAIYA SAYS: BOARD TIP

Word problem (linear ho ya quadratic) mein pehli line hamesha 'Let ... be x' likho. Equation banana hi aadhe marks hai. Answer ke saath units aur check bhi likho.



EXAM ALERT!

Linear + Quadratic milke board mein lagbhag 10-12 marks laate hain. Ek word problem (3-5 marks) inme se aana almost pakka hai!

MEME SAMJHO

Astra meme, maths inside



STUDENT WHO WRITES THE ROOTS WITHOUT PUTTING THEM BACK IN THE EQUATION

Roots nikaal ke check nahi kiye aur bolta hai pakka sahi hai... guarantee deta hai??

Asli baat: put each root back into the original equation (or check sum $= -\frac{b}{a}$ and product $= \frac{c}{a}$). Ten seconds of checking saves a whole question.

Day 11 Practice

20 board-pattern questions | solve first, then check

MCQ

1 A pair of linear equations is consistent when its lines are (1)

(a) parallel
(b) intersecting or coincident
(c) parallel or coincident
(d) always coincident

MCQ

2 The line $y = 5$ is (1)

(a) parallel to the y-axis (b) the x-axis itself
(c) parallel to the x-axis (d) through the origin

MCQ

3 The pair $2x - 3y + 4 = 0$ and $6x - 9y - p = 0$ has no solution when (1)

(a) $p = -12$ (b) $p = 12$ (c) $p \neq -12$ (d) $p \neq 12$

MCQ

4 The equation $x^2 - ax + 4 = 0$ has no real roots when (1)

(a) $a > 4$ (b) $-4 < a < 4$ (c) $a < -4$ (d) $a = \pm 4$

MCQ

5 If $x = \frac{3}{2}$ is a root of $2x^2 + px - 6 = 0$, then p and the other root are (1)

(a) $p = -1$, other root 2 (b) $p = 1$, other root 2
(c) $p = 1$, other root -2 (d) $p = 2$, other root -1

MCQ

6 The sum and product of the roots of $2x^2 - 9x + 4 = 0$ are (1)

(a) $\frac{9}{2}$ and 2 (b) $-\frac{9}{2}$ and 2
(c) 9 and 4 (d) $\frac{9}{2}$ and 4

ASSERTION-REASON

7 **Assertion (A):** The pair $kx + 2y - 5 = 0$ and $3x + y - 1 = 0$ has no solution when $k = 6$.
Reason (R): A pair has no solution when $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

8 **Assertion (A):** $x^2 - 2x + 2 = 0$ has two distinct real roots.
Reason (R): If $D < 0$, the equation has no real roots. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

VERY SHORT ANSWER

9 Find the discriminant of $4x^2 - 5 = 0$ and state the nature of its roots. (1)

SHORT ANSWER

10 Solve by substitution: $y - 3x = 2$ and $5x - 2y = -5$. (2)

SHORT ANSWER

11 Find k for which $kx + 3y = k - 3$ and $12x + ky = k$ have infinitely many solutions. (2)

SHORT ANSWER

12 A rectangular park has perimeter 50 m and area 150 m^2. Find its length and breadth. (2)

SHORT ANSWER

13 A taxi charges a fixed amount plus a rate per km. A 10 km ride costs Rs 105 and a 25 km ride costs Rs 240. What does a 30 km ride cost? (3)

SHORT ANSWER

14 If one root of $ax^2 + bx + c = 0$ is twice the other, prove that $2b^2 = 9ac$. (3)

SHORT ANSWER

15 A father's age is four times the sum of the ages of his two children. After 10 years, his age will be twice the sum of their ages. Find the father's present age. (3)

LONG ANSWER

16 If the length of a rectangle is increased by 2 m and its breadth decreased by 2 m, its area decreases by 28 m^2. If the length is decreased by 3 m and the breadth increased by 2 m, its area increases by 12 m^2. Find its dimensions. (4)

LONG ANSWER

17 **A square chessboard has 49 equal squares, each of area 4 cm^2, and a 2 cm wide border all around. Form a quadratic equation for the side x of the whole board and find x. (4)**

CASE-BASED

18 **Case study: Two gears Two circular gears touch each other externally. The distance between their centres is 12 cm and the sum of their areas is 74π cm^2. Let the radius of one gear be x cm. (4)**

(i) Write the quadratic equation in x. (1)

(ii) State the nature of its roots. (1)

(iii) Find the radii of the two gears. (2)

LONG ANSWER

19 **A boat goes 36 km upstream and returns. Its speed in still water is 15 km/h and the upstream trip takes 1 hour more than the downstream trip. Also solve: two numbers differ by 3 and their product is 180; find them (positive). (5)**

VERY SHORT ANSWER

20 **For what value of k does $2x^2 + kx + 8 = 0$ have equal roots? (1)**

Day 11 Answers with worked solutions

1 **(b)**
Consistent means at least one solution: intersecting (one) or coincident (infinitely many).

2 **(c)**
$y = 5$ has the same y-value at every point, so it is a horizontal line 5 units above the x-axis.

3 **(c)**
$\frac{6}{2} = \frac{-9}{-3} = 3$; no solution needs $\frac{-p}{4} \neq 3$, i.e. $p \neq -12$.

4 **(b)**
$D = a^2 - 16 < 0 \Rightarrow a^2 < 16 \Rightarrow -4 < a < 4$.

5 **(c)**
$2 \times \frac{9}{4} + \frac{3p}{2} - 6 = 0 \Rightarrow p = 1$.
$2x^2 + x - 6 = (2x - 3)(x + 2) = 0$, so the other root is -2.

6 **(a)**
Sum $= -\frac{b}{a} = \frac{9}{2}$; product $= \frac{c}{a} = \frac{4}{2} = 2$.

7 **(a)**
With $k = 6$: $\frac{6}{3} = \frac{2}{1} = 2$ and $\frac{-5}{-1} = 5 \neq 2$. A is true and R is the reason.

8 **(d)**
$D = 4 - 8 = -4 < 0$, so there are no real roots: A is false, R is true.

9 **Ans:** $D = 80$; two distinct real roots
$a = 4, b = 0, c = -5$: $D = 0 - 4(4)(-5) = 80 > 0$.

10 **Ans:** $x = 1, y = 5$
From the first: $y = 3x + 2$.
$5x - 2(3x + 2) = -5 \Rightarrow -x - 4 = -5 \Rightarrow x = 1$.
$y = 3(1) + 2 = 5$.

11 **Ans:** $k = 6$
Need $\frac{k}{12} = \frac{3}{k} = \frac{k-3}{k}$.
$\frac{k}{12} = \frac{3}{k} \Rightarrow k^2 = 36 \Rightarrow k = \pm 6$.
$\frac{3}{k} = \frac{k-3}{k} \Rightarrow k - 3 = 3 \Rightarrow k = 6$. So $k = 6$ ($k = -6$ fails the second condition).

12 **Ans:** 15 m and 10 m
$l + b = 25$, so $b = 25 - l$; $l(25 - l) = 150$.
$l^2 - 25l + 150 = 0 \Rightarrow (l - 15)(l - 10) = 0$.
Length 15 m, breadth 10 m.

13 **Ans:** Rs 285
Let the fixed charge be Rs x and the rate Rs y per km: $x + 10y = 105$, $x + 25y = 240$.
Subtract: $15y = 135 \Rightarrow y = 9$; $x = 15$.
30 km: $15 + 30 \times 9 = 285$.

14 **Ans:** Proved
Let the roots be α and 2α.
Sum: $3\alpha = -\frac{b}{a} \Rightarrow \alpha = -\frac{b}{3a}$.
Product: $2\alpha^2 = \frac{c}{a} \Rightarrow 2 \times \frac{b^2}{9a^2} = \frac{c}{a}$.
$\Rightarrow 2b^2 = 9ac$.

15 **Ans:** 60 years
Let the children's ages add up to s and the father's age be f: $f = 4s$.
After 10 years: $f + 10 = 2(s + 20)$ (each child gains 10 years).
$4s + 10 = 2s + 40 \Rightarrow s = 15$, so $f = 60$.

16 **Ans:** Length 18 m, breadth 6 m
$(l + 2)(b - 2) = lb - 28 \Rightarrow -2l + 2b - 4 = -28 \Rightarrow l - b = 12$.
$(l - 3)(b + 2) = lb + 12 \Rightarrow 2l - 3b - 6 = 12 \Rightarrow 2l - 3b = 18$.
Put $l = b + 12$: $2b + 24 - 3b = 18 \Rightarrow b = 6$, $l = 18$.
Check: $20 \times 4 = 80 = 108 - 28$ and $15 \times 8 = 120 = 108 + 12$.

17 **Ans:** $x^2 - 8x - 180 = 0$; $x = 18$ cm
Playing area $= 49 \times 4 = 196$ cm^2, so its side is 14 cm.
Board side x: playing side $= x - 4$, so $(x - 4)^2 = 196$.
$x^2 - 8x + 16 - 196 = 0 \Rightarrow x^2 - 8x - 180 = 0$.
$(x - 18)(x + 10) = 0 \Rightarrow x = 18$ cm.

18 **Ans:** $x^2 - 12x + 35 = 0$

$\pi x^2 + \pi(12 - x)^2 = 74\pi \Rightarrow 2x^2 - 24x + 70 = 0 \Rightarrow x^2 - 12x + 35 = 0.$

Ans: Two distinct real roots

$D = 144 - 140 = 4 > 0.$

Ans: 5 cm and 7 cm

$(x - 5)(x - 7) = 0 \Rightarrow x = 5$ or 7; the other radius is $12 - x$.

Radii: 5 cm and 7 cm.

19 **Ans:** Stream 3 km/h; numbers 12 and 15

Boat: $\frac{36}{15 - x} - \frac{36}{15 + x} = 1 \Rightarrow x^2 + 72x - 225 = 0 \Rightarrow x = 3$ km/h.

Numbers: $n(n + 3) = 180 \Rightarrow n^2 + 3n - 180 = 0 \Rightarrow (n + 15)(n - 12) = 0.$

Positive numbers: 12 and 15.

20 **Ans:** $k = \pm 8$

$D = k^2 - 64 = 0 \Rightarrow k = \pm 8.$

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Unique solution : $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$
2. No solution : $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$
3. Infinitely many : all three ratios equal
4. Quadratic formula : $x = \frac{-b \pm \sqrt{D}}{2a}$
5. Equal roots : $D = 0$
6. Sum and product of roots : $-\frac{b}{a}$ and $\frac{c}{a}$
7. One root twice the other : $2b^2 = 9ac$

DAY 11 COMPLETE?

- [ ] Recap read
- [ ] 20 questions solved
- [ ] Answers checked
- [ ] Weak topics noted

Next: Day 12, Arithmetic Progressions

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 12

ARITHMETIC PROGRESSIONS | CHAPTER 5 | PART 1 OF 3

What is an AP, and its nth Term

Arithmetic Progressions

Aaj ka target

- [ ] AP pehchaano: har baar same difference d
- [ ] 'Kaunsa term hai?' wale questions mein n natural number check karo
- [ ] $a_n=a+(n-1)d$ ko ulta-seedha lagana seekho
- [ ] Last se nth term aur 3-digit counting wale sums pakke karo

12.1 Arithmetic Progression (AP)

KEY TERM: ARITHMETIC PROGRESSION

A list of numbers in which each term (after the first) is obtained by adding a fixed number d to the previous term. d is the **common difference**; it can be positive, negative or zero.

GENERAL FORM AND COMMON DIFFERENCE

$$a,\ a+d,\ a+2d,\ a+3d,\ \ldots$$

$$d = a_2 - a_1 = a_3 - a_2 = \cdots = a_{k+1} - a_k$$

- $5, 9, 13, 17$: $d = 4$ (increasing AP).
- $20, 17, 14, 11$: $d = -3$ (decreasing AP).
- $-3, -3, -3$: $d = 0$ (constant AP).
- $1, 4, 9, 16$ is **not** an AP: differences 3, 5, 7 are not equal.



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

d hamesha **baad wala minus pehle wala**: $d = a_2 - a_1$. $20, 17, 14$ mein $d = 17 - 20 = -3$ hai, $+3$ nahi. Sign galat to poora answer galat!

12.2 The nth Term

NTH TERM (GENERAL TERM)

$$a_n = a + (n-1)d$$

nth term from the end (last term l):

$$l - (n-1)d$$

Three numbers a, b, c are in AP iff

$$2b = a + c$$

SOLVED EXAMPLE 1

Find the 25th term of the AP $5, 2, -1, -4, \ldots$

SOLUTION

1. $a = 5$, $d = 2 - 5 = -3$.
2. $a_{25} = 5 + 24 \times (-3) = 5 - 72 = -67$.

Answer: $a_{25} = -67$

SOLVED EXAMPLE 2

Which term of the AP $3, 8, 13, \ldots$ is 248? Is 302 a term of this AP?

SOLUTION

1. $a = 3, d = 5$. Put $a_n = 248$: $3 + (n-1)5 = 248 \Rightarrow n - 1 = 49 \Rightarrow n = 50$.
2. For 302: $3 + (n-1)5 = 302 \Rightarrow n - 1 = \dfrac{299}{5} = 59.8$.
3. n is not a natural number, so 302 is not a term.

Answer: 248 is the 50th term; 302 is not a term.

SOLVED EXAMPLE 3

The 5th term of an AP is 19 and the 11th term is 43. Find the AP.

SOLUTION

1. $a + 4d = 19$ and $a + 10d = 43$.
2. Subtract: $6d = 24 \Rightarrow d = 4$; then $a = 19 - 16 = 3$.

Answer: $3, 7, 11, 15, \ldots$



SHIVAM BHAIYA SAYS

Do terms diye hon to direct trick: $d = \dfrac{a_q - a_p}{q - p}$. Jaise $a_{11} = 43, a_5 = 19$: $d = \dfrac{24}{6} = 4$. Ek line mein d mil gaya!

SOLVED EXAMPLE 4

How many three-digit numbers are divisible by 7?

SOLUTION

1. The first is 105 and the last is 994, with $d = 7$.
2. $994 = 105 + (n-1)7 \Rightarrow n - 1 = 127 \Rightarrow n = 128$.

Answer: 128 numbers

SOLVED EXAMPLE 5

Find the 10th term from the end of the AP $4, 9, 14, \ldots, 254$.

SOLVED EXAMPLE 5

SOLUTION

1. Last term $l = 254$, $d = 5$.
2. 10th term from the end $= l - (n-1)d = 254 - 9 \times 5 = 209$.

Answer: 209



SHIVAM BHAIYA SAYS

Agar $a_n = pn + q$ form mein ho (jaise $a_n = 7 - 4n$), to woh pakka AP hai aur d = n ka coefficient $= -4$. Terms likhne ki zaroorat hi nahi.



SHIVAM BHAIYA SAYS: BOARD TIP

'Is X a term?' wale question mein n nikaal ke likho: 'n natural number nahi hai, isliye X term nahi hai.' Yeh conclusion line pe hi mark milta hai.



EXAM ALERT!

AP chapter ka weight acha hai: nth term ke 1-2 mark MCQ har saal, aur 'which term' ya 'how many numbers divisible by' wala 2-3 marker bhi common hai.

MEME SAMJHO

Astra meme, maths inside



REPORTER ASKS A DEER: AP KI EK LINE MEIN DEFINITION?

Reporter: AP kya hai? Main: har baar same d add karo, bas itna hi

Asli baat: a list is an AP only if $a_{k+1} - a_k$ is the same for every k. Check at least two differences, not just one.

Day 12 Practice

16 board-pattern questions | solve first, then check

MCQ

1 The common difference of the AP $\frac{1}{q}, \frac{1+q}{q}, \frac{1+2q}{q}, \ldots$ is (1)

(a) q (b) 1 (c) -1 (d) $\frac{1}{q}$

MCQ

2 The 11th term of the AP $-6, -\frac{7}{2}, -1, \ldots$ is (1)

(a) 19 (b) -19 (c) 21 (d) 16

MCQ

3 The nth term of the AP $a, 4a, 7a, \ldots$ is (1)

(a) $(3n+1)a$ (b) $3na$ (c) $(3n-2)a$ (d) $(n+3)a$

MCQ

4 The next term of $\sqrt{5}, \sqrt{20}, \sqrt{45}, \ldots$ is (1)

(a) $\sqrt{60}$ (b) $5\sqrt{5}$ (c) $\sqrt{70}$ (d) $4\sqrt{5}$

MCQ

5 The common difference of the AP whose nth term is $a_n = 7 - 4n$ is (1)

(a) 7 (b) -4 (c) 4 (d) 3

ASSERTION-REASON

6 **Assertion (A):** $2, 4, 8, 16, \ldots$ is not an AP.
Reason (R): In an AP, the difference between consecutive terms is constant. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

7 **Assertion (A):** 301 is a term of the AP $5, 11, 17, \ldots$
Reason (R): A number is a term of an AP only if n from $a_n = a + (n-1)d$ is a natural number. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

VERY SHORT ANSWER

8 In an AP with $d = 4$, find $a_{20} - a_{12}$. (1)

VERY SHORT ANSWER

9 If $x + 2$, $3x$ and $4x + 1$ are three consecutive terms of an AP, find x. (1)

SHORT ANSWER

10 Which term of the AP $21, 18, 15, \ldots$ is -81? Is 0 a term of this AP? (2)

SHORT ANSWER

11 How many terms are there in the AP $7, 11, 15, \ldots, 207$? (2)

SHORT ANSWER

12 If 9 times the 9th term of an AP equals 13 times its 13th term, show that its 22nd term is zero. (3)

SHORT ANSWER

13 How many numbers between 100 and 400 are divisible by 6? (3)

LONG ANSWER

14 The 8th term of an AP is 31 and its 15th term is 16 more than its 11th term. Find the AP and its 20th term. (4)

LONG ANSWER

15 For what value of n are the nth terms of the APs $63, 65, 67, \ldots$ and $3, 10, 17, \ldots$ equal? Find that common term. (4)

CASE-BASED

16 Case study: Auditorium seats In a school auditorium, the first row has 18 seats and each row after that has 4 more seats than the row in front of it. (4)

(i) How many seats are in the 10th row? (1)

(ii) Which row has 90 seats? (1)

(iii) The last row has 110 seats. How many rows are there? (2)

Day 12 Answers with worked solutions

1 (b)

$d = \frac{1+q}{q} - \frac{1}{q} = \frac{q}{q} = 1.$

2 (a)

$a = -6$, $d = \frac{5}{2}$; $a_{11} = -6 + 10 \times \frac{5}{2} = 19.$

3 (c)

$d = 3a$; $a_n = a + (n-1)3a = (3n-2)a.$

4 (d)

The terms are $\sqrt{5}, 2\sqrt{5}, 3\sqrt{5}$, so $d = \sqrt{5}$ and the next is $4\sqrt{5} = \sqrt{80}$.

5 (b)

$a_{n+1} - a_n = (7 - 4n - 4) - (7 - 4n) = -4.$

6 (a)

Differences are 2, 4, 8: not constant, so A is true, and R explains why.

7 (d)

$5 + 6(n-1) = 301 \Rightarrow n - 1 = \frac{296}{6}$, not a whole number. A is false; R is true.

8 **Ans:** 32

$a_{20} - a_{12} = (20 - 12)d = 8 \times 4 = 32.$

9 **Ans:** $x = 3$

$2(3x) = (x+2) + (4x+1) \Rightarrow 6x = 5x + 3 \Rightarrow x = 3$ (terms 5, 9, 13).

10 **Ans:** 35th term; yes, 0 is the 8th term

$a = 21, d = -3$: $21 - 3(n-1) = -81 \Rightarrow n - 1 = 34 \Rightarrow n = 35.$

$21 - 3(n-1) = 0 \Rightarrow n - 1 = 7 \Rightarrow n = 8$, a natural number, so 0 is the 8th term.

11 **Ans:** 51

$207 = 7 + (n-1)4 \Rightarrow n - 1 = 50 \Rightarrow n = 51.$

12 **Ans:** $a_{22} = 0$

$9(a + 8d) = 13(a + 12d) \Rightarrow 9a + 72d = 13a + 156d.$

$-4a = 84d \Rightarrow a = -21d.$

$a_{22} = a + 21d = -21d + 21d = 0.$

13 **Ans:** 50

First: 102, last: 396, $d = 6$.

$396 = 102 + (n-1)6 \Rightarrow n - 1 = 49 \Rightarrow n = 50.$

14 **Ans:** $3, 7, 11, \ldots$; $a_{20} = 79$

$a_{15} - a_{11} = 4d = 16 \Rightarrow d = 4.$

$a + 7d = 31 \Rightarrow a = 31 - 28 = 3.$

AP: $3, 7, 11, 15, \ldots$

$a_{20} = 3 + 19 \times 4 = 79.$

15 **Ans:** $n = 13$; the term is 87

nth term of the first AP: $63 + 2(n-1)$; of the second: $3 + 7(n-1)$.

$63 + 2(n-1) = 3 + 7(n-1) \Rightarrow 60 = 5(n-1) \Rightarrow n = 13.$

Common term $= 63 + 2 \times 12 = 87$ (check: $3 + 7 \times 12 = 87$).

16 **Ans:** 54

$a_{10} = 18 + 9 \times 4 = 54.$

Ans: 19th row

$18 + 4(n-1) = 90 \Rightarrow n - 1 = 18 \Rightarrow n = 19.$

Ans: 24 rows

$18 + 4(n-1) = 110 \Rightarrow 4(n-1) = 92 \Rightarrow n - 1 = 23 \Rightarrow n = 24.$

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. nth term : $a_n = a + (n-1)d$
2. Common difference : $d = a_2 - a_1$
3. nth term from the end : $l - (n-1)d$
4. Three terms in AP : $2b = a + c$

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. d of $a_n = 7 - 4n$: -4
2. Is 302 a term of 3, 8, 13...? : No, n is not natural
3. 3-digit numbers divisible by 7 : 128

DAY 12 COMPLETE?

- [ ] Notes read
- [ ] 16 questions solved
- [ ] Answers checked
- [ ] Mistakes noted

Next: Day 13, Arithmetic Progressions

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 13

ARITHMETIC PROGRESSIONS | CHAPTER 5 | PART 2 OF 3

Sum of First n Terms

Arithmetic Progressions

Aaj ka target

- [ ] S_n ke dono formulas yaad karo aur sahi wala chuno
- [ ] 'Kitne terms ka sum X hai' wale quadratic sums solve karo
- [ ] S_n diya ho to $a_n = S_n - S_{n-1}$ se term nikaalo
- [ ] Multiples ka sum (jaise 7 ke multiples) fast karo

13.1 Sum of the First n Terms

Young Gauss added $1 + 2 + \cdots + 100$ by pairing the first and last terms: each pair makes 101, and there are 50 pairs, so the sum is 5050. The same pairing idea gives the formula for any AP.

SUM FORMULAS

$$S_n = \frac{n}{2}\left[2a + (n-1)d\right]$$

When the last term $l = a_n$ is known:

$$S_n = \frac{n}{2}(a + l)$$

Sum of the first n natural numbers:

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}$$

nth term from the sums:

$$a_n = S_n - S_{n-1}$$

SOLVED EXAMPLE 1

Find the sum of the first 22 terms of the AP $8, 3, -2, \ldots$

SOLUTION

1. $a = 8, d = -5, n = 22$.
2. $S_{22} = \frac{22}{2}\left[16 + 21 \times (-5)\right] = 11(16 - 105) = 11 \times (-89)$.

Answer: $S_{22} = -979$

SOLVED EXAMPLE 2

How many terms of the AP $20, 17, 14, \ldots$ must be taken so that the sum is 77?

SOLUTION

1. $\frac{n}{2}\left[40 + (n-1)(-3)\right] = 77 \Rightarrow n(43 - 3n) = 154$.
2. $3n^2 - 43n + 154 = 0$; $D = 1849 - 1848 = 1$.
3. $n = \frac{43 \pm 1}{6} = 7$ or $\frac{22}{3}$.
4. n must be a natural number, so $n = 7$.

Answer: 7 terms



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

n ka answer fraction aaye to reject karo, number of terms kabhi fraction nahi hota. Par agar dono values natural numbers hon (jaise 7 aur 12) to dono sahi hain: beech ke negative terms cancel ho jaate hain.

SOLVED EXAMPLE 3

The sum of the first n terms of an AP is $S_n = 3n^2 + 5n$. Find its nth term and the AP.

SOLUTION

1. $a_n = S_n - S_{n-1} = (3n^2 + 5n) - \left[3(n-1)^2 + 5(n-1)\right]$.
2. $= 3n^2 + 5n - 3n^2 + 6n - 3 - 5n + 5 = 6n + 2$.
3. $a_1 = 8$, $a_2 = 14$, so $d = 6$.

Answer: $a_n = 6n + 2$; AP: $8, 14, 20, \ldots$



SHIVAM BHAIYA SAYS

Shortcut: $S_n = pn^2 + qn$ ho to $d = 2p$ aur $a_1 = S_1 = p + q$. Jaise $3n^2 + 5n$: $d = 6$, $a_1 = 8$. Board mein steps poore likho, par answer isse turant check kar lo.

SOLVED EXAMPLE 4

Find the sum of all multiples of 7 between 100 and 300.

SOLUTION

1. First multiple: 105, last: 294, $d = 7$.
2. $294 = 105 + (n-1)7 \Rightarrow n = 28$.
3. $S = \frac{28}{2}(105 + 294) = 14 \times 399$.

Answer: 5586

SOLVED EXAMPLE 5

The first and last terms of an AP are 7 and 49, and the sum of all its terms is 420. Find the number of terms and the common difference.

SOLVED EXAMPLE 5

SOLUTION

1. $S_n = \frac{n}{2}(a+l) \Rightarrow 420 = \frac{n}{2}(56) \Rightarrow n = 15.$
2. $49 = 7 + 14d \Rightarrow d = 3.$

Answer: $n = 15$, $d = 3$



SHIVAM BHAIYA SAYS: BOARD TIP

Formula choose karne ka rule: last term l diya hai to $\frac{n}{2}(a+l)$, warna $\frac{n}{2}[2a+(n-1)d]$. Formula pehle likho, phir values daalo: formula line pe bhi mark hai.



EXAM ALERT!

S_n wale questions AP ke sabse scoring hain. 'Sum of first n terms is given, find the AP' aur 'how many terms give the sum' 3 marks mein baar baar aaye hain.

MEME SAMJHO

Astra meme, maths inside



STUDENT WHO JUST USED GAUSS PAIRING

Jab 1 se 100 tak ka sum ek line mein 5050 aa gaya, calculator ki zaroorat hi nahi

Asli baat: $S_n = \frac{n}{2}(a+l)$. For $1+2+\cdots+100$: $\frac{100}{2}(1+100) = 5050$.

Day 13 Practice

15 board-pattern questions | solve first, then check

MCQ

1 The sum of the first 10 terms of $2, 7, 12, \ldots$ is (1)

(a) 245 **(b)** 235 **(c)** 255 **(d)** 490

MCQ

2 The sum of the first 25 odd natural numbers is (1)

(a) 600 **(b)** 650 **(c)** 625 **(d)** 576

MCQ

3 If $a = -4$ and $d = 3$, then S_8 is (1)

(a) 48 **(b)** 60 **(c)** -52 **(d)** 52

MCQ

4 If $S_n = 2n^2 + n$, then $a_n =$ (1)

(a) $4n+1$ **(b)** $4n-1$ **(c)** $2n+1$ **(d)** $4n$

MCQ

5 The sum of the first 16 terms of $12, 8, 4, \ldots$ is (1)

(a) 288 **(b)** -288 **(c)** -320 **(d)** -256

ASSERTION-REASON

6 **Assertion (A):** The sum of the first 50 natural numbers is 1275.
Reason (R): The sum of the first n natural numbers is $\frac{n(n+1)}{2}$. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

7 **Assertion (A):** The sum of the first 10 terms of $5, 3, 1, \ldots$ is -40.
Reason (R): The sum of the terms of an AP can never be negative. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

VERY SHORT ANSWER

8 Find the sum $4 + 8 + 12 + \cdots + 100$. (1)

SHORT ANSWER

9 Find the sum of the first 30 terms of the AP whose nth term is $3 + 2n$. (2)

SHORT ANSWER

10 How many terms of the AP $27, 24, 21, \ldots$ must be taken so that their sum is 126? Explain the double answer. (3)

SHORT ANSWER

11 The sum of the first n terms of an AP is $4n - n^2$. Find the first term, the common difference and the 10th term. (3)

SHORT ANSWER

12 Find the sum of all multiples of 4 between 10 and 250. (3)

LONG ANSWER

13 The sum of the first 5 terms of an AP is 35 and the sum of its first 12 terms is 168. Find the sum of its first n terms. (4)

LONG ANSWER

14 **The sum of the 3rd and 7th terms of an AP is 6 and the sum of its 5th and 9th terms is 14. Find the first three terms and the sum of the first 20 terms.** (5)

CASE-BASED

15 **Case study: Weekly savings Riya saves Rs 50 in the first week of the year and increases her saving by Rs 20 every week.** (4)

(i) How much does she save in the 10th week? (1)

(ii) What are her total savings in the first 10 weeks? (1)

(iii) In how many weeks will her total savings reach Rs 4800? (2)

Day 13 Answers with worked solutions

1 **(a)**

$S_{10} = 5(4 + 9 \times 5) = 5 \times 49 = 245.$

2 **(c)**

Sum of first n odd numbers $= n^2 = 25^2 = 625.$

3 **(d)**

$S_8 = 4\,[-8 + 7 \times 3] = 4 \times 13 = 52.$

4 **(b)**

$a_n = S_n - S_{n-1} = 2n^2 + n - \left[2(n-1)^2 + (n-1)\right] = 4n - 1.$

5 **(b)**

$S_{16} = 8\,[24 + 15 \times (-4)] = 8 \times (-36) = -288.$

6 **(a)**

$\frac{50 \times 51}{2} = 1275$: A is true and R gives it.

7 **(c)**

$S_{10} = 5\,[10 + 9 \times (-2)] = 5 \times (-8) = -40$: A is true. R is false (this very sum is negative).

8 **Ans:** 1300

$n = 25;\ S = \frac{25}{2}(4 + 100) = 1300.$

9 **Ans:** 1020

$a_1 = 5,\ a_{30} = 63.$

$S_{30} = \frac{30}{2}(5 + 63) = 15 \times 68 = 1020.$

10 **Ans:** 7 or 12 terms

$\frac{n}{2}\,[54 - 3(n-1)] = 126 \Rightarrow n(57 - 3n) = 252.$

$n^2 - 19n + 84 = 0 \Rightarrow (n-7)(n-12) = 0.$

Both work: terms 8 to 12 are $6, 3, 0, -3, -6$, which add up to 0, so $S_7 = S_{12} = 126.$

11 **Ans:** $a = 3,\ d = -2,\ a_{10} = -15$

$a_n = S_n - S_{n-1} = 4n - n^2 - \left[4(n-1) - (n-1)^2\right] = 5 - 2n.$

$a_1 = 3,\ a_2 = 1$, so $d = -2.$

$a_{10} = 5 - 20 = -15.$

12 **Ans:** 7800

First: 12, last: 248, $d = 4$: $248 = 12 + (n-1)4 \Rightarrow n = 60.$

$S = \frac{60}{2}(12 + 248) = 30 \times 260 = 7800.$

13 **Ans:** $S_n = n^2 + 2n$

$\frac{5}{2}(2a + 4d) = 35 \Rightarrow a + 2d = 7.$

$6(2a + 11d) = 168 \Rightarrow 2a + 11d = 28.$

Substitute $a = 7 - 2d$: $14 - 4d + 11d = 28 \Rightarrow d = 2,\ a = 3.$

$S_n = \frac{n}{2}\,[6 + 2(n-1)] = n(n+2) = n^2 + 2n.$

14 **Ans:** $-5, -3, -1;\ S_{20} = 280$

$(a + 2d) + (a + 6d) = 6 \Rightarrow 2a + 8d = 6.$

$(a + 4d) + (a + 8d) = 14 \Rightarrow 2a + 12d = 14.$

Subtract: $4d = 8 \Rightarrow d = 2;\ a = -5.$

First three terms: $-5, -3, -1.$

$S_{20} = 10\,[-10 + 19 \times 2] = 10 \times 28 = 280.$

15 **Ans:** Rs 230

$a_{10} = 50 + 9 \times 20 = 230.$

Ans: Rs 1400

$S_{10} = 5(100 + 9 \times 20) = 5 \times 280 = 1400.$

Ans: 20 weeks

$\frac{n}{2}\,[100 + 20(n-1)] = 4800 \Rightarrow n(40 + 10n) = 4800.$

$n^2 + 4n - 480 = 0 \Rightarrow (n+24)(n-20) = 0 \Rightarrow n = 20.$

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. S_n (with d) : $\frac{n}{2}[2a + (n-1)d]$
2. S_n (with last term) : $\frac{n}{2}(a + l)$
3. $1 + 2 + \cdots + n$: $\frac{n(n+1)}{2}$
4. Sum of first n odd numbers : n^2
5. nth term from S_n : $a_n = S_n - S_{n-1}$
6. $S_n = pn^2 + qn$ gives d : $2p$
7. Fraction value of n : reject it

DAY 13 COMPLETE? ☐ Notes read ☐ 15 questions solved ☐ Answers checked ☐ Mistakes noted

Next: Day 14, Arithmetic Progressions

DAY 14

ARITHMETIC PROGRESSIONS | CHAPTER 5 | PART 3 OF 3

AP Word Problems and Board Mix

Arithmetic Progressions

Aaj ka target

- [ ] Word problem padh ke decide karo: term chahiye ya sum
- [ ] Teen numbers ko a-d, a, a+d lo
- [ ] Logs, rows, prizes, salary wale sums practice karo
- [ ] Galat value (negative rows, fraction n) reject karna mat bhoolo

14.1 Term or Sum? Decoding Word Problems

The question asks for	Use
Amount in a particular week, row, year	$a_n = a + (n-1)d$
Total amount, total items, total distance	$S_n = \frac{n}{2}[2a + (n-1)d]$
How many weeks or rows to reach a total	Put S_n = total and solve the quadratic in n
Middle term of an AP with n terms (n odd)	The $\left(\frac{n+1}{2}\right)$th term

Choosing the right formula

SOLVED EXAMPLE 1

Aman's starting salary is Rs 15000 per month with a yearly increment of Rs 1200. In which year will his monthly salary be Rs 27000? What is the total of his monthly salaries over those years (one month taken per year)?

SOLUTION

1. $a = 15000$, $d = 1200$: $15000 + (n-1)1200 = 27000 \Rightarrow n - 1 = 10 \Rightarrow n = 11$.
2. Total $= \frac{11}{2}(15000 + 27000) = \frac{11}{2} \times 42000 = 231000$.

Answer: In the 11th year; total Rs 231000

SOLVED EXAMPLE 2

250 logs are stacked with 22 logs in the bottom row, 21 in the next row, 20 in the row after that, and so on. How many rows are there, and how many logs are in the top row?

SOLUTION

1. $a = 22, d = -1$: $\frac{n}{2}[44 - (n-1)] = 250 \Rightarrow n(45 - n) = 500$.
2. $n^2 - 45n + 500 = 0 \Rightarrow (n-20)(n-25) = 0$.
3. For $n = 25$ the top row would have $22 - 24 = -2$ logs, impossible; so $n = 20$.
4. Top row $= 22 - 19 = 3$ logs.

Answer: 20 rows; 3 logs in the top row



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Quadratic se do values aayi to dono ko situation mein check karo. Logs wale sawaal mein $n = 25$ pe top row mein -2 logs aa rahe the, isliye reject. Bina check kiye dono likh diye to mark katega.

SOLVED EXAMPLE 3

A contractor pays a penalty for delay: Rs 150 for the first day, Rs 190 for the second, Rs 230 for the third, and so on. How much does he pay for a delay of 25 days?

SOLUTION

1. $a = 150, d = 40, n = 25$.
2. $S_{25} = \frac{25}{2}[300 + 24 \times 40] = \frac{25}{2} \times 1260$.

Answer: Rs 15750

14.2 Choosing Terms Smartly

CONVENIENT CHOICES

Three numbers in AP:

$$a - d,\ a,\ a + d$$

Four numbers in AP:

$$a - 3d,\ a - d,\ a + d,\ a + 3d$$

(common difference $2d$)

SOLVED EXAMPLE 4

Three numbers in AP have sum 27 and product 648. Find them.

SOLUTION

1. Let them be $a - d, a, a + d$. Sum: $3a = 27 \Rightarrow a = 9$.
2. Product: $9(81 - d^2) = 648 \Rightarrow 81 - d^2 = 72 \Rightarrow d = \pm 3$.

Answer: $6, 9, 12$ (or $12, 9, 6$)

SOLVED EXAMPLE 5

Find the middle term of the AP $6, 13, 20, \ldots, 216$.

SOLUTION

1. $216 = 6 + (n-1)7 \Rightarrow n = 31$ (odd).
2. Middle term is the 16th term: $6 + 15 \times 7 = 111$.

Answer: 111



SHIVAM BHAIYA SAYS

Sum aur product diye hon to $a - d, a, a + d$ lo: sum mein d cut ho jaata hai aur a seedha mil jaata hai. Ek variable kam, aadha kaam kam!



SHIVAM BHAIYA SAYS: BOARD TIP

Word problem mein pehle likho: '$a = \ldots, d = \ldots, n = \ldots$'. Phir formula, phir answer units ke saath (Rs, rows, days). Examiner ko clarity dikhni chahiye.



EXAM ALERT!

AP ka case-study (seats, savings, stacking, jogging) lagbhag har saal aata hai, 4 marks ka. Term vs sum ka confusion hi sabse bada trap hai!

MEME SAMJHO

Astra meme, maths inside



STUDENT WHOSE ANSWER FOR N CAME OUT AS A FRACTION

Itni mehnat se solve kiya aur $n = 22/3$ aaya... matlab question ne bola 'yeh term hi nahi hai'

Asli baat: the number of terms must be a natural number. A fractional n means that value is not a term (or that sum is impossible), so reject it.

Day 14 Practice

15 board-pattern questions | solve first, then check

MCQ

1. Neha jogs 12 minutes in the first week and adds 5 minutes every week. In which week will she jog 72 minutes? (1)
   (a) 12th **(b)** 13th **(c)** 14th **(d)** 15th

MCQ

2. Three numbers in AP have sum 18. The middle number is (1)
   (a) 9 **(b)** 3 **(c)** 6 **(d)** 12

MCQ

3. The middle term of the AP $5, 9, 13, \ldots, 101$ is (1)
   (a) 53 **(b)** 49 **(c)** 57 **(d)** 51

MCQ

4. A garden has 30 plants in the first row, 28 in the second, and so on, with 2 plants in the last row. The number of rows is (1)
   (a) 14 **(b)** 16 **(c)** 15 **(d)** 30

MCQ

5. Rohan saves Rs 2 on day 1, Rs 4 on day 2, Rs 6 on day 3, and so on. After how many days will his total savings be Rs 240? (1)
   (a) 12 **(b)** 20 **(c)** 16 **(d)** 15

ASSERTION-REASON

6. **Assertion (A):** If three numbers in AP add up to 24, the middle number is 8.
   Reason (R): Three numbers in AP can be taken as $a - d, a, a + d$. (1)
   (a) Both A and R are true, and R is the correct explanation of A.
   (b) Both A and R are true, but R is not the correct explanation of A.
   (c) A is true, but R is false.
   (d) A is false, but R is true.

ASSERTION-REASON

7. **Assertion (A):** The 20th term of the AP $3, 8, 13, \ldots$ is 98.
   Reason (R): The nth term of an AP is $a + nd$. (1)
   (a) Both A and R are true, and R is the correct explanation of A.
   (b) Both A and R are true, but R is not the correct explanation of A.
   (c) A is true, but R is false.
   (d) A is false, but R is true.

VERY SHORT ANSWER

8. Find the sum of the first 15 multiples of 8. (1)

SHORT ANSWER

9. Rs 1600 is to be given as 8 cash prizes, each prize Rs 30 less than the one before it. Find the value of each prize. (2)

SHORT ANSWER

10 A man repays a loan of Rs 3250 by paying Rs 20 in the first month and increasing the payment by Rs 15 every month. In how many months will the loan be cleared? (3)

SHORT ANSWER

11 Three numbers in AP have sum 15 and product 80. Find the numbers. (3)

SHORT ANSWER

12 Which term of the AP $3, 15, 27, 39, \ldots$ is 120 more than its 21st term? (3)

LONG ANSWER

13 Job A offers Rs 20000 per month with a yearly increment of Rs 1500; Job B offers Rs 24000 per month with a yearly increment of Rs 1000. In which year will the two monthly salaries be equal? Taking one month's salary per year, find the total for each job up to that year. (4)

LONG ANSWER

14 The 17th term of an AP is 5 more than twice its 8th term, and its 11th term is 43. Find the nth term and the sum of the first 20 terms. (5)

CASE-BASED

15 Case study: Stacking cans A shopkeeper stacks cans in rows: 24 cans in the bottom row, and each row above has 2 cans fewer than the row below it. (4)

(i) If the top row has 4 cans, how many rows are there? (1)

(ii) How many cans are there in that stack? (1)

(iii) He has only 144 cans and starts with 24 in the bottom row. How many rows can he make? (2)

Day 14 Answers with worked solutions

1 **(b)**

$12 + 5(n-1) = 72 \Rightarrow n - 1 = 12 \Rightarrow n = 13$.

2 **(c)**

$(a-d) + a + (a+d) = 3a = 18 \Rightarrow a = 6$.

3 **(a)**

$101 = 5 + (n-1)4 \Rightarrow n = 25$; middle term $= a_{13} = 5 + 12 \times 4 = 53$.

4 **(c)**

$30 - 2(n-1) = 2 \Rightarrow n - 1 = 14 \Rightarrow n = 15$ (total plants $= \frac{15}{2}(30+2) = 240$).

5 **(d)**

$S_n = 2(1 + 2 + \cdots + n) = n(n+1) = 240 \Rightarrow n = 15$.

6 **(a)**

$3a = 24 \Rightarrow a = 8$: A is true, and R is the method that proves it.

7 **(c)**

$a_{20} = 3 + 19 \times 5 = 98$, so A is true. R is false: the nth term is $a + (n-1)d$.

8 **Ans:** 960

$8 + 16 + \cdots + 120 = 8(1 + 2 + \cdots + 15) = 8 \times 120 = 960$.

9 **Ans:** Rs 305, 275, 245, 215, 185, 155, 125, 95

$\frac{8}{2}[2a + 7(-30)] = 1600 \Rightarrow 2a - 210 = 400 \Rightarrow a = 305$.

Prizes: 305, 275, 245, 215, 185, 155, 125, 95 (the last is $305 - 7 \times 30 = 95$).

10 **Ans:** 20 months

$\frac{n}{2}[40 + 15(n-1)] = 3250 \Rightarrow n(15n + 25) = 6500$.

$3n^2 + 5n - 1300 = 0$; $D = 25 + 15600 = 15625 = 125^2$.

$n = \frac{-5 + 125}{6} = 20$ (reject the negative root).

11 **Ans:** $2, 5, 8$

Let them be $a - d, a, a + d$: $3a = 15 \Rightarrow a = 5$.

$5(25 - d^2) = 80 \Rightarrow d^2 = 9 \Rightarrow d = \pm 3$.

Numbers: $2, 5, 8$ (or $8, 5, 2$).

12 **Ans:** 31st term

$a_{21} = 3 + 20 \times 12 = 243$; required term $= 243 + 120 = 363$.

$3 + 12(n-1) = 363 \Rightarrow n - 1 = 30 \Rightarrow n = 31$.

(Quick check: $120 = 10d$, so it is $21 + 10 = 31$ st.)

13 **Ans:** 9th year; A: Rs 234000, B: Rs 252000

$20000 + 1500(n-1) = 24000 + 1000(n-1) \Rightarrow 500(n-1) = 4000 \Rightarrow n = 9$.

Job A: $S_9 = \frac{9}{2}[40000 + 8 \times 1500] = 4.5 \times 52000 = 234000$.

Job B: $S_9 = \frac{9}{2}[48000 + 8 \times 1000] = 4.5 \times 56000 = 252000$.

14 **Ans:** $a_n = 4n - 1$; $S_{20} = 820$

$a + 16d = 2(a + 7d) + 5 \Rightarrow a = 2d - 5$.

$a + 10d = 43 \Rightarrow 12d - 5 = 43 \Rightarrow d = 4$, $a = 3$.

$a_n = 3 + (n-1)4 = 4n - 1$.

$S_{20} = \frac{20}{2}[6 + 19 \times 4] = 10 \times 82 = 820$.

15 **Ans:** 11

$24 - 2(n-1) = 4 \Rightarrow n = 11.$

Ans: 154

$S_{11} = \frac{11}{2}(24 + 4) = 154.$

Ans: 9 rows

$\frac{n}{2}[48 - 2(n-1)] = 144 \Rightarrow n(25 - n) = 144.$

$n^2 - 25n + 144 = 0 \Rightarrow (n-9)(n-16) = 0.$

$n = 16$ would need $24 - 30 = -6$ cans in the top row, so $n = 9$.

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Asked for one week's value : use a_n
2. Asked for the total : use S_n
3. Three numbers in AP : $a - d, a, a + d$
4. Four numbers in AP : $a - 3d, a - d, a + d, a + 3d$
5. Middle term (n odd) : $\left(\frac{n+1}{2}\right)$th term
6. Negative rows or fraction n : reject
7. Penalty 150, 190, 230... for 25 days : Rs 15750

DAY 14 COMPLETE? ☐ Notes read ☐ 15 questions solved ☐ Answers checked ☐ Mistakes noted

Next: Day 15, Coordinate Geometry

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 15

COORDINATE GEOMETRY | CHAPTER 7 | PART 1 OF 2

Distance Formula

Coordinate Geometry

Aaj ka target

- [ ] Distance formula ko derive karke yaad karo, ratta nahi
- [ ] Triangle aur quadrilateral ka type distance se pehchano
- [ ] Collinear points ka test: do chhoti lengths ka sum = sabse badi
- [ ] Equidistant point waale sums: dono distance² barabar rakho

15.1 Quick Recap: The Cartesian Plane

- Every point is written as (x, y): x = abscissa (distance from the y-axis), y = ordinate (distance from the x-axis).
- Signs by quadrant: I $(+,+)$, II $(-,+)$, III $(-,-)$, IV $(+,-)$.
- A point on the x-axis is $(x, 0)$; a point on the y-axis is $(0, y)$. The origin is $O(0,0)$.
- Distance of $P(x, y)$ from the x-axis $= |y|$ and from the y-axis $= |x|$.

15.2 The Distance Formula

Join $P(x_1, y_1)$ and $Q(x_2, y_2)$ and drop perpendiculars to complete the right triangle PRQ. The legs are $x_2 - x_1$ and $y_2 - y_1$, so by Pythagoras theorem $PQ^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$.

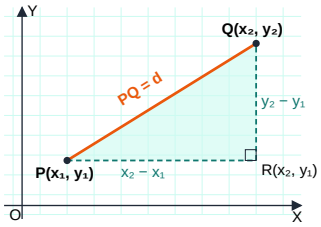


Fig. 15.1 The distance PQ is the hypotenuse of right triangle PRQ.

FORMULA BOX

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Distance of $P(x, y)$ from the origin:

$$OP = \sqrt{x^2 + y^2}$$

Order does not matter: $(x_2 - x_1)^2 = (x_1 - x_2)^2$.

SOLVED EXAMPLE 1

Find the distance between $A(-4, 6)$ and $B(2, -2)$.

SOLUTION

1. $x_2 - x_1 = 2 - (-4) = 6,\ y_2 - y_1 = -2 - 6 = -8$
2. $AB = \sqrt{6^2 + (-8)^2} = \sqrt{36 + 64} = \sqrt{100}$

Answer: $AB = 10$ units



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Sabse common galti: $(-8)^2$ ko -64 likh dena. Square hamesha positive hota hai! Aur minus minus ko dhyan se: $2 - (-4) = 6$, not -2.

15.3 Applications of the Distance Formula

1. COLLINEARITY: THREE POINTS A, B, C ARE COLLINEAR IF THE SUM OF THE TWO SHORTER DISTANCES EQUALS THE LONGEST ONE.

SOLVED EXAMPLE 2

Show that $A(1, 5)$, $B(2, 3)$ and $C(-2, 11)$ are collinear.

SOLUTION

1. $AB = \sqrt{1^2 + (-2)^2} = \sqrt{5}$
2. $BC = \sqrt{(-4)^2 + 8^2} = \sqrt{80} = 4\sqrt{5}$
3. $AC = \sqrt{(-3)^2 + 6^2} = \sqrt{45} = 3\sqrt{5}$
4. $AB + AC = \sqrt{5} + 3\sqrt{5} = 4\sqrt{5} = BC$

Answer: The points are collinear (A lies between B and C).

2. TYPE OF TRIANGLE AND QUADRILATERAL: FIND ALL SIDES (AND DIAGONALS FOR A QUADRILATERAL), THEN COMPARE.

Figure	What to show (with distances)
Isosceles triangle	Two sides equal
Equilateral triangle	All three sides equal
Right triangle	Longest side² = sum of squares of other two
Parallelogram	Opposite sides equal
Rectangle	Opposite sides equal AND diagonals equal
Rhombus	All four sides equal
Square	All four sides equal AND diagonals equal

Use this checklist in 'show that' questions.

3. EQUIDISTANT POINT: IF P IS EQUIDISTANT FROM A AND B, WRITE $PA^2 = PB^2$ (SQUARING REMOVES THE ROOTS).

SOLVED EXAMPLE 3

Find the point on the x-axis which is equidistant from $A(7, 6)$ and $B(-3, 4)$.

SOLUTION

1. Let the point be $P(x, 0)$ (it is on the x-axis).
2. $PA^2 = PB^2 \Rightarrow (x-7)^2 + 36 = (x+3)^2 + 16$
3. $x^2 - 14x + 85 = x^2 + 6x + 25 \Rightarrow 20x = 60$

Answer: $x = 3$, so the point is $(3, 0)$.



SHIVAM BHAIYA SAYS: BOARD TIP

Board mein 'show that ABCD is a square' aaye to sirf 4 sides equal dikhana kaafi nahi, woh rhombus bhi ho sakta hai. Diagonals bhi equal dikhao, tabhi poore marks milenge.

SOLVED EXAMPLE 4

Find a relation between x and y such that $P(x, y)$ is equidistant from $A(5, 1)$ and $B(-1, 5)$.

SOLUTION

1. $PA^2 = PB^2 \Rightarrow (x-5)^2 + (y-1)^2 = (x+1)^2 + (y-5)^2$
2. $-10x + 25 - 2y + 1 = 2x + 1 - 10y + 25$
3. $-12x + 8y = 0$

Answer: $3x = 2y$ (every point on this line is equidistant from A and B).



SHIVAM BHAIYA SAYS

Pythagorean triplets yaad rakho: (3, 4, 5), (5, 12, 13), (8, 15, 17), (7, 24, 25), (6, 8, 10). Differences 8 aur 15 dikhe to seedha 17 likho, calculation ka time bachega.



EXAM ALERT!

Distance formula har saal aata hai: 1 mark MCQ, 2 mark 'find x' ya 'equidistant point', aur 3 mark 'show that it is a square/rectangle'. Area of triangle wala formula 2026-27 syllabus se hata diya gaya hai, use mat padhna.



SHIVAM BHAIYA SAYS: BOARD TIP

Step marking hoti hai: pehle formula likho, phir values substitute karo, phir simplify. Seedha answer likhoge to marks kat sakte hain.

MEME SAMJHO — Astra meme, maths inside



JAB EXAM HALL MEIN DISTANCE FORMULA YAAD NA AAYE

Formula bhool gaye? Koi baat nahi, right triangle banao aur Pythagoras laga do. Formula khud ban jayega.

Asli baat: distance formula sirf Pythagoras hai: legs $x_2 - x_1$ aur $y_2 - y_1$, hypotenuse = PQ.

Day 15 Practice

16 board-pattern questions | solve first, then check

MCQ

1 The point on the x-axis which is equidistant from $(-6, 0)$ and $(10, 0)$ is: (1)

(a) $(4, 0)$ **(b)** $(0, 2)$ **(c)** $(2, 0)$ **(d)** $(-2, 0)$

MCQ

2 The distance between $(p, -q)$ and $(-p, q)$ is: (1)

(a) $\sqrt{p^2 + q^2}$ **(b)** $2\sqrt{p^2 + q^2}$ **(c)** $2(p + q)$ **(d)** $4\sqrt{p^2 + q^2}$

MCQ

3 The perimeter of the triangle with vertices $(0, 6)$, $(0, 0)$ and $(8, 0)$ is: (1)

(a) 14 units **(b)** 48 units
(c) 20 units **(d)** 24 units

MCQ

4 The distance of the point $(8, -15)$ from the origin is: (1)

(a) 17 units **(b)** 23 units
(c) 7 units **(d)** $\sqrt{161}$ units

MCQ

5 A circle has centre $(-4, 3)$ and passes through $(-1, -1)$. Its radius is: (1)

(a) 7 units **(b)** 5 units
(c) $\sqrt{7}$ units **(d)** 25 units

ASSERTION-REASON

6 **Assertion (A):** The distance of the point $(-5, 12)$ from the origin is 13 units.
Reason (R): The distance of a point $P(x, y)$ from the origin is $\sqrt{x^2 + y^2}$. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

7 **Assertion (A):** The point $P(-2,4)$ lies on the circle of radius 6 with centre $C(3,5)$.
Reason (R): A point lies on a circle if its distance from the centre equals the radius. (1)
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

VERY SHORT ANSWER

8 Find the distance of the point $(-7,2)$ from the y-axis. (1)

CASE-BASED

9 Read the passage and answer. Three friends stand in a park laid out on a square grid. Riya is at $A(1,2)$, Sam at $B(4,6)$ and Tara at $C(7,10)$, where 1 unit = 10 m. (4)

(i) Find the distance between Riya and Sam in units. (1)

(ii) Are the three friends standing in a straight line? Justify. (1)

(iii) Find the point on the x-axis that is equidistant from Riya and Sam. (2)

OR (iii) Find the point on the y-axis that is equidistant from Riya and Sam. (2)

SHORT ANSWER

10 Find the values of x for which the distance between $P(x,-1)$ and $Q(3,2)$ is 5 units. (2)

SHORT ANSWER

11 Find the point on the y-axis which is equidistant from $A(5,-2)$ and $B(-3,2)$. (2)

SHORT ANSWER

12 $A(3,\sqrt{3})$, $B(0,0)$ and $C(3,k)$ are the vertices of an equilateral triangle ABC. Find k. (3)

SHORT ANSWER

13 Show that the points $(1,7)$, $(4,2)$, $(-1,-1)$ and $(-4,4)$ are the vertices of a square. (3)

LONG ANSWER

14 Show that $A(0,-1)$, $B(5,4)$, $C(3,6)$ and $D(-2,1)$ are the vertices of a rectangle. Is it a square? (4)

LONG ANSWER

15 Find the point which is equidistant from $A(5,3)$, $B(5,-5)$ and $C(1,-5)$. Also find this common distance. (4)

LONG ANSWER

16 If $P(x,y)$ is equidistant from $A(a+b,b-a)$ and $B(a-b,a+b)$, prove that $bx=ay$. (4)

Day 15 Answers with worked solutions

1 **(c)**
Both points are on the x-axis, so the required point is their midpoint: $\left(\frac{-6+10}{2},0\right)=(2,0)$.

2 **(b)**
$\sqrt{(-2p)^2+(2q)^2}=\sqrt{4p^2+4q^2}=2\sqrt{p^2+q^2}$

3 **(d)**
Sides: 6, 8 and $\sqrt{8^2+6^2}=10$.
Perimeter $=6+8+10=24$ units.

4 **(a)**
$\sqrt{8^2+15^2}=\sqrt{289}=17$

5 **(b)**
$r=\sqrt{(-1+4)^2+(-1-3)^2}=\sqrt{9+16}=5$

6 **(a)**
$\sqrt{25+144}=13$, so A is true; R is the formula used, so R correctly explains A.

7 **(d)**
$CP=\sqrt{25+1}=\sqrt{26}\neq 6$, so A is false. R is a true statement.

8 **Ans:** 7 units
Distance from the y-axis $=|x|=|-7|=7$.

9 **Ans:** 5 units (50 m)
$AB=\sqrt{3^2+4^2}=5$

Ans: Yes
$BC=\sqrt{3^2+4^2}=5$, $AC=\sqrt{6^2+8^2}=10$
$AB+BC=10=AC$, so A, B, C are collinear.

Ans: $\left(\frac{47}{6},0\right)$
Let it be $(x,0)$: $(x-1)^2+4=(x-4)^2+36$
$-2x+5=-8x+52\Rightarrow 6x=47\Rightarrow x=\frac{47}{6}$

OR: Ans: $\left(0,\frac{47}{8}\right)$
Let it be $(0,y)$: $1+(y-2)^2=16+(y-6)^2$
$-4y+5=-12y+52\Rightarrow 8y=47\Rightarrow y=\frac{47}{8}$

10 **Ans:** $x=7$ or $x=-1$
$(x-3)^2+(2+1)^2=25$
$(x-3)^2=16\Rightarrow x-3=\pm 4$
$x=7$ or $x=-1$

11 **Ans:** $(0, -2)$

Let it be $P(0, y)$. $PA^2 = PB^2$:

$25 + (y+2)^2 = 9 + (y-2)^2$

$29 + 4y = 13 - 4y \Rightarrow 8y = -16 \Rightarrow y = -2$

12 **Ans:** $k = -\sqrt{3}$

$AB = \sqrt{9+3} = 2\sqrt{3}$

$BC = AB \Rightarrow 9 + k^2 = 12 \Rightarrow k = \pm\sqrt{3}$

$k = \sqrt{3}$ makes C = A, so $k = -\sqrt{3}$.

Check: $AC = |\sqrt{3} - (-\sqrt{3})| = 2\sqrt{3}$. All sides equal.

13 **Ans:** All sides $= \sqrt{34}$, both diagonals $= 2\sqrt{17}$, so it is a square.

Let A(1, 7), B(4, 2), C(-1, -1), D(-4, 4).

$AB = \sqrt{9+25} = \sqrt{34}$, $BC = \sqrt{25+9} = \sqrt{34}$,

$CD = \sqrt{9+25} = \sqrt{34}$, $DA = \sqrt{25+9} = \sqrt{34}$

$AC = \sqrt{4+64} = \sqrt{68}$, $BD = \sqrt{64+4} = \sqrt{68}$

Four equal sides and equal diagonals, so ABCD is a square.

14 **Ans:** Rectangle (opposite sides equal, diagonals equal); not a square since $AB \neq BC$.

$AB = \sqrt{25+25} = 5\sqrt{2}$, $BC = \sqrt{4+4} = 2\sqrt{2}$

$CD = \sqrt{25+25} = 5\sqrt{2}$, $DA = \sqrt{4+4} = 2\sqrt{2}$

Opposite sides are equal, so ABCD is a parallelogram.

$AC = \sqrt{9+49} = \sqrt{58}$, $BD = \sqrt{49+9} = \sqrt{58}$: diagonals equal, so it is a rectangle.

$AB \neq BC$, so it is not a square.

15 **Ans:** $(3, -1)$; distance $2\sqrt{5}$ units

Let $P(x, y)$. $PA^2 = PB^2$: $(y-3)^2 = (y+5)^2 \Rightarrow -6y + 9 = 10y + 25 \Rightarrow y = -1$

$PB^2 = PC^2$: $(x-5)^2 = (x-1)^2 \Rightarrow -10x + 25 = -2x + 1 \Rightarrow x = 3$

$P = (3, -1)$

$PA = \sqrt{(5-3)^2 + (3+1)^2} = \sqrt{20} = 2\sqrt{5}$

16 **Ans:** Proved: $bx = ay$

$PA^2 = PB^2$: $(x-a-b)^2 + (y-b+a)^2 = (x-a+b)^2 + (y-a-b)^2$

Expanding and cancelling x^2, y^2, a^2, b^2 terms that match on both sides:

LHS $-$ RHS $= -4bx + 4ay = 0$

$\Rightarrow bx = ay$

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Distance formula : $\sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$
2. Distance of (x, y) from origin : $\sqrt{x^2+y^2}$
3. Distance of $(-7, 2)$ from y-axis : 7 units
4. Collinear test : sum of two shorter = longest
5. Square vs rhombus : square also has equal diagonals
6. Equidistant point trick : $PA^2 = PB^2$
7. Point on x-axis looks like : $(x, 0)$

DAY 15 COMPLETE? ☐ Notes read ☐ 16 questions solved ☐ Answers checked ☐ Mistakes noted

Next: Day 16, Coordinate Geometry

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 16

COORDINATE GEOMETRY | CHAPTER 7 | PART 2 OF 2

Section Formula and Midpoint

Coordinate Geometry

Aaj ka target

- ☐ Section formula: m hamesha doosre point ke saath, yaad karo 'cross multiply'
- ☐ Ratio nikalna ho to k : 1 maan lo
- ☐ Trisection points = 1:2 aur 2:1 dono
- ☐ Parallelogram ka chautha vertex: diagonals ke midpoints same

16.1 Section Formula (Internal Division)

If a point $P(x, y)$ lies on segment AB and divides it internally in the ratio $m : n$ (that is, $AP : PB = m : n$), then its coordinates are given by the section formula.

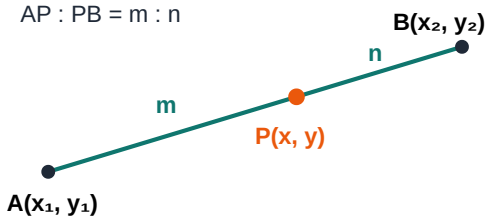


Fig. 16.1 P divides AB internally in the ratio m : n.

FORMULA BOX

$$P(x, y) = \left(\frac{mx_2 + nx_1}{m + n}, \frac{my_2 + ny_1}{m + n} \right)$$

Midpoint ($m = n$):

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Ratio $k : 1$:

$$P = \left(\frac{kx_2 + x_1}{k + 1}, \frac{ky_2 + y_1}{k + 1} \right)$$



SHIVAM BHAIYA SAYS

Cross wala pattern yaad rakho: m ko B (x_2) se multiply karo aur n ko A (x_1) se. Ratio ke paas wala point nahi, doosri taraf wala point multiply hota hai.

SOLVED EXAMPLE 1

Find the coordinates of the point which divides the join of $A(6, -5)$ and $B(3, 4)$ in the ratio $1 : 2$.

SOLUTION

1. $m = 1, n = 2, (x_1, y_1) = (6, -5), (x_2, y_2) = (3, 4)$
2. $x = \frac{1 \times 3 + 2 \times 6}{1 + 2} = \frac{15}{3} = 5$
3. $y = \frac{1 \times 4 + 2 \times (-5)}{3} = \frac{-6}{3} = -2$

Answer: $P(5, -2)$, which lies in Quadrant IV.

16.2 Finding the Ratio

1. Take the ratio as $k : 1$ (one unknown instead of two).
2. Write the coordinates of P using the section formula.
3. Equate the coordinate you know (or put P in the given line) and solve for k.
4. If k comes out as $\frac{p}{q}$, the ratio is $p : q$. A negative k means external division (not in syllabus questions).

- Point on the x-axis has $y = 0$: equate the y-coordinate to 0.
- Point on the y-axis has $x = 0$: equate the x-coordinate to 0.
- Point on a line $ax + by + c = 0$: substitute both coordinates into the line.

SOLVED EXAMPLE 2

In what ratio does the line $x - y - 2 = 0$ divide the segment joining $(3, -1)$ and $(8, 9)$?

SOLUTION

1. Let the ratio be $k : 1$. The point is $\left(\frac{8k+3}{k+1}, \frac{9k-1}{k+1}\right)$.
2. It lies on the line: $(8k + 3) - (9k - 1) - 2(k + 1) = 0$
3. $-3k + 2 = 0 \Rightarrow k = \frac{2}{3}$

Answer: The ratio is $2 : 3$.



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Line wale question mein '-2' ko bhi $(k + 1)$ se multiply karna mat bhoolna. Denominator hatate waqt har term ko multiply karo, warna k galat aayega.

16.3 Special Points

KEY TERM: POINTS OF TRISECTION

The two points that divide AB into three equal parts. P divides AB in $1 : 2$ and Q divides AB in $2 : 1$.

SOLVED EXAMPLE 3

Find the points of trisection of the segment joining $A(4,-1)$ and $B(-2,-3)$.

SOLUTION

1. P $(1:2)$: $x = \frac{1(-2)+2(4)}{3} = 2$, $y = \frac{1(-3)+2(-1)}{3} = -\frac{5}{3}$
2. Q $(2:1)$: $x = \frac{2(-2)+1(4)}{3} = 0$, $y = \frac{2(-3)+1(-1)}{3} = -\frac{7}{3}$

Answer: $P\left(2,-\frac{5}{3}\right)$ and $Q\left(0,-\frac{7}{3}\right)$

PARALLELOGRAM: THE DIAGONALS BISECT EACH OTHER, SO THE MIDPOINT OF AC = MIDPOINT OF BD. SHORTCUT: $D = A + C - B$ (COORDINATE-WISE).

SOLVED EXAMPLE 4

$A(-1,2)$, $B(5,4)$ and $C(7,-2)$ are three vertices of parallelogram ABCD. Find D.

SOLUTION

1. Midpoint of AC $= \left(\frac{-1+7}{2}, \frac{2-2}{2}\right) = (3,0)$
2. Midpoint of BD $= \left(\frac{5+x}{2}, \frac{4+y}{2}\right) = (3,0)$
3. $x = 1$, $y = -4$

Answer: $D(1,-4)$

ALSO KNOW: CENTROID

Centroid G of triangle with vertices $(x_1,y_1),(x_2,y_2),(x_3,y_3)$:

$$G = \left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$$

G divides each median in the ratio $2:1$ from the vertex.



EXAM ALERT!

Section formula se ratio nikalna (x-axis, y-axis ya line par) aur 'find the fourth vertex of a parallelogram' boards ke favourite hain. 2-3 marks pakke. Case study mein bhi midpoint aur section formula saath aate hain.



SHIVAM BHAIYA SAYS: BOARD TIP

Answer mein ratio hamesha simplest form mein likho ($\frac{3}{5}:1$ nahi, $3:5$ likho) aur yeh bhi likho ki 'internally'. Chhoti cheez, par marking scheme mein hoti hai.

MEME SAMJHO

Astra meme, maths inside



DOST: SECTION FORMULA MEIN M AUR N ULTA LIKH DO, ANSWER SAME AAYEGA Guarantee deta hai??

Asli baat: P divides AB in $m:n$ means $x = \frac{mx_2+nx_1}{m+n}$; m hamesha doosre point B ke saath. Ulta likha to doosra point aayega (sirf $m = n$ pe same).

Day 16 Practice

16 board-pattern questions | solve first, then check

MCQ

1 The point dividing the join of $(6,-5)$ and $(3,4)$ internally in the ratio $1:2$ lies in: (1)

(a) Quadrant I (b) Quadrant II
(c) Quadrant IV (d) Quadrant III

MCQ

2 The y-axis divides the segment joining $(4,-6)$ and $(-2,-4)$ in the ratio: (1)

(a) $1:2$ (b) $2:1$ (c) $3:2$ (d) $2:3$

MCQ

3 If $P\left(\frac{a}{2},4\right)$ is the midpoint of the segment joining $Q(-6,5)$ and $R(-2,3)$, then a equals: (1)

(a) -4 (b) 8 (c) -12 (d) -8

MCQ

4 One end of a diameter of a circle is $(3,-1)$ and the centre is $(-1,2)$. The other end is: (1)

(a) $(-5,5)$ (b) $(1,\frac{1}{2})$ (c) $(5,-5)$ (d) $(-4,3)$

MCQ

5 The x-axis divides the join of $(1,-6)$ and $(4,3)$ in the ratio: (1)

(a) $1:2$ (b) $2:1$ (c) $1:3$ (d) $3:1$

ASSERTION-REASON

6 **Assertion (A):** The point $(-3,k)$ divides the join of $(-5,4)$ and $(-2,3)$ in the ratio $2:1$.
Reason (R): The midpoint of a segment divides it in the ratio $1:1$. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

7 **Assertion (A):** $P(6,-2)$ is a point of trisection of the segment joining $A(8,-1)$ and $B(2,-4)$.
Reason (R): The points of trisection of AB divide it in the ratios $1:2$ and $2:1$. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

VERY SHORT ANSWER

8 Find the midpoint of the segment joining $(4,-7)$ and $(-2,3)$. (1)

CASE-BASED

9 Read the passage and answer. A school garden is a parallelogram ABCD drawn on a grid with $A(1,1)$, $B(7,2)$ and $C(9,6)$. A water tap is fixed where the two diagonals meet. (4)

(i) Find the coordinates of the water tap. (1)

(ii) Find the coordinates of D. (1)

(iii) A pole Q divides AB in the ratio $1:2$. Find Q. (2)

OR (iii) Find the length of the diagonal AC. (2)

SHORT ANSWER

10 Find the ratio in which $M(12,y)$ divides the segment joining $P(16,6)$ and $Q(10,21)$. Also find y. (2)

SHORT ANSWER

11 The centre of a circle is $(3,-2)$ and one end of a diameter is $B(1,5)$. Find the other end A. (2)

SHORT ANSWER

12 Find the points of trisection of the line segment joining $(-3,5)$ and $(6,-1)$. (3)

SHORT ANSWER

13 $A(-1,2)$, $B(a,1)$, $C(5,b)$ and $D(2,3)$ are the vertices of parallelogram ABCD. Find a and b, and show that all its sides are equal. (3)

LONG ANSWER

14 Find the ratio in which $P(x,3)$ divides the segment joining $A(11,6)$ and $B(3,-2)$. Hence find x. (4)

LONG ANSWER

15 P is the midpoint of the segment joining $A(-8,6)$ and $B(-2,2)$. P also lies on the segment joining $C(-8,1)$ and $D(-3,y)$. Find the ratio $CP:PD$ and the value of y. (5)

LONG ANSWER

16 Find the ratio in which the line $2x+y-4=0$ divides the segment joining $A(1,-3)$ and $B(4,6)$. Also find the point of division. (4)

Day 16 Answers with worked solutions

1 **(c)**
Point $=\left(\frac{3+12}{3},\frac{4-10}{3}\right)=(5,-2)$: x positive, y negative, so Quadrant IV.

2 **(b)**
Ratio $k:1$: $x=\frac{-2k+4}{k+1}=0\Rightarrow k=2$. Ratio $2:1$.

3 **(d)**
Midpoint $=(-4,4)$, so $\frac{a}{2}=-4\Rightarrow a=-8$.

4 **(a)**
Centre is the midpoint: other end $=(2(-1)-3,\ 2(2)-(-1))=(-5,5)$.

5 **(b)**
$y=\frac{3k-6}{k+1}=0\Rightarrow k=2$. Ratio $2:1$.

6 **(b)**
With ratio $k':1$: $\frac{-2k'-5}{k'+1}=-3\Rightarrow k'=2$, so A is true (and $k=\frac{10}{3}$).
R is true but has nothing to do with A, so R is not the explanation.

7 **(a)**
$1:2$ point $=\left(\frac{2+16}{3},\frac{-4-2}{3}\right)=(6,-2)$, so A is true and follows from R.

8 **Ans:** $(1,-2)$
$\left(\frac{4-2}{2},\frac{-7+3}{2}\right)=(1,-2)$

9 **Ans:** $\left(5,\frac{7}{2}\right)$
Diagonals bisect each other, so the tap is the midpoint of AC: $\left(\frac{1+9}{2},\frac{1+6}{2}\right)=\left(5,\frac{7}{2}\right)$
Ans: $D(3,5)$
$D=A+C-B=(1+9-7,\ 1+6-2)=(3,5)$
Ans: $Q\left(3,\frac{4}{3}\right)$
$x=\frac{1(7)+2(1)}{3}=3$
$y=\frac{1(2)+2(1)}{3}=\frac{4}{3}$
OR: Ans: $\sqrt{89}$ units
$AC=\sqrt{(9-1)^2+(6-1)^2}=\sqrt{64+25}=\sqrt{89}$

10 **Ans:** $2:1$; $y=16$
Let the ratio be $k:1$: $\frac{10k+16}{k+1}=12\Rightarrow 10k+16=12k+12\Rightarrow k=2$
$y=\frac{2(21)+1(6)}{3}=16$

11 **Ans:** $A(5,-9)$
Centre is the midpoint of AB: $\frac{x+1}{2}=3\Rightarrow x=5$
$\frac{y+5}{2}=-2\Rightarrow y=-9$

12 **Ans:** $(0,3)$ and $(3,1)$
$1:2$: $x=\frac{6+2(-3)}{3}=0$, $y=\frac{-1+10}{3}=3$
$2:1$: $x=\frac{12-3}{3}=3$, $y=\frac{-2+5}{3}=1$

13 **Ans:** $a = 2$, $b = 2$; every side $= \sqrt{10}$ (a rhombus)

Midpoint of AC = midpoint of BD: $\frac{-1+5}{2} = \frac{a+2}{2} \Rightarrow a = 2$

$\frac{2+b}{2} = \frac{1+3}{2} \Rightarrow b = 2$

$AB = \sqrt{3^2+1^2} = \sqrt{10}$, $BC = \sqrt{3^2+1^2} = \sqrt{10}$, $CD = \sqrt{3^2+1^2} = \sqrt{10}$, $DA = \sqrt{3^2+1^2} = \sqrt{10}$

14 **Ans:** $3:5$; $x = 8$

Ratio $k:1$: $\frac{-2k+6}{k+1} = 3 \Rightarrow -2k+6 = 3k+3 \Rightarrow k = \frac{3}{5}$

Ratio $= 3:5$

$x = \frac{3(3)+5(11)}{3+5} = \frac{64}{8} = 8$

15 **Ans:** $CP:PD = 3:2$; $y = 6$

$P = \left(\frac{-8-2}{2}, \frac{6+2}{2}\right) = (-5, 4)$

Let $CP:PD = k:1$: $\frac{-3k-8}{k+1} = -5 \Rightarrow -3k-8 = -5k-5 \Rightarrow k = \frac{3}{2}$

Ratio $= 3:2$

$\frac{3y+2(1)}{5} = 4 \Rightarrow 3y = 18 \Rightarrow y = 6$

16 **Ans:** $1:2$; the point is $(2, 0)$

Let the ratio be $k:1$. Point $= \left(\frac{4k+1}{k+1}, \frac{6k-3}{k+1}\right)$

It lies on the line: $2(4k+1)+(6k-3)-4(k+1) = 0$

$10k-5 = 0 \Rightarrow k = \frac{1}{2}$, ratio $1:2$

Point $= \left(\frac{4(\frac{1}{2})+1}{\frac{3}{2}}, \frac{6(\frac{1}{2})-3}{\frac{3}{2}}\right) = (2, 0)$

Check: $2(2)+0-4 = 0$ ✓

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Section formula x : $\frac{mx_2+nx_1}{m+n}$
2. Midpoint of $(4, -7)$ and $(-2, 3)$: $(1, -2)$
3. Ratio unknown? : take $k:1$
4. Point on x-axis : put $y = 0$
5. Trisection ratios : $1:2$ and $2:1$
6. Fourth vertex of parallelogram : $D = A + C - B$
7. Centroid : average of the three vertices

DAY 16 COMPLETE? ☐ Notes read ☐ 16 questions solved ☐ Answers checked ☐ Mistakes noted

Next: Day 17, Revision 3: Arithmetic Progressions + Coordinate Geometry

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 17

REVISION 3 | AP + COORDINATE GEOMETRY

Arithmetic Progressions + Coordinate Geometry

Revision 3

Aaj ka target

- [ ] AP ke 3 formulas (a_n, S_n dono forms) bina dekhe likho
- [ ] Distance, section, midpoint formula ek page pe likho
- [ ] 22 mixed questions timer laga ke solve karo (45 min)
- [ ] Galtiyan apni mistake list mein note karo

17.1 Arithmetic Progressions: One-Page Recap

AP FORMULAS

$$a_n = a + (n-1)d$$

$$S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a + l)$$

Three numbers in AP: $2b = a + c$. $a_n = S_n - S_{n-1}$

Sum of first n natural numbers $= \frac{n(n+1)}{2}$; of first n odd numbers $= n^2$

17.2 Coordinate Geometry: One-Page Recap

COORDINATE FORMULAS

Distance:

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Section $(m : n)$:

$$\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$$

Midpoint: $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$; Centroid: $\left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$

Common mistake	Correct way
Using $a + nd$ for the nth term	$a_n = a + (n-1)d$
Taking d as $a_1 - a_2$	$d = a_2 - a_1$ (later minus earlier)
Rejecting nothing when n comes out negative or fractional	Number of terms must be a positive integer
Writing $(-3)^2 = -9$ in the distance formula	Squares are never negative
Multiplying m with x_1 in the section formula	m goes with (x_2, y_2), the far point

Common mistake	Correct way
Calling 4 equal sides a square	Also show equal diagonals

Read this list once before solving.



SHIVAM BHAIYA SAYS

AP mein 'how many terms' aur coordinate mein 'find the ratio' dono mein ek hi trick: unknown ko ek variable (n ya k) maano, formula mein daalo, linear equation solve karo. Bas!



SHIVAM BHAIYA SAYS: BOARD TIP

Revision day pe naye sums mat dhundho. Pehle yeh 22 questions karo, jo galat ho unhe Day 12-16 ke notes se dobara padho. Ratta nahi, pattern samjho.



EXAM ALERT!

AP se 5-6 marks aur Coordinate Geometry se 5-6 marks: dono milake lagbhag 11-12 marks. AP mein case study aur 'sum of n terms' wala long answer pakka; coordinate mein ratio aur 'show that it is a ...' wale sums.

MEME SAMJHO

Astra meme, maths inside



REPORTER REVISION DAY PE POOCHH RAHA HAI: KYA YAAD HAI?

Deer: $a_n = a + (n-1)d$, $S_n = \frac{n}{2}[2a + (n-1)d]$, distance aur section formula. Bas, itna hi kaafi hai.

Asli baat: Revision 3 ke 4 formulas hi 80% questions solve kar dete hain. Inko roz ek baar likho.

Day 17 Practice

21 board-pattern questions | solve first, then check

MCQ

1 The first two terms of an AP are -4 and 3. Its 21st term is: (1)

(a) 143 (b) 136 (c) 140 (d) 129

MCQ

2 In an AP with common difference 4, the value of $a_{19} - a_{14}$ is: (1)

(a) 16 **(b)** 24 **(c)** 20 **(d)** 5

MCQ

3 If $k+1$, $3k$ and $4k+2$ are three consecutive terms of an AP, then k is: (1)

(a) 3 **(b)** 1 **(c)** 2 **(d)** 5

MCQ

4 For an AP with $a = -7$ and $d = 2$, the sum of the first 8 terms is: (1)

(a) -56 **(b)** 8 **(c)** -8 **(d)** 0

ASSERTION-REASON

5 **Assertion (A):** The sum of the first 15 odd natural numbers is 225.
Reason (R): The sum of the first n odd natural numbers is n^2. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SHORT ANSWER

6 Find the sum of the first 12 terms of the AP 11, 7, 3, ... (2)

SHORT ANSWER

7 The 8th term of an AP is -2 and its 20th term is -14. Find its 35th term. (3)

SHORT ANSWER

8 The coaches of a train carry passengers in AP: the first coach has 5 passengers and the last has 50. If the train carries 440 passengers in all, find the number of coaches and the common difference. (3)

LONG ANSWER

9 Garlands are made with flowers in AP: the smallest has 5 flowers and the largest 61. If 495 flowers are used in all, how many garlands are there and what is the difference between consecutive garlands? (4)

CASE-BASED

10 Read the passage and answer. A test car's speed rises by the same amount each time the driver presses the accelerator: 18 km/h after the first press, 36 km/h after the second, 54 km/h after the third, and so on. (4)

(i) Write the first term and common difference. (1)

(ii) After which press does the speed reach 180 km/h? (1)

(iii) After how many presses does the speed first exceed 300 km/h? (2)

OR (iii) Find the sum of the speeds after the first 10 presses. (2)

MCQ

11 The point on the y-axis equidistant from $A(2,5)$ and $B(5,6)$ is: (1)

(a) $(0,13)$ **(b)** $(16,0)$ **(c)** $(0,16)$ **(d)** $(0,-16)$

MCQ

12 The line $x + y - 5 = 0$ divides the segment joining $(1,1)$ and $(4,6)$ in the ratio: (1)

(a) $5:3$ **(b)** $3:5$ **(c)** $1:2$ **(d)** $2:3$

MCQ

13 The points $(-3,0)$, $(3,0)$ and $(0,4)$ form: (1)

(a) an equilateral triangle **(b)** a right triangle
(c) an isosceles triangle **(d)** no triangle

ASSERTION-REASON

14 **Assertion (A):** The origin is the only point equidistant from $(3,4)$ and $(-3,-4)$.
Reason (R): The midpoint of the segment joining $(3,4)$ and $(-3,-4)$ is the origin. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

VERY SHORT ANSWER

15 Find the distance between $(0,3)$ and $(4,0)$. (1)

SHORT ANSWER

16 $P(1,3)$ is equidistant from $A(4,k)$ and $B(k,6)$. Find k. (2)

SHORT ANSWER

17 Find the values of y for which the distance between $A(2,-3)$ and $B(8,y)$ is 10 units. (2)

SHORT ANSWER

18 $M(1,2)$ is the midpoint of the segment joining $A(-2,b)$ and $B(4,b+6)$. Find b. (2)

SHORT ANSWER

19 Find the ratio in which $P(2,y)$ divides the segment joining $A(4,-3)$ and $B(-6,7)$, and find y. (3)

CASE-BASED

20 **Read the passage and answer. On a village map drawn on a grid (1 unit = 100 m), the school is at $S(2,3)$ and the hospital at $H(8,11)$. A market is planned exactly halfway between them.** (4)

(i) Find the distance between the school and the hospital. (1)

(ii) Find the coordinates of the market. (1)

(iii) A bus stop B divides SH in the ratio $1:3$. Find B. (2)

OR (iii) How far is the bus stop from the school (in metres)? (2)

LONG ANSWER

21 **Show that $A(2,-1)$, $B(3,4)$, $C(-2,3)$ and $D(-3,-2)$ are the vertices of a rhombus. Is it a square?** (4)

Day 17 Answers with worked solutions

1 **(b)**

$a=-4$, $d=7$; $a_{21}=-4+20\times 7=136$

2 **(c)**

$a_{19}-a_{14}=(19-14)d=5\times 4=20$

3 **(a)**

$2(3k)=(k+1)+(4k+2)\Rightarrow 6k=5k+3\Rightarrow k=3$

4 **(d)**

$S_8=\frac{8}{2}[2(-7)+7(2)]=4[-14+14]=0$

5 **(a)**

$1+3+\cdots+29=\frac{15}{2}(1+29)=225=15^2$; R gives exactly this.

6 **Ans:** -132

$a=11$, $d=-4$, $n=12$

$S_{12}=\frac{12}{2}[22+11(-4)]=6\times(-22)=-132$

7 **Ans:** $a_{35}=-29$

$a+7d=-2$ and $a+19d=-14$

Subtract: $12d=-12\Rightarrow d=-1$, so $a=5$

$a_{35}=5+34(-1)=-29$

8 **Ans:** 16 coaches; $d=3$

$S_n=\frac{n}{2}(a+l)\Rightarrow 440=\frac{n}{2}(5+50)\Rightarrow n=16$

$50=5+15d\Rightarrow d=3$

9 **Ans:** 15 garlands; difference 4 flowers

$495=\frac{n}{2}(5+61)=33n\Rightarrow n=15$

$61=5+14d\Rightarrow d=4$

10 **Ans:** $a=18$, $d=18$

$36-18=54-36=18$

Ans: 10th press

$18+(n-1)18=180\Rightarrow 18n=180\Rightarrow n=10$

Ans: 17 presses

$a_n=18n>300\Rightarrow n>16.67$

Smallest integer: $n=17$ (speed 306 km/h).

OR: Ans: 990

$S_{10}=\frac{10}{2}[2(18)+9(18)]=5\times 198=990$

11 **(c)**

$4+(y-5)^2=25+(y-6)^2\Rightarrow -10y+29=-12y+61\Rightarrow y=16$

12 **(b)**

Point $\left(\frac{4k+1}{k+1},\frac{6k+1}{k+1}\right)$ on the line: $10k+2=5k+5\Rightarrow k=\frac{3}{5}$

13 **(c)**

Sides: 6, 5, 5 ($\sqrt{9+16}=5$). Two equal sides, and $5^2+5^2\neq 6^2$.

14 **(d)**

Every point on the perpendicular bisector ($3x+4y=0$) is equidistant, e.g. $(4,-3)$. So A is false; R is true.

15 **Ans:** 5 units

$\sqrt{16+9}=5$

16 **Ans:** $k=2$

$9+(k-3)^2=(k-1)^2+9$

$-6k+9=-2k+1\Rightarrow k=2$

17 **Ans:** $y=5$ or $y=-11$

$36+(y+3)^2=100\Rightarrow (y+3)^2=64$

$y+3=\pm 8\Rightarrow y=5$ or -11

18 **Ans:** $b=-1$

$\frac{b+b+6}{2}=2\Rightarrow 2b+6=4\Rightarrow b=-1$

(x-check: $\frac{-2+4}{2}=1$)

19 **Ans:** $1:4$; $y=-1$

$\frac{-6k+4}{k+1}=2\Rightarrow -6k+4=2k+2\Rightarrow k=\frac{1}{4}$, ratio $1:4$

$y=\frac{1(7)+4(-3)}{5}=-1$

20 **Ans:** 10 units (1 km)

$\sqrt{6^2+8^2}=10$

Ans: $(5,7)$

Midpoint $=\left(\frac{2+8}{2},\frac{3+11}{2}\right)=(5,7)$

Ans: $\left(\frac{7}{2},5\right)$

$x=\frac{1(8)+3(2)}{4}=\frac{7}{2}$, $y=\frac{1(11)+3(3)}{4}=5$

OR: Ans: 250 m

SB is $\frac{1}{4}$ of SH $=\frac{10}{4}=2.5$ units $=250$ m

21 **Ans:** All sides $\sqrt{26}$ (rhombus); diagonals $4\sqrt{2}$ and $6\sqrt{2}$ are unequal, so not a square.

$AB = \sqrt{1+25} = \sqrt{26}$, $BC = \sqrt{25+1} = \sqrt{26}$,

$CD = \sqrt{1+25} = \sqrt{26}$, $DA = \sqrt{25+1} = \sqrt{26}$

All four sides equal, so ABCD is a rhombus.

$AC = \sqrt{16+16} = 4\sqrt{2}$, $BD = \sqrt{36+36} = 6\sqrt{2}$

Diagonals unequal, so it is not a square.

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. nth term : $a + (n-1)d$
2. Sum with last term : $\frac{n}{2}(a+l)$
3. $2b = a + c$ means : a, b, c in AP
4. Distance from origin : $\sqrt{x^2+y^2}$
5. Section formula, m goes with : the far point (x_2, y_2)
6. Rhombus vs square : square has equal diagonals
7. Sum of first n odd numbers : n^2

DAY 17 COMPLETE? ☐ Recap read

☐ 21 questions solved ☐ Answers checked

☐ Mistake list updated Next: Day 18, Triangles

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 18

TRIANGLES | CHAPTER 6 | PART 1 OF 3

Similar Figures and the Basic Proportionality Theorem

Triangles

Aaj ka target

- [ ] Similar aur congruent ka fark clear karo
- [ ] BPT ka statement aur proof (5 marks wala) likh ke practice karo
- [ ] Converse of BPT se 'parallel hai' prove karna seekho
- [ ] Ratio wale sums: AD/DB = AE/EC seedha lagao

18.1 Similar Figures

- Two figures are **similar** if they have the same shape (not necessarily the same size). Congruent figures have the same shape AND size, so every congruent pair is similar, but not the other way round.
- All circles are similar; all squares are similar; all equilateral triangles are similar.
- Two polygons with the same number of sides are similar if (i) their corresponding angles are equal AND (ii) their corresponding sides are in the same ratio. Both conditions are needed for polygons.
- Example: a square and a rhombus of side 3 cm have proportional sides but unequal angles, so they are not similar. A square and a rectangle have equal angles but non-proportional sides, so they are not similar.

KEY TERM: SIMILAR TRIANGLES

$\triangle ABC \sim \triangle DEF$ means $\angle A = \angle D$, $\angle B = \angle E$, $\angle C = \angle F$ and $\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD}$. The order of letters tells you which vertices correspond.



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

$\triangle ABC \sim \triangle QPR$ likha ho to A ka partner Q, B ka P, C ka R. Ratio likhte waqt order follow karo: $\frac{AB}{QP} = \frac{BC}{PR} = \frac{AC}{QR}$. Sabse zyada marks yahin katte hain.

18.2 Basic Proportionality Theorem (Thales Theorem)

Statement: If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio.

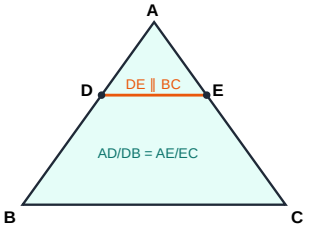


Fig. 18.1 In triangle ABC, DE is parallel to BC.

BPT AND ITS FORMS (DE ∥ BC)

$$\frac{AD}{DB} = \frac{AE}{EC}$$

$$\frac{AD}{AB} = \frac{AE}{AC} \qquad \frac{DB}{AB} = \frac{EC}{AC}$$

PROOF OF BPT (5 MARKS, LEARN IT STEP BY STEP)

1. **Given:** In $\triangle ABC$, $DE \parallel BC$, D on AB and E on AC. **To prove:** $\frac{AD}{DB} = \frac{AE}{EC}$.
2. **Construction:** Join BE and CD. Draw $EN \perp AB$ and $DM \perp AC$.
3. $\text{ar}(ADE) = \frac{1}{2} \times AD \times EN$ and $\text{ar}(BDE) = \frac{1}{2} \times DB \times EN$, so $\frac{\text{ar}(ADE)}{\text{ar}(BDE)} = \frac{AD}{DB}$ ...(1)
4. Similarly, $\frac{\text{ar}(ADE)}{\text{ar}(DEC)} = \frac{AE}{EC}$ ...(2)
5. $\triangle BDE$ and $\triangle DEC$ are on the same base DE and between the same parallels DE and BC, so $\text{ar}(BDE) = \text{ar}(DEC)$ ...(3)
6. From (1), (2) and (3): $\frac{AD}{DB} = \frac{AE}{EC}$. Hence proved.

SOLVED EXAMPLE 1

In $\triangle ABC$, $DE \parallel BC$. If $AD = x$, $DB = x - 3$, $AE = x + 3$ and $EC = x - 2$, find x.

SOLUTION

1. By BPT, $\frac{AD}{DB} = \frac{AE}{EC} \Rightarrow \frac{x}{x-3} = \frac{x+3}{x-2}$
2. $x(x-2) = (x-3)(x+3) \Rightarrow x^2 - 2x = x^2 - 9$
3. $-2x = -9$

Answer: $x = \frac{9}{2} = 4.5$



SHIVAM BHAIYA SAYS

Cross multiply karte hi x^2 dono taraf cancel ho jata hai, linear equation bachti hai. Agar x^2 cancel nahi hua, ek baar BPT ka ratio check karo, kahin AB ki jagah DB to nahi le liya.

18.3 Converse of BPT

If a line divides any two sides of a triangle in the same ratio, then the line is parallel to the third side. Use it whenever a question says 'show that DE is parallel to BC'.

SOLVED EXAMPLE 2

D and E lie on sides AB and AC of $\triangle ABC$ with $AD = 3$ cm, $DB = 4.5$ cm, $AE = 4$ cm and $EC = 6$ cm. Is $DE \parallel BC$?

SOLUTION

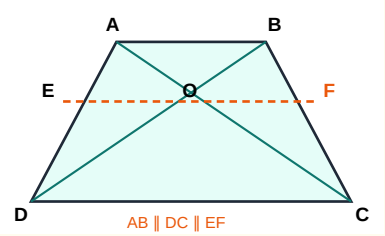
1. $\frac{AD}{DB} = \frac{3}{4.5} = \frac{2}{3}$
2. $\frac{AE}{EC} = \frac{4}{6} = \frac{2}{3}$
3. The ratios are equal, so by the converse of BPT, $DE \parallel BC$.

Answer: Yes, $DE \parallel BC$.

SPECIAL CASE: MID-POINT THEOREM. THE LINE JOINING THE MIDPOINTS OF TWO SIDES IS PARALLEL TO THE THIRD SIDE AND HALF OF IT.

SOLVED EXAMPLE 3

ABCD is a trapezium with $AB \parallel DC$. Its diagonals meet at O. Prove that $\frac{AO}{BO} = \frac{CO}{DO}$.



SOLUTION

1. Draw $EF \parallel AB$ through O, meeting AD at E and BC at F. Then $EF \parallel DC$ as well.
2. In $\triangle ADC$, $EO \parallel DC$: by BPT, $\frac{AE}{ED} = \frac{AO}{OC}$ ...(1)
3. In $\triangle DAB$, $EO \parallel AB$: by BPT, $\frac{AE}{ED} = \frac{BO}{OD}$ ...(2)
4. From (1) and (2): $\frac{AO}{OC} = \frac{BO}{OD} \Rightarrow \frac{AO}{BO} = \frac{CO}{DO}$

Answer: Hence proved.



EXAM ALERT!

BPT ka proof 3 ya 5 marks mein bahut baar aaya hai. Given, To prove, Construction, Proof: chaaron heading likhoge tabhi full marks. Areas of similar triangles aur Pythagoras ke proofs 2026-27 syllabus se hata diye gaye hain.



SHIVAM BHAIYA SAYS: BOARD TIP

Figure hamesha banao aur parallel lines par arrow marks lagao. Examiner ko figure dekh ke hi samajh aa jata hai ki aapko concept aata hai.

MEME SAMJHO — Astra meme, maths inside



JAB QUESTION KI FIGURE MEIN DE PARALLEL BC DIKH JAYE

BPT laga do, do line mein answer. Aaj ka din ban gaya!

Asli baat: DE ∥ BC in triangle ABC $\Rightarrow \frac{AD}{DB} = \frac{AE}{EC}$ (Basic Proportionality Theorem).

Day 18 Practice — 15 board-pattern questions | solve first, then check

MCQ

1 In $\triangle ABC$, $DE \parallel BC$ with D on AB and E on AC. If $AD = 3$ cm, $DB = 4$ cm and $EC = 2$ cm, then AE is: (1)

(a) 2.5 cm (b) 1.5 cm (c) $\frac{8}{3}$ cm (d) 6 cm

MCQ

2 In $\triangle ABC$, P on AB and Q on BC are such that $PQ \parallel AC$. If $BP = 5$ cm, $AP = 3$ cm and $BQ = 6$ cm, then BC is: (1)

(a) 9.6 cm (b) 3.6 cm (c) 10 cm (d) 8 cm

MCQ

3 Three parallel shelves cut the slanting side of a cabinet, 120 cm long, into parts in the ratio $1 : 2 : 3$. The longest part is: (1)

(a) 40 cm (b) 20 cm (c) 60 cm (d) 72 cm

MCQ

4 D and E are the midpoints of sides AB and AC of $\triangle ABC$. If $BC = 14$ cm, then DE is: (1)

(a) 28 cm (b) 14 cm (c) 3.5 cm (d) 7 cm

MCQ

5 Which pair of figures is ALWAYS similar? (1)

(a) Two rhombuses (b) Two rectangles
(c) Two isosceles triangles (d) Two squares

ASSERTION-REASON

6 **Assertion (A):** In $\triangle PQR$, E and F lie on PQ and PR with $PE = 1.5$ cm, $EQ = 3$ cm, $PF = 2$ cm and $FR = 4$ cm. Then $EF \parallel QR$.
Reason (R): If a line divides two sides of a triangle in the same ratio, it is parallel to the third side. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

7 **Assertion (A):** Every two congruent triangles are similar.
Reason (R): Similar triangles must always be equal in size. (1)
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

VERY SHORT ANSWER

8 Give an example of two figures that are similar but not congruent. (1)

CASE-BASED

9 Read the passage and answer. A triangular roof truss ABC has a horizontal beam DE parallel to the base BC, with D on AB and E on AC. On the drawing, $AD = 1.2$ m, $DB = 1.8$ m and $AE = 1.6$ m. (4)

(i) Find EC. (1)

(ii) Find the ratio $AD : AB$. (1)

(iii) If $BC = 7.5$ m, find the length of the beam DE. (2)

OR (iii) A second beam joins the midpoints of AB and AC. If $BC = 7.5$ m, how long is it? (2)

SHORT ANSWER

10 In $\triangle ABC$, $DE \parallel BC$. If $\frac{AD}{DB} = \frac{3}{5}$ and $AC = 5.6$ cm, find AE. (2)

SHORT ANSWER

11 E and F are points on sides PQ and PR of $\triangle PQR$ with $PE = 3$ cm, $QE = 4.5$ cm, $PF = 4$ cm and $RF = 6$ cm. State whether $EF \parallel QR$. (2)

SHORT ANSWER

12 In $\triangle ABC$, $DE \parallel BC$ with D on AB and E on AC. Show that $\frac{AD}{AB} = \frac{AE}{AC}$. (3)

SHORT ANSWER

13 In quadrilateral ABCD, a point L on diagonal AC is joined to M on AB and N on AD so that $LM \parallel CB$ and $LN \parallel CD$. Prove that $\frac{AM}{AB} = \frac{AN}{AD}$. (3)

LONG ANSWER

14 State and prove the Basic Proportionality Theorem. (5)

LONG ANSWER

15 ABCD is a trapezium with $AB \parallel DC$. E and F are points on the non-parallel sides AD and BC such that $EF \parallel AB$. Show that $\frac{AE}{ED} = \frac{BF}{FC}$. (4)

Day 18 Answers with worked solutions

1 **(b)**
By BPT, $\frac{AD}{DB} = \frac{AE}{EC} \Rightarrow AE = \frac{3\times2}{4} = 1.5$ cm

2 **(a)**
$\frac{BP}{BA} = \frac{BQ}{BC} \Rightarrow \frac{5}{8} = \frac{6}{BC} \Rightarrow BC = \frac{48}{5} = 9.6$ cm

3 **(c)**
Parts are $\frac{1}{6}, \frac{2}{6}, \frac{3}{6}$ of 120 cm, so the longest is $\frac{3}{6} \times 120 = 60$ cm.

4 **(d)**
By the mid-point theorem, $DE = \frac{1}{2}BC = 7$ cm.

5 **(d)**
All angles of every square are 90° and all sides are equal, so any two squares have proportional sides and equal angles. The others can differ in angles or side ratios.

6 **(a)**
$\frac{PE}{EQ} = \frac{1}{2} = \frac{PF}{FR}$, so by the converse of BPT (which is R), $EF \parallel QR$.

7 **(c)**
Congruent triangles have equal angles and sides in ratio 1 : 1, so they are similar: A is true.
Similar figures need only the same shape, not the same size: R is false.

8 **Ans:** Two circles of radii 2 cm and 5 cm (or any two squares of different sides).
Same shape, different size: similar but not congruent.

9 **Ans:** 2.4 m
By BPT, $\frac{AD}{DB} = \frac{AE}{EC} \Rightarrow EC = \frac{1.6\times1.8}{1.2} = 2.4$ m
Ans: $2 : 5$
$AB = 1.2 + 1.8 = 3$ m, so $\frac{AD}{AB} = \frac{1.2}{3} = \frac{2}{5}$
Ans: 3 m
$\triangle ADE \sim \triangle ABC$ (AA: common $\angle A$, corresponding angles $\angle ADE = \angle ABC$)
$\frac{DE}{BC} = \frac{AD}{AB} = \frac{2}{5} \Rightarrow DE = \frac{2}{5} \times 7.5 = 3$ m
OR: Ans: 3.75 m
By the mid-point theorem the beam is parallel to BC and half of it: $\frac{7.5}{2} = 3.75$ m

10 **Ans:** 2.1 cm
By BPT, $\frac{AE}{EC} = \frac{3}{5}$, so $AE = \frac{3}{8} \times AC$
$AE = \frac{3}{8} \times 5.6 = 2.1$ cm

11 **Ans:** Yes, $EF \parallel QR$
$\frac{PE}{EQ} = \frac{3}{4.5} = \frac{2}{3}$ and $\frac{PF}{FR} = \frac{4}{6} = \frac{2}{3}$
Equal ratios, so by the converse of BPT, $EF \parallel QR$.

12 **Ans:** Proved
By BPT, $\frac{AD}{DB} = \frac{AE}{EC}$, so $\frac{DB}{AD} = \frac{EC}{AE}$
Add 1 to both sides: $\frac{DB+AD}{AD} = \frac{EC+AE}{AE} \Rightarrow \frac{AB}{AD} = \frac{AC}{AE}$
Taking reciprocals, $\frac{AD}{AB} = \frac{AE}{AC}$.

13 **Ans:** Proved

In $\triangle ABC$, $LM \parallel CB$: by BPT, $\frac{AM}{AB} = \frac{AL}{AC}$ ...(1)

In $\triangle ACD$, $LN \parallel CD$: by BPT, $\frac{AN}{AD} = \frac{AL}{AC}$ ...(2)

From (1) and (2), $\frac{AM}{AB} = \frac{AN}{AD}$.

14 **Ans:** See the proof in the notes (areas of triangles on the same base between the same parallels).

Given: $\triangle ABC$, $DE \parallel BC$. To prove: $\frac{AD}{DB} = \frac{AE}{EC}$.

Construction: join BE, CD; draw $EN \perp AB$, $DM \perp AC$.

$\frac{\text{ar}(ADE)}{\text{ar}(BDE)} = \frac{\frac{1}{2}AD \times EN}{\frac{1}{2}DB \times EN} = \frac{AD}{DB}$

$\frac{\text{ar}(ADE)}{\text{ar}(DEC)} = \frac{\frac{1}{2}AE \times DM}{\frac{1}{2}EC \times DM} = \frac{AE}{EC}$

$\text{ar}(BDE) = \text{ar}(DEC)$ (same base DE, between parallels DE and BC)

Therefore $\frac{AD}{DB} = \frac{AE}{EC}$.

15 **Ans:** Proved

Join AC, meeting EF at G.

In $\triangle ADC$, $EG \parallel DC$: $\frac{AE}{ED} = \frac{AG}{GC}$ ...(1)

In $\triangle CAB$, $GF \parallel AB$: $\frac{CG}{GA} = \frac{CF}{FB}$, i.e. $\frac{AG}{GC} = \frac{BF}{FC}$ ...(2)

From (1) and (2), $\frac{AE}{ED} = \frac{BF}{FC}$.

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. BPT : DE ∥ BC gives AD/DB = AE/EC
2. Converse of BPT : same ratio means parallel
3. All squares similar? : Yes
4. All rhombuses similar? : No, angles may differ
5. Mid-point theorem : DE ∥ BC and DE = BC/2
6. Trapezium diagonals : AO/BO = CO/DO
7. Deleted in 2026-27 : areas of similar triangles, Pythagoras proofs

DAY 18 COMPLETE?

- [ ] Notes read
- [ ] 15 questions solved
- [ ] Answers checked
- [ ] Mistakes noted

Next: Day 19, Triangles

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 19

TRIANGLES | CHAPTER 6 | PART 2 OF 3

Criteria for Similarity of Triangles

Triangles

Aaj ka target

- [ ] AA, SSS, SAS teeno criteria ke statement yaad karo
- [ ] Correspondence (order of letters) sahi likhna seekho
- [ ] SAS mein angle 'included' hona chahiye, SSA bilkul nahi
- [ ] Shadow aur height wale sums AA se 2 minute mein

19.1 The Three Criteria

To prove two triangles similar we do not need all six conditions (3 angles + 3 side ratios). Any one of these criteria is enough.

Criterion	What you need	Conclusion
AA (or AAA)	Two angles of one triangle equal to two angles of the other	Triangles similar (third angles equal by angle sum)
SSS	All three pairs of corresponding sides in the same ratio	Triangles similar (angles equal)
SAS	One angle equal and the sides including it in the same ratio	Triangles similar

There is NO SSA or ASS criterion for similarity.

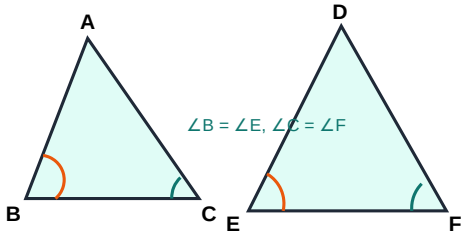


Fig. 19.1 Two pairs of equal angles: triangle ABC is similar to triangle DEF (AA).

IF TRIANGLE ABC IS SIMILAR TO TRIANGLE DEF

$$\angle A = \angle D,\ \angle B = \angle E,\ \angle C = \angle F$$

$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD} = \frac{\text{Perimeter}(ABC)}{\text{Perimeter}(DEF)}$$



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

SAS mein angle dono sides ke BEECH ka hona chahiye. Agar angle bahar ka hai (SSA), to similarity prove nahi hoti. Board mein yeh galti karne pe poore marks jaate hain.

19.2 Using AA: Heights and Shadows

At the same time of day the sun's rays are parallel, so a vertical pole and its shadow form a triangle similar to the one formed by a tower and its shadow (both have a right angle and the same angle of elevation of the sun).

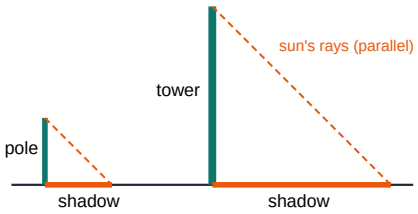


Fig. 19.2 Pole and tower with their shadows at the same time.

SOLVED EXAMPLE 1

A vertical pole 4.5 m high casts a shadow 3 m long. At the same time a tower casts a shadow 26 m long. Find the height of the tower.

SOLUTION

1. Both triangles have a right angle and equal angles of elevation of the sun, so they are similar (AA).
2. $\frac{\text{height of tower}}{\text{height of pole}} = \frac{\text{shadow of tower}}{\text{shadow of pole}}$
3. $h = \frac{4.5 \times 26}{3} = 39$

Answer: The tower is 39 m high.

SOLVED EXAMPLE 2 (SAS)

Segments BD and CE intersect at P with $PB = 4$ cm, $PD = 12$ cm, $PC = 5$ cm and $PE = 15$ cm. Show that $\triangle PBC \sim \triangle PDE$.

SOLUTION

1. $\frac{PB}{PD} = \frac{4}{12} = \frac{1}{3}$ and $\frac{PC}{PE} = \frac{5}{15} = \frac{1}{3}$
2. $\angle BPC = \angle DPE$ (vertically opposite angles), and it is included between these sides.
3. By SAS similarity, $\triangle PBC \sim \triangle PDE$.

Answer: Proved (SAS).

SOLVED EXAMPLE 3 (SSS)

The sides of one triangle are 3 cm, 4 cm and 6 cm, and of another 4.5 cm, 6 cm and 9 cm. Are they similar?

SOLUTION

1. Match smallest with smallest: $\frac{3}{4.5} = \frac{2}{3}$, $\frac{4}{6} = \frac{2}{3}$, $\frac{6}{9} = \frac{2}{3}$

SOLVED EXAMPLE 3 (SSS)

1.
2. All three ratios are equal.

Answer: Yes, similar by SSS (scale factor $\frac{2}{3}$).

SOLVED EXAMPLE 4

Diagonals AC and BD of a quadrilateral meet at O, and $\triangle ODC \sim \triangle OBA$. If $\angle BOC = 115°$ and $\angle CDO = 75°$, find $\angle DOC$, $\angle DCO$ and $\angle OAB$.

SOLUTION

1. $\angle DOC = 180° - 115° = 65°$ (linear pair, since BD is a straight line)
2. In $\triangle ODC$: $\angle DCO = 180° - 65° - 75° = 40°$
3. $\triangle ODC \sim \triangle OBA$ gives $\angle OAB = \angle OCD = 40°$

Answer: $65°$, $40°$, $40°$



SHIVAM BHAIYA SAYS

SSS check karte waqt sides ko chhote se bade order mein lagao: smallest/smallest, middle/middle, largest/largest. Order galat liya to similar triangles bhi 'not similar' lagenge.



EXAM ALERT!

'Prove that the triangles are similar' 2-3 marks mein har saal aata hai. Reason likhna zaroori hai: (AA), (SAS) ya (SSS). Case study mein shadow wale sums bahut common hain.



SHIVAM BHAIYA SAYS: BOARD TIP

Proof mein har step ke saamne bracket mein reason likho: (common angle), (vertically opposite angles), (alternate angles). Yahi reasons step marks dilaate hain.

MEME SAMJHO — Astra meme, maths inside



SSA SE SIMILARITY PROVE KAR DI, PHIR RESULT AAYA

Examiner ne poore marks kaat diye... SSA naam ka koi criterion hi nahi hai.

Asli baat: similarity criteria sirf AA, SSS aur SAS hain. SAS mein angle dono sides ke beech (included) hona chahiye.

Day 19 Practice

15 board-pattern questions | solve first, then check

MCQ

1. $\triangle ABC \sim \triangle QPR$. If $AC = 8$ cm, $BC = 6$ cm and $QR = 4$ cm, then PR is: (1)
   (a) 3 cm (b) 5.3 cm (c) 12 cm (d) 2 cm

MCQ

2. In $\triangle ABC$ and $\triangle DEF$, $\frac{AB}{DE} = \frac{BC}{FD}$. They will be similar if: (1)
   (a) $\angle B = \angle E$ (b) $\angle A = \angle D$ (c) $\angle B = \angle D$ (d) $\angle A = \angle F$

MCQ

3. The perimeters of two similar triangles are 36 cm and 24 cm. If a side of the first is 12 cm, the corresponding side of the second is: (1)
   (a) 18 cm (b) 8 cm (c) 6 cm (d) 16 cm

MCQ

4. $\triangle AHK \sim \triangle ABC$. If $AK = 12$ cm, $BC = 4.5$ cm and $HK = 9$ cm, then AC is: (1)
   (a) 24 cm (b) 4 cm (c) 6 cm (d) 8 cm

ASSERTION-REASON

5. **Assertion (A):** Two triangles in which two sides are proportional and a non-included angle is equal are always similar.
   Reason (R): In the SAS similarity criterion, the equal angle must be included between the proportional sides. (1)
   (a) Both A and R are true, and R is the correct explanation of A.
   (b) Both A and R are true, but R is not the correct explanation of A.
   (c) A is true, but R is false.
   (d) A is false, but R is true.

ASSERTION-REASON

6. **Assertion (A):** In $\triangle ABC$ and $\triangle DEF$, $\angle A = \angle D = 50°$, $\angle B = 60°$ and $\angle F = 70°$. Then $\triangle ABC \sim \triangle DEF$.
   Reason (R): If two angles of one triangle are equal to two angles of another, the triangles are similar. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

VERY SHORT ANSWER

7 **If $\triangle DEF \sim \triangle RPQ$, is it true that $\angle D = \angle R$ and $\angle F = \angle P$?** (1)

CASE-BASED

8 **Read the passage and answer. At 4 pm on a sunny day, a house 15 m tall casts a shadow 6 m long. At the same moment, a nearby tower and a pole also cast shadows on the level ground.** (4)

(i) The tower's shadow is 30 m long. Find its height. (1)

(ii) Which similarity criterion is used here? (1)

(iii) The pole's shadow is 4.8 m. Find the height of the pole. (2)

OR (iii) Find the length of the shadow of a 9 m tree at the same time. (2)

SHORT ANSWER

9 **In $\triangle PQR$, $\angle P = 50^\circ$ and $\angle Q = 30^\circ$. In $\triangle MST$, $\angle M = 100^\circ$ and $\angle S = 30^\circ$. Is $\triangle QPR \sim \triangle TSM$? Give the correct correspondence.** (2)

SHORT ANSWER

10 **$\triangle ABC \sim \triangle DEF$ with $DE = 4$ cm, $EF = 3$ cm, $DF = 3.5$ cm and $BC = 6$ cm. Find the perimeter of $\triangle ABC$.** (2)

SHORT ANSWER

11 **In $\triangle PQR$, $\angle QPR = 90^\circ$ and $PM \perp QR$. Prove that $PQ^2 = QM \times QR$.** (3)

SHORT ANSWER

12 **Segments AB and CD intersect at O such that $OA \times OB = OC \times OD$. Prove that $\angle A = \angle C$ and $\angle B = \angle D$ (angles of triangles AOD and COB).** (3)

LONG ANSWER

13 **ABCD is a parallelogram. P divides AB in the ratio $3:2$ and Q divides DC in the ratio $4:1$. PQ meets diagonal AC at O. Prove that $OC = \frac{1}{3}OA$.** (4)

LONG ANSWER

14 **CM and RN are medians of $\triangle ABC$ and $\triangle PQR$ respectively, and $\triangle ABC \sim \triangle PQR$. Prove that (i) $\triangle AMC \sim \triangle PNR$ (ii) $\frac{CM}{RN} = \frac{AB}{PQ}$.** (4)

SHORT ANSWER

15 **S and T are points on sides PQ and PR of $\triangle PQR$ such that $\angle P = \angle RTS$. Show that $\triangle RPQ \sim \triangle RTS$.** (2)

Day 19 Answers with worked solutions

1 **(a)**
A corresponds to Q, B to P, C to R, so $\frac{AC}{QR} = \frac{BC}{PR}$
$\frac{8}{4} = \frac{6}{PR} \Rightarrow PR = 3$ cm

2 **(c)**
For SAS the equal angle must be included: B lies between AB and BC, and D lies between DE and FD. So we need $\angle B = \angle D$.

3 **(b)**
Ratio of sides = ratio of perimeters: $\frac{12}{x} = \frac{36}{24} \Rightarrow x = 8$ cm

4 **(c)**
$\frac{AK}{AC} = \frac{HK}{BC} \Rightarrow \frac{12}{AC} = \frac{9}{4.5} \Rightarrow AC = 6$ cm

5 **(d)**
There is no SSA criterion, so A is false. R correctly states SAS.

6 **(a)**
$\angle C = 180^\circ - 110^\circ = 70^\circ = \angle F$ and $\angle A = \angle D$, so the triangles are similar by AA, which is R.

7 **Ans:** No. $\angle D = \angle R$ but $\angle F = \angle Q$.
Correspondence by order: D to R, E to P, F to Q.

8 **Ans:** 75 m
$\frac{h}{15} = \frac{30}{6} \Rightarrow h = 75$ m
Ans: AA
Both triangles have a right angle and the same angle of elevation of the sun.
Ans: 12 m
$\frac{h}{15} = \frac{4.8}{6} \Rightarrow h = \frac{15 \times 4.8}{6} = 12$ m
OR: Ans: 3.6 m
$\frac{s}{6} = \frac{9}{15} \Rightarrow s = \frac{54}{15} = 3.6$ m

9 **Ans:** No; the correct statement is $\triangle QPR \sim \triangle STM$.
$\angle R = 100^\circ$ and $\angle T = 50^\circ$.
Equal angles: $Q = S = 30^\circ$, $P = T = 50^\circ$, $R = M = 100^\circ$.
So $\triangle QPR \sim \triangle STM$ (AA), not $\triangle TSM$.

10 **Ans:** 21 cm
Scale factor $= \frac{BC}{EF} = \frac{6}{3} = 2$
Perimeter(DEF) $= 4 + 3 + 3.5 = 10.5$ cm
Perimeter(ABC) $= 2 \times 10.5 = 21$ cm

11 **Ans:** Proved
In $\triangle QMP$ and $\triangle QPR$: $\angle QMP = \angle QPR = 90^\circ$ and $\angle Q$ is common.
So $\triangle QMP \sim \triangle QPR$ (AA).
$\frac{QM}{QP} = \frac{QP}{QR} \Rightarrow PQ^2 = QM \times QR$

12 **Ans:** Proved

$OA \times OB = OC \times OD \Rightarrow \frac{OA}{OC} = \frac{OD}{OB}$

$\angle AOD = \angle COB$ (vertically opposite angles)

So $\triangle AOD \sim \triangle COB$ (SAS)

Hence $\angle OAD = \angle OCB$ and $\angle ODA = \angle OBC$, i.e. $\angle A = \angle C$, $\angle D = \angle B$.

13 **Ans:** Proved: $OA = 3 \times OC$

$AP = \frac{3}{5}AB$ and $CQ = \frac{1}{5}DC = \frac{1}{5}AB$ (since $DC = AB$).

In $\triangle APO$ and $\triangle CQO$: $\angle PAO = \angle QCO$ (alternate angles, $AB \parallel DC$) and $\angle AOP = \angle COQ$ (vertically opposite).

$\triangle APO \sim \triangle CQO$ (AA)

$\frac{OA}{OC} = \frac{AP}{CQ} = \frac{\frac{3}{5}AB}{\frac{1}{5}AB} = 3 \Rightarrow OC = \frac{1}{3}OA$

14 **Ans:** Proved

$\triangle ABC \sim \triangle PQR \Rightarrow \frac{AB}{PQ} = \frac{AC}{PR}$ and $\angle A = \angle P$

M, N are midpoints: $AM = \frac{AB}{2}$, $PN = \frac{PQ}{2}$, so $\frac{AM}{PN} = \frac{AB}{PQ} = \frac{AC}{PR}$

With $\angle A = \angle P$, $\triangle AMC \sim \triangle PNR$ (SAS) ... (i)

From (i), $\frac{CM}{RN} = \frac{AC}{PR} = \frac{AB}{PQ}$ ... (ii)

15 **Ans:** Proved (AA)

In $\triangle RPQ$ and $\triangle RTS$: $\angle RPQ = \angle RTS$ (given) and $\angle R$ is common.

So $\triangle RPQ \sim \triangle RTS$ (AA).

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Three similarity criteria : AA, SSS, SAS
2. SSA a criterion? : No
3. SAS angle must be : included between the two sides
4. Ratio of perimeters of similar triangles : equals ratio of sides
5. Shadow problems use : AA
6. $\triangle DEF \sim \triangle RPQ$, partner of F : Q
7. Right triangle with altitude PM : $PQ^2 = QM \times QR$

DAY 19 COMPLETE?

- ☐ Notes read
- ☐ 15 questions solved
- ☐ Answers checked
- ☐ Mistakes noted

Next: Day 20, Triangles

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 20

TRIANGLES | CHAPTER 6 | PART 3 OF 3

Similarity Proofs and Board Problems

Triangles

Aaj ka target

- [ ] Product form (AB x CD = PQ x RS) ko ratio mein badal ke triangles dhundho
- [ ] Parallel lines dikhe to alternate angles + AA
- [ ] Poles wala result 1/a + 1/b = 1/z derive karna seekho
- [ ] Case study mein har sub-part ka reason likho

20.1 How to Attack a Similarity Proof

1. Convert what is to be proved into a ratio. A product $AB \times CD = PQ \times RS$ becomes $\frac{AB}{PQ} = \frac{RS}{CD}$.
2. Pick the two triangles that contain the sides of the ratio (the numerators in one triangle, the denominators in the other).
3. Look for equal angles: right angles, a common angle, vertically opposite angles, alternate or corresponding angles from parallel lines, base angles of an isosceles triangle.
4. State the criterion (AA, SAS, SSS), then write the ratio of corresponding sides and finish.

You see in the figure	Use
Two lines crossing at a point	Vertically opposite angles
Parallel lines with a transversal	Alternate / corresponding angles
Perpendiculars	Right angles
AB = AC	Base angles $\angle B = \angle C$
Two triangles sharing a vertex	Common angle

Angle sources for AA.

SOLVED EXAMPLE 1

Diagonals AC and BD of quadrilateral ABCD meet at P, with $\angle BAP = \angle CDP = 90^\circ$. Prove that $AP \times PC = PB \times DP$.

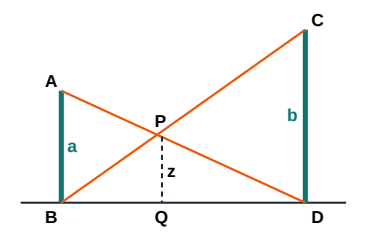
SOLUTION

1. In $\triangle APB$ and $\triangle DPC$: $\angle BAP = \angle CDP = 90^\circ$ and $\angle APB = \angle DPC$ (vertically opposite).
2. $\triangle APB \sim \triangle DPC$ (AA)
3. $\frac{AP}{DP} = \frac{PB}{PC} \Rightarrow AP \times PC = PB \times DP$

Answer: Hence proved.

SOLVED EXAMPLE 2 (THE POLES RESULT)

Two vertical poles AB = a and CD = b stand on level ground. AD and CB meet at P, and PQ = z is perpendicular to BD. Prove that $\frac{1}{a} + \frac{1}{b} = \frac{1}{z}$.



SOLUTION

1. $\triangle DQP \sim \triangle DBA$ (AA: right angles at Q and B, common $\angle D$), so $\frac{z}{a} = \frac{DQ}{DB}$ ...(1)
2. $\triangle BQP \sim \triangle BDC$ (AA: right angles at Q and D, common $\angle B$), so $\frac{z}{b} = \frac{BQ}{BD}$ ...(2)
3. Add (1) and (2): $\frac{z}{a} + \frac{z}{b} = \frac{DQ+BQ}{BD} = \frac{BD}{BD} = 1$
4. Divide by z: $\frac{1}{a} + \frac{1}{b} = \frac{1}{z}$

Answer: Hence proved. Note: z does not depend on the distance between the poles.



SHIVAM BHAIYA SAYS

Poles waala result yaad rakho: $z = \frac{ab}{a+b}$. MCQ mein seedha lagao: poles 6 m aur 3 m to crossing height $\frac{18}{9} = 2$ m. Distance chahe kitna bhi ho!

SOLVED EXAMPLE 3

In the figure-free statement: PQ and RS are perpendicular to QS at Q and S, and PR meets QS at M. If $PM = 4$ cm, $SM = 6$ cm and $MR = 9$ cm, find MQ.

SOLUTION

1. $\triangle PQM$ and $\triangle RSM$: $\angle Q = \angle S = 90^\circ$, $\angle PMQ = \angle RMS$ (vertically opposite), so $\triangle PQM \sim \triangle RSM$ (AA).
2. $\frac{PM}{RM} = \frac{QM}{SM} \Rightarrow \frac{4}{9} = \frac{QM}{6}$

Answer: $MQ = \frac{24}{9} = \frac{8}{3}$ cm



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Ratio likhte waqt dono triangles ke corresponding sides hi lo. Pehle similarity statement sahi order mein likho ($\triangle PQM \sim \triangle RSM$), phir letters ko position se match karo. Andaaze se ratio mat banao.



EXAM ALERT!

Triangles chapter se boards mein 6-8 marks tak aate hain: ek 3-5 mark proof (BPT ya similarity), ek case study (shadows, photo enlargement, folding table) aur 1-2 MCQ. Proofs ko figure ke saath 3 baar likh ke practice karo.



SHIVAM BHAIYA SAYS: BOARD TIP

Long answer mein pehle 'Given', 'To prove' aur figure, phir proof. Agar beech mein atak jao, tab bhi jo steps likhe hain unke marks milte hain, isliye khali mat chhodo.

MEME SAMJHO

Astra meme, maths inside



POORI RAAT AREAS OF SIMILAR TRIANGLES RATTA MAARA, PHIR PATA CHALA

Woh topic to 2026-27 syllabus se hi delete ho gaya hai...

Asli baat: is saal Triangles mein BPT, uska converse aur AA/SSS/SAS criteria hi focus hain. Pythagoras proof aur areas theorem deleted hain.

Day 20 Practice

15 board-pattern questions | solve first, then check

MCQ

1 In $\triangle PQR$, S lies on PQ and T on QR such that $\angle QTS = \angle QRP$. If $QT : TR = 3 : 2$ and $PR = 15$ cm, then ST is: (1)

(a) 6 cm **(b)** 10 cm **(c)** 9 cm **(d)** 7.5 cm

MCQ

2 In trapezium ABCD, $AD \parallel BC$ and the diagonals meet at O. If $AD = 5$ cm and $\frac{AO}{OC} = \frac{DO}{OB} = \frac{1}{3}$, then BC is: (1)

(a) 15 cm **(b)** 10 cm **(c)** $\frac{5}{3}$ cm **(d)** 20 cm

MCQ

3 Two vertical poles 6 m and 3 m high stand on level ground. The lines joining the top of each pole to the foot of the other cross at a height of: (1)

(a) 4.5 m **(b)** 2 m **(c)** 3 m **(d)** 1.5 m

MCQ

4 A square is drawn inside a right triangle with legs 12 cm and 6 cm so that one vertex of the square lies on the hypotenuse and two sides lie along the legs. The side of the square is: (1)

(a) 3 cm **(b)** 6 cm **(c)** 4.5 cm **(d)** 4 cm

MCQ

5 In $\triangle ABC$, D lies on AC and E on BC with $DE \parallel AB$. If $AB = 10$ cm, $BE = 3$ cm and $EC = 2$ cm, then DE is: (1)

(a) 6 cm **(b)** 4 cm **(c)** 5 cm **(d)** $\frac{20}{3}$ cm

ASSERTION-REASON

6 **Assertion (A):** In $\triangle ABC$, if $DE \parallel BC$ with D on AB and E on AC, then $\triangle ADE \sim \triangle ABC$.
Reason (R): BPT: a line parallel to one side divides the other two sides in the same ratio. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

7 **Assertion (A):** Two quadrilaterals whose corresponding angles are equal are always similar.
Reason (R): Two polygons are similar only if their corresponding angles are equal and their corresponding sides are proportional. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

VERY SHORT ANSWER

8 In $\triangle ABC$, D lies on AB such that $\angle ACB = \angle CDA$. If $AC = 8$ cm and $AD = 4$ cm, find AB. (1)

CASE-BASED

9 **Read the passage and answer. The two legs AB and CD of a folding table cross at O. The floor points A and C and the top points B and D are such that $AC \parallel BD$. In one position, $OB = x$ cm, $OD = (x + 2)$ cm, $OA = (2x + 3)$ cm and $OC = (4x + 3)$ cm, where O lies between A and B and between C and D.** (4)

(i) Name the criterion that makes $\triangle OAC \sim \triangle OBD$. (1)

(ii) Write the ratio of sides that follows. (1)

(iii) Find x and the length OC. (2)

OR (iii) If $x = 3$, find the ratio $BD : AC$. (2)

SHORT ANSWER

10 **PQ and RS are perpendicular to QS, and PR meets QS at M. If $PM = 4$ cm, $RM = 6$ cm and $SM = 7.5$ cm, find MQ.** (2)

SHORT ANSWER

11 **ABCD is a parallelogram and E lies on BC with $BE : EC = 1 : 2$. AE meets diagonal BD at F. Prove that $DF = 3\,BF$.** (3)

SHORT ANSWER

12 **In $\triangle ABC$, D is a point on BC such that $\angle ADC = \angle BAC$. Prove that $CA^2 = CB \times CD$.** (3)

SHORT ANSWER

13 **In $\triangle ABC$, $AB = AC$. E is a point on CB produced. $AD \perp BC$ and $EF \perp AC$. Prove that $\triangle ABD \sim \triangle ECF$ and hence $AB \times EF = AD \times EC$.** (3)

LONG ANSWER

14 **Two vertical poles 10 m and 15 m high stand 20 m apart on level ground. The line joining the top of each pole to the foot of the other cross at P. Find the height of P above the ground and the distance of its foot from the shorter pole.** (4)

LONG ANSWER

15 **Through the midpoint M of side CD of parallelogram ABCD, the line BM is drawn, meeting AC at L and AD produced at E. Prove that $EL = 2\,BL$.** (5)

Day 20 Answers with worked solutions

1 **(c)**

$\triangle QTS \sim \triangle QRP$ (AA: common $\angle Q$, $\angle QTS = \angle QRP$)

$\frac{ST}{PR} = \frac{QT}{QR} = \frac{3}{5} \Rightarrow ST = 9$ cm

2 **(a)**

$\triangle AOD \sim \triangle COB$ (AA, alternate angles)

$\frac{AD}{BC} = \frac{AO}{OC} = \frac{1}{3} \Rightarrow BC = 15$ cm

3 **(b)**

$z = \frac{ab}{a+b} = \frac{6\times3}{9} = 2$ m

4 **(d)**

Let the side be s. The small triangle on the 12 cm leg is similar to the whole: $\frac{12-s}{12} = \frac{s}{6}$

$6(12 - s) = 12s \Rightarrow 72 = 18s \Rightarrow s = 4$ cm

5 **(b)**

$\triangle CDE \sim \triangle CAB$ (AA)

$\frac{DE}{AB} = \frac{EC}{BC} = \frac{2}{5} \Rightarrow DE = 4$ cm

6 **(b)**

A is true by AA (common $\angle A$, corresponding angles). R is also true, but the similarity follows from AA, not from BPT, so R is not the explanation.

7 **(d)**

A rectangle and a square have equal angles but are not similar, so A is false. R is true.

8 **Ans:** 16 cm

$\triangle ADC \sim \triangle ACB$ (AA: common $\angle A$, $\angle CDA = \angle BCA$)

$\frac{AD}{AC} = \frac{AC}{AB} \Rightarrow AB = \frac{AC^2}{AD} = \frac{64}{4} = 16$ cm

9 **Ans:** AA

$\angle OAC = \angle OBD$ (alternate angles, $AC \parallel BD$) and $\angle AOC = \angle BOD$ (vertically opposite).

Ans: $\frac{OA}{OB} = \frac{OC}{OD} = \frac{AC}{BD}$

Corresponding sides of similar triangles are proportional.

Ans: $x = 3$; $OC = 15$ cm

$\frac{2x+3}{x} = \frac{4x+3}{x+2} \Rightarrow (2x + 3)(x + 2) = x(4x + 3)$

$2x^2 + 7x + 6 = 4x^2 + 3x \Rightarrow x^2 - 2x - 3 = 0 \Rightarrow$

$x = 3$ (x = -1 rejected)

$OC = 4(3) + 3 = 15$ cm

OR: Ans: $1 : 3$

$OB = 3$, $OA = 9$

$\frac{BD}{AC} = \frac{OB}{OA} = \frac{3}{9} = \frac{1}{3}$

10 **Ans:** MQ = 5 cm

$\angle Q = \angle S = 90°$ and $\angle PMQ = \angle RMS$ (vertically opposite), so $\triangle PQM \sim \triangle RSM$ (AA).

$\frac{QM}{SM} = \frac{PM}{RM} \Rightarrow QM = \frac{4\times7.5}{6} = 5$ cm

11 **Ans:** Proved

$AD \parallel BC$, so in $\triangle AFD$ and $\triangle EFB$: $\angle FAD = \angle FEB$ (alternate) and $\angle AFD = \angle EFB$ (vertically opposite).

$\triangle AFD \sim \triangle EFB$ (AA) $\Rightarrow \frac{BF}{DF} = \frac{BE}{AD}$

$AD = BC = 3\,BE$, so $\frac{BF}{DF} = \frac{1}{3} \Rightarrow DF = 3\,BF$

12 **Ans:** Proved

In $\triangle ADC$ and $\triangle BAC$: $\angle ADC = \angle BAC$ (given), $\angle C$ common.

$\triangle ADC \sim \triangle BAC$ (AA)

$\frac{CA}{CB} = \frac{CD}{CA} \Rightarrow CA^2 = CB \times CD$

13 **Ans:** Proved

$AB = AC \Rightarrow \angle ABD = \angle ECF$ (base angles; $\angle ACB$ is the same angle as $\angle ECF$).

$\angle ADB = \angle EFC = 90^\circ$

$\triangle ABD \sim \triangle ECF$ (AA)

$\frac{AB}{EC} = \frac{AD}{EF} \Rightarrow AB \times EF = AD \times EC$

14 **Ans:** Height 6 m; 8 m from the shorter pole

Let the feet be B (10 m pole) and D (15 m pole), Q the foot of the perpendicular from P, $PQ = z$, $BQ = d$.

$\triangle BQP \sim \triangle BDC$: $\frac{z}{15} = \frac{d}{20}$ ...(1)

$\triangle DQP \sim \triangle DBA$: $\frac{z}{10} = \frac{20-d}{20}$ ...(2)

Adding: $\frac{z}{15} + \frac{z}{10} = 1 \Rightarrow z = 6$ m

From (1): $d = \frac{20 \times 6}{15} = 8$ m

15 **Ans:** Proved

In $\triangle BMC$ and $\triangle EMD$: $MC = MD$, $\angle BMC = \angle EMD$ (vertically opposite), $\angle MCB = \angle MDE$ (alternate, $BC \parallel AE$). So $\triangle BMC \cong \triangle EMD$ (ASA) and $BC = DE$.

$AE = AD + DE = BC + BC = 2\,BC$

In $\triangle AEL$ and $\triangle CBL$: $\angle ALE = \angle CLB$ (vertically opposite), $\angle EAL = \angle BCL$ (alternate). So $\triangle AEL \sim \triangle CBL$ (AA).

$\frac{EL}{BL} = \frac{AE}{CB} = \frac{2\,BC}{BC} = 2 \Rightarrow EL = 2\,BL$

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Product to prove? : convert to a ratio first
2. Parallel lines give : alternate angles for AA
3. Poles crossing height : $z = \frac{ab}{a+b}$
4. $\angle ADC = \angle BAC$ gives : $CA^2 = CB \times CD$
5. Folding-table legs : AA via alternate + vertically opposite
6. Quadrilaterals with equal angles similar? : Not always
7. Ratio check : write the similarity statement in order first

DAY 20 COMPLETE?

- [ ] Notes read
- [ ] 15 questions solved
- [ ] Answers checked
- [ ] Mistakes noted

Next: Day 21, Circles

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 21

CIRCLES | CHAPTER 10 | PART 1 OF 2

Tangent to a Circle

Circles

Aaj ka target

- [ ] Secant aur tangent ka farak ek line mein bol pao
- [ ] Theorem 10.1 (tangent ⊥ radius) ka proof bina dekhe likho
- [ ] Tangent length = Pythagoras: 5 sums practice
- [ ] Concentric circles wala chord sum pakka karo

21.1 Secant, Tangent and Point of Contact

A line and a circle in the same plane can meet in three ways: they may have **no common point**, **exactly two common points** (the line is a **secant**), or **exactly one common point** (the line is a **tangent**). The common point of a tangent and the circle is called the **point of contact**.

KEY TERM: TANGENT

A tangent to a circle is a line that intersects the circle at exactly one point. It can be seen as the limiting position of a secant when its two points of intersection coincide.

- At any one point of a circle there is **exactly one** tangent.
- A circle has **infinitely many** tangents (one at every point).
- No tangent can be drawn from a point **inside** the circle; exactly **one** through a point **on** the circle; exactly **two** from a point **outside**.
- A circle can have at most **two parallel tangents**: the tangents at the two ends of a diameter.

Position of point P	Tangents through P
Inside the circle	0
On the circle	1
Outside the circle	2

Number of tangents through a point

21.2 Theorem 10.1: Tangent is Perpendicular to the Radius

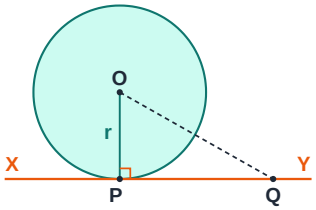


Fig. 21.1 Tangent XY at P is perpendicular to the radius OP; Q is any other point on XY

THEOREM 10.1

The tangent at any point of a circle is perpendicular to the radius through the point of contact.

$$OP \perp XY \Rightarrow \angle OPQ = 90^\circ$$

Tangent length from Q:

$$PQ = \sqrt{OQ^2 - r^2}$$

1. **Given:** a circle with centre O and a tangent XY at the point P.
2. **To prove:** $OP \perp XY$.
3. **Construction:** take any point Q on XY other than P and join OQ.
4. Q lies outside the circle (if Q were on or inside the circle, XY would meet the circle at two points and would be a secant, not a tangent).
5. So $OQ > OP$ (OP is a radius). This is true for **every** point Q on XY except P.
6. Hence OP is the **shortest** distance from O to the line XY, and the shortest segment from a point to a line is the perpendicular. So $OP \perp XY$.

- **Converse:** the line drawn through the end of a radius and perpendicular to it is a tangent.
- The perpendicular to a tangent at its point of contact passes through the centre.
- The line joining the centre to the point of contact is called the **normal** at that point.



SHIVAM BHAIYA SAYS

Jaise hi question mein 'tangent at P' dikhe, turant OP join karo aur wahan 90° ka box bana do. Aadha sum wahi ho gaya, ab Pythagoras laga do.

SOLVED EXAMPLE 1

A point Q is 25 cm from the centre O of a circle of radius 7 cm. Find the length of the tangent drawn from Q to the circle.

SOLUTION

1. Let the tangent touch the circle at P. Then $OP \perp PQ$ (Theorem 10.1), so $\triangle OPQ$ is right-angled at P.
2. $PQ^2 = OQ^2 - OP^2 = 25^2 - 7^2 = 625 - 49 = 576$
3. $PQ = \sqrt{576} = 24$ cm

Answer: Length of tangent $= 24$ cm

21.3 Concentric Circles and Parallel Tangents

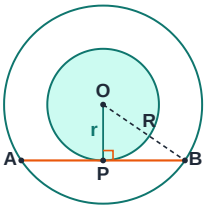


Fig. 21.2 Chord AB of the larger circle touches the smaller circle at P

If a chord AB of the larger of two concentric circles touches the smaller circle at P, then $OP \perp AB$ (tangent $\perp$ radius). A perpendicular from the centre bisects a chord, so **P is the midpoint of AB**.

CONCENTRIC CIRCLES (RADII R > R)

$$AB = 2\sqrt{R^2 - r^2}$$

Tangents at the ends of a diameter are parallel; distance between them $= 2r$ (the diameter).

SOLVED EXAMPLE 2

Two concentric circles have radii 13 cm and 5 cm. Find the length of the chord of the larger circle which touches the smaller circle.

SOLUTION

1. Let chord AB touch the smaller circle at P. Then $OP \perp AB$, $OP = 5$ cm, $OA = 13$ cm.
2. In right $\triangle OPA$: $AP = \sqrt{13^2 - 5^2} = \sqrt{144} = 12$ cm.
3. The perpendicular from the centre bisects the chord, so $AB = 2 \times AP = 24$ cm.

Answer: $AB = 24$ cm

SOLVED EXAMPLE 3 (PROOF)

Prove that the tangents drawn at the ends of a diameter of a circle are parallel.

SOLUTION

1. Let AB be a diameter of a circle with centre O, and let PQ and RS be the tangents at A and B.
2. By Theorem 10.1, $OA \perp PQ$, so $\angle OAP = 90^\circ$; and $OB \perp RS$, so $\angle OBS = 90^\circ$.
3. Hence $\angle PAB = \angle ABS = 90^\circ$, and these are alternate angles made by the transversal AB.
4. Alternate angles are equal, so $PQ \parallel RS$.

Answer: Tangents at the ends of a diameter are parallel. Hence proved.

21.4 Angle between a Tangent and a Chord

If AT is the tangent at A and AB is a chord with $\angle AOB = \theta$, then $\triangle OAB$ is isosceles (OA = OB), so $\angle OAB = 90^\circ - \frac{\theta}{2}$. Since $\angle OAT = 90^\circ$, we get $\angle BAT = \frac{\theta}{2}$: the angle between a tangent and a chord is **half the angle the chord subtends at the centre**.

SOLVED EXAMPLE 4

AB is a chord of a circle with centre O and AT is the tangent at A. If $\angle AOB = 110^\circ$, find $\angle BAT$.

SOLUTION

1. $OA = OB$ (radii), so $\angle OAB = \angle OBA = \frac{180^\circ - 110^\circ}{2} = 35^\circ$.
2. $\angle OAT = 90^\circ$ (tangent $\perp$ radius).
3. $\angle BAT = 90^\circ - 35^\circ = 55^\circ$

Answer: $\angle BAT = 55^\circ$ (half of 110°)



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Sabse common galti: tangent length nikalte waqt $OQ^2 + r^2$ kar dena. Yaad rakho, OQ hypotenuse hai (90° ke saamne), toh hamesha **minus**: $PQ^2 = OQ^2 - r^2$.



SHIVAM BHAIYA SAYS: BOARD TIP

Theorem 10.1 ka proof board mein 3 marks ka aata hai. Given, To prove, Construction, Proof, chaaron headings likho aur 'shortest distance is the perpendicular' wali line zaroor likhna, wahi step ke marks hain.



EXAM ALERT!

Circles se board mein lagbhag 6 to 8 marks aate hain. Theorem 10.1 ya 10.2 ka proof aur concentric circles wala chord sum har saal kisi na kisi form mein aata hai.

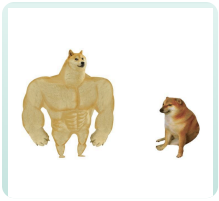


SHIVAM BHAIYA SAYS

Angle chase shortcut: tangent aur chord ke beech ka angle = centre pe chord ka angle ÷ 2. MCQ mein 10 second ka kaam.

MEME SAMJHO

Astra meme, maths inside



BUFF DOGE VS CRYING CHEEMS, A CIRCLES SUM

Buff doge: "Radius aur tangent milte hi 90°, Pythagoras laga do." Cheems: "Yeh 90° kahan se aaya?"

Asli baat: the radius to the point of contact is always perpendicular to the tangent (Theorem 10.1), so every tangent-length question is a right triangle with OQ as hypotenuse.

Day 21 Practice

16 board-pattern questions | solve first, then check

MCQ

1 PQ is a tangent at P to a circle with centre O and radius 6 cm. If $OQ = 10$ cm, then PQ is (1)

(a) 6 cm **(b)** $4\sqrt{2}$ cm **(c)** 10 cm **(d)** 8 cm

MCQ

2 Through a point lying on a circle, the number of tangents that can be drawn is (1)

(a) 0 **(b)** 1
(c) 2 **(d)** infinitely many

MCQ

3 Two concentric circles have radii 10 cm and 6 cm. The length of the chord of the larger circle that touches the smaller circle is (1)

(a) 16 cm **(b)** 8 cm **(c)** 12 cm **(d)** 20 cm

MCQ

4 From a point 13 cm away from the centre of a circle, the length of the tangent is 12 cm. The radius of the circle is (1)

(a) 7 cm **(b)** 25 cm **(c)** 5 cm **(d)** $\sqrt{313}$ cm

MCQ

5 AB is a chord of a circle with centre O and AT is the tangent at A. If $\angle AOB = 120°$, then $\angle BAT$ is (1)

(a) $30°$ **(b)** $60°$ **(c)** $120°$ **(d)** $90°$

ASSERTION-REASON

6 **Assertion (A):** A circle has infinitely many tangents.
Reason (R): The tangent at any point of a circle is perpendicular to the radius through the point of contact. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

7 **Assertion (A):** A line that meets a circle at exactly one point is a tangent to the circle.
Reason (R): The point of contact of a tangent lies inside the circle. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

VERY SHORT ANSWER

8 Find the distance between two parallel tangents of a circle of radius 4.5 cm. (1)

CASE-BASED

9 Football on a table. A football rests on a flat table. Seen from the side, the ball is a circle with centre O that touches the table edge line at P. A small marble Q also lies on the table line, away from P. (4)

(i) Compare OQ and OP. Give a reason. (1)

(ii) If the radius of the ball is 8 cm and $OQ = 17$ cm, find PQ. (1)

(iii) If $PQ = 10\sqrt{3}$ cm and $\angle POQ = 60°$, find the radius of the ball. (2)

OR (iii) If $OP = 6$ cm and $\angle PQO = 30°$, find OQ. (2)

SHORT ANSWER

10 Prove that the tangents drawn at the end points of a diameter of a circle are parallel. (2)

SHORT ANSWER

11 PQ is a tangent to a circle with centre O at the point Q, and QOR is a diameter. If $\angle POR = 130°$, find $\angle OPQ$. (2)

SHORT ANSWER

12 Two concentric circles have radii 8 cm and 5 cm. From a point P, a tangent PA of length 15 cm is drawn to the larger circle. Find the length of the tangent PB drawn from P to the smaller circle. (3)

SHORT ANSWER

13 A chord PQ of length 24 cm is drawn in a circle of radius 13 cm with centre O. The tangents at P and Q meet at T. Find the length TP. (3)

SHORT ANSWER

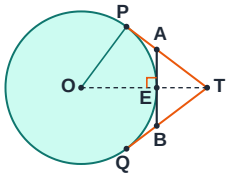
14 RS is a chord of a circle with centre O, parallel to the tangent at P, and it cuts the radius OP at its midpoint Q. If $RS = 18$ cm, find the radius of the circle. (3)

LONG ANSWER

15 Prove that the tangent at any point of a circle is perpendicular to the radius through the point of contact. Hence, find the radius of a circle if the tangent from a point 26 cm away from the centre is 24 cm long, and find the area of the triangle formed by the centre, the external point and the point of contact. (5)

LONG ANSWER

16 In the figure, O is the centre of a circle of radius 6 cm and T is a point with $OT = 10$ cm. TP and TQ are tangents to the circle. OT meets the circle at E, and the tangent at E meets TP at A and TQ at B. Find the length AB. (5)



Day 21 Answers with worked solutions

1 **(d)**
$OP \perp PQ$, so $PQ = \sqrt{10^2 - 6^2} = \sqrt{64} = 8$ cm.

2 **(b)**
At a point on the circle there is exactly one tangent.

3 **(a)**
Half-chord $= \sqrt{10^2 - 6^2} = 8$ cm, so chord $= 2 \times 8 = 16$ cm.

4 **(c)**
$r = \sqrt{13^2 - 12^2} = \sqrt{25} = 5$ cm.

5 **(b)**
$\angle OAB = \frac{180° - 120°}{2} = 30°$ and $\angle OAT = 90°$.
So $\angle BAT = 90° - 30° = 60°$.

6 **(b)**
A: true, there is one tangent at each of the infinitely many points of the circle.
R: true (Theorem 10.1), but it does not explain why the number of tangents is infinite. So (b).

7 **(c)**
A is the definition of a tangent, so it is true.
R is false: the point of contact lies **on** the circle.
So (c).

8 **Ans:** 9 cm
Parallel tangents touch at the ends of a diameter, so the distance between them is $2r = 9$ cm.

9 **Ans:** $OQ > OP$
$\angle OPQ = 90°$ (tangent $\perp$ radius), so OQ is the hypotenuse of right $\triangle OPQ$ and is longer than OP.

Ans: 15 cm
$PQ = \sqrt{17^2 - 8^2} = \sqrt{225} = 15$ cm.

Ans: 10 cm
In right $\triangle OPQ$: $\tan 60° = \frac{PQ}{OP}$.
$OP = \frac{10\sqrt{3}}{\sqrt{3}} = 10$ cm.

OR: Ans: 12 cm
$\sin 30° = \frac{OP}{OQ}$, so $OQ = \frac{6}{1/2} = 12$ cm.

10 **Ans:** Proof
Let AB be a diameter and PQ, RS the tangents at A and B.
$OA \perp PQ$ and $OB \perp RS$ (Theorem 10.1), so $\angle PAB = \angle ABS = 90°$.
These are equal alternate angles with transversal AB, hence $PQ \parallel RS$.

11 **Ans:** $40°$
QOR is a straight line, so $\angle POQ = 180° - 130° = 50°$.
$\angle OQP = 90°$ (tangent $\perp$ radius).
In $\triangle OPQ$: $\angle OPQ = 180° - 90° - 50° = 40°$.

12 **Ans:** $PB = 2\sqrt{66}$ cm (about 16.2 cm)

$OA \perp PA$, so $OP^2 = OA^2 + PA^2 = 64 + 225 = 289$, $OP = 17$ cm.

$OB \perp PB$, so $PB^2 = OP^2 - OB^2 = 289 - 25 = 264$.

$PB = \sqrt{264} = 2\sqrt{66} \approx 16.25$ cm.

13 **Ans:** $TP = \frac{156}{5} = 31.2$ cm

Let OT cut PQ at L. By symmetry $OT \perp PQ$ and $PL = 12$ cm.

$OL = \sqrt{13^2 - 12^2} = 5$ cm.

In right $\triangle OPT$ ($\angle OPT = 90^\circ$), PL is the altitude to the hypotenuse, so $\triangle OLP \sim \triangle OPT$ (AA).

$\frac{TP}{PL} = \frac{OP}{OL} \Rightarrow TP = \frac{12 \times 13}{5} = \frac{156}{5} = 31.2$ cm.

14 **Ans:** $6\sqrt{3}$ cm

The tangent at P is perpendicular to OP, and $RS \parallel$ tangent, so $OQ \perp RS$; hence $RQ = 9$ cm.

Let the radius be r. Then $OQ = \frac{r}{2}$ and $OR = r$.

$r^2 = \left(\frac{r}{2}\right)^2 + 9^2 \Rightarrow \frac{3r^2}{4} = 81 \Rightarrow r^2 = 108$.

$r = \sqrt{108} = 6\sqrt{3}$ cm.

15 **Ans:** Proof; radius 10 cm; area 120 cm²

Let XY be the tangent at P to the circle with centre O. Take any point Q on XY other than P and join OQ.

Q lies outside the circle (otherwise XY would be a secant), so $OQ > OP$ for every such Q.

So OP is the shortest segment from O to XY, hence $OP \perp XY$.

Radius: $r = \sqrt{26^2 - 24^2} = \sqrt{100} = 10$ cm.

Area $= \frac{1}{2} \times OP \times PQ = \frac{1}{2} \times 10 \times 24 = 120$ cm².

16 **Ans:** $AB = 6$ cm

$TP = \sqrt{10^2 - 6^2} = 8$ cm, and $TE = OT - OE = 10 - 6 = 4$ cm.

AB touches the circle at E, so $AB \perp OT$; by symmetry $AE = EB$.

A is outside the circle, so its tangents are equal: $AE = AP = x$ (say). Then $TA = 8 - x$.

In right $\triangle AET$: $TA^2 = AE^2 + TE^2 \Rightarrow (8 - x)^2 = x^2 + 16$.

$64 - 16x = 16 \Rightarrow x = 3$ cm.

$AB = 2x = 6$ cm.

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Tangent meets the circle at how many points? : exactly one
2. Tangents from a point inside the circle? : none
3. Angle between radius and tangent at the point of contact? : 90°
4. Tangent length from a point at distance d, radius r? : $\sqrt{d^2 - r^2}$
5. Chord of larger concentric circle touching smaller (R, r)? : $2\sqrt{R^2 - r^2}$
6. Maximum number of parallel tangents? : two (at ends of a diameter)
7. Angle between tangent and chord vs angle at centre? : half of it

DAY 21 COMPLETE? ☐ Notes read ☐ 16 questions solved ☐ Answers checked ☐ Mistakes noted

Next: Day 22, Circles

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 22

CIRCLES | CHAPTER 10 | PART 2 OF 2

Tangents from an External Point

Circles

Aaj ka target

- [ ] Theorem 10.2 (PA = PB) ka proof RHS se likho
- [ ] Angle APB + angle AOB = 180° wale sums fatafat
- [ ] Circumscribed quadrilateral: AB + CD = AD + BC pakka
- [ ] Incircle wale triangle mein tangent segments nikalna seekho

22.1 Theorem 10.2: Tangents from an External Point are Equal

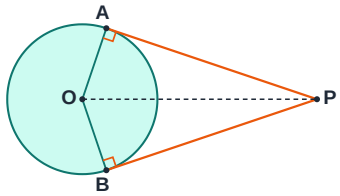


Fig. 22.1 PA and PB are tangents from the external point P

THEOREM 10.2

The lengths of the tangents drawn from an external point to a circle are equal.

$$PA = PB$$

1. **Given:** a circle with centre O; PA and PB are tangents from an external point P.
2. **To prove:** $PA = PB$.
3. **Construction:** join OA, OB and OP.
4. $\angle OAP = \angle OBP = 90^\circ$ (tangent $\perp$ radius, Theorem 10.1).
5. In right triangles OAP and OBP: $OA = OB$ (radii) and $OP = OP$ (common hypotenuse).
6. So $\triangle OAP \cong \triangle OBP$ (RHS), and hence $PA = PB$ (CPCT).

22.2 Results that Follow from Theorem 10.2

- $\angle APO = \angle BPO$: the centre lies on the **bisector of the angle** between the two tangents.
- $\angle AOP = \angle BOP$: OP also bisects $\angle AOB$.
- In quadrilateral OAPB two angles are 90°, so

$$\angle APB + \angle AOB = 180^\circ$$

- $\triangle PAB$ is isosceles, so $\angle PAB = \angle PBA = 90^\circ - \frac{1}{2}\angle APB$, and $\angle OAB = \frac{1}{2}\angle APB$.
- OP is the perpendicular bisector of the chord AB.

SOLVED EXAMPLE 1

PA and PB are tangents from P to a circle with centre O. If $\angle APB = 70^\circ$, find $\angle AOB$ and $\angle OAB$.

SOLUTION

1. $\angle OAP = \angle OBP = 90^\circ$, so in quadrilateral OAPB: $\angle AOB = 360^\circ - 90^\circ - 90^\circ - 70^\circ = 110^\circ$.
2. $OA = OB$, so $\angle OAB = \angle OBA = \frac{180^\circ - 110^\circ}{2} = 35^\circ$.

Answer: $\angle AOB = 110^\circ$, $\angle OAB = 35^\circ$



SHIVAM BHAIYA SAYS

Do tangents wale angle sums mein bas yeh yaad rakho: $\angle P + \angle O = 180^\circ$, aur $\angle OAB = \frac{1}{2}\angle P$. MCQ mein seedha number daalo, figure banane ki zaroorat bhi nahi.

22.3 Quadrilateral Circumscribing a Circle

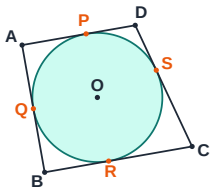


Fig. 22.2 Quadrilateral ABCD touches the circle at P, Q, R, S

SOLVED EXAMPLE 2 (PROOF)

A quadrilateral ABCD is drawn to circumscribe a circle. Prove that $AB + CD = AD + BC$.

SOLUTION

1. Let the sides AB, BC, CD, DA touch the circle at P, Q, R, S.
2. Tangents from an external point are equal: $AP = AS$, $BP = BQ$, $CR = CQ$, $DR = DS$.
3. Adding: $AP + BP + CR + DR = AS + BQ + CQ + DS$.
4. $(AP + BP) + (CR + DR) = (AS + DS) + (BQ + CQ)$, that is $AB + CD = AD + BC$.

Answer: $AB + CD = AD + BC$ (opposite sides add up equally). Hence proved.

- A **parallelogram** circumscribing a circle is a **rhombus** (use $AB = CD$, $AD = BC$ in the result above).
- A **rectangle** circumscribing a circle is a **square**.

22.4 Triangle with an Incircle

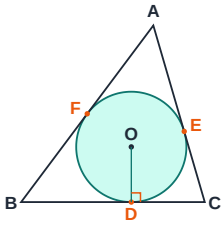


Fig. 22.3 Incircle of triangle ABC touching the sides at D, E, F

TANGENT SEGMENTS IN A TRIANGLE (S = SEMI-PERIMETER)

$$AF = AE = s - BC$$

$$BD = BF = s - CA$$

$$CD = CE = s - AB$$

Area of triangle $= r \times s$, where r is the inradius.

Right triangle with legs a, b and hypotenuse c:

$$r = \frac{a + b - c}{2}$$

SOLVED EXAMPLE 3

A circle is inscribed in $\triangle ABC$ with $AB = 14$ cm, $BC = 12$ cm and $CA = 10$ cm, touching BC at D, CA at E and AB at F. Find AF, BD and CE.

SOLUTION

1. Let $AF = AE = x$, $BD = BF = y$, $CD = CE = z$.
2. $x + y = 14$, $y + z = 12$, $z + x = 10$. Adding: $2(x + y + z) = 36$, so $x + y + z = 18$.
3. $x = 18 - 12 = 6$, $y = 18 - 10 = 8$, $z = 18 - 14 = 4$.

Answer: $AF = 6$ cm, $BD = 8$ cm, $CE = 4$ cm



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Incircle sums mein log seedha 'AF = AB ka half' le lete hain. Galat! Har vertex ke do tangent equal hote hain, teen variables x, y, z lo aur equations add karo.



SHIVAM BHAIYA SAYS: BOARD TIP

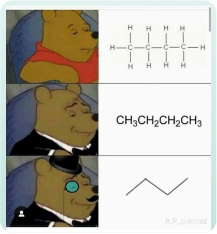
Theorem 10.2 ka proof (RHS congruence) aur 'AB + CD = AD + BC' dono board ke favourite 2 to 3 markers hain. Figure zaroor banao aur har equal pair ke aage reason likho: 'tangents from external point'.



EXAM ALERT!

Parallelogram circumscribing a circle is a rhombus: yeh proof pichhle kai saalon mein 3 marks ke liye aaya hai. Ek baar khud se likh ke dekho.

MEME SAMJHO — Astra meme, maths inside



WINNIE THE POOH, PLAIN VS FANCY, TANGENT LENGTHS

Normal: dono tangents alag alag Pythagoras se nikalo. Fancy: PA = PB, ek nikala, dono mil gaye.

Asli baat: tangents from one external point are equal (Theorem 10.2), so finding one tangent length gives you the other.

Day 22 Practice

16 board-pattern questions | solve first, then check

MCQ

1 TP and TQ are tangents from T to a circle with centre O. If $\angle POQ = 125°$, then $\angle PTQ$ equals (1)

(a) $65°$ **(b)** $55°$ **(c)** $125°$ **(d)** $35°$

MCQ

2 PA and PB are tangents to a circle with centre O. If $\angle APB = 60°$, then $\angle OAB$ is (1)

(a) $60°$ **(b)** $45°$ **(c)** $30°$ **(d)** $120°$

MCQ

3 A circle inscribed in $\triangle ABC$ touches AB at N, BC at L and CA at M. If $BN = 5$ cm and $CM = 7$ cm, then BC is (1)

(a) 10 cm **(b)** 14 cm **(c)** 2 cm **(d)** 12 cm

MCQ

4 Two tangents PQ and PR are drawn to a circle of radius 5 cm such that $\angle QPR = 90°$. The length of each tangent is (1)

(a) 5 cm **(b)** $5\sqrt{2}$ cm **(c)** 10 cm **(d)** 2.5 cm

MCQ

5 PA and PB are tangents to a circle and $\angle APB = 50°$. If Q is a point on the major arc AB, then $\angle AQB$ is (1)

(a) $130°$ (b) $65°$ (c) $50°$ (d) $25°$

ASSERTION-REASON

6 **Assertion (A):** If the length of one tangent from an external point P to a circle is 9 cm, the length of the other tangent from P is also 9 cm.
Reason (R): The lengths of the tangents drawn from an external point to a circle are equal. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

7 **Assertion (A):** If a quadrilateral ABCD circumscribes a circle, then $AB + BC = AD + DC$.
Reason (R): The tangents drawn from an external point to a circle are equal in length. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

VERY SHORT ANSWER

8 From a point P, tangents PA and PB are drawn to a circle, with $PA = 12$ cm. A third tangent touches the circle at E (on the minor arc AB) and meets PA at C and PB at D. Find the perimeter of $\triangle PCD$. (1)

CASE-BASED

9 Solar eclipse viewing. During an eclipse, a student draws the Moon's disc as a circle with centre O. From a point A outside it she draws two tangents AP and AQ touching the circle at P and Q. (4)

(i) If $\angle PAQ = 50°$, find $\angle POQ$. (1)

(ii) If $OP = 7$ cm and $OA = 25$ cm, find AP. (1)

(iii) If $\angle POQ = 100°$, find $\angle QAO$. (2)

OR (iii) If $AP = 12$ cm and $\angle PAQ = 60°$, find the length of the chord PQ. (2)

SHORT ANSWER

10 Two tangents PA and PB are drawn from an external point P to a circle of radius r with centre O. If $OP = 2r$, find $\angle APB$ and show that $\triangle APB$ is equilateral. (2)

SHORT ANSWER

11 PQ and PR are tangents to a circle with centre O, and $\angle RPQ = 40°$. S is a point on the major arc QR. Find $\angle RQP$ and $\angle RSQ$. (2)

SHORT ANSWER

12 A quadrilateral ABCD circumscribes a circle of radius 8 cm with centre O, touching AB at P, BC at Q, CD at R and DA at S. If $AD \perp AB$, $BC = 40$ cm and $CR = 28$ cm, find AB. (3)

SHORT ANSWER

13 Prove that a parallelogram circumscribing a circle is a rhombus. (3)

SHORT ANSWER

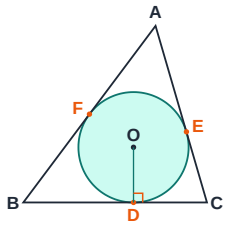
14 In $\triangle ABC$, $\angle B = 90°$, $AB = 15$ cm and $BC = 8$ cm. Find the diameter of the circle inscribed in the triangle. (3)

LONG ANSWER

15 Two tangents TP and TQ are drawn to a circle with centre O from an external point T. Prove that $\angle PTQ = 2\angle OPQ$. (4)

LONG ANSWER

16 A circle of radius 3 cm is inscribed in $\triangle ABC$ and touches BC at D such that $BD = 6$ cm and $DC = 4$ cm. Find the lengths of AB and AC. (5)



Day 22 Answers with worked solutions

1 (b)
$\angle PTQ + \angle POQ = 180°$, so $\angle PTQ = 55°$.

2 (c)
$\angle AOB = 120°$ and $OA = OB$, so $\angle OAB = \frac{180° - 120°}{2} = 30°$ (half of $\angle APB$).

3 (d)
$BL = BN = 5$ cm and $CL = CM = 7$ cm (tangents from B and from C).
$BC = BL + LC = 12$ cm.

4 (a)
OQPR has three right angles and $OQ = OR$, so it is a square; $PQ = OQ = 5$ cm.

5 (b)
$\angle AOB = 180° - 50° = 130°$.
Angle at the circumference is half the angle at the centre: $\angle AQB = 65°$.

6 **(a)**
Both are true and R (Theorem 10.2) directly explains A.

7 **(d)**
The correct result is $AB + CD = AD + BC$ (opposite sides), so A is false.
R is Theorem 10.2, true. So (d).

8 **Ans:** 24 cm
$CE = CA$ and $DE = DB$ (tangents from C and D).
Perimeter $= PC + CE + ED + DP = PC + CA + DB + DP = PA + PB = 2 \times 12 = 24$ cm.

9 **Ans:** 130°
$\angle POQ = 180^\circ - 50^\circ = 130^\circ$.

Ans: 24 cm
$AP = \sqrt{25^2 - 7^2} = \sqrt{576} = 24$ cm.

Ans: 40°
$\angle PAQ = 180^\circ - 100^\circ = 80^\circ$.
OA bisects $\angle PAQ$, so $\angle QAO = 40^\circ$.

OR: Ans: 12 cm
$AP = AQ$, so $\triangle APQ$ is isosceles with apex angle 60°; the base angles are also 60°.
So $\triangle APQ$ is equilateral and $PQ = 12$ cm.

10 **Ans:** $\angle APB = 60^\circ$; equilateral
In right $\triangle OAP$: $\sin \angle APO = \frac{OA}{OP} = \frac{r}{2r} = \frac{1}{2}$, so $\angle APO = 30^\circ$.
$\angle APB = 2 \times 30^\circ = 60^\circ$.
$PA = PB$, so $\angle PAB = \angle PBA = \frac{180^\circ - 60^\circ}{2} = 60^\circ$; all angles are 60°, so $\triangle APB$ is equilateral.

11 **Ans:** $\angle RQP = 70^\circ$, $\angle RSQ = 70^\circ$
$PQ = PR$, so $\angle RQP = \angle QRP = \frac{180^\circ - 40^\circ}{2} = 70^\circ$.
$\angle ROQ = 180^\circ - 40^\circ = 140^\circ$, so $\angle RSQ = \frac{1}{2} \times 140^\circ = 70^\circ$.

12 **Ans:** $AB = 20$ cm
$\angle A = 90^\circ$ and $OP \perp AB$, $OS \perp AD$, $OP = OS$, so APOS is a square: $AP = 8$ cm.
$CQ = CR = 28$ cm, so $BQ = 40 - 28 = 12$ cm, and $BP = BQ = 12$ cm.
$AB = AP + PB = 8 + 12 = 20$ cm.

13 **Ans:** Proof
Let parallelogram ABCD circumscribe a circle. Tangents from each vertex are equal, which gives $AB + CD = AD + BC$.
In a parallelogram $AB = CD$ and $AD = BC$, so $2AB = 2AD$, that is $AB = AD$.
So all four sides are equal ($AB = BC = CD = DA$) and ABCD is a rhombus.

14 **Ans:** 6 cm
$AC = \sqrt{15^2 + 8^2} = 17$ cm.
With the incircle touching AB and BC, the centre O and B form a square of side r, so the tangents from A and C are $15 - r$ and $8 - r$.
$AC = (15 - r) + (8 - r) = 17 \Rightarrow 23 - 2r = 17 \Rightarrow r = 3$ cm.
Diameter $= 6$ cm.

15 **Ans:** Proof
Let $\angle PTQ = \theta$. Since $TP = TQ$, $\triangle TPQ$ is isosceles: $\angle TPQ = \angle TQP = \frac{180^\circ - \theta}{2} = 90^\circ - \frac{\theta}{2}$.
$\angle OPT = 90^\circ$ (tangent $\perp$ radius).
$\angle OPQ = \angle OPT - \angle TPQ = 90^\circ - \left(90^\circ - \frac{\theta}{2}\right) = \frac{\theta}{2}$.
So $\angle PTQ = \theta = 2\angle OPQ$. Hence proved.

16 **Ans:** $AB = 12$ cm, $AC = 10$ cm
Let the circle touch AB at F and AC at E. Then $BF = BD = 6$, $CE = CD = 4$, and let $AF = AE = x$.
Sides: $BC = 10$, $AB = x + 6$, $AC = x + 4$; semi-perimeter $s = x + 10$.
Area $= r \times s = 3(x + 10)$.
By Heron's formula, area $= \sqrt{s(s-a)(s-b)(s-c)} = \sqrt{(x+10)(x)(4)(6)}$.
$9(x + 10)^2 = 24x(x + 10) \Rightarrow 9(x + 10) = 24x \Rightarrow 15x = 90 \Rightarrow x = 6$.
$AB = 12$ cm, $AC = 10$ cm. (Check: $s = 16$, area $= 48 = 3 \times 16$.)

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Tangents from an external point are? : equal in length
2. Which congruence rule proves PA = PB? : RHS
3. $\angle APB + \angle AOB =$? : 180°
4. $\angle OAB$ in terms of $\angle APB$? : half of $\angle APB$
5. Quadrilateral circumscribing a circle: ? : $AB + CD = AD + BC$
6. Parallelogram circumscribing a circle is a? : rhombus
7. Inradius of right triangle with legs a, b, hypotenuse c? : $\frac{a+b-c}{2}$

DAY 22 COMPLETE? ☐ Notes read ☐ 16 questions solved ☐ Answers checked ☐ Mistakes noted

Next: Day 23, Revision 4: Triangles + Circles

DAY 23

REVISION 4 | TRIANGLES + CIRCLES

Triangles + Circles

Revision 4

Aaj ka target

- [ ] BPT, converse aur AA/SAS/SSS ek page pe revise
- [ ] Circles ke dono theorems aur unke results zubaani
- [ ] 20 mixed questions, timer laga ke (40 min)
- [ ] Galtiyan mistake-notebook mein likho

23.1 Triangles at a Glance

BASIC PROPORTIONALITY THEOREM (BPT) AND ITS CONVERSE

If $DE \parallel BC$ in $\triangle ABC$ (D on AB, E on AC), then

$$\frac{AD}{DB} = \frac{AE}{EC}$$

Also $\frac{AD}{AB} = \frac{AE}{AC} = \frac{DE}{BC}$ (from $\triangle ADE \sim \triangle ABC$).

Converse: if $\frac{AD}{DB} = \frac{AE}{EC}$, then $DE \parallel BC$.

Criterion	What you need	Result
AA	two pairs of equal angles	triangles similar
SAS	one equal angle, including sides proportional	triangles similar
SSS	all three pairs of sides proportional	triangles similar

Similarity criteria (write the correspondence in the right order)

- Similar triangles: corresponding angles equal and corresponding sides in the same ratio.
- Ratio of perimeters (and of corresponding medians, altitudes) equals the ratio of sides.
- Areas of similar triangles is deleted from the 2026-27 syllabus.

23.2 Circles at a Glance

TWO THEOREMS AND THEIR RESULTS

Theorem 10.1: tangent $\perp$ radius at the point of contact. Tangent length $= \sqrt{d^2 - r^2}$.

Theorem 10.2: tangents from an external point are equal, $PA = PB$.

$\angle APB + \angle AOB = 180^\circ$, and $\angle OAB = \frac{1}{2}\angle APB$.

Quadrilateral circumscribing a circle: $AB + CD = AD + BC$.

Concentric circles: chord of larger touching smaller $= 2\sqrt{R^2 - r^2}$.

SOLVED EXAMPLE 1 (TRIANGLES)

In $\triangle ABC$, $DE \parallel BC$ with D on AB and E on AC. If $AD = x$, $DB = x - 2$, $AE = x + 2$ and $EC = x - 1$, find x.

SOLUTION

1. By BPT: $\frac{AD}{DB} = \frac{AE}{EC} \Rightarrow \frac{x}{x-2} = \frac{x+2}{x-1}$.
2. $x(x-1) = (x+2)(x-2) \Rightarrow x^2 - x = x^2 - 4 \Rightarrow x = 4$.
3. Check: $AD = 4$, $DB = 2$, $AE = 6$, $EC = 3$; both ratios are 2.

Answer: $x = 4$

SOLVED EXAMPLE 2 (CIRCLES)

From a point P, two tangents PA and PB are drawn to a circle with centre O. If $\angle APB = 70^\circ$ and Q is a point on the major arc AB, find $\angle AQB$.

SOLUTION

1. $\angle AOB = 180^\circ - 70^\circ = 110^\circ$.
2. The angle subtended by an arc at a point on the remaining part of the circle is half the angle at the centre: $\angle AQB = 55^\circ$.

Answer: $\angle AQB = 55^\circ$

MEME SAMJHO

Astra meme, maths inside



CAT CALMLY READING A BOOK, REVISION 4

Revision day: Triangles + Circles ek saath. Cat: bas BPT, 3 criteria aur 2 tangent theorems, ho jayega.

Asli baat: almost every board question from these two chapters uses one of BPT (or converse), AA/SAS/SSS, tangent $\perp$ radius, or equal tangents.

SHIVAM BHAIYA SAYS: BOARD TIP



Similarity proof mein correspondence ka order sabse important hai: $\triangle ABC \sim \triangle PQR$ likha toh A ka partner P, B ka Q, C ka R. Order galat toh marks kat jaate hain.



SHIVAM BHAIYA SAYS
Circle ke angle sums mein pehle saare 90° (radius-tangent) mark karo, phir isosceles triangles (OA = OB, PA = PB) dhoondo. 90% sums wahin khul jaate hain.



EXAM ALERT!
Triangles + Circles milake board mein lagbhag 15 marks. Ek 5-marker proof (BPT ya similarity) aur ek 3-marker tangent proof expected hai.

Day 23 Practice

22 board-pattern questions | solve first, then check

MCQ

1 In $\triangle ABC$, $DE \parallel BC$ with D on AB and E on AC. If $AE = 4$ cm, $EC = 6$ cm and $BC = 15$ cm, then DE is (1)

(a) 10 cm (b) 6 cm (c) 9 cm (d) 7.5 cm

MCQ

2 $\triangle ABC \sim \triangle PQR$ with $AB = 18$, $BC = 27$, $CA = 12$ and $PQ = 12$. Then QR and RP are (1)

(a) 18 and 8 (b) 24 and 8
(c) 18 and 6 (d) 20 and 8

MCQ

3 In $\triangle ABC$ and $\triangle DEF$, $\angle B = \angle E$, $\angle C = \angle F$ and $AB = \frac{1}{2}DE$. Then the two triangles are (1)

(a) congruent but not similar
(b) similar but not congruent
(c) neither similar nor congruent
(d) similar and congruent

MCQ

4 In $\triangle ABC$ and $\triangle PQR$, $\frac{AB}{PR} = \frac{BC}{RQ} = \frac{CA}{QP}$. Which statement is correct? (1)

(a) $\triangle PQR \sim \triangle ABC$ (b) $\triangle QRP \sim \triangle ABC$
(c) $\triangle PRQ \sim \triangle ABC$ (d) $\triangle RPQ \sim \triangle ABC$

MCQ

5 The length of the tangent from a point 15 cm away from the centre of a circle of radius 9 cm is (1)

(a) 6 cm (b) 12 cm (c) 24 cm (d) $3\sqrt{34}$ cm

MCQ

6 PT is a tangent at T to a circle with centre O. If $\angle TPO = 28°$, then $\angle TOP$ is (1)

(a) $62°$ (b) $28°$ (c) $152°$ (d) $118°$

MCQ

7 The maximum number of common tangents that two circles intersecting at two points can have is (1)

(a) 1 (b) 3 (c) 4 (d) 2

MCQ

8 ML is a tangent at M to a circle with centre O, and MN is a chord with $\angle LMN = 65°$. Then $\angle MON$ is (1)

(a) $65°$ (b) $115°$ (c) $130°$ (d) $25°$

ASSERTION-REASON

9 **Assertion (A):** If $DE \parallel BC$ in $\triangle ABC$, then $\frac{AD}{DB} = \frac{AE}{EC}$.
Reason (R): Two triangles are similar if their corresponding angles are equal. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

10 **Assertion (A):** If tangents XA and XB are drawn from X to a circle, and the tangent at a point R of the minor arc AB meets them at P and Q, then the perimeter of $\triangle XPQ = 2XA$.
Reason (R): A tangent can be drawn to a circle from a point inside it. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

VERY SHORT ANSWER

11 In $\triangle ABC$, $DE \parallel BC$. If $AE = 2.1$ cm, $BD = 6$ cm and $CE = 4.2$ cm, find AD. (1)

VERY SHORT ANSWER

12 A circle inscribed in $\triangle ABC$ touches AB at P, BC at Q and CA at R. If $AP = 5$ cm, $BP = 4$ cm and $AC = 12$ cm, find BC. (1)

CASE-BASED

13 Triangular bookshelf. A carpenter makes a triangular shelf ABC and fixes a straight plank DE parallel to the base BC, with D on AB and E on AC. (4)

(i) State the theorem that relates AD, DB, AE and EC here. (1)

(ii) If $AD = 4$ cm, $DB = 2$ cm and $AE = 6$ cm, find EC. (1)

(iii) If $AD = x$, $DB = x - 2$, $AE = x + 2$ and $EC = x - 1$, find x. (2)

OR (iii) On another shelf, $AD = 2$, $AB = 6$, $AE = 3$ and $AC = 9$ (all in dm). Is $DE \parallel BC$? Justify. (2)

CASE-BASED

14 **Mirror on the wall. A round mirror with centre O hangs on a wall. Two strings AP and AQ from a nail A touch the rim at P and Q, and $\angle PAQ = 60°$. Each string is 24 cm long ($AP = 24$ cm).** (4)

(i) Find $\angle POQ$. (1)

(ii) Find the length of PQ. (1)

(iii) Find $\angle OPQ$. (2)

OR (iii) Show that OA is the perpendicular bisector of PQ. (2)

SHORT ANSWER

15 **In $\triangle DEF$, P lies on DE and Q on DF with $DP = 5$ cm, $DE = 15$ cm, $DQ = 6$ cm and $QF = 12$ cm. Is $PQ \parallel EF$? Give a reason.** (2)

SHORT ANSWER

16 **A vertical pole 6 m tall casts a shadow 4 m long. At the same time a tower casts a shadow 28 m long. Find the height of the tower.** (2)

SHORT ANSWER

17 **Two concentric circles have centre O. A chord of length 80 cm of the larger circle touches the smaller circle, whose radius is 9 cm. Find the radius of the larger circle.** (2)

SHORT ANSWER

18 **Tangents PA and PB are drawn from P to a circle with centre O. If $PA = 7$ cm and $\angle APB = 60°$, find the length of the chord AB.** (2)

SHORT ANSWER

19 **Prove that if an acute angle of one right triangle is equal to an acute angle of another right triangle, the two triangles are similar.** (3)

SHORT ANSWER

20 **AB is a diameter of a circle with centre O. The tangent at a point T of the circle meets AB produced at P. If $\angle BAT = 25°$, find $\angle TPA$.** (3)

LONG ANSWER

21 **Prove that the tangents drawn at the ends of a chord of a circle make equal angles with the chord.** (4)

LONG ANSWER

22 **A circle touches the side BC of $\triangle ABC$ at P and touches AB and AC produced at Q and R respectively. Prove that $AQ = \frac{1}{2}$(perimeter of $\triangle ABC$).** (4)

Day 23 Answers with worked solutions

1 **(b)**
$\triangle ADE \sim \triangle ABC$, so $\frac{DE}{BC} = \frac{AE}{AC} = \frac{4}{10}$.
$DE = \frac{4}{10} \times 15 = 6$ cm.

2 **(a)**
Scale factor $= \frac{PQ}{AB} = \frac{12}{18} = \frac{2}{3}$.
$QR = \frac{2}{3} \times 27 = 18$, $RP = \frac{2}{3} \times 12 = 8$.

3 **(b)**
Two pairs of equal angles give similarity (AA), but the sides are in ratio 1 : 2, so they are not congruent.

4 **(c)**
B is common to AB and BC; R is common to PR and RQ, so $B \leftrightarrow R$. Then $A \leftrightarrow P$ and $C \leftrightarrow Q$.
So $\triangle ABC \sim \triangle PRQ$ (SSS), that is $\triangle PRQ \sim \triangle ABC$.

5 **(b)**
$\sqrt{15^2 - 9^2} = \sqrt{144} = 12$ cm.

6 **(a)**
$\angle OTP = 90°$, so $\angle TOP = 180° - 90° - 28° = 62°$.

7 **(d)**
Intersecting circles have exactly two common tangents (both direct).

8 **(c)**
$\angle OMN = 90° - 65° = 25° = \angle ONM$ (OM = ON).
$\angle MON = 180° - 50° = 130°$ (twice the tangent-chord angle).

9 **(b)**
A is BPT, true. R is the AAA criterion, true, but it is not the statement or proof of BPT. (b)

10 **(c)**
$PR = PA$ and $QR = QB$, so perimeter $= XA + XB = 2XA$: A is true.
No tangent can be drawn from an interior point: R is false. (c)

11 **Ans:** 3 cm
$\frac{AD}{DB} = \frac{AE}{EC} \Rightarrow AD = \frac{2.1 \times 6}{4.2} = 3$ cm.

12 **Ans:** 11 cm

$AR = AP = 5$, so $CR = 12 - 5 = 7 = CQ$;

$BQ = BP = 4$.

$BC = 4 + 7 = 11$ cm.

13 **Ans:** Basic Proportionality Theorem: $\frac{AD}{DB} = \frac{AE}{EC}$

A line parallel to one side of a triangle divides the other two sides in the same ratio.

Ans: 3 cm

$\frac{4}{2} = \frac{6}{EC} \Rightarrow EC = 3$ cm.

Ans: $x = 4$

$x(x-1) = (x+2)(x-2) \Rightarrow -x = -4 \Rightarrow x = 4$.

OR: Ans: Yes

$DB = 4$, $EC = 6$; $\frac{AD}{DB} = \frac{2}{4} = \frac{1}{2}$ and $\frac{AE}{EC} = \frac{3}{6} = \frac{1}{2}$.

Equal ratios, so $DE \parallel BC$ by the converse of BPT.

14 **Ans:** 120°

$\angle POQ = 180^\circ - 60^\circ = 120^\circ$.

Ans: 24 cm

$AP = AQ$ and $\angle PAQ = 60^\circ$, so $\triangle APQ$ is equilateral: $PQ = 24$ cm.

Ans: 30°

$\angle OPA = 90^\circ$ and $\angle APQ = 60^\circ$.

$\angle OPQ = 90^\circ - 60^\circ = 30^\circ$.

OR: Ans: Shown

$AP = AQ$ (tangents) and $OP = OQ$ (radii), so both A and O are equidistant from P and Q.

Hence both lie on the perpendicular bisector of PQ, so OA is that bisector.

15 **Ans:** Yes

$PE = 15 - 5 = 10$ cm; $\frac{DP}{PE} = \frac{5}{10} = \frac{1}{2}$ and $\frac{DQ}{QF} = \frac{6}{12} = \frac{1}{2}$.

Equal ratios, so $PQ \parallel EF$ (converse of BPT).

16 **Ans:** 42 m

The sun's rays make equal angles with the ground, and both pole and tower are vertical, so the two right triangles are similar (AA).

$\frac{h}{6} = \frac{28}{4} \Rightarrow h = 42$ m.

17 **Ans:** 41 cm

The radius to the point of contact is perpendicular to the chord and bisects it: half-chord $= 40$ cm.

$R = \sqrt{9^2 + 40^2} = \sqrt{1681} = 41$ cm.

18 **Ans:** 7 cm

$PA = PB$, so $\triangle PAB$ is isosceles with apex angle 60°; its base angles are 60° too.

So $\triangle PAB$ is equilateral and $AB = 7$ cm.

19 **Ans:** Proof

Let $\triangle ABC$ and $\triangle PQR$ have $\angle B = \angle Q = 90^\circ$ and $\angle A = \angle P$.

Then two pairs of angles are equal, so $\triangle ABC \sim \triangle PQR$ by AA (the third angles are then equal too).

20 **Ans:** 40°

$OA = OT$, so $\angle OTA = \angle OAT = 25^\circ$.

Exterior angle: $\angle TOP = 25^\circ + 25^\circ = 50^\circ$.

$\angle OTP = 90^\circ$ (tangent $\perp$ radius), so $\angle TPA = 180^\circ - 90^\circ - 50^\circ = 40^\circ$.

21 **Ans:** Proof

Let AB be a chord of a circle with centre O and let the tangents at A and B meet at P.

$PA = PB$ (tangents from an external point), so $\triangle PAB$ is isosceles.

Hence $\angle PAB = \angle PBA$: the tangents make equal angles with the chord AB.

22 **Ans:** Proof

Tangents from an external point are equal: $AQ = AR$, $BQ = BP$, $CP = CR$.

Perimeter $= AB + BC + CA = AB + (BP + PC) + CA = (AB + BQ) + (CR + CA) = AQ + AR$.

Since $AQ = AR$, perimeter $= 2AQ$, so $AQ = \frac{1}{2}$ (perimeter).

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. BPT in one line? : a line parallel to one side divides the other two sides in the same ratio
2. Converse of BPT gives? : the line is parallel to the third side
3. Three similarity criteria? : AA, SAS, SSS
4. Angle between radius and tangent? : 90°
5. Tangents from an external point? : equal
6. $\angle APB + \angle AOB$? : 180°
7. Circumscribed quadrilateral? : $AB + CD = AD + BC$

DAY 23 COMPLETE? ☐ Notes revised ☐ 22 questions solved ☐ Answers checked ☐ Mistakes noted

Next: Day 24, Introduction to Trigonometry

MY MISTAKES AND NOTES Write every mistake of this Day with the correct step

DAY 24

INTRODUCTION TO TRIGONOMETRY | CHAPTER 8 | PART 1 OF 3

Trigonometric Ratios

Introduction to Trigonometry

Aaj ka target

- [ ] Opposite, adjacent, hypotenuse: angle badlo toh side bhi badalti hai, yeh samjho
- [ ] Chhe ratios aur unke reciprocals zubaani
- [ ] Ek ratio diya ho toh baaki paanch Pythagoras se nikalo
- [ ] 5 'find the value' sums bina galti ke

24.1 Sides of a Right Triangle with respect to an Angle

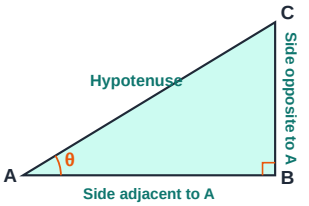


Fig. 24.1 In right $\triangle ABC$ ($\angle B = 90^\circ$), the sides named with respect to $\angle A$

In a right triangle the side opposite the right angle is the **hypotenuse** (always the longest side). For an acute angle A, the side facing A is the **side opposite** to A and the remaining side is the **side adjacent** to A. If we look from the other acute angle C, the opposite and adjacent sides **swap**; the hypotenuse stays the same.

24.2 The Six Trigonometric Ratios

TRIGONOMETRIC RATIOS OF $\angle A$

$$\sin A = \frac{\text{opp}}{\text{hyp}} = \frac{BC}{AC}$$

$$\cos A = \frac{\text{adj}}{\text{hyp}} = \frac{AB}{AC}$$

$$\tan A = \frac{\text{opp}}{\text{adj}} = \frac{BC}{AB}$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \quad \sec A = \frac{1}{\cos A}$$

$$\cot A = \frac{1}{\tan A}$$

$$\tan A = \frac{\sin A}{\cos A}, \quad \cot A = \frac{\cos A}{\sin A}$$

Ratio	Definition	Reciprocal of
$\sin A$	opp / hyp	$\operatorname{cosec} A$
$\cos A$	adj / hyp	$\sec A$
$\tan A$	opp / adj	$\cot A$

The three basic ratios and their reciprocals

- The ratios depend **only on the angle**, not on the size of the triangle (similar triangles have equal side ratios).
- For an acute angle: $0 < \sin A < 1$ and $0 < \cos A < 1$, so $\sec A > 1$ and $\operatorname{cosec} A > 1$. But $\tan A$ can be any positive number.
- $\sin A$ is one symbol: it is not 'sin times A'. $\sin^2 A$ means $(\sin A)^2$.



SHIVAM BHAIYA SAYS

Yaad karne ka desi tareeka: **'Pandit Badri Prasad Har Har Bole'** : sin = P/H, cos = B/H, tan = P/B (P = perpendicular, B = base, H = hypotenuse). Baaki teen bas ulta kar do.

24.3 Finding all Ratios from One Ratio

1. Draw a right triangle and mark the given angle.
2. Write the given ratio as a ratio of two sides, e.g. $\sin A = \frac{3}{5}$ means opposite $= 3k$, hypotenuse $= 5k$.
3. Find the third side with Pythagoras theorem.
4. Read off every other ratio from the triangle (the k cancels).

SOLVED EXAMPLE 1

If $\sin A = \frac{3}{5}$, find all the other trigonometric ratios of A.

SOLUTION

1. Let $BC = 3k$ and $AC = 5k$ (right angle at B).
2. $AB = \sqrt{(5k)^2 - (3k)^2} = \sqrt{16k^2} = 4k$.
3. $\cos A = \frac{4}{5}$, $\tan A = \frac{3}{4}$, $\operatorname{cosec} A = \frac{5}{3}$, $\sec A = \frac{5}{4}$, $\cot A = \frac{4}{3}$.

Answer: $\cos A = \frac{4}{5}$, $\tan A = \frac{3}{4}$, $\operatorname{cosec} A = \frac{5}{3}$, $\sec A = \frac{5}{4}$, $\cot A = \frac{4}{3}$

SOLVED EXAMPLE 2

If $15 \cot A = 8$, find $\sin A$ and $\sec A$.

SOLUTION

1. $\cot A = \frac{8}{15} = \frac{\text{adjacent}}{\text{opposite}}$: let $AB = 8k$, $BC = 15k$.

SOLVED EXAMPLE 2

1. $AC = \sqrt{64k^2 + 225k^2} = 17k$.
2. $\sin A = \frac{BC}{AC} = \frac{15}{17}$, $\sec A = \frac{AC}{AB} = \frac{17}{8}$.

Answer: $\sin A = \frac{15}{17}$, $\sec A = \frac{17}{8}$

SOLVED EXAMPLE 3

In $\triangle PQR$, right-angled at Q, $PQ = 12$ cm and $QR = 5$ cm. Find $\sin P$, $\cos P$, $\sin R$ and $\cos R$.

SOLUTION

1. $PR = \sqrt{12^2 + 5^2} = 13$ cm.
2. For $\angle P$: opposite $= QR = 5$, adjacent $= PQ = 12$. So $\sin P = \frac{5}{13}$, $\cos P = \frac{12}{13}$.
3. For $\angle R$: opposite $= PQ = 12$, adjacent $= QR = 5$. So $\sin R = \frac{12}{13}$, $\cos R = \frac{5}{13}$.

Answer: $\sin P = \cos R = \frac{5}{13}$ and $\cos P = \sin R = \frac{12}{13}$



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Sabse badi galti: angle R ke liye bhi wahi opposite side le lena jo angle P ke liye li thi. Har baar pehle angle pe ungli rakho, phir dekho uske saamne kaun si side hai.

SOLVED EXAMPLE 4

If $4\tan\theta = 3$, evaluate $\frac{4\sin\theta - \cos\theta}{4\sin\theta + \cos\theta}$.

SOLUTION

1. Divide numerator and denominator by $\cos\theta$: $\frac{4\tan\theta - 1}{4\tan\theta + 1}$.
2. $= \frac{3-1}{3+1} = \frac{2}{4} = \frac{1}{2}$

Answer: $\frac{1}{2}$



SHIVAM BHAIYA SAYS

Jab expression mein har term mein sin ya cos ek hi power mein ho aur tan diya ho, toh upar neeche cos se divide kar do. Triangle banane ki bhi zaroorat nahi, 2 line mein answer.



SHIVAM BHAIYA SAYS: BOARD TIP

Board mein triangle ka rough figure banake sides pe 3k, 4k, 5k likhna, examiner ko method dikhta hai aur step marks milte hain. Final answer mein k nahi bachna chahiye.



EXAM ALERT!

Trigonometry (Ch 8 + 9) milake board mein around 12 marks ka hai. 'Ek ratio diya hai, baaki nikalo' type 2 markers aur MCQ har saal aate hain.

MEME SAMJHO — Astra meme, maths inside



DIMAAG NAHI CHAL RAHA REACTION, FIRST TRIG SUM

Jab sirf sin θ = 3/5 diya ho aur baaki paanch ratios bhi chahiye.

Asli baat: draw a right triangle with opposite 3k and hypotenuse 5k; Pythagoras gives adjacent 4k, and all six ratios follow.

Day 24 Practice

16 board-pattern questions | solve first, then check

MCQ

1 If $\sin A = \frac{p}{q}$, then $\cos A$ equals (1)

(a) $\frac{q}{\sqrt{q^2-p^2}}$ (b) $\frac{\sqrt{q^2-p^2}}{q}$
(c) $\frac{p}{\sqrt{q^2-p^2}}$ (d) $\frac{\sqrt{q^2-p^2}}{p}$

MCQ

2 If $3\cot\theta = 2$, then $\frac{3\cos\theta - 2\sin\theta}{3\cos\theta + 2\sin\theta}$ is (1)

(a) 1 (b) $\frac{1}{2}$ (c) 0 (d) $\frac{5}{13}$

MCQ

3 In $\triangle XYZ$, right-angled at Y, $XY = 24$, $YZ = 7$ and $XZ = 25$. Then $\tan X$, $\operatorname{cosec} X$ and $\sec X$ are respectively (1)

(a) $\frac{24}{7}, \frac{25}{24}, \frac{25}{7}$ (b) $\frac{7}{25}, \frac{25}{7}, \frac{24}{25}$
(c) $\frac{7}{24}, \frac{25}{7}, \frac{25}{24}$ (d) $\frac{7}{24}, \frac{25}{24}, \frac{25}{7}$

MCQ

4 If $\cos\theta = \frac{4}{5}$, then $\frac{\sin\theta\tan\theta - 1}{2\tan^2\theta}$ equals (1)

(a) $-\frac{22}{45}$ (b) $\frac{22}{45}$ (c) $\frac{3}{160}$ (d) $-\frac{11}{20}$

MCQ

5 If $\sin\theta = \frac{5}{13}$, which of the following can be the sides of the right triangle? (1)

(a) 5, 12, 15 (b) 10, 24, 26
(c) 5, 13, 18 (d) 10, 13, 26

ASSERTION-REASON

6 **Assertion (A):** For an acute angle A, the value $\sin A = \frac{4}{3}$ is not possible.
Reason (R): In a right triangle the hypotenuse is the longest side. (1)
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

7 **Assertion (A):** For every acute angle A, $\sec A > 1$.
Reason (R): For an acute angle A, $0 < \cos A < 1$ and $\sec A = \frac{1}{\cos A}$. (1)
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

VERY SHORT ANSWER

8 If $\operatorname{cosec}\theta = \frac{13}{12}$, find $\cos\theta + \tan\theta$. (1)

CASE-BASED

9 Ladder against a lamp post. A ladder PR leans against a vertical lamp post PQ, with its foot R on level ground. The angle at Q is 90°, the top of the ladder is 12 m high ($PQ = 12$ m) and its foot is 5 m from the post ($QR = 5$ m). (4)
(i) Find $\cos R$. (1)
(ii) Find $\operatorname{cosec} P$. (1)
(iii) Find $\cot P - \operatorname{cosec} R$. (2)
OR (iii) Find $\tan P + \tan R$. (2)

SHORT ANSWER

10 If $4\cot A = 3$, check whether $\frac{1-\tan^2 A}{1+\tan^2 A} = \cos^2 A - \sin^2 A$. (2)

SHORT ANSWER

11 In $\triangle PQR$, right-angled at Q, $PQ = 8$ cm and $PR = 17$ cm. Find $\tan P - \cot R$. (2)

SHORT ANSWER

12 If $\cot\theta = \frac{5}{12}$, evaluate $\frac{(1+\sin\theta)(1-\sin\theta)}{(1+\cos\theta)(1-\cos\theta)}$. (2)

SHORT ANSWER

13 If $3\tan\theta = 4$, find the value of $\frac{3\sin\theta - \cos\theta + 1}{3\sin\theta + \cos\theta - 1}$. (3)

SHORT ANSWER

14 If $\sin\theta = \frac{3}{5}$, evaluate $\frac{\sin^2\theta - \cos^2\theta}{2\sin\theta\cos\theta} \times \frac{1}{\tan^2\theta}$. (3)

LONG ANSWER

15 A rectangular garden ABCD has $AD = 15$ m and $CD = 8$ m. A straight path runs along the diagonal AC, and $\angle CAD = \theta$. Find (i) $\tan\theta$, (ii) $\frac{1-\tan^2\theta}{1+\tan^2\theta}$, (iii) $2\sin\theta\cos\theta$, and (iv) verify that $\sin^2\theta + \cos^2\theta = 1$. (4)

LONG ANSWER

16 In $\triangle ABC$, right-angled at B, $AB = 7$ cm and $AC + BC = 49$ cm. Find the lengths of BC and AC, and then all six trigonometric ratios of $\angle A$. (5)

Day 24 Answers with worked solutions

1 **(b)**
Opposite $= p$, hypotenuse $= q$, adjacent $= \sqrt{q^2 - p^2}$, so $\cos A = \frac{\sqrt{q^2-p^2}}{q}$.

2 **(c)**
Divide by $\sin\theta$: $\frac{3\cot\theta - 2}{3\cot\theta + 2} = \frac{2-2}{2+2} = 0$.

3 **(c)**
For $\angle X$: opposite $= YZ = 7$, adjacent $= XY = 24$, hypotenuse $= 25$.
$\tan X = \frac{7}{24}$, $\operatorname{cosec} X = \frac{25}{7}$, $\sec X = \frac{25}{24}$.

4 **(a)**
$\sin\theta = \frac{3}{5}$, $\tan\theta = \frac{3}{4}$.
Numerator $= \frac{3}{5} \times \frac{3}{4} - 1 = -\frac{11}{20}$; denominator $= 2 \times \frac{9}{16} = \frac{9}{8}$.
Value $= -\frac{11}{20} \times \frac{8}{9} = -\frac{22}{45}$.

5 **(b)**
The sides must be in the ratio opposite : adjacent : hypotenuse $= 5 : 12 : 13$.
10, 24, 26 is $2 \times (5, 12, 13)$.

6 **(a)**
$\sin A = \frac{\text{opposite}}{\text{hypotenuse}}$ and the opposite side is shorter than the hypotenuse, so $\sin A < 1$. A is true and R explains it. (a)

7 **(a)**
The reciprocal of a number between 0 and 1 is greater than 1. Both true and R explains A. (a)

8 **Ans:** $\frac{181}{65}$
$\sin\theta = \frac{12}{13}$, adjacent $= \sqrt{13^2 - 12^2} = 5$.
$\cos\theta = \frac{5}{13}$, $\tan\theta = \frac{12}{5}$; sum $= \frac{25+156}{65} = \frac{181}{65}$.

9 **Ans:** $\frac{5}{13}$

$PR = \sqrt{12^2 + 5^2} = 13$ m; $\cos R = \frac{QR}{PR} = \frac{5}{13}$.

Ans: $\frac{13}{5}$

For $\angle P$ the opposite side is $QR = 5$: $\operatorname{cosec} P = \frac{PR}{QR} = \frac{13}{5}$.

Ans: $\frac{79}{60}$

$\cot P = \frac{PQ}{QR} = \frac{12}{5}$; $\operatorname{cosec} R = \frac{PR}{PQ} = \frac{13}{12}$.

$\frac{12}{5} - \frac{13}{12} = \frac{144-65}{60} = \frac{79}{60}$.

OR: Ans: $\frac{169}{60}$

$\tan P = \frac{5}{12}$, $\tan R = \frac{12}{5}$; sum $= \frac{25+144}{60} = \frac{169}{60}$.

10 **Ans:** Yes, both sides equal $-\frac{7}{25}$

$\cot A = \frac{3}{4}$: adjacent $3k$, opposite $4k$, hypotenuse $5k$. So $\tan A = \frac{4}{3}$, $\sin A = \frac{4}{5}$, $\cos A = \frac{3}{5}$.

$$\text{LHS} = \frac{1 - \frac{16}{9}}{1 + \frac{16}{9}} = \frac{-7/9}{25/9} = -\frac{7}{25}.$$

RHS $= \frac{9}{25} - \frac{16}{25} = -\frac{7}{25}$. LHS = RHS.

11 **Ans:** 0

$QR = \sqrt{17^2 - 8^2} = 15$ cm.

$\tan P = \frac{QR}{PQ} = \frac{15}{8}$ and $\cot R = \frac{QR}{PQ} = \frac{15}{8}$.

$\tan P - \cot R = 0$.

12 **Ans:** $\frac{25}{144}$

$$\text{Expression} = \frac{1 - \sin^2\theta}{1 - \cos^2\theta}.$$

With adjacent 5, opposite 12, hypotenuse 13: $1 - \sin^2\theta = 1 - \frac{144}{169} = \frac{25}{169}$ and $1 - \cos^2\theta = \frac{144}{169}$.

Value $= \frac{25}{144}$ (that is, $\cot^2\theta$).

13 **Ans:** $\frac{7}{5}$

$\tan\theta = \frac{4}{3}$: opposite $4k$, adjacent $3k$, hypotenuse $5k$.

$\sin\theta = \frac{4}{5}$, $\cos\theta = \frac{3}{5}$.

Numerator $= \frac{12}{5} - \frac{3}{5} + 1 = \frac{14}{5}$; denominator $= \frac{12}{5} + \frac{3}{5} - 1 = \frac{10}{5}$.

Value $= \frac{14}{10} = \frac{7}{5}$.

14 **Ans:** $-\frac{14}{27}$

$\cos\theta = \frac{4}{5}$, $\tan\theta = \frac{3}{4}$.

$$\frac{\sin^2\theta - \cos^2\theta}{2\sin\theta\cos\theta} = \frac{\frac{9}{25} - \frac{16}{25}}{\frac{24}{25}} = -\frac{7}{24}.$$

$\frac{1}{\tan^2\theta} = \frac{16}{9}$, so the value is $-\frac{7}{24} \times \frac{16}{9} = -\frac{14}{27}$.

15 **Ans:** (i) $\frac{8}{15}$ (ii) $\frac{161}{289}$ (iii) $\frac{240}{289}$ (iv) verified

In right $\triangle ADC$ ($\angle D = 90°$): $AC = \sqrt{15^2 + 8^2} = 17$ m.

(i) $\tan\theta = \frac{CD}{AD} = \frac{8}{15}$.

(ii) $$\frac{1 - \frac{64}{225}}{1 + \frac{64}{225}} = \frac{161}{289}.$$

(iii) $\sin\theta = \frac{8}{17}$, $\cos\theta = \frac{15}{17}$, so $2\sin\theta\cos\theta = \frac{240}{289}$.

(iv) $\frac{64}{289} + \frac{225}{289} = \frac{289}{289} = 1$.

16 **Ans:** $BC = 24$ cm, $AC = 25$ cm; $\sin A = \frac{24}{25}$, $\cos A = \frac{7}{25}$, $\tan A = \frac{24}{7}$, $\operatorname{cosec} A = \frac{25}{24}$, $\sec A = \frac{25}{7}$, $\cot A = \frac{7}{24}$

Let $BC = x$; then $AC = 49 - x$.

Pythagoras: $(49 - x)^2 = x^2 + 7^2 \Rightarrow 2401 - 98x = 49 \Rightarrow x = 24$.

$BC = 24$ cm, $AC = 25$ cm.

$\sin A = \frac{BC}{AC} = \frac{24}{25}$, $\cos A = \frac{AB}{AC} = \frac{7}{25}$, $\tan A = \frac{24}{7}$.

$\operatorname{cosec} A = \frac{25}{24}$, $\sec A = \frac{25}{7}$, $\cot A = \frac{7}{24}$.

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. sin A in words? : opposite / hypotenuse
2. Reciprocal of cos A? : sec A
3. tan A in terms of sin and cos? : $\frac{\sin A}{\cos A}$
4. Can sin A be 1.2 for an acute angle? : no, it is less than 1
5. If sin A = 3/5, cos A? : 4/5
6. Do the ratios change if the triangle is enlarged? : no, they depend only on the angle
7. $\sin^2 A$ means? : $(\sin A)^2$

DAY 24 COMPLETE? ☐ Notes read ☐ 16 questions solved ☐ Answers checked ☐ Mistakes noted

Next: Day 25, Introduction to Trigonometry

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 25

INTRODUCTION TO TRIGONOMETRY | CHAPTER 8 | PART 2 OF 3

Ratios of Specific Angles

Introduction to Trigonometry

Aaj ka target

- ☐ 0°, 30°, 45°, 60°, 90° ki poori table 1 minute mein likh pao
- ☐ 45° aur 30°/60° ki values triangle se derive karna seekho
- ☐ 'Find the value' expressions bina calculator ke
- ☐ sin(A+B), tan(A−B) wale 'find A and B' sums pakke

25.1 Ratios of 45°

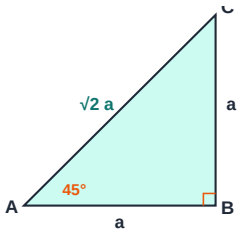


Fig. 25.1 Isosceles right triangle: legs a, hypotenuse $\sqrt{2}a$

In right $\triangle ABC$ with $\angle B = 90^\circ$ and $\angle A = 45^\circ$, we also have $\angle C = 45^\circ$, so $AB = BC = a$. By Pythagoras $AC = \sqrt{a^2 + a^2} = \sqrt{2}a$.

VALUES AT 45°

$$\sin 45^\circ = \cos 45^\circ = \frac{1}{\sqrt{2}}$$

$$\tan 45^\circ = \cot 45^\circ = 1$$

$$\sec 45^\circ = \operatorname{cosec} 45^\circ = \sqrt{2}$$

25.2 Ratios of 30° and 60°

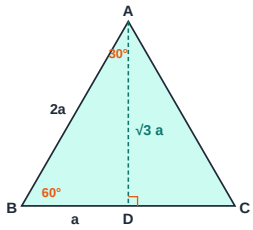


Fig. 25.2 Equilateral triangle of side 2a cut by the altitude AD

Take an equilateral $\triangle ABC$ of side $2a$ and draw the altitude AD. It bisects BC and $\angle A$, so $BD = a$, $\angle BAD = 30^\circ$, $\angle ABD = 60^\circ$, and $AD = \sqrt{(2a)^2 - a^2} = \sqrt{3}a$. In right $\triangle ABD$: $\sin 30^\circ = \frac{BD}{AB} = \frac{1}{2}$, $\cos 30^\circ = \frac{AD}{AB} = \frac{\sqrt{3}}{2}$, and for 60° the opposite and adjacent sides swap.

25.3 Ratios of 0° and 90°

As $\angle A$ shrinks towards 0°, the opposite side shrinks to 0 and the adjacent side becomes equal to the hypotenuse: $\sin 0^\circ = 0$, $\cos 0^\circ = 1$. As $\angle A$ grows towards 90°, the adjacent side shrinks to 0: $\sin 90^\circ = 1$, $\cos 90^\circ = 0$. Ratios with a zero denominator are **not defined**.

Ratio	0°	30°	45°	60°	90°
sin	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	N.D.
cosec	N.D.	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
sec	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	N.D.
cot	N.D.	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

Table of values (N.D. = not defined)



SHIVAM BHAIYA SAYS

Table yaad karne ki trick: sin row ke liye 0, 1, 2, 3, 4 likho, har ek ka $\sqrt{}$ lo aur 2 se divide karo: $\frac{\sqrt{0}}{2}, \frac{\sqrt{1}}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, \frac{\sqrt{4}}{2}$. cos row bas ulti. tan = sin ÷ cos. Ho gaya!

- As A increases from 0° to 90°: $\sin A$ increases from 0 to 1, $\cos A$ decreases from 1 to 0, $\tan A$ increases from 0 without bound.
- $\tan 90^\circ$, $\sec 90^\circ$, $\operatorname{cosec} 0^\circ$ and $\cot 0^\circ$ are not defined.

SOLVED EXAMPLE 1

Evaluate $\sin 60^\circ \cos 30^\circ + \sin 30^\circ \cos 60^\circ$.

SOLUTION

1. $= \frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2} + \frac{1}{2} \times \frac{1}{2}$
2. $= \frac{3}{4} + \frac{1}{4} = 1$

Answer: 1

SOLVED EXAMPLE 2

Evaluate $\dfrac{\sin 30^\circ + \tan 45^\circ - \operatorname{cosec} 60^\circ}{\sec 30^\circ + \cos 60^\circ + \cot 45^\circ}$.

SOLUTION

1. Numerator $= \frac{1}{2} + 1 - \frac{2}{\sqrt{3}} = \frac{3\sqrt{3}-4}{2\sqrt{3}}$; denominator $= \frac{2}{\sqrt{3}} + \frac{1}{2} + 1 = \frac{3\sqrt{3}+4}{2\sqrt{3}}$.
2. Value $= \dfrac{3\sqrt{3}-4}{3\sqrt{3}+4}$. Multiply by $\dfrac{3\sqrt{3}-4}{3\sqrt{3}-4}$:
3. $= \dfrac{27+16-24\sqrt{3}}{27-16} = \dfrac{43-24\sqrt{3}}{11}$

Answer: $\frac{43-24\sqrt{3}}{11}$

SOLVED EXAMPLE 3

In $\triangle ABC$, right-angled at B, $AB = 5$ cm and $\angle ACB = 30^\circ$. Find BC and AC.

SOLUTION

1. $\tan 30^\circ = \frac{AB}{BC} \Rightarrow \frac{1}{\sqrt{3}} = \frac{5}{BC} \Rightarrow BC = 5\sqrt{3}$ cm.
2. $\sin 30^\circ = \frac{AB}{AC} \Rightarrow \frac{1}{2} = \frac{5}{AC} \Rightarrow AC = 10$ cm.

Answer: $BC = 5\sqrt{3}$ cm, $AC = 10$ cm

SOLVED EXAMPLE 4

If $\tan(A+B) = \sqrt{3}$ and $\tan(A-B) = \frac{1}{\sqrt{3}}$, where $0^\circ < A+B \le 90^\circ$ and $A > B$, find A and B.

SOLUTION

1. $\tan(A+B) = \sqrt{3} = \tan 60^\circ \Rightarrow A+B = 60^\circ$.
2. $\tan(A-B) = \frac{1}{\sqrt{3}} = \tan 30^\circ \Rightarrow A-B = 30^\circ$.
3. Adding: $2A = 90^\circ$, $A = 45^\circ$; then $B = 15^\circ$.

Answer: $A = 45^\circ$, $B = 15^\circ$



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

$\sin(A+B) \neq \sin A + \sin B$! Jaise $\sin(30^\circ + 60^\circ) = \sin 90^\circ = 1$, par $\sin 30^\circ + \sin 60^\circ = \frac{1+\sqrt{3}}{2}$. Bracket ko kabhi tod ke mat likhna.



SHIVAM BHAIYA SAYS: BOARD TIP

Expression waale sums mein pehle har value substitute karke ek line likho, phir simplify. Answer mein denominator mein $\sqrt{3}$ bacha ho toh rationalise karna mat bhoolna.



EXAM ALERT!

Table of values se 1 mark ke MCQ pakke aate hain, aur 'find A and B' type 2 markers bhi. Note: complementary angles ($\sin(90^\circ - A) = \cos A$ type) 2026-27 syllabus se hata diye gaye hain.

MEME SAMJHO — Astra meme, maths inside



MR INCREDIBLE GETTING MORE AND MORE UNEASY, THE TAN ROW

tan 0° = 0, tan 30° = 1/√3, tan 45° = 1, tan 60° = √3, tan 90° = ...

Asli baat: $\tan 90^\circ = \frac{\sin 90^\circ}{\cos 90^\circ} = \frac{1}{0}$, which is not defined. tan keeps growing as the angle approaches 90°.

Day 25 Practice

17 board-pattern questions | solve first, then check

MCQ

1 If $x \sin 30^\circ \tan 45^\circ = \tan 60^\circ \cos 30^\circ$, then x equals (1)

(a) $\sqrt{3}$ (b) 3 (c) $\frac{3}{2}$ (d) 1

MCQ

2 If $\cos(A+B) = 0$ and $\sin(A-B) = \frac{1}{2}$, with A and B acute and $A > B$, then (1)

(a) $A = 45^\circ, B = 45^\circ$ (b) $A = 30^\circ, B = 60^\circ$
(c) $A = 60^\circ, B = 30^\circ$ (d) $A = 75^\circ, B = 15^\circ$

MCQ

3 If $\cos(2x - 10^\circ) = \frac{1}{2}$, where $2x - 10^\circ$ is acute, then x is (1)

(a) 25° (b) 30° (c) 35° (d) 40°

MCQ

4 $\dfrac{1 - \tan^2 30^\circ}{1 + \tan^2 30^\circ}$ is equal to (1)

(a) $\sin 60^\circ$ (b) $\cos 60^\circ$ (c) $\tan 60^\circ$ (d) $\cos 30^\circ$

MCQ

5 If $\cos A = \frac{1}{2}$ for an acute angle A, then $\operatorname{cosec} A$ is (1)

(a) 2 (b) $\frac{1}{2}$ (c) $\frac{2}{\sqrt{3}}$ (d) $\sqrt{3}$

ASSERTION-REASON

6 **Assertion (A):** In right $\triangle ABC$ with $\angle B = 90^\circ$, if $\tan A = 1$ then $2 \sin A \cos A = 1$.
Reason (R): $\tan 45^\circ = \sin 45^\circ = \cos 45^\circ = \frac{1}{\sqrt{2}}$. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

7 **Assertion (A):** $\tan 90^\circ$ **is not defined.**
Reason (R): $\cos 90^\circ = 0$. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

VERY SHORT ANSWER

8 **Evaluate** $\cos^2 45^\circ + 3\tan^2 30^\circ - \sin^2 90^\circ$. (1)

CASE-BASED

9 **Wheelchair ramp. A school builds a straight ramp 20 m long from the ground up to a stage. The ramp makes an angle of 30° with the level ground.** (4)

(i) Find the height of the stage. (1)

(ii) Find the horizontal distance covered by the ramp. (1)

(iii) If the same 20 m ramp were set at 60°, how much higher would the stage be? (2)

OR (iii) A second ramp rises 4 m over a horizontal distance of 4 m. Find the angle it makes with the ground. (2)

SHORT ANSWER

10 **Evaluate** $\dfrac{4\cos^2 60^\circ + 3\sec^2 30^\circ - 2\tan^2 45^\circ}{\sin^2 30^\circ + \cos^2 30^\circ}$. (2)

SHORT ANSWER

11 **If** $\tan(A+B) = 1$ **and** $\tan(A-B) = \frac{1}{\sqrt{3}}$, **where** $0^\circ < A+B \le 90^\circ$ **and** $A > B$, **find A and B.** (2)

SHORT ANSWER

12 **Find the acute angle** θ **if** $\sin\theta - \sqrt{3}\cos\theta = 0$. (2)

SHORT ANSWER

13 **Evaluate** $(\cos^4 30^\circ + \operatorname{cosec}^4 60^\circ) - 3(\tan^2 45^\circ - \sin^2 30^\circ)$. (3)

SHORT ANSWER

14 **Find x if** $2\sec^2 45^\circ + x\cos^2 30^\circ - \cot^2 30^\circ = 2$. (3)

SHORT ANSWER

15 **In** $\triangle ABC$, **right-angled at B,** $AB = 6$ **cm and** $\angle ACB = 30^\circ$. **Find the lengths of BC and AC.** (3)

LONG ANSWER

16 **For** $A = 30^\circ$, **verify that (i)** $\sin 2A = 2\sin A\cos A$, **(ii)** $\cos 2A = \cos^2 A - \sin^2 A = 1 - 2\sin^2 A$, **(iii)** $\tan 2A = \dfrac{2\tan A}{1-\tan^2 A}$. (4)

LONG ANSWER

17 Evaluate $\dfrac{4\tan^2 45^\circ + 3\sec^2 30^\circ + \cot^2 60^\circ + 5\cos^2 90^\circ}{\operatorname{cosec} 30^\circ + \sec 60^\circ - \cot^2 30^\circ}$. (4)

Day 25 Answers with worked solutions

1 **(b)**
$x \times \frac{1}{2} \times 1 = \sqrt{3} \times \frac{\sqrt{3}}{2} = \frac{3}{2}$, so $x = 3$.

2 **(c)**
$A + B = 90^\circ$ and $A - B = 30^\circ$, so $A = 60^\circ$, $B = 30^\circ$.

3 **(c)**
$\cos 60^\circ = \frac{1}{2}$, so $2x - 10^\circ = 60^\circ$ and $x = 35^\circ$.

4 **(b)**
$\dfrac{1-\frac{1}{3}}{1+\frac{1}{3}} = \dfrac{2/3}{4/3} = \frac{1}{2} = \cos 60^\circ$.

5 **(c)**
$A = 60^\circ$, so $\operatorname{cosec} A = \frac{1}{\sin 60^\circ} = \frac{2}{\sqrt{3}}$.

6 **(c)**
$A = 45^\circ$: $2 \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} = 1$, so A is true.
R is false because $\tan 45^\circ = 1$, not $\frac{1}{\sqrt{2}}$. (c)

7 **(a)**
$\tan 90^\circ = \frac{\sin 90^\circ}{\cos 90^\circ} = \frac{1}{0}$, not defined; R is the reason. (a)

8 **Ans:** $\frac{1}{2}$
$\frac{1}{2} + 3 \times \frac{1}{3} - 1 = \frac{1}{2}$.

9 **Ans:** 10 m
Height $= 20\sin 30^\circ = 20 \times \frac{1}{2} = 10$ m.
Ans: $10\sqrt{3}$ m
Distance $= 20\cos 30^\circ = 20 \times \frac{\sqrt{3}}{2} = 10\sqrt{3}$ m (about 17.32 m).
Ans: $(10\sqrt{3} - 10)$ m, about 7.32 m
New height $= 20\sin 60^\circ = 10\sqrt{3}$ m.
Increase $= 10\sqrt{3} - 10 \approx 7.32$ m.
OR: Ans: 45°
$\tan\theta = \frac{4}{4} = 1 = \tan 45^\circ$, so $\theta = 45^\circ$.

10 **Ans:** 3
Numerator $= 4 \times \frac{1}{4} + 3 \times \frac{4}{3} - 2 \times 1 = 1 + 4 - 2 = 3$.
Denominator $= \frac{1}{4} + \frac{3}{4} = 1$. Value $= 3$.

11 **Ans:** $A = 37.5^\circ$, $B = 7.5^\circ$
$A + B = 45^\circ$ and $A - B = 30^\circ$.
$2A = 75^\circ \Rightarrow A = 37.5^\circ$, $B = 7.5^\circ$.

12 **Ans:** 60°

$\sin\theta = \sqrt{3}\cos\theta \Rightarrow \tan\theta = \sqrt{3} = \tan 60^\circ$.

$\theta = 60^\circ$.

13 **Ans:** $\frac{13}{144}$

$\cos^4 30^\circ = \left(\frac{3}{4}\right)^2 = \frac{9}{16}$; $\operatorname{cosec}^4 60^\circ = \left(\frac{4}{3}\right)^2 = \frac{16}{9}$.

$3(\tan^2 45^\circ - \sin^2 30^\circ) = 3\left(1 - \frac{1}{4}\right) = \frac{9}{4}$.

Value $= \frac{81+256-324}{144} = \frac{13}{144}$.

14 **Ans:** $x = \frac{4}{3}$

$2 \times 2 + x \times \frac{3}{4} - 3 = 2$.

$1 + \frac{3x}{4} = 2 \Rightarrow \frac{3x}{4} = 1 \Rightarrow x = \frac{4}{3}$.

15 **Ans:** $BC = 6\sqrt{3}$ cm, $AC = 12$ cm

$\tan 30^\circ = \frac{AB}{BC} \Rightarrow BC = \frac{6}{1/\sqrt{3}} = 6\sqrt{3}$ cm.

$\sin 30^\circ = \frac{AB}{AC} \Rightarrow AC = \frac{6}{1/2} = 12$ cm.

16 **Ans:** All three verified

$2A = 60^\circ$.

(i) $\sin 60^\circ = \frac{\sqrt{3}}{2}$; $2 \times \frac{1}{2} \times \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2}$. Equal.

(ii) $\cos 60^\circ = \frac{1}{2}$; $\frac{3}{4} - \frac{1}{4} = \frac{1}{2}$; $1 - 2 \times \frac{1}{4} = \frac{1}{2}$. Equal.

(iii) $\tan 60^\circ = \sqrt{3}$; $\frac{2/\sqrt{3}}{1 - 1/3} = \frac{2/\sqrt{3}}{2/3} = \frac{3}{\sqrt{3}} = \sqrt{3}$.

Equal.

17 **Ans:** $\frac{25}{3}$

Numerator $= 4(1) + 3\left(\frac{4}{3}\right) + \frac{1}{3} + 5(0) = 4 + 4 + \frac{1}{3} = \frac{25}{3}$.

Denominator $= 2 + 2 - 3 = 1$.

Value $= \frac{25}{3}$.

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. sin 30°? : $\frac{1}{2}$
2. cos 30°? : $\frac{\sqrt{3}}{2}$
3. tan 60°? : $\sqrt{3}$
4. sin 45° = cos 45° = ? : $\frac{1}{\sqrt{2}}$
5. tan 90°? : not defined
6. As A goes 0° to 90°, cos A? : decreases from 1 to 0
7. Is sin(A + B) = sin A + sin B? : no

DAY 25 COMPLETE? ☐ Notes read ☐ 17 questions solved ☐ Answers checked ☐ Mistakes noted

Next: Day 26, Introduction to Trigonometry

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 26

INTRODUCTION TO TRIGONOMETRY | CHAPTER 8 | PART 3 OF 3

Trigonometric Identities

Introduction to Trigonometry

Aaj ka target

- [ ] Teeno identities ka proof right triangle se likho
- [ ] Har ratio ko kisi ek ratio (jaise cot A) mein express karna seekho
- [ ] 'Prove that' sums: LHS ko simplify karke RHS tak pahuncho
- [ ] sec + tan aur sec − tan reciprocal wali trick pakki

26.1 The Three Identities

An equation involving trigonometric ratios of an angle is a **trigonometric identity** if it is true for **all** values of the angle for which the ratios are defined.

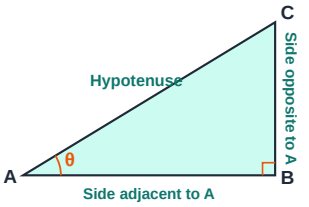


Fig. 26.1 Right $\triangle ABC$: $AB^2 + BC^2 = AC^2$

1. In right $\triangle ABC$ (right angle at B): $AB^2 + BC^2 = AC^2$.
2. Divide by AC^2: $\left(\frac{AB}{AC}\right)^2 + \left(\frac{BC}{AC}\right)^2 = 1$, that is $\cos^2 A + \sin^2 A = 1$ (true for $0^\circ \le A \le 90^\circ$).
3. Divide by AB^2: $1 + \left(\frac{BC}{AB}\right)^2 = \left(\frac{AC}{AB}\right)^2$, that is $1 + \tan^2 A = \sec^2 A$ $(0^\circ \le A < 90^\circ)$.
4. Divide by BC^2: $\left(\frac{AB}{BC}\right)^2 + 1 = \left(\frac{AC}{BC}\right)^2$, that is $1 + \cot^2 A = \operatorname{cosec}^2 A$ $(0^\circ < A \le 90^\circ)$.

IDENTITIES TO REMEMBER

$$\sin^2 A + \cos^2 A = 1$$

$$1 + \tan^2 A = \sec^2 A$$

$$1 + \cot^2 A = \operatorname{cosec}^2 A$$

Useful forms: $\sec^2 A - \tan^2 A = 1$, so $(\sec A + \tan A)(\sec A - \tan A) = 1$; similarly $(\operatorname{cosec} A + \cot A)(\operatorname{cosec} A - \cot A) = 1$.



SHIVAM BHAIYA SAYS

Teeno identity ek hi line se: sin² + cos² = 1 ko pehle $\cos^2$ se divide karo (tan, sec wali identity), phir $\sin^2$ se divide karo (cot, cosec wali). Ek yaad, teen free!

26.2 Expressing Ratios in Terms of One Ratio

SOLVED EXAMPLE 1

Express $\sin A$, $\sec A$ and $\tan A$ in terms of $\cot A$.

SOLUTION

1. $\operatorname{cosec}^2 A = 1 + \cot^2 A \Rightarrow \sin A = \dfrac{1}{\sqrt{1 + \cot^2 A}}$.
2. $\tan A = \dfrac{1}{\cot A}$.
3. $\sec^2 A = 1 + \tan^2 A = 1 + \dfrac{1}{\cot^2 A} = \dfrac{\cot^2 A + 1}{\cot^2 A}$, so $\sec A = \dfrac{\sqrt{1 + \cot^2 A}}{\cot A}$.

Answer: $\sin A = \frac{1}{\sqrt{1+\cot^2 A}}$, $\tan A = \frac{1}{\cot A}$, $\sec A = \frac{\sqrt{1+\cot^2 A}}{\cot A}$

26.3 How to Prove an Identity

1. Start from the more complicated side (usually LHS) and work towards the other side. Never move terms across the = sign as if solving an equation.
2. Convert everything to sin and cos when stuck.
3. Take LCM of fractions; look for $1 - \sin^2 A = \cos^2 A$, $\sec^2 A - 1 = \tan^2 A$ and similar.
4. Multiply by a conjugate such as $(1 - \sin A)$ or $(\sec A + \tan A)$ when a sum or difference sits in the denominator.

SOLVED EXAMPLE 2

Prove that $\dfrac{\cos A}{1 + \sin A} + \dfrac{1 + \sin A}{\cos A} = 2 \sec A$.

SOLUTION

1. LHS $= \dfrac{\cos^2 A + (1 + \sin A)^2}{\cos A(1 + \sin A)}$
2. $= \dfrac{\cos^2 A + 1 + 2\sin A + \sin^2 A}{\cos A(1 + \sin A)} = \dfrac{2 + 2\sin A}{\cos A(1 + \sin A)}$
3. $= \dfrac{2(1 + \sin A)}{\cos A(1 + \sin A)} = \dfrac{2}{\cos A} = 2 \sec A =$ RHS

Answer: Hence proved.

SOLVED EXAMPLE 3

Prove that $\dfrac{\sin\theta - \cos\theta + 1}{\sin\theta + \cos\theta - 1} = \dfrac{1}{\sec\theta - \tan\theta}$

SOLVED EXAMPLE 3

.

SOLUTION

1. Divide numerator and denominator of LHS by $\cos\theta$: LHS $= \dfrac{\tan\theta - 1 + \sec\theta}{\tan\theta + 1 - \sec\theta}$.
2. Write the 1 in the numerator as $\sec^2\theta - \tan^2\theta$: numerator $= (\tan\theta + \sec\theta) - (\sec^2\theta - \tan^2\theta) = (\sec\theta + \tan\theta)(1 - \sec\theta + \tan\theta)$.
3. So LHS $= \dfrac{(\sec\theta + \tan\theta)(\tan\theta - \sec\theta + 1)}{\tan\theta - \sec\theta + 1} = \sec\theta + \tan\theta$.
4. Since $(\sec\theta + \tan\theta)(\sec\theta - \tan\theta) = 1$, $\sec\theta + \tan\theta = \dfrac{1}{\sec\theta - \tan\theta} =$ RHS.

Answer: Hence proved.

SOLVED EXAMPLE 4

If $\sec\theta + \tan\theta = p$, find $\sec\theta$ and $\tan\theta$ in terms of p.

SOLUTION

1. $(\sec\theta + \tan\theta)(\sec\theta - \tan\theta) = 1$, so $\sec\theta - \tan\theta = \frac{1}{p}$.
2. Adding: $2\sec\theta = p + \frac{1}{p} \Rightarrow \sec\theta = \dfrac{p^2+1}{2p}$.
3. Subtracting: $2\tan\theta = p - \frac{1}{p} \Rightarrow \tan\theta = \dfrac{p^2-1}{2p}$.

Answer: $\sec\theta = \frac{p^2+1}{2p}$, $\tan\theta = \frac{p^2-1}{2p}$



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Identity prove karte waqt dono side pe ek saath kaam karke beech mein 'milana' risky hai, aur RHS ko LHS mein shift karna toh bilkul galat. Ek side lo, simplify karte jao jab tak doosri side na ban jaaye.



SHIVAM BHAIYA SAYS: BOARD TIP

3 marks ka identity question har saal pakka hai. Har step alag line mein likho aur jahan identity lagayi wahan likh do, jaise '(using $1 + \tan^2 A = \sec^2 A$)'. Last line mein 'LHS = RHS, hence proved' zaroor.



EXAM ALERT!

Board paper mein identities se ek 3 marker ya 5 marker 'prove that' lagbhag har saal aata hai, aur 'sec θ + tan θ = p' type sum bhi favourite hai.

MEME SAMJHO — Astra meme, maths inside



"KYA TAPLEEK HAI AAPKO?" REACTION, AN IDENTITY QUESTION

Student ne $\sin^2\theta$ ko $\sin(\theta^2)$ likh diya. Identity: "Kya tapleek hai aapko?"

Asli baat: $\sin^2\theta$ means $(\sin\theta)^2$, the square of the ratio. $\sin(\theta^2)$ is a completely different thing.

Day 26 Practice

17 board-pattern questions | solve first, then check

MCQ

1 $(1 + \tan^2 A)(1 - \sin^2 A)$ is equal to (1)

(a) $\sec^2 A$ (b) 0 (c) $\sin^2 A$ (d) 1

MCQ

2 If $\operatorname{cosec}\theta - \cot\theta = \frac{1}{4}$, then $\operatorname{cosec}\theta + \cot\theta$ is (1)

(a) $\frac{1}{4}$ (b) 4 (c) $\frac{15}{4}$ (d) 2

MCQ

3 $9\sec^2 A - 9\tan^2 A$ equals (1)

(a) 1 (b) 9 (c) 0 (d) -9

MCQ

4 If $\sin\theta + \cos\theta = \sqrt{2}$ for an acute angle θ, then $\sin^4\theta + \cos^4\theta$ equals (1)

(a) 1 (b) $\frac{1}{4}$ (c) $\frac{1}{2}$ (d) $\sqrt{2}$

MCQ

5 $\dfrac{1 - \cos A}{\sin A}$ is equal to (1)

(a) $\dfrac{\sin A}{1 + \cos A}$ (b) $\dfrac{1 + \cos A}{\sin A}$

(c) $\dfrac{\sin A}{1 - \cos A}$ (d) $\dfrac{\cos A}{1 + \sin A}$

ASSERTION-REASON

6 **Assertion (A):** If $x = 3\sin^2\theta$ and $y = 3\cos^2\theta + 2$, then $x + y = 5$.

Reason (R): $\sin^2\theta + \cos^2\theta = 1$ for every angle θ. (1)

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

ASSERTION-REASON

7 **Assertion (A):** $\sin(A+B) = \sin A + \sin B$ **for all acute angles A and B.**
Reason (R): **For every acute angle A,** $1 + \tan^2 A = \sec^2 A$. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

VERY SHORT ANSWER

8 **Simplify** $\dfrac{1+\cot^2 A}{1+\tan^2 A}$. (1)

CASE-BASED

9 **Tower and its shadow. A vertical tower PQ stands on level ground with its foot at Q. A rope is tied from the top P to a peg R on the ground.** $PQ = 24$ **m,** $QR = 7$ **m and** $\angle PQR = 90°$. (4)

(i) Find $\cos R$. (1)

(ii) Find $\cot P$. (1)

(iii) Find $\operatorname{cosec} P - \cot P$ and verify that $1 + \cot^2 P = \operatorname{cosec}^2 P$. (2)

OR (iii) Verify that $\sec^2 R - \tan^2 R = 1$. (2)

SHORT ANSWER

10 **Prove that** $\tan^4\theta + \tan^2\theta = \sec^4\theta - \sec^2\theta$. (2)

SHORT ANSWER

11 **If** $5\sin^2 A + 2\cos^2 A = 3$ **for an acute angle A, show that** $\tan A = \frac{1}{\sqrt{2}}$. (2)

SHORT ANSWER

12 **Prove that** $\dfrac{\tan\theta}{1-\cot\theta} + \dfrac{\cot\theta}{1-\tan\theta} = 1 + \sec\theta\operatorname{cosec}\theta$. (3)

SHORT ANSWER

13 **Prove that** $\dfrac{\cot A + \operatorname{cosec} A - 1}{\cot A - \operatorname{cosec} A + 1} = \dfrac{1+\cos A}{\sin A}$. (3)

SHORT ANSWER

14 **Prove that** $\dfrac{\tan A}{\sec A - 1} + \dfrac{\tan A}{\sec A + 1} = 2\operatorname{cosec} A$. (3)

LONG ANSWER

15 **If** $\operatorname{cosec}\theta + \cot\theta = q$, **prove that** $\cos\theta = \dfrac{q^2-1}{q^2+1}$. (4)

LONG ANSWER

16 **If** $\sin\theta + \cos\theta = p$ **and** $\sec\theta + \operatorname{cosec}\theta = q$, **prove that** $q(p^2-1) = 2p$. (4)

LONG ANSWER

17 **Prove that** $\dfrac{\sin A - 2\sin^3 A}{2\cos^3 A - \cos A} = \tan A$. (4)

Day 26 Answers with worked solutions

1 **(d)**
$= \sec^2 A \times \cos^2 A = 1$.

2 **(b)**
$(\operatorname{cosec}\theta + \cot\theta)(\operatorname{cosec}\theta - \cot\theta) = 1$, so the value is 4.

3 **(b)**
$9(\sec^2 A - \tan^2 A) = 9 \times 1 = 9$.

4 **(c)**
Squaring: $1 + 2\sin\theta\cos\theta = 2 \Rightarrow \sin\theta\cos\theta = \frac{1}{2}$.
$\sin^4\theta + \cos^4\theta = (\sin^2\theta + \cos^2\theta)^2 - 2\sin^2\theta\cos^2\theta = 1 - \frac{1}{2} = \frac{1}{2}$.

5 **(a)**
Multiply by $\frac{1+\cos A}{1+\cos A}$: $\dfrac{1-\cos^2 A}{\sin A(1+\cos A)} = \dfrac{\sin^2 A}{\sin A(1+\cos A)} = \dfrac{\sin A}{1+\cos A}$.

6 **(a)**
$x + y = 3(\sin^2\theta + \cos^2\theta) + 2 = 3 + 2 = 5$, using R. (a)

7 **(d)**
A is false: $\sin(30° + 60°) = 1$ but $\sin 30° + \sin 60° = \frac{1+\sqrt{3}}{2}$.
R is a true identity. (d)

8 **Ans:** $\cot^2 A$
$= \dfrac{\operatorname{cosec}^2 A}{\sec^2 A} = \dfrac{\cos^2 A}{\sin^2 A} = \cot^2 A$.

9 **Ans:** $\frac{7}{25}$
$PR = \sqrt{24^2 + 7^2} = 25$ m; $\cos R = \frac{QR}{PR} = \frac{7}{25}$.
Ans: $\frac{24}{7}$
For $\angle P$: adjacent $= PQ = 24$, opposite $= QR = 7$; $\cot P = \frac{24}{7}$.
Ans: $\frac{1}{7}$; verified
$\operatorname{cosec} P = \frac{25}{7}$, so $\operatorname{cosec} P - \cot P = \frac{1}{7}$.
$1 + \frac{576}{49} = \frac{625}{49} = \left(\frac{25}{7}\right)^2$.
OR: Ans: Verified
$\sec R = \frac{25}{7}$, $\tan R = \frac{24}{7}$.
$\frac{625}{49} - \frac{576}{49} = \frac{49}{49} = 1$.

10 **Ans:** Proof
LHS $= \tan^2\theta(\tan^2\theta + 1) = \tan^2\theta\sec^2\theta$.
$= (\sec^2\theta - 1)\sec^2\theta = \sec^4\theta - \sec^2\theta =$ RHS.

11 **Ans:** Shown
$5\sin^2 A + 2(1 - \sin^2 A) = 3 \Rightarrow 3\sin^2 A = 1 \Rightarrow \sin^2 A = \frac{1}{3}$.
$\cos^2 A = \frac{2}{3}$, so $\tan^2 A = \frac{1/3}{2/3} = \frac{1}{2}$ and $\tan A = \frac{1}{\sqrt{2}}$.

12 **Ans:** Proof

Let $t = \tan\theta$. LHS $= \dfrac{t}{1-\frac{1}{t}} + \dfrac{1/t}{1-t} = \dfrac{t^2}{t-1} - \dfrac{1}{t(t-1)} = \dfrac{t^3-1}{t(t-1)}$.

$= \dfrac{(t-1)(t^2+t+1)}{t(t-1)} = \dfrac{t^2+t+1}{t} = 1 + t + \frac{1}{t}$.

$t + \frac{1}{t} = \dfrac{\sin^2\theta + \cos^2\theta}{\sin\theta\cos\theta} = \sec\theta\,\mathrm{cosec}\,\theta$.

So LHS $= 1 + \sec\theta\,\mathrm{cosec}\,\theta =$ RHS.

13 **Ans:** Proof

Write the 1 in the numerator as $\mathrm{cosec}^2 A - \cot^2 A$.

Numerator $= (\cot A + \mathrm{cosec}\,A) - (\mathrm{cosec}\,A - \cot A)(\mathrm{cosec}\,A + \cot A) = (\mathrm{cosec}\,A + \cot A)(1 - \mathrm{cosec}\,A + \cot A)$.

The second factor equals the denominator, so LHS $= \mathrm{cosec}\,A + \cot A = \dfrac{1}{\sin A} + \dfrac{\cos A}{\sin A} = \dfrac{1+\cos A}{\sin A} =$ RHS.

14 **Ans:** Proof

LHS $= \tan A \cdot \dfrac{(\sec A + 1) + (\sec A - 1)}{\sec^2 A - 1} = \dfrac{2\tan A \sec A}{\tan^2 A} = \dfrac{2\sec A}{\tan A}$.

$= \dfrac{2/\cos A}{\sin A/\cos A} = \dfrac{2}{\sin A} = 2\,\mathrm{cosec}\,A =$ RHS.

15 **Ans:** Proof

$\mathrm{cosec}^2\theta - \cot^2\theta = 1$, so $\mathrm{cosec}\,\theta - \cot\theta = \frac{1}{q}$.

Adding: $\mathrm{cosec}\,\theta = \dfrac{q^2+1}{2q}$. Subtracting: $\cot\theta = \dfrac{q^2-1}{2q}$.

$\cos\theta = \dfrac{\cot\theta}{\mathrm{cosec}\,\theta} = \dfrac{q^2-1}{2q} \times \dfrac{2q}{q^2+1} = \dfrac{q^2-1}{q^2+1}$.

16 **Ans:** Proof

$p^2 - 1 = \sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta - 1 = 2\sin\theta\cos\theta$.

$q = \dfrac{1}{\cos\theta} + \dfrac{1}{\sin\theta} = \dfrac{\sin\theta + \cos\theta}{\sin\theta\cos\theta} = \dfrac{p}{\sin\theta\cos\theta}$.

$q(p^2-1) = \dfrac{p}{\sin\theta\cos\theta} \times 2\sin\theta\cos\theta = 2p$.

17 **Ans:** Proof

LHS $= \dfrac{\sin A(1 - 2\sin^2 A)}{\cos A(2\cos^2 A - 1)}$.

$1 - 2\sin^2 A = 1 - 2(1 - \cos^2 A) = 2\cos^2 A - 1$.

So LHS $= \dfrac{\sin A}{\cos A} = \tan A =$ RHS.

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. $\sin^2A + \cos^2A$ = ? : 1
2. $1 + \tan^2A$ = ? : $\sec^2A$
3. $1 + \cot^2A$ = ? : cosec^2A
4. sec A – tan A if sec A + tan A = 3? : $\frac{1}{3}$
5. sin A in terms of cot A? : $\frac{1}{\sqrt{1+\cot^2 A}}$
6. First move when stuck in a proof? : convert to sin and cos
7. $1 - \sin^2 A$ = ? : $\cos^2 A$

DAY 26 COMPLETE? ☐ Notes read ☐ 17 questions solved ☐ Answers checked ☐ Mistakes noted

Next: Day 27, Some Applications of Trigonometry

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 27

SOME APPLICATIONS OF TRIGONOMETRY | CHAPTER 9 | PART 1 OF 2

Heights and Distances: Angle of Elevation

Some Applications of Trigonometry

Aaj ka target

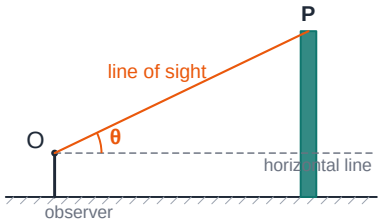
- [ ] Line of sight, horizontal aur angle of elevation clear karo
- [ ] 30°, 45°, 60° ke tan, sin, cos zubaani
- [ ] Har sawal mein pehle figure, phir tan wala ratio
- [ ] 15 sawal khud se, answer baad mein milana

27.1 Line of Sight and Angle of Elevation

In heights and distances we find the height of a tower or the distance of an object **without measuring it directly**. Every problem reduces to one or two **right triangles**, and the trigonometric ratios of 30°, 45° and 60° do the rest.

KEY TERMS

Line of sight: the line drawn from the eye of the observer to the point being viewed.
Horizontal line: the level line through the observer's eye.
Angle of elevation: the angle between the line of sight and the horizontal when the object is **above** the horizontal level (you raise your head).



Angle of elevation θ of the top P, seen from the eye O.

RATIOS YOU WILL USE

$$\tan\theta = \frac{\text{height}}{\text{distance}}$$

$$\sin\theta = \frac{\text{height}}{\text{slant}} \qquad \cos\theta = \frac{\text{distance}}{\text{slant}}$$

Height $= d\tan\theta$, distance $= h\cot\theta$, slant (string, ladder, wire) $= \dfrac{h}{\sin\theta}$

θ	30°	45°	60°
$\sin\theta$	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$
$\cos\theta$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$
$\tan\theta$	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$

Use $\sqrt{3} \approx 1.732$ only when the question asks for a decimal.

27.2 The 4-step method

1. Draw the figure: vertical object, horizontal ground, the right angle at the foot.
2. Mark the given angle at the observer's eye (or at the point on the ground) and write the known length.
3. Pick the ratio that joins the known side and the unknown side (usually $\tan$ for height and distance).
4. Solve, rationalise $\sqrt{3}$ in the denominator, and write the unit.



SHIVAM BHAIYA SAYS
Height aur distance dono diye/maange hain? Seedha **tan**. String, ladder, wire ki length aayi? **sin** (height ke saath) ya **cos** (distance ke saath).

SOLVED EXAMPLE 1

From a point on the ground 40 m away from the foot of a tower, the angle of elevation of its top is 60°. Find the height of the tower.

SOLUTION

1. Let AB be the tower, C the point. $BC = 40$ m, $\angle ACB = 60^\circ$.
2. $\tan 60^\circ = \dfrac{AB}{BC} \Rightarrow \sqrt{3} = \dfrac{AB}{40}$
3. $AB = 40\sqrt{3} \approx 69.28$ m

Answer: $40\sqrt{3}$ m ≈ 69.28 m

SOLVED EXAMPLE 2

A kite is flying at a height of 75 m. Its string, assumed straight, makes an angle of 60° with the ground. Find the length of the string.

SOLUTION

1. Height $= 75$ m is opposite the 60°

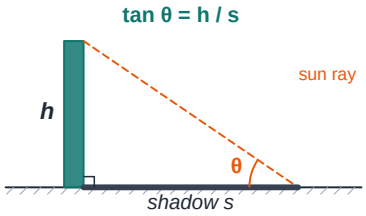
SOLVED EXAMPLE 2

1. angle; the string is the hypotenuse.
2. $\sin 60^\circ = \dfrac{75}{L} \Rightarrow L = \dfrac{75}{\sqrt{3}/2} = \dfrac{150}{\sqrt{3}}$
3. $L = 50\sqrt{3} \approx 86.6$ m

Answer: $50\sqrt{3}$ m

27.3 Shadows and the Sun's Altitude

The **altitude of the sun** is the angle of elevation of the sun. A vertical pole of height h casting a shadow s gives $\tan\theta = \dfrac{h}{s}$. As the sun rises, θ increases and the shadow **shortens**.



Tower of height h with shadow s: tan θ = h / s.

- Shadow = height $\Rightarrow \theta = 45^\circ$
- Height : shadow $= \sqrt{3} : 1 \Rightarrow \theta = 60^\circ$
- Height : shadow $= 1 : \sqrt{3} \Rightarrow \theta = 30^\circ$
- Height and distance both scaled by the same factor: the angle does **not** change.

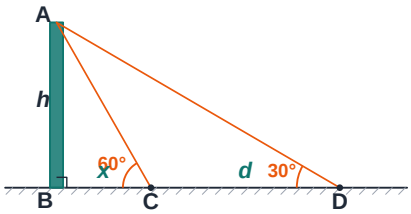
SOLVED EXAMPLE 3 (TWO SHADOWS)

The shadow of a tower is 40 m longer when the sun's altitude is 30° than when it is 60°. Find the height of the tower.

SOLUTION

1. Let height $= h$. Shadow at 60°: $h\cot 60^\circ = \dfrac{h}{\sqrt{3}}$. Shadow at 30°: $h\cot 30^\circ = h\sqrt{3}$.
2. $h\sqrt{3} - \dfrac{h}{\sqrt{3}} = 40 \Rightarrow \dfrac{3h - h}{\sqrt{3}} = 40 \Rightarrow \dfrac{2h}{\sqrt{3}} = 40$
3. $h = 20\sqrt{3} \approx 34.64$ m

Answer: $20\sqrt{3}$ m



Two observation points C and D on the same side of the tower AB.

SOLVED EXAMPLE 4 (BROKEN TREE)

A tree breaks due to a storm; the broken part bends so that its top touches the ground 12 m from the foot, making an angle of 30° with the ground. Find the original height of the tree.

SOLUTION

1. Standing part AB, broken part AC (hypotenuse), $BC = 12$ m, $\angle ACB = 30^\circ$.
2. $AB = 12\tan 30^\circ = \dfrac{12}{\sqrt{3}} = 4\sqrt{3}$ m
3. $AC = \dfrac{12}{\cos 30^\circ} = \dfrac{24}{\sqrt{3}} = 8\sqrt{3}$ m
4. Height $= AB + AC = 12\sqrt{3} \approx 20.78$ m

Answer: $12\sqrt{3}$ m ≈ 20.78 m



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Broken tree mein sabse common galti: sirf standing part likh ke ruk jaana. Original height = **standing part + broken part**, dono jodo.



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Ladder ka angle **wall ke saath** diya hai ya **ground ke saath**, dhyaan se padho. 60° with the wall matlab 30° with the ground.



SHIVAM BHAIYA SAYS: BOARD TIP

Observer ki height (jaise 1.5 m) di ho to tower ki height mein woh **add** karo, aur angle eye level se naapo. Figure mein eye level ki dotted line zaroor kheecho, step marks milte hain.



EXAM ALERT!

Heights and Distances se board mein har saal ek **4-5 mark** ka long answer ya case study aata hai. Figure ke bina answer adhoora maana jaata hai.

MEME SAMJHO — Astra meme, maths inside



STUDENT WHO HAS LEARNT THE 30-45-60 TABLE

Tower ki height naapne upar chadhna kyon? Neeche khade raho, tan laga do.

Asli baat: height $= d\tan\theta$. Only the horizontal distance and the angle of elevation are needed.

Day 27 Practice

16 board-pattern questions | solve first, then check

MCQ

1 A point on the ground is $12\sqrt{3}$ m from the foot of a 12 m high pole. The angle of elevation of the top of the pole from that point is (1)

(a) 60° **(b)** 45° **(c)** 30° **(d)** 90°

MCQ

2 The length of the shadow of a vertical pole is $\sqrt{3}$ times its height. The sun's altitude is (1)

(a) 45° **(b)** 30° **(c)** 60° **(d)** 15°

MCQ

3 A kite is at a height of 40 m and its taut string is 80 m long. The angle the string makes with the ground is (1)

(a) 60° **(b)** 45° **(c)** 90° **(d)** 30°

MCQ

4 A 14 m long ladder leans against a wall, making an angle of 60° **with the wall**. How high up the wall does it reach? (1)

(a) $7\sqrt{3}$ m **(b)** 7 m **(c)** 14 m **(d)** $\frac{7}{\sqrt{3}}$ m

MCQ

5 If the height of a tower and its distance from the point of observation are both doubled, the angle of elevation of its top (1)

(a) is doubled **(b)** is halved
(c) does not change **(d)** becomes 90°

ASSERTION-REASON

6 **Assertion (A):** A vertical pole 9 m high casts a shadow $3\sqrt{3}$ m long, so the sun's elevation is 60°.

Reason (R): The angle of elevation θ of the sun satisfies $\tan\theta = \frac{\text{height}}{\text{shadow}}$. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

7 **Assertion (A):** As the shadow of a tower gets longer, the sun's angle of elevation increases.

Reason (R): For a tower of height h and shadow s, $\tan\theta = \frac{h}{s}$. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

CASE-BASED

8 Read the case and answer. Riya stands at a point P on level ground, 30 m from the foot B of her school building AB. She measures the angle of elevation of the top A as 30°. A flagpole AF stands on the roof. (4)

(i) Find the height of the building. (1)

(ii) Riya walks 20 m towards the building. Find the new angle of elevation of A. (1)

(iii) From P, the angle of elevation of the top F of the flagpole is 45°. Find the length of the flagpole. (2)

(iii) OR: Find the length of the building's shadow when the sun's altitude is 60°. (2)

VERY SHORT ANSWER

9 An observer 1.6 m tall stands 23.4 m from a tower 25 m high. Find the angle of elevation of the top of the tower from his eye. (1)

SHORT ANSWER

10 A slide in a park is inclined at 30° to the ground and its top is 2.5 m high. Find the length of the slide. (2)

SHORT ANSWER

11 A tree breaks in a storm and its top touches the ground 9 m from its foot, the broken part making 30° with the ground. Find the original height of the tree. (3)

SHORT ANSWER

12 The tops of two poles of heights 18 m and 11 m are joined by a taut wire that makes an angle of 30° with the horizontal. Find the length of the wire. (3)

SHORT ANSWER

13 The shadow of a tower is 30 m longer when the sun's altitude is 30° than when it is 60°. Find the height of the tower. (3)

LONG ANSWER

14 A statue 2 m tall stands on a pedestal. From a point on the ground, the angle of elevation of the top of the statue is 60° and that of the top of the pedestal is 45°. Find the height of the pedestal. (5)

LONG ANSWER

15 The angle of elevation of the top of a hill from the foot of a 40 m tower is 60°, and the angle of elevation of the top of the tower from the foot of the hill is 30°. Find the height of the hill. (4)

LONG ANSWER

16 Two poles of equal height stand on opposite sides of an 80 m wide road. From a point on the road between them, the angles of elevation of their tops are 60° and 30°. Find the height of the poles and the distances of the point from the poles. (5)

Day 27 Answers with worked solutions

1 **(c)**

$\tan\theta = \dfrac{12}{12\sqrt{3}} = \dfrac{1}{\sqrt{3}} \Rightarrow \theta = 30^\circ$

2 **(b)**

$\tan\theta = \dfrac{h}{\sqrt{3}h} = \dfrac{1}{\sqrt{3}} \Rightarrow \theta = 30^\circ$

3 **(d)**

$\sin\theta = \dfrac{40}{80} = \dfrac{1}{2} \Rightarrow \theta = 30^\circ$

4 **(b)**

Angle with the wall is 60°, so height $= 14\cos 60^\circ = 14 \times \dfrac{1}{2} = 7$ m.

5 **(c)**

$\tan\theta = \dfrac{2h}{2d} = \dfrac{h}{d}$, so θ is unchanged.

6 **(a)**

$\tan\theta = \dfrac{9}{3\sqrt{3}} = \sqrt{3} \Rightarrow \theta = 60^\circ$. A is true, R is true and is the reason used.

7 **(d)**

A larger s makes $\tan\theta$ smaller, so θ decreases. A is false, R is true.

8 **Ans:** $10\sqrt{3}$ m

$AB = 30\tan 30^\circ = \dfrac{30}{\sqrt{3}} = 10\sqrt{3}$ m

Ans: 60°

New distance $= 10$ m. $\tan\theta = \dfrac{10\sqrt{3}}{10} = \sqrt{3} \Rightarrow \theta = 60^\circ$

Ans: $30 - 10\sqrt{3} \approx 12.68$ m

$BF = 30\tan 45^\circ = 30$ m

$AF = BF - AB = 30 - 10\sqrt{3} \approx 30 - 17.32 = 12.68$ m

OR: Ans: 10 m

Shadow $= AB\cot 60^\circ = \dfrac{10\sqrt{3}}{\sqrt{3}} = 10$ m

9 **Ans:** 45°

Height above eye level $= 25 - 1.6 = 23.4$ m

$\tan\theta = \dfrac{23.4}{23.4} = 1 \Rightarrow \theta = 45^\circ$

10 **Ans:** 5 m

$\sin 30^\circ = \dfrac{2.5}{L} \Rightarrow L = \dfrac{2.5}{1/2} = 5$ m

11 **Ans:** $9\sqrt{3} \approx 15.59$ m

Standing part $= 9\tan 30^\circ = 3\sqrt{3}$ m

Broken part $= \dfrac{9}{\cos 30^\circ} = \dfrac{18}{\sqrt{3}} = 6\sqrt{3}$ m

Height $= 3\sqrt{3} + 6\sqrt{3} = 9\sqrt{3} \approx 15.59$ m

12 **Ans:** 14 m

Vertical gap $= 18 - 11 = 7$ m, opposite the 30° angle.

$\sin 30^\circ = \dfrac{7}{L} \Rightarrow L = 14$ m

13 **Ans:** $15\sqrt{3} \approx 25.98$ m

$h\cot 30^\circ - h\cot 60^\circ = 30 \Rightarrow h\sqrt{3} - \dfrac{h}{\sqrt{3}} = 30$

$\dfrac{2h}{\sqrt{3}} = 30 \Rightarrow h = 15\sqrt{3} \approx 25.98$ m

14 **Ans:** $(\sqrt{3}+1) \approx 2.73$ m

Let pedestal height $= p$, distance from the point $= x$.

Top of pedestal: $\tan 45^\circ = \dfrac{p}{x} \Rightarrow x = p$

Top of statue: $\tan 60^\circ = \dfrac{p+2}{x} \Rightarrow \sqrt{3}p = p + 2$

$p = \dfrac{2}{\sqrt{3}-1} = \dfrac{2(\sqrt{3}+1)}{2} = \sqrt{3}+1 \approx 2.73$ m

15 **Ans:** 120 m

Let the distance between the feet be d.

From the hill's foot: $\tan 30^\circ = \dfrac{40}{d} \Rightarrow d = 40\sqrt{3}$ m

From the tower's foot: $\tan 60^\circ = \dfrac{H}{d} \Rightarrow H = 40\sqrt{3} \times \sqrt{3} = 120$ m

16 **Ans:** Height $20\sqrt{3} \approx 34.64$ m; distances 20 m and 60 m

Let the height be h and the point be x m from the first pole.

$$\tan 60^\circ = \frac{h}{x} \Rightarrow h = \sqrt{3}x$$

$$\tan 30^\circ = \frac{h}{80 - x} \Rightarrow h = \frac{80 - x}{\sqrt{3}}$$

$$\sqrt{3}x = \frac{80 - x}{\sqrt{3}} \Rightarrow 3x = 80 - x \Rightarrow x = 20 \text{ m}$$

$h = 20\sqrt{3} \approx 34.64$ m; other distance $= 60$ m

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Angle of elevation is measured from : the horizontal, upward to the line of sight
2. Height from distance d and angle θ : $d\tan\theta$
3. String/ladder from height h and angle θ : $\frac{h}{\sin\theta}$
4. Shadow equal to height means sun at : 45°
5. Height : shadow $= \sqrt{3} : 1$ means : 60°
6. Longer shadow means the sun is : lower (smaller angle)
7. Broken tree height : standing part + broken part

DAY 27 COMPLETE? ☐ Notes read ☐ 16 questions solved ☐ Answers checked ☐ Mistakes noted

Next: Day 28, Some Applications of Trigonometry

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 28

SOME APPLICATIONS OF TRIGONOMETRY | CHAPTER 9 | PART 2 OF 2

Angle of Depression and Two-Triangle Problems

Some Applications of Trigonometry

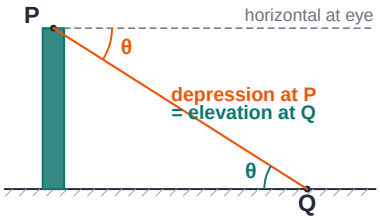
Aaj ka target

- ☐ Angle of depression ko alternate angle se ground pe utaaro
- ☐ Building + tower aur speed wale sawal step by step
- ☐ Same side vs opposite side: cot wala shortcut
- ☐ 16 sawal, figure ke saath, phir answer milao

28.1 Angle of Depression

KEY TERM: ANGLE OF DEPRESSION

The angle between the line of sight and the horizontal when the object is **below** the horizontal level of the eye (you lower your head), e.g. a boat seen from the top of a lighthouse.



Angle of depression at P equals the angle of elevation at Q.

- The horizontal at the eye is **parallel** to the ground, so the angle of depression at P = the angle of elevation of P from Q (**alternate angles**).
- So mark the angle **at the object on the ground** and solve the right triangle exactly as on Day 27.
- The angle of depression is never measured from the vertical tower.

SOLVED EXAMPLE 1

From the top of a 75 m high lighthouse, the angle of depression of a ship is 30°. How far is the ship from the foot of the lighthouse?

SOLUTION

1. Angle of elevation of the top from the ship $= 30^\circ$ (alternate angles).
2. $\tan 30^\circ = \dfrac{75}{d} \Rightarrow d = 75\sqrt{3}$
3. $d \approx 129.9$ m

Answer: $75\sqrt{3} \approx 129.9$ m



SHIVAM BHAIYA SAYS

Depression diya hai? Horizontal line kheecho, angle ko alternate angle se **neeche object pe** le aao. Ab sawal elevation jaisa hi hai.

28.2 Two-Triangle Problems

When two angles are given, there are two right triangles sharing the **same height** h (or the same distance). Write one equation from each triangle and eliminate the common unknown.

SHORTCUTS (TOWER OF HEIGHT H)

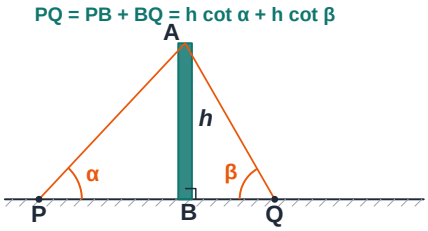
Two points on the **same side**, angles $\alpha < \beta$:

$$d = h(\cot\alpha - \cot\beta)$$

Two points on **opposite sides**:

$$PQ = h(\cot\alpha + \cot\beta)$$

For 30° and 60°: $\cot 30^\circ - \cot 60^\circ = \sqrt{3} - \dfrac{1}{\sqrt{3}} = \dfrac{2}{\sqrt{3}}$



Points P and Q on opposite sides of the tower AB.

SOLVED EXAMPLE 2 (SAME SIDE)

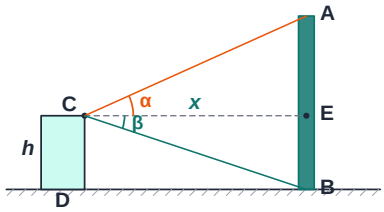
From the top of a 60 m tower, the angles of depression of two cars on the same side of the tower, in line with its foot, are 60° and 30°. Find the distance between the cars.

SOLUTION

1. Near car: $d_1 = 60\cot 60^\circ = \dfrac{60}{\sqrt{3}} = 20\sqrt{3}$ m
2. Far car: $d_2 = 60\cot 30^\circ = 60\sqrt{3}$ m
3. Distance $= 60\sqrt{3} - 20\sqrt{3} = 40\sqrt{3} \approx 69.28$ m

Answer: $40\sqrt{3} \approx 69.28$ m

28.3 Building and Tower



From the top C of building CD: elevation α of A and depression β of B. CE is horizontal, so EB = CD.

SOLVED EXAMPLE 3

From the top of a 12 m high building, the angle of elevation of the top of a tower is 60° and the angle of depression of its foot is 45°. Find the height of the tower.

SOLUTION

1. Draw horizontal CE. $EB = CD = 12$ m.
2. Depression 45°: $\tan 45^\circ = \dfrac{EB}{CE} \Rightarrow CE = 12$ m (the gap between them).
3. Elevation 60°: $AE = CE \tan 60^\circ = 12\sqrt{3}$ m
4. Tower $AB = AE + EB = 12\sqrt{3} + 12 = 12(\sqrt{3}+1) \approx 32.78$ m

Answer: $12(\sqrt{3}+1) \approx 32.78$ m

28.4 Speed and Time Problems

SOLVED EXAMPLE 4

A car moves on a straight road towards a tower. The angle of depression of the car from the top of the tower changes from 30° to 60° in 12 minutes. How much more time will the car take to reach the tower?

SOLUTION

1. Let the tower be h. At 30° the car is $h\sqrt{3}$ away; at 60° it is $\dfrac{h}{\sqrt{3}}$ away.
2. Distance in 12 min $= h\sqrt{3} - \dfrac{h}{\sqrt{3}} = \dfrac{2h}{\sqrt{3}}$
3. Remaining distance $= \dfrac{h}{\sqrt{3}}$, which is half of $\dfrac{2h}{\sqrt{3}}$.
4. At the same speed, time $= \dfrac{12}{2} = 6$ minutes

Answer: 6 minutes



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Speed wale sawal mein **units** pe dhyaan: m/s ko km/h banane ke liye $\times \dfrac{18}{5}$. Seconds aur minutes mix mat karo.



SHIVAM BHAIYA SAYS: BOARD TIP

Cloud aur uski reflection wale sawal mein observer lake se h m upar hai: cloud ki height H ho to upar ka gap $H - h$ aur reflection tak ka gap $H + h$. Yahi do equations banti hain.



SHIVAM BHAIYA SAYS

30° aur 60° ke combination mein near distance aur far distance ka ratio hamesha $1 : 3$ hota hai. Answer check karne ka mast tareeka.



EXAM ALERT!

Board ka favourite: **two angles, one height** (same side ya opposite side), aur speed nikalna. 5 mark ke long answer mein figure + do equations + rationalise, teeno pe marks hain.

MEME SAMJHO — Astra meme, maths inside



STUDENT SEEING THE ANGLE OF DEPRESSION MARKED AT THE BOAT

Upar wala angle neeche kaise aa gaya? Alternate angles, bhai.

Asli baat: The horizontal through the eye is parallel to the ground, so the angle of depression equals the angle of elevation measured at the object.

Day 28 Practice

16 board-pattern questions | solve first, then check

MCQ

1 From the top of a 120 m tower, the angle of depression of a car is 30°. The distance of the car from the foot of the tower is (1)

(a) $40\sqrt{3}$ m **(b)** 120 m **(c)** $120\sqrt{3}$ m **(d)** 60 m

MCQ

2 The angle of depression of a point Q from a point P above it is always equal to (1)

(a) the angle of elevation of P from Q
(b) 90° minus the angle of elevation of P from Q
(c) twice the angle of elevation of P from Q
(d) the angle at P between PQ and the vertical

MCQ

3 From the top of a 21 m building, the angle of elevation of the top of a tower is 30° and the angle of depression of its foot is 45°. The height of the tower is (1)

(a) $7(3+\sqrt{3})$ m **(b)** $21\sqrt{3}$ m
(c) $7(3-\sqrt{3})$ m **(d)** 42 m

MCQ

4 From the top of a 45 m pole, the angle of depression of a bike is 60°. Its distance from the foot of the pole is (1)

(a) $45\sqrt{3}$ m **(b)** $15\sqrt{3}$ m **(c)** 15 m **(d)** 90 m

MCQ

5 An observer on top of a 50 m cliff sees a boat at an angle of depression of 45°. The boat is at a distance of (1)

(a) $50\sqrt{3}$ m **(b)** $\frac{50}{\sqrt{3}}$ m **(c)** 25 m **(d)** 50 m

ASSERTION-REASON

6 **Assertion (A):** From the top of a 30 m tower the angle of depression of a point on the ground is 30°; the point is $30\sqrt{3}$ m from the foot.
Reason (R): The angle of depression of a point from the top equals the angle of elevation of the top from that point. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

7 **Assertion (A):** A man standing 3 m above a lake sees a cloud at an angle of elevation equal to the angle of depression of its reflection in the lake.
Reason (R): The reflection of the cloud is as far below the water surface as the cloud is above it. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

CASE-BASED

8 Read the case and answer. A footbridge over a river is 20 m above the water. From a point P on the bridge, the angles of depression of the two banks A and B, on opposite sides, are 60° and 30° respectively. Let O be the point on the water surface directly below P. (4)

(i) Find the distance OA. (1)

(ii) Find the distance OB. (1)

(iii) Find the width AB of the river. (2)

(iii) OR: Find the length of the line of sight PB. (2)

VERY SHORT ANSWER

9 From the top of a 64 m tower, the angle of depression of a car is 45°. How far is the car from the tower? (1)

SHORT ANSWER

10 From a window 8 m above the ground, the angle of depression of a point P on the ground is 30°. From P, the angle of elevation of the top of the building opposite the window, on the same vertical line, is 60°. Find the height of the building. (3)

SHORT ANSWER

11 Two men are on opposite sides of a 60 m high building. They see its top at angles of elevation 30° and 60°. Find the distance between the men. (3)

SHORT ANSWER

12 From the top of a 120 m cliff, the angle of depression of a boat changes from 60° to 45° in 2 minutes as it sails away. Find the speed of the boat in m/min. (Use $\sqrt{3}=1.732$) (3)

LONG ANSWER

13 From a point 20 m above a lake, the angle of elevation of a cloud is 30° and the angle of depression of its reflection in the lake is 60°. Find the height of the cloud above the lake. (5)

LONG ANSWER

14 An aeroplane flying horizontally at a height of $1200\sqrt{3}$ m is seen at an angle of elevation of 60°. After 10 seconds the angle becomes 30°. Find the speed of the plane in km/h. (5)

LONG ANSWER

15 From the top of a tall building, the angles of depression of the top and the bottom of a 10 m high building opposite are 30° and 45°. Find the height of the tall building. (4)

LONG ANSWER

16 **A 1.5 m tall girl sees a balloon, moving horizontally at 61.5 m above the ground, at an angle of elevation of 60°. Some time later the angle is 30°. Find the distance the balloon travelled.** (5)

Day 28 Answers with worked solutions

1 **(c)**

$\tan 30^\circ = \dfrac{120}{d} \Rightarrow d = 120\sqrt{3} \approx 207.85$ m

2 **(a)**

The horizontal at P is parallel to the ground, so the two angles are alternate angles and are equal.

3 **(a)**

Depression 45°: gap between them $= 21$ m.

Part above eye level $= 21\tan 30^\circ = 7\sqrt{3}$ m

Tower $= 21 + 7\sqrt{3} = 7(3+\sqrt{3})$ m

4 **(b)**

$d = 45\cot 60^\circ = \dfrac{45}{\sqrt{3}} = 15\sqrt{3}$ m

5 **(d)**

$\tan 45^\circ = 1 = \dfrac{50}{d} \Rightarrow d = 50$ m

6 **(a)**

Elevation from the point is 30°, so $d = 30\cot 30^\circ = 30\sqrt{3}$ m. A is true and R is the reason used.

7 **(d)**

Because the eye is 3 m above the water, the vertical gaps to the cloud $(H-3)$ and to its reflection $(H+3)$ differ, so the two angles are not equal. A is false; R is true.

8 **Ans:** $\dfrac{20}{\sqrt{3}} = \dfrac{20\sqrt{3}}{3} \approx 11.55$ m

$OA = 20\cot 60^\circ = \dfrac{20}{\sqrt{3}}$ m

Ans: $20\sqrt{3} \approx 34.64$ m

$OB = 20\cot 30^\circ = 20\sqrt{3}$ m

Ans: $\dfrac{80\sqrt{3}}{3} \approx 46.19$ m

$AB = OA + OB = \dfrac{20}{\sqrt{3}} + 20\sqrt{3} = \dfrac{20+60}{\sqrt{3}} = \dfrac{80}{\sqrt{3}}$

$= \dfrac{80\sqrt{3}}{3} \approx 46.19$ m

OR: Ans: 40 m

$\sin 30^\circ = \dfrac{20}{PB} \Rightarrow PB = 40$ m

9 **Ans:** 64 m

$d = 64\cot 45^\circ = 64$ m

10 **Ans:** 24 m

Distance of P from the building $= 8\cot 30^\circ = 8\sqrt{3}$ m

Height $= 8\sqrt{3}\tan 60^\circ = 8\sqrt{3} \times \sqrt{3} = 24$ m

11 **Ans:** $80\sqrt{3} \approx 138.56$ m

Distances: $60\cot 30^\circ = 60\sqrt{3}$ and $60\cot 60^\circ = 20\sqrt{3}$

Total $= 80\sqrt{3} \approx 138.56$ m

12 **Ans:** ≈ 25.36 m/min

At 60°: $d_1 = \dfrac{120}{\sqrt{3}} = 40\sqrt{3} \approx 69.28$ m; at 45°: $d_2 = 120$ m

Distance moved $= 120 - 40\sqrt{3} \approx 50.72$ m

Speed $= \dfrac{50.72}{2} \approx 25.36$ m/min

13 **Ans:** 40 m

Let the cloud be H m above the lake and x m away horizontally.

Elevation: $\tan 30^\circ = \dfrac{H-20}{x} \Rightarrow x = \sqrt{3}(H-20)$

Reflection is H m below the surface: $\tan 60^\circ = \dfrac{H+20}{x} \Rightarrow x = \dfrac{H+20}{\sqrt{3}}$

$3(H-20) = H+20 \Rightarrow 2H = 80 \Rightarrow H = 40$ m

14 **Ans:** 864 km/h

Horizontal distances: $1200\sqrt{3}\cot 60^\circ = 1200$ m and $1200\sqrt{3}\cot 30^\circ = 3600$ m

Distance flown $= 3600 - 1200 = 2400$ m in 10 s

Speed $= 240$ m/s $= 240 \times \dfrac{18}{5} = 864$ km/h

15 **Ans:** $15 + 5\sqrt{3} \approx 23.66$ m

Let the tall building be H and the gap x.

Depression 45° of the bottom: $x = H$.

Depression 30° of the top: $\tan 30^\circ = \dfrac{H-10}{x} = \dfrac{H-10}{H}$

$H = \sqrt{3}(H-10) \Rightarrow H(\sqrt{3}-1) = 10\sqrt{3} \Rightarrow H = \dfrac{10\sqrt{3}}{\sqrt{3}-1}$

$H = \dfrac{10\sqrt{3}(\sqrt{3}+1)}{2} = 15 + 5\sqrt{3} \approx 23.66$ m

16 **Ans:** $40\sqrt{3} \approx 69.28$ m

Height above her eye $= 61.5 - 1.5 = 60$ m

Horizontal distances: $60\cot 60^\circ = 20\sqrt{3}$ and $60\cot 30^\circ = 60\sqrt{3}$

Distance travelled $= 60\sqrt{3} - 20\sqrt{3} = 40\sqrt{3} \approx 69.28$ m

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Angle of depression is measured from : the horizontal at the eye, downward
2. Depression at P equals : elevation of P seen from the object (alternate angles)

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Same side, angles $\alpha < \beta$: $d = h(\cot\alpha - \cot\beta)$
2. Opposite sides : $PQ = h(\cot\alpha + \cot\beta)$
3. m/s to km/h : multiply by $\frac{18}{5}$
4. Cloud-reflection : gaps are $H - h$ and $H + h$
5. Building and tower, depression 45° of the foot : gap = building height

DAY 28 COMPLETE?

- [ ] Notes read
- [ ] 16 questions solved
- [ ] Answers checked
- [ ] Mistakes noted

Next: Day 29, Revision 5: Trigonometry (Chapters 8 and 9)

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 29

REVISION 5 | CHAPTERS 8 AND 9

Trigonometry (Chapters 8 and 9)

Revision 5

Aaj ka target

- [ ] Ratios, values table aur 3 identities ek baar likh ke dekho
- [ ] Identity proofs: ek side se shuru, dusri tak pahuncho
- [ ] Heights and distances: figure + tan, do angle wale pakke
- [ ] 20 sawal mixed, timer laga ke 60 min

29.1 Chapter 8 in one box

RATIOS AND IDENTITIES

$$\sin\theta = \frac{P}{H},\ \cos\theta = \frac{B}{H},\ \tan\theta = \frac{P}{B} = \frac{\sin\theta}{\cos\theta}$$

$$\csc\theta = \frac{1}{\sin\theta},\ \sec\theta = \frac{1}{\cos\theta},\ \cot\theta = \frac{1}{\tan\theta}$$

$$\sin^2\theta + \cos^2\theta = 1,\quad 1 + \tan^2\theta = \sec^2\theta,\quad 1$$

θ	0°	30°	45°	60°	90°
sin	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	not defined

Standard values (sin increases, cos decreases from 0° to 90°).



SHIVAM BHAIYA SAYS

Proof mein atak gaye? Sab kuch **sin aur cos** mein badlo, LCM lo, phir $\sin^2 + \cos^2 = 1$ lagao. 90% proofs yahin khul jaate hain.



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

$\sin^2\theta$ ka matlab $(\sin\theta)^2$ hai, $\sin(\theta^2)$ nahi. Aur $\sin(A+B) \neq \sin A + \sin B$.

29.2 Chapter 9 in one box

HEIGHTS AND DISTANCES

Height $= d\tan\theta$, distance $= h\cot\theta$, slant $= \frac{h}{\sin\theta}$

Same side (angles $\alpha < \beta$): $d = h(\cot\alpha - \cot\beta)$;

opposite sides: $d = h(\cot\alpha + \cot\beta)$

Angle of depression at the top = angle of elevation at the object (alternate angles).



SHIVAM BHAIYA SAYS: BOARD TIP

Heights and distances mein figure, given values aur 'Let height = h' likhna mat bhoolo. Final answer mein unit aur $\sqrt{3} = 1.732$ tabhi lagao jab poocha ho.



EXAM ALERT!

Ch 8 + Ch 9 milake board paper mein lagbhag **12 marks**: ek identity proof (2-3 marks), values wala MCQ, aur heights and distances ka 4-5 mark long answer ya case study.

Day 29 Practice

20 board-pattern questions | solve first, then check

MCQ

1 If $\cos A = \frac{3}{5}$, then $\tan A + \cot A$ equals (1)

(a) $\frac{7}{12}$ (b) $\frac{25}{12}$ (c) $\frac{12}{25}$ (d) $\frac{5}{3}$

MCQ

2 If $4\tan A = 3$, the value of $\frac{5\sin A + 2\cos A}{5\sin A - 2\cos A}$ is (1)

(a) $\frac{23}{7}$ (b) $\frac{7}{23}$ (c) 3 (d) $\frac{17}{7}$

MCQ

3 If $\cos A + \cos^2 A = 1$, then $\sin^2 A + \sin^4 A$ equals (1)

(a) 0 (b) 2 (c) 1 (d) $\frac{1}{2}$

MCQ

4 If $\sin\theta - \cos\theta = 0$, then $\sin^4\theta + \cos^4\theta$ is (1)

(a) 1 (b) $\frac{3}{4}$ (c) $\frac{1}{4}$ (d) $\frac{1}{2}$

MCQ

5 A kite's string is 100 m long and makes 30° with the ground. The height of the kite is (1)

(a) $50\sqrt{3}$ m (b) 50 m
(c) 100 m (d) $\frac{100}{\sqrt{3}}$ m

MCQ

6 The ratio of the height of a tower to the length of its shadow is $\sqrt{3} : 1$. The sun's altitude is (1)

(a) 30° (b) 45° (c) 60° (d) 90°

ASSERTION-REASON

7 **Assertion (A):** $\csc\theta - \cot\theta$ and $\csc\theta + \cot\theta$ are reciprocals of each other.
Reason (R): $\csc^2\theta - \cot^2\theta = 1$. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

8 **Assertion (A):** $\cos^4 A - \sin^4 A = 2\cos^2 A - 1$.
Reason (R): $1 + \cos^2 A = \sin^2 A$. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

9 **Assertion (A):** As an observer walks towards a tower, the angle of elevation of its top increases.
Reason (R): For a tower of height h seen from distance d, $\tan\theta = \frac{h}{d}$, which decreases as d decreases. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

VERY SHORT ANSWER

10 Evaluate $\frac{4\tan^2 60^\circ}{1 + \tan^2 30^\circ}$. (1)

VERY SHORT ANSWER

11 If $(1 + \cos A)(1 - \cos A) = \frac{1}{4}$ and A is acute, find $\sec A$. (1)

SHORT ANSWER

12 If $\tan A = \frac{5}{12}$, find $\frac{1}{\sin A} + \frac{1}{\cos A}$. (2)

SHORT ANSWER

13 Prove that $\frac{1}{1 - \sin\theta} + \frac{1}{1 + \sin\theta} = 2\sec^2\theta$. (2)

SHORT ANSWER

14 If $\sec\theta = x + \frac{1}{4x}$, prove that $\sec\theta + \tan\theta = 2x$ or $\frac{1}{2x}$. (3)

SHORT ANSWER

15 From the top of a 30 m high building, the angle of elevation of the top of a lighthouse is 30° and the angle of depression of its foot is 60°. Find the height of the lighthouse and its distance from the building. (3)

SHORT ANSWER

16 An aeroplane flying at 3000 m passes vertically above another. At that instant, from a point on the ground, their angles of elevation are 60° and 45°. Find the vertical distance between them. (3)

LONG ANSWER

17 Prove that $(1 + \cot A - \csc A)(1 + \tan A + \sec A) = 2$. (4)

LONG ANSWER

18 Two ships are on opposite sides of a lighthouse. From its top the angles of depression of the ships are 30° and 45°. If the ships are 200 m apart, find the height of the lighthouse. (5)

LONG ANSWER

19 From a point X on the ground the angle of elevation of the top of a tower is 60°. From a point Y, 30 m vertically above X, the angle of elevation of the top is 45°. Find the height of the tower. (5)

CASE-BASED

20 Read the case and answer. A school builds a straight wheelchair ramp 2 m long that rises to a door 1 m above the ground. (4)

(i) Find the angle the ramp makes with the ground. (1)

(ii) Find the horizontal distance covered by the ramp. (1)

(iii) Find $\tan\theta + \cot\theta$ for this ramp. (2)

(iii) OR: A second ramp at the same angle is 4 m long. How high does it rise? (2)

Day 29 Answers with worked solutions

1 **(b)**

$\sin A = \frac{4}{5}$, $\tan A = \frac{4}{3}$, $\cot A = \frac{3}{4}$

Sum $= \frac{16+9}{12} = \frac{25}{12}$

2 **(a)**

Divide by $\cos A$: $\frac{5\tan A + 2}{5\tan A - 2} = \frac{15/4 + 2}{15/4 - 2} = \frac{23/4}{7/4} = \frac{23}{7}$

3 **(c)**

$\cos A = 1 - \cos^2 A = \sin^2 A$

$\sin^2 A + \sin^4 A = \cos A + \cos^2 A = 1$

4 **(d)**

$\theta = 45°$, so $\sin^4 + \cos^4 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$

5 **(b)**

$h = 100 \sin 30° = 50$ m

6 **(c)**

$\tan\theta = \sqrt{3} \Rightarrow \theta = 60°$

7 **(a)**

$(\csc\theta - \cot\theta)(\csc\theta + \cot\theta) = \csc^2\theta - \cot^2\theta = 1$, so R explains A.

8 **(c)**

$\cos^4 A - \sin^4 A = (\cos^2 A - \sin^2 A)(1) = \cos^2 A - (1 - \cos^2 A) = 2\cos^2 A - 1$: A is true.

R is false: the identity is $\sin^2 A + \cos^2 A = 1$.

9 **(c)**

Smaller d makes $\frac{h}{d}$ larger, so θ increases: A is true.

R wrongly says $\tan\theta$ decreases: R is false.

10 **Ans:** 9

$\frac{4 \times 3}{1 + \frac{1}{3}} = \frac{12}{4/3} = 9$

11 **Ans:** $\frac{2}{\sqrt{3}}$

$1 - \cos^2 A = \sin^2 A = \frac{1}{4} \Rightarrow \sin A = \frac{1}{2} \Rightarrow A = 30°$

$\sec 30° = \frac{2}{\sqrt{3}}$

12 **Ans:** $\frac{221}{60}$

Sides 5, 12, hypotenuse 13: $\sin A = \frac{5}{13}$, $\cos A = \frac{12}{13}$

$\frac{13}{5} + \frac{13}{12} = \frac{156+65}{60} = \frac{221}{60}$

13 **Ans:** Proved

LHS $= \frac{(1+\sin\theta) + (1-\sin\theta)}{1 - \sin^2\theta} = \frac{2}{\cos^2\theta}$

$= 2\sec^2\theta =$ RHS

14 **Ans:** Proved

$\tan^2\theta = \sec^2\theta - 1 = x^2 + \frac{1}{2} + \frac{1}{16x^2} - 1 = \left(x - \frac{1}{4x}\right)^2$

$\tan\theta = \pm\left(x - \frac{1}{4x}\right)$

With $+$: $\sec\theta + \tan\theta = 2x$; with $-$: $\sec\theta + \tan\theta = \frac{1}{2x}$

15 **Ans:** 40 m; $10\sqrt{3} \approx 17.32$ m

Distance $= 30\cot 60° = 10\sqrt{3}$ m

Part above the building's top $= 10\sqrt{3}\tan 30° = 10$ m

Lighthouse $= 30 + 10 = 40$ m

16 **Ans:** $1000(3 - \sqrt{3}) \approx 1268$ m

Horizontal distance of the point $= 3000\cot 60° = 1000\sqrt{3}$ m

Lower plane's height $= 1000\sqrt{3}\tan 45° = 1000\sqrt{3}$ m

Gap $= 3000 - 1000\sqrt{3} = 1000(3 - \sqrt{3}) \approx 1268$ m

17 **Ans:** Proved

In sin and cos:

$\left(\frac{\sin A + \cos A - 1}{\sin A}\right)\left(\frac{\cos A + \sin A + 1}{\cos A}\right)$

$= \frac{(\sin A + \cos A)^2 - 1}{\sin A \cos A} = \frac{1 + 2\sin A\cos A - 1}{\sin A\cos A}$

$= 2 =$ RHS

18 **Ans:** $100(\sqrt{3} - 1) \approx 73.2$ m

Distances: $h\cot 30° = h\sqrt{3}$ and $h\cot 45° = h$

$h\sqrt{3} + h = 200 \Rightarrow h = \frac{200}{\sqrt{3}+1} = 100(\sqrt{3} - 1)$

$h \approx 73.2$ m

19 **Ans:** $15(3 + \sqrt{3}) \approx 70.98$ m

Let the tower be h and the horizontal distance x.

From X: $h = x\sqrt{3}$. From Y: $h - 30 = x$.

$x\sqrt{3} - x = 30 \Rightarrow x = \frac{30}{\sqrt{3}-1} = 15(\sqrt{3}+1)$

$h = 15(\sqrt{3}+1)\sqrt{3} = 15(3 + \sqrt{3}) \approx 70.98$ m

20 **Ans:** $30°$

$\sin\theta = \frac{1}{2} \Rightarrow \theta = 30°$

Ans: $\sqrt{3}$ m

$\sqrt{2^2 - 1^2} = \sqrt{3}$ m

Ans: $\frac{4}{\sqrt{3}}$

$\tan 30° + \cot 30° = \frac{1}{\sqrt{3}} + \sqrt{3} = \frac{4}{\sqrt{3}} = \frac{4\sqrt{3}}{3}$

OR: Ans: 2 m

$h = 4\sin 30° = 2$ m

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. $\sin^2\theta + \cos^2\theta$: 1
2. $\sec^2\theta - \tan^2\theta$: 1
3. $\tan 45^\circ$: 1
4. $\sin 60^\circ$: $\frac{\sqrt{3}}{2}$
5. Height from distance and angle : $d\tan\theta$
6. Depression at top equals : elevation at the object
7. Opposite-side distance : $h(\cot\alpha + \cot\beta)$
8. Proof strategy : convert to sin and cos

DAY 29 COMPLETE?

- [ ] Box revised
- [ ] 20 questions solved
- [ ] Answers checked
- [ ] Mistakes noted

Next: Day 30, Areas Related to Circles

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 30

AREAS RELATED TO CIRCLES | CHAPTER 11 | PART 1 OF 2

Arc Length and Area of a Sector

Areas Related to Circles

Aaj ka target

- ☐ Circle ke 2 formula 2 minute mein revise
- ☐ Arc length aur sector area: bas θ/360 ka khel
- ☐ Clock aur pizza wale sawal pakke
- ☐ 16 sawal, π=22/7 ya 3.14 jo diya ho wahi

30.1 Quick Review: Circumference and Area

CIRCLE (RADIUS R)

$$C = 2\pi r = \pi d \qquad A = \pi r^2$$

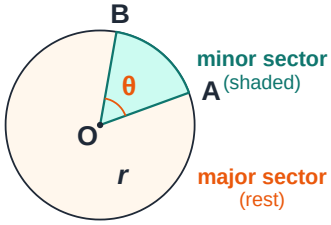
Use $\pi = \frac{22}{7}$ when r is a multiple of 7, else the value the question gives (3.14).

In the 2026-27 syllabus only **sectors and segments** (and figures made from them) are tested in this chapter; the circle formulas above are just the base you need.

30.2 Sector and Arc

KEY TERMS

Arc: a part of the circumference. **Sector**: the region between two radii and the arc they cut off. The smaller one is the **minor sector** (angle θ); the rest of the circle is the **major sector** (angle $360^\circ - \theta$).



Minor sector OAB with angle θ; the unshaded part is the major sector.

SECTOR FORMULAS (ANGLE Θ IN DEGREES)

$$\text{Arc length } l = \frac{\theta}{360^\circ} \times 2\pi r$$

$$\text{Area of sector} = \frac{\theta}{360^\circ} \times \pi r^2 = \frac{1}{2} l\, r$$

Major sector $= \pi r^2 -$ minor sector. Perimeter of a sector $= 2r + l$.



SHIVAM BHAIYA SAYS

Sector = circle ka **fraction**. $90^\circ = \frac{1}{4}$, $60^\circ = \frac{1}{6}$, $45^\circ = \frac{1}{8}$, $120^\circ = \frac{1}{3}$, $30^\circ = \frac{1}{12}$. Pehle poora circle nikalo, phir fraction lagao.

SOLVED EXAMPLE 1

Find the length of the arc and the area of the sector of a circle of radius 14 cm whose angle is 45°.

SOLUTION

1. 45° is $\frac{1}{8}$ of 360°.
2. Arc $= \frac{1}{8} \times 2 \times \frac{22}{7} \times 14 = \frac{88}{8} = 11$ cm
3. Area $= \frac{1}{8} \times \frac{22}{7} \times 14 \times 14 = \frac{616}{8} = 77$ cm^2

Answer: Arc 11 cm, area 77 cm^2

SOLVED EXAMPLE 2 (AREA FROM PERIMETER)

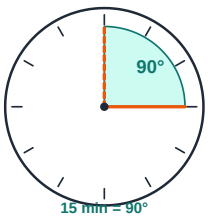
The perimeter of a sector of radius 7.5 cm is 31 cm. Find its area.

SOLUTION

1. Perimeter $= 2r + l \Rightarrow 31 = 15 + l \Rightarrow l = 16$ cm
2. Area $= \frac{1}{2} l\, r = \frac{1}{2} \times 16 \times 7.5 = 60$ cm^2

Answer: 60 cm^2

30.3 Clock Problems



In 15 minutes the minute hand sweeps a quadrant (90°).

- Minute hand turns 360° in 60 min, i.e. 6° **per minute**.
- Hour hand turns 360° in 12 h, i.e. 0.5° **per minute**.
- Area swept = sector with radius = length of the hand; distance moved by the tip = arc length.

SOLVED EXAMPLE 3

The minute hand of a clock is 14 cm long. Find the area swept by it in 5 minutes.

SOLUTION

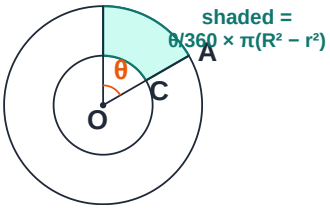
1. Angle $= 5 \times 6^\circ = 30^\circ$
2. Area $= \frac{30}{360} \times \frac{22}{7} \times 14 \times 14 = \frac{616}{12}$

SOLVED EXAMPLE 3

1. $= 51.33$ cm^2 (approx.)

Answer: $\frac{154}{3} \approx 51.33$ cm^2

30.4 Sector of a Ring (Concentric Circles)



Region between two concentric arcs with the same angle θ.

SOLVED EXAMPLE 4

Two concentric circles have radii 21 cm and 10.5 cm. Find the area of the region between them that lies inside a 30° angle at the centre.

SOLUTION

1. Area $= \frac{30}{360} \times \pi\,(R^2 - r^2)$
2. $= \frac{1}{12} \times \frac{22}{7} \times (441 - 110.25) = \frac{1}{12} \times \frac{22}{7} \times 330.75$
3. $= 86.625$ cm^2

Answer: 86.625 cm^2



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Arc length mein $2\pi r$ aur area mein πr^2. Exam mein log arc ke liye πr^2 laga dete hain, unit bhi dekho: arc cm mein, area cm^2 mein.



SHIVAM BHAIYA SAYS: BOARD TIP

Answer mein π ki value wahi lo jo question mein di hai. 3.14 diya hai aur aapne $\frac{22}{7}$ lagaya to answer match nahi karega aur marks kat sakte hain.



EXAM ALERT!

Sector area, arc length aur clock hand wale sawal 1-2 mark mein almost har saal aate hain. Formula box ek baar likh ke hi solve karo.

MEME SAMJHO — Astra meme, maths inside



STUDENT WHO MULTIPLIES BY $\frac{\theta}{360}$ EVERY TIME

Sector ka formula? Bas poore circle ka $\frac{\theta}{360}$ wala hissa. Guarantee wala shortcut.

Asli baat: A sector of angle θ is exactly $\frac{\theta}{360}$ of the circle, so both its arc and its area are that fraction of the full circle's.

Day 30 Practice

16 board-pattern questions | solve first, then check

MCQ

1 The length of an arc of a circle of radius 21 cm subtending 120° at the centre is (1)

(a) 22 cm (b) 44 cm (c) 66 cm (d) 462 cm

MCQ

2 The circumference of a circle is 88 cm. The area of one of its quadrants is (1)

(a) 616 cm² (b) 308 cm² (c) 77 cm² (d) 154 cm²

MCQ

3 A sector of a circle of radius r has arc length l. Its area is (1)

(a) lr (b) $\frac{1}{2}lr$ (c) $2lr$ (d) l^2r

MCQ

4 A sector has arc length 4π cm and area 12π cm^2. The radius of the circle is (1)

(a) 3 cm (b) 6 cm (c) 4 cm (d) 12 cm

MCQ

5 The minute hand of a clock is 21 cm long. The distance moved by its tip in 10 minutes is (1)

(a) 22 cm (b) 11 cm (c) 44 cm (d) 66 cm

ASSERTION-REASON

6 **Assertion (A):** The area of a sector of angle 90° in a circle of radius 7 cm is 38.5 cm^2.
Reason (R): The area of a sector of angle θ is $\frac{\theta}{360^\circ} \times \pi r^2$. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

7 **Assertion (A):** In a circle of radius 9 cm, an arc subtending 40° at the centre has length 2π cm.
Reason (R): The length of an arc of angle θ is $\frac{\theta}{360^\circ} \times \pi r^2$. (1)
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

CASE-BASED

8 Read the case and answer. A circular pizza of diameter 28 cm is cut into 8 equal slices by cuts through the centre. (Use $\pi = \frac{22}{7}$) (4)
(i) What angle does each slice make at the centre? (1)
(ii) Find the length of the crust (arc) of one slice. (1)
(iii) Find the area of the top of one slice. (2)
(iii) OR: Find the perimeter of one slice. (2)

VERY SHORT ANSWER

9 Find the area of a sector of radius 12 cm and angle 30°. (Use $\pi = 3.14$) (1)

SHORT ANSWER

10 A 44 cm long wire is bent into an arc of a circle of radius 42 cm. Find the angle the arc subtends at the centre. (2)

SHORT ANSWER

11 The minute hand of a clock is 10.5 cm long. Find the area swept by it between 7:05 pm and 7:25 pm. (3)

SHORT ANSWER

12 The perimeter of a sector of a circle of radius 14 cm is 50 cm. Find the area of the sector and its angle. (3)

SHORT ANSWER

13 Two concentric circles have radii 7 cm and 3.5 cm. Find the area of the part of the ring that lies between two radii making an angle of 60°. (3)

LONG ANSWER

14 Sector A has radius 6 cm and angle 120°; sector B has radius 18 cm and angle 40°. Compare their arc lengths and their areas (leave answers in terms of π). (4)

LONG ANSWER

15 The hour hand and minute hand of a clock are 6 cm and 8 cm long. Find the total distance travelled by their tips in one day. (Use $\pi = \frac{22}{7}$) (5)

LONG ANSWER

16 A horse is tied with a 7 m rope to a corner of a square field of side 20 m. Find the area it can graze. Find the increase in grazing area if the rope is made 14 m long. (4)

Day 30 Answers with worked solutions

1 **(b)**
$l = \frac{120}{360} \times 2 \times \frac{22}{7} \times 21 = \frac{1}{3} \times 132 = 44$ cm

2 **(d)**
$2\pi r = 88 \Rightarrow r = 14$ cm
Quadrant $= \frac{1}{4} \times \frac{22}{7} \times 196 = 154$ cm^2

3 **(b)**
Area $= \frac{\theta}{360}\pi r^2 = \frac{1}{2}\left(\frac{\theta}{360}2\pi r\right) r = \frac{1}{2}lr$

4 **(b)**
Area $= \frac{1}{2}lr \Rightarrow 12\pi = \frac{1}{2} \times 4\pi \times r \Rightarrow r = 6$ cm

5 **(a)**
Angle $= 10 \times 6^\circ = 60^\circ$
Arc $= \frac{1}{6} \times 2 \times \frac{22}{7} \times 21 = 22$ cm

6 **(a)**
$\frac{1}{4} \times \frac{22}{7} \times 49 = 38.5$ cm^2. A is true and follows from R.

7 **(c)**
$l = \frac{40}{360} \times 2\pi \times 9 = 2\pi$ cm, so A is true.
R gives the sector area, not the arc length, so R is false.

8 **Ans:** 45°
$\frac{360^\circ}{8} = 45^\circ$
Ans: 11 cm
$l = \frac{1}{8} \times 2 \times \frac{22}{7} \times 14 = 11$ cm
Ans: 77 cm^2
Area $= \frac{1}{8} \times \frac{22}{7} \times 14 \times 14 = \frac{616}{8} = 77$ cm^2
OR: Ans: 39 cm
Perimeter $= 2r + l = 28 + 11 = 39$ cm

9 **Ans:** 37.68 cm^2
$\frac{30}{360} \times 3.14 \times 144 = \frac{452.16}{12} = 37.68$ cm^2

10 **Ans:** 60°
$44 = \frac{\theta}{360} \times 2 \times \frac{22}{7} \times 42 = \frac{\theta}{360} \times 264$
$\theta = \frac{44 \times 360}{264} = 60^\circ$

11 **Ans:** 115.5 cm^2

Time $= 20$ min, angle $= 20 \times 6^\circ = 120^\circ$

Area $= \frac{1}{3} \times \frac{22}{7} \times 10.5 \times 10.5 = \frac{1}{3} \times 346.5 = 115.5$ cm^2

12 **Ans:** 154 cm^2; 90°

$l = 50 - 2 \times 14 = 22$ cm

Area $= \frac{1}{2} \times 22 \times 14 = 154$ cm^2

$\theta = \frac{22}{2\pi \times 14} \times 360 = \frac{22}{88} \times 360 = 90^\circ$

13 **Ans:** 19.25 cm^2

Area $= \frac{60}{360} \times \frac{22}{7} \times (49 - 12.25)$

$= \frac{1}{6} \times \frac{22}{7} \times 36.75 = 19.25$ cm^2

14 **Ans:** Arcs equal (4π cm each); areas 12π and 36π cm^2 (B is 3 times A)

Arc A $= \frac{120}{360} \times 2\pi \times 6 = 4\pi$; arc B $= \frac{40}{360} \times 2\pi \times 18 = 4\pi$

Area A $= \frac{1}{3} \times \pi \times 36 = 12\pi$; area B $= \frac{1}{9} \times \pi \times 324 = 36\pi$

Equal arcs, but B's area is 3 times A's (area $= \frac{1}{2}lr$ grows with r).

15 **Ans:** $408\pi \approx 1282.29$ cm

In 24 h the minute hand makes 24 rounds: $24 \times 2\pi \times 8 = 384\pi$ cm

The hour hand makes 2 rounds: $2 \times 2\pi \times 6 = 24\pi$ cm

Total $= 408\pi = 408 \times \frac{22}{7} \approx 1282.29$ cm

16 **Ans:** 38.5 m^2; increase 115.5 m^2

The horse grazes a quadrant: $\frac{1}{4} \times \frac{22}{7} \times 7^2 = 38.5$ m^2

With 14 m: $\frac{1}{4} \times \frac{22}{7} \times 14^2 = 154$ m^2 (still inside the field since $14 < 20$)

Increase $= 154 - 38.5 = 115.5$ m^2

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Arc length : $\frac{\theta}{360} \times 2\pi r$
2. Sector area : $\frac{\theta}{360} \times \pi r^2 = \frac{1}{2}lr$
3. Perimeter of a sector : $2r + l$
4. Minute hand per minute : 6°
5. Hour hand per minute : 0.5°
6. Quadrant means angle : 90°, one-fourth of the circle
7. Ring sector area : $\frac{\theta}{360}\pi(R^2 - r^2)$

DAY 30 COMPLETE? ☐ Notes read ☐ 16 questions solved ☐ Answers checked ☐ Mistakes noted

Next: Day 31, Areas Related to Circles

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 31

AREAS RELATED TO CIRCLES | CHAPTER 11 | PART 2 OF 2

Segments and Combination Figures

Areas Related to Circles

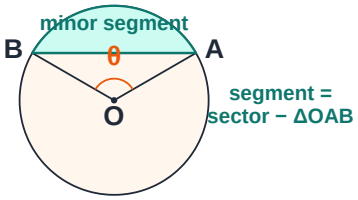
Aaj ka target

- [ ] Segment = sector minus triangle, ek line mein yaad
- [ ] 60°, 90°, 120° wale triangles ka area pakka
- [ ] Shaded region: total minus unshaded ka funda
- [ ] 16 sawal, figure dekh ke plan banao phir solve

31.1 Segment of a Circle

KEY TERM: SEGMENT

The region between a **chord** and its arc. The smaller one is the **minor segment**; the larger one is the **major segment**.



Minor segment = sector OAB minus triangle OAB.

SEGMENT FORMULAS

$$\text{Minor segment} = \frac{\theta}{360^\circ}\pi r^2 - \text{ar}(\triangle OAB)$$

$$\text{Major segment} = \pi r^2 - \text{minor segment}$$

θ	Triangle OAB	Why
60°	$\frac{\sqrt{3}}{4}r^2$	equilateral (OA = OB, angle 60°)
90°	$\frac{1}{2}r^2$	right triangle, legs r, r
120°	$\frac{\sqrt{3}}{4}r^2$	base $r\sqrt{3}$, height $\frac{r}{2}$

Area of triangle OAB for the angles that come in boards.



SHIVAM BHAIYA SAYS

120° wala triangle: centre se chord pe perpendicular daalo. Height $= \frac{r}{2}$, aadha chord $= \frac{\sqrt{3}}{2}r$. Area $= \frac{\sqrt{3}}{4}r^2$, bilkul 60° wale jaisa!

SOLVED EXAMPLE 1 (90°)

A chord of a circle of radius 20 cm subtends a right angle at the centre. Find the areas of the minor and major segments. (Use $\pi = 3.14$)

SOLUTION

1. Sector $= \frac{1}{4} \times 3.14 \times 400 = 314$ cm^2
2. Triangle $= \frac{1}{2} \times 20 \times 20 = 200$ cm^2
3. Minor segment $= 314 - 200 = 114$ cm^2
4. Major segment $= 3.14 \times 400 - 114 = 1256 - 114 = 1142$ cm^2

Answer: 114 cm^2 and 1142 cm^2

SOLVED EXAMPLE 2 (60°)

Find the area of the minor segment of a circle of radius 12 cm cut off by a chord subtending 60° at the centre. (Use $\pi = 3.14$, $\sqrt{3} = 1.73$)

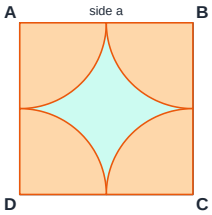
SOLUTION

1. Sector $= \frac{1}{6} \times 3.14 \times 144 = 75.36$ cm^2
2. Triangle is equilateral: $\frac{\sqrt{3}}{4} \times 144 = 36\sqrt{3} = 62.28$ cm^2
3. Minor segment $= 75.36 - 62.28 = 13.08$ cm^2

Answer: 13.08 cm^2

31.2 Combination Figures: the Shaded-Region Method

1. Name every simple shape in the figure: square, rectangle, triangle, sectors, semicircles.
2. Shaded area = (area of the outer shape) minus (areas of the unshaded parts), or the sum of the shaded parts.
3. Look for angle sums: arcs at the 3 vertices of a triangle make 180° (a semicircle); at the 4 vertices of a quadrilateral, 360° (a full circle).



Four quadrants of radius a/2 at the corners of a square of side a.

SOLVED EXAMPLE 3

From each corner of a square of side 14 cm, a quadrant of radius 7 cm is cut off. Find the area of the remaining part.

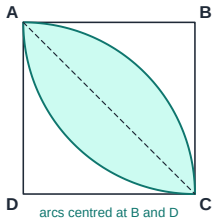
SOLUTION

1. Four quadrants of radius 7 = one full circle: $\frac{22}{7}$

SOLVED EXAMPLE 3

1. $\times 49 = 154$ cm^2
2. Square $= 14^2 = 196$ cm^2
3. Remaining $= 196 - 154 = 42$ cm^2

Answer: 42 cm^2



Leaf (lens) shape made by two quadrant arcs inside a square.

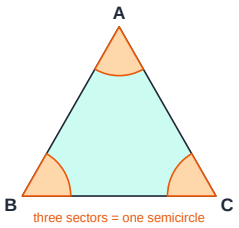
SOLVED EXAMPLE 4 (LEAF DESIGN)

In a square ABCD of side 14 cm, quadrants are drawn with centres B and D, each of radius 14 cm. Find the area of the leaf-shaped region common to both.

SOLUTION

1. Each half of the leaf is a segment of a 90° sector of radius 14.
2. One segment $= \frac{1}{4} \times \frac{22}{7} \times 196 - \frac{1}{2} \times 14 \times 14 = 154 - 98 = 56$ cm^2
3. Leaf $= 2 \times 56 = 112$ cm^2

Answer: 112 cm^2



Sectors at the three vertices of a triangle add up to a semicircle.

SOLVED EXAMPLE 5

An equilateral triangle has side 12 cm. With each vertex as centre, a sector of radius 6 cm is cut away. Find the area left. (Use $\pi = 3.14$, $\sqrt{3} = 1.73$)

SOLUTION

1. Triangle $= \frac{\sqrt{3}}{4} \times 144 = 36\sqrt{3} = 62.28$ cm^2
2. Three sectors (60° each, total 180°) $= \frac{1}{2} \times 3.14 \times 36 = 56.52$ cm^2
3. Area left $= 62.28 - 56.52 = 5.76$ cm^2

Answer: 5.76 cm^2



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Segment nikalte waqt triangle ko **subtract** karna bhool jaate hain, ya major segment mein triangle ghata dete hain. Major segment = poora circle minus minor segment.



SHIVAM BHAIYA SAYS: BOARD TIP

Shaded region ke sawal mein pehli line mein plan likho: *Shaded area = area of square minus area of 4 quadrants*. Answer galat bhi ho jaaye to method ke marks milte hain.



EXAM ALERT!

Segment + combination figure ka 3 mark ya 5 mark ka sawal board mein bahut common hai, aksar case study (sprinkler, brooch, garden design) ban ke aata hai.

MEME SAMJHO — Astra meme, maths inside



INTERVIEWER ASKING A TOPPER ABOUT SEGMENTS

Segment ka area kaise nikala? Simple: sector minus triangle. Itna hi.

Asli baat: Minor segment = sector minus the triangle formed by the two radii and the chord; major segment = whole circle minus the minor one.

Day 31 Practice

16 board-pattern questions | solve first, then check

MCQ

1 The area of the minor segment of a circle of radius r cut off by a chord that subtends a right angle at the centre is (1)

(a) $\frac{r^2}{4}(\pi - 2)$ (b) $\frac{\pi r^2}{4}$
(c) $\frac{r^2}{2}(\pi - 1)$ (d) $r^2(\pi - 2)$

MCQ

2 A chord of a circle of radius 8 cm subtends a right angle at the centre. The length of the chord is (1)

(a) 8 cm (b) $8\sqrt{3}$ cm (c) $8\sqrt{2}$ cm (d) 16 cm

MCQ

3 With the three vertices of a triangle as centres, arcs of radius 7 cm are drawn inside it. The total area of the three sectors is (1)

(a) 154 cm² (b) 38.5 cm²
(c) 231 cm² (d) 77 cm²

MCQ

4 With the four vertices of a quadrilateral as centres, sectors of radius 7 cm are drawn inside it. Their total area is (1)

(a) 154 cm² (b) 77 cm² (c) 308 cm² (d) 49 cm²

ASSERTION-REASON

5 **Assertion (A):** In a circle of radius 14 cm, the minor segment cut off by a chord subtending 90° at the centre has area 56 cm^2.
Reason (R): Area of the minor segment = area of the sector minus area of the triangle formed by the radii and the chord. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

6 **Assertion (A):** Arcs of radius 7 cm drawn at the three vertices of any triangle always enclose sectors of total area 77 cm^2.
Reason (R): The sum of the angles of a triangle is 180°. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

CASE-BASED

7 Read the case and answer. A square lawn ABCD has side 40 m. Two sprinklers at opposite corners A and C each water a quarter circle of radius 40 m inside the lawn. (Use $\pi = 3.14$) (4)

(i) Find the area watered by the sprinkler at A. (1)

(ii) Find the area of triangle ABD. (1)

(iii) Find the area watered by both sprinklers. (2)

(iii) OR: Find the perimeter of the region watered by both sprinklers. (2)

VERY SHORT ANSWER

8 A chord of a circle of radius 7 cm subtends 60° at the centre. Find the length of the chord. (1)

SHORT ANSWER

9 Find the area of the minor segment of a circle of radius 14 cm when the angle at the centre is 120°. (Use $\pi = \frac{22}{7}$, $\sqrt{3} = 1.73$) (3)

SHORT ANSWER

10 ABCD is a square of side 7 cm. Quadrants of radius 7 cm are drawn with centres B and D. Find the area of the region common to the two quadrants. (3)

SHORT ANSWER

11 Sectors of radius 5 cm are cut from each corner of an equilateral triangle of side 10 cm. Find the area of the remaining part. (Use $\pi = 3.14$, $\sqrt{3} = 1.73$) (3)

SHORT ANSWER

12 A chord of length $6\sqrt{2}$ cm subtends a right angle at the centre of a circle. Find the difference between the areas of the major and minor segments. (Use $\pi = 3.14$) (3)

LONG ANSWER

13 A chord of a circle of radius 20 cm subtends 60° at the centre. Find the areas of the minor and major segments. (Use $\pi = 3.14$, $\sqrt{3} = 1.73$) (5)

LONG ANSWER

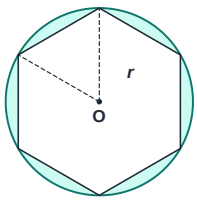
14 A square tile of side 28 cm has four quadrants of radius 14 cm drawn at its corners and a circle of diameter 14 cm at its centre, all painted. Find the unpainted area. (4)

LONG ANSWER

15 A triangular field has sides 13 m, 14 m and 15 m. A goat is tied at each corner with a 3.5 m rope. Find the area of the field the goats cannot graze. (4)

LONG ANSWER

16 A regular hexagon is inscribed in a circle of radius 21 cm. Find the total area of the six segments between the hexagon and the circle. (Use $\sqrt{3} = 1.73$) (5)



Day 31 Answers with worked solutions

1 (a)

$$\frac{\pi r^2}{4} - \frac{r^2}{2} = \frac{r^2}{4}(\pi - 2)$$

2 **(c)**
The triangle is right-angled with legs 8 and 8: chord $= \sqrt{64+64} = 8\sqrt{2}$ cm

3 **(d)**
Angles add to $180°$, so the sectors make a semicircle: $\frac{1}{2} \times \frac{22}{7} \times 49 = 77$ cm^2

4 **(a)**
Angles of a quadrilateral add to $360°$: one full circle $= \frac{22}{7} \times 49 = 154$ cm^2

5 **(a)**
$154 - 98 = 56$ cm^2; A is true and is obtained from R.

6 **(a)**
Total angle $180°$ gives a semicircle of radius 7: 77 cm^2. R explains A.

7 **Ans:** 1256 m^2
$\frac{1}{4} \times 3.14 \times 40^2 = 1256$ m^2
Ans: 800 m^2
$\frac{1}{2} \times 40 \times 40 = 800$ m^2
Ans: 912 m^2
The common region is two segments, each $= 1256 - 800 = 456$ m^2
Common area $= 2 \times 456 = 912$ m^2
OR: Ans: 125.6 m
It is bounded by two quarter arcs of radius 40: $2 \times \frac{1}{4} \times 2 \times 3.14 \times 40 = 125.6$ m

8 **Ans:** 7 cm
OA = OB and the angle is $60°$, so triangle OAB is equilateral: chord $= 7$ cm

9 **Ans:** ≈ 120.56 cm^2
Sector $= \frac{1}{3} \times \frac{22}{7} \times 196 = 205.33$ cm^2
Triangle $= \frac{\sqrt{3}}{4} \times 196 = 49\sqrt{3} = 84.77$ cm^2
Segment $= 205.33 - 84.77 = 120.56$ cm^2

10 **Ans:** 28 cm^2
One segment $= \frac{1}{4} \times \frac{22}{7} \times 49 - \frac{1}{2} \times 7 \times 7 = 38.5 - 24.5 = 14$ cm^2
Common region $= 2 \times 14 = 28$ cm^2

11 **Ans:** 4 cm^2
Triangle $= \frac{\sqrt{3}}{4} \times 100 = 25\sqrt{3} = 43.25$ cm^2
Three sectors $= \frac{1}{2} \times 3.14 \times 25 = 39.25$ cm^2
Remaining $= 43.25 - 39.25 = 4$ cm^2

12 **Ans:** 92.52 cm^2
$r\sqrt{2} = 6\sqrt{2} \Rightarrow r = 6$ cm
Minor $= \frac{1}{4} \times 3.14 \times 36 - \frac{1}{2} \times 36 = 28.26 - 18 = 10.26$ cm^2
Major $= 113.04 - 10.26 = 102.78$ cm^2; difference $= 92.52$ cm^2

13 **Ans:** 36.33 cm^2 and 1219.67 cm^2
Sector $= \frac{1}{6} \times 3.14 \times 400 = 209.33$ cm^2
Triangle (equilateral) $= \frac{\sqrt{3}}{4} \times 400 = 100\sqrt{3} = 173$ cm^2
Minor segment $= 209.33 - 173 = 36.33$ cm^2
Major segment $= 1256 - 36.33 = 1219.67$ cm^2

14 **Ans:** 14 cm^2
Square $= 28^2 = 784$ cm^2
Four quadrants of radius 14 = one circle: $\frac{22}{7} \times 196 = 616$ cm^2
Central circle (r = 7) $= 154$ cm^2
Unpainted $= 784 - 616 - 154 = 14$ cm^2

15 **Ans:** 64.75 m^2
$s = 21$; area $= \sqrt{21 \times 8 \times 7 \times 6} = 84$ m^2 (Heron's formula)
Grazed parts are three sectors with angles adding to $180°$: $\frac{1}{2} \times \frac{22}{7} \times 3.5^2 = 19.25$ m^2
Ungrazed $= 84 - 19.25 = 64.75$ m^2

16 **Ans:** ≈ 241.6 cm^2
The hexagon is six equilateral triangles of side 21: $6 \times \frac{\sqrt{3}}{4} \times 441 = 6 \times 190.73 = 1144.4$ cm^2
Circle $= \frac{22}{7} \times 441 = 1386$ cm^2
Six segments $= 1386 - 1144.4 \approx 241.6$ cm^2

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Minor segment : sector minus triangle OAB
2. Major segment : $\pi r^2 -$ minor segment
3. Triangle OAB for $90°$: $\frac{1}{2}r^2$
4. Triangle OAB for $60°$ or $120°$: $\frac{\sqrt{3}}{4}r^2$
5. Arcs at 3 vertices of a triangle : total = semicircle
6. Arcs at 4 vertices of a quadrilateral : total = full circle
7. Leaf in a square (side a) : $2\left(\frac{\pi a^2}{4} - \frac{a^2}{2}\right)$

DAY 31 COMPLETE? ☐ Notes read ☐ 16 questions solved ☐ Answers checked ☐ Mistakes noted

Next: Day 32, Surface Areas and Volumes

DAY 32

SURFACE AREAS AND VOLUMES | CHAPTER 12 | PART 1 OF 3

Surface Area of Combined Solids

Surface Areas and Volumes

Aaj ka target

- [ ] Cuboid, cube, cylinder, cone, sphere, hemisphere ke CSA aur TSA formulas ratta nahi, samajh ke yaad karo
- [ ] Slant height $l=\sqrt{r^2+h^2}$ har cone sawal mein pehle nikalo
- [ ] Combined solid mein sirf BAHAR dikhne wali surface jodo
- [ ] 16 sawal khud se, phir answers milao (lagbhag 90 min)

32.1 The six basic solids (quick refresh)

Every combined-solid question is built from six basic shapes. Learn the curved (lateral) surface area (CSA) and the total surface area (TSA) of each; the combined solid is just a careful sum of the right pieces.

Solid	CSA / LSA	TSA
Cuboid $l \times b \times h$	$2h(l+b)$	$2(lb+bh+hl)$
Cube, edge a	$4a^2$	$6a^2$
Cylinder r, h	$2\pi rh$	$2\pi r(r+h)$
Cone r, h, slant l	πrl	$\pi r(l+r)$
Sphere r	$4\pi r^2$	$4\pi r^2$
Hemisphere r	$2\pi r^2$	$3\pi r^2$

Surface area formulas (use $\pi = \frac{22}{7}$ unless told 3.14)

MUST-KNOW LINKS

$$l = \sqrt{r^2 + h^2} \quad \text{(cone)}$$

Diagonal of cuboid $= \sqrt{l^2+b^2+h^2}$, diagonal of cube $= \sqrt{3}\,a$

KEY TERM: COMBINED SOLID

A solid made by joining two or more basic solids (for example a cone on a hemisphere, a cone on a cylinder, a cylinder with hemispherical ends). Its surface area is the area of the **outer surface that can be touched**; the faces where the solids are glued together are hidden and are not counted.

32.2 Rule for surface area of a combined solid

1. Draw the solid and name each part (cone, cylinder, hemisphere...).
2. For each part, decide which surfaces are visible: usually a CSA, sometimes a base.
3. Add the visible areas only. Never add the TSA of each part.
4. If something is **scooped out** (a hemisphere or cone hollowed from a solid), its curved surface is now visible and gets ADDED, while the circle it removed gets SUBTRACTED.

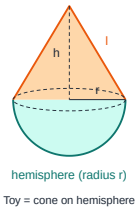
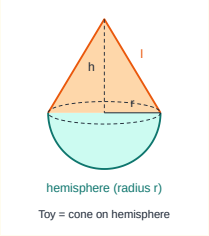


Fig. 32.1 A toy: cone on a hemisphere. TSA = $\pi rl + 2\pi r^2$

SOLVED EXAMPLE 1 (TOY)

A toy is a cone of radius 3.5 cm mounted on a hemisphere of the same radius. The total height of the toy is 15.5 cm. Find its total surface area.



SOLUTION

1. Height of cone $h = 15.5 - 3.5 = 12$ cm.
2. Slant height $l = \sqrt{3.5^2 + 12^2} = \sqrt{12.25 + 144} = \sqrt{156.25} = 12.5$ cm.
3. CSA of cone $= \pi rl = \frac{22}{7} \times 3.5 \times 12.5 = 137.5\ \text{cm}^2$.
4. CSA of hemisphere $= 2\pi r^2 = 2 \times \frac{22}{7} \times 12.25 = 77\ \text{cm}^2$.
5. The flat circles where they join are hidden, so TSA $= 137.5 + 77$.

Answer: TSA of the toy $= 214.5\ \text{cm}^2$

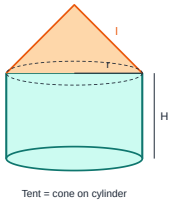


Fig. 32.2 A tent: cone on a cylinder (no floor canvas)

SOLVED EXAMPLE 2 (TENT CANVAS)

A tent is a cylinder of diameter 7 m and height 3 m, surmounted by a cone of the same base with slant height 5 m. Find the canvas used (no floor) and its cost at Rs 500 per m^2.

SOLUTION

1. $r = 3.5$ m. CSA of cylinder $= 2\pi rh = 2 \times \frac{22}{7} \times 3.5 \times 3 = 66\ \text{m}^2$.
2. CSA of cone $= \pi rl = \frac{22}{7} \times 3.5 \times 5 = 55\ \text{m}^2$.
3. Canvas $= 66 + 55 = 121\ \text{m}^2$ (the floor is not covered).
4. Cost $= 121 \times 500$.

Answer: Canvas $= 121\ \text{m}^2$, cost = Rs 60,500



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Sabse common galti: cone ka TSA ($\pi r(l + r)$) + hemisphere ka TSA ($3\pi r^2$) jod dena. Jo circle chipak gaya woh dikhta hi nahi, toh **sirf CSA + CSA**. Pehle diagram banao, phir socho kaunsa face bahar hai.

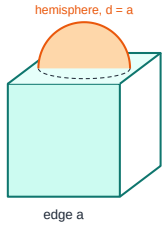


Fig. 32.3 Hemisphere on a cube: one circle hidden, curved dome added

SOLVED EXAMPLE 3 (CUBE WITH DOME)

A cubical block of edge 7 cm is surmounted by a hemisphere of the largest possible diameter. Find the surface area of the solid.

SOLUTION

1. Largest diameter = edge, so $r = 3.5$ cm.
2. Visible = (6 faces of cube) $-$ (circle covered) $+$ (curved dome).
3. $= 6a^2 - \pi r^2 + 2\pi r^2 = 6a^2 + \pi r^2$.
4. $= 6 \times 49 + \frac{22}{7} \times 12.25 = 294 + 38.5$.

Answer: Surface area $= 332.5\ \text{cm}^2$



SHIVAM BHAIYA SAYS

Shortcut yaad rakho: **cube + hemisphere =** $6a^2 + \pi r^2$ (ek circle gaya, dome aaya, net $+\pi r^2$). Aur **capsule (cylinder + do hemispheres) =** $2\pi rL$, jahan L poori length hai. Kyunki $2\pi r(L - 2r) + 4\pi r^2 = 2\pi rL$. MCQ mein 10 second ka kaam!

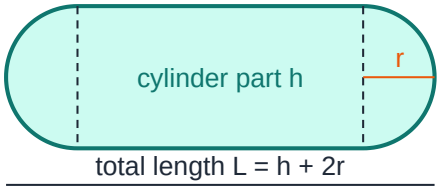


Fig. 32.4 Capsule: cylinder with two hemispherical ends

MEME SAMJHO — Astra meme, maths inside



FRIEND SAYS: JOIN TWO 5 CM CUBES AND THE SURFACE AREA SIMPLY DOUBLES TO 300

Do cube jode toh area seedha double ho jayega? Guarantee deta hai??

Asli baat: Asli baat: Joining hides one face of each cube ($2 \times 25 = 50\ \text{cm}^2$). The new cuboid is $10 \times 5 \times 5$: $2(50 + 25 + 50) = 250\ \text{cm}^2$.

32.3 Scooped-out solids

When a hemisphere (or cone) is hollowed out of a solid, the inside curved surface becomes part of the outside. Example: a cylinder of height h and radius r with a hemisphere scooped from each end has surface area $2\pi rh + 2(2\pi r^2)$: the two flat ends are removed and two hemispherical bowls appear.



EXAM ALERT!

Combined solids se board mein har saal 3 ya 5 marks ka sawal aata hai (tent, toy, bird-bath, wooden article). Units likhna mat bhoolo: area mein cm^2 ya m^2. Aur π kaunsa lena hai, question mein dekh lo.



SHIVAM BHAIYA SAYS: BOARD TIP

Answer sheet mein har part ka naam likh ke formula likho: 'CSA of cone $= \pi rl$'. Agar final calculation galat bhi ho gayi, step marks bach jaate hain.

Day 32 Practice

16 board-pattern questions | solve first, then check

MCQ

1 The total surface area of a solid hemisphere of radius 14 cm is (1)

(a) $1232\ \text{cm}^2$ (b) $1848\ \text{cm}^2$
(c) $2464\ \text{cm}^2$ (d) $616\ \text{cm}^2$

MCQ

2 Two cubes, each of edge 5 cm, are joined end to end. The surface area of the resulting cuboid is (1)

(a) $300\ \text{cm}^2$ (b) $275\ \text{cm}^2$ (c) $250\ \text{cm}^2$ (d) $200\ \text{cm}^2$

MCQ

3 A solid is a cylinder of radius r and height h with a hemisphere of radius r attached at each end. Its total surface area is (1)

(a) $2\pi rh + 3\pi r^2$ (b) $2\pi rh + 4\pi r^2$
(c) $2\pi r(r+h) + 4\pi r^2$ (d) $2\pi rh + 2\pi r^2$

MCQ

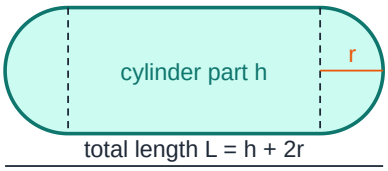
4 A cone and a cylinder have equal base radii and equal curved surface areas. The ratio of the slant height of the cone to the height of the cylinder is (1)

(a) $1:2$ (b) $1:1$ (c) $2:1$ (d) $\sqrt{2}:1$

MCQ

5 A medicine capsule has two hemispherical ends of radius 0.35 cm and is 3 cm long overall. Its surface area is (1)

(a) $6.6\ \text{cm}^2$ (b) $3.3\ \text{cm}^2$ (c) $7.7\ \text{cm}^2$ (d) $5.5\ \text{cm}^2$



ASSERTION-REASON

6 **Assertion (A):** Two cubes of edge 4 cm are joined face to face. The surface area of the new solid is $160\ \text{cm}^2$.
Reason (R): The total surface area of a cuboid of dimensions l, b, h is $2(lb + bh + hl)$. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

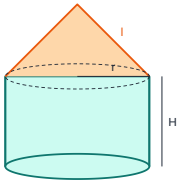
ASSERTION-REASON

7 **Assertion (A):** The total surface area of a top made of a cone on a hemisphere is CSA of hemisphere + CSA of cone.
Reason (R): The flat circular faces where the two solids are joined are not part of the outer surface. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

CASE-BASED

8 Case study: A camping tent A family tent is shaped as a cylinder of diameter 14 ft and height 5 ft with a conical top of the same base. The vertical height of the cone is 24 ft. The floor is not covered with canvas. (Use $\pi = \frac{22}{7}$) (4)



Tent = cone on cylinder

(i) Find the curved surface area of the cylindrical part. (1)

(ii) Find the slant height of the conical part. (1)

(iii) Find the total canvas needed for the tent. (2)

(iii) OR: Find the cost of the canvas at Rs 20 per ft^2. (2)

VERY SHORT ANSWER

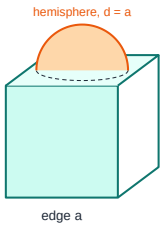
9 Two identical solid hemispheres of radius 7 cm are joined along their bases. Find the total surface area of the new solid. (1)

SHORT ANSWER

10 A toy is a cone of radius 3.5 cm on a hemisphere of the same radius. The curved surface areas of the cone and the hemisphere are equal. Find the ratio of the radius to the height of the cone. (2)

SHORT ANSWER

11 A cubical block of side 10 cm is surmounted by a hemisphere of the largest possible diameter. Find the total surface area of the solid and the cost of painting it at Rs 5 per 100 cm^2. (Use $\pi = 3.14$) (3)

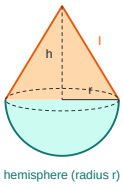


SHORT ANSWER

12 A wooden article is made by scooping out a hemisphere from each end of a solid cylinder of height 10 cm and base radius 3.5 cm. Find the total surface area of the article. (3)

SHORT ANSWER

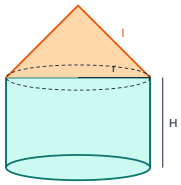
13 A solid toy is a cone of height 4 cm on a hemisphere of radius 3 cm. Find its total surface area in terms of π. (3)



Toy = cone on hemisphere

LONG ANSWER

14 A tent maker supplies 1000 tents for flood relief. Each tent has a cylindrical part of radius 2.8 m and height 3.5 m, topped by a cone of the same radius and height 2.1 m. The canvas costs Rs 100 per m^2, and 40 schools share the cost equally. Find the canvas per tent and each school's share. (5)



Tent = cone on cylinder

LONG ANSWER

15 A bird-bath is a hollow cylinder of height 1.45 m and radius 30 cm with a hemispherical depression at the top. Find its total outer-plus-inner surface area (curved surface of the cylinder and the hemispherical bowl). (4)

MCQ

16 A solid is a cylinder of radius r with a hemisphere of radius r at each end. Its total surface area equals that of a sphere of radius $2r$. The height of the cylindrical part is (1)

(a) $5r$ **(b)** $6r$ **(c)** $7r$ **(d)** $4r$

Day 32 Answers with worked solutions

1 **(b)**
TSA $= 3\pi r^2 = 3 \times \frac{22}{7} \times 196 = 1848$ cm^2.

2 **(c)**
Cuboid $10 \times 5 \times 5$: $2(50 + 25 + 50) = 250$ cm^2.

3 **(b)**
Visible: CSA of cylinder $2\pi rh$ + two hemispherical CSAs $2 \times 2\pi r^2$.

4 **(c)**
$\pi rl = 2\pi rh \Rightarrow l = 2h$, so $l : h = 2 : 1$.

5 **(a)**
Capsule SA $= 2\pi r(L - 2r) + 4\pi r^2 = 2\pi rL = 2 \times \frac{22}{7} \times 0.35 \times 3 = 6.6$ cm^2.

6 **(a)**
New cuboid $8 \times 4 \times 4$: $2(32 + 16 + 32) = 160$. A is true, R is the formula used, so R explains A.

7 **(a)**
Both true and R is exactly why only the curved surfaces are added.

8 **Ans:** 220 ft^2
$2\pi rh = 2 \times \frac{22}{7} \times 7 \times 5 = 220$ ft^2.
Ans: 25 ft
$l = \sqrt{7^2 + 24^2} = \sqrt{625} = 25$ ft.
Ans: 770 ft^2
CSA of cone $= \pi rl = \frac{22}{7} \times 7 \times 25 = 550$ ft^2.
Canvas $= 220 + 550 = 770$ ft^2.
OR: Ans: Rs 15,400
Canvas $= 220 + 550 = 770$ ft^2.
Cost $= 770 \times 20 =$ Rs 15,400.

9 **Ans:** 616 cm^2
The result is a sphere: $4\pi r^2 = 4 \times \frac{22}{7} \times 49 = 616$ cm^2 (not $6\pi r^2$).

10 **Ans:** $r : h = 1 : \sqrt{3}$
$\pi rl = 2\pi r^2 \Rightarrow l = 2r$.
$h = \sqrt{l^2 - r^2} = \sqrt{4r^2 - r^2} = \sqrt{3}\,r$.
So $r : h = 1 : \sqrt{3}$ (the value 3.5 is not needed).

11 **Ans:** 678.5 cm^2; Rs 33.93 (about)
$r = 5$ cm. SA $= 6a^2 + \pi r^2 = 600 + 3.14 \times 25 = 600 + 78.5 = 678.5$ cm^2.
Cost $= \frac{678.5}{100} \times 5 =$ Rs 33.925, about Rs 33.93.

12 **Ans:** 374 cm^2
Visible: CSA of cylinder + CSAs of two hemispherical hollows.
$2\pi rh = 2 \times \frac{22}{7} \times 3.5 \times 10 = 220$ cm^2.
$2 \times 2\pi r^2 = 4 \times \frac{22}{7} \times 12.25 = 154$ cm^2.
TSA $= 220 + 154 = 374$ cm^2.

13 **Ans:** 33π cm^2
$l = \sqrt{3^2 + 4^2} = 5$ cm.
TSA $= \pi rl + 2\pi r^2 = 15\pi + 18\pi = 33\pi$ cm^2.

14 **Ans:** 92.4 m^2 per tent; Rs 2,31,000 per school

Slant height $l = \sqrt{2.8^2 + 2.1^2} = \sqrt{12.25} = 3.5$ m.

CSA of cylinder $= 2\pi rh = 2 \times \frac{22}{7} \times 2.8 \times 3.5 = 61.6\ m^2$.

CSA of cone $= \pi rl = \frac{22}{7} \times 2.8 \times 3.5 = 30.8\ m^2$.

Canvas per tent $= 92.4\ m^2$; for 1000 tents $= 92400\ m^2$.

Total cost $= 92400 \times 100 =$ Rs 92,40,000; each school pays $92{,}40{,}000 \div 40 =$ Rs 2,31,000.

15 **Ans:** $33000\ cm^2 = 3.3\ m^2$

Work in cm: $h = 145$, $r = 30$.

Area $= 2\pi rh + 2\pi r^2 = 2\pi r(h + r)$.

$= 2 \times \frac{22}{7} \times 30 \times 175 = 33000\ cm^2$.

$= 3.3\ m^2$.

16 **(b)**

TSA of solid $= 2\pi rh + 4\pi r^2$; sphere of radius $2r$: $16\pi r^2$.

$2\pi rh = 12\pi r^2 \Rightarrow h = 6r$.

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. CSA of cone : πrl, with $l = \sqrt{r^2 + h^2}$
2. TSA of hemisphere : $3\pi r^2$
3. Two cubes of edge a joined : SA $= 10a^2$
4. Cube + largest hemisphere : $6a^2 + \pi r^2$
5. Capsule of length L : SA $= 2\pi rL$
6. Two hemispheres joined : a sphere, $4\pi r^2$
7. Tent canvas : CSA cylinder + CSA cone (no floor)

DAY 32 COMPLETE? ☐ Notes read ☐ 16 questions solved ☐ Answers checked ☐ Mistakes noted

Next: Day 33, Surface Areas and Volumes

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 33

SURFACE AREAS AND VOLUMES | CHAPTER 12 | PART 2 OF 3

Volume of Combined Solids

Surface Areas and Volumes

Aaj ka target

- [ ] Saare volume formulas ek page pe likh ke yaad karo
- [ ] 'Kitne marbles/cones banenge' type: bada volume / chhota volume
- [ ] Joda hua solid: volume JODO; khokhla (cavity): volume GHATAO
- [ ] 16 sawal khud se, phir answers milao (lagbhag 90 min)

33.1 Volume formulas

Solid	Volume
Cuboid	lbh
Cube	a^3
Cylinder	$\pi r^2 h$
Cone	$\frac{1}{3}\pi r^2 h$
Sphere	$\frac{4}{3}\pi r^3$
Hemisphere	$\frac{2}{3}\pi r^3$

Volumes of the basic solids

KEY IDEA: VOLUME ALWAYS ADDS

Unlike surface area, **no part of the volume gets hidden** when solids are joined. Volume of a combined solid = sum of the volumes of its parts. If a part is **removed** (a cavity or hollow), subtract its volume.

RATIOS WORTH REMEMBERING (SAME RADIUS R, HEIGHT $H = R$)

$$V_{\text{cyl}} : V_{\text{cone}} : V_{\text{hemi}} = \pi r^3 : \tfrac{1}{3}\pi r^3 : \tfrac{2}{3}\pi r^3 = 3 : 1 : 2$$

Same base and height: cylinder $= 3\times$ cone

Radius $\times k$: sphere volume $\times k^3$, surface area $\times k^2$

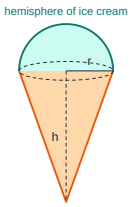


Fig. 33.1 Ice-cream cone: cone + hemisphere

SOLVED EXAMPLE 1 (TOY VOLUME)

A toy is a cone of radius 3.5 cm on a hemisphere of the same radius; the total height is 15.5 cm. Find the volume of the toy.

SOLUTION

1. Cone height $= 15.5 - 3.5 = 12$ cm.
2. $V_{\text{cone}} = \frac{1}{3} \times \frac{22}{7} \times 12.25 \times 12 = 154 \text{ cm}^3$.
3. $V_{\text{hemi}} = \frac{2}{3} \times \frac{22}{7} \times (3.5)^3 = \frac{539}{6} \approx 89.83 \text{ cm}^3$.
4. Total $= 154 + 89.83$.

Answer: Volume $\approx 243.83 \text{ cm}^3$



SHIVAM BHAIYA SAYS

$\pi = \frac{22}{7}$ aur $r = 3.5$ ya 7 ho toh pehle πr^2 nikal lo: $r = 3.5 \Rightarrow \pi r^2 = 38.5$, $r = 7 \Rightarrow \pi r^2 = 154$. Ye do numbers yaad kar lo, aadhe sawal mental math se ho jaayenge.

SOLVED EXAMPLE 2 (GULAB JAMUN SYRUP)

Each gulab jamun is a cylinder with two hemispherical ends, 5 cm long overall and 2.8 cm in diameter. It contains sugar syrup up to 30% of its volume. How much syrup is in 50 gulab jamuns?

SOLUTION

1. $r = 1.4$ cm; cylindrical part $= 5 - 2 \times 1.4 = 2.2$ cm.
2. $V = \pi r^2(2.2) + \frac{4}{3}\pi r^3 = \frac{22}{7}(1.96 \times 2.2 + \frac{4}{3} \times 2.744)$
3. $= \frac{22}{7}(4.312 + 3.659) \approx 25.05 \text{ cm}^3$ each.
4. Syrup $= 0.3 \times 50 \times 25.05 \approx 375.8 \text{ cm}^3$.

Answer: About 376 cm^3 of syrup



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Capsule ya gulab jamun mein 'length' poori hoti hai. Cylinder ki height $= L - 2r$ lo, poori L nahi. Ye galti karke log 5 cm wale sawal mein 7.8 cm ka solid bana dete hain!

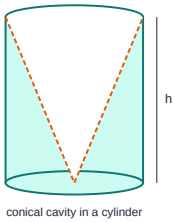


Fig. 33.2 Cavity: subtract the removed volume

SOLVED EXAMPLE 3 (WATER LEFT OVER)

A solid made of a cone (radius 30 cm, height 60 cm) on a hemisphere (radius 30 cm) is placed upright in a right circular cylinder of radius 30 cm and height 90 cm, which was full of water, so that it touches the bottom. Find the water left in the cylinder.

SOLUTION

1. $V_{\text{cyl}} = \pi(30)^2(90) = 81000\pi$.
2. $V_{\text{solid}} = \frac{1}{3}\pi(30)^2(60) + \frac{2}{3}\pi(30)^3 = 18000\pi + 18000\pi = 36000\pi$.
3. Total height of solid $= 60 + 30 = 90$ cm, so it is fully under water.
4. Water left $= 81000\pi - 36000\pi = 45000\pi = 45000 \times \frac{22}{7}$.

Answer: $\approx 141428.6\ \text{cm}^3 \approx 0.1414\ \text{m}^3$

MEME SAMJHO

Astra meme, maths inside



STUDENT LEARNING THAT A CYLINDER HOLDS EXACTLY THREE CONES OF THE SAME BASE AND HEIGHT

Same base, same height, aur cylinder mein poore 3 cone ka paani aa jaata hai? Acha!?

Asli baat: Asli baat: $\pi r^2 h = 3 \times \frac{1}{3}\pi r^2 h$. That is why a cylinder with a cone-shaped cavity of the same base and height keeps $\frac{2}{3}$ of its volume.

33.2 How many pieces? (counting questions)

1. Find the volume of the big container (what is being filled or melted).
2. Find the volume of one small object.
3. Number $= \dfrac{\text{big volume}}{\text{small volume}}$ (if a fraction is left empty, multiply the big volume by the filled fraction first).



EXAM ALERT!

Volume + surface area mila ke 5 marks ka sawal board ka favourite hai (toy, gulab jamun, pen stand, water left). Answer ke saath unit cm^3 likho; litre maanga ho toh $1000\ \text{cm}^3 = 1$ L yaad rakho.



SHIVAM BHAIYA SAYS: BOARD TIP

Jab bhi solid paani mein daalo, check karo ki poora doobta hai ya nahi (solid ki height $\leq$ cylinder ki height). Ek line likh do: 'solid is completely immersed'. Examiner ko ye line pasand hai.

Day 33 Practice

16 board-pattern questions | solve first, then check

MCQ

1 A cylinder, a cone and a hemisphere have the same base radius r and the same height r. The ratio of their volumes is (1)

(a) $1:2:3$ (b) $3:1:2$ (c) $2:1:3$ (d) $3:2:1$

MCQ

2 If the radius of a sphere is tripled, its volume becomes (1)

(a) 3 times (b) 9 times
(c) 6 times (d) 27 times

MCQ

3 A metal sphere of volume Q is dropped into a cuboidal tank whose base area is P and which is partly filled with water. The water level rises by (1)

(a) $\frac{P}{Q}$ (b) PQ (c) $\frac{Q}{P}$ (d) $P+Q$

MCQ

4 Two cones have equal heights and their radii are in the ratio $4:5$. The ratio of their volumes is (1)

(a) $4:5$ (b) $16:25$ (c) $64:125$ (d) $2:\sqrt{5}$

MCQ

5 The volume of the largest sphere that can be carved out of a cube of edge 14 cm is (1)

(a) $1437\frac{1}{3}\ \text{cm}^3$ (b) $2744\ \text{cm}^3$
(c) $11498\frac{2}{3}\ \text{cm}^3$ (d) $718\frac{2}{3}\ \text{cm}^3$

ASSERTION-REASON

6 **Assertion (A):** A hollow cube of internal edge 22 cm can be completely filled with exactly 2541 spherical balls of diameter 2 cm (ignoring gaps).
Reason (R): Number of balls $= \dfrac{\text{volume of cube}}{\text{volume of one ball}}$. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

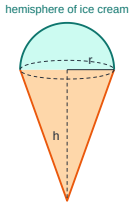
ASSERTION-REASON

7 **Assertion (A):** If the radius of a sphere is halved, its volume becomes one-fourth.
Reason (R): The volume of a sphere is proportional to the cube of its radius. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

CASE-BASED

8 Case study: Ice-cream cones An ice-cream parlour fills cones of radius 3 cm and height 9 cm completely with ice cream and puts a hemisphere of ice cream of the same radius on top. The ice cream comes in a cylindrical tub of radius 6 cm and height 15 cm. (4)



(i) Find the volume of ice cream in the conical part, in terms of π. (1)

(ii) Find the volume of the hemispherical top, in terms of π. (1)

(iii) How many such cones can be filled from one full tub? (2)

(iii) OR: Find the total ice cream in one cone (use $\pi = \frac{22}{7}$). (2)

VERY SHORT ANSWER

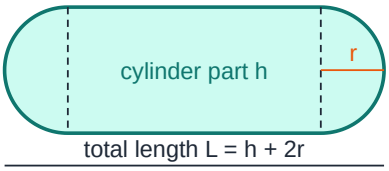
9 The volume and the total surface area of a solid hemisphere are numerically equal. Find its diameter. (2)

SHORT ANSWER

10 The height of a cylinder equals its radius, and its volume is $84\frac{6}{7}\ \text{cm}^3$. Find the height. (2)

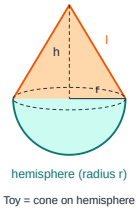
SHORT ANSWER

11 A capsule is 2.1 cm long overall with hemispherical ends of diameter 0.7 cm. Find the medicine it can hold. (2)



SHORT ANSWER

12 A toy is a cone on a hemisphere of diameter 7 cm. The volume of the toy is $308\ \text{cm}^3$. Find the total height of the toy. (3)

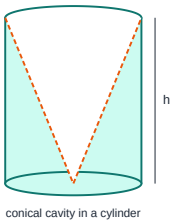


SHORT ANSWER

13 A heap of rice is a cone of diameter 10.5 m and height 3 m. Find the volume of rice and the canvas needed to just cover it. (3)

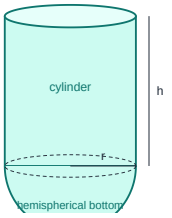
LONG ANSWER

14 From a solid cylinder of height 14 cm and base diameter 12 cm, a cone of the same height and same base is hollowed out. Find the volume and the total surface area of the remaining solid. (5)



LONG ANSWER

15 A vessel is a hollow cylinder with a hollow hemispherical bottom. The diameter is 14 cm and the total height is 13 cm. Find the capacity of the vessel. (4)



MCQ

16 Two cubes, each of volume $64\ \text{cm}^3$, are joined end to end. The surface area of the resulting cuboid is (1)

(a)	(b)	(c)	(d)
$192\ \text{cm}^2$	$128\ \text{cm}^2$	$176\ \text{cm}^2$	$160\ \text{cm}^2$

Day 33 Answers with worked solutions

1 (b)
$\pi r^3 : \frac{1}{3}\pi r^3 : \frac{2}{3}\pi r^3 = 3 : 1 : 2$.

2 (d)
$V \propto r^3$, so $3^3 = 27$ times.

3 (c)
Rise $\times$ base area = volume displaced, so rise $= \frac{Q}{P}$.

4 (b)
$V \propto r^2$ for equal h: $16 : 25$.

5 (a)
$r = 7$ cm. $V = \frac{4}{3} \times \frac{22}{7} \times 343 = \frac{4312}{3} = 1437\frac{1}{3}\ \text{cm}^3$.

6 (a)
$22^3 = 10648$; one ball $= \frac{4}{3} \times \frac{22}{7} \times 1 = \frac{88}{21}$.
$10648 \times \frac{21}{88} = 2541$. Both true, R explains A.

7 (d)
$V \propto r^3$: halving r gives $\frac{1}{8}$, not $\frac{1}{4}$. A false, R true.

8 **Ans:** $27\pi\ \text{cm}^3$
$\frac{1}{3}\pi(3)^2(9) = 27\pi$.
Ans: $18\pi\ \text{cm}^3$
$\frac{2}{3}\pi(3)^3 = 18\pi$.
Ans: 12
One cone uses $27\pi + 18\pi = 45\pi$.
Tub $= \pi(6)^2(15) = 540\pi$.
Number $= \frac{540\pi}{45\pi} = 12$.
OR: Ans: $\approx 141.43\ \text{cm}^3$
$45\pi = 45 \times \frac{22}{7} = \frac{990}{7} \approx 141.43\ \text{cm}^3$.

9 **Ans:** 9 units
$\frac{2}{3}\pi r^3 = 3\pi r^2 \Rightarrow r = \frac{9}{2}$.
Diameter $= 9$ units.

10 **Ans:** 3 cm
$\pi r^2 h = \pi r^3 = \frac{594}{7}$.
$r^3 = \frac{594}{7} \times \frac{7}{22} = 27 \Rightarrow r = h = 3$ cm.

11 **Ans:** $\approx 0.72\ \text{cm}^3$
$r = 0.35$, cylinder part $= 2.1 - 0.7 = 1.4$ cm.
$V = \pi r^2(1.4) + \frac{4}{3}\pi r^3 = \frac{22}{7}(0.1715 + 0.0572) \approx 0.719\ \text{cm}^3$.

12 **Ans:** 20.5 cm
$V_{\text{hemi}} = \frac{2}{3} \times \frac{22}{7} \times (3.5)^3 = \frac{539}{6}$.
$V_{\text{cone}} = 308 - \frac{539}{6} = \frac{1309}{6}$.
$\frac{1}{3} \times 38.5 \times h = \frac{1309}{6} \Rightarrow h = 17$ cm.
Total height $= 17 + 3.5 = 20.5$ cm.

13 **Ans:** $86.625\ \text{m}^3$; about $99.77\ \text{m}^2$
$r = 5.25$. $V = \frac{1}{3} \times \frac{22}{7} \times 27.5625 \times 3 = 86.625\ \text{m}^3$.
$l = \sqrt{5.25^2 + 3^2} = \sqrt{36.5625} \approx 6.047$ m.
Canvas $= \pi r l \approx \frac{22}{7} \times 5.25 \times 6.047 \approx 99.77\ \text{m}^2$.

14 **Ans:** Volume $= 1056\ \text{cm}^3$; TSA $\approx 928.4\ \text{cm}^2$
$r = 6, h = 14$. Remaining volume $= \pi r^2 h - \frac{1}{3}\pi r^2 h = \frac{2}{3} \times \frac{22}{7} \times 36 \times 14 = 1056\ \text{cm}^3$.
Slant height $l = \sqrt{36 + 196} = \sqrt{232} \approx 15.23$ cm.
TSA = CSA of cylinder + top circle + CSA of cone (inside).
$= 2\pi r h + \pi r^2 + \pi r l = 528 + 113.14 + 287.2 \approx 928.4\ \text{cm}^2$.

15 **Ans:** $\approx 1642.67\ \text{cm}^3$
$r = 7$; cylinder height $= 13 - 7 = 6$ cm.
$V = \pi r^2(6) + \frac{2}{3}\pi r^3 = 154 \times 6 + \frac{2}{3} \times \frac{22}{7} \times 343$.
$= 924 + 718.67 = 1642.67\ \text{cm}^3$.

16 (d)
Edge $= 4$ cm; cuboid $8 \times 4 \times 4$: $2(32 + 16 + 32) = 160\ \text{cm}^2$.

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Volume of cone : $\frac{1}{3}\pi r^2 h$
2. Volume of hemisphere : $\frac{2}{3}\pi r^3$
3. Cylinder : cone : hemisphere (same r, $h = r$) : $3 : 1 : 2$
4. Radius doubled : volume 8 times
5. Cavity : subtract its volume
6. Number of balls : big volume / small volume
7. $1\ \text{L} : 1000\ \text{cm}^3$

DAY 33 COMPLETE? ☐ Notes read ☐ 16 questions solved ☐ Answers checked ☐ Mistakes noted

Next: Day 34, Surface Areas and Volumes

MY MISTAKES AND NOTES Write every mistake of this Day with the correct step

DAY 34

SURFACE AREAS AND VOLUMES | CHAPTER 12 | PART 3 OF 3

Mixed Board Problems

Surface Areas and Volumes

Aaj ka target

- [ ] Har sawal mein pehle decide karo: area maanga hai ya volume
- [ ] Ratio wale MCQ formula ke power se 10 second mein
- [ ] Units ek jaise karo (mm, cm, m) calculation se PEHLE
- [ ] 16 board-level sawal, timer lagake (lagbhag 100 min)

34.1 Area or volume? Read the verb

The question talks about	You need
canvas, paint, polish, cloth, sheet, wrapping	Surface area (usually CSA)
capacity, water, milk, metal, wood left, melted	Volume
cost of painting / canvas	Area $\times$ rate
number of objects that fit / can be made	Big volume $\div$ small volume

Decoding the wording

UNITS

$1\text{ m} = 100\text{ cm}$, $1\text{ cm} = 10\text{ mm}$

$1\text{ m}^2 = 10^4\text{ cm}^2$, $1\text{ m}^3 = 10^6\text{ cm}^3$

$1\text{ L} = 1000\text{ cm}^3$, $1\text{ m}^3 = 1000\text{ L}$

CUBOID SHORTCUT

$$(l+b+h)^2 = l^2+b^2+h^2+2(lb+bh+hl)$$

So TSA $= (l+b+h)^2 - (\text{diagonal})^2$

SOLVED EXAMPLE 1 (RATIO GIVEN)

The radius and height of a cone are in the ratio $3:4$ and its volume is 301.44 cm^3. Find its total surface area. (Use $\pi = 3.14$)

SOLUTION

1. Let $r = 3k$, $h = 4k$. Volume $= \frac{1}{3} \times 3.14 \times 9k^2 \times 4k = 37.68k^3$.
2. $37.68k^3 = 301.44 \Rightarrow k^3 = 8 \Rightarrow k = 2$, so $r = 6$, $h = 8$.
3. $l = \sqrt{36+64} = 10$ cm.
4. TSA $= \pi r(l+r) = 3.14 \times 6 \times 16$.

Answer: TSA $= 301.44\text{ cm}^2$ (same number as the volume, a coincidence)

SOLVED EXAMPLE 2 (CSA AS A FRACTION OF TSA)

The total surface area of a solid cylinder is 924 cm^2 and its curved surface area is two-thirds of its total surface area. Find its volume.

SOLUTION

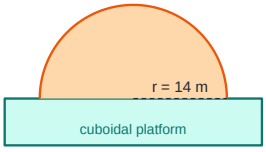
1. $2\pi rh = \frac{2}{3} \times 2\pi r(r+h) \Rightarrow 3h = 2r + 2h \Rightarrow h = 2r$.
2. TSA $= 2\pi r(r+2r) = 6\pi r^2 = 924 \Rightarrow r^2 = 49$, $r = 7$ cm, $h = 14$ cm.
3. $V = \pi r^2 h = \frac{22}{7} \times 49 \times 14$.

Answer: Volume $= 2156\text{ cm}^3$



SHIVAM BHAIYA SAYS

Ratio MCQ ka formula: **length ka ratio k ho toh area ka k^2, volume ka k^3.** Ulta bhi: volume $64:27$ diya hai toh pehle cube root $(4:3)$, phir square $(16:9)$ = surface area ratio.



Stupa: hemisphere on a cuboid

Fig. 34.1 A stupa-like structure: hemisphere on a cuboid

MEME SAMJHO

Astra meme, maths inside



STUDENT IN THE EXAM TRYING TO COPY THE ANSWER FOR THE TENT CANVAS

Bagal wale ne tent ka TSA likha, maine bhi copy kiya... floor bhi jod diya usne aur maine bhi

Asli baat: Asli baat: Canvas of a tent = CSA of cylinder + CSA of cone. The floor is not canvas, so never add πr^2 unless the question says the base is covered.



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Wire ya pipe wale sawal mein diameter mm mein hota hai aur baaki cm mein. Pehle sab cm mein badlo: 3 mm $= 0.3$ cm. Units mix kiye toh answer 10 ya 100 guna galat aayega.



EXAM ALERT!
Case-study mein is chapter ke real-life solids aate hain: stupa, tent, ice cream, coins, cans. (i), (ii) seedhe formula, (iii) 2 marks mein do formula jodna. Pura case 4 marks ka hai, chhodna mat.



SHIVAM BHAIYA SAYS: BOARD TIP
Final answer ke neeche box banao aur unit likho. Agar decimal lamba aa raha hai (99.77...), do decimal tak round karke 'approx' likh do. $\pi = \frac{22}{7}$ wale sawal mein 7 ke multiples dekh ke pehle cancel karo.

Day 34 Practice

15 board-pattern questions | solve first, then check

MCQ

1 The sum of the length, breadth and height of a cuboid is 19 cm and its diagonal is 11 cm. Its total surface area is (1)

(a) $361\ \text{cm}^2$ (b) $121\ \text{cm}^2$ (c) $240\ \text{cm}^2$ (d) $480\ \text{cm}^2$

MCQ

2 A hemisphere of radius r is scooped out from the top of a solid cylinder of radius r and height h. The volume of the remaining solid is (1)

(a) $\frac{\pi r^2}{3}(3h - 2r)$ (b) $\frac{\pi r^2}{3}(3h + 2r)$
(c) $\pi r^2(h - r)$ (d) $\frac{\pi r^2}{3}(2h - 3r)$

MCQ

3 The volume of a sphere is $36\pi\ \text{cm}^3$. Its surface area is (1)

(a) $9\pi\ \text{cm}^2$ (b) $36\pi\ \text{cm}^2$ (c) $27\pi\ \text{cm}^2$ (d) $12\pi\ \text{cm}^2$

MCQ

4 A semicircular sheet of radius 14 cm is bent into a cone. The base radius of the cone is (1)

(a) 14 cm (b) 28 cm (c) 3.5 cm (d) 7 cm

MCQ

5 The ratio of the total surface areas of a solid sphere and a solid hemisphere of the same radius is (1)

(a) $2:1$ (b) $4:3$ (c) $3:2$ (d) $1:1$

ASSERTION-REASON

6 **Assertion (A):** A right triangle with legs 3 cm and 4 cm is revolved about the 4 cm leg; the solid formed has volume $12\pi\ \text{cm}^3$.
Reason (R): Revolving a right triangle about one leg gives a cone whose radius is the other leg and whose height is that leg. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

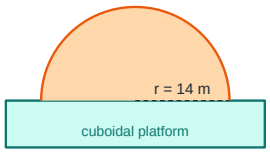
ASSERTION-REASON

7 **Assertion (A):** A sphere whose surface area is $154\ \text{cm}^2$ has diameter 7 cm.
Reason (R): The surface area of a sphere of radius r is $4\pi r^2$. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

CASE-BASED

8 Case study: A stupa A heritage park builds a model stupa: a solid hemispherical dome of radius 14 m resting on a cuboidal platform 30 m long, 30 m wide and 4 m high. (Use $\pi = \frac{22}{7}$) (4)



Stupa: hemisphere on a cuboid

(i) Find the curved surface area of the dome. (1)
(ii) Find the volume of the platform. (1)
(iii) Find the volume of the dome. (2)
(iii) OR: Find the cost of painting the dome at Rs 10 per m^2. (2)

VERY SHORT ANSWER

9 A cone has the same base radius as a cylinder and three times its height. Find the ratio of their volumes. (2)

SHORT ANSWER

10 A copper wire of diameter 3 mm is wound evenly around a cylinder of height 15 cm and radius 7 cm, covering its curved surface completely. Find the length of the wire. (2)

SHORT ANSWER

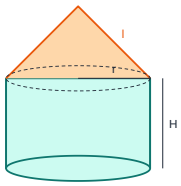
11 The largest possible sphere is carved out of a wooden cube of edge 21 cm. Find the volume of wood left. (3)

SHORT ANSWER

12 Cylindrical cans of radius 7 cm and height 14 cm are packed upright in a cubical box of internal edge 28 cm. Find the capacity of one can in litres, the number of cans that fit, and the empty space in the box. (3)

LONG ANSWER

13 A military tent of height 8.25 m is a cylinder of diameter 30 m and height 5.5 m surmounted by a cone. Find the length of canvas 1.5 m wide needed to make it (no floor). (5)



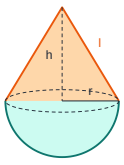
Tent = cone on cylinder

LONG ANSWER

14 A pen stand is a wooden cuboid $15\ \text{cm} \times 10\ \text{cm} \times 3.5\ \text{cm}$ with four conical depressions, each of radius 0.5 cm and depth 1.4 cm, to hold pens. Find the volume of wood in the stand. (4)

LONG ANSWER

15 A solid toy is a hemisphere of diameter 6 cm with a cone of height 3 cm and the same base on it. A right circular cylinder just encloses the toy. Find the difference between the volume of the cylinder and the volume of the toy. (4)



hemisphere (radius r)

Toy = cone on hemisphere

Day 34 Answers with worked solutions

1 **(c)**
TSA $= (l+b+h)^2 - d^2 = 361 - 121 = 240\ \text{cm}^2$.

2 **(a)**
$\pi r^2 h - \frac{2}{3}\pi r^3 = \frac{\pi r^2}{3}(3h - 2r)$.

3 **(b)**
$\frac{4}{3}\pi r^3 = 36\pi \Rightarrow r^3 = 27,\ r = 3$.
SA $= 4\pi(9) = 36\pi\ \text{cm}^2$.

4 **(d)**
Arc length $= \pi(14) =$ base circumference $2\pi r \Rightarrow r = 7$ cm (slant height = 14 cm).

5 **(b)**
$4\pi r^2 : 3\pi r^2 = 4 : 3$.

6 **(a)**
Cone $r = 3,\ h = 4$: $V = \frac{1}{3}\pi(9)(4) = 12\pi$. Both true, R explains A.

7 **(a)**
$4 \times \frac{22}{7}r^2 = 154 \Rightarrow r^2 = 12.25,\ r = 3.5,\ d = 7$ cm.
R is the formula used.

8 **Ans:** $1232\ \text{m}^2$
$2\pi r^2 = 2 \times \frac{22}{7} \times 196 = 1232\ \text{m}^2$.
Ans: $3600\ \text{m}^3$
$30 \times 30 \times 4 = 3600\ \text{m}^3$.
Ans: $\approx 5749.33\ \text{m}^3$
$\frac{2}{3}\pi r^3 = \frac{2}{3} \times \frac{22}{7} \times 2744 = \frac{17248}{3} \approx 5749.33\ \text{m}^3$.
OR: Ans: Rs 12,320
CSA $= 1232\ \text{m}^2$; cost $= 1232 \times 10 =$ Rs 12,320.

9 **Ans:** $1 : 1$
$\frac{1}{3}\pi r^2(3h) : \pi r^2 h = 1 : 1$.

10 **Ans:** 22 m
Number of turns $= \frac{150\ \text{mm}}{3\ \text{mm}} = 50$.
Length of one turn $= 2\pi r = 2 \times \frac{22}{7} \times 7 = 44$ cm.
Length $= 50 \times 44 = 2200$ cm $= 22$ m.

11 **Ans:** $4410\ \text{cm}^3$
$r = 10.5$ cm. Sphere $= \frac{4}{3} \times \frac{22}{7} \times (10.5)^3 = 4851\ \text{cm}^3$.
Cube $= 21^3 = 9261\ \text{cm}^3$.
Wood left $= 9261 - 4851 = 4410\ \text{cm}^3$.

12 **Ans:** 2.156 L; 8 cans; $4704\ \text{cm}^3$
Can $= \frac{22}{7} \times 49 \times 14 = 2156\ \text{cm}^3 = 2.156$ L.
Each can is $14 \times 14 \times 14$ cm in its packing space, so $2 \times 2 \times 2 = 8$ cans fit.
Empty $= 28^3 - 8 \times 2156 = 21952 - 17248 = 4704\ \text{cm}^3$.

13 **Ans:** 825 m
Cone: $r = 15,\ h = 8.25 - 5.5 = 2.75,\ l = \sqrt{225 + 7.5625} = 15.25$ m.
CSA of cylinder $= 2\pi rh = 2 \times \frac{22}{7} \times 15 \times 5.5 \approx 518.57\ \text{m}^2$.
CSA of cone $= \pi rl = \frac{22}{7} \times 15 \times 15.25 \approx 718.93\ \text{m}^2$.
Total $= 1237.5\ \text{m}^2$; length $= \frac{1237.5}{1.5} = 825$ m.

14 **Ans:** $\approx 523.53\ \text{cm}^3$
Cuboid $= 15 \times 10 \times 3.5 = 525\ \text{cm}^3$.
One cone $= \frac{1}{3} \times \frac{22}{7} \times 0.25 \times 1.4 = \frac{11}{30}\ \text{cm}^3$.
Wood $= 525 - 4 \times \frac{11}{30} = 525 - 1.47 \approx 523.53\ \text{cm}^3$.

15 **Ans:** $27\pi \approx 84.86\ \text{cm}^3$

$r = 3$. Cylinder: radius 3, height $3 + 3 = 6$: 54π.

Toy $= \frac{1}{3}\pi(9)(3) + \frac{2}{3}\pi(27) = 9\pi + 18\pi = 27\pi$.

Difference $= 27\pi = \frac{594}{7} \approx 84.86\ \text{cm}^3$.

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Canvas, paint : surface area
2. Water, metal, wood : volume
3. $1\ \text{m}^3$: 1000 L
4. Cuboid TSA from sum and diagonal : $(l + b + h)^2 - d^2$
5. Semicircle of radius R bent into cone : $l = R$, $r = R/2$
6. Volume ratio $64 : 27$: SA ratio $16 : 9$
7. Right triangle revolved about a leg : cone

DAY 34 COMPLETE?

- [ ] Notes read
- [ ] 16 questions solved
- [ ] Answers checked
- [ ] Mistakes noted

Next: Day 35, Revision 6: Mensuration (Chapters 11 and 12)

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 35

REVISION 6 | CHAPTERS 11 AND 12

Mensuration (Chapters 11 and 12)

Revision 6

Aaj ka target

- [ ] Sector, arc, segment ke formulas ek baar likh ke dekho
- [ ] Saare solids ke CSA, TSA, volume ek table mein
- [ ] Area vs volume: question ka verb pehchano
- [ ] 20 mixed sawal timer ke saath (lagbhag 100 min)

35.1 Areas Related to Circles: one-box summary

SECTOR AND SEGMENT (ANGLE AT THE CENTRE)

Arc length $l = \frac{\theta}{360^\circ} \times 2\pi r$

Area of sector $= \frac{\theta}{360^\circ} \times \pi r^2 = \frac{1}{2}lr$

Area of minor segment = area of sector $-$ area of $\triangle OAB$

$\triangle OAB$: $\theta = 60^\circ \Rightarrow \frac{\sqrt{3}}{4}r^2$; $\theta = 90^\circ \Rightarrow \frac{1}{2}r^2$; $\theta = 120^\circ \Rightarrow \frac{\sqrt{3}}{4}r^2$

Major segment = $\pi r^2 -$ minor segment

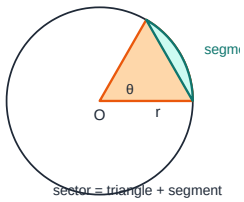


Fig. 35.1 Sector, chord and segment

35.2 Surface Areas and Volumes: one-table summary

Solid	CSA	TSA	Volume
Cuboid	$2h(l+b)$	$2(lb+bh+hl)$	lbh
Cube	$4a^2$	$6a^2$	a^3
Cylinder	$2\pi rh$	$2\pi r(r+h)$	$\pi r^2 h$
Cone	πrl	$\pi r(l+r)$	$\frac{1}{3}\pi r^2 h$
Sphere	$4\pi r^2$	$4\pi r^2$	$\frac{4}{3}\pi r^3$
Hemisphere	$2\pi r^2$	$3\pi r^2$	$\frac{2}{3}\pi r^3$

Every mensuration formula of Chapter 12

- Combined solid, **area**: add only the visible surfaces (joined faces are hidden).
- Combined solid, **volume**: always add; cavity means subtract.
- Scooped hemisphere: add its curved surface, subtract the circle it removed.
- Frustum and conversion of solids are not in the 2026-27 syllabus.

SOLVED EXAMPLE 1 (SEGMENT)

A chord of a circle of radius 21 cm subtends 60° at the centre. Find the area of the minor segment. (Use $\pi = \frac{22}{7}$, $\sqrt{3} = 1.73$)

SOLUTION

1. Sector $= \frac{60}{360} \times \frac{22}{7} \times 441 = 231\ \text{cm}^2$.
2. $\triangle OAB$ is equilateral: $\frac{\sqrt{3}}{4} \times 441 = \frac{1.73 \times 441}{4} \approx 190.73\ \text{cm}^2$.
3. Segment $= 231 - 190.73$.

Answer: $\approx 40.27\ \text{cm}^2$

SOLVED EXAMPLE 2 (COMBINED VOLUME)

A toy is a cylinder of radius 3.5 cm and height 10 cm with a cone of the same radius and height 12 cm on top. Find its volume.

SOLUTION

1. $\pi r^2 = \frac{22}{7} \times 12.25 = 38.5$.
2. Cylinder $= 38.5 \times 10 = 385\ \text{cm}^3$; cone $= \frac{1}{3} \times 38.5 \times 12 = 154\ \text{cm}^3$.
3. Total $= 385 + 154$.

Answer: $539\ \text{cm}^3$



SHIVAM BHAIYA SAYS

$\theta = 90^\circ$ wala sector = circle ka $\frac{1}{4}$, $\theta = 60^\circ$ = $\frac{1}{6}$, $\theta = 120^\circ$ = $\frac{1}{3}$. Fraction pehchano, calculation aadhi ho jaayegi. Ghadi ki minute hand 5 minute mein 30° ghoomti hai = $\frac{1}{12}$.

SHIVAM BHAIYA SAYS: GALTI MAT KARNA



Top 3 galtiyan: (1) segment mein triangle ghatana bhool jaana, (2) combined solid ke area mein joined circle bhi jod dena, (3) diameter diya ho aur usi ko radius maan lena. Har sawal ke start mein 'r = ?' alag se likho.

MEME SAMJHO

Astra meme, maths inside



STUDENT AFTER FINISHING BOTH MENSURATION CHAPTERS IN ONE REVISION DAY

Sector, segment, cone, cylinder, hemisphere... sab ek din mein revise? Presentation is done, ab bas 20 sawal aur!

Asli baat: Asli baat: Chapters 11 and 12 together usually carry about 10 marks in the board paper. One formula table plus 20 mixed questions is the fastest revision.



EXAM ALERT!

Board paper mein Areas + SA&V se milake lagbhag 10 marks aate hain: ek case study (wiper, stupa, tent jaisa) aur ek 3 ya 5 marks ka sawal. π ki value question mein dekh ke hi lo.

Day 35 Practice

18 board-pattern questions | solve first, then check

MCQ

1 The area of a sector of angle 90° in a circle of radius 14 cm is (1)

(a) $308\ \text{cm}^2$ (b) $154\ \text{cm}^2$ (c) $616\ \text{cm}^2$ (d) $77\ \text{cm}^2$

MCQ

2 The length of the arc of a sector of angle 60° in a circle of radius 21 cm is (1)

(a) 44 cm (b) 11 cm (c) 22 cm (d) 66 cm

MCQ

3 The minute hand of a clock is 14 cm long. The area it sweeps in 5 minutes is (1)

(a) $51.33\ \text{cm}^2$ (b) $102.67\ \text{cm}^2$
(c) $25.67\ \text{cm}^2$ (d) $616\ \text{cm}^2$

MCQ

4 The curved surface area of a hemisphere of radius 21 cm is (1)

(a) $1386\ \text{cm}^2$ (b) $4158\ \text{cm}^2$
(c) $2772\ \text{cm}^2$ (d) $5544\ \text{cm}^2$

MCQ

5 The volume of a cone of radius 7 cm and height 24 cm is (1)

(a) $3696\ \text{cm}^3$ (b) $1232\ \text{cm}^3$
(c) $616\ \text{cm}^3$ (d) $550\ \text{cm}^3$

MCQ

6 Two cubes of edge 10 cm are joined end to end. The surface area of the new solid is (1)

(a) $1200\ \text{cm}^2$ (b) $1000\ \text{cm}^2$
(c) $800\ \text{cm}^2$ (d) $600\ \text{cm}^2$

ASSERTION-REASON

7 **Assertion (A):** If the arc of a sector of a circle of radius 14 cm is 22 cm long, the angle of the sector is 90°.
Reason (R): Arc length $= \frac{\theta}{360^\circ} \times 2\pi r$. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

8 **Assertion (A):** The volume of a cylinder whose radius and height are both 7 cm is $1078\ \text{cm}^3$.
Reason (R): The volume of a cylinder is $2\pi rh$. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

CASE-BASED

9 **Case study: Car wipers A car has two wipers that do not overlap. Each wiper blade is 25 cm long and sweeps through an angle of 115°. (Use $\pi = \frac{22}{7}$) (4)**

(i) What shape does one wiper sweep? (1)

(ii) Find the length of the arc traced by the tip of one blade. (1)

(iii) Find the total area cleaned by both wipers in one sweep. (2)

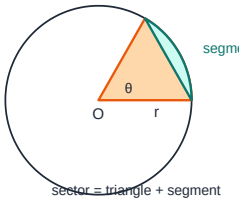
(iii) OR: Find the area of the sector swept by one wiper if the angle were 90°. (2)

VERY SHORT ANSWER

10 **Find the volume of a hemisphere of radius 3 cm in terms of π. (1)**

SHORT ANSWER

11 **Find the area of the minor and major segments of a circle of radius 10 cm cut off by a chord subtending a right angle at the centre. (Use $\pi = 3.14$) (2)**



SHORT ANSWER

12 **A horse is tied by a 5 m rope to one corner of a square field of side 15 m. Find the area it can graze, and the increase if the rope is 10 m long. (Use $\pi = 3.14$) (3)**

SHORT ANSWER

13 **Four quadrants of radius 7 cm are drawn with centres at the corners of a square of side 14 cm. Find the area of the square not covered by the quadrants. (3)**

SHORT ANSWER

14 **A conical tent is 24 m high and has a base radius of 7 m. Find the canvas required and its cost at Rs 70 per m^2. (3)**

SHORT ANSWER

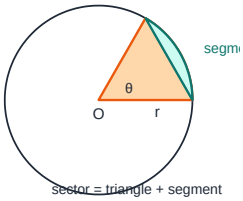
15 **A toy is a cylinder of radius 3.5 cm and height 10 cm with a cone of the same radius and height 12 cm on top. Find its total surface area. (3)**

LONG ANSWER

16 **From a solid cylinder of height 12 cm and diameter 10 cm, a hemisphere of radius 5 cm is scooped out from each end. Find the volume of the remaining solid and its total surface area. (5)**

LONG ANSWER

17 **A chord of a circle of radius 12 cm subtends an angle of 120° at the centre. Find the area of the minor segment. (Use $\pi = 3.14$, $\sqrt{3} = 1.73$) (4)**



MCQ

18 **A cylinder, a cone and a hemisphere have the same base radius r and the same height r. Which has the largest curved surface area? (1)**

(a) Cylinder
(b) Cone
(c) Hemisphere
(d) Cylinder and hemisphere (equal)

Day 35 Answers with worked solutions

1 **(b)**
$\frac{1}{4} \times \frac{22}{7} \times 196 = 154$ cm^2.

2 **(c)**
$\frac{60}{360} \times 2 \times \frac{22}{7} \times 21 = 22$ cm.

3 **(a)**
5 minutes $= 30^\circ$.
Area $= \frac{30}{360} \times \frac{22}{7} \times 196 = \frac{154}{3} \approx 51.33$ cm^2.

4 **(c)**
$2\pi r^2 = 2 \times \frac{22}{7} \times 441 = 2772$ cm^2.

5 **(b)**
$\frac{1}{3} \times \frac{22}{7} \times 49 \times 24 = 1232$ cm^3.

6 **(b)**
Cuboid $20 \times 10 \times 10$: $2(200 + 100 + 200) = 1000$ cm^2.

7 **(a)**
$2\pi r = 88$; $\frac{22}{88} \times 360^\circ = 90^\circ$. R is the formula used.

8 **(c)**
$\pi r^2 h = \frac{22}{7} \times 49 \times 7 = 1078$: A true.
R is false: $2\pi rh$ is the curved surface area, not the volume.

9 **Ans:** A sector of radius 25 cm and angle 115°

The blade rotates about one end, so it sweeps a sector.

Ans: ≈ 50.2 cm

$\frac{115}{360} \times 2 \times \frac{22}{7} \times 25 = \frac{12650}{252} \approx 50.2$ cm.

Ans: $\approx 1254.96\ \text{cm}^2$

One sector $= \frac{115}{360} \times \frac{22}{7} \times 625 \approx 627.48\ \text{cm}^2$.

Both $= 2 \times 627.48 \approx 1254.96\ \text{cm}^2$.

OR: Ans: $\approx 491.07\ \text{cm}^2$

$\frac{1}{4} \times \frac{22}{7} \times 625 = \frac{13750}{28} \approx 491.07\ \text{cm}^2$.

10 **Ans:** $18\pi\ \text{cm}^3$

$\frac{2}{3}\pi(27) = 18\pi$.

11 **Ans:** Minor $28.5\ \text{cm}^2$; major $285.5\ \text{cm}^2$

Sector $= \frac{1}{4} \times 3.14 \times 100 = 78.5$; triangle $= \frac{1}{2} \times 10 \times 10 = 50$.

Minor $= 28.5\ \text{cm}^2$; major $= 314 - 28.5 = 285.5\ \text{cm}^2$.

12 **Ans:** $19.625\ \text{m}^2$; increase $58.875\ \text{m}^2$

The grazed region is a quadrant: $\frac{1}{4} \times 3.14 \times 25 = 19.625\ \text{m}^2$.

With 10 m: $\frac{1}{4} \times 3.14 \times 100 = 78.5\ \text{m}^2$.

Increase $= 78.5 - 19.625 = 58.875\ \text{m}^2$.

13 **Ans:** $42\ \text{cm}^2$

The four quadrants together make one full circle of radius 7: $\frac{22}{7} \times 49 = 154\ \text{cm}^2$.

Uncovered $= 196 - 154 = 42\ \text{cm}^2$.

14 **Ans:** $550\ \text{m}^2$; Rs 38,500

$l = \sqrt{49 + 576} = 25$ m.

Canvas $= \pi rl = \frac{22}{7} \times 7 \times 25 = 550\ \text{m}^2$.

Cost $= 550 \times 70 =$ Rs 38,500.

15 **Ans:** $396\ \text{cm}^2$

$l = \sqrt{12.25 + 144} = 12.5$ cm.

CSA of cone $= \frac{22}{7} \times 3.5 \times 12.5 = 137.5$; CSA of cylinder $= 2 \times \frac{22}{7} \times 3.5 \times 10 = 220$; base $= 38.5$.

TSA $= 137.5 + 220 + 38.5 = 396\ \text{cm}^2$.

16 **Ans:** Volume $\approx 419.05\ \text{cm}^3$; TSA $\approx 691.43\ \text{cm}^2$

Volume $= \pi r^2 h - 2 \times \frac{2}{3}\pi r^3 = \pi(300 - 166.67) = \frac{400}{3}\pi \approx 419.05\ \text{cm}^3$.

TSA $= 2\pi rh + 2 \times 2\pi r^2 = 2\pi(5)(12) + 4\pi(25) = 120\pi + 100\pi = 220\pi$.

$= 220 \times \frac{22}{7} \approx 691.43\ \text{cm}^2$.

17 **Ans:** $\approx 88.44\ \text{cm}^2$

Sector $= \frac{120}{360} \times 3.14 \times 144 = 150.72\ \text{cm}^2$.

Triangle $OAB = \frac{1}{2}r^2 \sin 120^\circ = \frac{\sqrt{3}}{4} \times 144 = 36\sqrt{3} \approx 62.28\ \text{cm}^2$.

Segment $= 150.72 - 62.28 = 88.44\ \text{cm}^2$.

18 **(d)**

Cylinder: $2\pi r^2$; hemisphere: $2\pi r^2$; cone: $\pi r\sqrt{2}r \approx 1.41\pi r^2$.

So the cylinder and hemisphere tie.

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Arc length : $\frac{\theta}{360} \times 2\pi r$
2. Sector area : $\frac{\theta}{360} \times \pi r^2$
3. Segment : sector − triangle
4. Minute hand in 5 min : 30°
5. Cone : CSA πrl, volume $\frac{1}{3}\pi r^2 h$
6. Hemisphere : CSA $2\pi r^2$, TSA $3\pi r^2$, volume $\frac{2}{3}\pi r^3$
7. Joined solids : area of visible parts, volume adds

DAY 35 COMPLETE? ☐ Formulas revised ☐ 18 questions solved ☐ Answers checked ☐ Mistakes noted

Next: Day 36, Statistics

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 36

STATISTICS | CHAPTER 13 | PART 1 OF 3

Mean of Grouped Data

Statistics

Aaj ka target

- [ ] Class mark x_i nikalna 2 second ka kaam banao
- [ ] Direct aur assumed-mean dono method se ek hi sawal karke dekho
- [ ] Table saaf banao: column headings ke saath, Total row ke saath
- [ ] 15 sawal khud se, phir answers milao (lagbhag 90 min)

36.1 Class mark and the idea of a grouped mean

In grouped data we do not know the exact values inside a class, so we assume every observation of a class sits at its middle, the **class mark**. Then the mean is the weighted average of the class marks.

FORMULA BOX

Class mark: $x_i = \dfrac{\text{lower limit} + \text{upper limit}}{2}$

Direct method:

$$\overline{x} = \frac{\sum f_i x_i}{\sum f_i}$$

Assumed-mean method:

$$\overline{x} = a + \frac{\sum f_i d_i}{\sum f_i}, \quad d_i = x_i - a$$

Step-deviation method:

$$\overline{x} = a + h\,\frac{\sum f_i u_i}{\sum f_i}, \quad u_i = \frac{x_i - a}{h}$$

KEY TERM: ASSUMED MEAN A

Any convenient class mark, usually the one in the middle of the table with a large frequency. It only makes the arithmetic smaller; all three methods give the **same** mean.

SOLVED EXAMPLE 1 (DIRECT METHOD)

A survey of 20 houses recorded the number of plants in each house. Find the mean number of plants per house.
Plants: 0-2, 2-4, 4-6, 6-8, 8-10, 10-12, 12-14; Houses: 2, 3, 1, 5, 4, 3, 2.

SOLUTION

1. Make the table with class marks 1, 3, 5, ... and products $f_i x_i$ (table below).
2. $\sum f_i = 20$, $\sum f_i x_i = 146$.
3. $\overline{x} = \frac{146}{20} = 7.3$.

Answer: Mean $= 7.3$ plants per house

Class	x_i	f_i	$f_i x_i$
0-2	1	2	2
2-4	3	3	9
4-6	5	1	5
6-8	7	5	35
8-10	9	4	36
10-12	11	3	33
12-14	13	2	26
Total		20	146

Direct method table

SOLVED EXAMPLE 2 (ASSUMED-MEAN METHOD)

Daily wages of 50 workers: Rs 100-120: 10, 120-140: 14, 140-160: 8, 160-180: 8, 180-200: 10. Find the mean daily wage.

SOLUTION

1. Take $a = 150$ (middle class mark). $d_i = x_i - 150$: $-40, -20, 0, 20, 40$.
2. $\sum f_i d_i = -400 - 280 + 0 + 160 + 400 = -120$.
3. $\overline{x} = 150 + \frac{-120}{50} = 150 - 2.4$.

Answer: Mean daily wage = Rs 147.60

Class	x_i	f_i	$d_i = x_i - 150$	$f_i d_i$
100-120	110	10	-40	-400
120-140	130	14	-20	-280
140-160	150	8	0	0
160-180	170	8	20	160
180-200	190	10	40	400
Total		50		-120

Assumed-mean table ($a = 150$)



SHIVAM BHAIYA SAYS

Assumed mean hamesha beech wali class mark lo jiski frequency badi ho. Phir d_i chhote aate hain aur aadhe zero ke paas. Answer check: mean hamesha pehli class ke lower limit aur aakhri class ke upper limit ke beech hi aayega.

SOLVED EXAMPLE 3 (STEP-DEVIATION METHOD)

Daily expenditure on food of 25 households: Rs 100-150: 5, 150-200: 4, 200-250: 12, 250-300: 3, 300-350: 1. Find the mean expenditure.

SOLUTION

1. Class size $h = 50$; take $a = 225$. $u_i = \frac{x_i - 225}{50}$: $-2, -1, 0, 1, 2$.
2. $\sum f_i u_i = -10 - 4 + 0 + 3 + 2 = -9$.
3. $\overline{x} = 225 + 50 \times \frac{-9}{25} = 225 - 18$.

Answer: Mean = Rs 207

Class	x_i	f_i	$u_i = \frac{x_i - 225}{50}$	$f_i u_i$
100-150	125	5	-2	-10
150-200	175	4	-1	-4
200-250	225	12	0	0
250-300	275	3	1	3
300-350	325	1	2	2
Total		25		-9

Step-deviation table ($a = 225, h = 50$)



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Step-deviation ke end mein h se multiply karna bhool jaate hain: $225 + \frac{-9}{25}$ likh ke 224.64 aa jaata hai. Formula mein $a + h \times \bar{u}$ hai, h mat chhodna!

MEME SAMJHO Astra meme, maths inside



STUDENT READING THE QUESTION 'FIND THE MEAN' WITH A 7-ROW FREQUENCY TABLE

Teacher: method apni marzi se chuno. Main: direct, assumed, step-deviation... answer toh teeno ka same aayega na?

Asli baat: Asli baat: Yes. All three methods are the same formula rearranged, so the mean is identical. Choose the one with the easiest arithmetic (step-deviation when all d_i share a factor h).

36.2 Missing frequency questions

1. Write the unknown frequency as p (or f_1, f_2) in the table.
2. Write $\sum f_i$ and $\sum f_i x_i$ (or $\sum f_i d_i$) in terms of p.

1. Put them in the mean formula, equate to the given mean, solve the linear equation.
2. If two frequencies are missing, use $\sum f_i = N$ as the second equation.



EXAM ALERT!
Mean of grouped data 3 marks ka pakka sawal hai, aur 'missing frequency' wala version bahut baar aaya hai. Table ke bina sirf answer likha toh marks nahi milenge.



SHIVAM BHAIYA SAYS: BOARD TIP
Board mein method khud chuno, par jo method use kiya uska naam ek line mein likh do ('Using assumed mean method, a = 150'). Table mein Total row zaroor banao.

Day 36 Practice

15 board-pattern questions | solve first, then check

MCQ

1 The class mark of the class 10-25 is (1)
(a) 15 (b) 17.5 (c) 12.5 (d) 35

MCQ

2 For a grouped distribution, $\sum f_i = 40$ and $\sum f_i x_i = 1260$. The mean is (1)
(a) 31.5 (b) 32 (c) 30.5 (d) 1220

MCQ

3 In the step-deviation method, u_i is (1)
(a) $x_i - a$ (b) $\frac{x_i + a}{h}$
(c) $\frac{x_i - a}{h}$ (d) $h(x_i - a)$

MCQ

4 If the assumed mean is 47.5, $\sum f_i d_i = -35$ and $\sum f_i = 70$, the mean is (1)
(a) 48 (b) 47 (c) 47.5 (d) 46.5

ASSERTION-REASON

5 **Assertion (A):** The class mark of the class 20-30 is 25.
Reason (R): The class mark of a class is the average of its upper and lower limits. (1)
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

6 **Assertion (A):** The mean found by the assumed-mean method depends on which class mark is chosen as a.
Reason (R): In the assumed-mean method, $d_i = x_i - a$. (1)
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

CASE-BASED

7 Case study: Pocket money A class teacher recorded the daily pocket money (in Rs) of the children of a locality:
11-13: 3, 13-15: 6, 15-17: 9, 17-19: 13, 19-21: f, 21-23: 5, 23-25: 4.
The mean pocket money is Rs 18. (4)
(i) Write the class marks of the classes. (1)
(ii) Taking $a = 18$, find $\sum f_i d_i$ in terms of f. (1)
(iii) Find the missing frequency f. (2)
(iii) OR: If $f = 8$, find $\sum f_i x_i$. (2)

SHORT ANSWER

8 Find the mean of the distribution: 10-20: 4, 20-30: 6, 30-40: 10, 40-50: 6, 50-60: 4. (2)

SHORT ANSWER

9 The mean of the distribution below is 54. Find p.
0-20: 5, 20-40: 8, 40-60: p, 60-80: 12, 80-100: 7. (3)

SHORT ANSWER

10 Find the mean of the following data by the assumed-mean method.
0-10: 7, 10-20: 10, 20-30: 15, 30-40: 8, 40-50: 6, 50-60: 4. (3)

SHORT ANSWER

11 Using the step-deviation method, find the mean of 25 households' daily expenditure: 100-150: 4, 150-200: 5, 200-250: 12, 250-300: 2, 300-350: 2. (3)

LONG ANSWER

12 The mean of the following distribution of 50 observations is 31. Find f_1 and f_2.
0-10: 5, 10-20: f_1, 20-30: 10, 30-40: f_2, 40-50: 7, 50-60: 8. (5)

LONG ANSWER

13 Find the mean number of plants per house: 0-2: 1, 2-4: 2, 4-6: 1, 6-8: 5, 8-10: 6, 10-12: 2, 12-14: 3. Which method did you use and why? (4)

MCQ

14 The mean of a grouped distribution is 50. If every class mark is increased by 5 and frequencies stay the same, the new mean is (1)

(a) 50 **(b)** 45 **(c)** 55 **(d)** 250

VERY SHORT ANSWER

15 If $\sum f_i u_i = 12$, $\sum f_i = 40$, $a = 35$ and $h = 10$, find the mean. (1)

Day 36 Answers with worked solutions

1 **(b)**
$x = \frac{10+25}{2} = 17.5$.

2 **(a)**
$\overline{x} = \frac{1260}{40} = 31.5$.

3 **(c)**
By definition $u_i = \frac{x_i - a}{h}$.

4 **(b)**
$47.5 + \frac{-35}{70} = 47.5 - 0.5 = 47$.

5 **(a)**
$\frac{20+30}{2} = 25$; R gives the rule used.

6 **(d)**
Every choice of a gives the same mean, so A is false. R is the true definition.

7 **Ans:** 12, 14, 16, 18, 20, 22, 24
Each class mark is the midpoint, e.g. $\frac{11+13}{2} = 12$.
Ans: $2f - 16$
$d_i = -6, -4, -2, 0, 2, 4, 6$.
$\sum f_i d_i = -18 - 24 - 18 + 0 + 2f + 20 + 24 = 2f - 16$.
Ans: $f = 8$
Mean $= 18 + \frac{2f-16}{40+f} = 18 \Rightarrow 2f - 16 = 0$.
$f = 8$.
OR: Ans: 864
$\sum f_i = 48$; mean 18, so $\sum f_i x_i = 48 \times 18 = 864$.

8 **Ans:** 35
Class marks 15, 25, 35, 45, 55; $\sum f_i = 30$.
$\sum f_i x_i = 60 + 150 + 350 + 270 + 220 = 1050$.
Mean $= \frac{1050}{30} = 35$ (the data is symmetric about 35, a quick check).

9 **Ans:** $p = 8$
Class marks 10, 30, 50, 70, 90.
$\sum f_i = 32 + p$, $\sum f_i x_i = 50 + 240 + 50p + 840 + 630 = 1760 + 50p$.
$\frac{1760+50p}{32+p} = 54 \Rightarrow 1760 + 50p = 1728 + 54p \Rightarrow 4p = 32$.
$p = 8$.

10 **Ans:** 26.6
$a = 25$, $d_i = -20, -10, 0, 10, 20, 30$.
$\sum f_i d_i = -140 - 100 + 0 + 80 + 120 + 120 = 80$, $\sum f_i = 50$.
$\overline{x} = 25 + \frac{80}{50} = 26.6$.

11 **Ans:** Rs 211
$a = 225$, $h = 50$, $u_i = -2, -1, 0, 1, 2$.
$\sum f_i u_i = -8 - 5 + 0 + 2 + 4 = -7$.
$\overline{x} = 225 + 50 \times \frac{-7}{25} = 225 - 14 = 211$.

12 **Ans:** $f_1 = 9$, $f_2 = 11$
$5 + f_1 + 10 + f_2 + 7 + 8 = 50 \Rightarrow f_1 + f_2 = 20$.
$\sum f_i x_i = 25 + 15f_1 + 250 + 35f_2 + 315 + 440 = 1030 + 15f_1 + 35f_2$.
$\frac{1030+15f_1+35f_2}{50} = 31 \Rightarrow 15f_1 + 35f_2 = 520 \Rightarrow 3f_1 + 7f_2 = 104$.
With $f_1 = 20 - f_2$: $60 + 4f_2 = 104 \Rightarrow f_2 = 11$, $f_1 = 9$.

13 **Ans:** 8.1
Class marks 1, 3, 5, 7, 9, 11, 13 are small, so the direct method is easy.
$\sum f_i = 20$, $\sum f_i x_i = 1 + 6 + 5 + 35 + 54 + 22 + 39 = 162$.
$\overline{x} = \frac{162}{20} = 8.1$.

14 **(c)**
Adding 5 to every x_i adds 5 to the mean.

15 **Ans:** 38
$35 + 10 \times \frac{12}{40} = 35 + 3 = 38$.

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Class mark : (lower + upper)/2
2. Direct : $\sum f_i x_i / \sum f_i$
3. Assumed mean : $a + \sum f_i d_i / \sum f_i$
4. Step deviation : $a + h \sum f_i u_i / \sum f_i$
5. Do the methods give different means? : No, same mean
6. Missing frequency : make an equation in p

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Mean always lies : between the first lower limit and the last upper limit

DAY 36 COMPLETE?

- [ ] Notes read
- [ ] 15 questions solved
- [ ] Answers checked
- [ ] Mistakes noted

Next: Day 37, Statistics

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 37

STATISTICS | CHAPTER 13 | PART 2 OF 3

Mode of Grouped Data

Statistics

Aaj ka target

- [ ] Modal class pehchano: sabse badi frequency wali class
- [ ] Empirical relation 3,Median=Mode+2,Mean ratt lo
- [ ] f_1, f_0, f_2 ka matlab galat mat karna
- [ ] 15 sawal khud se, phir answers milao (lagbhag 80 min)

37.1 Mode of grouped data

The mode is the value that occurs most often. In grouped data we first find the **modal class** (the class with the highest frequency) and then locate the mode inside it with a formula.

FORMULA BOX

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$$

l = lower limit of modal class, h = class size

f_1 = frequency of modal class, f_0 = class **before** it, f_2 = class **after** it

SOLVED EXAMPLE 1 (RUNS SCORED)

The runs scored by top batsmen in ODIs are: 3000-4000: 4, 4000-5000: 16, 5000-6000: 9, 6000-7000: 7, 7000-8000: 6, 8000-9000: 3, 9000-10000: 10, 10000-11000: 1. Find the mode.

SOLUTION

1. Highest frequency is 16, so the modal class is 4000-5000.
2. $l = 4000$, $h = 1000$, $f_1 = 16$, $f_0 = 4$, $f_2 = 9$.
3. Mode $= 4000 + \frac{16-4}{32-4-9} \times 1000 = 4000 + \frac{12}{19} \times 1000$.
4. $= 4000 + 631.58$.

Answer: Mode ≈ 4631.58 runs



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Denominator mein $2f_1 - f_0 - f_2$ hai, $f_1 - f_0 - f_2$ nahi. Aur f_0 hamesha modal class se PEHLE wali class ki frequency hai, sabse pehli class ki nahi. Ye do galtiyan sabse zyada marks katwati hain.

SOLVED EXAMPLE 2 (MODE AND MEAN TOGETHER)

Ages (in years) of patients admitted to a hospital in a year: 5-15: 6, 15-25: 12, 25-35: 20, 35-45: 24, 45-55: 14, 55-65: 4. Find the mode and the mean and compare them.

SOLUTION

1. Modal class 35-45: $l = 35$, $f_1 = 24$, $f_0 = 20$, $f_2 = 14$, $h = 10$.
2. Mode $= 35 + \frac{24-20}{48-20-14} \times 10 = 35 + \frac{4}{14} \times 10 \approx 37.86$ years.
3. Mean: class marks 10, 20, 30, 40, 50, 60; $\sum f_i x_i = 60 + 240 + 600 + 960 + 700 + 240 = 2800$, $\sum f_i = 80$.
4. Mean $= \frac{2800}{80} = 35$ years.

Answer: Mode ≈ 37.86 years: most patients are about 38; average age is 35 years

KEY RELATION: EMPIRICAL FORMULA

For moderately skewed data,

$$3\,\text{Median} = \text{Mode} + 2\,\text{Mean}$$

or Mode $= 3$ Median $- 2$ Mean. Use it when two of the three are given.



SHIVAM BHAIYA SAYS

Empirical formula yaad karne ka tarika: **'3 Me = Mo + 2 Mi'**, yaani teen Median = ek Mode + do Mean. Check: agar teeno barabar ho (x), toh $3x = x + 2x$. Sahi!

MEME SAMJHO

Astra meme, maths inside



STUDENT WHO FOUND THE MODAL CLASS BUT USED THE FIRST CLASS'S FREQUENCY AS F_0

f_0 matlab pehli class ki frequency... nahi? Modal class se just pehle wali? Hey Maa, Mataji!

Asli baat: Asli baat: f_0 and f_2 are the neighbours of the modal class. If the modal class is the first class, take $f_0 = 0$; if it is the last, take $f_2 = 0$.



EXAM ALERT!

Mode wala sawal 2-3 marks ka aata hai, aur case-study mein 'modal class' ya 'upper limit of modal class' 1 mark ka MCQ ban ke aata hai. Formula likh ke values substitute karo, tabhi full marks.



SHIVAM BHAIYA SAYS: BOARD TIP

Pehle ek line mein likho: 'Maximum frequency = 16, so modal class is 4000-5000'. Phir l, h, f_1, f_0, f_2 ki values alag alag likho, phir formula. Examiner ko step dikhna chahiye.

Day 37 Practice

15 board-pattern questions | solve first, then check

MCQ

1 The modal class of the distribution 0-10: 3, 10-20: 9, 20-30: 15, 30-40: 11, 40-50: 8 is (1)

(a) 10-20 **(b)** 30-40 **(c)** 20-30 **(d)** 40-50

MCQ

2 For a data set, mean = 24 and median = 26. Using the empirical relation, the mode is (1)

(a) 28 **(b)** 30 **(c)** 25 **(d)** 22

MCQ

3 In the mode formula, f_0 denotes the frequency of (1)

(a) the modal class
(b) the first class
(c) the class preceding the modal class
(d) the class succeeding the modal class

MCQ

4 For the distribution 0-20: 6, 20-40: 8, 40-60: 10, 60-80: 12, 80-100: 6, 100-120: 5, 120-140: 3, the mode is (1)

(a) 60 **(b)** 65 **(c)** 70 **(d)** 62.5

ASSERTION-REASON

5 **Assertion (A):** If the modal class is 40-50 with $f_1 = 12$, $f_0 = 8$, $f_2 = 10$, then the mode is about 46.67.

Reason (R): Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

6 **Assertion (A):** If the mean of a data set is 5 and its median is 6, then its mode is 8.

Reason (R): For moderately skewed data, Mode $= 3$ Median $- 2$ Mean. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

CASE-BASED

7 Case study: Class test marks The marks (out of 100) of 50 students in a class test are: 0-10: 2, 10-20: 3, 20-30: 4, 30-40: 6, 40-50: 7, 50-60: 3, 60-70: 8, 70-80: 7, 80-90: 6, 90-100: 4. (4)

(i) Write the modal class and its upper limit. (1)

(ii) How many students scored less than 40? (1)

(iii) Find the mode of the marks. (2)

(iii) OR: Find the class mark of the modal class and the class size. (2)

VERY SHORT ANSWER

8 Find the mode of a distribution whose modal class is 20-30 with $f_1 = 15$, $f_0 = 10$, $f_2 = 10$. (1)

SHORT ANSWER

9 The mode of the distribution below is 25. Find x. 0-10: 5, 10-20: x, 20-30: 15, 30-40: 10, 40-50: 6. (3)

SHORT ANSWER

10 The mode and mean of a data set are 34.3 and 32.2. Find the median using the empirical relation. (2)

SHORT ANSWER

11 Find the mode of the ages of patients: 5-15: 6, 15-25: 11, 25-35: 21, 35-45: 23, 45-55: 14, 55-65: 5. (3)

LONG ANSWER

12 For the distribution 0-10: 4, 10-20: 6, 20-30: 10, 30-40: 16, 40-50: 8, 50-60: 6, find the mean and the mode. Then use the empirical relation to estimate the median. (5)

LONG ANSWER

13 The runs scored by 40 top batsmen are given: 3000-4000: 4, 4000-5000: 18, 5000-6000: 9, 6000-7000: 7, 7000-8000: 6, 8000-9000: 3, 9000-10000: 10, 10000-11000: 1. Find the mode. (4)

MCQ

14 If the modal class is the first class of a distribution, then in the mode formula we take (1)

(a) $f_2 = 0$ (b) $f_0 = 0$ (c) $f_1 = 0$ (d) $h = 0$

VERY SHORT ANSWER

15 Write the empirical relation between mean, median and mode. (1)

Day 37 Answers with worked solutions

1 **(c)**
Highest frequency 15 belongs to 20-30.

2 **(b)**
Mode $= 3(26) - 2(24) = 78 - 48 = 30$.

3 **(c)**
f_0 is the class just before the modal class.

4 **(b)**
Modal class 60-80: $l = 60$, $f_1 = 12$, $f_0 = 10$, $f_2 = 6$, $h = 20$.
Mode $= 60 + \frac{2}{24-16} \times 20 = 60 + 5 = 65$.

5 **(a)**
$40 + \frac{4}{24-18} \times 10 = 40 + 6.67 = 46.67$. R is the formula used.

6 **(a)**
$3(6) - 2(5) = 18 - 10 = 8$. Both true, R explains A.

7 **Ans:** 60-70; upper limit 70
Highest frequency 8 is for 60-70.
Ans: 15
$2 + 3 + 4 + 6 = 15$.
Ans: ≈ 68.33
$l = 60$, $f_1 = 8$, $f_0 = 3$, $f_2 = 7$, $h = 10$.
Mode $= 60 + \frac{5}{16-10} \times 10 = 60 + 8.33 = 68.33$.
OR: Ans: 65; 10
Class mark $= \frac{60+70}{2} = 65$; class size $= 70 - 60 = 10$.

8 **Ans:** 25
Mode $= 20 + \frac{5}{30-20} \times 10 = 25$ (symmetric neighbours put the mode at the class middle).

9 **Ans:** $x = 10$
Mode 25 lies in 20-30, so that is the modal class: $l = 20$, $f_1 = 15$, $f_0 = x$, $f_2 = 10$.
$25 = 20 + \frac{15-x}{30-x-10} \times 10 \Rightarrow 5(20 - x) = 10(15 - x)$
.
$100 - 5x = 150 - 10x \Rightarrow 5x = 50 \Rightarrow x = 10$.

10 **Ans:** 32.9
$3\,\text{Median} = 34.3 + 2(32.2) = 98.7$.
$\text{Median} = 32.9$.

11 **Ans:** ≈ 36.8 years
Modal class 35-45: $l = 35$, $f_1 = 23$, $f_0 = 21$, $f_2 = 14$, $h = 10$.
Mode $= 35 + \frac{2}{46-35} \times 10 = 35 + \frac{20}{11} \approx 36.82$ years.

12 **Ans:** Mean 32.2; mode ≈ 34.29; median ≈ 32.9
Class marks 5, 15, ..., 55; $\sum f_i = 50$, $\sum f_i x_i = 20 + 90 + 250 + 560 + 360 + 330 = 1610$; mean $= 32.2$.
Modal class 30-40: $l = 30$, $f_1 = 16$, $f_0 = 10$, $f_2 = 8$: mode $= 30 + \frac{6}{32-18} \times 10 = 30 + 4.29 = 34.29$.
Median $\approx \frac{34.29+2(32.2)}{3} = \frac{98.69}{3} \approx 32.9$.
(Direct calculation gives 33.125; the empirical value is an estimate.)

13 **Ans:** ≈ 4608.7 runs
Modal class 4000-5000: $l = 4000$, $f_1 = 18$, $f_0 = 4$, $f_2 = 9$, $h = 1000$.
Mode $= 4000 + \frac{14}{36-13} \times 1000 = 4000 + \frac{14000}{23} \approx 4608.7$.

14 **(b)**
There is no class before it, so $f_0 = 0$.

15 **Ans:** $3\,\text{Median} = \text{Mode} + 2\,\text{Mean}$
Standard relation for moderately skewed data.

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Modal class : class with the highest frequency
2. Mode formula : $l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$
3. f_0 : frequency of the class before the modal class
4. Modal class is first : take $f_0 = 0$
5. Empirical relation : $3\,\text{Median} = \text{Mode} + 2\,\text{Mean}$
6. Mean 24, median 26 : mode 30
7. Mode tells : the most common value

DAY 37 COMPLETE? ☐ Notes read ☐ 15 questions solved ☐ Answers checked ☐ Mistakes noted

Next: Day 38, Statistics

DAY 38

STATISTICS | CHAPTER 13 | PART 3 OF 3

Median of Grouped Data

Statistics

Aaj ka target

- [ ] Cumulative frequency (cf) column bina galti ke banao
- [ ] N/2 se median class pakdo
- [ ] 'Less than' table ko normal frequency table mein badalna seekho
- [ ] 15 sawal khud se, phir answers milao (lagbhag 90 min)

38.1 Cumulative frequency and the median class

The median is the middle value. For grouped data, add a **cumulative frequency (cf)** column: the running total of the frequencies. The class whose cf first reaches or crosses $\frac{N}{2}$ is the **median class**.

FORMULA BOX

$$\text{Median} = l + \left(\frac{\frac{N}{2} - cf}{f} \right) \times h$$

l = lower limit of median class, $N = \sum f_i$, h = class size

cf = cumulative frequency of the class **before** the median class, f = frequency of the median class

SOLVED EXAMPLE 1 (WEIGHTS)

Weights (kg) of 30 students: 40-45: 2, 45-50: 3, 50-55: 8, 55-60: 6, 60-65: 6, 65-70: 3, 70-75: 2. Find the median weight.

SOLUTION

1. cf column: 2, 5, 13, 19, 25, 28, 30. $N = 30$, $\frac{N}{2} = 15$.
2. cf first crosses 15 at 19, so the median class is 55-60.
3. $l = 55$, $cf = 13$, $f = 6$, $h = 5$.
4. Median $= 55 + \frac{15-13}{6} \times 5 = 55 + \frac{10}{6}$.

Answer: Median ≈ 56.67 kg

Weight (kg)	f	cf
40-45	2	2
45-50	3	5
50-55	8	13
55-60	6	19 (median class)
60-65	6	25
65-70	3	28
70-75	2	30

Cumulative frequency table



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Formula mein cf median class ka nahi, uske **pehle wali class** ka hota hai. Example mein 19 nahi, 13 lena hai. Aur $\frac{N}{2}$ lo, $\frac{N+1}{2}$ nahi: grouped data mein $\frac{N}{2}$ hi chalta hai.

38.2 'Less than' and 'more than' tables

Sometimes data is given as cumulative counts, e.g. 'less than 10: 3, less than 20: 9, ...'. Convert it back to class frequencies by subtracting consecutive values, then proceed as usual. (Drawing ogives is not in the 2026-27 syllabus; only this conversion is needed.)

SOLVED EXAMPLE 2 (MISSING FREQUENCIES)

The median of 60 observations is 30. Find x and y.

0-10: 5, 10-20: x, 20-30: 20, 30-40: 15, 40-50: y, 50-60: 5.

SOLUTION

1. $5 + x + 20 + 15 + y + 5 = 60 \Rightarrow x + y = 15$.
2. $\frac{N}{2} = 30$. The median 30 lies in the class 20-30 (at its upper end), so take $l = 20$, $cf = 5 + x$, $f = 20$, $h = 10$.
3. $30 = 20 + \frac{30-(5+x)}{20} \times 10 \Rightarrow 20 = 25 - x \Rightarrow x = 5$.
4. Then $y = 15 - 5 = 10$.

Answer: $x = 5$, $y = 10$

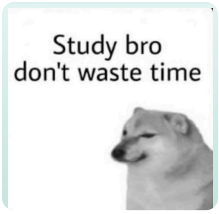


SHIVAM BHAIYA SAYS

Missing frequency + median wale sawal mein do equation banti hain: ek $\sum f = N$ se, doosri median formula se. Median kis class mein hai woh value dekh ke batao (median 30 aaye toh 20-30 ka upper end bhi ho sakta hai, toh 20-30 class lo aur check karo ki cf sahi baith raha hai).

MEME SAMJHO

Astra meme, maths inside



CHEEMS REMINDING A STUDENT BEFORE THE BOARD EXAM

Study bro: mean ke liye $\sum fx$, mode ke liye f_1, f_0, f_2, median ke liye cf column. Teeno table alag!

Asli baat: Asli baat: Mean needs class marks, mode needs only the modal class and its two neighbours, median needs the cumulative frequency column.



EXAM ALERT!

Median ka 3 ya 5 marks ka sawal har saal aata hai, aksar missing frequency (x, y) ke saath. cf column galat hua toh poora answer galat, isliye har row dobara jod ke check karo: last cf $= N$.



SHIVAM BHAIYA SAYS: BOARD TIP

Long answer mein mean, median, mode teeno maange jaa sakte hain. Ek hi table mein x_i, f_ix_i aur cf ke columns bana do, time bachega.

Day 38 Practice

14 board-pattern questions | solve first, then check

MCQ

1 For the distribution 0-10: 4, 10-20: 7, 20-30: 12, 30-40: 9, 40-50: 8, the median class is (1)

(a) 10-20 **(b)** 20-30 **(c)** 30-40 **(d)** 40-50

MCQ

2 In the median formula, cf stands for the cumulative frequency of (1)

(a) the median class
(b) the class preceding the median class
(c) the last class
(d) the modal class

MCQ

3 For the distribution 0-20: 7, 20-40: 11, 40-60: 20, 60-80: 15, 80-100: 7, the sum of the lower limits of the median class and the modal class is (1)

(a) 80 **(b)** 60 **(c)** 100 **(d)** 120

MCQ

4 For a data set, mode = 46 and mean = 40. The median by the empirical relation is (1)

(a) 43 **(b)** 42 **(c)** 44 **(d)** 41

ASSERTION-REASON

5 **Assertion (A):** For a grouped distribution with $N = 30$, the median class is the class whose cumulative frequency first exceeds 15.
Reason (R): The median class is located using $\frac{N}{2}$. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

6 **Assertion (A):** The cumulative frequency of the last class always equals the total number of observations.
Reason (R): Cumulative frequency is the running total of the frequencies. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

CASE-BASED

7 Case study: Life of LED bulbs A company tested 400 LED bulbs and recorded their lifetimes (hours): 1500-2000: 14, 2000-2500: 56, 2500-3000: 60, 3000-3500: 86, 3500-4000: 74, 4000-4500: 62, 4500-5000: 48. (4)

(i) Find the median class. (1)

(ii) Find the modal class. (1)

(iii) Find the median lifetime. (2)

(iii) OR: Find the mode of the lifetimes. (2)

VERY SHORT ANSWER

8 In a grouped distribution, $N = 50$, the median class is 30-40, $cf = 22$ and $f = 10$. Find the median. (1)

SHORT ANSWER

9 Convert to a frequency table and find the median: marks less than 10: 3, less than 20: 9, less than 30: 17, less than 40: 27, less than 50: 40. (3)

SHORT ANSWER

10 Find the median weight of 30 students: 40-45: 2, 45-50: 3, 50-55: 9, 55-60: 6, 60-65: 6, 65-70: 2, 70-75: 2. (3)

LONG ANSWER

11 The median of the distribution below is 30 and the total frequency is 60. Find x and y.
0-10: 5, 10-20: x, 20-30: 20, 30-40: 15, 40-50: y, 50-60: 5. (5)

LONG ANSWER

12 For the distribution 0-20: 7, 20-40: 11, 40-60: 20, 60-80: 15, 80-100: 7, find the mean, median and mode, and verify the empirical relation approximately. (5)

MCQ

13 The median of a grouped distribution can be read from the class whose cumulative frequency is (1)

(a) the largest
(b) just greater than or equal to $\frac{N}{2}$
(c) equal to the highest frequency
(d) the smallest

VERY SHORT ANSWER

14 Which measure of central tendency needs a cumulative frequency column? (1)

Day 38 Answers with worked solutions

1 **(b)**
$N = 40$, $\frac{N}{2} = 20$; cf: 4, 11, 23, ... first crosses 20 at 20-30.

2 **(b)**
cf is the cf of the class just before the median class.

3 **(a)**
$N = 60$, $\frac{N}{2} = 30$, cf 7, 18, 38: median class 40-60.
Modal class (f = 20) is also 40-60. Sum $= 40 + 40 = 80$.

4 **(b)**
$3\,\text{Median} = 46 + 2(40) = 126 \Rightarrow \text{Median} = 42$.

5 **(a)**
$\frac{N}{2} = 15$; R gives the rule used in A.

6 **(a)**
The running total ends at $\sum f_i = N$. R explains A.

7 **Ans:** 3000-3500
$\frac{N}{2} = 200$; cf: 14, 70, 130, 216 ... first crosses 200 at 3000-3500.
Ans: 3000-3500
Highest frequency 86.
Ans: ≈ 3406.98 hours
$l = 3000$, $cf = 130$, $f = 86$, $h = 500$.
Median $= 3000 + \frac{200-130}{86} \times 500 = 3000 + \frac{35000}{86} \approx 3406.98$ hours.
OR: Ans: ≈ 3342.11 hours
$l = 3000$, $f_1 = 86$, $f_0 = 60$, $f_2 = 74$, $h = 500$.
Mode $= 3000 + \frac{26}{172-134} \times 500 = 3000 + \frac{13000}{38} \approx 3342.11$ hours.

8 **Ans:** 33
Median $= 30 + \frac{25-22}{10} \times 10 = 33$.

9 **Ans:** 33
Frequencies: 0-10: 3, 10-20: 6, 20-30: 8, 30-40: 10, 40-50: 13. $N = 40$, $\frac{N}{2} = 20$.
Median class 30-40 (cf 17 before it, $f = 10$).
Median $= 30 + \frac{20-17}{10} \times 10 = 33$.

10 **Ans:** ≈ 55.83 kg
cf: 2, 5, 14, 20, 26, 28, 30; $N = 30$, $\frac{N}{2} = 15$.
Median class 55-60: $l = 55$, $cf = 14$, $f = 6$, $h = 5$.
Median $= 55 + \frac{15-14}{6} \times 5 = 55 + 0.83 \approx 55.83$ kg.

11 **Ans:** $x = 5$, $y = 10$
$x + y = 15$.
With median class 20-30: $30 = 20 + \frac{30-(5+x)}{20} \times 10 \Rightarrow x = 5$, $y = 10$.
Check: cf before 20-30 is 10, and $10 + 20 = 30 = \frac{N}{2}$, so the median is exactly 30.

12 **Ans:** Mean ≈ 51.33, median 52, mode ≈ 52.86
Mean: class marks 10, 30, 50, 70, 90; $\sum f_i x_i = 70 + 330 + 1000 + 1050 + 630 = 3080$; mean $= \frac{3080}{60} \approx 51.33$.
Median: $\frac{N}{2} = 30$, median class 40-60, $cf = 18$, $f = 20$: $40 + \frac{12}{20} \times 20 = 52$.
Mode: $40 + \frac{20-11}{40-11-15} \times 20 = 40 + \frac{9}{14} \times 20 \approx 52.86$.
Check: Mode + 2 Mean $\approx 52.86 + 102.67 = 155.5$ and 3 Median $= 156$: close, so the relation holds approximately.

13 **(b)**
Median class: cf first reaches or crosses $\frac{N}{2}$.

14 **Ans:** Median
Only the median formula uses cf.

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Median formula : $l + \frac{\frac{N}{2} - cf}{f} \times h$
2. cf in formula : of the class before the median class
3. Median class : cf first reaches $\frac{N}{2}$
4. Last cf : equals N
5. Less-than table to frequencies : subtract consecutive values
6. Ogive drawing : not in 2026-27 syllabus
7. Three tables : mean (x_i), mode (modal class), median (cf)

DAY 38 COMPLETE?

- [ ] Notes read
- [ ] 14 questions solved
- [ ] Answers checked
- [ ] Mistakes noted

Next: Day 39, Probability

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 39

PROBABILITY | CHAPTER 14 | PART 1 OF 2

Theoretical Probability

Probability

Aaj ka target

- [ ] P(E) = favourable / total, bina soche likhna aa jaye
- [ ] 16 sawal lagao, har answer simplest form mein
- [ ] Sure, impossible aur complementary events clear karo
- [ ] 0 se 1 ke bahar answer aaya to wahin ruk ke check karo

39.1 The language of probability

An **experiment** is an action whose result is uncertain (tossing a coin, throwing a die, drawing a card). Each possible result is an **outcome**. Outcomes are **equally likely** when none has a better chance than another (a fair coin, an unbiased die, a well-shuffled deck, a ball drawn 'at random').

KEY TERM: EVENT

An **event** E is a collection of outcomes, e.g. 'getting an even number' on a die = {2, 4, 6}. An event with exactly one outcome is an **elementary event**.

FORMULA BOX: THEORETICAL (CLASSICAL) PROBABILITY

$$P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$$

Outcomes must be **equally likely**.

$0 \le P(E) \le 1$ for every event

$P(\text{sure event}) = 1, \quad P(\text{impossible event}) = 0$

$P(E) + P(\overline{E}) = 1 \Rightarrow P(\overline{E}) = 1 - P(E)$

Sum of probabilities of all elementary events $= 1$



SHIVAM BHAIYA SAYS

Har question mein pehle **total** likho, phir **favourable** gino. Total galat = poora answer galat. 'Between 1 and 30' mein 1 aur 30 shamil nahi hote, 'from 1 to 30' mein hote hain.

39.2 Sure, impossible and complementary events

- **Sure event**: always happens. Getting a number less than 7 on a die: $P = \frac{6}{6} = 1$.
- **Impossible event**: never happens. Getting 8 on a die: $P = \frac{0}{6} = 0$.
- **Complement** $\overline{E}$ ('not E') happens exactly when E does not. If $P(E) = 0.36$ then $P(\overline{E}) = 0.64$.
- A probability can be a fraction, a decimal or a percent, but never negative and never more than 1: so $\frac{7}{6}$, 1.5 and -0.2 are impossible values.

SOLVED EXAMPLE 1

A number is chosen at random from 1 to 30. Find the probability that it is a prime number.

SOLUTION

1. Total outcomes = 30.
2. Primes from 1 to 30: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, so favourable = 10.
3. $P(\text{prime}) = \frac{10}{30} = \frac{1}{3}$

Answer: $\frac{1}{3}$

SOLVED EXAMPLE 2

A lot of 25 bulbs contains 5 defective ones. One bulb is drawn at random. (i) Find P(defective). (ii) The bulb drawn is found to be good and is **not** put back. Find the probability that the next bulb drawn is not defective.

SOLUTION

1. (i) $P(\text{defective}) = \frac{5}{25} = \frac{1}{5}$
2. (ii) After removing one good bulb: 24 bulbs left, of which $20 - 1 = 19$ are good.
3. $P(\text{good}) = \frac{19}{24}$

Answer: (i) $\frac{1}{5}$ (ii) $\frac{19}{24}$

SOLVED EXAMPLE 3

The probability that an egg in a tray is rotten is 0.04. How many good eggs are there in a lot of 500 eggs?

SOLUTION

1. Expected rotten eggs $= 0.04 \times 500 = 20$
2. Good eggs $= 500 - 20 = 480$
3. Check: $P(\text{good}) = \frac{480}{500} = 0.96 = 1 - 0.04$

Answer: 480 good eggs

Type of event	Example	P
Sure	Number ≤ 6 on a die	1
Impossible	Sum 13 on two dice	0
Elementary	Getting 4 on a die	$\frac{1}{6}$
Complement	Not getting 4	$\frac{5}{6}$

Quick classification



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

'Not put back' wale sawal mein dono cheezein ghatao: **total bhi 1 kam** aur jo nikla uski **category bhi 1 kam**. Sirf total ghatana sabse common galti hai.



EXAM ALERT!

Probability se board mein lagbhag **5 marks** aate hain: 1-2 MCQ, ek case study ya 2-3 mark ka sawal. Asaan marks hain, bas total galat mat karna aur answer **simplest form** mein likhna.

MEME SAMJHO — Astra meme, maths inside



TEACHER: 'DIE PE 7 SE CHHOTA NUMBER AAYEGA.' STUDENT:

Guarantee deta hai?? Haan beta, P = 6/6 = 1, sure event hai!

Asli baat: a sure event has probability 1 and an impossible event has probability 0; every other event lies strictly between them.



SHIVAM BHAIYA SAYS: BOARD TIP

Answer ke saath ek line likho: 'Total outcomes = ..., favourable outcomes = ...'. Board marking scheme mein yahi step ka mark hota hai, sirf final fraction likhne se poore marks nahi milte.

Day 39 Practice

16 board-pattern questions | solve first, then check

MCQ

1 If the probability of an event E is 0.12, the probability of 'not E' is (1)

(a) 0.12 (b) 0.88 (c) 1.12 (d) 0.78

MCQ

2 Which of the following **cannot** be the probability of an event? (1)

(a) $\frac{2}{3}$ (b) 15% (c) $\frac{7}{6}$ (d) 0.7

MCQ

3 A letter of the English alphabet is chosen at random. The probability that it is one of the letters of the word 'PROBABILITY' is (1)

(a) $\frac{11}{26}$ (b) $\frac{9}{26}$ (c) $\frac{7}{26}$ (d) $\frac{1}{2}$

MCQ

4 A number is picked at random from the first 60 natural numbers. The probability that it is a multiple of both 3 and 4 is (1)

(a) $\frac{1}{12}$ (b) $\frac{1}{6}$ (c) $\frac{7}{12}$ (d) $\frac{1}{4}$

MCQ

5 The probability that a randomly picked egg from a lot of 600 is rotten is 0.035. The number of rotten eggs in the lot is (1)

(a) 35 (b) 21 (c) 17 (d) 24

MCQ

6 A student guesses answers and gets x correct out of y questions. If the probability of a wrong guess is $\frac{3}{5}$, then (1)

(a) $3x = 5y$ (b) $5x = 3y$ (c) $5x = 2y$ (d) $2x = 5y$

ASSERTION-REASON

7 **Assertion (A):** If the probability of a team winning a match is 0.36, the probability of it not winning is 0.64.
Reason (R): For any event E, $P(E) + P(\overline{E}) = 1$. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

8 **Assertion (A):** A number is chosen at random from 1 to 17. The probability that it is odd is $\frac{1}{2}$.
Reason (R): The probability of any event lies between 0 and 1, both inclusive. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

CASE-BASED

9 **At a school fair, the 'Hook-a-Duck' stall has 100 rubber ducks floating in a tank. Each duck has a letter under it: N (no prize), S (small prize), M (medium prize) or L (large prize), in the ratio $N : S : M : L = 9 : 6 : 4 : 1$. A player hooks one duck at random.** (4)

(i) How many ducks carry each letter, and what is the probability of winning the large prize? (1)

(ii) Find the probability of winning some prize. (1)

(iii) Two players have already hooked ducks marked N and kept them out. Find the probability that the third player wins a small prize. (2)

(iii) OR The first player hooks an S duck and keeps it. Which result is now most likely for the next player, and with what probability? (2)

VERY SHORT ANSWER

10 **A bag has 4 red, 6 black and 5 white balls. A ball is drawn at random. Find the probability that it is not black.** (2)

SHORT ANSWER

11 **An integer x is chosen at random from $-4, -3, \ldots, 3, 4$. Find the probability that $x^2 \leq 4$.** (2)

SHORT ANSWER

12 **In a class of 30 students, grades were: A 5, B 9, C 8, D 3, E 5. One student is picked at random. Find the probability that the student did not get grade A, B or C.** (2)

SHORT ANSWER

13 **An integer is chosen at random between 60 and 90. Find the probability that it is (i) a prime number (ii) divisible by 6.** (3)

SHORT ANSWER

14 **A batch of 200 shirts has 172 good shirts, 18 with minor defects and 10 with major defects. Ravi, a trader, accepts only good shirts; Seema rejects only shirts with major defects. One shirt is drawn at random. Find the probability that it is acceptable to (i) Ravi (ii) Seema.** (3)

LONG ANSWER

15 **A housing society planted 800 saplings: 180 neem, 220 peepal, 70 creepers, 190 fruit plants and 140 flowering plants. One sapling is chosen at random. Find the probability that it is (i) a fruit or a flowering plant (ii) a neem or a peepal (iii) not a creeper.** (4)

LONG ANSWER

16 **A box has 30 balls: 6 orange, 9 white and the rest blue or yellow. The probability of drawing a yellow ball is twice that of drawing a blue ball. Find (i) the number of blue and yellow balls (ii) P(not blue) (iii) P(blue or white) (iv) P(neither yellow nor white).** (5)

Day 39 Answers with worked solutions

1 **(b)**
$P(\overline{E}) = 1 - 0.12 = 0.88$

2 **(c)**
A probability never exceeds 1, and $\frac{7}{6} > 1$.

3 **(b)**
Distinct letters: P, R, O, B, A, I, L, T, Y = 9.
$P = \frac{9}{26}$

4 **(a)**
Multiples of both 3 and 4 = multiples of 12: 12, 24, 36, 48, 60, i.e. 5.
$P = \frac{5}{60} = \frac{1}{12}$

5 **(b)**
$0.035 \times 600 = 21$

6 **(c)**
$P(\text{correct}) = 1 - \frac{3}{5} = \frac{2}{5} = \frac{x}{y}$
So $5x = 2y$.

7 **(a)**
$1 - 0.36 = 0.64$, so A is true; R is the rule used, so R explains A.

8 **(d)**
Odd numbers from 1 to 17: 9, so $P = \frac{9}{17}$, not $\frac{1}{2}$: A is false.
R is a true statement.

9 **Ans:** N 45, S 30, M 20, L 5; $P(L) = \frac{1}{20}$

Total parts $= 20$, so one part $= 5$ ducks: N 45, S 30, M 20, L 5.

$P(L) = \frac{5}{100} = \frac{1}{20}$

Ans: $\frac{11}{20}$

Prize ducks $= 30 + 20 + 5 = 55$

$P = \frac{55}{100} = \frac{11}{20}$

Ans: $\frac{15}{49}$

Ducks left $= 98$; small-prize ducks are still 30.

$P(S) = \frac{30}{98} = \frac{15}{49}$

OR: Ans: No prize, $\frac{5}{11}$

Left: N 45, S 29, M 20, L 5; total 99.

N is the largest group: $P(N) = \frac{45}{99} = \frac{5}{11}$

10 **Ans:** $\frac{3}{5}$

Total $= 4 + 6 + 5 = 15$; not black $= 4 + 5 = 9$.

$P = \frac{9}{15} = \frac{3}{5}$

11 **Ans:** $\frac{5}{9}$

Total integers $= 9$.

$x^2 \leq 4$ for $x = -2, -1, 0, 1, 2$: 5 values.

$P = \frac{5}{9}$

12 **Ans:** $\frac{4}{15}$

Students with D or E $= 3 + 5 = 8$.

$P = \frac{8}{30} = \frac{4}{15}$

13 **Ans:** (i) $\frac{7}{29}$ (ii) $\frac{4}{29}$

'Between' excludes 60 and 90: integers 61 to 89, total 29.

Primes: 61, 67, 71, 73, 79, 83, 89 = 7, so $P = \frac{7}{29}$

Divisible by 6: 66, 72, 78, 84 = 4, so $P = \frac{4}{29}$.

14 **Ans:** (i) 0.86 (ii) 0.95

(i) $P = \frac{172}{200} = 0.86$

(ii) Acceptable to Seema $= 172 + 18 = 190$, so $P = \frac{190}{200} = 0.95$

15 **Ans:** (i) $\frac{33}{80}$ (ii) $\frac{1}{2}$ (iii) $\frac{73}{80}$

Check total: $180 + 220 + 70 + 190 + 140 = 800$.

(i) $\frac{190 + 140}{800} = \frac{330}{800} = \frac{33}{80}$

(ii) $\frac{180 + 220}{800} = \frac{400}{800} = \frac{1}{2}$

(iii) $1 - \frac{70}{800} = \frac{730}{800} = \frac{73}{80}$

16 **Ans:** (i) blue 5, yellow 10 (ii) $\frac{5}{6}$ (iii) $\frac{7}{15}$ (iv) $\frac{11}{30}$

Blue + yellow $= 30 - 6 - 9 = 15$. Let blue $= b$; then yellow $= 2b$ (same total, so twice the probability means twice the count).

$3b = 15 \Rightarrow b = 5$, yellow $= 10$.

(ii) $P(\text{not blue}) = \frac{25}{30} = \frac{5}{6}$

(iii) $P(\text{blue or white}) = \frac{5 + 9}{30} = \frac{7}{15}$

(iv) Neither yellow nor white = orange or blue $= 6 + 5 = 11$: $P = \frac{11}{30}$

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Formula of P(E) : favourable outcomes / total equally likely outcomes
2. Range of any probability : 0 to 1 (both included)
3. P(sure event) : 1
4. P(impossible event) : 0
5. P(not E) : 1 - P(E)
6. Sum of probabilities of all elementary events : 1
7. 'Between 60 and 90' : 61 to 89, 29 numbers
8. Can 7/6 or -0.2 be a probability? : No

DAY 39 COMPLETE? ☐ Notes read ☐ 16 questions solved ☐ Answers checked ☐ Mistakes noted

Next: Day 40, Probability

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 40

PROBABILITY | CHAPTER 14 | PART 2 OF 2

Coins, Dice, Cards and Mixed Problems

Probability

Aaj ka target

- [ ] Do coins (4), teen coins (8), do dice (36) ke outcomes muh-zubaani
- [ ] 52 cards ka poora breakup yaad karo
- [ ] Two-dice grid khud bana ke 5 sawal karo
- [ ] 16 sawal lagao, case study time se

40.1 Coins: list the sample space first

Experiment	Outcomes	Total
1 coin	H, T	2
2 coins	HH, HT, TH, TT	4
3 coins	HHH, HHT, HTH, THH, HTT, THT, TTH, TTT	8
n coins		2^n

Sample spaces for coins

HT and TH are **different** outcomes (first coin H, second T, and the other way round). Missing one of them is the classic reason for a wrong answer like $P(\text{one head}) = \frac{1}{3}$; the right value is $\frac{2}{4} = \frac{1}{2}$.

SOLVED EXAMPLE 1

Hanif tosses a coin three times. He wins if all three tosses show the same face, otherwise he loses. Find the probability that he loses.

SOLUTION

1. Total outcomes $= 2^3 = 8$.
2. Win: HHH, TTT, so P(win) $= \frac{2}{8} = \frac{1}{4}$.
3. P(lose) $= 1 - \frac{1}{4} = \frac{3}{4}$

Answer: $\frac{3}{4}$

40.2 Two dice: the 36-cell grid

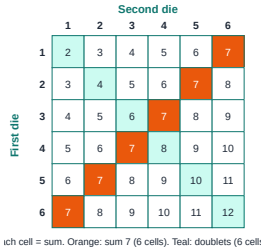


Fig. Sample space of two dice thrown together (36 outcomes)

- Total outcomes $= 6 \times 6 = 36$; a pair (a, b) means first die a, second die b.
- Sums 2 and 12 occur once each, 7 occurs 6 times (most likely). Number of ways for sum s: $s - 1$ for $s \le 7$, and $13 - s$ for $s \ge 7$.
- Doublets $(1,1), \ldots, (6,6)$: 6 outcomes, $P = \frac{1}{6}$.
- 'At least one 6' $= 36 - 25 = 11$ outcomes (25 have no 6).

SOLVED EXAMPLE 2

Two dice are thrown together. Find the probability that (i) the sum is 8 (ii) the sum is 13 (iii) the sum is at most 12.

SOLUTION

1. (i) Sum 8: (2,6), (3,5), (4,4), (5,3), (6,2) = 5 outcomes, $P = \frac{5}{36}$.
2. (ii) Largest possible sum is 12, so sum 13 is impossible: $P = 0$.
3. (iii) Every sum is at most 12: sure event, $P = 1$.

Answer: $\frac{5}{36}$, 0, 1



SHIVAM BHAIYA SAYS

'At least one' wala sawal seedha ginne ki jagah ulta socho: **P(at least one 6) = 1 - P(koi 6 nahi) = 1 - 25/36 = 11/36.** Coins mein bhi: P(at least one head) = 1 - P(TT) = 3/4.

40.3 A deck of 52 playing cards

Group	Count	Details
Suits	4 × 13	Spades, clubs (black); hearts, diamonds (red)
Red / Black	26 / 26	2 red suits, 2 black suits
Face cards	12	King, Queen, Jack of each suit
Aces	4	one per suit (not face cards)
Each rank	4	e.g. 4 kings: 2 red, 2 black

Deck breakup

SOLVED EXAMPLE 3

One card is drawn from a well-shuffled deck of 52 cards. Find the probability that it is neither a red card nor a queen.

SOLUTION

1. Red cards = 26; black queens = 2 (the red queens are already counted among the red cards).
2. Red or queen $= 26 + 2 = 28$; neither $= 52 - 28 = 24$.
3. $P = \frac{24}{52} = \frac{6}{13}$

Answer: $\frac{6}{13}$



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Ace face card nahi hota! Face cards sirf K, Q, J = 12. Aur 'red or king' ginte waqt red kings do baar mat gino: 26 + 2 = 28, 30 nahi.

40.4 Mixed problems

- **Leap year** has 366 days = 52 weeks + 2 days. The 2 extra days can be (Sun, Mon), (Mon, Tue), ..., (Sat, Sun): 7 equally likely pairs. P(53 Sundays) $= \frac{2}{7}$; for a non-leap year (1 extra day) it is $\frac{1}{7}$.
- **Cards removed**: first find how many cards are left, then count in what is left.
- **Families with 3 children**: same 8 outcomes as 3 coins (B/G instead of H/T).



EXAM ALERT!

Two dice aur cards ke sawal board ke case study aur 2-3 mark section ke favourite hain. Grid aur deck table **answer sheet par bana do**, examiner ko counting dikhti hai aur step marks pakke ho jaate hain.

MEME SAMJHO — Astra meme, maths inside



DO DICE PHENKE AUR DOST BOLA: 'SUM 13 AAYA!'

Hain? Ye kaise ho gaya? Maximum to 6 + 6 = 12 hi hota hai!

Asli baat: the sum of two dice lies between 2 and 12, so P(sum 13) = 0, an impossible event.



SHIVAM BHAIYA SAYS: BOARD TIP

Dice ke sawal mein favourable outcomes ko **pairs mein likh do**, jaise (2,6), (3,5)... Sirf '5 outcomes' likhne par examiner step ka mark kaat sakta hai.

Day 40 Practice

16 board-pattern questions | solve first, then check

MCQ

1 Three coins are tossed together. The probability of getting at most one head is (1)

(a) $\frac{3}{8}$ (b) $\frac{1}{8}$ (c) $\frac{7}{8}$ (d) $\frac{1}{2}$

MCQ

2 Two dice are thrown simultaneously. The probability of getting a doublet is (1)

(a) $\frac{1}{36}$ (b) $\frac{1}{6}$ (c) $\frac{1}{3}$ (d) $\frac{5}{6}$

MCQ

3 A card is drawn at random from a well-shuffled deck of 52 cards. The probability that it is a black king is (1)

(a) $\frac{1}{13}$ (b) $\frac{1}{52}$ (c) $\frac{1}{26}$ (d) $\frac{3}{26}$

MCQ

4 A die has its six faces marked P, Q, R, P, S, P. It is thrown once. The probability of getting P is (1)

(a) $\frac{1}{6}$ (b) $\frac{1}{3}$ (c) $\frac{2}{3}$ (d) $\frac{1}{2}$

MCQ

5 Rohan tosses a coin three times and wins only if he gets exactly two heads. The probability that he loses is (1)

(a) $\frac{3}{8}$ (b) $\frac{5}{8}$ (c) $\frac{1}{4}$ (d) $\frac{3}{4}$

ASSERTION-REASON

6 **Assertion (A):** The probability that a leap year chosen at random has 53 Sundays is $\frac{2}{7}$.

Reason (R): The probability that a non-leap year chosen at random has 53 Sundays is $\frac{2}{7}$. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

7 **Assertion (A):** When two coins are tossed together, the probability of getting at least one head is $\frac{3}{4}$.

Reason (R): When two coins are tossed, the outcomes HH, HT, TH, TT are equally likely. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

CASE-BASED

8 In a board-game evening, two friends throw a pair of dice in every round and score points depending on the numbers that come up. Use the 36-outcome sample space. (4)

(i) Find the probability that the difference of the two numbers is 2. (1)

(ii) Find the probability that the product of the numbers is greater than 20. (1)

(iii) Find the probability that 5 does not appear on either die. (2)

(iii) OR Find the probability that the sum of the numbers is a prime number. (2)

VERY SHORT ANSWER

9 One card is drawn from a well-shuffled deck of 52 cards. Find the probability that it is neither a spade nor a queen. (2)

SHORT ANSWER

10 The five cards 9, 10, jack, queen and king of hearts are shuffled face down and one is picked. (i) Find P(king). (ii) The king is drawn and kept aside. Another card is picked from the rest. Find P(it is the 9) and P(it is a king). (3)

SHORT ANSWER

11 Two dice are thrown at the same time. Find the probability that the sum of the numbers is greater than 8. (2)

SHORT ANSWER

12 All the queens, jacks and clubs are removed from a deck of 52 cards. A card is drawn at random from the remaining cards. Find the probability that it is (i) a face card (ii) a red card (iii) a black card. (3)

SHORT ANSWER

13 Farah and Sheena each have four tokens numbered 1, 2, 3, 4. Each picks one of her own tokens at random. Farah wins if the sum of the two numbers is prime, otherwise Sheena wins. Who has the better chance? Justify. (3)

LONG ANSWER

14 A family has three children. Assuming a boy and a girl are equally likely, find the probability that the family has (i) exactly two girls (ii) at least one boy (iii) at most one girl. (4)

LONG ANSWER

15 Two dice are thrown together. Find the probability that (i) the sum is 7 (ii) the product is 6 (iii) at least one die shows 5 (iv) the sum is a multiple of 4. (5)

MCQ

16 A card is drawn from a deck of 52 cards. The probability that it is a face card or a black card is (1)

(a) $\frac{8}{13}$ **(b)** $\frac{19}{26}$ **(c)** $\frac{7}{13}$ **(d)** $\frac{9}{13}$

Day 40 Answers with worked solutions

1 **(d)**
At most one head: TTT, HTT, THT, TTH = 4 of 8.
$P = \frac{4}{8} = \frac{1}{2}$

2 **(b)**
Doublets: (1,1) to (6,6) = 6 outcomes.
$P = \frac{6}{36} = \frac{1}{6}$

3 **(c)**
Black kings: king of spades, king of clubs = 2.
$P = \frac{2}{52} = \frac{1}{26}$

4 **(d)**
P appears on 3 of the 6 faces.
$P = \frac{3}{6} = \frac{1}{2}$

5 **(b)**
Exactly two heads: HHT, HTH, THH = 3 of 8.
P(lose) $= 1 - \frac{3}{8} = \frac{5}{8}$

6 **(c)**
Leap year: 2 extra days, 7 equally likely pairs, 2 contain Sunday: A true.
Non-leap year: 1 extra day, so $P = \frac{1}{7}$: R false.

7 **(a)**
At least one head: HH, HT, TH = 3 of 4 equally likely outcomes, so A is true and R is exactly why.

8 **Ans:** $\frac{2}{9}$

Difference 2: (1,3), (3,1), (2,4), (4,2), (3,5), (5,3), (4,6), (6,4) = 8.

$P = \frac{8}{36} = \frac{2}{9}$

Ans: $\frac{1}{6}$

Product > 20: (4,6), (6,4), (5,5), (5,6), (6,5), (6,6) = 6.

$P = \frac{6}{36} = \frac{1}{6}$

Ans: $\frac{25}{36}$

Each die can show 1, 2, 3, 4 or 6: $5 \times 5 = 25$ outcomes.

$P = \frac{25}{36}$

OR: Ans: $\frac{5}{12}$

Prime sums: 2 (1 way), 3 (2), 5 (4), 7 (6), 11 (2) = 15.

$P = \frac{15}{36} = \frac{5}{12}$

9 **Ans:** $\frac{9}{13}$

Spade or queen $= 13 + 3 = 16$ (queen of spades counted once).

Neither $= 52 - 16 = 36$, so $P = \frac{36}{52} = \frac{9}{13}$.

10 **Ans:** (i) $\frac{1}{5}$ (ii) $\frac{1}{4}$, 0

(i) One king among 5 cards: $P = \frac{1}{5}$.

(ii) 4 cards remain: $P(9) = \frac{1}{4}$.

No king is left, so $P(\text{king}) = 0$.

11 **Ans:** $\frac{5}{18}$

Sum 9: 4 ways, 10: 3, 11: 2, 12: 1, total 10.

$P = \frac{10}{36} = \frac{5}{18}$

12 **Ans:** (i) $\frac{1}{11}$ (ii) $\frac{2}{3}$ (iii) $\frac{1}{3}$

Removed: 4 queens + 4 jacks + 13 clubs - 2 (queen and jack of clubs counted twice) = 19; left = 33.

(i) Face cards left: kings of spades, hearts, diamonds = 3, $P = \frac{3}{33} = \frac{1}{11}$.

(ii) Hearts left 11, diamonds left 11: $P = \frac{22}{33} = \frac{2}{3}$.

(iii) Black cards left are spades only: 11, $P = \frac{11}{33} = \frac{1}{3}$.

13 **Ans:** Farah, $\frac{9}{16}$ against $\frac{7}{16}$

Total outcomes $= 4 \times 4 = 16$.

Prime sums: 2 (1 way), 3 (2), 5 (4), 7 (2) = 9, so P(Farah) $= \frac{9}{16}$.

P(Sheena) $= 1 - \frac{9}{16} = \frac{7}{16}$, so Farah has the better chance.

14 **Ans:** (i) $\frac{3}{8}$ (ii) $\frac{7}{8}$ (iii) $\frac{1}{2}$

Sample space: BBB, BBG, BGB, GBB, BGG, GBG, GGB, GGG (8).

(i) Exactly two girls: BGG, GBG, GGB, $P = \frac{3}{8}$.

(ii) At least one boy $= 1 - P(GGG) = 1 - \frac{1}{8} = \frac{7}{8}$.

(iii) At most one girl: BBB, BBG, BGB, GBB, $P = \frac{4}{8} = \frac{1}{2}$.

15 **Ans:** (i) $\frac{1}{6}$ (ii) $\frac{1}{9}$ (iii) $\frac{11}{36}$ (iv) $\frac{1}{4}$

Total 36 outcomes.

(i) (1,6), (2,5), (3,4), (4,3), (5,2), (6,1): $\frac{6}{36} = \frac{1}{6}$

(ii) (1,6), (2,3), (3,2), (6,1): $\frac{4}{36} = \frac{1}{9}$

(iii) No 5 on either die: 25 outcomes, so $1 - \frac{25}{36} = \frac{11}{36}$

(iv) Sum 4 (3 ways), 8 (5 ways), 12 (1 way) = 9: $\frac{9}{36} = \frac{1}{4}$

16 **(a)**

Black cards 26 + red face cards 6 = 32.

$P = \frac{32}{52} = \frac{8}{13}$

TONIGHT'S 2-MINUTE RECALL Answer from memory, then check the notes

1. Outcomes with 3 coins : 8
2. Outcomes with 2 dice : 36
3. Most likely sum with two dice : 7 (6 ways)
4. Face cards in a deck : 12 (K, Q, J of each suit)
5. Is an ace a face card? : No
6. P(at least one 6 with two dice) : 11/36
7. P(53 Sundays in a leap year) : 2/7
8. P(sum 13 with two dice) : 0

DAY 40 COMPLETE? ☐ Notes read ☐ 16 questions solved ☐ Answers checked ☐ Mistakes noted

Next: Day 41, Revision 7: Statistics + Probability

DAY 41

REVISION 7 | STATISTICS + PROBABILITY

Statistics + Probability

Revision 7

Aaj ka target

- [ ] Mean, mode, median ke teeno formulas likh ke yaad karo
- [ ] Ek hi table se teeno nikaalne ki practice
- [ ] Probability ke 'at least' aur 'removed' wale sawal
- [ ] 20 sawal time laga ke, galtiyan note karo

41.1 Statistics in one box

GROUPED DATA FORMULAS

Class mark $x_i = \dfrac{\text{lower limit} + \text{upper limit}}{2}$

Direct method: $\overline{x} = \dfrac{\sum f_i x_i}{\sum f_i}$

Assumed mean: $\overline{x} = a + \dfrac{\sum f_i d_i}{\sum f_i}$, $d_i = x_i - a$

Mode $= l + \left(\dfrac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) h$

Median $= l + \left(\dfrac{\frac{n}{2} - cf}{f}\right) h$

$3\,\text{Median} = \text{Mode} + 2\,\text{Mean}$

- **Modal class**: the class with the highest frequency; f_0, f_2 are the frequencies of the classes just before and after it.
- **Median class**: the first class whose cumulative frequency is $\geq \dfrac{n}{2}$; cf is the cumulative frequency of the class **before** it.
- Classes must be continuous (exclusive form) before using the mode or median formula.

SOLVED EXAMPLE 1 (ONE TABLE, THREE ANSWERS)

Find the mean, mode and median of the data: classes 0-10, 10-20, 20-30, 30-40, 40-50 with frequencies 5, 8, 12, 10, 5.

SOLUTION

1. Class marks 5, 15, 25, 35, 45; $\sum f = 40$, $\sum fx = 25 + 120 + 300 + 350 + 225 = 1020$.
2. Mean $= \dfrac{1020}{40} = 25.5$
3. Modal class 20-30: $l = 20, f_1 = 12, f_0 = 8, f_2 = 10, h = 10$. Mode $= 20 + \dfrac{4}{24 - 18} \times 10 = 20 + 6.67 = 26.67$
4. Cumulative frequencies 5, 13, 25, 35, 40; $\dfrac{n}{2} = 20$, so median class 20-30 with $cf = 13$, $f = 12$.
5. Median $= 20 + \dfrac{20 - 13}{12} \times 10 = 20 + 5.83 = 25.83$

Answer: Mean 25.5, Mode 26.67, Median 25.83 (approx.)



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Median formula mein cf **median class se pehle wali class** ka cumulative frequency hai, median class ka nahi. Aur mode mein $2f_1 - f_0 - f_2$, yahan sign ki galti sabse common hai.

41.2 Probability in one box

PROBABILITY

$P(E) = \dfrac{\text{favourable outcomes}}{\text{total outcomes}}$, $0 \leq P(E) \leq 1$

$P(\overline{E}) = 1 - P(E)$

Coins 2^n; two dice 36; deck 52 (26 red, 26 black, 12 face cards, 4 aces)

SOLVED EXAMPLE 2

A bag has 16 balls, some white and the rest red. P(white) $= \dfrac{3}{4}$. (i) How many red balls are there? (ii) How many white balls must be added so that P(white) becomes $\dfrac{5}{6}$?

SOLUTION

1. (i) White $= \dfrac{3}{4} \times 16 = 12$, red $= 4$.
2. (ii) Add w white balls: $\dfrac{12 + w}{16 + w} = \dfrac{5}{6}$
3. $72 + 6w = 80 + 5w \Rightarrow w = 8$

Answer: (i) 4 red (ii) 8 white balls



SHIVAM BHAIYA SAYS

Balls **add ya remove** hue to total bhi badlega. Equation banao: (naya favourable) / (naya total) = di hui probability, phir cross multiply. Do line mein khatam.



EXAM ALERT!

Statistics + Probability milke board mein lagbhag **11 marks**: grouped data ka ek 5-mark (missing frequency ya median) aur probability ka case study bahut common hai.

MEME SAMJHO

Astra meme, maths inside



JAB 'AT LEAST ONE' WALA LAMBA COUNTING SAWAL AAYE:

Sab gino mat. 1 minus P(ek bhi nahi) kar do!

Asli baat: P(at least one) = 1 - P(none). With two dice, P(at least one 6) = 1 - 25/36 = 11/36.

Day 41 Practice

18 board-pattern questions | solve first, then check

MCQ

1 For a distribution, mode = 18 and median = 15. Using the empirical relation, the mean is (1)

(a) 16.5 **(b)** 13.5 **(c)** 12 **(d)** 21

MCQ

2 The class mark of the class 10-25 is (1)

(a) 15 **(b)** 12.5 **(c)** 17.5 **(d)** 35

MCQ

3 Using the assumed mean method with $a = 45$, $\sum f_i = 50$ and $\sum f_i d_i = -30$, the mean is (1)

(a) 44.4 **(b)** 45.6 **(c)** 44.6 **(d)** 39

MCQ

4 Cards numbered 5 to 40 are put in a box and one is drawn at random. The probability that its number is a multiple of 6 is (1)

(a) $\frac{1}{6}$ **(b)** $\frac{5}{36}$ **(c)** $\frac{1}{5}$ **(d)** $\frac{7}{36}$

MCQ

5 A bowl has 4 red and 3 blue marbles. Which of these changes increase the probability of drawing a red marble? (i) add one red marble (ii) remove one blue marble (iii) add one red and one blue marble (iv) remove one red marble (1)

(a) (i) and (iii) only **(b)** (i), (ii) and (iii)
(c) (i) and (ii) only **(d)** (ii) and (iv) only

ASSERTION-REASON

6 **Assertion (A):** The median class of a grouped distribution is the class whose cumulative frequency is just greater than or equal to $\frac{n}{2}$.

Reason (R): The median divides the data into two halves, each containing $\frac{n}{2}$ observations. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

7 **Assertion (A):** A bag has 7 white, 3 red and 2 blue balls. The probability of drawing a white ball is $\frac{7}{12}$.

Reason (R): For any event E, $P(E) = 1 - P(\overline{E})$. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

CASE-BASED

8 A spinning wheel at a fair is divided into 10 equal sectors carrying the numbers 3, 49, 6, 11, 4, 15, 25, 17, 36, 8. A player spins it once and the arrow stops on one number. (4)

(i) Find the probability that the number is prime. (1)

(ii) Find the probability that the number is a perfect square. (1)

(iii) Find the probability that the number is a multiple of 3. (2)

(iii) OR Find the probability that the number is odd. (2)

CASE-BASED

9 The daily study hours (in minutes) of 40 students were grouped: 0-10: 5, 10-20: 8, 20-30: 12, 30-40: 10, 40-50: 5. (Use the table from Solved Example 1 only as a check; work it yourself.) (4)

(i) What is the modal class? (1)

(ii) What is the median class? (1)

(iii) Find the mean study time. (2)

VERY SHORT ANSWER

10 A box has only blue, black and green marbles. P(blue) $= \frac{1}{4}$, P(black) $= \frac{1}{3}$ and there are 10 green marbles. Find the total number of marbles. (2)

SHORT ANSWER

11 A lot of 180 pens contains 24 defective pens. Neha buys a pen only if it is good. One pen is taken out at random. Find the probability that she buys it. (2)

SHORT ANSWER

12 Find the mean of the data by the assumed mean method: classes 10-20, 20-30, 30-40, 40-50, 50-60 with frequencies 6, 10, 14, 12, 8. (3)

SHORT ANSWER

13 The mean of the following distribution is 54. Find the missing frequency p: classes 0-20, 20-40, 40-60, 60-80, 80-100 with frequencies 7, p, 10, 9, 13. (3)

SHORT ANSWER

14 Find the mode of the data: classes 0-10, 10-20, 20-30, 30-40, 40-50 with frequencies 5, 8, 12, 10, 5. (3)

LONG ANSWER

15 The median of the following data of 50 observations is 27.5. Find x and y: classes 0-10, 10-20, 20-30, 30-40, 40-50, 50-60 with frequencies 4, x, 16, 12, y, 6. (5)

LONG ANSWER

16 In a survey of 200 students about their favourite sport: football 40, cricket 56, basketball 30, tennis 20, badminton 24, swimming 30. One student is chosen at random. Find the probability that the student likes (i) swimming or badminton (ii) neither football nor cricket (iii) cricket or tennis. (4)

MCQ

17 If $P(E) + P(\overline{E}) = k$, then $k^2 + 4$ equals (1)

(a) 4 **(b)** 5 **(c)** 6 **(d)** 8

SHORT ANSWER

18 A digital clock shows time in hours and minutes (00:00 to 23:59). Find the probability that the tens digit of the minutes shown at a random moment is 4. (2)

Day 41 Answers with worked solutions

1 **(b)**

$3(15) = 18 + 2\overline{x} \Rightarrow 2\overline{x} = 27 \Rightarrow \overline{x} = 13.5$

2 **(c)**

$\frac{10 + 25}{2} = 17.5$

3 **(a)**

$\overline{x} = 45 + \frac{-30}{50} = 45 - 0.6 = 44.4$

4 **(a)**

Total cards $= 36$; multiples of 6: 6, 12, 18, 24, 30, 36 = 6.

$P = \frac{6}{36} = \frac{1}{6}$

5 **(c)**

Now $P(R) = \frac{4}{7} \approx 0.571$.

(i) $\frac{5}{8} = 0.625$ up; (ii) $\frac{4}{6} = 0.667$ up; (iii) $\frac{5}{9} = 0.556$ down; (iv) $\frac{3}{6} = 0.5$ down.

6 **(a)**

Both true; the $\frac{n}{2}$-th observation lies in that class precisely because the median splits the data into two equal halves.

7 **(b)**

$P(W) = \frac{7}{12}$: A true. R is a true rule, but A is found directly by counting, not from R.

8 **Ans:** $\frac{3}{10}$

Primes: 3, 11, 17.

$P = \frac{3}{10}$

Ans: $\frac{2}{5}$

Perfect squares: 49, 4, 25, 36.

$P = \frac{4}{10} = \frac{2}{5}$

Ans: $\frac{2}{5}$

Multiples of 3: 3, 6, 15, 36 = 4.

$P = \frac{4}{10} = \frac{2}{5}$

OR: Ans: $\frac{3}{5}$

Odd: 3, 49, 11, 15, 25, 17 = 6.

$P = \frac{6}{10} = \frac{3}{5}$

9 **Ans:** 20-30

Highest frequency 12 belongs to 20-30.

Ans: 20-30

$\frac{n}{2} = 20$; cumulative frequencies 5, 13, 25: first ≥ 20 is 25, class 20-30.

Ans: 25.5 minutes

$\sum fx = 1020$, $\sum f = 40$

$\overline{x} = \frac{1020}{40} = 25.5$

10 **Ans:** 24

P(green) $= 1 - \frac{1}{4} - \frac{1}{3} = \frac{5}{12}$

$\frac{5}{12} \times N = 10 \Rightarrow N = 24$

11 **Ans:** $\frac{13}{15}$

Good pens $= 180 - 24 = 156$

$P = \frac{156}{180} = \frac{13}{15}$

12 **Ans:** 36.2

Class marks 15, 25, 35, 45, 55; take $a = 35$, so $d_i = -20, -10, 0, 10, 20$.

$\sum f_i d_i = -120 - 100 + 0 + 120 + 160 = 60$, $\sum f_i = 50$

$\overline{x} = 35 + \frac{60}{50} = 36.2$

13 **Ans:** $p = 11$

Class marks 10, 30, 50, 70, 90.

$\sum f = 39 + p$, $\sum fx = 70 + 30p + 500 + 630 + 1170 = 2370 + 30p$

$\frac{2370 + 30p}{39 + p} = 54 \Rightarrow 2370 + 30p = 2106 + 54p \Rightarrow 24p = 264 \Rightarrow p = 11$

14 **Ans:** 26.67 (approx.)

Modal class 20-30: $l = 20$, $f_1 = 12$, $f_0 = 8$, $f_2 = 10$, $h = 10$.

Mode $= 20 + \frac{12 - 8}{24 - 8 - 10} \times 10 = 20 + \frac{40}{6} = 26.67$

15 **Ans:** $x = 9$, $y = 3$

$4 + x + 16 + 12 + y + 6 = 50 \Rightarrow x + y = 12$

Median 27.5 lies in 20-30: $l = 20$, $f = 16$, $h = 10$, $cf = 4 + x$, $\frac{n}{2} = 25$.

$27.5 = 20 + \frac{25 - (4 + x)}{16} \times 10 \Rightarrow 7.5 \times 1.6 = 21 - x \Rightarrow 12 = 21 - x \Rightarrow x = 9$

$y = 12 - 9 = 3$

16 **Ans:** (i) $\frac{27}{100}$ (ii) $\frac{13}{25}$ (iii) $\frac{19}{50}$

Check total: $40 + 56 + 30 + 20 + 24 + 30 = 200$.

(i) $\frac{30 + 24}{200} = \frac{54}{200} = \frac{27}{100}$

(ii) $\frac{200 - 96}{200} = \frac{104}{200} = \frac{13}{25}$

(iii) $\frac{56 + 20}{200} = \frac{76}{200} = \frac{19}{50}$

17 **(b)**

$k = 1$, so $k^2 + 4 = 5$.

18 **Ans:** $\frac{1}{6}$

Tens digit of minutes can be 0, 1, 2, 3, 4 or 5, each equally likely (10 minutes each in every hour).

$P = \frac{1}{6}$

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Mean (direct) : sum of fx / sum of f
2. Mode formula : $l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$
3. Median formula : $l + \frac{n/2 - cf}{f} \times h$
4. cf in the median formula : cumulative frequency of the class before the median class
5. Empirical relation : 3 Median = Mode + 2 Mean
6. P(at least one) : 1 - P(none)
7. P(not E) : 1 - P(E)

DAY 41 COMPLETE?

- ☐ Both formula boxes read
- ☐ 18 questions solved
- ☐ Answers checked
- ☐ Mistakes noted

Next: Day 42, Formula Day: All 14 chapters on one sheet

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 42

FORMULA DAY | ALL 14 CHAPTERS

All 14 chapters on one sheet

Formula Day

Aaj ka target

- [ ] 14 chapters ke formulas padh ke likho
- [ ] Trig table 2 min mein khud banao
- [ ] 25 rapid quiz sawal bina dekhe
- [ ] Atka formula 3 baar likho

Every formula and key result of all 14 chapters (NCERT 2026-27 syllabus), chapter by chapter. Read it once fully today, then use it as your last-week sheet before the board exam. Cover the right side and say each formula aloud.

42.1 Ch 1 Real Numbers

REAL NUMBERS

FTA: every composite number is a product of primes, unique apart from order.

HCF = product of the **smallest** powers of common primes; LCM = product of the **greatest** powers of all primes.

$$\text{HCF}(a,b) \times \text{LCM}(a,b) = a \times b$$

(two numbers only)

If p is prime and p divides a^2, then p divides a. So $\sqrt{2}, \sqrt{3}, \sqrt{5}$ are irrational; rational + irrational = irrational.

42.2 Ch 2 Polynomials

ZEROS AND COEFFICIENTS

ax^2+bx+c: $\alpha+\beta = -\dfrac{b}{a}$, $\alpha\beta = \dfrac{c}{a}$

Quadratic with given zeros: $x^2 - (\alpha+\beta)x + \alpha\beta$

Graph of $y = p(x)$: number of zeros = number of points where it meets the x-axis (quadratic: at most 2).



SHIVAM BHAIYA SAYS

Sum mein **minus** b/a, product mein seedha c/a. Yaad rakho: '**Sum se pehle minus, product seedha**'.

42.3 Ch 3 Pair of Linear Equations

For $a_1x+b_1y+c_1=0$, $a_2x+b_2y+c_2=0$. Methods: graphical, substitution, elimination.

Ratios	Lines	Solutions
$\dfrac{a_1}{a_2} \neq \dfrac{b_1}{b_2}$	Intersecting	Exactly one (consistent)
$\dfrac{a_1}{a_2} = \dfrac{b_1}{b_2} = \dfrac{c_1}{c_2}$	Coincident	Infinitely many
$\dfrac{a_1}{a_2} = \dfrac{b_1}{b_2} \neq \dfrac{c_1}{c_2}$	Parallel	None (inconsistent)

42.4 Ch 4 Quadratic Equations

QUADRATIC FORMULA

$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}, \quad D = b^2-4ac$$

$D>0$: two distinct real roots; $D=0$: two equal roots $-\dfrac{b}{2a}$; $D<0$: no real roots

42.5 Ch 5 Arithmetic Progressions

AP

$a_n = a + (n-1)d$

$S_n = \dfrac{n}{2}[2a+(n-1)d] = \dfrac{n}{2}(a+l)$

$a_n = S_n - S_{n-1}$; three terms in AP: $a-d,\ a,\ a+d$

nth term from the end $= l-(n-1)d$



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

a_n aur S_n mein confuse mat hona: '**term**' pucha to a_n, '**sum / total**' pucha to S_n. Aur n hamesha positive integer hona chahiye.

42.6 Ch 6 Triangles

SIMILARITY

BPT: if $DE \parallel BC$ in $\triangle ABC$, then $\dfrac{AD}{DB} = \dfrac{AE}{EC}$.

Converse also true.

Similarity criteria: **AAA (AA), SSS, SAS**

Similar triangles: corresponding angles equal, corresponding sides in the same ratio.

42.7 Ch 7 Coordinate Geometry

DISTANCE AND SECTION

$$d = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

Distance from origin: $\sqrt{x^2+y^2}$

DISTANCE AND SECTION

Section $(m:n)$: $\left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}\right)$

Mid-point: $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$



SHIVAM BHAIYA SAYS

Section formula mein **cross multiply**: m jaata hai doosre point (x_2) ke saath, n pehle point (x_1) ke saath.

42.8 Ch 8 Introduction to Trigonometry

RATIOS AND IDENTITIES

$\sin\theta = \frac{P}{H}$, $\cos\theta = \frac{B}{H}$, $\tan\theta = \frac{P}{B}$

$\csc\theta = \frac{1}{\sin\theta}$, $\sec\theta = \frac{1}{\cos\theta}$, $\cot\theta = \frac{1}{\tan\theta}$, $\tan\theta = \frac{\sin\theta}{\cos\theta}$

$\sin^2\theta + \cos^2\theta = 1$

$1 + \tan^2\theta = \sec^2\theta$

$1 + \cot^2\theta = \csc^2\theta$

θ	sin	cos	tan
0°	0	1	0
30°	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
45°	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	1
60°	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
90°	1	0	not defined

Trig values table (cosec, sec, cot are the reciprocals)



SHIVAM BHAIYA SAYS

Sin table trick: $\sqrt{0}, \sqrt{1}, \sqrt{2}, \sqrt{3}, \sqrt{4}$ likho, sabko 2 se divide karo: 0, 1/2, $1/\sqrt{2}$, $\sqrt{3}/2$, 1. Cos = sin ki row **ulti**. Tan = sin / cos. 'Pandit Badri Prasad Har Har Bole': P/H, B/H, P/B.

42.9 Ch 9 Some Applications of Trigonometry

HEIGHTS AND DISTANCES

$\tan\theta = \frac{\text{height}}{\text{horizontal distance}}$

Angle of elevation (looking up) = angle of depression (looking down) between the same two points (alternate angles).

Useful: $\tan 30^\circ = \frac{1}{\sqrt{3}}$, $\tan 45^\circ = 1$, $\tan 60^\circ = \sqrt{3}$; $\sqrt{3} \approx 1.732$

42.10 Ch 10 Circles

TANGENTS

Tangent at any point is **perpendicular** to the radius through the point of contact.

Lengths of the two tangents from an external point are **equal**.

Tangent length from a point at distance d from centre: $\sqrt{d^2 - r^2}$

Number of tangents: inside point 0, on circle 1, outside 2.

42.11 Ch 11 Areas Related to Circles

SECTOR AND SEGMENT (ANGLE Θ, RADIUS R)

Area of sector $= \frac{\theta}{360^\circ} \times \pi r^2$

Length of arc $= \frac{\theta}{360^\circ} \times 2\pi r$

Area of minor segment = area of sector $-$ area of $\triangle$; for $\theta = 60^\circ$: triangle area $= \frac{\sqrt{3}}{4}r^2$, for $\theta = 90^\circ$: $\frac{1}{2}r^2$

Major sector / segment = $\pi r^2 -$ minor one; circumference $2\pi r$, area πr^2

42.12 Ch 12 Surface Areas and Volumes

Solid	CSA	TSA	Volume
Cuboid	$2h(l+b)$	$2(lb+bh+hl)$	lbh
Cube	$4a^2$	$6a^2$	a^3
Cylinder	$2\pi rh$	$2\pi r(r+h)$	$\pi r^2 h$
Cone	πrl	$\pi r(l+r)$	$\frac{1}{3}\pi r^2 h$
Sphere	$4\pi r^2$	$4\pi r^2$	$\frac{4}{3}\pi r^3$
Hemisphere	$2\pi r^2$	$3\pi r^2$	$\frac{2}{3}\pi r^3$

*Cone: slant height $l=\sqrt{r^2+h^2}$. Combined solid: add only the **visible** curved surfaces; volumes simply add.*



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Combination solid ki surface area mein joint wali base **mat jodo**, sirf bahar dikhne wali surfaces. Volume mein sab jod do.

42.13 Ch 13 Statistics

GROUPED DATA

Direct method: $\overline{x}=\frac{\sum f_i x_i}{\sum f_i}$

Assumed mean: $\overline{x}=a+\frac{\sum f_i d_i}{\sum f_i}$, $d_i=x_i-a$

Mode $=l+\left(\frac{f_1-f_0}{2f_1-f_0-f_2}\right)\times h$

Median $=l+\left(\frac{\frac{n}{2}-cf}{f}\right)\times h$

Empirical relation: $3\,\text{Median}=\text{Mode}+2\,\text{Mean}$



SHIVAM BHAIYA SAYS

Mode ke liye **sabse badi frequency** wali class, median ke liye **n/2 wali cumulative frequency** ki class. Median mein cf = us class se **pehle** wali class ki cumulative frequency.

42.14 Ch 14 Probability

PROBABILITY

$P(E)=\frac{\text{favourable outcomes}}{\text{total outcomes}}$, $0\le P(E)\le 1$

$P(\overline{E})=1-P(E)$; sure event 1, impossible event 0

Coins: 2^n outcomes; two dice: 36; deck: 52 cards, 12 face cards, 4 aces



EXAM ALERT!

Board paper ka bada hissa seedha formulas par chalta hai. Har formula ke saath **units** (cm, cm^2, cm^3) aur $\pi=\frac{22}{7}$ ya 3.14 jo question bole wahi lo.

MEME SAMJHO

Astra meme, maths inside



EXAM SE PEHLI RAAT, FORMULA SHEET KE SAATH:

Ek baar mic pe bol ke sunao: 'Volume of cone = 1/3 pi r squared h'. Bolke padha, pakka yaad!

Asli baat: saying a formula aloud and writing it once is a quick way to fix it in memory; cone volume is one third of the cylinder with the same base and height.

Day 42 Practice

5 quick questions | answer from memory

VERY SHORT ANSWER

1 Set A: Algebra (Ch 1 to 5) (Q1 to Q5)

(i) HCF of 84 and 126 is 42. Their LCM is

(ii) Sum and product of the zeros of $x^2-7x+12$

(iii) Condition for $a_1x+b_1y+c_1=0$, $a_2x+b_2y+c_2=0$ to have no solution

(iv) Discriminant of $2x^2-5x+3=0$ and nature of roots

(v) 10th term of the AP 3, 7, 11, ... (5)

VERY SHORT ANSWER

2 Set B: AP, Triangles, Coordinate Geometry (Q6 to Q10)

(i) Sum of the first 20 terms of the AP 3, 7, 11, ...

(ii) In $\triangle ABC$, $DE \parallel BC$, $AD = 2$, $DB = 3$, $AE = 4$. Find EC.

(iii) Distance between $(2, 3)$ and $(5, 7)$

(iv) Mid-point of $(2, 3)$ and $(8, 11)$

(v) Point dividing $(2, 1)$ and $(8, 7)$ internally in the ratio $1 : 2$ (5)

VERY SHORT ANSWER

3 Set C: Trigonometry (Ch 8, 9) (Q11 to Q15)

(i) $\sin^2 35^\circ + \cos^2 35^\circ$

(ii) $\sec^2\theta - \tan^2\theta$

(iii) Value of $\tan 60^\circ \times \cot 60^\circ$

(iv) $2\sin 30^\circ + \cos 60^\circ$

(v) A tower casts a shadow 30 m long when the sun's elevation is 60°. Height of the tower (5)

VERY SHORT ANSWER

4 Set D: Circles, Areas, Solids (Q16 to Q20)

(i) Angle between a tangent and the radius at the point of contact

(ii) Tangent length from a point 13 cm from the centre of a circle of radius 5 cm

(iii) Area of a sector of angle 60° in a circle of radius 7 cm ($\pi = \frac{22}{7}$)

(iv) Length of the arc of a sector of angle 90°, radius 14 cm

(v) TSA of a cylinder with $r = 7$ cm, $h = 10$ cm (5)

VERY SHORT ANSWER

5 Set E: Solids, Statistics, Probability (Q21 to Q25)

(i) Volume of a hemisphere of radius 21 cm

(ii) Slant height of a cone with $r = 6$ cm, $h = 8$ cm

(iii) Mode = 24 and Mean = 21 for a data set. Median (empirical relation)

(iv) Probability of getting a prime number on one throw of a die

(v) If $P(E) = 0.27$, find $P(\text{not } E)$ (5)

Day 42 Answers with worked solutions

1 **Ans:** (i) 252; (ii) 7 and 12; (iii) $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$; (iv) $D = 1$, two distinct real roots; (v) 39

(i) $\text{LCM} = \frac{84 \times 126}{42} = 252$

(ii) $-\frac{b}{a} = 7$, $\frac{c}{a} = 12$

(iii) Parallel lines never meet.

(iv) $D = 25 - 24 = 1 > 0$

(v) $a_{10} = 3 + 9 \times 4 = 39$

2 **Ans:** (i) 820; (ii) 6; (iii) 5; (iv) $(5, 7)$; (v) $(4, 3)$

(i) $S_{20} = 10\,[6 + 19 \times 4] = 820$

(ii) BPT: $\frac{2}{3} = \frac{4}{EC} \Rightarrow EC = 6$

(iii) $\sqrt{3^2 + 4^2} = 5$

(iv) $\left(\frac{10}{2}, \frac{14}{2}\right)$

(v) $\left(\frac{8+4}{3}, \frac{7+2}{3}\right)$

3 **Ans:** (i) 1; (ii) 1; (iii) 1; (iv) $\frac{3}{2}$; (v) $30\sqrt{3}$ m

(i) Identity $\sin^2\theta + \cos^2\theta = 1$

(ii) From $1 + \tan^2\theta = \sec^2\theta$

(iii) $\cot\theta = \frac{1}{\tan\theta}$

(iv) $2 \times \frac{1}{2} + \frac{1}{2}$

(v) $h = 30\tan 60^\circ = 30\sqrt{3}$

4 **Ans:** (i) 90°; (ii) 12 cm; (iii) $\frac{77}{3}$ cm^2; (iv) 22 cm; (v) 748 cm^2

(i) Tangent is perpendicular to the radius.

(ii) $\sqrt{169 - 25} = 12$

(iii) $\frac{60}{360} \times \frac{22}{7} \times 49 = \frac{77}{3}$

(iv) $\frac{1}{4} \times 2 \times \frac{22}{7} \times 14 = 22$

(v) $2 \times \frac{22}{7} \times 7 \times 17 = 748$

5 **Ans:** (i) 19404 cm^3; (ii) 10 cm; (iii) 22; (iv) $\frac{1}{2}$; (v) 0.73

(i) $\frac{2}{3} \times \frac{22}{7} \times 9261 = 19404$

(ii) $l = \sqrt{36 + 64} = 10$

(iii) $3M = 24 + 42 = 66 \Rightarrow M = 22$

(iv) Primes 2, 3, 5: $\frac{3}{6}$

(v) $1 - 0.27$

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. HCF x LCM : product of the two numbers
2. Sum of zeros of $ax^2 + bx + c$: $-\frac{b}{a}$
3. nth term of an AP : $a + (n-1)d$
4. Distance formula : $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
5. $\sin^2\theta + \cos^2\theta$: 1
6. Area of a sector : $\frac{\theta}{360^\circ} \times \pi r^2$
7. Volume of a cone : $\frac{1}{3}\pi r^2 h$
8. Empirical relation : 3 Median = Mode + 2 Mean

DAY 42 COMPLETE?

- [ ] Formula sheet read
- [ ] 25 quiz questions done
- [ ] Trig table written from memory
- [ ] Weak formulas noted

Next: Day 43, Board Practice: Number Systems + Algebra I (Ch 1 to 3)

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 43

BOARD PRACTICE 1 | CH 1 TO 3

Number Systems + Algebra I (Ch 1 to 3)

Board Practice 1

Aaj ka target

- [ ] Board pattern table padho, traps yaad karo
- [ ] 29 board-style sawal 90 min mein, bina chapter khole
- [ ] Har galti ko trap ke naam ke saath mistakes page pe likho
- [ ] Proof aur graph wale sawal poore steps mein likho

43.1 Board pattern for this unit

Number Systems carries **6 marks** and Algebra (Ch 2 to 5) carries **20 marks** out of 80. Real Numbers, Polynomials and Linear Equations together usually bring 14 to 16 of those 26 marks.

Chapter	What the board asks	Typical marks
Real Numbers	HCF/LCM MCQ, HCF × LCM, word problem (HCF or LCM), irrationality proof	1 + 2 + 3
Polynomials	sum/product of zeroes, find k, form the polynomial, zeroes from a graph, AR	1 + 1 + 2 or 3
Linear Equations	condition on k (unique / none / infinite), word problem, graphical method	1 + 3 + 5 or case

Ch 1 to 3 in the board paper

- **Trap 1:** HCF uses the **smallest** powers of common primes; LCM uses the **largest** powers of all primes.
- **Trap 2:** HCF × LCM = product works for **two** numbers only, never for three.
- **Trap 3:** 'No solution' needs $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$: always check the third ratio after finding k.
- **Trap 4:** Sum of zeroes is $-\frac{b}{a}$ (minus sign!), product is $\frac{c}{a}$.
- **Trap 5:** In irrationality proofs, write 'assume rational', the rearranged form and 'contradiction' clearly: each line carries a mark.



SHIVAM BHAIYA SAYS: BOARD TIP

Word problem mein pehle **'Let ... be x'** likho, phir equations ko (1), (2) number do aur end mein **units ke saath answer** box karo. Board ki marking scheme mein equation banane ka alag mark hota hai, isliye sirf answer likhne se poore marks nahi milte.



EXAM ALERT!

Graph wale 5-marker mein dono lines ki table (kam se kam 2-3 points) aur intersection point ke coordinates likhna zaroori hai. Bina table ke seedha graph = marks katenge.

43.2 How to attempt today

Sit for **about 90 minutes** without the chapters. Attempt Sections A to E in order, mark the questions you guessed, then check the worked solutions. Every wrong answer goes on your mistakes page with the trap it fell into.

Day 43 Practice

29 board-pattern questions | solve first, then check

SECTION A | MCQ AND ASSERTION-REASON (1 MARK EACH)

MCQ

1 The smallest natural number that is exactly divisible by every integer from 1 to 12 is (1)

(a) 2520 **(b)** 5040 **(c)** 27720 **(d)** 55440

MCQ

2 If $a = p^4q^2$ and $b = p^2q^5$, where p, q are primes, and $\text{HCF}(a,b) = p^mq^n$, $\text{LCM}(a,b) = p^rq^s$, then $(m+n)(r+s)$ equals (1)

(a) 30 **(b)** 36 **(c)** 40 **(d)** 45

MCQ

3 If $x = m^2n^3$ and $y = m^4n$, where m, n are primes, then HCF(x,y) is (1)

(a) m^2n **(b)** m^4n^3 **(c)** mn **(d)** m^2n^3

MCQ

4 The HCF of 28 and 77 can be written as $28m - 77$. The value of m is (1)

(a) 1 **(b)** 2 **(c)** 4 **(d)** 3

MCQ

5 If $a = 2^3 \times 3^2 \times 5$ and $b = 2^2 \times 3^4 \times 7$, then HCF (a,b) is (1)

(a) 36 **(b)** 72 **(c)** 180 **(d)** 12

MCQ

6 For the polynomial $kx^2 + 6x + 2k$, the sum of the zeroes equals their product. Then k is (1)

(a) 3 (b) -2 (c) -3 (d) $\frac{1}{3}$

MCQ

7 A quadratic polynomial whose zeroes are 3 and -7 is (1)

(a) $x^2 - 4x - 21$ (b) $x^2 + 4x - 21$

(c) $x^2 + 4x + 21$ (d) $x^2 - 10x + 21$

MCQ

8 The pair $2x + 3y = 7$ and $6x + ky - 4 = 0$ has a unique solution when (1)

(a) $k = 9$ (b) $k \neq 6$ (c) $k = 6$ (d) $k \neq 9$

MCQ

9 The pair of equations $kx + 2y = 3$ and $8x + ky = 5$ has no solution for (1)

(a) $k = \pm 4$ (b) $k = 4$ only

(c) $k = -4$ only (d) no value of k

MCQ

10 The pair $2x + y = 3$ and $(3k - 1)x + (k + 1)y = 2k + 1$ is inconsistent when k equals (1)

(a) 1 (b) 2 (c) 3 (d) -3

ASSERTION-REASON

11 **Assertion (A):** $2058 = 2 \times 3 \times 7^3$.

Reason (R): A composite number has at least one factor other than 1 and itself. (1)

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

ASSERTION-REASON

12 **Assertion (A):** The polynomial $x^2 + 2x + 7$ has two real zeroes.

Reason (R): A quadratic polynomial can have at most two zeroes. (1)

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

ASSERTION-REASON

13 **Assertion (A):** The pair $2x - 3y = 4$ and $4x - 6y = 8$ has a unique solution.

Reason (R): If $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$, the pair of linear equations is inconsistent. (1)

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

SECTION B | VERY SHORT ANSWER (2 MARKS EACH)

VERY SHORT ANSWER

14 The product of two numbers is 20160 and their HCF is 12. Find their LCM. Also give one possible pair of such numbers. (2)

VERY SHORT ANSWER

15 The HCF of two numbers is 18 and their LCM is 540. If one number is 90, find the other. (2)

VERY SHORT ANSWER

16 Simplify $\frac{3\sqrt{12} + 2\sqrt{27}}{4\sqrt{3}}$ and state whether the result is rational or irrational. (2)

VERY SHORT ANSWER

17 If α and β are the zeroes of $x^2 - 6x + 2$, find the value of $\frac{1}{\alpha} + \frac{1}{\beta} - \alpha\beta$. (2)

VERY SHORT ANSWER

18 Find the value of k for which the pair $x + 3y = 4$ and $(k - 3)x + (k + 1)y = k + 3$ has infinitely many solutions. (2)

SECTION C | SHORT ANSWER (3 MARKS EACH)

SHORT ANSWER

19 Find the HCF and LCM of 336 and 54 by prime factorisation and verify that HCF $\times$ LCM $=$ product of the numbers. (3)

SHORT ANSWER

20 Given that $\sqrt{3}$ is irrational, prove that $5 - 2\sqrt{3}$ is irrational. (3)

SHORT ANSWER

21 Two sections of 156 and 208 students are to be seated in rows. Every row must have the same number of students and only students of one section. Find the maximum number of students in a row and the number of rows needed. (3)

SHORT ANSWER

22 **Find the zeroes of $4x^2 - 9x + 2$ and verify the relationship between the zeroes and the coefficients. Hence write a quadratic polynomial whose zeroes are the reciprocals of these zeroes.** (3)

SHORT ANSWER

23 **Find the zeroes of $6x^2 - 7x - 3$ and verify the relationship between the zeroes and its coefficients.** (3)

SHORT ANSWER

24 **In a two-digit number, the tens digit is twice the units digit. If 27 is subtracted from the number, the digits are reversed. Find the number.** (3)

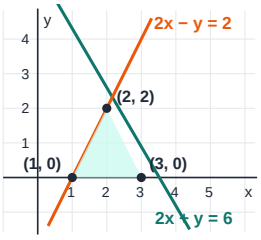
SECTION D | LONG ANSWER (5 MARKS EACH)

LONG ANSWER

25 **Two places A and B are 100 km apart on a highway. One car starts from A and another from B at the same time. If they move in the same direction, they meet in 10 hours. If they move towards each other, they meet in 1 hour. Find the speeds of the two cars.** (5)

LONG ANSWER

26 **Solve graphically: $2x + y = 6$ and $2x - y = 2$. Find the coordinates of the vertices of the triangle formed by these lines and the x-axis, and find its area.** (5)



LONG ANSWER

27 **If α and β are the zeroes of $p(x) = 2x^2 - 5x - 3$, then without finding the zeroes, find (i) $\alpha^2 + \beta^2$ and (ii) a quadratic polynomial whose zeroes are $2\alpha + 1$ and $2\beta + 1$.** (5)

SECTION E | CASE-BASED (4 MARKS EACH)

CASE-BASED

28 **Read the passage and answer the questions that follow. Three traffic signals on a city road change colour every 48 seconds, 72 seconds and 108 seconds respectively. All three changed together at 7:00:00 AM.** (4)

(i) Write 108 as a product of its prime factors. (1)

(ii) Find the HCF of 48 and 72. (1)

(iii) At what time will all three change together again? (2)

OR (iii) How many times after 7:00 AM will they change together within the next one hour? (2)

CASE-BASED

29 **Read the passage and answer the questions that follow. At a school fair, the Sharma family bought 3 adult and 2 child tickets for Rs 390, and the Verma family bought 2 adult and 3 child tickets for Rs 360. Let an adult ticket cost Rs x and a child ticket Rs y.** (4)

(i) Write the pair of linear equations for this situation. (1)

(ii) Is this pair consistent? Give a reason. (1)

(iii) Find the cost of each ticket. (2)

OR (iii) Find the total cost for 4 adults and 5 children. (2)

Day 43 Answers with worked solutions

1 **(c)**
LCM of 1 to 12 needs the highest prime powers up to 12: 2^3 (from 8), 3^2 (from 9), 5, 7, 11.
LCM $= 8 \times 9 \times 5 \times 7 \times 11 = 27720$.

2 **(b)**
HCF takes the smaller powers: p^2q^2, so $m + n = 4$.
LCM takes the larger powers: p^4q^5, so $r + s = 9$.
$(m + n)(r + s) = 4 \times 9 = 36$.

3 **(a)**
Common primes m and n at their smaller powers: m^2 and n^1.
HCF $= m^2n$.

4 **(d)**
$28 = 2^2 \times 7$, $77 = 7 \times 11$, so HCF $= 7$.
$28m - 77 = 7 \Rightarrow 28m = 84 \Rightarrow m = 3$.

5 **(a)**
Common primes are 2 and 3; smaller powers 2^2 and 3^2.
HCF $= 4 \times 9 = 36$.

6 **(c)**
Sum $= -\frac{6}{k}$, product $= \frac{2k}{k} = 2$.
$-\frac{6}{k} = 2 \Rightarrow k = -3$.

7 **(b)**
$p(x) = x^2 - (\text{sum})x + \text{product} = x^2 - (-4)x + (-21)$.
$= x^2 + 4x - 21$.

8 **(d)**

Unique solution needs $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$, i.e. $\frac{2}{6} \neq \frac{3}{k}$.

$\frac{1}{3} \neq \frac{3}{k} \Rightarrow k \neq 9$.

9 **(a)**

No solution: $\frac{k}{8} = \frac{2}{k} \neq \frac{3}{5}$.

$k^2 = 16 \Rightarrow k = \pm 4$.

Check $\frac{2}{k} \neq \frac{3}{5}$: for $k = 4$, $\frac{1}{2} \neq \frac{3}{5}$; for $k = -4$, $-\frac{1}{2} \neq \frac{3}{5}$. Both work, so $k = \pm 4$.

10 **(c)**

Inconsistent: $\frac{3k-1}{2} = \frac{k+1}{1} \neq \frac{2k+1}{3}$.

$3k - 1 = 2k + 2 \Rightarrow k = 3$.

Check: ratio $= \frac{8}{2} = 4$ and $\frac{2k+1}{3} = \frac{7}{3} \neq 4$. So $k = 3$.

11 **(b)**

A: $2 \times 3 \times 343 = 2058$, true.

R is a true statement (definition of composite), but it does not explain why this particular factorisation is correct; that comes from the FTA. So (b).

12 **(d)**

$D = b^2 - 4ac = 4 - 28 = -24 < 0$, so the graph never meets the x-axis: no real zeroes. A is false.

R is true. So (d).

13 **(d)**

Ratios: $\frac{2}{4} = \frac{-3}{-6} = \frac{4}{8} = \frac{1}{2}$: all equal, so the lines coincide and there are infinitely many solutions. A is false.

R is the correct condition for inconsistency, so R is true. Answer (d).

14 **Ans:** LCM $= 1680$; e.g. 48 and 420

HCF $\times$ LCM $=$ product of the two numbers.

LCM $= \frac{20160}{12} = 1680$.

Write the numbers as $12a, 12b$ with a, b coprime and $ab = \frac{20160}{144} = 140 = 4 \times 35$: so 48 and 420 (check: HCF 12, LCM 1680).

15 **Ans:** 108

Other number $= \frac{\text{HCF} \times \text{LCM}}{90} = \frac{18 \times 540}{90} = 108$.

Check: $90 = 2 \times 3^2 \times 5$, $108 = 2^2 \times 3^3$; HCF $= 2 \times 3^2 = 18$, LCM $= 2^2 \times 3^3 \times 5 = 540$. Correct.

16 **Ans:** 3, rational

$3\sqrt{12} = 3 \times 2\sqrt{3} = 6\sqrt{3}$ and $2\sqrt{27} = 2 \times 3\sqrt{3} = 6\sqrt{3}$.

Numerator $= 12\sqrt{3}$, so the value $= \frac{12\sqrt{3}}{4\sqrt{3}} = 3$.

3 can be written as $\frac{3}{1}$, so it is rational.

17 **Ans:** 1

$\alpha + \beta = 6$, $\alpha\beta = 2$.

$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha+\beta}{\alpha\beta} = \frac{6}{2} = 3$.

Required value $= 3 - 2 = 1$.

18 **Ans:** $k = 5$

Infinitely many solutions: $\frac{k-3}{1} = \frac{k+1}{3} = \frac{k+3}{4}$.

From the first two: $3k - 9 = k + 1 \Rightarrow k = 5$.

Check with the third: $\frac{5+3}{4} = 2$ and $\frac{5-3}{1} = 2$.

All ratios equal 2, so $k = 5$.

19 **Ans:** HCF 6, LCM 3024

$336 = 2^4 \times 3 \times 7$ and $54 = 2 \times 3^3$.

HCF $= 2^1 \times 3^1 = 6$ (common primes, smaller powers).

LCM $= 2^4 \times 3^3 \times 7 = 16 \times 27 \times 7 = 3024$.

Verify: $6 \times 3024 = 18144$ and $336 \times 54 = 18144$. Verified.

20 **Ans:** Proof

Assume, to the contrary, that $5 - 2\sqrt{3}$ is rational. Then $5 - 2\sqrt{3} = \frac{a}{b}$ where a, b are integers, $b \neq 0$.

Rearranging: $2\sqrt{3} = 5 - \frac{a}{b} \Rightarrow \sqrt{3} = \frac{5b-a}{2b}$.

Since a, b are integers, $\frac{5b-a}{2b}$ is rational, so $\sqrt{3}$ would be rational.

This contradicts the fact that $\sqrt{3}$ is irrational.

Hence $5 - 2\sqrt{3}$ is irrational.

21 **Ans:** 52 per row; 7 rows

The number per row must divide both 156 and 208, and be the largest such: HCF.

$156 = 2^2 \times 3 \times 13$, $208 = 2^4 \times 13$, so HCF $= 2^2 \times 13 = 52$.

Rows $= \frac{156}{52} + \frac{208}{52} = 3 + 4 = 7$.

22 **Ans:** Zeroes $2, \frac{1}{4}$; $k(2x^2 - 9x + 4)$

$4x^2 - 9x + 2 = 4x^2 - 8x - x + 2 = 4x(x-2) - 1(x-2) = (4x-1)(x-2)$. Zeroes: 2 and $\frac{1}{4}$.

Sum $= 2 + \frac{1}{4} = \frac{9}{4} = -\frac{(-9)}{4}$ and product $= \frac{1}{2} = \frac{2}{4}$. Verified.

Reciprocals: $\frac{1}{2}$ and 4. Sum $= \frac{9}{2}$, product $= 2$.

Polynomial: $x^2 - \frac{9}{2}x + 2$, i.e. $k(2x^2 - 9x + 4)$, $k \neq 0$.

23 **Ans:** $\frac{3}{2}, -\frac{1}{3}$

Split $-7x$: $6 \times (-3) = -18 = -9 \times 2$. So $6x^2 - 9x + 2x - 3 = 3x(2x-3) + 1(2x-3) = (3x+1)(2x-3)$.

Zeroes: $\frac{3}{2}$ and $-\frac{1}{3}$.

Sum $= \frac{3}{2} - \frac{1}{3} = \frac{7}{6} = -\frac{b}{a}$. Product $= -\frac{1}{2} = \frac{-3}{6} = \frac{c}{a}$. Verified.

24 **Ans:** 63

Let the units digit be u; tens digit $= 2u$.

Number $= 10(2u) + u = 21u$.

Reversed number $= 10u + 2u = 12u$.

$21u - 27 = 12u \Rightarrow 9u = 27 \Rightarrow u = 3$.

Number $= 63$. Check: $63 - 27 = 36$, which is 63 reversed.

25 **Ans:** 55 km/h and 45 km/h

Let the speeds be x km/h (car from A) and y km/h (car from B), $x > y$.

Same direction: the faster car gains 100 km in 10 h, so $10x - 10y = 100 \Rightarrow x - y = 10$ ...(1)

Towards each other: together they cover 100 km in 1 h, so $x + y = 100$ ...(2)

Adding (1) and (2): $2x = 110 \Rightarrow x = 55$. Then $y = 45$.

Speeds: 55 km/h and 45 km/h.

26 **Ans:** $x = 2, y = 2$; vertices $(3,0), (1,0), (2,2)$; area 2 sq units

Table for $2x + y = 6$: $(0,6), (3,0), (2,2)$. Table for $2x - y = 2$: $(0,-2), (1,0), (2,2)$.

Plot both lines (figure). They intersect at $(2,2)$, so $x = 2, y = 2$.

They meet the x-axis at $(3,0)$ and $(1,0)$.

Triangle vertices: $(1,0), (3,0), (2,2)$. Base $= 3 - 1 = 2$, height $= 2$.

Area $= \frac{1}{2} \times 2 \times 2 = 2$ square units.

27 **Ans:** (i) $\frac{37}{4}$ (ii) $k(x^2 - 7x)$

$\alpha + \beta = \frac{5}{2}$, $\alpha\beta = -\frac{3}{2}$.

(i) $\alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta = \frac{25}{4} + 3 = \frac{37}{4}$.

(ii) New sum $= 2(\alpha+\beta) + 2 = 5 + 2 = 7$.

New product $= (2\alpha+1)(2\beta+1) = 4\alpha\beta + 2(\alpha+\beta) + 1 = -6 + 5 + 1 = 0$.

Required polynomial: $x^2 - 7x + 0 = x^2 - 7x$, or $k(x^2 - 7x)$. (Check: zeroes of p are $3, -\frac{1}{2}$, giving 7 and 0.)

28 **Ans:** $2^2 \times 3^3$

$108 = 4 \times 27 = 2^2 \times 3^3$.

Ans: 24

$48 = 2^4 \times 3$, $72 = 2^3 \times 3^2$; HCF $= 2^3 \times 3 = 24$.

Ans: 7:07:12 AM

LCM$(48, 72, 108)$: $2^4 \times 3^3 = 432$ s.

432 s = 7 min 12 s, so next together at 7:07:12 AM.

OR: Ans: 8 times

They change together every 432 s.

$\frac{3600}{432} = 8.33$, so 8 times (at 432 s, 864 s, ..., 3456 s).

29 **Ans:** $3x + 2y = 390$, $2x + 3y = 360$

Sharma: $3x + 2y = 390$; Verma: $2x + 3y = 360$.

Ans: Yes, unique solution

$\frac{3}{2} \neq \frac{2}{3}$, so the lines intersect: consistent with a unique solution.

Ans: Adult Rs 90, child Rs 60

Add: $5x + 5y = 750 \Rightarrow x + y = 150$. Subtract: $x - y = 30$.

So $x = 90$, $y = 60$.

OR: Ans: Rs 660

With $x = 90$, $y = 60$ (solve as above): $4(90) + 5(60) = 360 + 300 = 660$.

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. HCF takes which powers of common primes? : the smallest
2. LCM of $2^3 3^2$ and $2^2 3^4$? : $2^3 3^4 = 648$
3. Sum and product of zeroes of $ax^2 + bx + c$? : $-b/a$ and c/a
4. Condition for infinitely many solutions? : $a_1/a_2 = b_1/b_2 = c_1/c_2$
5. Polynomial with zeroes 3 and -7? : $x^2 + 4x - 21$
6. $\alpha^2 + \beta^2$ in terms of sum and product? : $(\alpha+\beta)^2 - 2\alpha\beta$

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Two lines with $a_1/a_2 \neq b_1/b_2$ are? : intersecting (unique solution)

DAY 43 COMPLETE?

- [ ] Board pattern read
- [ ] 29 questions attempted
- [ ] Solutions checked
- [ ] Traps noted

Next: Day 44, Board Practice: Algebra II (Ch 4 and 5)

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 44

BOARD PRACTICE 2 | CH 4 AND 5

Algebra II (Ch 4 and 5)

Board Practice 2

Aaj ka target

- [ ] Discriminant aur a_n, S_n formula zubaani
- [ ] 29 board-style sawal 90 min mein
- [ ] Har word problem mein galat root ka reason likho
- [ ] Galtiyan trap ke naam ke saath note karo

44.1 Board pattern for this unit

Quadratic Equations and Arithmetic Progressions are the heavy end of the 20-mark Algebra unit. Together they usually bring **10 to 12 marks**: MCQs on the discriminant and nth term, one 3-mark word problem and very often a 5-marker or a case study.

Chapter	What the board asks	Typical marks
Quadratic Equations	nature of roots / find k (MCQ, AR), factorisation or formula, word problem (speed, stream, tap, area, ages)	1 + 2 + 5
Arithmetic Progressions	a_n, S_n, three terms in AP, 'which term', $S_n \to a_n$, case study (rows, savings, seats)	1 + 3 + case

Ch 4 and 5 in the board paper

- **Trap 1:** Equal roots means $D = 0$, real roots means $D \geq 0$ (not just $D > 0$).
- **Trap 2:** In word problems, reject the negative or impossible root and **write the reason** ('speed cannot be negative').
- **Trap 3:** $a_n = S_n - S_{n-1}$, not $S_n - S_{n+1}$; check with $a_1 = S_1$.
- **Trap 4:** Counting terms: number of terms $= \frac{l - a}{d} + 1$. The '+1' gets forgotten.
- **Trap 5:** With $d < 0$, put the minus sign in brackets: $11 \times (-1.5)$, not $11 - 1.5$.



SHIVAM BHAIYA SAYS: BOARD TIP

Quadratic word problem ka final step: dono roots likho, phir ek ko **reason ke saath reject** karo. Marking scheme mein equation banane, solve karne aur sahi root chunne ke alag-alag marks hote hain, toh teeno steps dikhne chahiye.



EXAM ALERT!

AP case study lagbhag har saal aata hai (rows, savings, seats). a_n aur S_n dono formula aur 'number of terms' wala trick ready rakho.

44.2 How to attempt today

Give yourself **90 minutes**. Do Section A in 15 minutes, then the word problems with full steps. Mark every question where you had to guess the method; those are your revision list for Day 55.

Day 44 Practice

29 board-pattern questions | solve first, then check

SECTION A | MCQ AND ASSERTION-REASON (1 MARK EACH)

MCQ

1 The values of k for which $3x^2 - kx + 3k = 0$ has two equal roots are (1)

(a) 0 only (b) 0 or 36 (c) 36 only (d) ± 6

MCQ

2 The values of y satisfying $y - 4 = \frac{21}{y}$ are (1)

(a) $-7, 3$ (b) $7, 3$ (c) $-7, -3$ (d) $7, -3$

MCQ

3 The equation $3x^2 - 2x + 5 = 0$ has (1)

(a) no real roots (b) two equal real roots
(c) two distinct real roots (d) more than two real roots

MCQ

4 If $3x^2 - 12x + c = 0$ has equal roots, then c is (1)

(a) 4 (b) 36 (c) 12 (d) -12

MCQ

5 If $x = 3$ is a root of $2x^2 - kx + m = 0$ and $m = 6$, then k equals (1)

(a) 8 (b) 6 (c) -8 (d) 4

MCQ

6 Two APs have the same common difference. Their first terms are 5 and -6. The difference between their 10th terms is (1)

(a) 99 (b) 11 (c) 1 (d) -1

MCQ

7 The 9th term of the AP $\sqrt{3}, 4\sqrt{3}, 7\sqrt{3}, \ldots$ is (1)

(a) $27\sqrt{3}$ (b) $24\sqrt{3}$ (c) $28\sqrt{3}$ (d) $25\sqrt{3}$

MCQ

8 In an AP, $d = -3$, $n = 7$ and $a_n = 0$. The first term a is (1)

(a) 21 (b) -18 (c) 18 (d) 15

MCQ

9 If $2p$, 13 and $4p + 2$ are three consecutive terms of an AP, then p is (1)

(a) 4 (b) 3 (c) 5 (d) $\frac{11}{3}$

MCQ

10 If $k + 2$, $4k - 6$ and $3k - 2$ are in AP, then k is (1)

(a) 2 (b) 3 (c) 4 (d) -3

ASSERTION-REASON

11 **Assertion (A):** For $3x^2 - 5x - 2 = 0$, the sum of the roots is $\frac{5}{3}$ and the product is $-\frac{2}{3}$.

Reason (R): For $ax^2 + bx + c = 0$, sum of roots $= -\frac{b}{a}$ and product $= \frac{c}{a}$. (1)

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

ASSERTION-REASON

12 **Assertion (A):** The equation $x^2 - 6x + 9 = 0$ has two distinct real roots.

Reason (R): A quadratic equation has two equal real roots when its discriminant is zero. (1)

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

ASSERTION-REASON

13 **Assertion (A):** -20 is a term of the AP $7, 4, 1, \ldots$

Reason (R): The nth term of an AP is $a_n = a + (n - 1)d$. (1)

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

SECTION B | VERY SHORT ANSWER (2 MARKS EACH)

VERY SHORT ANSWER

14 Find the nature of the roots of $2x^2 - 6x + 3 = 0$. If real, find them. (2)

VERY SHORT ANSWER

15 The 4th term of an AP is 11 and the 9th term is 26. Find the AP. (2)

VERY SHORT ANSWER

16 Find the sum of the first 12 terms of the AP $20, 18.5, 17, \ldots$ (2)

VERY SHORT ANSWER

17 Find the roots of $\sqrt{3}x^2 + 10x + 7\sqrt{3} = 0$ by factorisation. (2)

VERY SHORT ANSWER

18 How many three-digit numbers are divisible by 7? (2)

SECTION C | SHORT ANSWER (3 MARKS EACH)

SHORT ANSWER

19 A train covers 480 km at a uniform speed. Had the speed been 8 km/h more, it would have taken 2 hours less. Find the original speed. (3)

SHORT ANSWER

20 The sum of a number and its reciprocal is $\frac{26}{5}$. Find the number. (3)

SHORT ANSWER

21 The sum of the first n terms of an AP is $S_n = 3n^2 + 2n$. Find its nth term and the 15th term. (3)

SHORT ANSWER

22 Find the sum of all multiples of 7 lying between 100 and 300. (3)

SHORT ANSWER

23 Find the value of k for which $kx^2 - 2\sqrt{5}x + 4 = 0$ has equal roots. Also find the roots. (3)

SHORT ANSWER

24 The 4th term of an AP is zero. Prove that its 25th term is three times its 11th term. (3)

SECTION D | LONG ANSWER (5 MARKS EACH)

LONG ANSWER

25 The sum of the first 7 terms of an AP is 49 and the sum of the first 17 terms is 289. Find the sum of its first n terms. (5)

LONG ANSWER

26 **A motorboat has a speed of 15 km/h in still water. It goes 30 km upstream and returns to the starting point, taking 1 hour 30 minutes more upstream than downstream. Find the speed of the stream.** (5)

LONG ANSWER

27 **Two taps together fill a tank in 6 hours. The larger tap alone fills it 5 hours faster than the smaller tap. Find the time each tap takes alone.** (5)

SECTION E | CASE-BASED (4 MARKS EACH)

CASE-BASED

28 **Read the passage and answer the questions that follow. In a shoe shop, pairs of shoes are displayed in rows: 4 pairs in the first row, 7 in the second, 10 in the third, and so on, each row having 3 more pairs than the row before.** (4)

(i) How many pairs are there in the 15th row? (1)

(ii) What is the difference between the number of pairs in the 17th and 10th rows? (1)

(iii) How many rows are needed to display 246 pairs? (2)

OR (iii) How many pairs are there in the first 20 rows altogether? (2)

CASE-BASED

29 **Read the passage and answer the questions that follow. A rectangular community garden has an area of 640 m^2. Its length is 24 m more than its breadth. Let the breadth be x m.** (4)

(i) Write the quadratic equation for this situation. (1)

(ii) Find the discriminant and state the nature of the roots. (1)

(iii) Find the length and breadth of the garden. (2)

OR (iii) Find the perimeter of the garden. (2)

Day 44 Answers with worked solutions

1 **(b)**
Equal roots: $D = b^2 - 4ac = k^2 - 4(3)(3k) = k^2 - 36k = 0$.
$k(k - 36) = 0 \Rightarrow k = 0$ or $k = 36$.

2 **(d)**
Multiply by y ($y \neq 0$): $y^2 - 4y - 21 = 0$.
$(y - 7)(y + 3) = 0 \Rightarrow y = 7$ or $y = -3$.

3 **(a)**
$D = (-2)^2 - 4(3)(5) = 4 - 60 = -56 < 0$.
So it has no real roots.

4 **(c)**
Equal roots: $D = 144 - 12c = 0$.
$c = 12$.

5 **(a)**
Put $x = 3$: $2(9) - 3k + 6 = 0$.
$24 = 3k \Rightarrow k = 8$.

6 **(b)**
$a_{10} = a + 9d$ for both, with the same d.
Difference $= (5 + 9d) - (-6 + 9d) = 11$. (It is 11 for every term.)

7 **(d)**
$a = \sqrt{3}$, $d = 3\sqrt{3}$.
$a_9 = \sqrt{3} + 8(3\sqrt{3}) = 25\sqrt{3}$.

8 **(c)**
$a_n = a + (n - 1)d \Rightarrow 0 = a + 6(-3)$.
$a = 18$.

9 **(a)**
Middle term is the average: $2 \times 13 = 2p + (4p + 2)$.
$26 = 6p + 2 \Rightarrow p = 4$. (Terms 8, 13, 18.)

10 **(b)**
$2(4k - 6) = (k + 2) + (3k - 2) \Rightarrow 8k - 12 = 4k$.
$k = 3$ (terms 5, 6, 7).

11 **(a)**
Roots: $(3x + 1)(x - 2) = 0 \Rightarrow x = 2, -\frac{1}{3}$. Sum $\frac{5}{3}$, product $-\frac{2}{3}$: A true.
R is the general result that gives A directly. So (a).

12 **(d)**
$D = 36 - 36 = 0$, so the roots are equal ($x = 3, 3$), not distinct. A is false.
R is true. Answer (d).

13 **(a)**
$7 + (n - 1)(-3) = -20 \Rightarrow n - 1 = 9 \Rightarrow n = 10$, a natural number, so A is true.
A is checked using exactly the formula in R, so (a).

14 **Ans:** Distinct real; $\frac{3 \pm \sqrt{3}}{2}$
$D = 36 - 24 = 12 > 0$: two distinct real roots.
$x = \frac{6 \pm \sqrt{12}}{4} = \frac{6 \pm 2\sqrt{3}}{4} = \frac{3 \pm \sqrt{3}}{2}$.

15 **Ans:** 2, 5, 8, 11, ...
$a + 3d = 11$ and $a + 8d = 26$.
Subtract: $5d = 15 \Rightarrow d = 3$; then $a = 2$.
AP: 2, 5, 8, 11, ...

16 **Ans:** 141

$a = 20$, $d = -1.5$, $n = 12$.

$S_{12} = \frac{12}{2}[2(20) + 11(-1.5)] = 6(40 - 16.5) = 6 \times 23.5 = 141$.

17 **Ans:** $-\sqrt{3}$, $-\frac{7}{\sqrt{3}}$

Product $ac = \sqrt{3} \times 7\sqrt{3} = 21 = 3 \times 7$, and $3 + 7 = 10$.

$\sqrt{3}x^2 + 3x + 7x + 7\sqrt{3} = \sqrt{3}x(x + \sqrt{3}) + 7(x + \sqrt{3}) = (x + \sqrt{3})(\sqrt{3}x + 7)$.

Roots: $x = -\sqrt{3}$ and $x = -\frac{7}{\sqrt{3}} = -\frac{7\sqrt{3}}{3}$.

18 **Ans:** 128

Smallest: 105, largest: 994. AP with $a = 105$, $d = 7$.

$994 = 105 + (n-1)7 \Rightarrow n - 1 = 127 \Rightarrow n = 128$.

19 **Ans:** 40 km/h

Let the speed be x km/h. Time $= \frac{480}{x}$.

$\frac{480}{x} - \frac{480}{x+8} = 2 \Rightarrow \frac{480 \times 8}{x(x+8)} = 2$.

$x^2 + 8x - 1920 = 0 \Rightarrow (x + 48)(x - 40) = 0$.

Speed cannot be negative, so $x = 40$ km/h.

20 **Ans:** 5 or $\frac{1}{5}$

Let the number be x: $x + \frac{1}{x} = \frac{26}{5}$.

$5x^2 - 26x + 5 = 0 \Rightarrow 5x^2 - 25x - x + 5 = 0 \Rightarrow (5x - 1)(x - 5) = 0$.

$x = 5$ or $x = \frac{1}{5}$.

21 **Ans:** $a_n = 6n - 1$; $a_{15} = 89$

$a_n = S_n - S_{n-1} = 3n^2 + 2n - [3(n-1)^2 + 2(n-1)]$.

$= 3(2n - 1) + 2 = 6n - 1$.

$a_{15} = 6(15) - 1 = 89$. (Check: $a_1 = S_1 = 5$.)

22 **Ans:** 5586

Multiples: 105, 112, ..., 294. $a = 105$, $d = 7$, $l = 294$.

$294 = 105 + (n-1)7 \Rightarrow n = 28$.

$S = \frac{n}{2}(a + l) = 14 \times 399 = 5586$.

23 **Ans:** $k = \frac{5}{4}$; roots $\frac{4\sqrt{5}}{5}, \frac{4\sqrt{5}}{5}$

Equal roots: $D = (2\sqrt{5})^2 - 16k = 20 - 16k = 0 \Rightarrow k = \frac{5}{4}$.

Equal root $= -\frac{b}{2a} = \frac{2\sqrt{5}}{2k} = \frac{\sqrt{5}}{5/4} = \frac{4\sqrt{5}}{5}$.

24 **Ans:** Proof

$a_4 = a + 3d = 0 \Rightarrow a = -3d$.

$a_{25} = a + 24d = 21d$ and $a_{11} = a + 10d = 7d$.

So $a_{25} = 3 \times 7d = 3a_{11}$. Proved.

25 **Ans:** $S_n = n^2$

$S_7 = \frac{7}{2}(2a + 6d) = 49 \Rightarrow a + 3d = 7$ ...(1)

$S_{17} = \frac{17}{2}(2a + 16d) = 289 \Rightarrow a + 8d = 17$ ...(2)

Subtract: $5d = 10 \Rightarrow d = 2$, so $a = 1$.

$S_n = \frac{n}{2}[2 + (n-1)2] = \frac{n}{2}(2n) = n^2$.

26 **Ans:** 5 km/h

Let the stream speed be s km/h. Upstream speed $15 - s$, downstream $15 + s$.

$\frac{30}{15-s} - \frac{30}{15+s} = \frac{3}{2} \Rightarrow \frac{60s}{225 - s^2} = \frac{3}{2}$.

$120s = 675 - 3s^2 \Rightarrow s^2 + 40s - 225 = 0$.

$(s + 45)(s - 5) = 0$; speed is positive, so $s = 5$ km/h.

27 **Ans:** Smaller 15 h, larger 10 h

Let the smaller tap take x h; the larger takes $x - 5$ h.

In 1 hour: $\frac{1}{x} + \frac{1}{x-5} = \frac{1}{6}$.

$6(2x - 5) = x(x - 5) \Rightarrow x^2 - 17x + 30 = 0 \Rightarrow (x - 15)(x - 2) = 0$.

$x = 2$ gives $x - 5 < 0$, rejected. So $x = 15$: smaller 15 h, larger 10 h.

28 **Ans:** 46

$a_{15} = 4 + 14 \times 3 = 46$.

Ans: 21

$a_{17} - a_{10} = 7d = 21$.

Ans: 12 rows

$S_n = \frac{n}{2}[8 + 3(n-1)] = \frac{n(3n+5)}{2} = 246$.

$3n^2 + 5n - 492 = 0 \Rightarrow (n - 12)(3n + 41) = 0 \Rightarrow n = 12$.

OR: Ans: 650

$S_{20} = \frac{20}{2}[8 + 19 \times 3] = 10 \times 65 = 650$.

29 **Ans:** $x^2 + 24x - 640 = 0$

$x(x + 24) = 640$.

Ans: 3136; two distinct real roots

$D = 24^2 + 4 \times 640 = 576 + 2560 = 3136 = 56^2 > 0$.

Ans: Breadth 16 m, length 40 m

$x = \frac{-24 \pm 56}{2} = 16$ or -40.

Breadth cannot be negative: 16 m; length 40 m.

OR: Ans: 112 m

Solving as above, breadth 16 m and length 40 m.

Perimeter $= 2(40 + 16) = 112$ m.

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Discriminant of $ax^2+bx+c=0$? : b^2-4ac
2. Equal roots condition? : $D=0$
3. Quadratic formula? : $x=\dfrac{-b\pm\sqrt{D}}{2a}$
4. nth term of an AP? : $a+(n-1)d$
5. Sum of n terms? : $\dfrac{n}{2}[2a+(n-1)d]$ or $\dfrac{n}{2}(a+l)$
6. a_n from S_n? : S_n-S_{n-1}
7. a,b,c in AP means? : $2b=a+c$

DAY 44 COMPLETE?

- [ ] Board pattern read
- [ ] 29 questions attempted
- [ ] Solutions checked
- [ ] Traps noted

Next: Day 45, Board Practice: Coordinate Geometry + Triangles + Circles

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 45

BOARD PRACTICE 3 | CH 6, 7 AND 10

Coordinate Geometry + Triangles + Circles

Board Practice 3

Aaj ka target

- [ ] Section formula, BPT aur tangent theorems ek baar bol ke dohrao
- [ ] 30 board-style sawal 95 min mein
- [ ] Har geometry sawal ka rough figure banao
- [ ] Proof ke har step ka reason likho

45.1 Board pattern for this unit

Coordinate Geometry carries **6 marks**; Geometry (Triangles + Circles) carries **15 marks**. Expect one theorem proof (BPT or a tangent theorem), quick MCQs on distance and section formula, and a case study on either similar triangles or coordinates.

Chapter	What the board asks	Typical marks
Coordinate Geometry	distance, section formula, ratio by an axis or a line, mid-point, 4th vertex, equidistant point	1 + 2 + 3
Triangles	BPT and converse, similarity criteria (AA, SAS, SSS), BPT proof, case on shelves or shadows	1 + 3 + 5
Circles	tangent $\perp$ radius, equal tangents, angle between tangents, incircle of a triangle or quadrilateral	1 + 2 + 3

Ch 6, 7 and 10 in the board paper

- **Trap 1:** In the section formula the ratio $m : n$ multiplies the **far** point first: $x = \dfrac{mx_2 + nx_1}{m + n}$.
- **Trap 2:** A point on the x-axis has $y = 0$; on the y-axis $x = 0$. Students swap them.
- **Trap 3:** BPT is $\dfrac{AD}{DB} = \dfrac{AE}{EC}$, but for DE use similarity: $\dfrac{DE}{BC} = \dfrac{AD}{AB}$ (whole side, not DB).
- **Trap 4:** Angle between two tangents and the angle at the centre are **supplementary** ($\angle APB + \angle AOB = 180^\circ$).
- **Trap 5:** Theorem proofs need Given, To prove, Construction, Proof with reasons. Missing 'Construction' costs a mark.



SHIVAM BHAIYA SAYS: BOARD TIP

Theorem proof mein figure zaroor banao aur har step ke aage reason bracket mein likho, jaise (radius $\perp$ tangent) ya (RHS). Board examiner pehle figure aur 'Given / To prove' dekhta hai, aadhe marks wahi se aate hain.



EXAM ALERT!

BPT ka proof aur 'tangents from an external point are equal' wala theorem, dono mein se ek lagbhag har paper mein hota hai. Dono ko figure ke saath likh ke practice karo.

45.2 How to attempt today

Take **95 minutes**. Draw a rough figure for every geometry question before you solve it, even when one is not given. Check every coordinate answer by putting it back into the condition.

Day 45 Practice

30 board-pattern questions | solve first, then check

SECTION A | MCQ AND ASSERTION-REASON (1 MARK EACH)

MCQ

1 $P(2,5)$, $Q(-1,3)$ and $R(-4,-1)$ are three vertices of parallelogram $PQRS$, taken in order. The fourth vertex S is (1)

(a) $(1,-1)$ **(b)** $(-7,1)$ **(c)** $(-1,1)$ **(d)** $(1,3)$

MCQ

2 The point $P(k,0)$ divides the join of $A(4,-3)$ and $B(-5,6)$ in the ratio $1:2$. The value of k is (1)

(a) 1 **(b)** -1 **(c)** 2 **(d)** $\dfrac{1}{3}$

MCQ

3 The mid-point of the segment joining $A(8,-4)$ and $B(k,6)$ is (a,b). If $a-2b=10$, then k is (1)

(a) 12 **(b)** 16 **(c)** 20 **(d)** 24

MCQ

4 If the distance between $(k,7)$ and $(2,3)$ is 5 units, then k is (1)

(a) 5 only **(b)** -1 only **(c)** ± 5 **(d)** 5 or -1

MCQ

5 In $\triangle ABC$, D on AB and E on AC with $DE \parallel BC$. If $AD=p$, $DB=q$ and $DE=m$, then BC equals (1)

(a) $\frac{mp}{p+q}$ **(b)** $\frac{mq}{p}$

(c) $\frac{m(p+q)}{p}$ **(d)** $\frac{m(p+q)}{q}$

MCQ

6 $\triangle ABC \sim \triangle PQR$ with $AB=12$ cm, $BC=15$ cm, $CA=9$ cm and $PQ=8$ cm. Then $QR+RP$ is (1)

(a) 14 cm **(b)** 16 cm **(c)** 18 cm **(d)** 24 cm

MCQ

7 A circle is inscribed in $\triangle ABC$, touching AB at P, BC at Q and CA at R. If $AP=5$ cm, $BP=4$ cm and $AC=12$ cm, then BC is (1)

(a) 11 cm **(b)** 10 cm **(c)** 12 cm **(d)** 9 cm

MCQ

8 From a point Q, 17 cm from the centre O of a circle of radius 8 cm, a tangent QP is drawn. The length QP is (1)

(a) 9 cm **(b)** $\sqrt{353}$ cm **(c)** 12 cm **(d)** 15 cm

MCQ

9 Two tangents drawn from a point P to a circle of radius 6 cm are inclined to each other at 60°. The distance OP is (1)

(a) 6 cm **(b)** $6\sqrt{3}$ cm **(c)** 12 cm **(d)** $4\sqrt{3}$ cm

MCQ

10 PA and PB are tangents to a circle with centre O, and $\angle APB=70^\circ$. If Q is a point on the major arc AB, then $\angle AQB$ is (1)

(a) 110° **(b)** 55° **(c)** 35° **(d)** 70°

ASSERTION-REASON

11 **Assertion (A):** The mid-point of a line segment is the only point that is equidistant from its two end points.
Reason (R): Every point on the perpendicular bisector of a line segment is equidistant from its end points. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

12 **Assertion (A):** If $\triangle ABC \sim \triangle DEF$ and $\frac{AB}{DE}=\frac{2}{3}$, then the ratio of their perimeters is $2:3$.
Reason (R): The corresponding sides of similar triangles are proportional. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

13 **Assertion (A):** From an external point P, tangents PA and PB are drawn to a circle. Then $PA=PB$.
Reason (R): A tangent can be drawn to a circle from a point inside the circle. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B | VERY SHORT ANSWER (2 MARKS EACH)

VERY SHORT ANSWER

14 Find the ratio in which the x-axis divides the segment joining $(-3,4)$ and $(5,-6)$. Also find the point of division. (2)

VERY SHORT ANSWER

15 Find the ratio in which the line $x+y=6$ divides the segment joining $A(-1,2)$ and $B(3,6)$. (2)

VERY SHORT ANSWER

16 In $\triangle ABC$, $DE \parallel BC$ with D on AB and E on AC. If $AE=2.1$ cm, $EC=6.3$ cm and $DB=7.5$ cm, find AD. (2)

VERY SHORT ANSWER

17 In $\triangle PQR$, A lies on PQ and B on PR with $PA = 6$ cm, $PQ = 15$ cm, $PB = 4$ cm and $BR = 6$ cm. Is $AB \parallel QR$? Justify. (2)

VERY SHORT ANSWER

18 Two tangents PA and PB are drawn to a circle from an external point P such that $PA = 9$ cm and $\angle APB = 60°$. Find the length of chord AB. (2)

SECTION C | SHORT ANSWER (3 MARKS EACH)

SHORT ANSWER

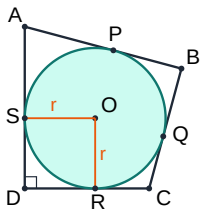
19 A circle is inscribed in $\triangle ABC$ with $AB = 14$ cm, $BC = 10$ cm and $CA = 12$ cm, touching AB, BC, CA at D, E, F respectively. Find AD, BE and CF. (3)

SHORT ANSWER

20 AB is a chord of length 24 cm of a circle of radius 15 cm. The tangents at A and B meet at P. Find the length PA. (3)

SHORT ANSWER

21 A circle with centre O is inscribed in quadrilateral $ABCD$, touching AB, BC, CD, DA at P, Q, R, S, and $\angle ADC = 90°$. If $BC = 40$ cm, $CD = 30$ cm and $BP = 26$ cm, find the radius of the circle. (3)



SHORT ANSWER

22 Find the point on the x-axis which is equidistant from $(2, -5)$ and $(-2, 9)$. (3)

SHORT ANSWER

23 In $\triangle ABC$, $DE \parallel BC$ with D on AB and E on AC. If $AD = x$, $DB = x - 3$, $AE = x + 3$ and $EC = x - 1$, find x. (3)

SHORT ANSWER

24 Show that $A(2, 1)$, $B(5, 3)$, $C(3, 6)$ and $D(0, 4)$ are the vertices of a square. (3)

SECTION D | LONG ANSWER (5 MARKS EACH)

LONG ANSWER

25 (a) Prove that the lengths of the tangents drawn from an external point to a circle are equal. (b) A quadrilateral $ABCD$ is drawn to circumscribe a circle. Prove that $AB + CD = AD + BC$. (5)

LONG ANSWER

26 State and prove the Basic Proportionality Theorem. Using it, find AE if in $\triangle ABC$, $DE \parallel BC$, $\frac{AD}{DB} = \frac{3}{5}$ and $AC = 5.6$ cm. (5)

LONG ANSWER

27 (a) Find the ratio in which $P(-4, 6)$ divides the segment joining $A(-6, 10)$ and $B(3, -8)$. (b) Find the point on the y-axis which is equidistant from A and B. (5)

SECTION E | CASE-BASED (4 MARKS EACH)

CASE-BASED

28 Read the passage and answer the questions that follow. A carpenter makes a triangular shelf ABC with a support bar DE parallel to the base BC, where D lies on AB and E on AC. (4)

(i) Write the relation given by the Basic Proportionality Theorem for this shelf. (1)

(ii) If $AD = x + 4$, $DB = 3x + 16$, $AE = x$ and $EC = 3x + 2$ (in cm), find x. (1)

(iii) In another shelf, $AD = 3$ cm, $DB = 6$ cm, $AE = 4$ cm and $EC = 8$ cm. Is $DE \parallel BC$? Find $DE : BC$. (2)

OR (iii) If $DE \parallel BC$, $AD = 2.4$ cm, $AB = 7.2$ cm and $BC = 9$ cm, find DE. (2)

CASE-BASED

29 Read the passage and answer the questions that follow. A lever AB is drawn on a grid with $A(-2, 1)$ and $B(10, 7)$. The fulcrum P divides AB in the ratio $1 : 2$ from A. (4)

(i) Find the length of the lever AB. (1)

(ii) Find the coordinates of the fulcrum P. (1)

(iii) Find the distance of P from the mid-point M of AB. (2)

OR (iii) Find the ratio in which the y-axis divides AB. (2)

CASE-BASED

30 **Read the passage and answer the questions that follow. A model of a glass pyramid face is a triangle ABC. Inside it, a smaller triangle DEF is fixed with $DE \parallel AB$ and $EF \parallel BC$, so that $\triangle DEF \sim \triangle ABC$. Here $AB = 15$ cm, $BC = 21$ cm and $DE = 10$ cm.** (4)

(i) Which similarity criterion shows $\triangle DEF \sim \triangle ABC$? (1)

(ii) Find the scale factor $\frac{DE}{AB}$. (1)

(iii) Find EF. (2)

OR (iii) If the perimeter of $\triangle ABC$ is 54 cm, find the perimeter of $\triangle DEF$. (2)

Day 45 Answers with worked solutions

1 **(c)**

Diagonals of a parallelogram bisect each other: midpoint of PR = midpoint of QS.

$S = P + R - Q = (2 - 4 + 1,\ 5 - 1 - 3) = (-1, 1)$.

2 **(a)**

Section formula with $m : n = 1 : 2$: $x = \frac{1(-5) + 2(4)}{3} = 1$, $y = \frac{1(6) + 2(-3)}{3} = 0$.

So $k = 1$.

3 **(b)**

$a = \frac{8 + k}{2}$, $b = \frac{-4 + 6}{2} = 1$.

$\frac{8 + k}{2} - 2 = 10 \Rightarrow 8 + k = 24 \Rightarrow k = 16$.

4 **(d)**

$(k - 2)^2 + (7 - 3)^2 = 25 \Rightarrow (k - 2)^2 = 9$.

$k - 2 = \pm 3 \Rightarrow k = 5$ or $k = -1$.

5 **(c)**

$\triangle ADE \sim \triangle ABC$ (AA), so $\frac{DE}{BC} = \frac{AD}{AB} = \frac{p}{p + q}$.

$BC = \frac{m(p + q)}{p}$.

6 **(b)**

Scale factor $= \frac{PQ}{AB} = \frac{8}{12} = \frac{2}{3}$.

$QR = 15 \times \frac{2}{3} = 10$, $RP = 9 \times \frac{2}{3} = 6$; sum $= 16$ cm.

7 **(a)**

Tangents from a point are equal: $AR = AP = 5$, so $CR = 12 - 5 = 7$.

$CQ = CR = 7$ and $BQ = BP = 4$, so $BC = 7 + 4 = 11$ cm.

8 **(d)**

$OP \perp QP$, so $QP = \sqrt{17^2 - 8^2} = \sqrt{225}$.

$QP = 15$ cm.

9 **(c)**

OP bisects the $60°$ angle, so in right $\triangle OAP$, $\angle OPA = 30°$.

$\sin 30° = \frac{OA}{OP} \Rightarrow OP = \frac{6}{1/2} = 12$ cm.

10 **(b)**

In quadrilateral $OAPB$, $\angle OAP = \angle OBP = 90°$, so $\angle AOB = 180° - 70° = 110°$.

Angle at the circumference $= \frac{1}{2} \times 110° = 55°$.

11 **(d)**

Every point on the perpendicular bisector is equidistant, so there are infinitely many such points, not only the mid-point. A is false.

R is true. Answer (d).

12 **(a)**

All sides are in the ratio $2 : 3$, so the sums (perimeters) are also in the ratio $2 : 3$. A is true.

R is the reason A holds. So (a).

13 **(c)**

A is the standard theorem: tangents from an external point are equal. True.

No tangent can be drawn from a point inside a circle, so R is false. Answer (c).

14 **Ans:** $2 : 3$; $\left(\frac{1}{5}, 0\right)$

Let the ratio be $k : 1$. The point on the x-axis has $y = 0$: $\frac{-6k + 4}{k + 1} = 0 \Rightarrow k = \frac{2}{3}$. Ratio $2 : 3$.

$x = \frac{2(5) + 3(-3)}{5} = \frac{1}{5}$. Point $\left(\frac{1}{5}, 0\right)$.

15 **Ans:** $5 : 3$

Let the ratio be $k : 1$. Point $= \left(\frac{3k - 1}{k + 1}, \frac{6k + 2}{k + 1}\right)$.

It lies on $x + y = 6$: $\frac{9k + 1}{k + 1} = 6 \Rightarrow 9k + 1 = 6k + 6 \Rightarrow k = \frac{5}{3}$.

Ratio $5 : 3$ (point $(1.5, 4.5)$, and $1.5 + 4.5 = 6$).

16 **Ans:** 2.5 cm

By BPT, $\frac{AD}{DB} = \frac{AE}{EC}$.

$\frac{AD}{7.5} = \frac{2.1}{6.3} = \frac{1}{3} \Rightarrow AD = 2.5$ cm.

17 **Ans:** Yes

$AQ = 15 - 6 = 9$ cm. $\frac{PA}{AQ} = \frac{6}{9} = \frac{2}{3}$ and $\frac{PB}{BR} = \frac{4}{6} = \frac{2}{3}$.

The sides are divided in the same ratio, so by the converse of BPT, $AB \parallel QR$.

18 **Ans:** 9 cm

$PA = PB$ (tangents from an external point), so $\triangle PAB$ is isosceles with $\angle PAB = \angle PBA = \frac{180° - 60°}{2} = 60°$.

So $\triangle PAB$ is equilateral and $AB = PA = 9$ cm.

19 **Ans:** $AD = 8$, $BE = 6$, $CF = 4$ cm

Let $AD = AF = x$, $BD = BE = y$, $CE = CF = z$ (equal tangents).

$x + y = 14$, $y + z = 10$, $z + x = 12$. Adding: $2(x + y + z) = 36 \Rightarrow x + y + z = 18$.

$x = 18 - 10 = 8$, $y = 18 - 12 = 6$, $z = 18 - 14 = 4$.

20 **Ans:** 20 cm

Let OP meet AB at M. $OP \perp AB$ and $AM = 12$ cm, so $OM = \sqrt{15^2 - 12^2} = 9$ cm.

In right $\triangle OAP$ (right angle at A), $AM \perp OP$, so $\triangle AMO \sim \triangle PMA$: $AM^2 = OM \times MP \Rightarrow 144 = 9\,MP \Rightarrow MP = 16$ cm.

$PA = \sqrt{AM^2 + MP^2} = \sqrt{144 + 256} = 20$ cm.

21 **Ans:** 16 cm

$BQ = BP = 26$, so $CQ = 40 - 26 = 14$ cm, and $CR = CQ = 14$ cm.

$DR = CD - CR = 30 - 14 = 16$ cm.

$OSDR$ has three right angles ($\angle D$ and the two radius-tangent angles) and $OS = OR$, so it is a square: $r = DR = 16$ cm.

22 **Ans:** $(-7, 0)$

Let the point be $(x, 0)$. $(x - 2)^2 + 25 = (x + 2)^2 + 81$.

$-4x + 29 = 4x + 85 \Rightarrow -8x = 56 \Rightarrow x = -7$.

Point: $(-7, 0)$.

23 **Ans:** $x = 9$

By BPT: $\dfrac{x}{x-3} = \dfrac{x+3}{x-1}$.

$x(x - 1) = (x + 3)(x - 3) \Rightarrow x^2 - x = x^2 - 9 \Rightarrow x = 9$.

Check: $AD = 9, DB = 6, AE = 12, EC = 8$; $\dfrac{9}{6} = \dfrac{12}{8}$.

24 **Ans:** Proof

$AB = \sqrt{9 + 4} = \sqrt{13}$, $BC = \sqrt{4 + 9} = \sqrt{13}$, $CD = \sqrt{9 + 4} = \sqrt{13}$, $DA = \sqrt{4 + 9} = \sqrt{13}$.

Diagonals: $AC = \sqrt{1 + 25} = \sqrt{26}$, $BD = \sqrt{25 + 1} = \sqrt{26}$.

All four sides equal and the diagonals equal, so $ABCD$ is a square.

25 **Ans:** Proof

(a) Let PQ, PR be tangents from P to the circle with centre O. Join OQ, OR, OP.

$\angle OQP = \angle ORP = 90^\circ$ (radius $\perp$ tangent), $OQ = OR$ (radii), OP common.

So $\triangle OQP \cong \triangle ORP$ (RHS), hence $PQ = PR$ (CPCT).

(b) Let the sides AB, BC, CD, DA touch the circle at P, Q, R, S. By (a): $AP = AS$, $BP = BQ$, $CR = CQ$, $DR = DS$.

Adding: $AP + BP + CR + DR = AS + BQ + CQ + DS \Rightarrow AB + CD = AD + BC$.

26 **Ans:** Proof; $AE = 2.1$ cm

Statement: If a line is drawn parallel to one side of a triangle to meet the other two sides in distinct points, the other two sides are divided in the same ratio.

Given: $DE \parallel BC$ in $\triangle ABC$. **To prove:** $\dfrac{AD}{DB} = \dfrac{AE}{EC}$. **Construction:** join BE, CD; draw $EN \perp AB$ and $DM \perp AC$.

$\dfrac{ar(ADE)}{ar(BDE)} = \dfrac{\frac{1}{2}AD \times EN}{\frac{1}{2}DB \times EN} = \dfrac{AD}{DB}$ and $\dfrac{ar(ADE)}{ar(DEC)} = \dfrac{AE}{EC}$.

$\triangle BDE$ and $\triangle DEC$ are on the same base DE between the same parallels, so $ar(BDE) = ar(DEC)$. Hence $\dfrac{AD}{DB} = \dfrac{AE}{EC}$.

Application: $\dfrac{AE}{EC} = \dfrac{3}{5} \Rightarrow AE = \dfrac{3}{8} \times 5.6 = 2.1$ cm.

27 **Ans:** (a) $2 : 7$ (b) $\left(0, \dfrac{7}{4}\right)$

(a) Let the ratio be $k : 1$. x: $\dfrac{3k - 6}{k + 1} = -4 \Rightarrow 3k - 6 = -4k - 4 \Rightarrow k = \dfrac{2}{7}$.

Check y: $\dfrac{-8(2/7) + 10}{9/7} = \dfrac{54/7}{9/7} = 6$. So the ratio is $2 : 7$.

(b) Let the point be $(0, y)$: $(0 + 6)^2 + (y - 10)^2 = (0 - 3)^2 + (y + 8)^2$.

$36 + y^2 - 20y + 100 = 9 + y^2 + 16y + 64 \Rightarrow 63 = 36y \Rightarrow y = \dfrac{7}{4}$.

Point: $\left(0, \dfrac{7}{4}\right)$.

28 **Ans:** $\frac{AD}{DB} = \frac{AE}{EC}$

Since $DE \parallel BC$, BPT gives $\frac{AD}{DB} = \frac{AE}{EC}$.

Ans: $x = 4$

$(x+4)(3x+2) = x(3x+16) \Rightarrow 3x^2 + 14x + 8 = 3x^2 + 16x \Rightarrow x = 4$.

Ans: Yes; $1:3$

$\frac{AD}{DB} = \frac{3}{6} = \frac{1}{2} = \frac{4}{8} = \frac{AE}{EC}$, so $DE \parallel BC$ (converse of BPT).

Then $\triangle ADE \sim \triangle ABC$, so $\frac{DE}{BC} = \frac{AD}{AB} = \frac{3}{9} = \frac{1}{3}$.

OR: Ans: 3 cm

$\triangle ADE \sim \triangle ABC$ (AA), so $\frac{DE}{BC} = \frac{AD}{AB} = \frac{2.4}{7.2} = \frac{1}{3}$.

$DE = 3$ cm.

29 **Ans:** $6\sqrt{5}$ units

$AB = \sqrt{12^2 + 6^2} = \sqrt{180} = 6\sqrt{5}$.

Ans: $(2, 3)$

$P = \left(\frac{1(10) + 2(-2)}{3}, \frac{1(7) + 2(1)}{3}\right) = (2, 3)$.

Ans: $\sqrt{5}$ units

$M = \left(\frac{-2 + 10}{2}, \frac{1 + 7}{2}\right) = (4, 4)$.

$PM = \sqrt{(4-2)^2 + (4-3)^2} = \sqrt{5}$.

OR: Ans: $1:5$

Let the ratio be $k:1$: $\frac{10k - 2}{k + 1} = 0 \Rightarrow k = \frac{1}{5}$.

Ratio $1:5$.

30 **Ans:** AA

Parallel sides give equal corresponding angles, so the AA criterion applies.

Ans: $\frac{2}{3}$

$\frac{10}{15} = \frac{2}{3}$.

Ans: 14 cm

$\frac{EF}{BC} = \frac{DE}{AB} = \frac{2}{3} \Rightarrow EF = \frac{2}{3} \times 21 = 14$ cm.

OR: Ans: 36 cm

Perimeters of similar triangles are in the ratio of corresponding sides: $\frac{2}{3} \times 54 = 36$ cm.

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Distance formula? : $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
2. Section formula, ratio $m:n$? : $\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$
3. Point on x-axis has? : $y = 0$
4. BPT says? : $\frac{AD}{DB} = \frac{AE}{EC}$ when $DE \parallel BC$
5. Similarity criteria? : AA, SAS, SSS
6. Tangent and radius at the point of contact? : perpendicular
7. $\angle APB$ and $\angle AOB$ for two tangents? : add up to $180°$
8. Tangents from an external point? : equal in length

DAY 45 COMPLETE? ☐ Board pattern read ☐ 30 questions attempted ☐ Solutions checked ☐ Traps noted

Next: Day 46, Board Practice: Trigonometry (Ch 8 and 9)

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 46

BOARD PRACTICE 4 | CH 8 AND 9

Trigonometry (Ch 8 and 9)

Board Practice 4

Aaj ka target

- [ ] Standard values ka table yaad se likho
- [ ] Har heights sawal ka figure pehle banao
- [ ] 30 board-style sawal 90 min mein
- [ ] Identity proofs ek side se hi karo

46.1 Board pattern for this unit

Trigonometry (Ch 8 and 9) carries **12 marks**. Introduction to Trigonometry brings identities and standard values; Applications brings one heights-and-distances problem (3 or 5 marks) and often a case study.

Chapter	What the board asks	Typical marks
Introduction to Trigonometry	ratios from one given ratio, standard values (0° to 90°), evaluate, prove an identity	1 + 2 + 3
Applications of Trigonometry	elevation/depression with one or two observations, shadows, two towers, moving boat or car	3 or 5 + case

Ch 8 and 9 in the board paper

- **Trap 1:** Angle of depression is measured from the **horizontal** at the eye, never from the vertical; it equals the angle of elevation from the object.
- **Trap 2:** 'Angle with the wall' and 'angle with the ground' are different: draw the figure and mark which side is adjacent.
- **Trap 3:** $\sin^2\theta$ means $(\sin\theta)^2$, not $\sin(\theta^2)$.
- **Trap 4:** In identities, work on one side only (usually the harder one); never cross-multiply across the '=' you are proving.
- **Trap 5:** Leave answers as $20\sqrt{3}$ unless the question gives $\sqrt{3} = 1.732$; then give the decimal too.



SHIVAM BHAIYA SAYS: BOARD TIP

Heights and distances mein figure hi aadha answer hai: horizontal line, angle aur right angle saaf mark karo, phir har triangle ke liye ek tan wali equation likho. Do unknown ho toh do equations, aur ek ko doosre mein substitute karo. Figure ke bina examiner step marks dene mein hichkichata hai.



EXAM ALERT!

5-mark heights and distances mein lagbhag hamesha **do angles** hote hain (30 aur 60, ya 30 aur 45). Dono equations alag likho, numbering karo (1), (2), tabhi poore marks.

46.2 How to attempt today

Take **90 minutes**. Write the standard values table from memory before you start, then cover it. For every heights question, draw the figure first and label the unknowns.

Day 46 Practice

30 board-pattern questions | solve first, then check

SECTION A | MCQ AND ASSERTION-REASON (1 MARK EACH)

MCQ

1 If $\tan A = \sqrt{3}$ and $\sin B = \frac{1}{2}$, where A and B are acute angles, then $\cos(A - B)$ equals (1)

(a) 0 (b) $\frac{1}{2}$ (c) $\frac{\sqrt{3}}{2}$ (d) 1

MCQ

2 If $4\sec\theta - 5 = 0$, then $\cot\theta$ is (1)

(a) $\frac{4}{3}$ (b) $\frac{3}{4}$ (c) $\frac{5}{3}$ (d) $\frac{3}{5}$

MCQ

3 If $\tan\theta = \frac{1}{\sqrt{5}}$, then $\frac{\csc^2\theta - \sec^2\theta}{\csc^2\theta + \sec^2\theta}$ equals (1)

(a) $\frac{1}{2}$ (b) $\frac{3}{4}$ (c) $\frac{1}{3}$ (d) $\frac{2}{3}$

MCQ

4 If $\sec\theta = 2$ (θ acute), then $\frac{1 + \tan\theta}{\sin\theta}$ equals (1)

(a) $2\sqrt{2}$ (b) $\frac{2\sqrt{3}+6}{3}$ (c) $1 + \sqrt{3}$ (d) $\frac{\sqrt{3}+3}{3}$

MCQ

5 If $\tan\theta + \cot\theta = 2$ (θ acute), then $\sin^6\theta + \cos^6\theta$ is (1)

(a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) 1 (d) $\frac{1}{8}$

MCQ

6 If $\frac{\cos\theta+\sin\theta}{\cos\theta-\sin\theta}=\frac{\sqrt{3}+1}{\sqrt{3}-1}$ (θ acute), then θ is (1)

(a) 60° (b) 45° (c) 30° (d) 0°

MCQ

7 A 12 m long wire is tied from the top of a 6 m high pole to a peg on the level ground. The angle the wire makes with the ground is (1)

(a) 60° (b) 30° (c) 45° (d) 15°

MCQ

8 The shadow of a vertical pole is $5\sqrt{3}$ m long when the Sun's elevation is 30°. The height of the pole is (1)

(a) 15 m (b) $5\sqrt{3}$ m (c) 10 m (d) 5 m

MCQ

9 From the top of a 60 m high cliff, the angle of depression of a car on the road is 60°. The distance of the car from the foot of the cliff is (1)

(a) $20\sqrt{3}$ m (b) $60\sqrt{3}$ m (c) 30 m (d) $10\sqrt{3}$ m

MCQ

10 From the mid-point of the line joining the feet of two towers of heights h_1 and h_2, the angles of elevation of their tops are 60° and 30° respectively. Then $h_1 : h_2$ is (1)

(a) $1:3$ (b) $\sqrt{3}:1$ (c) $3:1$ (d) $1:\sqrt{3}$

ASSERTION-REASON

11 **Assertion (A):** A 16 m long ladder making an angle of 60° with the wall reaches a height of $8\sqrt{3}$ m on the wall.

Reason (R): In a right triangle, $\cos\theta=\frac{\text{adjacent side}}{\text{hypotenuse}}$. (1)

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

ASSERTION-REASON

12 **Assertion (A):** $\sin^2\theta+\cos^2\theta=1$ for every acute angle θ.

Reason (R): In a right triangle, (perpendicular)^2 + (base)^2 = (hypotenuse)^2. (1)

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

ASSERTION-REASON

13 **Assertion (A):** For an acute angle θ, the value of $\sec\theta$ cannot be $\frac{3}{4}$.

Reason (R): For an acute angle θ, $0<\cos\theta<1$, so $\sec\theta>1$. (1)

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

SECTION B | VERY SHORT ANSWER (2 MARKS EACH)

VERY SHORT ANSWER

14 In $\triangle ABC$, right-angled at B, $\tan A=\sqrt{3}$. Find the values of $\sin A\cos C+\cos A\sin C$ and $\cos A\cos C-\sin A\sin C$. (2)

VERY SHORT ANSWER

15 Evaluate $(1+\cot^2\theta)(1-\cos\theta)(1+\cos\theta)$. (2)

VERY SHORT ANSWER

16 If $x=a\sec\theta$ and $y=b\tan\theta$, prove that $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$. (2)

VERY SHORT ANSWER

17 A kite is flying at a height of 45 m above the ground. The string makes an angle of 60° with the ground. Find the length of the string, assuming no slack. (2)

VERY SHORT ANSWER

18 Evaluate $\frac{2\cos^2 45^\circ+3\sec^2 30^\circ-\cot^2 45^\circ}{\sin^2 60^\circ+\cos^2 60^\circ}$. (2)

SECTION C | SHORT ANSWER (3 MARKS EACH)

SHORT ANSWER

19 Prove that $1+\frac{\tan^2\theta}{1+\sec\theta}=\sec\theta$. (3)

SHORT ANSWER

20 Prove that $(\sin A+\csc A)^2+(\cos A+\sec A)^2=7+\tan^2 A+\cot^2 A$. (3)

SHORT ANSWER

21 If $x\cos\theta-y\sin\theta=a$ and $x\sin\theta+y\cos\theta=b$, prove that $x^2+y^2=a^2+b^2$. (3)

SHORT ANSWER

22 If $21\tan A=20$, find $\sin A$ and $\sec A$, and verify that $\sin^2 A+\cos^2 A=1$. (3)

SHORT ANSWER

23 From a point on one bank of a river, the angle of elevation of the top of a tree on the opposite bank is 60°. After walking 30 m straight back from the bank, the angle becomes 30°. Find the height of the tree and the width of the river. (3)

SHORT ANSWER

24 From a point on a bridge 10 m above a river, the angles of depression of the two banks on opposite sides are 30° and 60°. Find the width of the river. (3)

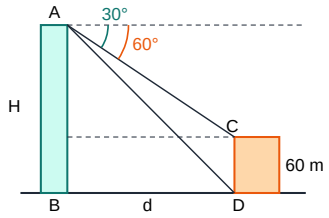
SHORT ANSWER

25 Two poles stand 30 m apart on level ground. The angle of elevation of the top of the first pole from the foot of the second is 60°, and that of the top of the second from the foot of the first is 30°. Find the difference between their heights. (3)

SECTION D | LONG ANSWER (5 MARKS EACH)

LONG ANSWER

26 From the top of a tower, the angles of depression of the top and the bottom of a 60 m high building are 30° and 60° respectively. Find the height of the tower and the distance between the tower and the building. (5)



LONG ANSWER

27 A boat is moving away from a 75 m high lighthouse. The angle of depression of the boat from the top of the lighthouse changes from 60° to 30° in 2 minutes. Find the speed of the boat in metres per minute. (5)

LONG ANSWER

28 A balloon is seen from two windows of a building, one 3 m and the other 9 m above the ground. The angles of elevation of the balloon from these windows are 60° and 30° respectively. Find the height of the balloon above the ground. (5)

SECTION E | CASE-BASED (4 MARKS EACH)

CASE-BASED

29 Read the passage and answer the questions that follow. A temple gate frame is modelled as right triangle ABC, right-angled at B, with $AB = 24$ m and $BC = 7$ m. (4)

(i) Find $\sin A$. (1)

(ii) Find $\cos C$. (1)

(iii) Find the value of $\tan A - \cot C$. (2)

OR (iii) Find the value of $\csc^2 C - \cot^2 C$. (2)

CASE-BASED

30 Read the passage and answer the questions that follow. A guard on top of a 180 m high watchtower at the coast sees a boat coming towards the tower. The angle of depression of the boat is 30° at first and 45° a little later. (Take $\sqrt{3} = 1.732$.) (4)

(i) The angle of depression from the guard equals which angle at the boat? (1)

(ii) Find the distance of the boat from the foot of the tower when the angle is 30°. (1)

(iii) How far did the boat travel between the two observations? (2)

OR (iii) If the boat took 1 minute between the observations, find its speed in m/min. (2)

Day 46 Answers with worked solutions

1 **(c)**

$\tan A = \sqrt{3} \Rightarrow A = 60^\circ$; $\sin B = \frac{1}{2} \Rightarrow B = 30^\circ$.

$\cos(A - B) = \cos 30^\circ = \frac{\sqrt{3}}{2}$.

2 **(a)**

$\sec\theta = \frac{5}{4} \Rightarrow \cos\theta = \frac{4}{5}$, so $\sin\theta = \frac{3}{5}$ (3-4-5 triangle).

$\cot\theta = \frac{\cos\theta}{\sin\theta} = \frac{4}{3}$.

3 **(d)**

$\cot^2\theta = 5 \Rightarrow \csc^2\theta = 1 + 5 = 6$; $\sec^2\theta = 1 + \frac{1}{5} = \frac{6}{5}$.

Value $= \frac{6 - \frac{6}{5}}{6 + \frac{6}{5}} = \frac{24/5}{36/5} = \frac{2}{3}$.

4 **(b)**

$\sec\theta = 2 \Rightarrow \theta = 60^\circ$: $\tan\theta = \sqrt{3}$, $\sin\theta = \frac{\sqrt{3}}{2}$.

Value $= \frac{1 + \sqrt{3}}{\sqrt{3}/2} = \frac{2(1 + \sqrt{3})}{\sqrt{3}} = \frac{2\sqrt{3} + 6}{3}$.

5 **(a)**

$\tan\theta + \dfrac{1}{\tan\theta} = 2 \Rightarrow (\tan\theta - 1)^2 = 0 \Rightarrow \theta = 45^\circ$.

$\sin^6 45^\circ + \cos^6 45^\circ = 2\left(\dfrac{1}{\sqrt{2}}\right)^6 = 2 \times \dfrac{1}{8} = \dfrac{1}{4}$.

6 **(c)**

Divide numerator and denominator by $\cos\theta$: $\dfrac{1+\tan\theta}{1-\tan\theta}$. Divide the right side by $\sqrt{3}$: $\dfrac{1+\frac{1}{\sqrt{3}}}{1-\frac{1}{\sqrt{3}}}$.

So $\tan\theta = \dfrac{1}{\sqrt{3}} \Rightarrow \theta = 30^\circ$.

7 **(b)**

$\sin\theta = \dfrac{\text{height}}{\text{wire}} = \dfrac{6}{12} = \dfrac{1}{2}$.

$\theta = 30^\circ$.

8 **(d)**

$\tan 30^\circ = \dfrac{h}{5\sqrt{3}}$.

$h = 5\sqrt{3} \times \dfrac{1}{\sqrt{3}} = 5$ m.

9 **(a)**

Angle of elevation of the cliff top from the car $= 60^\circ$ (alternate angles).

$\tan 60^\circ = \dfrac{60}{d} \Rightarrow d = \dfrac{60}{\sqrt{3}} = 20\sqrt{3}$ m.

10 **(c)**

Let the half distance be d. $h_1 = d\tan 60^\circ = \sqrt{3}d$, $h_2 = d\tan 30^\circ = \dfrac{d}{\sqrt{3}}$.

$h_1 : h_2 = \sqrt{3} : \dfrac{1}{\sqrt{3}} = 3 : 1$.

11 **(d)**

The angle with the wall is at the top, and the wall height is the side adjacent to it: height $= 16\cos 60^\circ = 8$ m, not $8\sqrt{3}$ m. A is false.

R is true. Answer (d).

12 **(a)**

Dividing $p^2 + b^2 = h^2$ by h^2 gives $\sin^2\theta + \cos^2\theta = 1$. So A is true and R is exactly why. (a)

13 **(a)**

$\dfrac{3}{4} < 1$, and R shows $\sec\theta > 1$. Both true and R explains A. (a)

14 **Ans:** 1 and 0

$\tan A = \sqrt{3} \Rightarrow A = 60^\circ$, so $C = 90^\circ - 60^\circ = 30^\circ$.

$\sin A\cos C + \cos A\sin C = \dfrac{\sqrt{3}}{2} \times \dfrac{\sqrt{3}}{2} + \dfrac{1}{2} \times \dfrac{1}{2} = \dfrac{3}{4} + \dfrac{1}{4} = 1$.

$\cos A\cos C - \sin A\sin C = \dfrac{1}{2} \times \dfrac{\sqrt{3}}{2} - \dfrac{\sqrt{3}}{2} \times \dfrac{1}{2} = 0$.

15 **Ans:** 1

$(1-\cos\theta)(1+\cos\theta) = 1 - \cos^2\theta = \sin^2\theta$.

$1 + \cot^2\theta = \csc^2\theta$.

Product $= \csc^2\theta \times \sin^2\theta = 1$.

16 **Ans:** Proof

$\dfrac{x}{a} = \sec\theta$ and $\dfrac{y}{b} = \tan\theta$.

$\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = \sec^2\theta - \tan^2\theta = 1$.

17 **Ans:** $30\sqrt{3}$ m

$\sin 60^\circ = \dfrac{45}{l}$.

$l = \dfrac{45}{\sqrt{3}/2} = \dfrac{90}{\sqrt{3}} = 30\sqrt{3}$ m (≈ 51.96 m).

18 **Ans:** 4

$2\cos^2 45^\circ = 2 \times \dfrac{1}{2} = 1$; $3\sec^2 30^\circ = 3 \times \dfrac{4}{3} = 4$; $\cot^2 45^\circ = 1$.

Numerator $= 1 + 4 - 1 = 4$; denominator $= 1$ (identity).

Value $= 4$.

19 **Ans:** Proof

$\tan^2\theta = \sec^2\theta - 1 = (\sec\theta - 1)(\sec\theta + 1)$.

LHS $= 1 + \dfrac{(\sec\theta - 1)(\sec\theta + 1)}{1 + \sec\theta} = 1 + \sec\theta - 1 = \sec\theta =$ RHS.

20 **Ans:** Proof

LHS $= \sin^2 A + \csc^2 A + 2\sin A\csc A + \cos^2 A + \sec^2 A + 2\cos A\sec A$.

$= (\sin^2 A + \cos^2 A) + \csc^2 A + \sec^2 A + 2 + 2$ (since $\sin A\csc A = \cos A\sec A = 1$).

$= 1 + (1 + \cot^2 A) + (1 + \tan^2 A) + 4 = 7 + \tan^2 A + \cot^2 A =$ RHS.

21 **Ans:** Proof

Square and add: $a^2 + b^2 = (x\cos\theta - y\sin\theta)^2 + (x\sin\theta + y\cos\theta)^2$.

$= x^2(\cos^2\theta + \sin^2\theta) + y^2(\sin^2\theta + \cos^2\theta) - 2xy\sin\theta\cos\theta + 2xy\sin\theta\cos\theta$.

$= x^2 + y^2$. Hence $x^2 + y^2 = a^2 + b^2$.

22 **Ans:** $\sin A = \dfrac{20}{29}$, $\sec A = \dfrac{29}{21}$

$\tan A = \dfrac{20}{21}$: take opposite $= 20k$, adjacent $= 21k$; hypotenuse $= \sqrt{400 + 441}\,k = 29k$.

$\sin A = \dfrac{20}{29}$, $\cos A = \dfrac{21}{29}$, $\sec A = \dfrac{29}{21}$.

$\sin^2 A + \cos^2 A = \dfrac{400 + 441}{841} = 1$. Verified.

23 **Ans:** Height $15\sqrt{3}$ m; width 15 m

Let the width be x m and the height h m.

$\tan 60^\circ = \dfrac{h}{x} \Rightarrow h = \sqrt{3}x$.

$\tan 30^\circ = \dfrac{h}{x+30} \Rightarrow h = \dfrac{x+30}{\sqrt{3}}$.

$\sqrt{3}x = \dfrac{x+30}{\sqrt{3}} \Rightarrow 3x = x + 30 \Rightarrow x = 15$.

Width $= 15$ m, height $= 15\sqrt{3} \approx 25.98$ m.

24 **Ans:** $\frac{40\sqrt{3}}{3}$ m ≈ 23.09 m

Distance to the first bank $= \frac{10}{\tan 30^\circ} = 10\sqrt{3}$ m.

Distance to the second bank $= \frac{10}{\tan 60^\circ} = \frac{10}{\sqrt{3}}$ m.

Width $= 10\sqrt{3} + \frac{10}{\sqrt{3}} = \frac{30+10}{\sqrt{3}} = \frac{40\sqrt{3}}{3} \approx 23.09$ m.

25 **Ans:** $20\sqrt{3}$ m ≈ 34.64 m

Height of first pole $= 30\tan 60^\circ = 30\sqrt{3}$ m.

Height of second pole $= 30\tan 30^\circ = 10\sqrt{3}$ m.

Difference $= 20\sqrt{3} \approx 34.64$ m.

26 **Ans:** Tower 90 m; distance $30\sqrt{3}$ m ≈ 51.96 m

Let the tower $AB = H$ m and the distance $BD = d$ m (figure). The building $CD = 60$ m.

From the bottom D: $\tan 60^\circ = \frac{H}{d} \Rightarrow H = \sqrt{3}d$ ...(1)

From the top C: the vertical drop is $H - 60$, so $\tan 30^\circ = \frac{H-60}{d} \Rightarrow H - 60 = \frac{d}{\sqrt{3}}$ ...(2)

From (1), $d = \frac{H}{\sqrt{3}}$, so $H - 60 = \frac{H}{3} \Rightarrow \frac{2H}{3} = 60 \Rightarrow H = 90$ m.

$d = \frac{90}{\sqrt{3}} = 30\sqrt{3} \approx 51.96$ m.

27 **Ans:** $25\sqrt{3} \approx 43.3$ m/min

First position: distance $= \frac{75}{\tan 60^\circ} = \frac{75}{\sqrt{3}} = 25\sqrt{3}$ m.

Second position: distance $= \frac{75}{\tan 30^\circ} = 75\sqrt{3}$ m.

Distance moved $= 75\sqrt{3} - 25\sqrt{3} = 50\sqrt{3}$ m in 2 minutes.

Speed $= \frac{50\sqrt{3}}{2} = 25\sqrt{3} \approx 43.3$ m/min.

28 **Ans:** 12 m

Let the balloon be H m high at horizontal distance d m from the building.

From the lower window: $\tan 60^\circ = \frac{H-3}{d} \Rightarrow H - 3 = \sqrt{3}d$ ...(1)

From the upper window: $\tan 30^\circ = \frac{H-9}{d} \Rightarrow H - 9 = \frac{d}{\sqrt{3}}$ ...(2)

Divide (1) by (2): $\frac{H-3}{H-9} = 3 \Rightarrow H - 3 = 3H - 27 \Rightarrow H = 12$ m.

(Then $d = \frac{9}{\sqrt{3}} = 3\sqrt{3}$ m.)

29 **Ans:** $\frac{7}{25}$

$AC = \sqrt{24^2 + 7^2} = 25$ m. $\sin A = \frac{BC}{AC} = \frac{7}{25}$.

Ans: $\frac{7}{25}$

$\cos C = \frac{BC}{AC} = \frac{7}{25}$.

Ans: 0

$\tan A = \frac{BC}{AB} = \frac{7}{24}$, $\cot C = \frac{BC}{AB} = \frac{7}{24}$.

$\tan A - \cot C = 0$.

OR: Ans: 1

$\csc C = \frac{25}{24}$, $\cot C = \frac{7}{24}$.

$\frac{625 - 49}{576} = \frac{576}{576} = 1$ (as the identity says).

30 **Ans:** The angle of elevation of the guard

Alternate angles between the horizontal at the guard and the ground line are equal.

Ans: $180\sqrt{3} \approx 311.76$ m

$d_1 = \frac{180}{\tan 30^\circ} = 180\sqrt{3} \approx 311.76$ m.

Ans: $180(\sqrt{3} - 1) \approx 131.76$ m

At 45°: $d_2 = \frac{180}{\tan 45^\circ} = 180$ m.

Distance $= 180\sqrt{3} - 180 = 180(\sqrt{3} - 1) \approx 131.76$ m.

OR: Ans: ≈ 131.76 m/min

Distance $= 180(\sqrt{3} - 1) \approx 131.76$ m in 1 minute, so speed ≈ 131.76 m/min.

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. $\sin^2\theta + \cos^2\theta$? : 1
2. $\sec^2\theta - \tan^2\theta$? : 1
3. $\csc^2\theta - \cot^2\theta$? : 1
4. $\tan 60^\circ$? : $\sqrt{3}$
5. $\sin 30^\circ$ and $\cos 30^\circ$? : $\frac{1}{2}$ and $\frac{\sqrt{3}}{2}$
6. Angle of depression equals? : the angle of elevation from the object (alternate angles)
7. Shadow equals height when the Sun is at? : 45°
8. Height from distance d and elevation θ? : $d\tan\theta$

DAY 46 COMPLETE? ☐ Board pattern read ☐ 30 questions attempted ☐ Solutions checked ☐ Traps noted

Next: Day 47, Board Practice: Mensuration + Statistics + Probability

DAY 47

BOARD PRACTICE 5 | CH 11 TO 14

Mensuration + Statistics + Probability

Board Practice 5

Aaj ka target

- [ ] Sector, segment aur saare solids ke formula ek page pe likho
- [ ] 30 board-style sawal 95 min mein
- [ ] Statistics mein poori table banao
- [ ] Probability mein total outcomes pehle gino

47.1 Board pattern for this unit

Mensuration (Areas Related to Circles + Surface Areas and Volumes) carries **10 marks**, and Statistics + Probability carry **11 marks**. These 21 marks are the most scoring part of the paper: the formulas are fixed and the board rarely twists them.

Chapter	What the board asks	Typical marks
Areas Related to Circles	sector, arc, segment, ring sector, clock hand, grazing animal, combination figure	1 + 2 + 3
Surface Areas and Volumes	combined solids (cone/hemisphere/cylinder/cube), TSA vs CSA, volume of remaining solid	1 + 3 + 5
Statistics	mean (direct, assumed mean), mode, median, missing frequency, empirical relation	1 + 3 + 5 or case
Probability	dice, cards, coins, bags, numbered cards, letters of a word	1 + 2 + case

Ch 11 to 14 in the board paper

- **Trap 1:** In combined solids, the joined faces are **not** part of the surface area: add only what you can touch.
- **Trap 2:** Perimeter of a sector $=$ arc $+2r$; students forget the two radii.
- **Trap 3:** In the mode formula f_0 is the class **before** the modal class and f_2 the class **after**; in the median formula cf is the cumulative frequency of the class **before** the median class.
- **Trap 4:** 1 is neither prime nor composite; a deck has 12 face cards (J, Q, K), and aces are not face cards.
- **Trap 5:** Use the π value the question gives ($\frac{22}{7}$ or 3.14) and write units: cm, cm^2, cm^3.



SHIVAM BHAIYA SAYS: BOARD TIP

Statistics ke 5-marker mein pehle **poori table** banao (class, f, x, d ya cf) aur formula likh ke values substitute karo. Board ki marking mein table ke marks alag hain. Calculation galat ho jaye tab bhi table aur formula ke step marks mil jaate hain.



EXAM ALERT!

Combined solid wala sawal (cone + hemisphere, cube + hemisphere) aur missing frequency wala median, dono 5-mark ke favourite hain. Dono ka ek-ek poora sawal aaj time lagakar karo.

47.2 How to attempt today

Take **95 minutes**. Write the formulas for sector, segment, all solids, mean, mode and median on a rough sheet first. Draw the table before every statistics calculation.

Day 47 Practice

30 board-pattern questions | solve first, then check

SECTION A | MCQ AND ASSERTION-REASON (1 MARK EACH)

MCQ

1. Three sectors, each of radius 14 cm, have central angles $50°$, $70°$ and $60°$. Their total area is ($\pi = \frac{22}{7}$) (1)

(a) 154 cm^2 (b) 308 cm^2 (c) 616 cm^2 (d) 462 cm^2

MCQ

2 The area of a semicircle is 77 cm^2. Its perimeter is ($\pi = \frac{22}{7}$) (1)

(a) 22 cm (b) 29 cm (c) 44 cm (d) 36 cm

MCQ

3 A chord of length 10.5 cm subtends an angle of 60° at the centre of a circle. The length of the minor arc is ($\pi = \frac{22}{7}$) (1)

(a) 11 cm (b) 22 cm (c) 5.5 cm (d) 33 cm

MCQ

4 The volumes of two spheres are in the ratio $125 : 8$. The ratio of their surface areas is (1)

(a) $5 : 2$ (b) $125 : 8$ (c) $25 : 4$ (d) $15 : 4$

MCQ

5 The volume of a solid hemisphere is numerically equal to its total surface area. Its radius is (1)

(a) 3 units (b) 4.5 units
(c) 6 units (d) 2 units

MCQ

6 Two dice are thrown together. The probability that the number 6 does not appear on either die is (1)

(a) $\frac{25}{36}$ (b) $\frac{11}{36}$ (c) $\frac{5}{6}$ (d) $\frac{1}{36}$

MCQ

7 An integer x is chosen at random from $-4, -3, \ldots, 3, 4$. The probability that $x^2 \leq 4$ is (1)

(a) $\frac{4}{9}$ (b) $\frac{1}{3}$ (c) $\frac{2}{3}$ (d) $\frac{5}{9}$

MCQ

8 For a data set, the mean is 24 and the median is 26. Using the empirical relation, the mode is (1)

(a) 28 (b) 22 (c) 30 (d) 25

MCQ

9 For the distribution: 0-10 (5), 10-20 (9), 20-30 (14), 30-40 (11), 40-50 (6), the modal class is (1)

(a) 10-20 (b) 20-30 (c) 30-40 (d) 40-50

MCQ

10 A bag has 6 red, 9 green and 5 white balls. A ball is drawn at random. The probability that it is neither green nor red is (1)

(a) $\frac{1}{4}$ (b) $\frac{3}{4}$ (c) $\frac{9}{20}$ (d) $\frac{3}{10}$

ASSERTION-REASON

11 **Assertion (A):** If the radius of a cylinder is halved and its height is doubled, its volume becomes half.
Reason (R): The volume of a cylinder is $\pi r^2 h$. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

12 **Assertion (A):** The area of a sector of angle 90° of a circle of radius 14 cm is 154 cm^2.
Reason (R): Area of a sector of angle θ is $\frac{\theta}{360^\circ} \times \pi r^2$. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

13 **Assertion (A):** The probability of getting a prime number when a die is thrown once is $\frac{1}{2}$.
Reason (R): 1 is a prime number. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B | VERY SHORT ANSWER (2 MARKS EACH)

VERY SHORT ANSWER

14 The minute hand of a clock is 14 cm long. Find the area swept by it in 15 minutes. ($\pi = \frac{22}{7}$) (2)

VERY SHORT ANSWER

15 Two concentric circles have radii 14 cm and 21 cm. Find the area of the region between them cut off by two radii making an angle of 72°. ($\pi = \frac{22}{7}$) (2)

VERY SHORT ANSWER

16 The total surface area of a solid hemisphere is 1848 cm^2. Find its volume. ($\pi = \frac{22}{7}$) (2)

VERY SHORT ANSWER

17 A letter is chosen at random from the letters of the word EXAMINATION. Find the probability that it is (i) a vowel, (ii) the letter N. (2)

VERY SHORT ANSWER

18 Find the mean of the following data by the direct method: 0-10 (4), 10-20 (6), 20-30 (10), 30-40 (6), 40-50 (4). (2)

SECTION C | SHORT ANSWER (3 MARKS EACH)

SHORT ANSWER

19 Find the area of the minor segment of a circle of radius 14 cm when the angle of the corresponding sector is 60°. ($\pi = \frac{22}{7}$, $\sqrt{3} = 1.73$) (3)

SHORT ANSWER

20 A horse is tied to a peg at one corner of a 30 m by 25 m rectangular field with a 21 m rope. Find (i) the area of the field the horse can graze, (ii) the increase in grazing area if the rope were 28 m long. ($\pi = \frac{22}{7}$) (3)

SHORT ANSWER

21 A solid is a cube of edge 10 cm with a hemisphere of diameter 7 cm placed on one face. Find the total surface area of the solid. ($\pi = \frac{22}{7}$) (3)

SHORT ANSWER

22 Cards numbered 2 to 61 are put in a box and one card is drawn at random. Find the probability that the number is (i) a perfect square, (ii) a multiple of 7, (iii) a prime less than 15. (3)

SHORT ANSWER

23 Find the mode of the data: 10-20 (6), 20-30 (10), 30-40 (16), 40-50 (12), 50-60 (6). (3)

SHORT ANSWER

24 From a solid cylinder of radius 4 cm and height 9 cm, a conical cavity of radius 2 cm and depth 3 cm is drilled out at the top. Find the volume of the remaining solid and the ratio of the remaining volume to the volume removed. ($\pi = \frac{22}{7}$) (3)

SECTION D | LONG ANSWER (5 MARKS EACH)

LONG ANSWER

25 A toy is a cone of radius 3.5 cm mounted on a hemisphere of the same radius. The total height of the toy is 15.5 cm. Find the total surface area and the volume of the toy. ($\pi = \frac{22}{7}$) (5)

LONG ANSWER

26 The median of the following data is 28 and the total frequency is 40. Find x and y.
0-10 (5), 10-20 (x), 20-30 (10), 30-40 (y), 40-50 (6), 50-60 (4) (5)

LONG ANSWER

27 A Pac-Man shaped badge is a circle of which a 90° sector (the mouth) has been cut away. The area of the badge is 48π cm^2. Find (i) the radius, (ii) the length of ribbon needed to go once around its boundary, (iii) the area of the piece that was cut away. ($\pi = 3.14$) (5)

SECTION E | CASE-BASED (4 MARKS EACH)

CASE-BASED

28 Read the passage and answer the questions that follow. A hand fan is made in the shape of a sector of a circle of radius 21 cm with a central angle of 120°. Its curved edge is decorated with lace. ($\pi = \frac{22}{7}$, $\sqrt{3} = 1.73$) (4)

(i) Find the length of lace needed for the curved edge. (1)

(ii) Find the area of the fan. (1)

(iii) The region between the curved edge and the chord joining its ends is painted. Find the painted area. (2)

OR (iii) Find the total length of the fan's boundary (lace plus the two straight edges). (2)

CASE-BASED

29 Read the passage and answer the questions that follow. For a card game, the four kings are removed from a well-shuffled deck of 52 cards. One card is then drawn at random from the remaining cards. (4)

(i) Find the probability that it is a red card. (1)

(ii) Find the probability that it is a face card. (1)

(iii) Find the probability that it is a black jack or a black queen. (2)

OR (iii) Find the probability that it is neither a heart nor a queen. (2)

CASE-BASED

30 **Read the passage and answer the questions that follow. A teacher records the time (in minutes) that 40 students spent on homework one evening: 20-30 (4), 30-40 (8), 40-50 (12), 50-60 (10), 60-70 (6).** (4)

(i) Find the modal class. (1)

(ii) Find the median class. (1)

(iii) Find the mean time using the assumed mean method. (2)

OR (iii) Find the median time. (2)

Day 47 Answers with worked solutions

1 **(b)**
Total angle $= 50^\circ + 70^\circ + 60^\circ = 180^\circ$, so the total area is half a circle.
$\frac{1}{2} \times \frac{22}{7} \times 14 \times 14 = 308$ cm^2.

2 **(d)**
$\frac{1}{2}\pi r^2 = 77 \Rightarrow r^2 = 49 \Rightarrow r = 7$ cm.
Perimeter $= \pi r + 2r = 22 + 14 = 36$ cm.

3 **(a)**
A chord subtending 60° forms an equilateral triangle with the radii, so $r = 10.5$ cm.
Arc $= \frac{60}{360} \times 2 \times \frac{22}{7} \times 10.5 = 11$ cm.

4 **(c)**
$\frac{r_1^3}{r_2^3} = \frac{125}{8} \Rightarrow \frac{r_1}{r_2} = \frac{5}{2}$.
Surface areas: $\frac{r_1^2}{r_2^2} = \frac{25}{4}$.

5 **(b)**
$\frac{2}{3}\pi r^3 = 3\pi r^2$.
$r = \frac{9}{2} = 4.5$ units.

6 **(a)**
Outcomes with no 6: $5 \times 5 = 25$ out of 36.
$P = \frac{25}{36}$.

7 **(d)**
Total integers: 9. Favourable: $-2, -1, 0, 1, 2$, that is 5.
$P = \frac{5}{9}$.

8 **(c)**
Mode $= 3$ Median $- 2$ Mean $= 78 - 48$.
Mode $= 30$.

9 **(b)**
The modal class is the class with the highest frequency.
Highest frequency 14 is for 20-30.

10 **(a)**
Total $= 20$; neither green nor red means white: 5.
$P = \frac{5}{20} = \frac{1}{4}$.

11 **(a)**
New volume $= \pi \left(\frac{r}{2}\right)^2 (2h) = \frac{1}{2}\pi r^2 h$. A is true.
The formula in R gives A directly. (a)

12 **(a)**
$\frac{90}{360} \times \frac{22}{7} \times 196 = 154$ cm^2. A true, and it comes from R. (a)

13 **(c)**
Primes on a die: 2, 3, 5, so $P = \frac{3}{6} = \frac{1}{2}$. A is true.
1 is neither prime nor composite, so R is false. (c)

14 **Ans:** 154 cm^2
In 60 minutes the hand sweeps 360°, so in 15 minutes it sweeps 90°.
Area $= \frac{90}{360} \times \frac{22}{7} \times 14 \times 14 = 154$ cm^2.

15 **Ans:** 154 cm^2
Required area $= \frac{72}{360} \times \pi(R^2 - r^2)$.
$= \frac{1}{5} \times \frac{22}{7} \times (441 - 196) = \frac{1}{5} \times \frac{22}{7} \times 245 = 154$ cm^2.

16 **Ans:** $\frac{17248}{3} \approx 5749.33$ cm^3
TSA $= 3\pi r^2 = 1848 \Rightarrow r^2 = \frac{1848 \times 7}{66} = 196 \Rightarrow r = 14$ cm.
Volume $= \frac{2}{3}\pi r^3 = \frac{2}{3} \times \frac{22}{7} \times 2744 = \frac{17248}{3} \approx 5749.33$ cm^3.

17 **Ans:** (i) $\frac{6}{11}$ (ii) $\frac{2}{11}$
The word has 11 letters: E, X, A, M, I, N, A, T, I, O, N.
(i) Vowels: E, A, I, A, I, O = 6, so $P = \frac{6}{11}$.
(ii) N appears twice, so $P = \frac{2}{11}$.

18 **Ans:** 25
Class marks x_i: 5, 15, 25, 35, 45. $f_i x_i$: 20, 90, 250, 210, 180.
$\sum f_i = 30$, $\sum f_i x_i = 750$.
Mean $= \frac{750}{30} = 25$.

19 **Ans:** ≈ 17.90 cm^2

Area of sector $= \frac{60}{360} \times \frac{22}{7} \times 196 = \frac{308}{3} \approx 102.67$ cm^2.

The triangle is equilateral (two radii, angle $60°$): area $= \frac{\sqrt{3}}{4} \times 196 = 49\sqrt{3} \approx 84.77$ cm^2.

Segment $= 102.67 - 84.77 \approx 17.90$ cm^2.

20 **Ans:** (i) 346.5 m^2 (ii) 269.5 m^2

The horse grazes a quadrant (the corner angle is $90°$, and 21 m and 28 m are both less than 25 m).

(i) $\frac{1}{4} \times \frac{22}{7} \times 21^2 = 346.5$ m^2.

(ii) $\frac{1}{4} \times \frac{22}{7} \times (28^2 - 21^2) = \frac{1}{4} \times \frac{22}{7} \times 343 = 269.5$ m^2.

21 **Ans:** 638.5 cm^2

TSA = surface of cube − base circle covered + curved surface of hemisphere.

$= 6(10)^2 - \pi r^2 + 2\pi r^2 = 600 + \pi r^2$ with $r = 3.5$.

$= 600 + \frac{22}{7} \times 12.25 = 600 + 38.5 = 638.5$ cm^2.

22 **Ans:** (i) $\frac{1}{10}$ (ii) $\frac{2}{15}$ (iii) $\frac{1}{10}$

Total cards $= 61 - 2 + 1 = 60$.

(i) Perfect squares: 4, 9, 16, 25, 36, 49 = 6, so $P = \frac{6}{60} = \frac{1}{10}$.

(ii) Multiples of 7: 7, 14, ..., 56 = 8, so $P = \frac{8}{60} = \frac{2}{15}$.

(iii) Primes less than 15: 2, 3, 5, 7, 11, 13 = 6, so $P = \frac{1}{10}$.

23 **Ans:** 36

Modal class 30-40: $l = 30$, $f_1 = 16$, $f_0 = 10$, $f_2 = 12$, $h = 10$.

Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 30 + \frac{6}{32 - 22} \times 10$.

$= 30 + 6 = 36$.

24 **Ans:** 440 cm^3; 35 : 1

Cylinder: $\pi r^2 h = \pi \times 16 \times 9 = 144\pi$.

Cone: $\frac{1}{3}\pi \times 4 \times 3 = 4\pi$.

Remaining $= 140\pi = 140 \times \frac{22}{7} = 440$ cm^3.

Ratio $= 140\pi : 4\pi = 35 : 1$.

25 **Ans:** TSA 214.5 cm^2; volume ≈ 243.83 cm^3

Height of cone $= 15.5 - 3.5 = 12$ cm; slant height $l = \sqrt{12^2 + 3.5^2} = \sqrt{156.25} = 12.5$ cm.

TSA $= \pi rl + 2\pi r^2 = \frac{22}{7} \times 3.5 \times 12.5 + 2 \times \frac{22}{7} \times 12.25 = 137.5 + 77 = 214.5$ cm^2.

Volume of cone $= \frac{1}{3} \times \frac{22}{7} \times 12.25 \times 12 = 154$ cm^3.

Volume of hemisphere $= \frac{2}{3} \times \frac{22}{7} \times 42.875 \approx 89.83$ cm^3.

Total volume ≈ 243.83 cm^3.

26 **Ans:** $x = 7$, $y = 8$

Total: $5 + x + 10 + y + 6 + 4 = 40 \Rightarrow x + y = 15$ ...(1)

Median 28 lies in 20-30: $l = 20$, $\frac{n}{2} = 20$, $cf = 5 + x$, $f = 10$, $h = 10$.

$28 = 20 + \frac{20 - (5 + x)}{10} \times 10 = 35 - x \Rightarrow x = 7$.

From (1), $y = 8$. (Check: cumulative frequencies 5, 12, 22, so 20-30 is indeed the median class.)

27 **Ans:** (i) 8 cm (ii) ≈ 53.68 cm (iii) 50.24 cm^2

(i) The badge is a $270°$ sector: $\frac{270}{360}\pi r^2 = 48\pi \Rightarrow r^2 = 64 \Rightarrow r = 8$ cm.

(ii) Boundary = major arc + two radii $= \frac{270}{360} \times 2\pi \times 8 + 16 = 12\pi + 16$.

$= 37.68 + 16 = 53.68$ cm.

(iii) Cut piece $= \frac{90}{360} \times 3.14 \times 64 = 50.24$ cm^2.

28 **Ans:** 44 cm

Arc $= \frac{120}{360} \times 2 \times \frac{22}{7} \times 21 = 44$ cm.

Ans: 462 cm^2

$\frac{120}{360} \times \frac{22}{7} \times 441 = 462$ cm^2.

Ans: ≈ 271.27 cm^2

Drop a perpendicular OM from the centre to the chord AB. It bisects the $120°$ angle, so in right $\triangle OMA$: $OM = 21\cos 60° = 10.5$ cm and $AM = 21\sin 60° = \frac{21\sqrt{3}}{2}$ cm.

Area of $\triangle OAB = \frac{1}{2} \times AB \times OM = \frac{1}{2} \times 21\sqrt{3} \times 10.5 = \frac{441\sqrt{3}}{4} \approx 190.73$ cm^2.

Painted segment $= 462 - 190.73 \approx 271.27$ cm^2.

OR: Ans: 86 cm

Boundary = arc $+ 2r = 44 + 42 = 86$ cm.

29 **Ans:** $\frac{1}{2}$

48 cards remain; red cards $= 26 - 2 = 24$.
$P = \frac{24}{48} = \frac{1}{2}$.

Ans: $\frac{1}{6}$

Face cards left: 4 jacks + 4 queens $= 8$. $P = \frac{8}{48} = \frac{1}{6}$.

Ans: $\frac{1}{12}$

Black jacks 2 and black queens 2: 4 cards.
$P = \frac{4}{48} = \frac{1}{12}$.

OR: Ans: $\frac{11}{16}$

Hearts left: 12 (king removed). Queens not hearts: 3. So 15 cards are a heart or a queen.
$P = \frac{48 - 15}{48} = \frac{33}{48} = \frac{11}{16}$.

30 **Ans:** 40-50

Highest frequency 12 is for 40-50.

Ans: 40-50

$\frac{n}{2} = 20$; cumulative frequencies 4, 12, 24, ..., so the median class is 40-50.

Ans: 46.5 min

Take $a = 45$. $d_i = x_i - a$: $-20, -10, 0, 10, 20$; $f_i d_i$: $-80, -80, 0, 100, 120$.

$\sum f_i d_i = 60$; mean $= 45 + \frac{60}{40} = 46.5$ min.

OR: Ans: ≈ 46.67 min

$l = 40$, $cf = 12$, $f = 12$, $h = 10$: median $= 40 + \frac{20 - 12}{12} \times 10 = 40 + 6.67 = 46.67$ min.

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Area of a sector of angle θ? : $\frac{\theta}{360^\circ}\pi r^2$
2. Length of an arc? : $\frac{\theta}{360^\circ} \times 2\pi r$
3. Segment area? : sector area $-$ triangle area
4. TSA of a solid hemisphere? : $3\pi r^2$
5. Volume of a cone? : $\frac{1}{3}\pi r^2 h$
6. Mode formula? : $l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$
7. Median formula? : $l + \frac{n/2 - cf}{f} \times h$
8. Empirical relation? : Mode $= 3$ Median $- 2$ Mean

DAY 47 COMPLETE?

- ☐ Board pattern read
- ☐ 30 questions attempted
- ☐ Solutions checked
- ☐ Traps noted

Next: Day 48, Board Practice: Case-Based and Assertion-Reason Marathon

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 48

BOARD PRACTICE 6 | ALL CHAPTERS

Case-Based and Assertion-Reason Marathon

Board Practice 6

Aaj ka target

- [ ] AR ka 3-step method yaad karo: A, phir R, phir 'kya R wajah hai?'
- [ ] 12 AR sawal 15 min mein
- [ ] 11 case studies, har ek lagbhag 7 min
- [ ] Har case ka chapter pehle pehchano

48.1 Board pattern: case studies and Assertion-Reason

Every board paper has **3 case-based questions** (Section E, 4 marks each: 1 + 1 + 2, with an internal choice in the 2-mark part) and **2 Assertion-Reason questions** (Section A, 1 mark each). That is 14 marks that reward a calm, step-by-step method more than speed.

Format	How it is marked	Your method
Case study (4 marks)	(i) 1 mark, (ii) 1 mark, (iii) 2 marks or its OR	Underline the numbers in the passage, write the formula, then substitute
Assertion-Reason (1 mark)	only the option letter	Judge A alone, judge R alone, then ask: does R explain A?

Section A and Section E formats

1. **AR step 1:** Is A true? Test it with a quick calculation or a counter-example.
2. **AR step 2:** Is R true on its own, as a general statement?
3. **AR step 3:** Only if both are true, ask whether R is the reason for A: yes gives (a), no gives (b). A true and R false is (c); A false and R true is (d).
4. **Case step:** The passage is the data sheet. Most cases are a standard chapter question dressed as a story; name the chapter first, then the formula.

- **Trap 1:** In AR, a false A can never be (a) or (b), however correct R looks.
- **Trap 2:** 'Always', 'only' and 'every' in A usually hide a counter-example (try $p = 2$, $\theta = 45°$, $k = 0$).
- **Trap 3:** In cases, part (iii) often needs the answer of (ii); if you get (ii) wrong, still show the method in (iii) for step marks.
- **Trap 4:** Attempt only ONE of (iii) and its OR. If you attempt both, the board checks only the first one.



SHIVAM BHAIYA SAYS: BOARD TIP

Case study mein passage ko kahani mat samjho, data sheet samjho: numbers underline karo aur pehle socho ye kaunsa chapter hai (AP, trig, probability...). Har part ka answer units ke saath likho. 2-mark wale part mein formula + substitution + answer, teeno lines dikhni chahiye.

MEME SAMJHO Astra meme, maths inside



THANOS BABA, AFTER SOLVING AN ASSERTION-REASON QUESTION

Pehle A check, phir R check, phir dekho R hi A ki wajah hai ya nahi. Perfectly balanced, jaise har AR hona chahiye.

Asli baat: AR has a fixed order: test A alone, test R alone, and only then decide whether R explains A. A false A rules out (a), (b) and (c) at once.



EXAM ALERT!

Section E ke 3 case studies 12 marks ke hain. Board inhe paper ke end mein deta hai, isliye time plan mein inke liye kam se kam 20 minute pehle se bachake rakho.

48.2 How to attempt today

Part 1: 12 AR questions in **15 minutes**. Part 2: 11 case studies in about **75 minutes** (about 7 minutes each). In every case, attempt part (iii) first as written, then try its OR as extra practice.

MY MISTAKES AND NOTES Write every mistake of this Day with the correct step

Day 48 Practice

23 board-pattern questions | solve first, then check

PART 1 | ASSERTION-REASON MARATHON (12 QUESTIONS, 1 MARK EACH)

ASSERTION-REASON

1. **Assertion (A):** $n = 5^2 \times 11 \times p$, where p is a prime other than 5 and 11, is always an odd number.
Reason (R): $5^2 \times 11$ is an odd number. (1)
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

2. **Assertion (A):** $\sqrt{5}$ is an irrational number.
Reason (R): If m is a natural number that is not a perfect square, then $\sqrt{m}$ is irrational. (1)
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

3. **Assertion (A):** The degree of the zero polynomial is not defined.
Reason (R): The degree of a non-zero constant polynomial is 0. (1)
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

4. **Assertion (A):** The pair $3x + y = 5$ and $6x + ky = 7$ has no solution when $k = 2$.
Reason (R): A pair of linear equations having infinitely many solutions is called consistent. (1)
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

5. **Assertion (A):** $(x-2)^2 + 1 = 2x - 3$ is a quadratic equation.
Reason (R): An equation of the form $ax^2 + bx + c = 0$ with $a \neq 0$ is a quadratic equation. (1)
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

6. **Assertion (A):** 0 is a term of the AP $31, 28, 25, \ldots$
Reason (R): The nth term of an AP is $a_n = a + (n-1)d$. (1)
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

7. **Assertion (A):** The point $(0, -7)$ lies on the y-axis.
Reason (R): The x-coordinate of every point on the y-axis is zero. (1)
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

8. **Assertion (A):** If in $\triangle ABC$ and $\triangle PQR$, $\frac{AB}{PQ} = \frac{BC}{QR}$, then $\triangle ABC \sim \triangle PQR$.
Reason (R): For SAS similarity, the included angles must also be equal. (1)
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

9. **Assertion (A):** The tangents drawn at the ends of a diameter of a circle are parallel.
Reason (R): A tangent to a circle is perpendicular to the radius through the point of contact. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

10 **Assertion (A):** For an acute angle θ, the greatest value of $\sin\theta + \cos\theta$ is 1.
Reason (R): The greatest value of each of $\sin\theta$ and $\cos\theta$ is 1. (1)
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

11 **Assertion (A):** An 18 m long rope tied from the top of a vertical pole to the ground makes an angle of 30° with the ground, so the pole is 9 m high.
Reason (R): $\sin\theta = \dfrac{\text{hypotenuse}}{\text{perpendicular}}$. (1)
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

12 **Assertion (A):** A cone of height 8 cm and base diameter 30 cm has a slant height of 23 cm.
Reason (R): The slant height of a cone is $l = \sqrt{h^2 + r^2}$. (1)
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

PART 2 | CASE-STUDY MARATHON (11 CASES, 4 MARKS EACH)

CASE-BASED

13 **Read the passage and answer the questions that follow. For a class party, samosas come in packs of 12 and chutney cups in packs of 18. The organisers want exactly one chutney cup per samosa, with no samosa or cup left over.** (4)
(i) Find the least number of samosas they can buy. (1)
(ii) How many packs of each must they buy? (1)
(iii) The party hall floor is 18 m by 24 m. Find the side of the largest square tile that fits exactly, and the number of tiles. (2)
OR (iii) If samosas came in packs of 15 and chutney in packs of 20, how many packs of each would be needed? (2)

CASE-BASED

14 **Read the passage and answer the questions that follow. A ball thrown in a park follows the curve $y = p(x) = -x^2 + 6x - 5$, where x is the horizontal distance in metres. The ball is thrown from ground level and lands back on the ground.** (4)
(i) What is the shape of the path, and does it open upward or downward? (1)
(ii) Find the zeroes of $p(x)$. (1)
(iii) Verify the relationship between the zeroes and the coefficients. (2)
OR (iii) Write a quadratic polynomial whose zeroes are 2 and 8. (2)

CASE-BASED

15 **Read the passage and answer the questions that follow. Ankit starts saving: Rs 50 in the first week, and every week he saves Rs 20 more than the week before.** (4)
(i) How much does he save in the 10th week? (1)
(ii) How much does he save in total in the first 10 weeks? (1)
(iii) In which week does he save Rs 410? (2)
OR (iii) How much does he save in total in the first 12 weeks? (2)

CASE-BASED

16 **Read the passage and answer the questions that follow. A taxi charges a fixed amount plus a charge per kilometre. A 10 km ride costs Rs 105 and a 15 km ride costs Rs 155. Let the fixed charge be Rs x and the rate Rs y per km.** (4)
(i) Write the pair of linear equations. (1)
(ii) Find the charge per kilometre. (1)
(iii) Find the fixed charge and the fare for a 25 km ride. (2)
OR (iii) How far can a person travel for Rs 205? (2)

CASE-BASED

17 **Read the passage and answer the questions that follow. On a school map drawn on a grid, the library is at $L(2, 3)$ and the canteen at $C(8, 11)$ (1 unit = 10 m).** (4)

(i) Find the distance between the library and the canteen on the map. (1)

(ii) A water cooler is at the mid-point of LC. Find its coordinates. (1)

(iii) A notice board P divides LC in the ratio $2:3$ from L. Find P. (2)

OR (iii) Find the point on the x-axis equidistant from L and C. (2)

CASE-BASED

18 **Read the passage and answer the questions that follow. A field in the shape of trapezium $ABCD$ ($AB \parallel DC$) is drawn on a map with $AB = 6$ cm and $DC = 15$ cm. Its diagonals meet at O. Scale: 1 cm represents 20 m.** (4)

(i) Which criterion shows $\triangle AOB \sim \triangle COD$? (1)

(ii) Find $AO : OC$. (1)

(iii) If $OB = 4$ cm on the map, find OD on the map and its actual length. (2)

OR (iii) Find the actual lengths of the parallel sides. (2)

CASE-BASED

19 **Read the passage and answer the questions that follow. A decorative quadrilateral frame $ABCD$ is made so that a circular plate touches all four sides: AB at P, BC at Q, CD at R and DA at S. Here $BP = 4$ cm, $DS = 6$ cm and $BC = 10$ cm.** (4)

(i) Which property of tangents is used to solve such problems? (1)

(ii) Find CD. (1)

(iii) If the perimeter of $ABCD$ is 50 cm, find AS. (2)

OR (iii) Prove that $AB + CD = AD + BC$. (2)

CASE-BASED

20 **Read the passage and answer the questions that follow. A sailboat has a vertical mast AB of height 6 m. Two ropes run from the top A to points C and D on the deck, with B, C, D on a straight line, $BC = 2$ m and $BD = 4$ m.** (4)

(i) Find $\sec D$. (1)

(ii) Find $\csc C$. (1)

(iii) Find $\tan D + \cot C$. (2)

OR (iii) Find $\sin^2 C + \cos^2 D$. (2)

CASE-BASED

21 **Read the passage and answer the questions that follow. An air traffic controller watches a plane flying horizontally at a height of $1500\sqrt{3}$ m. The angle of elevation of the plane changes from 60° to 30° in 20 seconds.** (4)

(i) Find the horizontal distance of the plane from the controller when the angle is 60°. (1)

(ii) Find the horizontal distance when the angle is 30°. (1)

(iii) Find the speed of the plane in km/h. (2)

OR (iii) Find the line-of-sight distance from the controller to the plane when the angle is 30°. (2)

CASE-BASED

22 **Read the passage and answer the questions that follow. A medicine capsule is a cylinder with a hemisphere at each end. Its total length is 20 mm and its diameter is 6 mm. ($\pi = \frac{22}{7}$)** (4)

(i) Find the length of the cylindrical part. (1)

(ii) Find the curved surface area of the cylindrical part. (1)

(iii) Find the volume of the capsule. (2)

OR (iii) Find the total surface area of the capsule. (2)

CASE-BASED

23 **Read the passage and answer the questions that follow. A rectangular garden is 20 m by 14 m. It has a circular pond of radius 3.5 m and two flower beds, each a quadrant of radius 7 m at a corner; the rest is grass. A ball falls at a random point in the garden. ($\pi = \frac{22}{7}$)** (4)

(i) Find the area of the pond. (1)

(ii) Find the probability that the ball falls in the pond. (1)

(iii) Find the probability that the ball falls on the grass. (2)

OR (iii) Find the probability that the ball does not fall in the pond. (2)

Day 48 Answers with worked solutions

1 **(d)**

If $p = 2$, then $n = 550$, which is even. So A is false.

$5^2 \times 11 = 275$ is odd, so R is true. Answer (d).

2 **(a)**

5 is not a perfect square, so by R, $\sqrt{5}$ is irrational. Both true and R explains A. (a)

3 **(b)**
Both statements are true (standard facts), but the degree of a non-zero constant does not explain why the zero polynomial has no degree. (b)

4 **(b)**
For $k = 2$: $\frac{3}{6} = \frac{1}{2} = \frac{1}{2} \neq \frac{5}{7}$, so the lines are parallel and there is no solution. A is true.
R is a true definition but says nothing about the no-solution case. (b)

5 **(a)**
Expanding: $x^2 - 4x + 5 = 2x - 3 \Rightarrow x^2 - 6x + 8 = 0$, of the form in R with $a = 1 \neq 0$. (a)

6 **(d)**
$31 + (n-1)(-3) = 0 \Rightarrow n - 1 = \frac{31}{3} \Rightarrow n = \frac{34}{3}$, not a natural number. A is false.
R is true. (d)

7 **(a)**
Its x-coordinate is 0, which by R puts it on the y-axis. (a)

8 **(d)**
Two sides in proportion are not enough: without $\angle B = \angle Q$ (or the third ratio) the triangles need not be similar. A is false.
R is true. (d)

9 **(a)**
Both tangents are perpendicular to the same line (the diameter), so they are parallel. R gives exactly this. (a)

10 **(d)**
At $\theta = 45^\circ$, $\sin\theta + \cos\theta = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \sqrt{2} > 1$.
A is false.
R is true. (d)

11 **(c)**
Height $= 18\sin 30^\circ = 9$ m, so A is true.
$\sin\theta = \frac{\text{perpendicular}}{\text{hypotenuse}}$, so R is false. (c)

12 **(d)**
$r = 15$ cm, so $l = \sqrt{64 + 225} = \sqrt{289} = 17$ cm, not 23 cm. A is false.
R is true. (d)

13 **Ans:** 36
LCM$(12, 18)$: $12 = 2^2 \times 3$, $18 = 2 \times 3^2$, LCM $= 2^2 \times 3^2 = 36$.
Ans: 3 packs of samosas, 2 packs of chutney
$\frac{36}{12} = 3$ and $\frac{36}{18} = 2$.
Ans: 6 m; 12 tiles
Largest side $=$ HCF$(18, 24) = 6$ m.
Tiles $= \frac{18}{6} \times \frac{24}{6} = 3 \times 4 = 12$.
OR: Ans: 4 and 3
LCM$(15, 20) = 60$; $\frac{60}{15} = 4$ packs of samosas, $\frac{60}{20} = 3$ packs of chutney.

14 **Ans:** A parabola opening downward
$p(x)$ is quadratic with $a = -1 < 0$, so its graph is a parabola opening downward.
Ans: 1 and 5
$-x^2 + 6x - 5 = -(x-1)(x-5)$, so the zeroes are 1 and 5.
Ans: Sum 6, product 5
Sum $= 1 + 5 = 6 = -\frac{b}{a} = -\frac{6}{-1}$.
Product $= 5 = \frac{c}{a} = \frac{-5}{-1}$. Verified.
OR: Ans: $k(x^2 - 10x + 16)$
Sum $= 10$, product $= 16$, so $x^2 - 10x + 16$, or any non-zero multiple.

15 **Ans:** Rs 230
$a_{10} = 50 + 9 \times 20 = 230$.
Ans: Rs 1400
$S_{10} = \frac{10}{2}[100 + 9 \times 20] = 5 \times 280 = 1400$.
Ans: 19th week
$50 + (n-1)20 = 410 \Rightarrow n - 1 = 18 \Rightarrow n = 19$.
OR: Ans: Rs 1920
$S_{12} = \frac{12}{2}[100 + 11 \times 20] = 6 \times 320 = 1920$.

16 **Ans:** $x + 10y = 105$, $x + 15y = 155$
Fare $= x + (\text{distance})\,y$.
Ans: Rs 10
Subtracting: $5y = 50 \Rightarrow y = 10$.
Ans: Rs 5; Rs 255
$x = 105 - 100 = 5$.
Fare for 25 km $= 5 + 25 \times 10 = 255$.
OR: Ans: 20 km
$5 + 10d = 205 \Rightarrow d = 20$ km.

17 **Ans:** 10 units (100 m)

$LC = \sqrt{6^2 + 8^2} = 10$ units $= 100$ m.

Ans: $(5, 7)$

$\left(\frac{2+8}{2}, \frac{3+11}{2}\right) = (5, 7)$.

Ans: $\left(\frac{22}{5}, \frac{31}{5}\right)$

$P = \left(\frac{2(8)+3(2)}{5}, \frac{2(11)+3(3)}{5}\right) = \left(\frac{22}{5}, \frac{31}{5}\right)$.

OR: Ans: $\left(\frac{43}{3}, 0\right)$

$(x-2)^2 + 9 = (x-8)^2 + 121 \Rightarrow -4x + 13 = -16x + 185 \Rightarrow 12x = 172 \Rightarrow x = \frac{43}{3}$.

18 **Ans:** AA

$\angle OAB = \angle OCD$ and $\angle OBA = \angle ODC$ (alternate angles, $AB \parallel DC$), so AA.

Ans: $2 : 5$

$\frac{AO}{OC} = \frac{AB}{CD} = \frac{6}{15} = \frac{2}{5}$.

Ans: 10 cm; 200 m

$\frac{OB}{OD} = \frac{2}{5} \Rightarrow OD = 10$ cm.

Actual length $= 10 \times 20 = 200$ m.

OR: Ans: 120 m and 300 m

$AB = 6 \times 20 = 120$ m, $DC = 15 \times 20 = 300$ m.

19 **Ans:** Tangents from an external point are equal

Lengths of the tangents drawn from an external point to a circle are equal.

Ans: 12 cm

$BQ = BP = 4$, so $CQ = 6 = CR$; $DR = DS = 6$. $CD = 6 + 6 = 12$ cm.

Ans: 9 cm

$AB + CD = AD + BC$, so each pair sums to 25 cm.

$AB = 25 - 12 = 13$ cm; $AP = 13 - 4 = 9$ cm, and $AS = AP = 9$ cm.

OR: Ans: Proof

$AP = AS$, $BP = BQ$, $CR = CQ$, $DR = DS$ (equal tangents).

Adding: $(AP + BP) + (CR + DR) = (AS + DS) + (BQ + CQ)$, i.e. $AB + CD = AD + BC$.

20 **Ans:** $\frac{\sqrt{13}}{2}$

$AD = \sqrt{36 + 16} = 2\sqrt{13}$ m; $\sec D = \frac{AD}{BD} = \frac{2\sqrt{13}}{4} = \frac{\sqrt{13}}{2}$.

Ans: $\frac{\sqrt{10}}{3}$

$AC = \sqrt{36 + 4} = 2\sqrt{10}$ m; $\csc C = \frac{AC}{AB} = \frac{2\sqrt{10}}{6} = \frac{\sqrt{10}}{3}$.

Ans: $\frac{11}{6}$

$\tan D = \frac{AB}{BD} = \frac{6}{4} = \frac{3}{2}$; $\cot C = \frac{BC}{AB} = \frac{2}{6} = \frac{1}{3}$.

Sum $= \frac{3}{2} + \frac{1}{3} = \frac{11}{6}$.

OR: Ans: $\frac{157}{130}$

$\sin^2 C = \frac{36}{40} = \frac{9}{10}$; $\cos^2 D = \frac{16}{52} = \frac{4}{13}$.

Sum $= \frac{117 + 40}{130} = \frac{157}{130}$.

21 **Ans:** 1500 m

$d_1 = \frac{1500\sqrt{3}}{\tan 60^\circ} = 1500$ m.

Ans: 4500 m

$d_2 = \frac{1500\sqrt{3}}{\tan 30^\circ} = 1500 \times 3 = 4500$ m.

Ans: 540 km/h

Distance covered $= 4500 - 1500 = 3000$ m in 20 s, so 150 m/s.

$150 \times \frac{18}{5} = 540$ km/h.

OR: Ans: $3000\sqrt{3}$ m

$\sin 30^\circ = \frac{1500\sqrt{3}}{s} \Rightarrow s = 3000\sqrt{3} \approx 5196$ m.

22 **Ans:** 14 mm

$20 - 2 \times 3 = 14$ mm.

Ans: 264 mm^2

$2\pi rh = 2 \times \frac{22}{7} \times 3 \times 14 = 264$ mm^2.

Ans: ≈ 509.14 mm^3

$V = \pi r^2 h + \frac{4}{3}\pi r^3 = \pi(9 \times 14 + 36) = 162\pi$.

$= 162 \times \frac{22}{7} = \frac{3564}{7} \approx 509.14$ mm^3.

OR: Ans: ≈ 377.14 mm^2

TSA $= 2\pi rh + 4\pi r^2 = 84\pi + 36\pi = 120\pi = \frac{2640}{7} \approx 377.14$ mm^2.

23 **Ans:** 38.5 m^2

$\pi r^2 = \frac{22}{7} \times 12.25 = 38.5$ m^2.

Ans: $\frac{11}{80}$

Garden area $= 280$ m^2; $P = \frac{38.5}{280} = \frac{11}{80}$.

Ans: $\frac{47}{80}$

Flower beds $= 2 \times \frac{1}{4} \times \frac{22}{7} \times 49 = 77$ m^2.

Grass $= 280 - 38.5 - 77 = 164.5$ m^2; $P = \frac{164.5}{280} = \frac{47}{80}$.

OR: Ans: $\frac{69}{80}$

$1 - \frac{11}{80} = \frac{69}{80}$.

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. A true, R true, R explains A? : option (a)
2. A true, R true, R does not explain A? : option (b)
3. A true, R false? : option (c)
4. A false, R true? : option (d)
5. Marks split in a case study? : 1 + 1 + 2
6. Attempt both (iii) and its OR? : No, only one is checked
7. First thing to do in a case passage? : name the chapter and underline the data

DAY 48 COMPLETE?

- ☐ AR method read
- ☐ 12 AR questions done
- ☐ 11 case studies done
- ☐ Solutions checked

Next: Day 49, Mock Test 1: Full Paper, 80 marks, 3 hours

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 49

MOCK TEST 1 | FULL PAPER | 80 MARKS

Mock Test 1: Full Board Paper

Mock Test 1

Aaj ka target

- [ ] Ek hi sitting mein poora paper, 3 ghante ka timer laga ke
- [ ] Pehle 15 minute sirf paper padhna, likhna nahi
- [ ] Har section ke time ka plan follow karna
- [ ] Answers kal (Day 50) check karna, aaj nahi

49.1 How to sit a mock

1. Clear the table: pen, pencil, scale, eraser, a watch and rough sheets. Phone in another room.
2. Set a 3-hour timer and start only when you are ready to go without a break.
3. Use the first 15 minutes to read the whole paper. Mark easy questions with a tick and lengthy ones with a star.
4. Suggested time plan: Section A 30 min, B 20 min, C 40 min, D 50 min, E 25 min, and 15 min to revise.
5. Leave a question you are stuck on after 3 minutes, mark it, and return to it at the end.
6. When the timer stops, stop writing. Mark yourself tomorrow with the Day 50 solutions.



SHIVAM BHAIYA SAYS: BOARD TIP

Mock ko asli board exam ki tarah do. Beech mein solution dekh liya to mock ka faayda khatam. Galti aaj ho jaaye, kal fix karenge, board mein nahi honi chahiye.

INSTRUCTIONS

Time: 3 hours | Maximum marks: 80 | 38 questions, all compulsory | Sections A (20 × 1), B (5 × 2), C (6 × 3), D (4 × 5), E (3 × 4); internal choice where OR is given; draw figures where needed; use $\pi = \frac{22}{7}$ unless stated.

Day 49 Practice

Mock paper | 80 marks | 3 hours | attempt in one sitting

SECTION A | MCQ AND ASSERTION-REASON (20 MARKS)

MCQ

1 In a factor tree, $z = 2 \times y$, $y = 5 \times x$ and $x = 3 \times 11$. The value of z is (1)

(a) 165 (b) 330 (c) 110 (d) 660

MCQ

2 For any two positive integers a and b, the value of $\frac{\text{HCF}(a,b) \times \text{LCM}(a,b)}{a \times b}$ is (1)

(a) 0 (b) a (c) 1 (d) ab

MCQ

3 The quadratic polynomial whose zeroes have sum -4 and product -21 is (1)

(a) $x^2 - 4x - 21$ (b) $x^2 + 4x + 21$
(c) $x^2 - 4x + 21$ (d) $x^2 + 4x - 21$

MCQ

4 The pair of equations $5x - 2y = 7$ and $15x - 6y = 12$ represents two lines which are (1)

(a) intersecting at one point (b) coincident
(c) parallel (d) perpendicular

MCQ

5 Which of the following equations has the sum of its roots equal to 4? (1)

(a) $2x^2 - 4x + 1 = 0$ (b) $-x^2 + 4x - 3 = 0$
(c) $3x^2 - 4x + 4 = 0$ (d) $x^2 + 4x + 1 = 0$

MCQ

6 The 12th term from the end of the AP $5, 9, 13, \ldots, 185$ is (1)

(a) 141 (b) 137 (c) 145 (d) 49

MCQ

7 A quadratic polynomial, the sum of whose zeroes is 0 and one zero is 5, is (1)

(a) $x^2 + 25$ (b) $x^2 - 5x$
(c) $x^2 - 25$ (d) $x^2 - 10x + 25$

MCQ

8 $AOBC$ is a rectangle with vertices $A(0, 5)$, $O(0, 0)$ and $B(12, 0)$. The length of its diagonal is (1)

(a) 17 (b) 13 (c) 12 (d) $\sqrt{119}$

MCQ

9 A circle has centre at the origin O and passes through $(5, 0)$. Which of these points does **not** lie in the interior of the circle? (1)

(a) $(2, 3)$ **(b)** $(-4, 1)$
(c) $(3, -4)$ **(d)** $(1, -4.5)$

MCQ

10 In $\triangle KLM$ and $\triangle PQR$, $\frac{KL}{PQ} = \frac{LM}{QR}$. Which extra information is enough to conclude $\triangle KLM \sim \triangle PQR$? (1)

(a) $\angle K = \angle P$ only **(b)** $\angle L = \angle Q$, or $\frac{KM}{PR} = \frac{KL}{PQ}$
(c) $\angle M = \angle R$ only **(d)** $KM = PR$

MCQ

11 From a point P, 17 cm away from the centre O of a circle of radius 8 cm, tangents PQ and PR are drawn. The area of quadrilateral $PQOR$ is (1)

(a) $60\ \text{cm}^2$ **(b)** $136\ \text{cm}^2$ **(c)** $68\ \text{cm}^2$ **(d)** $120\ \text{cm}^2$

MCQ

12 Two tangents PA and PB drawn from an external point P to a circle of radius 6 cm are perpendicular to each other. The length of each tangent is (1)

(a) $6\sqrt{2}$ cm **(b)** 6 cm **(c)** 3 cm **(d)** 12 cm

MCQ

13 The angles A, B, C of a triangle are in the ratio $1 : 2 : 3$. The value of $\frac{\sec A}{\text{cosec } B} - \frac{\tan A}{\cot B}$ is (1)

(a) 0 **(b)** 1 **(c)** $\sqrt{3}$ **(d)** $\frac{1}{\sqrt{3}}$

MCQ

14 If $\sec A + \tan A = 3$, then $\sec A$ equals (1)

(a) $\frac{4}{3}$ **(b)** $\frac{5}{3}$ **(c)** $\frac{3}{5}$ **(d)** $\frac{10}{3}$

MCQ

15 At a certain time the shadow of a vertical pole is $\sqrt{3}$ times its height. The angle of elevation of the sun is (1)

(a) $60°$ **(b)** $45°$ **(c)** $30°$ **(d)** $90°$

MCQ

16 For a distribution, the mode is 45 and the mean is 27. Using the empirical relation, the median is (1)

(a) 36 **(b)** 33 **(c)** 39 **(d)** 30

MCQ

17 One card is drawn at random from a well-shuffled deck of 52 cards. The probability that it is a face card of spades is (1)

(a) $\frac{3}{13}$ **(b)** $\frac{1}{13}$ **(c)** $\frac{3}{52}$ **(d)** $\frac{3}{26}$

MCQ

18 In a lucky draw, the probability of a coupon winning a prize is 0.04. If 2500 coupons are issued, the number of prizes is (1)

(a) 40 **(b)** 250 **(c)** 100 **(d)** 1000

ASSERTION-REASON

19 **Assertion (A):** Since 8 and 15 are co-prime, 8^3 and 15^3 are also co-prime.
Reason (R): If a and b have no common prime factor, then a^n and b^n also have no common prime factor. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

20 **Assertion (A):** If $x + 1$, $2x + 3$ and $4x + 1$ are three consecutive terms of an AP, then $x = 4$.
Reason (R): Three numbers a, b, c are in AP if and only if $2b = a + c$. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B | VERY SHORT ANSWER (5 X 2 = 10 MARKS)

VERY SHORT ANSWER

21 Find the value of k for which the pair of equations $(k - 2)x + 2y = k$ and $kx + ky = 8$ has infinitely many solutions. (2)

VERY SHORT ANSWER

22 In $\triangle ABC$, D is a point on AB and E is a point on AC such that $\angle ADE = \angle ACB$. Prove that $\triangle ADE \sim \triangle ACB$ and hence $AD \times AB = AE \times AC$. (2)

VERY SHORT ANSWER

23 A tangent PR at the point P of a circle with centre O makes an angle of $35°$ with the chord PQ. Find $\angle POQ$.

OR

Prove that the tangents drawn at the ends of a diameter of a circle are parallel. (2)

VERY SHORT ANSWER

24 If $\tan\theta = \frac{1}{\sqrt{3}}$, find the value of $\frac{2\sec\theta}{1+\tan^2\theta}$.

OR

Evaluate: $\frac{\tan 45° + 2\sin 30°}{\sec 60°}$. (2)

VERY SHORT ANSWER

25 The area of a sector of a circle of radius 21 cm is $231\ \text{cm}^2$. Find the length of the arc of the sector. (2)

SECTION C | SHORT ANSWER (6 X 3 = 18 MARKS)

SHORT ANSWER

26 Given that $\sqrt{3}$ is irrational, prove that $5 - 2\sqrt{3}$ is irrational.

OR

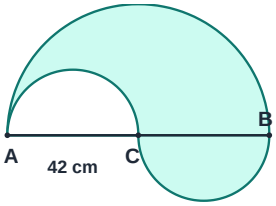
Find the HCF and LCM of 120 and 156 by prime factorisation and verify that HCF $\times$ LCM $=$ product of the numbers. (3)

SHORT ANSWER

27 Find the zeroes of the quadratic polynomial $6x^2 - 7x - 3$ and verify the relationship between the zeroes and the coefficients. (3)

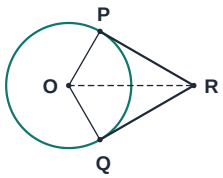
SHORT ANSWER

28 In the figure, AB is a diameter of length 42 cm and C is its midpoint. A semicircle is drawn on AB above it. Semicircles are drawn on AC (inside the big semicircle) and on CB (below AB). Find the area of the shaded region. (Use $\pi = \frac{22}{7}$) (3)



SHORT ANSWER

29 Two tangents RP and RQ are drawn from an external point R to a circle with centre O. If $\angle PRQ = 120°$, prove that $OR = PR + RQ$. (3)



∠PRQ = 120°

SHORT ANSWER

30 Prove that $\sqrt{\frac{\sec\theta - 1}{\sec\theta + 1}} + \sqrt{\frac{\sec\theta + 1}{\sec\theta - 1}} = 2\,\text{cosec}\,\theta$.

OR

If $x = p\sec\theta$ and $y = q\tan\theta$, prove that $\frac{x^2}{p^2} - \frac{y^2}{q^2} = 1$. (3)

SHORT ANSWER

31 Cards numbered 1 to 40 are put in a box and one card is drawn at random. Find the probability that the number on the card is (i) divisible by 2 or 3, (ii) a prime number, (iii) a perfect square. (3)

SECTION D | LONG ANSWER (4 X 5 = 20 MARKS)

LONG ANSWER

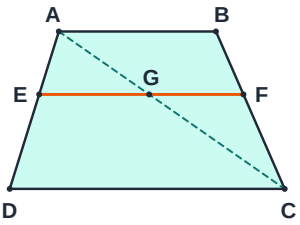
32 A plane left 40 minutes late. To reach its destination 1600 km away on time, it had to increase its usual speed by 400 km/h. Find its usual speed.

OR

The sum of the reciprocals of Ravi's ages (in years) 2 years ago and 4 years from now is $\frac{1}{4}$. Find his present age. (5)

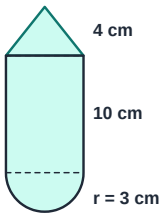
LONG ANSWER

33 $ABCD$ is a trapezium with $AB \parallel DC$. A line through E on AD and F on BC is parallel to AB. (i) Prove that $\frac{AE}{ED} = \frac{BF}{FC}$. (ii) If $AE = 4$ cm, $ED = 6$ cm, $BF = (x+1)$ cm and $FC = (2x-1)$ cm, find x and the length BC. (5)



LONG ANSWER

34 A toy rocket is made of a cylinder of radius 3 cm and height 10 cm, with a cone of the same radius and height 4 cm on top and a hemisphere of the same radius at the bottom. Find (i) the volume of the toy, (ii) its total surface area. (Use $\pi = 3.14$) (5)



LONG ANSWER

35 Find the mean and the mode of the following distribution.
Class: 0-10, 10-20, 20-30, 30-40, 40-50, 50-60
Frequency: 5, 8, 12, 15, 6, 4
OR
Find the median of the same distribution. (5)

SECTION E | CASE-BASED (3 X 4 = 12 MARKS)

CASE-BASED

36 Case Study 1: Meera starts a reading habit. On day 1 she reads 24 pages, and every day after that she reads 3 pages more than the day before. (4)

(i) How many pages does she read on day 12? (1)

(ii) On which day does she read 60 pages? (1)

(iii) Find the total number of pages she reads in the first 15 days. (2)

OR (iii) Find the total number of pages she reads in the first 20 days. (2)

CASE-BASED

37 Case Study 2: A garden designer marks a rectangular flower bed on a grid. Three corners are $A(1,2)$, $B(7,4)$ and $C(9,-2)$, and $ABCD$ is to be a parallelogram. (4)

(i) Find the midpoint of the diagonal AC. (1)

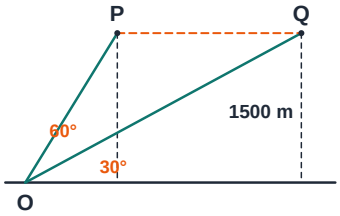
(ii) Find the length AB. (1)

(iii) Find the coordinates of the fourth corner D. (2)

OR (iii) Show that the bed $ABCD$ (with $D(3,-4)$) is in fact a rectangle. (2)

CASE-BASED

38 Case Study 3: A plane flies horizontally at a height of 1500 m. From a point on the ground, the angle of elevation of the plane is $60°$; after 20 seconds the angle of elevation becomes $30°$. (4)



(i) Find the horizontal distance of the plane from the observer when the angle is $60°$. (1)

(ii) Find the horizontal distance when the angle is $30°$. (1)

(iii) Find the distance flown in 20 s and the speed of the plane. (2)

OR (iii) After how many more seconds (same speed) will the plane be at a horizontal distance of $1500\sqrt{3} + 1000\sqrt{3}$ m, and what will the angle's tangent be then? (2)

ANSWERS AND FULL WORKED SOLUTIONS FOR THIS PAPER ARE ON DAY 50. CHECK ONLY AFTER YOU FINISH THE WHOLE PAPER.

DAY 49 COMPLETE?

☐ Paper done in one sitting ☐ Timer followed
☐ All 38 attempted ☐ Not checked yet (Day 50)

Next: Day 50, Mock 1 Review: Solutions and Mistake Fixing

MY MISTAKES AND NOTES Write every mistake of this Day with the correct step

DAY 50

MOCK 1 REVIEW | SOLUTIONS AND MISTAKE FIXING

Mock 1 Review: Mark, Analyse, Fix

Mock 1 Review

Aaj ka target

- [ ] Mock 1 khud check karo, board ki tarah step marks do
- [ ] Har galti mistake table mein likho
- [ ] Top traps padho
- [ ] 10 fix-it sawal solve karo

50.1 How to mark yourself

1. Keep a red pen. Check each answer against the worked solutions printed below, question by question.
2. MCQ and AR: 1 mark only if the letter matches.
3. 2, 3 and 5 mark questions: give step marks the way the board does: correct formula or setup, correct working, correct final answer with units. A right answer with no working gets at most half.
4. Case-based: mark each part separately (1 + 1 + 2).
5. Fill the score table, then the mistake table, then solve the fix-it questions.

Section	Max	My marks
A (Q1-20)	20	
B (Q21-25)	10	
C (Q26-31)	18	
D (Q32-35)	20	
E (Q36-38)	12	
Total	80	

Fill in after checking. 60+ is on track, 70+ is strong.

50.2 Top traps in Mock 1



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Q9 mein $(3, -4)$ circle ke **upar** hai, andar nahi. Interior ka matlab distance radius se **kam**; barabar ho to point circle par hai.



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Q32 (plane) mein $x = -1200$ bhi aata hai. Speed negative nahi ho sakti, isliye likh ke reject karo: board mein is line ka bhi mark hota hai.



SHIVAM BHAIYA SAYS

Q28 mein dono chhote semicircle same size ke hain, ek ghata, ek juda: cancel. Bada calculation karne se pehle figure dekh ke socho, time bachega.



SHIVAM BHAIYA SAYS: BOARD TIP

Q33 mein BPT seedha nahi lagta: pehle AC join karo, phir $\triangle ADC$ aur $\triangle CAB$ mein do baar BPT. Construction line nahi likhi to step marks kat jaate hain.



EXAM ALERT!

Mode formula mein denominator $2f_1 - f_0 - f_2$ hai. $f_1 - f_0 - f_2$ likhne wali galti har saal hoti hai.

50.3 Mistake analysis

Q no.	Chapter	Type of mistake	Fix (one line)
		concept / calculation / reading / time	

Type of mistake: concept (did not know), calculation (silly slip), reading (misread the question), time (did not reach it).

Day 50 Practice

10 board-pattern questions | solve first, then check

FIX-IT QUESTIONS (MOST-MISSED TOPICS OF MOCK 1)

MCQ

1 In the factor tree of 1260: $1260 = 2 \times 630$, $630 = 2 \times p$, $p = 3 \times q$, $q = 3 \times 35$. The values of p and q are (1)

(a) $p = 315$, $q = 105$ (b) $p = 630$, $q = 210$
(c) $p = 315$, $q = 35$ (d) $p = 105$, $q = 35$

MCQ

2 If the zeroes of $ax^2 + bx + c$ ($c \neq 0$) are equal, then (1)

(a) c and a have opposite signs
(b) c and b have opposite signs
(c) c and a have the same sign
(d) c and b have the same sign

MCQ

3 If p is a prime number, the quadratic equation whose roots are the two positive factors of p is (1)

(a) $x^2 - px + 1 = 0$ (b) $x^2 - (p+1)x + p = 0$
(c) $x^2 + (p+1)x + p = 0$ (d) $x^2 - (p-1)x - p = 0$

VERY SHORT ANSWER

4 Write a linear equation which, together with $3x - 7y = 4$, forms a pair of coincident lines. Justify your answer. (2)

VERY SHORT ANSWER

5 Is 184 a term of the AP $3, 7, 11, \ldots$? If not, which term of the AP is 183? (2)

VERY SHORT ANSWER

6 PQ is a chord of a circle with centre O and PT is the tangent at P. If $\angle QPT = 70^\circ$, find $\angle OPQ$ and $\angle POQ$. (2)

VERY SHORT ANSWER

7 Prove that $\cos^2\theta + \dfrac{1}{1 + \cot^2\theta} = 1$. (2)

VERY SHORT ANSWER

8 From the top of a 30 m high tower, the angle of depression of a ball on the ground is 60°. How far is the ball from the foot of the tower? (Use $\sqrt{3} = 1.732$) (2)

VERY SHORT ANSWER

9 Find the perimeter of a sector of a circle of radius 21 cm with central angle 60°. (Use $\pi = \dfrac{22}{7}$) (2)

VERY SHORT ANSWER

10 Two dice are thrown together. Find the probability that the product of the numbers on the top faces is 8. (2)

Day 50 Answers with worked solutions

1 **(a)**
$p = 630 \div 2 = 315$, $q = 315 \div 3 = 105$ (and $105 = 3 \times 35$ checks).

2 **(c)**
Equal zeroes: $b^2 = 4ac$. Since $b^2 \geq 0$ and $c \neq 0$, $ac > 0$, so a and c have the same sign.

3 **(b)**
The positive factors of a prime p are 1 and p.
Sum $= p + 1$, product $= p$.
Equation: $x^2 - (p+1)x + p = 0$.

4 **Ans:** e.g. $6x - 14y = 8$
Multiply the whole equation by any non-zero number, say 2: $6x - 14y = 8$.
$\dfrac{a_1}{a_2} = \dfrac{3}{6} = \dfrac{1}{2}$, $\dfrac{b_1}{b_2} = \dfrac{-7}{-14} = \dfrac{1}{2}$, $\dfrac{c_1}{c_2} = \dfrac{4}{8} = \dfrac{1}{2}$: all equal, so the lines coincide.

5 **Ans:** No; 183 is the 46th term
$a_n = 3 + (n-1)4 = 184 \Rightarrow n - 1 = \dfrac{181}{4}$, not a whole number, so 184 is not a term.
$3 + (n-1)4 = 183 \Rightarrow n - 1 = 45 \Rightarrow n = 46$.

6 **Ans:** 20° and 140°
$OP \perp PT$, so $\angle OPQ = 90^\circ - 70^\circ = 20^\circ$.
$OP = OQ$, so $\angle OQP = 20^\circ$ and $\angle POQ = 180^\circ - 40^\circ = 140^\circ$.

7 **Ans:** Proof
$1 + \cot^2\theta = \operatorname{cosec}^2\theta$, so $\dfrac{1}{1 + \cot^2\theta} = \sin^2\theta$.
LHS $= \cos^2\theta + \sin^2\theta = 1 =$ RHS.

8 **Ans:** $10\sqrt{3} \approx 17.32$ m
Angle of depression = angle of elevation from the ball $= 60^\circ$ (alternate angles).
$\tan 60^\circ = \dfrac{30}{d} \Rightarrow d = \dfrac{30}{\sqrt{3}} = 10\sqrt{3} \approx 17.32$ m.

9 **Ans:** 64 cm
Arc $= \dfrac{60}{360} \times 2 \times \dfrac{22}{7} \times 21 = 22$ cm.
Perimeter = arc $+ 2r = 22 + 42 = 64$ cm.

10 **Ans:** $\dfrac{1}{18}$
Total outcomes $= 36$. Product 8: $(2, 4)$ and $(4, 2)$, so 2 outcomes.
$P = \dfrac{2}{36} = \dfrac{1}{18}$.

Mock Test 1: worked solutions

marks as the board gives them

Mock Test 1 answers

1 **(b)**
$x = 33$, $y = 5 \times 33 = 165$, $z = 2 \times 165 = 330$.

2 **(c)**
For two numbers, $\text{HCF} \times \text{LCM} = a \times b$, so the ratio is 1.

3 **(d)**
$x^2 - (\text{sum})x + \text{product} = x^2 - (-4)x + (-21) = x^2 + 4x - 21$.

4 **(c)**
$\frac{a_1}{a_2} = \frac{5}{15} = \frac{1}{3}$, $\frac{b_1}{b_2} = \frac{-2}{-6} = \frac{1}{3}$, $\frac{c_1}{c_2} = \frac{7}{12}$.
$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$, so the lines are parallel (no solution).

5 **(b)**
Sum of roots $= -\frac{b}{a}$.
(a) 2, (b) $-\frac{4}{-1} = 4$, (c) $\frac{4}{3}$, (d) -4. Only (b) gives 4.

6 **(a)**
From the end the AP is $185, 181, 177, \ldots$ with $d = -4$.
12th term from end $= 185 + 11(-4) = 185 - 44 = 141$.

7 **(c)**
Other zero $= 0 - 5 = -5$, product $= -25$.
Polynomial $= x^2 - 0x - 25 = x^2 - 25$.

8 **(b)**
$C = (12, 5)$. Diagonals of a rectangle are equal: $AB = \sqrt{(12-0)^2 + (0-5)^2} = \sqrt{169} = 13$.

9 **(c)**
Radius $= 5$, so a point is inside if its distance from O is less than 5, i.e. $x^2 + y^2 < 25$.
$(2, 3)$: 13; $(-4, 1)$: 17; $(1, -4.5)$: 21.25; $(3, -4)$: 25, which is ON the circle, not inside.

10 **(b)**
$\angle L$ and $\angle Q$ are the angles included between the proportional sides, so $\angle L = \angle Q$ gives SAS similarity.
If the third pair of sides is also in the same ratio, SSS similarity holds. Either one is enough.

11 **(d)**
$OQ \perp PQ$, so $PQ = \sqrt{17^2 - 8^2} = \sqrt{225} = 15$ cm.
Area $= 2 \times \frac{1}{2} \times OQ \times PQ = 8 \times 15 = 120$ cm^2.

12 **(b)**
$\angle OAP = \angle OBP = \angle APB = 90°$, so $OAPB$ is a rectangle with $OA = OB$: a square.
Hence $PA = OA = 6$ cm.

13 **(a)**
$A = 30°$, $B = 60°$, $C = 90°$.
$\frac{\sec 30°}{\text{cosec}\, 60°} = \frac{2/\sqrt{3}}{2/\sqrt{3}} = 1$ and $\frac{\tan 30°}{\cot 60°} = \frac{1/\sqrt{3}}{1/\sqrt{3}} = 1$.
Value $= 1 - 1 = 0$.

14 **(b)**
$\sec^2 A - \tan^2 A = 1 \Rightarrow (\sec A - \tan A)(\sec A + \tan A) = 1 \Rightarrow \sec A - \tan A = \frac{1}{3}$.
Adding: $2 \sec A = 3 + \frac{1}{3} = \frac{10}{3}$, so $\sec A = \frac{5}{3}$.

15 **(c)**
$\tan\theta = \frac{h}{\sqrt{3}h} = \frac{1}{\sqrt{3}}$, so $\theta = 30°$.

16 **(b)**
$3\,\text{Median} = \text{Mode} + 2\,\text{Mean} = 45 + 54 = 99$, so Median $= 33$.

17 **(c)**
Face cards of spades: king, queen, jack, i.e. 3 cards.
$P = \frac{3}{52}$.

18 **(c)**
$P = \frac{\text{prizes}}{\text{coupons}} \Rightarrow \text{prizes} = 0.04 \times 2500 = 100$.

19 **(a)**
$8^3 = 2^9$ and $15^3 = 3^3 \times 5^3$ have no common prime, so A is true.
By the Fundamental Theorem of Arithmetic, raising to a power does not create new prime factors, so R is true and explains A.

20 **(a)**
$2(2x + 3) = (x + 1) + (4x + 1) \Rightarrow 4x + 6 = 5x + 2 \Rightarrow x = 4$ (terms $5, 11, 17$, $d = 6$). A is true.
R is the standard condition and is exactly what was used, so (a).

21 **Ans:** $k = 4$
For infinitely many solutions: $\frac{k-2}{k} = \frac{2}{k} = \frac{k}{8}$.
From $\frac{2}{k} = \frac{k}{8}$: $k^2 = 16$, so $k = 4$ or $k = -4$.
From $\frac{k-2}{k} = \frac{2}{k}$: $k - 2 = 2$, so $k = 4$.
Check $k = 4$: $2x + 2y = 4$ and $4x + 4y = 8$, the same line. Hence $k = 4$.

22 **Ans:** Proof

In $\triangle ADE$ and $\triangle ACB$: $\angle A = \angle A$ (common) and $\angle ADE = \angle ACB$ (given).

So $\triangle ADE \sim \triangle ACB$ (AA similarity).

Corresponding sides are proportional: $\frac{AD}{AC} = \frac{AE}{AB}$.

Cross-multiplying, $AD \times AB = AE \times AC$.

Hence proved.

23 **Ans:** $\angle POQ = 70^\circ$

$OP \perp PR$, so $\angle OPQ = 90^\circ - 35^\circ = 55^\circ$.

$OP = OQ$ (radii), so $\angle OQP = \angle OPQ = 55^\circ$.

$\angle POQ = 180^\circ - 55^\circ - 55^\circ = 70^\circ$.

OR: Let AB be a diameter, l the tangent at A and m the tangent at B. $OA \perp l$ and $OB \perp m$, so $\angle OAl = 90^\circ$ and $\angle OBm = 90^\circ$.

These are alternate angles made by the transversal AB with l and m, and they are equal, so $l \parallel m$.

24 **Ans:** $\sqrt{3}$; OR: 1

$\tan\theta = \frac{1}{\sqrt{3}} \Rightarrow \theta = 30^\circ$, $\sec\theta = \frac{2}{\sqrt{3}}$.

$1 + \tan^2\theta = 1 + \frac{1}{3} = \frac{4}{3}$.

Value $= \frac{2 \times \frac{2}{\sqrt{3}}}{\frac{4}{3}} = \frac{4}{\sqrt{3}} \times \frac{3}{4} = \frac{3}{\sqrt{3}} = \sqrt{3}$.

OR: $\frac{1 + 2 \times \frac{1}{2}}{2} = \frac{2}{2} = 1$.

25 **Ans:** 22 cm

Area of sector $= \frac{\theta}{360^\circ}\pi r^2$ and arc $l = \frac{\theta}{360^\circ}2\pi r$, so area $= \frac{1}{2}lr$.

$231 = \frac{1}{2} \times l \times 21 \Rightarrow l = \frac{462}{21} = 22$ cm.

(Check: $\theta = \frac{231 \times 360}{\frac{22}{7} \times 441} = 60^\circ$.)

26 **Ans:** Proof; OR: HCF 12, LCM 1560

Assume $5 - 2\sqrt{3}$ is rational: $5 - 2\sqrt{3} = \frac{p}{q}$, p, q integers, $q \neq 0$.

Then $2\sqrt{3} = 5 - \frac{p}{q} = \frac{5q - p}{q}$, so $\sqrt{3} = \frac{5q - p}{2q}$.

The right side is rational (integers, $2q \neq 0$), so $\sqrt{3}$ would be rational. This contradicts that $\sqrt{3}$ is irrational.

Hence $5 - 2\sqrt{3}$ is irrational.

OR: $120 = 2^3 \times 3 \times 5$, $156 = 2^2 \times 3 \times 13$.

HCF $= 2^2 \times 3 = 12$ (lowest powers of common primes); LCM $= 2^3 \times 3 \times 5 \times 13 = 1560$ (highest powers of all primes).

HCF $\times$ LCM $= 12 \times 1560 = 18720$ and $120 \times 156 = 18720$. Verified.

27 **Ans:** $\frac{3}{2}$ and $-\frac{1}{3}$

Split the middle term: $6x^2 - 9x + 2x - 3 = 3x(2x - 3) + 1(2x - 3) = (2x - 3)(3x + 1)$.

Zeroes: $x = \frac{3}{2}$ and $x = -\frac{1}{3}$.

Sum $= \frac{3}{2} - \frac{1}{3} = \frac{7}{6}$ and $-\frac{b}{a} = -\frac{-7}{6} = \frac{7}{6}$.

Verified.

Product $= \frac{3}{2} \times \left(-\frac{1}{3}\right) = -\frac{1}{2}$ and $\frac{c}{a} = \frac{-3}{6} = -\frac{1}{2}$. Verified.

28 **Ans:** 693 cm^2

Shaded area = big semicircle − semicircle on AC + semicircle on CB.

The two small semicircles have the same radius 10.5 cm, so they cancel.

Shaded area = area of big semicircle $= \frac{1}{2}\pi R^2 = \frac{1}{2} \times \frac{22}{7} \times 21 \times 21 = 693$ cm^2.

29 **Ans:** Proof

Join OP, OQ, OR. $OP \perp RP$ and $OQ \perp RQ$.

$\triangle OPR \cong \triangle OQR$ (RHS: $OP = OQ$, OR common), so $\angle ORP = \angle ORQ = 60^\circ$ and $PR = RQ$.

In right $\triangle OPR$: $\cos 60^\circ = \frac{PR}{OR} \Rightarrow \frac{1}{2} = \frac{PR}{OR} \Rightarrow OR = 2PR$.

Since $PR = RQ$, $OR = PR + PR = PR + RQ$. Hence proved.

30 **Ans:** Proof

LHS $= \frac{(\sec\theta - 1) + (\sec\theta + 1)}{\sqrt{(\sec\theta + 1)(\sec\theta - 1)}} = \frac{2\sec\theta}{\sqrt{\sec^2\theta - 1}}$.

$\sec^2\theta - 1 = \tan^2\theta$, so LHS $= \frac{2\sec\theta}{\tan\theta} = \frac{2/\cos\theta}{\sin\theta/\cos\theta} = \frac{2}{\sin\theta} = 2\,\mathrm{cosec}\,\theta =$ RHS (for $0^\circ < \theta < 90^\circ$).

OR: $\frac{x}{p} = \sec\theta$, $\frac{y}{q} = \tan\theta$.

$\frac{x^2}{p^2} - \frac{y^2}{q^2} = \sec^2\theta - \tan^2\theta = 1$. Hence proved.

31 **Ans:** (i) $\frac{27}{40}$ (ii) $\frac{3}{10}$ (iii) $\frac{3}{20}$

Total outcomes $= 40$.

(i) Multiples of 2: 20, multiples of 3: 13, multiples of 6 (counted twice): 6. Favourable $= 20 + 13 - 6 = 27$, $P = \frac{27}{40}$.

(ii) Primes up to 40: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, i.e. 12. $P = \frac{12}{40} = \frac{3}{10}$.

(iii) Squares: 1, 4, 9, 16, 25, 36, i.e. 6. $P = \frac{6}{40} = \frac{3}{20}$.

32 **Ans:** 800 km/h; OR: 8 years

Let the usual speed be x km/h. $\frac{1600}{x} - \frac{1600}{x+400} = \frac{40}{60} = \frac{2}{3}$.

$\frac{1600 \times 400}{x(x+400)} = \frac{2}{3} \Rightarrow x^2 + 400x - 960000 = 0$.

$D = 160000 + 3840000 = 4000000 = 2000^2$, so $x = \frac{-400 \pm 2000}{2} = 800$ or -1200.

Speed cannot be negative, so the usual speed is 800 km/h.

OR: Let present age be x. $\frac{1}{x-2} + \frac{1}{x+4} = \frac{1}{4} \Rightarrow 4(2x+2) = (x-2)(x+4)$.

$8x + 8 = x^2 + 2x - 8 \Rightarrow x^2 - 6x - 16 = 0 \Rightarrow (x-8)(x+2) = 0$.

$x = 8$ ($x = -2$ rejected). Present age $= 8$ years.

33 **Ans:** $x = 5$, $BC = 15$ cm

Join AC, meeting EF at G. Then $EG \parallel DC$ and $GF \parallel AB$.

In $\triangle ADC$, $EG \parallel DC$, so by BPT $\frac{AE}{ED} = \frac{AG}{GC}$ ... (1)

In $\triangle CAB$, $GF \parallel AB$, so by BPT $\frac{CG}{GA} = \frac{CF}{FB}$, i.e. $\frac{AG}{GC} = \frac{BF}{FC}$ ... (2)

From (1) and (2), $\frac{AE}{ED} = \frac{BF}{FC}$. Hence proved.

(ii) $\frac{4}{6} = \frac{x+1}{2x-1} \Rightarrow 4(2x-1) = 6(x+1) \Rightarrow 8x - 4 = 6x + 6 \Rightarrow x = 5$.

$BF = 6$ cm, $FC = 9$ cm, so $BC = 15$ cm (check: $\frac{4}{6} = \frac{6}{9}$).

34 **Ans:** (i) $376.8\ \text{cm}^3$ (ii) $292.02\ \text{cm}^2$

$r = 3$ cm, cylinder $h = 10$ cm, cone height $h_1 = 4$ cm, slant height $l = \sqrt{3^2 + 4^2} = 5$ cm.

(i) Volume $= \pi r^2 h + \frac{1}{3}\pi r^2 h_1 + \frac{2}{3}\pi r^3 = \pi(90 + 12 + 18) = 120\pi$.

$= 120 \times 3.14 = 376.8\ \text{cm}^3$.

(ii) TSA = CSA of cylinder + CSA of cone + CSA of hemisphere $= 2\pi rh + \pi rl + 2\pi r^2$.

$= \pi(60 + 15 + 18) = 93\pi = 93 \times 3.14 = 292.02\ \text{cm}^2$. (The joined circular faces are inside, so they are not counted.)

35 **Ans:** Mean 29.2, Mode 32.5; OR: Median 30

Class marks x_i: 5, 15, 25, 35, 45, 55. $f_i x_i$: 25, 120, 300, 525, 270, 220.

$\sum f_i = 50$, $\sum f_i x_i = 1460$. Mean $= \frac{1460}{50} = 29.2$.

Modal class 30-40 (highest frequency 15): $l = 30$, $f_1 = 15$, $f_0 = 12$, $f_2 = 6$, $h = 10$.

Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 30 + \frac{3}{30 - 18} \times 10 = 30 + 2.5 = 32.5$.

OR: Cumulative frequencies: 5, 13, 25, 40, 46, 50. $\frac{n}{2} = 25$, so the median class is 20-30 (first $cf \geq 25$), $l = 20$, $cf = 13$, $f = 12$, $h = 10$.

Median $= 20 + \frac{25 - 13}{12} \times 10 = 20 + 10 = 30$.

36 **Ans:** (i) 57

$a = 24$, $d = 3$: $a_{12} = 24 + 11 \times 3 = 57$.

Ans: (ii) Day 13

$24 + 3(n-1) = 60 \Rightarrow 3(n-1) = 36 \Rightarrow n = 13$.

Ans: (iii) 675

$S_{15} = \frac{15}{2}[2 \times 24 + 14 \times 3] = \frac{15}{2} \times 90 = 675$.

OR: Ans: (iii) 1050

$a_{20} = 24 + 19 \times 3 = 81$; $S_{20} = \frac{20}{2}(24 + 81) = 10 \times 105 = 1050$.

37 **Ans:** (i) $(5, 0)$

Midpoint $= \left(\frac{1+9}{2}, \frac{2-2}{2}\right) = (5, 0)$.

Ans: (ii) $2\sqrt{10}$ units

$AB = \sqrt{(7-1)^2 + (4-2)^2} = \sqrt{40} = 2\sqrt{10}$.

Ans: (iii) $D(3, -4)$

Diagonals of a parallelogram bisect each other, so the midpoint of BD is $(5, 0)$.

$\frac{7+x}{2} = 5 \Rightarrow x = 3$; $\frac{4+y}{2} = 0 \Rightarrow y = -4$. $D = (3, -4)$.

OR: Ans: (iii) Diagonals equal

$AC = \sqrt{8^2 + (-4)^2} = \sqrt{80}$ and $BD = \sqrt{(3-7)^2 + (-4-4)^2} = \sqrt{80}$.

A parallelogram with equal diagonals is a rectangle.

38 **Ans:** (i) $500\sqrt{3}$ m

$\tan 60^\circ = \dfrac{1500}{x_1} \Rightarrow x_1 = \dfrac{1500}{\sqrt{3}} = 500\sqrt{3}$ m.

Ans: (ii) $1500\sqrt{3}$ m

$\tan 30^\circ = \dfrac{1500}{x_2} \Rightarrow x_2 = 1500\sqrt{3}$ m.

Ans: (iii) $1000\sqrt{3}$ m; $50\sqrt{3} \approx 86.6$ m/s

Distance $= x_2 - x_1 = 1500\sqrt{3} - 500\sqrt{3} = 1000\sqrt{3} \approx 1732$ m.

Speed $= \dfrac{1000\sqrt{3}}{20} = 50\sqrt{3} \approx 86.6$ m/s (about 311.8 km/h).

OR: Ans: (iii) 20 s; $\tan\theta = \dfrac{3}{5\sqrt{3}} = \dfrac{\sqrt{3}}{5}$

It must fly another $1000\sqrt{3}$ m at $50\sqrt{3}$ m/s: 20 s.

$\tan\theta = \dfrac{1500}{2500\sqrt{3}} = \dfrac{3}{5\sqrt{3}} = \dfrac{\sqrt{3}}{5}$.

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Point inside circle condition : distance from centre less than radius
2. Speed equation gives a negative root : reject it in writing
3. BPT in a trapezium : join a diagonal, then apply BPT in two triangles
4. Mode : $l + \dfrac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$
5. Equal zeroes of $ax^2 + bx + c$: $b^2 = 4ac$
6. Sector perimeter : arc length plus $2r$

DAY 50 COMPLETE?

- ☐ Mock 1 marked
- ☐ Mistake table filled
- ☐ 10 fix-its solved
- ☐ Weak chapters noted

Next: Day 51, Mock Test 2: Full Paper, 80 marks, 3 hours

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 51

MOCK TEST 2 | FULL PAPER | 80 MARKS

Mock Test 2: Full Board Paper

Mock Test 2

Aaj ka target

- [ ] Mock 1 ki galtiyon ki list ek baar padh ke baitho
- [ ] 3 ghante ka timer, 15 minute reading
- [ ] Section D aur E pe pura time do
- [ ] Answers Day 52 pe check karna



SHIVAM BHAIYA SAYS

Is baar target simple hai: Mock 1 se zyada marks, aur Mock 1 wali galti dobara nahi. Paper shuru karne se pehle apni mistake list 2 minute dekh lo.

INSTRUCTIONS

Time: 3 hours | Maximum marks: 80 | 38 questions, all compulsory | Sections A (20 × 1), B (5 × 2), C (6 × 3), D (4 × 5), E (3 × 4); internal choice where OR is given; draw figures where needed; use $\pi = \frac{22}{7}$ unless stated.

Day 51 Practice

Mock paper | 80 marks | 3 hours | attempt in one sitting

SECTION A | MCQ AND ASSERTION-REASON (20 MARKS)

MCQ

1 The HCF of 14, 15 and 121 is (1)

(a) 7 (b) 11 (c) 1 (d) 5

MCQ

2 The least number that is divisible by all the numbers from 1 to 7 is (1)

(a) 210 (b) 420 (c) 840 (d) 5040

MCQ

3 A quadratic polynomial whose zeroes have sum $\sqrt{2}$ and product $\frac{1}{3}$ is (1)

(a) $3x^2 - 3\sqrt{2}x + 1$ (b) $3x^2 + 3\sqrt{2}x + 1$
(c) $x^2 - \sqrt{2}x + 3$ (d) $3x^2 - \sqrt{2}x + 1$

MCQ

4 Which equation forms a pair of coincident lines with $3x - 4y = 7$? (1)

(a) $6x - 8y = 7$ (b) $-6x + 8y = -14$
(c) $6x + 8y = 14$ (d) $3x - 4y = 14$

MCQ

5 Which of the following equations has -3 as a root? (1)

(a) $x^2 + 2x + 3 = 0$ (b) $2x^2 + 5x - 3 = 0$
(c) $x^2 - 3x = 0$ (d) $3x^2 - x - 3 = 0$

MCQ

6 How many two-digit numbers are divisible by 7? (1)

(a) 12 (b) 14 (c) 13 (d) 15

MCQ

7 The discriminant of the quadratic equation $3x^2 - 2x + \frac{1}{3} = 0$ is (1)

(a) 0 (b) 8 (c) -8 (d) 4

MCQ

8 The distance between the points $A(0, 9)$ and $B(-9, 0)$ is (1)

(a) 9 units (b) 18 units
(c) $9\sqrt{2}$ units (d) $3\sqrt{2}$ units

MCQ

9 How many points lie in the third quadrant at a distance of 7 units from the origin? (1)

(a) none (b) exactly one
(c) exactly two (d) infinitely many

MCQ

10 Diagonals AC and BD of quadrilateral $ABCD$ meet at P, with $PA = 8$, $PB = 4$, $PC = 3$, $PD = 6$ (all cm). If $\angle APB = 70^\circ$ and $\angle CDP = 45^\circ$, then $\angle PBA$ is (1)

(a) 45° (b) 65° (c) 70° (d) 110°

MCQ

11 PQ is a chord of a circle with centre O and PT is the tangent at P. If $\angle POQ = 136^\circ$, then $\angle TPQ$ is (1)

(a) 44° (b) 22° (c) 68° (d) 136°

MCQ

12 Two concentric circles have radii 17 cm and 8 cm. The length of the chord of the larger circle which touches the smaller circle is (1)

(a) 15 cm (b) 25 cm (c) 30 cm (d) 9 cm

MCQ

13 If $\tan A = \sqrt{3}$ and $\sin B = \frac{1}{2}$, where A and B are acute, then $A - B$ equals (1)

(a) 0° (b) 30° (c) 60° (d) 90°

MCQ

14 If $\operatorname{cosec}\theta = 3x$ and $\cot\theta = \frac{3}{x}$, then the value of $x^2 - \frac{1}{x^2}$ is (1)

(a) $\frac{1}{3}$ (b) 9 (c) 1 (d) $\frac{1}{9}$

MCQ

15 The angle of elevation of the sun is 60°. The length of the shadow of a 45 m tall tower is (1)

(a) $45\sqrt{3}$ m (b) $15\sqrt{3}$ m (c) 15 m (d) $\frac{45}{2}$ m

MCQ

16 For the classes 10-20, 20-30, 30-40, 40-50, 50-60 with frequencies 6, 9, 15, 12, 8, the median class is (1)

(a) 20-30 (b) 30-40 (c) 40-50 (d) 10-20

MCQ

17 A number is chosen at random from 1 to 50. The probability that it is a prime number is (1)

(a) $\frac{1}{4}$ (b) $\frac{3}{10}$ (c) $\frac{7}{25}$ (d) $\frac{8}{25}$

MCQ

18 A bag has 4 red, 7 white and 9 green balls. One ball is drawn at random. The probability that it is **not** white is (1)

(a) $\frac{7}{20}$ (b) $\frac{9}{20}$ (c) $\frac{13}{20}$ (d) $\frac{11}{20}$

ASSERTION-REASON

19 **Assertion (A):** The HCF of the smallest prime number and the smallest composite number is 2.
Reason (R): The HCF of two numbers always divides their LCM. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

20 **Assertion (A):** $(x-2)^2 + 1 = 2x - 3$ is a quadratic equation.
Reason (R): An equation of the form $ax^2 + bx + c = 0$ with $a \neq 0$ is a quadratic equation. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B | VERY SHORT ANSWER (5 X 2 = 10 MARKS)

VERY SHORT ANSWER

21 Find the values of p for which the quadratic equation $2x^2 + px + 8 = 0$ has two equal real roots.
OR
Find the roots of $2x^2 - x - 6 = 0$ by factorisation. (2)

VERY SHORT ANSWER

22 A vertical pole 6 m long casts a shadow 4 m long on the ground. At the same time a tower casts a shadow 28 m long. Using similar triangles, find the height of the tower. (2)

VERY SHORT ANSWER

23 PA and PB are tangents from an external point P to a circle with centre O. If $\angle PAB = 65^\circ$, find $\angle APB$ and $\angle AOB$. (2)

VERY SHORT ANSWER

24 Find K if $(1 + \tan^2\theta)(1 - \sin\theta)(1 + \sin\theta) = K$.
OR
Show that $(1 + \cot^2\theta)(1 - \cos\theta)(1 + \cos\theta) = 1$. (2)

VERY SHORT ANSWER

25 The surface area of a sphere is 2464 cm^2. Find its volume. (Use $\pi = \frac{22}{7}$) (2)

SECTION C | SHORT ANSWER (6 X 3 = 18 MARKS)

SHORT ANSWER

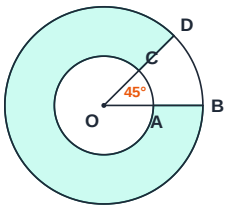
26 Traffic lights at three different crossings change after every 36 seconds, 60 seconds and 84 seconds respectively. If they all change together at 8 a.m., at what time will they next change together?
OR
Given that $\sqrt{5}$ is irrational, prove that $7 - 3\sqrt{5}$ is irrational. (3)

SHORT ANSWER

27 Four years from now, a father will be three times as old as his son. Four years ago, he was five times as old as his son. Find their present ages. (3)

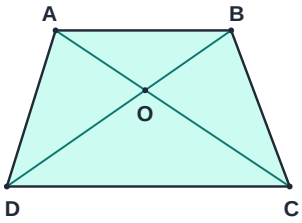
SHORT ANSWER

28 In the figure, $OA = 14$ cm and $OB = 28$ cm are radii of two concentric circles with centre O, and $\angle AOC = 45°$. Find the area of the shaded region, which is the part of the ring **outside** the $45°$ sector. (Use $\pi = \frac{22}{7}$) (3)



SHORT ANSWER

29 $ABCD$ is a trapezium with $AB \parallel DC$ whose diagonals meet at O. Show that $\frac{AO}{OC} = \frac{BO}{OD}$. If $AO = 3x - 1$, $OC = 5x - 3$, $BO = 2x + 1$ and $OD = 6x - 5$, find x. (3)



SHORT ANSWER

30 If $\cos\theta + \sin\theta = \sqrt{2}\cos\theta$, prove that $\cos\theta - \sin\theta = \sqrt{2}\sin\theta$.

OR

Prove that $\frac{\cos A}{1 - \sin A} + \frac{1 - \sin A}{\cos A} = 2\sec A$. (3)

SHORT ANSWER

31 The mean of the following distribution is 52. Find the missing frequency f.

Class: 0-20, 20-40, 40-60, 60-80, 80-100

Frequency: 6, 9, f, 12, 7 (3)

SECTION D | LONG ANSWER (4 X 5 = 20 MARKS)

LONG ANSWER

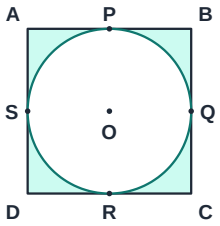
32 The sum of the first 6 terms of an AP is 66 and the sum of its next 6 terms is 210. Find the AP and its 20th term.

OR

How many terms of the AP $45, 41, 37, \ldots$ must be taken so that their sum is 273? Show that only one value is possible. (5)

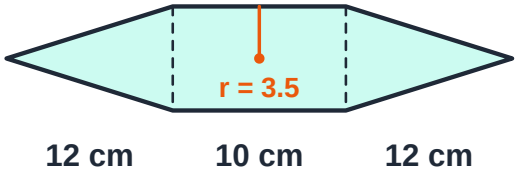
LONG ANSWER

33 Prove that the lengths of the tangents drawn from an external point to a circle are equal. Using this, prove that a rectangle circumscribing a circle is a square. (5)



LONG ANSWER

34 A wooden toy (a 'gilli') is made of a cylinder of length 10 cm with a cone of height 12 cm attached at each end. The common radius is 3.5 cm. Find the volume of wood and the total surface area of the toy. (Use $\pi = \frac{22}{7}$) (5)



LONG ANSWER

35 Find the median and the mode of the following data.

Class: 10-20, 20-30, 30-40, 40-50, 50-60, 60-70

Frequency: 4, 7, 12, 18, 6, 3 (5)

SECTION E | CASE-BASED (3 X 4 = 12 MARKS)

CASE-BASED

36 Case Study 1: A rectangular kitchen garden has its length 7 m more than its breadth, and its area is 330 m^2. (4)

(i) Taking the breadth as x m, write the quadratic equation for the area. (1)

(ii) Find the discriminant of this equation. (1)

(iii) Find the length and breadth of the garden. (2)

OR (iii) A path 2 m wide is laid all around the outside of the garden. Find the area of the path. (2)

CASE-BASED

37 **Case Study 2: On a town map drawn on a grid, a school is at $A(2, 3)$ and a library at $B(8, 11)$ (1 unit $=$ 100 m).** (4)

(i) Find the distance between the school and the library on the map. (1)

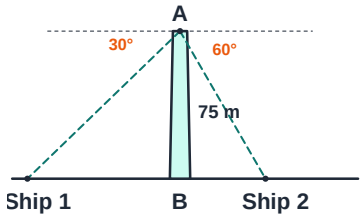
(ii) A bus stop is exactly midway between them. Find its coordinates. (1)

(iii) A park P lies on AB with $AP : PB = 1 : 3$. Find its coordinates. (2)

OR (iii) Find the point on the x-axis that is equidistant from A and B. (2)

CASE-BASED

38 **Case Study 3: A lighthouse is 75 m high. From its top, the angles of depression of two ships on opposite sides of it, in line with its foot, are $30°$ and $60°$.** (4)



(i) How far is the ship seen at $60°$ from the foot of the lighthouse? (1)

(ii) How far is the ship seen at $30°$ from the foot? (1)

(iii) Find the distance between the two ships. (Use $\sqrt{3} = 1.732$) (2)

OR (iii) If both ships were on the same side of the lighthouse, how far apart would they be? (2)

ANSWERS AND FULL WORKED SOLUTIONS FOR THIS PAPER ARE ON DAY 52. CHECK ONLY AFTER YOU FINISH THE WHOLE PAPER.

DAY 51 COMPLETE? ☐ Paper done in one sitting ☐ Timer followed ☐ All 38 attempted ☐ Not checked yet (Day 52)

Next: Day 52, Mock 2 Review: Solutions and Mistake Fixing

MY MISTAKES AND NOTES Write every mistake of this Day with the correct step

DAY 52

MOCK 2 REVIEW | SOLUTIONS AND MISTAKE FIXING

Mock 2 Review: Mark, Compare, Fix

Mock 2 Review

Aaj ka target

- [ ] Mock 2 board ki tarah check karo
- [ ] Jo galti dono mock mein hui, use red mein likho
- [ ] Mock 1 se score compare karo
- [ ] 10 fix-it sawal solve karo

52.1 How to mark yourself

1. Keep a red pen. Check each answer against the worked solutions printed below, question by question.
2. MCQ and AR: 1 mark only if the letter matches.
3. 2, 3 and 5 mark questions: give step marks the way the board does: correct formula or setup, correct working, correct final answer with units. A right answer with no working gets at most half.
4. Case-based: mark each part separately (1 + 1 + 2).
5. Fill the score table, then the mistake table, then solve the fix-it questions.

Section	Max	My marks
A (Q1-20)	20	
B (Q21-25)	10	
C (Q26-31)	18	
D (Q32-35)	20	
E (Q36-38)	12	
Total	80	

Fill in after checking. 60+ is on track, 70+ is strong.

	Mock 1	Mock 2
Total (80)		
Section A (20)		
Sections D+E (32)		

	Mock 1	Mock 2
Questions left		

Compare: the aim is a higher Mock 2 score and fewer repeated mistakes.

52.2 Top traps in Mock 2



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Q29 (trapezium) mein $x = \frac{1}{2}$ bhi aata hai, par tab OC negative ho jaata hai. Har length positive honi chahiye: check karke hi answer likho.



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Q35 median mein cf **median class se pehle wali** class ka lena hai (23), median class ka (41) nahi.



SHIVAM BHAIYA SAYS

Q32 OR mein $n = 10.5$ bhi aata hai. Terms ki ginti natural number hi hogi, isliye likh ke reject karo, phir $n = 13$.



SHIVAM BHAIYA SAYS: BOARD TIP

Case study (Q38) mein ships opposite side pe hain to distances **jodo**, same side pe ho to **ghatao**. Pehle rough figure banao, phir likho.

52.3 Mistake analysis

Q no.	Chapter	Type of mistake	Fix (one line)
		concept / calculation / reading / time	

Type of mistake: concept (did not know), calculation (silly slip), reading (misread the question), time (did not reach it).

Day 52 Practice

10 board-pattern questions | solve first, then check

FIX-IT QUESTIONS (MOST-MISSED TOPICS OF MOCK 2)

VERY SHORT ANSWER

1 If $(x-2)$ is a factor of x^3+ax^2+b and $a+b=4$, find a and b. (2)

VERY SHORT ANSWER

2 Find k so that the lines $3x-4y+7=0$ and $6x+ky+14=0$ are coincident. (2)

MCQ

3 The pair $a_1x+b_1y+c_1=0$, $a_2x+b_2y+c_2=0$ is inconsistent when (1)

(a) $\frac{a_1}{a_2}\neq\frac{b_1}{b_2}$ (b) $\frac{a_1}{a_2}=\frac{b_1}{b_2}=\frac{c_1}{c_2}$

(c) $\frac{a_1}{a_2}=\frac{b_1}{b_2}\neq\frac{c_1}{c_2}$ (d) $\frac{b_1}{b_2}=\frac{c_1}{c_2}\neq\frac{a_1}{a_2}$

VERY SHORT ANSWER

4 Solve by factorisation: $x^2+2x-35=0$. (2)

VERY SHORT ANSWER

5 Find the sum of all odd numbers between 100 and 200. (2)

VERY SHORT ANSWER

6 AOB is a diameter of a circle with centre O and AC is the tangent at A. If $\angle BOC=125^\circ$, find $\angle ACO$. (2)

SHORT ANSWER

7 If $a\cot\theta+b\,\mathrm{cosec}\,\theta=p$ and $b\cot\theta+a\,\mathrm{cosec}\,\theta=q$, prove that $p^2-q^2=b^2-a^2$. (3)

SHORT ANSWER

8 Two boats approach a cliff along the same line. From the top of the cliff, the angles of depression of the boats are 30° and 45°, and the boats are 60 m apart. Find the height of the cliff. (Use $\sqrt{3}=1.732$) (3)

VERY SHORT ANSWER

9 Find the area of a quadrant of a circle of diameter 42 cm. (Use $\pi=\frac{22}{7}$) (2)

VERY SHORT ANSWER

10 A number is chosen at random from 1 to 30. Find the probability that it is a multiple of 4 or 6. (2)

Day 52 Answers with worked solutions

1 **Ans:** $a=-4$, $b=8$

$(x-2)$ is a factor, so $p(2)=0$: $8+4a+b=0\Rightarrow 4a+b=-8$.

With $a+b=4$: subtracting, $3a=-12\Rightarrow a=-4$, so $b=8$.

Check: $p(2)=8-16+8=0$.

2 **Ans:** $k=-8$

Coincident: $\frac{3}{6}=\frac{-4}{k}=\frac{7}{14}$.

$\frac{-4}{k}=\frac{1}{2}\Rightarrow k=-8$.

3 **(c)**

Inconsistent means no solution: parallel lines, so $\frac{a_1}{a_2}=\frac{b_1}{b_2}\neq\frac{c_1}{c_2}$.

4 **Ans:** $x=5$ or $x=-7$

$x^2+7x-5x-35=x(x+7)-5(x+7)=(x-5)(x+7)=0$.

$x=5$ or $x=-7$.

5 **Ans:** 7500

AP: $101,103,\ldots,199$; $199=101+(n-1)2\Rightarrow n=50$.

$S_{50}=\frac{50}{2}(101+199)=25\times 300=7500$.

6 **Ans:** 35°

$\angle AOC=180^\circ-125^\circ=55^\circ$ (linear pair).

$\angle OAC=90^\circ$ (radius $\perp$ tangent), so $\angle ACO=180^\circ-90^\circ-55^\circ=35^\circ$.

7 **Ans:** Proof

$p^2=a^2\cot^2\theta+2ab\cot\theta\,\mathrm{cosec}\,\theta+b^2\mathrm{cosec}^2\theta$.

$q^2=b^2\cot^2\theta+2ab\cot\theta\,\mathrm{cosec}\,\theta+a^2\mathrm{cosec}^2\theta$.

$p^2-q^2=a^2(\cot^2\theta-\mathrm{cosec}^2\theta)+b^2(\mathrm{cosec}^2\theta-\cot^2\theta)=-a^2+b^2$, since $\mathrm{cosec}^2\theta-\cot^2\theta=1$.

8 **Ans:** $30(\sqrt{3}+1)\approx 81.96$ m

Let height $=h$. Nearer boat (at 45°) is h m from the foot; farther boat (at 30°) is $h\sqrt{3}$ m from the foot.

$h\sqrt{3}-h=60\Rightarrow h=\frac{60}{\sqrt{3}-1}=\frac{60(\sqrt{3}+1)}{2}=30(\sqrt{3}+1)$.

$h\approx 30\times 2.732=81.96$ m.

9 **Ans:** $346.5\ \mathrm{cm}^2$

$r=21$ cm. Quadrant $=\frac{1}{4}\pi r^2=\frac{1}{4}\times\frac{22}{7}\times 441=346.5\ \mathrm{cm}^2$.

10 **Ans:** $\frac{1}{3}$

Multiples of 4: 7; multiples of 6: 5; multiples of 12 (counted twice): 2.

Favourable $= 7 + 5 - 2 = 10$; $P = \frac{10}{30} = \frac{1}{3}$.

Mock Test 2: worked solutions

marks as the board gives them

Mock Test 2 answers

1 **(c)**

$14 = 2 \times 7$, $15 = 3 \times 5$, $121 = 11^2$: no prime is common to all three, so HCF $= 1$.

2 **(b)**

LCM$(1, \ldots, 7)$ takes the highest prime powers: $2^2 \times 3 \times 5 \times 7 = 420$.

3 **(a)**

$k\left(x^2 - \sqrt{2}x + \frac{1}{3}\right)$; with $k = 3$: $3x^2 - 3\sqrt{2}x + 1$.

4 **(b)**

Coincident lines need $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$.

$-6x + 8y = -14$ is -2 times $3x - 4y = 7$, so all three ratios equal $-\frac{1}{2}$.

5 **(b)**

Substitute $x = -3$: (a) $9 - 6 + 3 = 6$; (b) $18 - 15 - 3 = 0$; (c) $9 + 9 = 18$; (d) $27 + 3 - 3 = 27$. Only (b) gives 0.

6 **(c)**

AP: $14, 21, \ldots, 98$ with $d = 7$.

$98 = 14 + (n-1)7 \Rightarrow n - 1 = 12 \Rightarrow n = 13$.

7 **(a)**

$D = b^2 - 4ac = 4 - 4 \times 3 \times \frac{1}{3} = 4 - 4 = 0$ (equal roots).

8 **(c)**

$AB = \sqrt{(-9-0)^2 + (0-9)^2} = \sqrt{162} = 9\sqrt{2}$.

9 **(d)**

All points with $x^2 + y^2 = 49$, $x < 0$, $y < 0$ qualify: the whole quarter arc of the circle of radius 7 in the third quadrant. So infinitely many.

10 **(b)**

$\frac{PA}{PD} = \frac{8}{6} = \frac{4}{3}$ and $\frac{PB}{PC} = \frac{4}{3}$, with $\angle APB = \angle DPC = 70^\circ$ (vertically opposite).

So $\triangle APB \sim \triangle DPC$ (SAS), giving $\angle PAB = \angle PDC = 45^\circ$.

$\angle PBA = 180^\circ - 70^\circ - 45^\circ = 65^\circ$.

11 **(c)**

$OP = OQ$, so $\angle OPQ = \frac{180^\circ - 136^\circ}{2} = 22^\circ$.

$OP \perp PT$, so $\angle TPQ = 90^\circ - 22^\circ = 68^\circ$.

12 **(c)**

The radius of the small circle is perpendicular to the chord at the point of contact, which is also its midpoint.

Half chord $= \sqrt{17^2 - 8^2} = 15$ cm, so the chord $= 30$ cm.

13 **(b)**

$\tan A = \sqrt{3} \Rightarrow A = 60^\circ$; $\sin B = \frac{1}{2} \Rightarrow B = 30^\circ$.

$A - B = 30^\circ$.

14 **(d)**

$\operatorname{cosec}^2\theta - \cot^2\theta = 1 \Rightarrow 9x^2 - \frac{9}{x^2} = 1$.

So $x^2 - \frac{1}{x^2} = \frac{1}{9}$.

15 **(b)**

$\tan 60^\circ = \frac{45}{s} \Rightarrow s = \frac{45}{\sqrt{3}} = 15\sqrt{3}$ m.

16 **(b)**

$n = 50$, $\frac{n}{2} = 25$. Cumulative frequencies: 6, 15, 30, 42, 50.

The first $cf \geq 25$ is 30, so the median class is 30-40.

17 **(b)**

Primes up to 50: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, i.e. 15.

$P = \frac{15}{50} = \frac{3}{10}$.

18 **(c)**

Total $= 20$. Not white $= 4 + 9 = 13$. $P = \frac{13}{20}$.

19 **(b)**

Smallest prime $= 2$, smallest composite $= 4$, HCF $= 2$: A is true.

R is also true (HCF divides each number, and each number divides the LCM), but R does not explain why the HCF here is 2.

20 **(a)**

Expanding: $x^2 - 4x + 5 = 2x - 3 \Rightarrow x^2 - 6x + 8 = 0$, with $a = 1 \neq 0$. A is true.

R is the definition used to decide this, so R explains A.

21 **Ans:** $p = \pm 8$; OR: $2, -\frac{3}{2}$

Equal roots: $D = b^2 - 4ac = 0$.

$p^2 - 4 \times 2 \times 8 = 0 \Rightarrow p^2 = 64 \Rightarrow p = 8$ or $p = -8$.

OR: $2x^2 - 4x + 3x - 6 = 2x(x-2) + 3(x-2) = (x-2)(2x+3) = 0$.

$x = 2$ or $x = -\frac{3}{2}$.

22 **Ans:** 42 m

Sun's rays make the same angle with the ground at the same time, and both objects are vertical, so the two right triangles are similar (AA).

$\frac{\text{height of tower}}{28} = \frac{6}{4} \Rightarrow \text{height} = \frac{6 \times 28}{4} = 42$ m.

23 **Ans:** $\angle APB = 50^\circ$, $\angle AOB = 130^\circ$

$PA = PB$ (tangents from an external point), so $\angle PBA = \angle PAB = 65^\circ$ and $\angle APB = 180^\circ - 130^\circ = 50^\circ$.

In quadrilateral $OAPB$, $\angle OAP = \angle OBP = 90^\circ$, so $\angle AOB = 360^\circ - 90^\circ - 90^\circ - 50^\circ = 130^\circ$.

24 **Ans:** $K = 1$

$(1 - \sin\theta)(1 + \sin\theta) = 1 - \sin^2\theta = \cos^2\theta$ and $1 + \tan^2\theta = \sec^2\theta$.

$K = \sec^2\theta \times \cos^2\theta = 1$.

OR: $(1 - \cos\theta)(1 + \cos\theta) = \sin^2\theta$ and $1 + \cot^2\theta = \text{cosec}^2\theta$; product $= 1$.

25 **Ans:** $11498\frac{2}{3}$ cm^3

$4\pi r^2 = 2464 \Rightarrow r^2 = \frac{2464 \times 7}{4 \times 22} = 196 \Rightarrow r = 14$ cm.

$V = \frac{4}{3}\pi r^3 = \frac{4}{3} \times \frac{22}{7} \times 2744 = \frac{34496}{3} = 11498\frac{2}{3}$ cm^3.

26 **Ans:** 8:21 a.m.; OR: proof

Required time $=$ LCM$(36, 60, 84)$.

$36 = 2^2 \times 3^2$, $60 = 2^2 \times 3 \times 5$, $84 = 2^2 \times 3 \times 7$.

LCM $= 2^2 \times 3^2 \times 5 \times 7 = 1260$ s $= 21$ minutes.

They change together again at 8:21 a.m.

OR: Suppose $7 - 3\sqrt{5} = \frac{p}{q}$ (p, q integers, $q \neq 0$).

Then $\sqrt{5} = \frac{7q - p}{3q}$, which is rational.

This contradicts that $\sqrt{5}$ is irrational, so $7 - 3\sqrt{5}$ is irrational.

27 **Ans:** Son 12 years, father 44 years

Let the present ages be father x, son y.

$x + 4 = 3(y + 4) \Rightarrow x - 3y = 8$ ... (1)

$x - 4 = 5(y - 4) \Rightarrow x - 5y = -16$ ... (2)

(1) $-$ (2): $2y = 24 \Rightarrow y = 12$; then $x = 8 + 36 = 44$.

Check: in 4 years, $48 = 3 \times 16$; 4 years ago, $40 = 5 \times 8$.

28 **Ans:** 1617 cm^2

Area of ring $= \pi(R^2 - r^2) = \frac{22}{7}(784 - 196) = \frac{22}{7} \times 588 = 1848$ cm^2.

Part of the ring inside the 45° sector $= \frac{45}{360} \times 1848 = 231$ cm^2.

Shaded area $= 1848 - 231 = 1617$ cm^2.

29 **Ans:** $x = 2$

In $\triangle AOB$ and $\triangle COD$: $\angle OAB = \angle OCD$ and $\angle OBA = \angle ODC$ (alternate angles, $AB \parallel DC$).

So $\triangle AOB \sim \triangle COD$ (AA), giving $\frac{AO}{OC} = \frac{BO}{OD}$.

$\frac{3x - 1}{5x - 3} = \frac{2x + 1}{6x - 5} \Rightarrow (3x - 1)(6x - 5) = (2x + 1)(5x - 3)$.

$18x^2 - 21x + 5 = 10x^2 - x - 3 \Rightarrow 8x^2 - 20x + 8 = 0 \Rightarrow 2x^2 - 5x + 2 = 0 \Rightarrow x = 2$ or $\frac{1}{2}$.

$x = \frac{1}{2}$ makes $OC = -\frac{1}{2} < 0$, so $x = 2$.

30 **Ans:** Proof

$\sin\theta = (\sqrt{2} - 1)\cos\theta \Rightarrow \cos\theta = \frac{\sin\theta}{\sqrt{2} - 1} = (\sqrt{2} + 1)\sin\theta$ (rationalising).

So $\cos\theta - \sin\theta = \sqrt{2}\sin\theta + \sin\theta - \sin\theta = \sqrt{2}\sin\theta$. Proved.

OR: LHS $= \frac{\cos^2 A + (1 - \sin A)^2}{(1 - \sin A)\cos A} = \frac{\cos^2 A + 1 - 2\sin A + \sin^2 A}{(1 - \sin A)\cos A}$.

$= \frac{2 - 2\sin A}{(1 - \sin A)\cos A} = \frac{2(1 - \sin A)}{(1 - \sin A)\cos A} = \frac{2}{\cos A} = 2\sec A$.

31 **Ans:** $f = 16$

Class marks: 10, 30, 50, 70, 90. $\sum f_i = 34 + f$.

$\sum f_i x_i = 60 + 270 + 50f + 840 + 630 = 1800 + 50f$.

$\frac{1800 + 50f}{34 + f} = 52 \Rightarrow 1800 + 50f = 1768 + 52f \Rightarrow 2f = 32 \Rightarrow f = 16$.

32 **Ans:** AP $1, 5, 9, \ldots$; $a_{20} = 77$. OR: 13 terms

$S_6 = 3(2a + 5d) = 66 \Rightarrow 2a + 5d = 22$ ... (1)

$S_{12} = 66 + 210 = 276 = 6(2a + 11d) \Rightarrow 2a + 11d = 46$ ... (2)

(2) $-$ (1): $6d = 24 \Rightarrow d = 4$; then $2a = 2 \Rightarrow a = 1$. AP: $1, 5, 9, 13, \ldots$

$a_{20} = 1 + 19 \times 4 = 77$.

OR: $a = 45$, $d = -4$: $\frac{n}{2}[90 + (n-1)(-4)] = 273 \Rightarrow n(94 - 4n) = 546 \Rightarrow 2n^2 - 47n + 273 = 0$.

$D = 47^2 - 8 \times 273 = 2209 - 2184 = 25$, so $n = \frac{47 \pm 5}{4} = 13$ or 10.5.

n must be a natural number, so $n = 13$ (only one value).

33 **Ans:** Proof

Given: circle with centre O, external point P, tangents PA and PB. **To prove:** $PA = PB$.

Join OA, OB, OP. $\angle OAP = \angle OBP = 90^\circ$ (radius $\perp$ tangent).

In right $\triangle OAP$ and $\triangle OBP$: $OA = OB$ (radii), OP common. So $\triangle OAP \cong \triangle OBP$ (RHS) and $PA = PB$ (CPCT).

Application: Let rectangle $ABCD$ touch the circle at P, Q, R, S on AB, BC, CD, DA.

$AP = AS$, $BP = BQ$, $CR = CQ$, $DR = DS$ (tangents from the same point).

Adding: $(AP + BP) + (CR + DR) = (AS + DS) + (BQ + CQ) \Rightarrow AB + CD = AD + BC$.

In a rectangle $AB = CD$ and $AD = BC$, so $2AB = 2BC \Rightarrow AB = BC$. A rectangle with adjacent sides equal is a square.

34 **Ans:** Volume $693\ \text{cm}^3$, TSA $495\ \text{cm}^2$

$r = 3.5$, cylinder $h = 10$, each cone $h_1 = 12$.

Slant $l = \sqrt{12^2 + 3.5^2} = \sqrt{156.25} = 12.5$ cm.

Volume $= \pi r^2 h + 2 \times \frac{1}{3}\pi r^2 h_1 = \pi r^2 (10 + 8) = \frac{22}{7} \times 12.25 \times 18 = 693\ \text{cm}^3$.

TSA = CSA of cylinder + 2 CSA of cone $= 2\pi rh + 2\pi rl = 2\pi r(h + l)$.

$= 2 \times \frac{22}{7} \times 3.5 \times (10 + 12.5) = 22 \times 22.5 = 495\ \text{cm}^2$.

35 **Ans:** Median ≈ 41.11, Mode ≈ 43.33

$n = 50$, $\frac{n}{2} = 25$. Cumulative frequencies: 4, 11, 23, 41, 47, 50.

Median class 40-50 (first $cf \geq 25$): $l = 40$, $cf = 23$, $f = 18$, $h = 10$.

Median $= 40 + \frac{25 - 23}{18} \times 10 = 40 + \frac{20}{18} = 41.11$ (approx.).

Modal class 40-50: $f_1 = 18$, $f_0 = 12$, $f_2 = 6$.

Mode $= 40 + \frac{18 - 12}{36 - 12 - 6} \times 10 = 40 + \frac{60}{18} = 43.33$ (approx.).

36 **Ans:** (i) $x^2 + 7x - 330 = 0$

$x(x + 7) = 330 \Rightarrow x^2 + 7x - 330 = 0$.

Ans: (ii) 1369

$D = 7^2 + 4 \times 330 = 49 + 1320 = 1369 = 37^2$.

Ans: (iii) 22 m by 15 m

$x = \frac{-7 \pm 37}{2} = 15$ or -22; breadth cannot be negative, so $x = 15$ m and length $= 22$ m.

OR: Ans: (iii) $164\ \text{m}^2$

Outer rectangle: $(22 + 4) \times (15 + 4) = 26 \times 19 = 494\ \text{m}^2$.

Path $= 494 - 330 = 164\ \text{m}^2$.

37 **Ans:** (i) 10 units (1 km)

$AB = \sqrt{6^2 + 8^2} = 10$ units $= 1000$ m.

Ans: (ii) $(5, 7)$

$\left(\frac{2 + 8}{2}, \frac{3 + 11}{2}\right) = (5, 7)$.

Ans: (iii) $(3.5, 5)$

$P = \left(\frac{1 \times 8 + 3 \times 2}{4}, \frac{1 \times 11 + 3 \times 3}{4}\right) = \left(\frac{14}{4}, \frac{20}{4}\right) = (3.5, 5)$.

OR: Ans: (iii) $\left(\frac{43}{3}, 0\right)$

Let it be $(x, 0)$: $(x - 2)^2 + 9 = (x - 8)^2 + 121$.

$-4x + 13 = -16x + 185 \Rightarrow 12x = 172 \Rightarrow x = \frac{43}{3}$.

38 **Ans:** (i) $25\sqrt{3}$ m

$\tan 60^\circ = \frac{75}{d_1} \Rightarrow d_1 = \frac{75}{\sqrt{3}} = 25\sqrt{3} \approx 43.3$ m.

Ans: (ii) $75\sqrt{3}$ m

$\tan 30^\circ = \frac{75}{d_2} \Rightarrow d_2 = 75\sqrt{3} \approx 129.9$ m.

Ans: (iii) $100\sqrt{3} \approx 173.2$ m

Ships are on opposite sides, so distance $= d_1 + d_2 = 25\sqrt{3} + 75\sqrt{3} = 100\sqrt{3}$.

$= 100 \times 1.732 = 173.2$ m.

OR: Ans: (iii) $50\sqrt{3} \approx 86.6$ m

Distance $= d_2 - d_1 = 75\sqrt{3} - 25\sqrt{3} = 50\sqrt{3} = 86.6$ m.

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Factor theorem : $(x - a)$ is a factor if $p(a) = 0$
2. Coincident lines : all three ratios equal
3. Inconsistent pair : $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$
4. Median cf : of the class before the median class
5. Opposite sides in heights and distances : add the distances
6. Lengths from an equation : reject a value that gives a negative length

DAY 52 COMPLETE? ☐ Mock 2 marked ☐ Compared with Mock 1 ☐ 10 fix-its solved ☐ Repeated mistakes circled

Next: Day 53, Mock Test 3: Full Paper, 80 marks, 3 hours

DAY 53

MOCK TEST 3 | FULL PAPER | 80 MARKS

Mock Test 3: Full Paper, 80 marks, 3 hours

Mock Test 3

Aaj ka target

- [ ] Timer lagao: 3 ghante, ek hi baar mein poora paper
- [ ] Har step likho, sirf answer nahi
- [ ] Section A 25 min mein khatam, phir B aur C
- [ ] Answers sirf Day 54 pe, paper khatam hone ke baad

Instructions: 3 hours, 80 marks, 38 questions in 5 sections: A (20 × 1), B (5 × 2), C (6 × 3), D (4 × 5), E (3 case studies x 4). All questions are compulsory; internal choice (OR) in Q25, Q29, Q31, Q32, Q34, Q35 and the last part of each case study. Use $\pi = \frac{22}{7}$ unless stated.



SHIVAM BHAIYA SAYS: BOARD TIP

Paper shuru karne se pehle 15 minute sirf padho: har section mein jo sawal pakke aate hain unpe tick lagao, aur OR wale sawal mein wahi choose karo jiska pehla step turant dimaag mein aa jaye. Phir Section A se seedha shuru.

Day 53 Practice

Mock paper | 80 marks | 3 hours | attempt in one sitting

SECTION A | MCQ AND AR, 1 MARK EACH

MCQ

1 The ratio of the LCM to the HCF of the smallest odd composite number and the smallest odd prime number is (1)

(a) $1:3$ (b) $9:1$ (c) $3:1$ (d) $1:1$

MCQ

2 If $a = x^4y^3$ and $b = x^2y^6$, where x and y are distinct primes, then LCM(a, b) is (1)

(a) x^2y^3 (b) x^6y^9 (c) x^4y^6 (d) x^4y^3

MCQ

3 A quadratic polynomial whose zeroes are $2+\sqrt{5}$ and $2-\sqrt{5}$ is (1)

(a) $x^2 + 4x - 1$ (b) $x^2 - 4x + 1$ (c) $x^2 - 4x - 1$ (d) $x^2 - 4x - 5$

MCQ

4 The graph of $y = p(x)$ for a polynomial $p(x)$ cuts the x-axis at exactly three points and the y-axis at one point. The number of zeroes of $p(x)$ is (1)

(a) 1 (b) 3 (c) 4 (d) 2

MCQ

5 The pair of equations $x + (k-1)y = 3$ and $(k-1)x + 16y = 2k + 2$ has infinitely many solutions when k equals (1)

(a) -3 (b) 5 or -3 (c) 4 (d) 5

MCQ

6 The roots of the equation $4x^2 - 12x + 9 = 0$ are (1)

(a) real and distinct
(b) real and equal
(c) not real
(d) one positive and one negative

MCQ

7 The 15th term of an AP whose first term is $2m$ and common difference is $-m$ is (1)

(a) $-12m$ (b) $-13m$ (c) $16m$ (d) $-14m$

MCQ

8 The point P which divides the line segment joining A(−3, 4) and B(6, −5) internally in the ratio 1 : 2 lies on (1)

(a) the x-axis (b) the y-axis
(c) quadrant I (d) quadrant IV

MCQ

9 In $\triangle DEF$ and $\triangle LMN$, $\frac{DE}{MN} = \frac{EF}{NL} = \frac{FD}{LM}$. Then (1)

(a) $\triangle DEF \sim \triangle LMN$ (b) $\triangle DEF \sim \triangle MNL$
(c) $\triangle DEF \sim \triangle NLM$ (d) $\triangle EFD \sim \triangle LMN$

MCQ

10 From a point Q, the length of the tangent to a circle is 21 cm and the distance of Q from the centre is 29 cm. The radius of the circle is (1)

(a) 20 cm (b) 8 cm (c) 25 cm (d) 22 cm

MCQ

11 If $\cos A = \frac{1}{2}$ (A acute), then $3\tan^2 A - 2\sec^2 A$ equals (1)

(a) 7 (b) -1 (c) 1 (d) 0

MCQ

12 When the angle of elevation of the sun is $30°$, a vertical pole 12 m high casts a shadow of length (1)

(a) $4\sqrt{3}$ m (b) 12 m (c) $12\sqrt{3}$ m (d) $6\sqrt{3}$ m

MCQ

13 The area of a circle is numerically equal to twice its circumference. Its radius is (1)

(a) 2 units (b) 4 units (c) 8 units (d) 1 unit

MCQ

14 The perimeter of a quadrant of a circle of radius r is (1)

(a) $\frac{\pi r}{2}$ (b) $r\left(\frac{\pi}{4}+2\right)$

(c) $r\left(\frac{\pi}{2}+2\right)$ (d) $\frac{\pi r}{2}+r$

MCQ

15 A cone and a cylinder have equal base radii and equal volumes. The ratio of the height of the cone to the height of the cylinder is (1)

(a) 1 : 3 (b) 3 : 1 (c) 1 : 1 (d) 2 : 3

MCQ

16 The surface area of a sphere is 1386 cm². Its diameter is (use $\pi = \frac{22}{7}$) (1)

(a) 21 cm (b) 10.5 cm (c) 14 cm (d) 42 cm

MCQ

17 For a grouped distribution, $\sum f_i = 25$ and $\sum f_i d_i = -50$, where $d_i = x_i - 45$. The mean is (1)

(a) 43 (b) 47 (c) 45 (d) 41

MCQ

18 Cards numbered 21 to 50 are put in a box and one card is drawn at random. The probability that it shows a prime number is (1)

(a) $\frac{7}{30}$ (b) $\frac{1}{5}$ (c) $\frac{4}{15}$ (d) $\frac{7}{29}$

ASSERTION-REASON

19 **Assertion (A):** $\sqrt{49}$ is a rational number.
Reason (R): The square root of every positive integer is irrational. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

20 **Assertion (A):** In an AP with common difference 4, $a_{25} - a_{18} = 28$.
Reason (R): For an AP, $a_m - a_n = (m-n)d$. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B | 2 MARKS EACH

VERY SHORT ANSWER

21 If the zeroes of the quadratic polynomial $(k+2)x^2 + 11x + 3k$ are reciprocals of each other, find the value of k. (2)

VERY SHORT ANSWER

22 A(1, −2), B(5, 1) and C(4, 6) are three vertices of a parallelogram ABCD, taken in order. Find the coordinates of the fourth vertex D. (2)

VERY SHORT ANSWER

23 If $\sec\theta = \frac{13}{5}$ (θ acute), find the value of $\frac{1-\cos^2\theta}{1+\cos^2\theta}$. (2)

VERY SHORT ANSWER

24 Two concentric circles have radii 37 cm and 12 cm. Find the length of the chord of the larger circle that touches the smaller circle. (2)

VERY SHORT ANSWER

25 The length of an arc of a circle of radius 10.5 cm is 33 cm. Find the area of the sector bounded by this arc.

OR

The minute hand of a clock is 14 cm long. Find the area swept by it from 9:10 am to 9:30 am. (Use $\pi = \frac{22}{7}$) (2)

SECTION C | 3 MARKS EACH

SHORT ANSWER

26 Prove that $\sqrt{5}$ is irrational. Hence show that $\frac{3+2\sqrt{5}}{7}$ is irrational. (3)

SHORT ANSWER

27 For what values of p and q will the pair of linear equations $(2p+1)x + 3y = 9$ and $4x + (q-2)y = 6$ have infinitely many solutions? (3)

SHORT ANSWER

28 In $\triangle ABC$, the altitudes AD (on BC) and CE (on AB) intersect at P. Prove that (i) $\triangle ABD \sim \triangle CBE$ (ii) $\triangle PDC \sim \triangle BEC$. (3)

SHORT ANSWER

29 Prove that $\frac{\sin^3 A - \cos^3 A}{\sin A - \cos A} = 1 + \sin A \cos A$.

OR

Prove that $\frac{\tan\theta}{1-\cot\theta} + \frac{\cot\theta}{1-\tan\theta} = 1 + \sec\theta \operatorname{cosec}\theta$. (3)

SHORT ANSWER

30 The points A(−2, y) and B(4, 5) lie on a circle with centre O(1, −2y). Find the possible values of y. (3)

SHORT ANSWER

31 Prove that the tangents drawn at the ends of a chord of a circle make equal angles with the chord.

OR

From an external point P, tangents PA and PB are drawn to a circle, with PA = 15 cm. A third tangent touches the circle at Q (on the minor arc AB) and meets PA at C and PB at D. If CQ = DQ = 4 cm, find PC + PD. (3)

SECTION D | 5 MARKS EACH

LONG ANSWER

32 The diagonal of a rectangular field is 16 m more than its shorter side, and the longer side is 14 m more than the shorter side. Find the sides of the field.

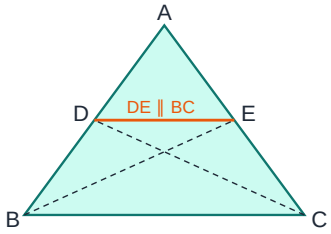
OR

The difference of the squares of two positive numbers is 180, and the square of the smaller number is 8 times the larger. Find the two numbers. (5)

LONG ANSWER

33 (a) In $\triangle ABC$, D and E are points on AB and AC such that $DE \parallel BC$. Prove that $\triangle ADE \sim \triangle ABC$ and hence $\frac{AD}{AB} = \frac{DE}{BC}$.

(b) A girl 1.2 m tall walks away from the base of a lamp-post 4.8 m high at a speed of 1.5 m/s. Find the length of her shadow after 3 seconds. (5)



LONG ANSWER

34 From the top of a building 10 m high, the angle of elevation of the top of a tower is 60° and the angle of depression of its foot is 30°. Find the height of the tower and its distance from the building.

OR

From a point 50 m above a lake, the angle of elevation of a cloud is 30° and the angle of depression of its reflection in the lake is 60°. Find the height of the cloud above the lake. (5)

LONG ANSWER

35 The median of the following distribution is 25 and the total frequency is 70. Find x and y.

Class	0-10	10-20	20-30	30-40	40-50	50-60
f	6	x	22	16	y	4

OR

Find the mean of the following data by the assumed-mean method.

Class	10-20	20-30	30-40	40-50	50-60
f	6	10	15	11	8

(5)

SECTION E | CASE-BASED, 4 MARKS EACH

CASE-BASED

36 **Case study 1.** A school auditorium has 20 seats in the first row. Each row has 2 seats more than the row in front of it, and there are 30 rows in all. (4)

(i) How many seats are there in the 10th row? (1)

(ii) Which row has 60 seats? (1)

(iii) Find the total number of seats in the auditorium. (2)

OR (iii) Find the number of seats in the last 10 rows. (2)

CASE-BASED

37 **Case study 2.** At a school fair, a stall has a spinner with 8 equal sectors numbered 1 to 8, and a bag of 24 balls of which 6 are black and the rest white. A player spins once and also draws one ball at random. (4)

(i) Find the probability that the spinner stops on an even number. (1)

(ii) Find the probability that the spinner stops on a number greater than 6. (1)

(iii) Find the probability that the ball drawn is not black. (2)

OR (iii) How many more black balls must be added to the bag so that the probability of drawing a black ball becomes $\frac{2}{5}$? (2)

CASE-BASED

38 **Case study 3.** A shopkeeper stacks 32 identical coins, each of diameter 2.8 cm and thickness 0.25 cm, one on top of another to form a cylindrical column. (Use $\pi = \frac{22}{7}$) (4)

(i) Find the height of the stack. (1)

(ii) Find the volume of one coin. (1)

(iii) Find the volume of the whole stack. (2)

OR (iii) Find the total surface area of the stack. (2)

ANSWERS AND FULL WORKED SOLUTIONS FOR THIS PAPER ARE ON DAY 54. CHECK ONLY AFTER YOU FINISH THE WHOLE PAPER.

DAY 53 COMPLETE? ☐ Paper done in 3 hours ☐ No answers peeked ☐ Rough work kept ☐ Marks totalled on Day 54

Next: Day 54, Mock 3 Review: Solutions and Mistake Fixing

MY MISTAKES AND NOTES Write every mistake of this Day with the correct step

DAY 54

MOCK TEST 3 REVIEW | SOLUTIONS | MISTAKE FIXING

Mock 3 Review: Solutions and Mistake Fixing

Mock 3 Review

Aaj ka target

- [ ] Mock 3 khud check karo, step marks ke saath
- [ ] Har galti ko C, S ya T mein daalo
- [ ] Top 5 traps dhyan se padho
- [ ] 10 fix-it sawal bina dekhe solve karo

54.1 How to mark yourself

1. Keep your answer sheet beside the solutions below. Mark one question at a time; do not just look at the final answer.
2. MCQ and AR: 1 mark only if the letter matches. No part marks.
3. 2, 3 and 5-markers: give step marks the way the board does. Correct formula or set-up gets about a third of the marks, correct working the next part, the final answer with units the last part.
4. Proofs: full marks only if Given, To prove, Construction (if any) and the reason for each step are written.
5. Case studies: mark each part separately; for the OR part, mark only the one you attempted.
6. Total your marks section by section and write them in the table below.

Section	Max marks	My marks	Time I took
A (Q1 to Q20)	20		
B (Q21 to Q25)	10		
C (Q26 to Q31)	18		
D (Q32 to Q35)	20		
E (Q36 to Q38)	12		
Total	80		

Your score	What it means	Next step
70 to 80	Board ready	Only revise your C mistakes
55 to 69	Strong, a few gaps	Redo every wrong question tomorrow
40 to 54	Concepts shaky in 2 or 3 chapters	Re-read those chapters' notes, then the fix-it set
Below 40	Needs a reset	Formula Day (Day 42) first, then the learn days of your weakest unit

Score bands for Mock Test 3

54.2 Top traps in Mock 3



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Q5 mein $(k-1)^2 = 16$ se k = 5 ya -3 aaya, aur bahut bachche dono likh dete hain. Infinitely many solutions ke liye **teeno** ratios barabar hone chahiye; -3 daalo to tisra ratio match nahi karta. Har value ko teesre ratio mein zaroor check karo.



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Q9 aur Q28 jaise sawal mein similar triangles ka naam likhte waqt order hi sab kuch hai. $\frac{DE}{MN}$ ka matlab D se M aur E se N. Order galat likha to examiner seedha mark kaat deta hai, chahe logic sahi ho.



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Q34 ke OR (cloud aur reflection) mein reflection lake ke neeche h metre hai, to observer se uski doori $h + 50$ hai, $h - 50$ nahi. Figure banao, lake ki line aur observer ki height alag dikhao, phir dono tan likho.

54.3 My mistake analysis

Q no.	Topic	Marks lost	Mistake type (C / S / T)
Q5	Pair of equations: infinite solutions		
Q26	Irrationality proof		
Q30	Distance formula, circle		
Q32	Quadratic word problem		
Q33	Similar triangles, lamp-post shadow		
Q34	Heights and distances		
Q35	Median, missing frequency		
Q36 to Q38	Case studies		
Other			

C = concept not clear, S = silly or calculation slip, T = ran out of time. Fill it honestly; every C goes into your weak-spot list for Days 55 to 57.



SHIVAM BHAIYA SAYS: BOARD TIP

Q35 median wale sawal mein pehle total frequency se $x + y$ ka equation likho, phir median formula. Formula mein cf hamesha median class se **pehle** wali class ka cumulative frequency hota hai. Aur Q32 mein negative speed ko reject karte waqt ek line ka reason likho: "speed cannot be negative". Isi line ka bhi mark hota hai.

Now solve the 10 fix-it questions below. They hit the same topics as the most-missed questions of Mock 3, with fresh numbers. Do them without looking at the solutions, then check.

Day 54 Practice

10 board-pattern questions | solve first, then check

FIX-IT SET: THE TOPICS MOST OFTEN LOST IN MOCK 3

SHORT ANSWER

1. Express 4620 and 1386 as products of their prime factors, and hence find their HCF and LCM. (3)

VERY SHORT ANSWER

2. The zeroes of $x^2 + (k-3)x - 10$ are negatives of each other. Find k and the zeroes. (2)

VERY SHORT ANSWER

3. Solve $x - \frac{1}{x} = \frac{3}{2}$, $x \neq 0$. (2)

SHORT ANSWER

4. In what ratio does the point P(x, 3) divide the line segment joining A(10, 7) and B(2, −5)? Also find x. (3)

VERY SHORT ANSWER

5. A tangent PT is drawn from a point P to a circle of radius 16 cm with centre O, touching it at T. If OP = 34 cm, find PT. (2)

VERY SHORT ANSWER

6. If $\sqrt{3}\tan\theta = 3\sin\theta$ (θ acute), find $\sin^2\theta - \cos^2\theta$. (2)

SHORT ANSWER

7. A man on top of a tower watches a car moving towards the tower at uniform speed. The angle of depression of the car changes from 30° to 60° in 18 seconds. How much more time will the car take to reach the foot of the tower? (3)

SHORT ANSWER

8. A circular pizza of diameter 42 cm is cut into 6 equal slices (sectors). Find the area of the top of one slice and the perimeter of one slice. (Use $\pi = \frac{22}{7}$) (3)

SHORT ANSWER

9 The mean of the following distribution is 36. Find the missing frequency p.

Class	10-20	20-30	30-40	40-50	50-60
f	5	8	p	12	5

(3)

VERY SHORT ANSWER

10 Two dice are thrown together. Find the probability that the numbers on them differ by 2. (2)

Day 54 Answers with worked solutions

1 **Ans:** HCF 462, LCM 13860

$4620 = 2^2 \times 3 \times 5 \times 7 \times 11$; $1386 = 2 \times 3^2 \times 7 \times 11$.

HCF = product of the smallest powers of common primes $= 2 \times 3 \times 7 \times 11 = 462$.

LCM = product of the greatest powers of all primes $= 2^2 \times 3^2 \times 5 \times 7 \times 11 = 13860$.

Check: $462 \times 13860 = 6403320 = 4620 \times 1386$.

2 **Ans:** k = 3; zeroes $\pm\sqrt{10}$

If the zeroes are α and $-\alpha$, their sum is 0: $-(k-3) = 0 \Rightarrow k = 3$.

Polynomial: $x^2 - 10$, zeroes $\sqrt{10}$ and $-\sqrt{10}$.

3 **Ans:** $x = 2$ or $x = -\frac{1}{2}$

Multiply by 2x: $2x^2 - 2 = 3x \Rightarrow 2x^2 - 3x - 2 = 0$.

$(2x+1)(x-2) = 0 \Rightarrow x = 2$ or $x = -\frac{1}{2}$.

4 **Ans:** 1 : 2; $x = \frac{22}{3}$

Let the ratio be k : 1. y-coordinate: $\frac{k(-5)+7}{k+1} = 3 \Rightarrow -5k + 7 = 3k + 3 \Rightarrow k = \frac{1}{2}$.

Ratio = 1 : 2.

$x = \frac{1(2)+2(10)}{3} = \frac{22}{3}$.

5 **Ans:** 30 cm

OT $\perp$ PT, so $PT^2 = OP^2 - OT^2 = 1156 - 256 = 900$.

PT = 30 cm.

6 **Ans:** $\frac{1}{3}$

$\sqrt{3}\frac{\sin\theta}{\cos\theta} = 3\sin\theta$; $\sin\theta \neq 0$, so $\cos\theta = \frac{\sqrt{3}}{3} = \frac{1}{\sqrt{3}}$.

$\cos^2\theta = \frac{1}{3}$, $\sin^2\theta = \frac{2}{3}$; answer $= \frac{2}{3} - \frac{1}{3} = \frac{1}{3}$.

7 **Ans:** 9 seconds

Let the tower height be h. At 30° the car is $h\cot 30^\circ = h\sqrt{3}$ away; at 60° it is $\frac{h}{\sqrt{3}}$ away.

Distance covered in 18 s $= h\sqrt{3} - \frac{h}{\sqrt{3}} = \frac{2h}{\sqrt{3}}$.

Remaining distance $\frac{h}{\sqrt{3}}$ is half of that, so it takes $\frac{18}{2} = 9$ s.

8 **Ans:** 231 cm²; 64 cm

r = 21 cm, angle of each slice $= 60^\circ$.

Area $= \frac{60}{360} \times \frac{22}{7} \times 441 = 231$ cm².

Arc $= \frac{60}{360} \times 2 \times \frac{22}{7} \times 21 = 22$ cm; perimeter $= 22 + 21 + 21 = 64$ cm.

9 **Ans:** p = 10

Class marks: 15, 25, 35, 45, 55. $\sum f_i x_i = 75 + 200 + 35p + 540 + 275 = 1090 + 35p$; $\sum f_i = 30 + p$.

$\frac{1090+35p}{30+p} = 36 \Rightarrow 1090 + 35p = 1080 + 36p \Rightarrow p = 10$.

10 **Ans:** $\frac{2}{9}$

Favourable: (1,3), (3,1), (2,4), (4,2), (3,5), (5,3), (4,6), (6,4), so 8 outcomes out of 36.

P $= \frac{8}{36} = \frac{2}{9}$.

Mock Test 3: worked solutions

marks as the board gives them

Mock Test 3 answers

1 **(c)**

Smallest odd composite number = 9, smallest odd prime = 3.

$9 = 3^2$, $3 = 3$, so HCF = 3 and LCM = 9.

LCM : HCF = 9 : 3 = 3 : 1.

2 **(c)**

LCM = product of the highest powers of each prime: x^4 and y^6.

LCM $= x^4y^6$. (HCF would be x^2y^3.)

3 **(c)**

Sum of zeroes $= (2+\sqrt{5}) + (2-\sqrt{5}) = 4$; product $= 4 - 5 = -1$.

Polynomial $= x^2 - (\text{sum})x + \text{product} = x^2 - 4x - 1$.

4 **(b)**

The zeroes of $p(x)$ are the x-coordinates of the points where the graph meets the x-axis.

It meets the x-axis at 3 points, so there are 3 zeroes. The y-axis crossing does not give a zero.

5 **(d)**

Infinitely many solutions: $\frac{1}{k-1} = \frac{k-1}{16} = \frac{3}{2k+2}$.

From the first two: $(k-1)^2 = 16$, so $k = 5$ or $k = -3$.

k = 5: ratios $\frac{1}{4}, \frac{4}{16}, \frac{3}{12}$ all equal $\frac{1}{4}$. Works.

k = -3: $\frac{-4}{16} = -\frac{1}{4}$ but $\frac{3}{-4} = -\frac{3}{4}$. Fails. So k = 5.

6 **(b)**

$D = b^2 - 4ac = 144 - 4(4)(9) = 144 - 144 = 0$, so the roots are real and equal.

(Indeed $(2x-3)^2 = 0$, so $x = \frac{3}{2}, \frac{3}{2}$.)

7 **(a)**

$a_{15} = a + 14d = 2m + 14(-m) = 2m - 14m = -12m$.

8 **(b)**

Section formula with $m_1 : m_2 = 1 : 2$: $x = \frac{1(6)+2(-3)}{3} = 0$, $y = \frac{1(-5)+2(4)}{3} = 1$.

P = (0, 1); x-coordinate 0, so P lies on the y-axis.

9 **(b)**

Read the correspondence from the ratios: DE with MN gives D to M, E to N; EF with NL gives F to L.

So D, E, F correspond to M, N, L and $\triangle DEF \sim \triangle MNL$ (SSS similarity).

10 **(a)**

Radius is perpendicular to the tangent at the point of contact, so $r^2 + 21^2 = 29^2$.

$r^2 = 841 - 441 = 400$, so $r = 20$ cm.

11 **(c)**

$\cos A = \frac{1}{2}$ gives $A = 60°$, so $\tan^2 A = 3$ and $\sec^2 A = 4$.

$3(3) - 2(4) = 9 - 8 = 1$.

12 **(c)**

$\tan 30° = \frac{12}{s} \Rightarrow s = \frac{12}{1/\sqrt{3}} = 12\sqrt{3}$ m.

13 **(b)**

$\pi r^2 = 2(2\pi r)$, so $r = 4$ units.

14 **(c)**

Perimeter = arc + 2 radii $= \frac{90}{360}(2\pi r) + 2r = \frac{\pi r}{2} + 2r = r\left(\frac{\pi}{2} + 2\right)$.

15 **(b)**

$\frac{1}{3}\pi r^2 h_1 = \pi r^2 h_2 \Rightarrow h_1 = 3h_2$, so $h_1 : h_2 = 3 : 1$.

16 **(a)**

$4 \times \frac{22}{7} \times r^2 = 1386$, so $r^2 = \frac{1386 \times 7}{88} = 110.25$ and $r = 10.5$ cm.

Diameter $= 2r = 21$ cm.

17 **(a)**

Assumed-mean method: $\bar{x} = a + \frac{\sum f_i d_i}{\sum f_i} = 45 + \frac{-50}{25} = 45 - 2 = 43$.

18 **(a)**

Total cards = 50 − 21 + 1 = 30.

Primes from 21 to 50: 23, 29, 31, 37, 41, 43, 47, so 7 favourable.

P = $\frac{7}{30}$.

19 **(c)**

A is true: $\sqrt{49} = 7 = \frac{7}{1}$.

R is false: square roots of perfect squares (1, 4, 9, 49, ...) are rational. So (c).

20 **(a)**

$a_m - a_n = [a + (m-1)d] - [a + (n-1)d] = (m-n)d$, so R is true.

$a_{25} - a_{18} = 7 \times 4 = 28$, so A is true and follows from R. Answer (a).

21 **Ans:** $k = 1$

Let the zeroes be α and $\frac{1}{\alpha}$. Then product of zeroes $= \alpha \times \frac{1}{\alpha} = 1$.

Product of zeroes $= \frac{c}{a} = \frac{3k}{k+2}$.

$\frac{3k}{k+2} = 1 \Rightarrow 3k = k + 2 \Rightarrow k = 1$.

Check: polynomial becomes $3x^2 + 11x + 3$, product of zeroes $= \frac{3}{3} = 1$.

22 **Ans:** D(0, 3)

Let D = (x, y). Diagonals of a parallelogram bisect each other, so midpoint of AC = midpoint of BD.

Midpoint of AC $= \left(\frac{1+4}{2}, \frac{-2+6}{2}\right) = \left(\frac{5}{2}, 2\right)$.

Midpoint of BD $= \left(\frac{5+x}{2}, \frac{1+y}{2}\right)$.

$\frac{5+x}{2} = \frac{5}{2} \Rightarrow x = 0$; $\frac{1+y}{2} = 2 \Rightarrow y = 3$. So D = (0, 3).

23 **Ans:** $\frac{72}{97}$

$\cos\theta = \frac{1}{\sec\theta} = \frac{5}{13}$, so $\cos^2\theta = \frac{25}{169}$.

$\frac{1 - \frac{25}{169}}{1 + \frac{25}{169}} = \frac{144/169}{194/169} = \frac{144}{194} = \frac{72}{97}$.

24 **Ans:** 70 cm

Let chord AB of the larger circle touch the smaller circle at P. Then OP = 12 cm and OP ⊥ AB (radius is perpendicular to the tangent).

In right $\triangle OPA$: $AP = \sqrt{OA^2 - OP^2} = \sqrt{1369 - 144} = \sqrt{1225} = 35$ cm.

The perpendicular from the centre bisects the chord, so AB = 2 × 35 = 70 cm.

25 **Ans:** 173.25 cm² ; OR $\frac{616}{3} \approx 205.33$ cm²

Area of sector $= \frac{1}{2} \times \text{arc length} \times r = \frac{1}{2} \times 33 \times 10.5 = 173.25$ cm².

(Check: $\theta = \frac{33 \times 360}{2 \times \frac{22}{7} \times 10.5} = 180°$, a semicircle; $\frac{1}{2} \times \frac{22}{7} \times 10.5^2 = 173.25$.)

OR: In 20 minutes the minute hand turns $\frac{20}{60} \times 360° = 120°$.

Area $= \frac{120}{360} \times \frac{22}{7} \times 14^2 = \frac{1}{3} \times 616 = \frac{616}{3} \approx 205.33$ cm².

26 **Ans:** Proof

Suppose $\sqrt{5}$ is rational: $\sqrt{5} = \frac{a}{b}$, a and b coprime integers, $b \neq 0$.

Squaring: $5b^2 = a^2$, so 5 divides a^2, hence 5 divides a (5 is prime). Write a = 5c.

Then $5b^2 = 25c^2$, so $b^2 = 5c^2$; 5 divides b^2, hence 5 divides b.

So 5 is a common factor of a and b, contradicting that they are coprime. Hence $\sqrt{5}$ is irrational.

Now suppose $\frac{3+2\sqrt{5}}{7} = r$ is rational. Then $\sqrt{5} = \frac{7r-3}{2}$, which is rational (rationals are closed under +, −, ×, ÷).

This contradicts $\sqrt{5}$ irrational. Hence $\frac{3+2\sqrt{5}}{7}$ is irrational.

27 **Ans:** $p = \frac{5}{2}$, $q = 4$

For infinitely many solutions: $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$.

$\frac{2p+1}{4} = \frac{3}{q-2} = \frac{9}{6} = \frac{3}{2}$.

$\frac{2p+1}{4} = \frac{3}{2} \Rightarrow 2p+1 = 6 \Rightarrow p = \frac{5}{2}$.

$\frac{3}{q-2} = \frac{3}{2} \Rightarrow q-2 = 2 \Rightarrow q = 4$.

28 **Ans:** Proof (AA similarity both times)

(i) In $\triangle ABD$ and $\triangle CBE$: $\angle ADB = \angle CEB = 90°$ (altitudes) and $\angle ABD = \angle CBE$ (common angle B).

So $\triangle ABD \sim \triangle CBE$ (AA criterion).

(ii) In $\triangle PDC$ and $\triangle BEC$: $\angle PDC = \angle BEC = 90°$.

$\angle PCD = \angle BCE$ (same angle at C, as P lies on CE).

So $\triangle PDC \sim \triangle BEC$ (AA criterion).

29 **Ans:** Proof

Use $a^3 - b^3 = (a-b)(a^2+ab+b^2)$: LHS $= \frac{(\sin A - \cos A)(\sin^2 A + \sin A \cos A + \cos^2 A)}{\sin A - \cos A}$.

$= \sin^2 A + \cos^2 A + \sin A \cos A = 1 + \sin A \cos A$ = RHS.

OR: Let $t = \tan\theta$, so $\cot\theta = \frac{1}{t}$. LHS $= \frac{t}{1-\frac{1}{t}} + \frac{1/t}{1-t} = \frac{t^2}{t-1} - \frac{1}{t(t-1)} = \frac{t^3-1}{t(t-1)}$.

$= \frac{(t-1)(t^2+t+1)}{t(t-1)} = \frac{t^2+t+1}{t} = t+1+\frac{1}{t} = 1 + \tan\theta + \cot\theta$.

$\tan\theta + \cot\theta = \frac{\sin^2\theta + \cos^2\theta}{\sin\theta\cos\theta} = \sec\theta\,\text{cosec}\,\theta$. So LHS = RHS.

30 **Ans:** $y = 5$ or $y = -1$

OA = OB (radii), so $OA^2 = OB^2$.

$OA^2 = (-2-1)^2 + (y+2y)^2 = 9 + 9y^2$.

$OB^2 = (4-1)^2 + (5+2y)^2 = 9 + 25 + 20y + 4y^2$.

$9y^2 = 25 + 20y + 4y^2 \Rightarrow 5y^2 - 20y - 25 = 0 \Rightarrow y^2 - 4y - 5 = 0$.

$(y-5)(y+1) = 0$, so $y = 5$ or $y = -1$.

31 **Ans:** Proof ; OR PC + PD = 22 cm

Let AB be a chord and let the tangents at A and B meet at P.

Tangents from an external point are equal, so PA = PB; $\triangle PAB$ is isosceles.

Angles opposite equal sides are equal: $\angle PAB = \angle PBA$, i.e. the tangents make equal angles with the chord. (If AB is a diameter, the tangents are parallel and both angles are $90°$.)

OR: Tangents from C: CA = CQ = 4 cm; tangents from D: DB = DQ = 4 cm; and PB = PA = 15 cm.

PC = PA − CA = 15 − 4 = 11 cm; PD = PB − DB = 15 − 4 = 11 cm.

PC + PD = 22 cm.

32 **Ans:** 10 m and 24 m ; OR 18 and 12

Let the shorter side be x m. Then the longer side is $(x+14)$ m and the diagonal is $(x+16)$ m.

By Pythagoras: $x^2+(x+14)^2=(x+16)^2 \Rightarrow x^2+x^2+28x+196=x^2+32x+256$.

$x^2-4x-60=0 \Rightarrow (x-10)(x+6)=0 \Rightarrow x=10$ or -6.

A length cannot be negative, so x = 10. Sides: 10 m and 24 m (diagonal 26 m; check $100+576=676=26^2$).

OR: Let the larger number be x and the smaller y. $x^2-y^2=180$ and $y^2=8x$.

$x^2-8x-180=0 \Rightarrow (x-18)(x+10)=0 \Rightarrow x=18$ (positive).

$y^2=144 \Rightarrow y=12$. The numbers are 18 and 12. (Check: 324 − 144 = 180.)

33 **Ans:** (a) proof (b) 1.5 m

(a) $DE \parallel BC$ with AB as transversal: $\angle ADE=\angle ABC$ (corresponding angles).

Similarly with AC as transversal: $\angle AED=\angle ACB$; and $\angle A$ is common.

So $\triangle ADE \sim \triangle ABC$ (AA criterion), and corresponding sides are proportional: $\frac{AD}{AB}=\frac{AE}{AC}=\frac{DE}{BC}$.

(b) Distance walked in 3 s $=1.5\times 3=4.5$ m.

Let the shadow be s m. The lamp-post and the girl are both vertical and the light ray is common, so the two right triangles are similar (AA).

$\frac{1.2}{s}=\frac{4.8}{4.5+s} \Rightarrow 4.8s=5.4+1.2s \Rightarrow 3.6s=5.4 \Rightarrow s=1.5$ m.

34 **Ans:** Tower 40 m, distance $10\sqrt{3}\approx 17.32$ m ; OR 100 m

Let AB be the building (AB = 10 m), CD the tower, and BD = x the distance between their feet. Draw AE ∥ BD meeting CD at E; then ED = 10 m, AE = x.

Depression of the foot D is 30°, so in $\triangle ABD$: $\tan 30^\circ=\frac{10}{x} \Rightarrow x=10\sqrt{3}$ m.

In $\triangle AEC$: $\tan 60^\circ=\frac{CE}{x} \Rightarrow CE=10\sqrt{3}\times\sqrt{3}=30$ m.

Height of tower = CE + ED = 30 + 10 = 40 m; distance $=10\sqrt{3}\approx 17.32$ m.

OR: Let the cloud be h m above the lake and x m away horizontally. The observer is 50 m above the lake.

Elevation: $\tan 30^\circ=\frac{h-50}{x} \Rightarrow x=\sqrt{3}(h-50)$.

The reflection is h m below the lake, i.e. $h+50$ m below the observer: $\tan 60^\circ=\frac{h+50}{x} \Rightarrow h+50=\sqrt{3}x=3(h-50)$.

$2h=200 \Rightarrow h=100$ m.

35 **Ans:** x = 18, y = 4 ; OR mean = 36

Total: $6+x+22+16+y+4=70 \Rightarrow x+y=22$.

Median 25 lies in class 20-30: $l=20$, $h=10$, $f=22$, cf before it $=6+x$, $\frac{N}{2}=35$.

Median $=l+\frac{\frac{N}{2}-cf}{f}\times h$: $25=20+\frac{35-6-x}{22}\times 10$.

$5=\frac{(29-x)\times 10}{22} \Rightarrow 29-x=11 \Rightarrow x=18$, and $y=22-18=4$.

OR: Class marks x_i = 15, 25, 35, 45, 55. Take $a=35$, $d_i=x_i-35$ = −20, −10, 0, 10, 20.

f_id_i = −120, −100, 0, 110, 160; $\sum f_id_i=50$, $\sum f_i=50$.

Mean $=a+\frac{\sum f_id_i}{\sum f_i}=35+\frac{50}{50}=36$.

36 **Ans:** 38

$a=20$, $d=2$; $a_{10}=20+9\times 2=38$.

Ans: 21st row

$20+(n-1)2=60 \Rightarrow n-1=20 \Rightarrow n=21$.

Ans: 1470

$S_{30}=\frac{30}{2}[2(20)+29(2)]=15\times 98=1470$.

OR: Ans: 690

Last 10 rows are rows 21 to 30: $a_{21}=60$, $a_{30}=20+29\times 2=78$.

Sum $=\frac{10}{2}(60+78)=690$. (Check: $S_{30}-S_{20}=1470-780=690$.)

37 **Ans:** $\frac{1}{2}$

Even numbers: 2, 4, 6, 8, so 4 of 8; P $= \frac{4}{8} = \frac{1}{2}$.

Ans: $\frac{1}{4}$

Numbers greater than 6: 7, 8; P $= \frac{2}{8} = \frac{1}{4}$.

Ans: $\frac{3}{4}$

White balls = 24 − 6 = 18.

P(not black) $= \frac{18}{24} = \frac{3}{4}$ (or $1 - \frac{6}{24}$).

OR: Ans: 6

Let n black balls be added: $\frac{6+n}{24+n} = \frac{2}{5}$.

$30 + 5n = 48 + 2n \Rightarrow 3n = 18 \Rightarrow n = 6$.

38 **Ans:** 8 cm

Height $= 32 \times 0.25 = 8$ cm.

Ans: 1.54 cm³

Each coin is a cylinder with $r = 1.4$ cm, $h = 0.25$ cm.

$V = \pi r^2 h = \frac{22}{7} \times 1.96 \times 0.25 = 1.54$ cm³.

Ans: 49.28 cm³

Stack is a cylinder: $r = 1.4$, $h = 8$. $V = \frac{22}{7} \times 1.96 \times 8 = 49.28$ cm³ (= 32 × 1.54).

OR: Ans: 82.72 cm²

TSA $= 2\pi r(h + r) = 2 \times \frac{22}{7} \times 1.4 \times (8 + 1.4) = 8.8 \times 9.4 = 82.72$ cm².

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Infinitely many solutions condition : $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$, all three
2. Area of a sector from arc length l : $\frac{1}{2}lr$
3. Median formula : $l + \frac{\frac{N}{2} - cf}{f} \times h$, cf of the class before
4. Tangent length from P : $\sqrt{OP^2 - r^2}$
5. Reflection of a cloud h m above a lake, seen from 50 m up : it is $h + 50$ m below the observer
6. BPT : $DE \parallel BC \Rightarrow \frac{AD}{DB} = \frac{AE}{EC}$
7. Empirical relation : Mode = 3 Median − 2 Mean

DAY 54 COMPLETE?

- ☐ Mock 3 marked
- ☐ Mistake table filled
- ☐ 10 fix-its solved
- ☐ Weak topics noted

Next: Day 55, Weak-Spot Sprint: Algebra Traps (Ch 1 to 5)

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 55

WEAK-SPOT SPRINT 1 | ALGEBRA | CHAPTERS 1 TO 5

Algebra Traps (Ch 1 to 5)

Weak-Spot Sprint

Aaj ka target

- [ ] 12 algebra traps padho: galat vs sahi dono dekho
- [ ] Har trap ke saath apni purani galti mistakes page pe likho
- [ ] 18 trap-questions, timer 50 min
- [ ] Jo galat hua, usi trap ka box dobara padho

Board papers lose most algebra marks not on hard questions but on small, repeated slips: a sign, a forgotten root, a condition not checked. Each box below shows a mini example done the **wrong** way (as students actually write it) and the **right** way (as the examiner's marking scheme wants it).

55.1 Real Numbers and Polynomials traps

TRAP 1: HCF X LCM = PRODUCT FOR THREE NUMBERS

Find the LCM of 6, 10 and 15 given that their HCF is 1.

SOLUTION

1. **WRONG:** LCM $= \dfrac{6 \times 10 \times 15}{1} = 900$.
2. **RIGHT:** HCF $\times$ LCM = product holds only for **TWO** numbers. Use prime factors: $6 = 2 \times 3$, $10 = 2 \times 5$, $15 = 3 \times 5$.
3. **RIGHT:** LCM $= 2 \times 3 \times 5 = 30$ (highest power of each prime).

Answer: LCM = 30, not 900

TRAP 2: 'IRRATIONAL BECAUSE IT HAS A ROOT' IS NOT A PROOF

Prove that $3 + 2\sqrt{5}$ is irrational.

SOLUTION

1. **WRONG:** $\sqrt{5}$ is irrational, so $3 + 2\sqrt{5}$ is irrational. (0 marks: nothing is proved.)
2. **RIGHT:** Assume $3 + 2\sqrt{5} = r$, a rational number. Then $\sqrt{5} = \dfrac{r - 3}{2}$.
3. **RIGHT:** RHS is rational (rationals are closed under subtraction and division by a non-zero number), so $\sqrt{5}$ would be rational.
4. **RIGHT:** This contradicts the fact that $\sqrt{5}$ is irrational. Hence $3 + 2\sqrt{5}$ is irrational.

Answer: Proof by contradiction, with the conclusion written

TRAP 3: COUNTING ZEROES FROM A GRAPH

The graph of $y = p(x)$ cuts the x-axis at 2 points, touches it at 1 more point and cuts the y-axis once. How many zeroes?

SOLUTION

1. **WRONG:** 2 + 1 + 1 = 4 zeroes (counted the y-axis crossing too).
2. **RIGHT:** Zeroes are the x-coordinates where the graph **MEETS THE X-AXIS**, cutting or touching. The y-axis crossing is $p(0)$, not a zero.
3. **RIGHT:** Zeroes $= 2 + 1 = 3$.

Answer: 3 zeroes

TRAP 4: SIGN OF THE SUM OF ZEROES

Sum and product of zeroes of $2x^2 - 5x + 3$.

SOLUTION

1. **WRONG:** Sum $= \dfrac{-5}{2}$, product $= \dfrac{3}{2}$.
2. **RIGHT:** $a = 2$, $b = -5$, $c = 3$. Sum $= -\dfrac{b}{a} = -\dfrac{-5}{2} = \dfrac{5}{2}$.
3. **RIGHT:** Product $= \dfrac{c}{a} = \dfrac{3}{2}$.

Answer: Sum $\frac{5}{2}$, product $\frac{3}{2}$

TRAP 5: FORMING A POLYNOMIAL FROM SUM AND PRODUCT

Find a quadratic polynomial whose zeroes have sum -3 and product 2.

SOLUTION

1. **WRONG:** $x^2 - 3x + 2$ (put the sum in directly).
2. **RIGHT:** $p(x) = k\,[x^2 - (\text{sum})x + \text{product}] = k\,[x^2 - (-3)x + 2]$.
3. **RIGHT:** $= k(x^2 + 3x + 2)$; taking $k = 1$: $x^2 + 3x + 2$. Check: zeroes $-1, -2$, sum -3 ✓.

Answer: $x^2 + 3x + 2$

55.2 Linear equations traps

TRAP 6: 'NO SOLUTION' VALUE THAT IS REALLY 'INFINITELY MANY'

For what value of k do $2x + 3y = 5$ and $4x + ky = 10$ have no solution?

TRAP 6: 'NO SOLUTION' VALUE THAT IS REALLY 'INFINITELY MANY'

SOLUTION

1. **WRONG:** $\frac{2}{4} = \frac{3}{k} \Rightarrow k = 6$. So $k = 6$.
2. **RIGHT:** No solution needs $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$.
3. **RIGHT:** $k = 6$ gives $\frac{2}{4} = \frac{3}{6} = \frac{1}{2}$, but also $\frac{c_1}{c_2} = \frac{5}{10} = \frac{1}{2}$, so the lines coincide (infinitely many solutions).
4. **RIGHT:** Hence there is **NO** value of k for which the pair is inconsistent.

Answer: No such k exists

TRAP 7: READING COEFFICIENTS BEFORE REARRANGING

Classify the pair $x = 2y + 3$ and $3x - 6y = 9$.

SOLUTION

1. **WRONG:** $a_1 = 1,\ b_1 = 2$: $\frac{1}{3} \neq \frac{2}{-6}$, so a unique solution.
2. **RIGHT:** First write both in the form $ax + by + c = 0$: $x - 2y - 3 = 0$ and $3x - 6y - 9 = 0$.
3. **RIGHT:** $\frac{a_1}{a_2} = \frac{1}{3}$, $\frac{b_1}{b_2} = \frac{-2}{-6} = \frac{1}{3}$, $\frac{c_1}{c_2} = \frac{-3}{-9} = \frac{1}{3}$: all equal.
4. **RIGHT:** The lines are coincident: infinitely many solutions.

Answer: Coincident lines

55.3 Quadratic equations traps

TRAP 8: DIVIDING BY X LOSES A ROOT

Solve $3x^2 = 12x$.

SOLUTION

1. **WRONG:** Divide by x: $3x = 12$, so $x = 4$.
2. **RIGHT:** Never divide by an expression that can be zero. $3x^2 - 12x = 0 \Rightarrow 3x(x - 4) = 0$.
3. **RIGHT:** $x = 0$ or $x = 4$.

Answer: $x = 0,\ 4$

TRAP 9: DISCRIMINANT WITH A NEGATIVE CONSTANT

Find k so that $2x^2 - 4x - k = 0$ has real roots.

SOLUTION

1. **WRONG:** $D = 16 - 8k \geq 0 \Rightarrow k \leq 2$.
2. **RIGHT:** $a = 2,\ b = -4,\ c = -k$. $D = b^2 - 4ac = 16 - 4(2)(-k) = 16 + 8k$.
3. **RIGHT:** Real roots: $16 + 8k \geq 0 \Rightarrow k \geq -2$.

Answer: $k \geq -2$

TRAP 10: KEEPING A ROOT THE PROBLEM REJECTS

A rectangular park has length 3 m more than its breadth and area $108\ \text{m}^2$. Find the breadth.

SOLUTION

1. **WRONG:** $x(x + 3) = 108 \Rightarrow x^2 + 3x - 108 = 0 \Rightarrow x = 9$ or $x = -12$. Breadth $= 9$ or -12 m.
2. **RIGHT:** $(x + 12)(x - 9) = 0 \Rightarrow x = 9$ or $x = -12$.
3. **RIGHT:** A length cannot be negative, so **REJECT** $x = -12$ and write the reason.
4. **RIGHT:** Breadth $= 9$ m, length $= 12$ m.

Answer: Breadth 9 m (reject −12 with reason)

55.4 Arithmetic Progression traps

TRAP 11: DIFFERENCE OF TWO TERMS

In an AP with $d = -4$, find $a_{25} - a_{17}$.

SOLUTION

1. **WRONG:** $(25 - 17 + 1)d = 9 \times (-4) = -36$.
2. **RIGHT:** $a_{25} - a_{17} = [a + 24d] - [a + 16d] = 8d$.
3. **RIGHT:** $= 8 \times (-4) = -32$.

Answer: -32

TRAP 12: USING S_n AS IF IT WERE a_n

If $S_n = 2n^2 + 3n$, find the nth term.

SOLUTION

1. **WRONG:** $a_n = 2n^2 + 3n$.
2. **RIGHT:** $a_n = S_n - S_{n-1} = (2n^2 + 3n) - [2(n - 1)^2 + 3(n - 1)]$.
3. **RIGHT:** $= 4n + 1$. Check: $a_1 = S_1 = 5$ ✓.

Answer: $a_n = 4n + 1$



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Linear pair mein k nikalne ke baad **teeno ratios** mein wapas daal ke dekho. Agar c wala ratio bhi barabar aa gaya to woh 'parallel' nahi, 'coincident' hai. Board isi pe 1 mark kaatta hai.



SHIVAM BHAIYA SAYS: BOARD TIP

Quadratic word problem mein negative root ko chupchaap mat chhodo. Ek line likho: '$x = -12$ rejected, as breadth cannot be negative'. Marking scheme mein is line ka mark hota hai.



SHIVAM BHAIYA SAYS

$\alpha + \beta = -\frac{b}{a}$ likhne se pehle a, b, c ko **sign ke saath** alag line mein likh lo: $a = 2,\ b = -5,\ c = 3$. Sign wali 80% galtiyan yahin ruk jaati hain.



EXAM ALERT!

Algebra (Ch 1 to 5) board paper mein **20 marks** ka hai, sabse bada unit. In 12 traps mein se kam se kam 3 to har saal kisi na kisi MCQ ya AR mein aate hain.

MEME SAMJHO — Astra meme, maths inside



JAB POORA QUADRATIC SAHI SOLVE KIYA, PAR LAST MEIN NEGATIVE ROOT REJECT KARNA BHOOL GAYE

Poore steps sahi... aur answer mein 'speed = -48 km/h' bhi likh diya. Ouch, mera dil (aur mera 1 mark)!

Asli baat: speed, length, age, number of items kabhi negative nahi hote. Root reject karo aur reason ek line mein likho.

Day 55 Practice

18 board-pattern questions | solve first, then check

SECTION A | MCQ AND ASSERTION-REASON (1 MARK EACH)

MCQ

1 If the HCF of two positive integers is 1, the two numbers are called (1)

(a) composite numbers (b) co-prime numbers
(c) both prime numbers (d) even numbers

MCQ

2 The product of the HCF and the LCM of 12, 18 and 30 is (1)

(a) 6480 (b) 180 (c) 1080 (d) 36

MCQ

3 Which of the following is a rational number? (1)

(a) $\sqrt{2} + \sqrt{3}$ (b) $(2 + \sqrt{3})(2 - \sqrt{3})$
(c) $3\sqrt{5}$ (d) $2 - \sqrt{3}$

ASSERTION-REASON

4 **Assertion (A):** The sum of two irrational numbers is always irrational.
Reason (R): $(3 + \sqrt{2}) + (3 - \sqrt{2}) = 6$. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

MCQ

5 How many quadratic polynomials have 3 and -7 as their zeroes? (1)

(a) exactly one (b) exactly two
(c) exactly three (d) infinitely many

MCQ

6 The sum and the product of the zeroes of $3x^2 + 7x - 6$ are respectively (1)

(a) $\frac{7}{3},\ -2$ (b) $-\frac{7}{3},\ 2$
(c) $-\frac{7}{3},\ -2$ (d) $\frac{7}{3},\ 2$

ASSERTION-REASON

7 **Assertion (A):** If $3 - \sqrt{2}$ is a zero of a quadratic polynomial with rational coefficients, then $3 + \sqrt{2}$ is its other zero.
Reason (R): Irrational zeroes of a quadratic polynomial with rational coefficients occur in conjugate pairs $a \pm \sqrt{b}$. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

MCQ

8 The graph of the equation $x = 5$ in the Cartesian plane is a straight line (1)

(a) parallel to the x-axis
(b) passing through the origin
(c) parallel to the y-axis
(d) along the line $y = x$

MCQ

9 The value of k for which $kx(x-3)+9=0$ has two equal roots is (1)
(a) 0 (b) 0 or 4 (c) 4 (d) 9

MCQ

10 The 10th term from the end of the AP $7, 11, 15, \ldots, 147$ is (1)
(a) 43 (b) 111 (c) 115 (d) 107

SECTION B | SHORT ANSWERS (2 AND 3 MARKS)

VERY SHORT ANSWER

11 Find the value(s) of k for which the pair $2x + 3y = 5$ and $4x + ky = 10$ has no solution. (2)

VERY SHORT ANSWER

12 Solve $x - \frac{6}{x} = 1, x \neq 0$. (2)

VERY SHORT ANSWER

13 Solve $5x^2 = 20x$. (2)

SHORT ANSWER

14 Find the value of k for which the pair $kx + 2y = k - 2$ and $8x + ky = k$ has no solution. (3)

SHORT ANSWER

15 The sum of the first n terms of an AP is $S_n = 3n^2 + 2n$. Find its nth term and its 10th term. (3)

SHORT ANSWER

16 Is 184 a term of the AP $3, 8, 13, \ldots$? Justify. (3)

SECTION C | LONG ANSWER AND CASE STUDY

LONG ANSWER

17 A train covers 480 km at a uniform speed. Had the speed been 8 km/h more, it would have taken 2 hours less. Find the speed of the train. (5)

CASE-BASED

18 A school's morning bells ring every 18 minutes (Block A) and every 24 minutes (Block B). Both rang together at 8:00 AM. The art room floor, 6 m 72 cm long and 4 m 32 cm wide, is to be covered with identical square tiles, as large as possible, with no tile cut. (4)

(i) After how many minutes will both bells next ring together? (1)

(ii) At what time will they next ring together? (1)

(iii) Find the side of the largest square tile and the number of tiles needed. (2)

Day 55 Answers with worked solutions

1 (b)
HCF 1 means co-prime. They need not be prime: 8 and 9 are co-prime but both are composite.

2 (c)
$12 = 2^2 \times 3$, $18 = 2 \times 3^2$, $30 = 2 \times 3 \times 5$.
HCF $= 2 \times 3 = 6$, LCM $= 2^2 \times 3^2 \times 5 = 180$.
HCF $\times$ LCM $= 1080$. (6480 is the product of the three numbers; the rule HCF x LCM = product works only for two numbers.)

3 (b)
$(2+\sqrt{3})(2-\sqrt{3}) = 4 - 3 = 1$, rational. The product of two irrationals can be rational; the other three are irrational.

4 (d)
R is a true statement, and it is exactly a counter-example: two irrationals with a rational sum. So A is false. Answer (d).

5 (d)
Every $k(x-3)(x+7) = k(x^2+4x-21)$ with $k \neq 0$ has these zeroes, so there are infinitely many.

6 (c)
$a = 3,\ b = 7,\ c = -6$. Sum $= -\frac{b}{a} = -\frac{7}{3}$, product $= \frac{c}{a} = \frac{-6}{3} = -2$.

7 (a)
Sum $= 6$ and product $= 9 - 2 = 7$ are rational only when the pair is $3 \pm \sqrt{2}$; the polynomial is $x^2 - 6x + 7$.
Both are true and R explains A. Answer (a).
Note: the rule needs **rational** coefficients.

8 (c)
All points $(5, y)$ have the same x-coordinate: a vertical line 5 units right of the y-axis, i.e. parallel to the y-axis (trap: students pick x-axis because the equation has x).

9 (c)
$kx^2 - 3kx + 9 = 0$: $D = 9k^2 - 36k = 9k(k-4) = 0 \Rightarrow k = 0$ or 4.
At $k = 0$ the equation becomes $9 = 0$, not a quadratic, so reject it. $k = 4$.

10 (b)
From the end the AP runs $147, 143, \ldots$ with $d = -4$.
10th term from end $= l - (n-1)d = 147 - 9 \times 4 = 111$. (43 is the 10th term from the start.)

11 **Ans:** No value of k

No solution requires $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$.

$\frac{2}{4} = \frac{3}{k} \Rightarrow k = 6$.

But $\frac{c_1}{c_2} = \frac{5}{10} = \frac{1}{2} = \frac{a_1}{a_2}$, so at $k = 6$ the lines coincide.

For $k \neq 6$ the lines intersect. Hence no value of k makes the pair inconsistent.

12 **Ans:** $x = 3$ or $x = -2$

Multiply by x: $x^2 - 6 = x \Rightarrow x^2 - x - 6 = 0 \Rightarrow (x - 3)(x + 2) = 0$.

$x = 3$ or $x = -2$. Both are non-zero, so **both** are valid. Trap: reject a negative root only when the context (length, speed, age) forbids it.

13 **Ans:** $x = 0$ or $x = 4$

$5x^2 - 20x = 0 \Rightarrow 5x(x - 4) = 0$.

$x = 0$ or $x = 4$. (Dividing both sides by x would lose the root $x = 0$.)

14 **Ans:** $k = -4$

No solution: $\frac{k}{8} = \frac{2}{k} \neq \frac{k-2}{k}$.

$\frac{k}{8} = \frac{2}{k} \Rightarrow k^2 = 16 \Rightarrow k = \pm 4$.

$k = 4$: $\frac{b_1}{b_2} = \frac{2}{4} = \frac{1}{2}$ and $\frac{c_1}{c_2} = \frac{2}{4} = \frac{1}{2}$, equal, so the lines coincide. Reject.

$k = -4$: $\frac{b_1}{b_2} = -\frac{1}{2}$, $\frac{c_1}{c_2} = \frac{-6}{-4} = \frac{3}{2} \neq -\frac{1}{2}$ ✓.

Hence $k = -4$.

15 **Ans:** $a_n = 6n - 1$, $a_{10} = 59$

$a_n = S_n - S_{n-1} = (3n^2 + 2n) - [3(n-1)^2 + 2(n-1)]$.

$= 3(2n - 1) + 2 = 6n - 1$.

$a_{10} = 6(10) - 1 = 59$. (Not $S_{10} = 320$.) Check: $a_1 = S_1 = 5$ ✓.

16 **Ans:** No

$a = 3$, $d = 5$. Suppose $a_n = 184$: $3 + (n-1)5 = 184$.

$(n-1)5 = 181 \Rightarrow n = \frac{186}{5} = 37.2$.

n must be a positive integer, so 184 is **not** a term of the AP.

17 **Ans:** 40 km/h

Let the speed be x km/h. Time $= \frac{480}{x}$ h; with the faster speed, $\frac{480}{x+8}$ h.

$\frac{480}{x} - \frac{480}{x+8} = 2 \Rightarrow \frac{480 \times 8}{x(x+8)} = 2$.

$x^2 + 8x - 1920 = 0 \Rightarrow (x + 48)(x - 40) = 0$.

$x = 40$ or $x = -48$. Speed cannot be negative, so $x = -48$ is rejected.

Speed of the train $= 40$ km/h. Check: $12 - 10 = 2$ h ✓.

18 **Ans:** 72 minutes

Together again after LCM(18, 24). $18 = 2 \times 3^2$, $24 = 2^3 \times 3$, LCM $= 2^3 \times 3^2 = 72$ min.

Ans: 9:12 AM

8:00 AM + 72 min = 9:12 AM.

Ans: 48 cm, 126 tiles

Largest tile side = HCF(672, 432) cm. $672 = 2^5 \times 3 \times 7$, $432 = 2^4 \times 3^3$, HCF $= 2^4 \times 3 = 48$ cm.

Tiles $= \frac{672}{48} \times \frac{432}{48} = 14 \times 9 = 126$.

Trap: 'largest tile / greatest piece' needs HCF; 'next time together' needs LCM.

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. HCF x LCM = product works for : only two numbers
2. Sum of zeroes of $ax^2 + bx + c$: $-b/a$ (write signs first)
3. Parallel lines condition : $a_1/a_2 = b_1/b_2 \neq c_1/c_2$
4. k makes all three ratios equal : coincident, not parallel
5. Dividing $3x^2 = 12x$ by x : loses root $x = 0$
6. D for $2x^2 - 4x - k$: $16 + 8k$
7. $a_{25} - a_{17}$: $8d$
8. a_n from S_n : $S_n - S_{n-1}$

DAY 55 COMPLETE? ☐ 12 traps read ☐ 18 questions solved ☐ Answers checked ☐ My traps noted

Next: Day 56, Weak-Spot Sprint: Geometry, Coordinate and Trigonometry Traps

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 56

WEAK-SPOT SPRINT 2 | GEOMETRY, COORDINATE AND TRIGONOMETRY

Geometry, Coordinate and Trigonometry Traps

Weak-Spot Sprint

Aaj ka target

- ☐ 12 traps: coordinate, triangles, circles, trig, heights
- ☐ Har figure khud bana ke label karo
- ☐ 18 trap-questions, timer 50 min
- ☐ Galat wale trap ka box dobara padho

Geometry, coordinate geometry and trigonometry together carry about **33 marks** in the board paper. The traps here are almost never about not knowing the formula; they are about the wrong order of points, a sign lost while squaring, an identity used where it does not apply, or a figure drawn wrongly.

56.1 Coordinate Geometry traps

TRAP 1: LOSING THE SIGN BEFORE SQUARING

Find the distance between $A(5,-1)$ and $B(-3,5)$.

SOLUTION

1. **WRONG:** $AB=\sqrt{(5-3)^2+(-1-5)^2}=\sqrt{4+36}=\sqrt{40}$ (dropped the minus of -3).
2. **RIGHT:** $x_2-x_1=-3-5=-8$, $y_2-y_1=5-(-1)=6$.
3. **RIGHT:** $AB=\sqrt{(-8)^2+6^2}=\sqrt{64+36}=10$ units.

Answer: 10 units

TRAP 2: SWAPPING M AND N IN THE SECTION FORMULA

Find the point dividing $A(1,3)$ and $B(10,9)$ internally in the ratio $1:2$.

SOLUTION

1. **WRONG:** $\left(\frac{2\times 10+1\times 1}{3},\frac{2\times 9+1\times 3}{3}\right)=(7,7)$.
2. **RIGHT:** $m:n=1:2$ goes with $(x_1,y_1)=A$, $(x_2,y_2)=B$: $P=\left(\frac{mx_2+nx_1}{m+n},\frac{my_2+ny_1}{m+n}\right)$.
3. **RIGHT:** $P=\left(\frac{1(10)+2(1)}{3},\frac{1(9)+2(3)}{3}\right)=(4,5)$. Check: P is nearer A, as $1:2$ says.

Answer: (4, 5)

TRAP 3: POINT ON THE X-AXIS MEANS Y = 0

In what ratio does the x-axis divide the segment joining $A(3,-4)$ and $B(-2,6)$?

SOLUTION

1. **WRONG:** Put $x=0$: $\frac{-2k+3}{k+1}=0\Rightarrow k=\frac{3}{2}$.
2. **RIGHT:** A point on the **X-AXIS** has $y=0$. Take ratio $k:1$: $\frac{6k-4}{k+1}=0\Rightarrow k=\frac{2}{3}$.
3. **RIGHT:** Ratio $2:3$; the point is $\left(\frac{-2(2/3)+3}{5/3},0\right)=(1,0)$.

Answer: 2 : 3, at (1, 0)

TRAP 4: FOURTH VERTEX OF A PARALLELOGRAM

$A(1,2)$, $B(4,3)$, $C(6,6)$ are three vertices of parallelogram $ABCD$ (in order). Find D.

SOLUTION

1. **WRONG:** Mid-point of AB = mid-point of CD, so $D=A+B-C=(-1,-1)$.
2. **RIGHT:** In $ABCD$ the **DIAGONALS** are AC and BD, and they bisect each other.
3. **RIGHT:** Mid-point of $AC=\left(\frac{7}{2},4\right)$ = mid-point of $BD=\left(\frac{4+x}{2},\frac{3+y}{2}\right)$.
4. **RIGHT:** $x=3$, $y=5$, so $D(3,5)$.

Answer: D(3, 5)

56.2 Triangles and Circles traps

TRAP 5: SIMILARITY IN THE WRONG ORDER

In $\triangle ABC$ and $\triangle PQR$, $\angle A=\angle Q$ and $\angle B=\angle R$. Write the similarity and the ratio of sides.

SOLUTION

1. **WRONG:** $\triangle ABC\sim\triangle PQR$, so $\frac{AB}{PQ}=\frac{BC}{QR}$.
2. **RIGHT:** Match the equal angles in order: $A\leftrightarrow Q$, $B\leftrightarrow R$, $C\leftrightarrow P$.
3. **RIGHT:** $\triangle ABC\sim\triangle QRP$ (AA), so $\frac{AB}{QR}=\frac{BC}{RP}=\frac{CA}{PQ}$.

Answer: $\triangle ABC\sim\triangle QRP$

TRAP 6: BPT WITH THE WHOLE SIDE

In $\triangle ABC$, $DE\parallel BC$, $AD=3$ cm, $DB=5$ cm, $AE=4.5$ cm. Find EC.

TRAP 6: BPT WITH THE WHOLE SIDE

SOLUTION

1. **WRONG:** $\frac{AD}{AB} = \frac{AE}{EC} \Rightarrow \frac{3}{8} = \frac{4.5}{EC} \Rightarrow EC = 12$ cm.
2. **RIGHT:** BPT: $\frac{AD}{DB} = \frac{AE}{EC}$ (part : part), or $\frac{AD}{AB} = \frac{AE}{AC}$ (part : whole). Never mix.
3. **RIGHT:** $\frac{3}{5} = \frac{4.5}{EC} \Rightarrow EC = 7.5$ cm.

Answer: EC = 7.5 cm

TRAP 7: ANGLE BETWEEN TANGENTS

A chord AB subtends 110° at the centre O. Tangents at A and B meet at P. Find $\angle APB$.

SOLUTION

1. **WRONG:** $\angle APB = \angle AOB = 110^\circ$.
2. **RIGHT:** $\angle OAP = \angle OBP = 90^\circ$ (radius $\perp$ tangent), so in quadrilateral $OAPB$: $\angle APB = 360^\circ - 90^\circ - 90^\circ - 110^\circ$.
3. **RIGHT:** $\angle APB = 70^\circ$ (the two angles are supplementary).

Answer: 70°

TRAP 8: TANGENT LENGTH WITH A PLUS SIGN

P is 17 cm from the centre of a circle of radius 8 cm. Find the length of the tangent from P.

SOLUTION

1. **WRONG:** $PT = \sqrt{17^2 + 8^2} = \sqrt{353}$ cm.
2. **RIGHT:** $OT \perp PT$, and OP is the **HYPOTENUSE**: $PT^2 = OP^2 - OT^2$.
3. **RIGHT:** $PT = \sqrt{289 - 64} = \sqrt{225} = 15$ cm.

Answer: 15 cm

56.3 Trigonometry and Heights traps

TRAP 9: IDENTITY ON TWO DIFFERENT ANGLES

Evaluate $\sin^2 30^\circ + \cos^2 60^\circ$ and $\sin(30^\circ + 60^\circ)$.

SOLUTION

1. **WRONG:** $\sin^2 30^\circ + \cos^2 60^\circ = 1$ (identity) and $\sin(30^\circ + 60^\circ) = \sin 30^\circ + \sin 60^\circ$.
2. **RIGHT:** $\sin^2\theta + \cos^2\theta = 1$ needs the **SAME** angle. Here $\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$.
3. **RIGHT:** $\sin(30^\circ + 60^\circ) = \sin 90^\circ = 1$, while $\sin 30^\circ + \sin 60^\circ = \frac{1 + \sqrt{3}}{2} \neq 1$.

Answer: $\frac{1}{2}$ and 1

TRAP 10: SEC IS 1/COS, NOT 1/SIN

If $\cos\theta = \frac{12}{13}$ (θ acute), find $\sec\theta$ and $\csc\theta$.

SOLUTION

1. **WRONG:** $\sec\theta = \frac{1}{\sin\theta} = \frac{13}{5}$.
2. **RIGHT:** $\sin\theta = \sqrt{1 - \frac{144}{169}} = \frac{5}{13}$.
3. **RIGHT:** $\sec\theta = \frac{1}{\cos\theta} = \frac{13}{12}$, $\csc\theta = \frac{1}{\sin\theta} = \frac{13}{5}$. (s goes with c, c goes with s.)

Answer: $\sec\theta = \frac{13}{12}$, $\csc\theta = \frac{13}{5}$

TRAP 11: FORGETTING THE OBSERVER'S HEIGHT

A girl 1.5 m tall stands 28.5 m from a tower and sees its top at an elevation of 45°. Find the height of the tower.

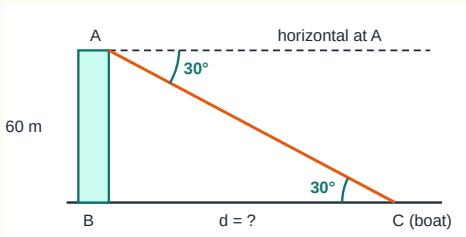
SOLUTION

1. **WRONG:** $h = 28.5\tan 45^\circ = 28.5$ m.
2. **RIGHT:** The angle is measured from her **EYE**, 1.5 m above the ground: height above eye level $= 28.5\tan 45^\circ = 28.5$ m.
3. **RIGHT:** Tower $= 28.5 + 1.5 = 30$ m.

Answer: 30 m

TRAP 12: ANGLE OF DEPRESSION AT THE WRONG PLACE

From the top of a 60 m cliff, the angle of depression of a boat is 30°. How far is the boat from the foot of the cliff?



SOLUTION

1. **WRONG:** Put 30° at the top inside the triangle: $d = 60\tan 30^\circ = 20\sqrt{3}$ m.
2. **RIGHT:** Depression is measured from the **HORIZONTAL AT THE EYE**; by alternate angles, the elevation at the boat is 30°.
3. **RIGHT:** $\tan 30^\circ = \frac{60}{d} \Rightarrow d = 60\sqrt{3} \approx 103.9$ m.

Answer: $60\sqrt{3}$ m



SHIVAM BHAIYA SAYS: BOARD TIP

Heights and distances mein **figure ke 1 mark** hote hain. Horizontal line dotted kheencho, angle of depression usi horizontal se dikhao, aur 'Let height = h m' pehli line mein likho.



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Similarity likhte waqt vertices ka order equal angles se match karo. $\triangle ABC \sim \triangle QRP$ likha hai to AB ka partner QR hai, PQ nahi. Order galat, poora ratio galat.



SHIVAM BHAIYA SAYS

Section formula mein ek check: ratio $1:2$ hai to point **pehle point ke paas** aayega. Answer doosre point ke paas aa gaya, matlab m aur n ulte lag gaye.

MEME SAMJHO — Astra meme, maths inside



JAB TOWER KI HEIGHT NIKAAL LI AUR TAB YAAD AAYA KI OBSERVER BHI 1.5 M LAMBA HAI

Answer: tower = 28.5 m. Phir dekha, ladki ki aankh zameen pe nahi, 1.5 m upar hai. Oops!

Asli baat: angle aankh se naapa jaata hai. Jo height tan se nikli, usme observer ki height jodo.

Day 56 Practice

18 board-pattern questions | solve first, then check

SECTION A | MCQ AND ASSERTION-REASON (1 MARK EACH)

MCQ

1 $A(-2,5)$ and $B(-2,-7)$ are two points. Which point lies **on the segment** AB? (1)

(a) $(-2,8)$ **(b)** $(5,-2)$ **(c)** $(-2,-3)$ **(d)** $(2,-3)$

MCQ

2 The distance between $P(-4,7)$ and $Q(2,-1)$ is (1)

(a) 14 units **(b)** $2\sqrt{5}$ units
(c) 10 units **(d)** $\sqrt{20}$ units

MCQ

3 In $\triangle ABC$, $D(2,5)$ and $E(-4,-3)$ are the mid-points of AB and AC. The length of BC is (1)

(a) 10 units **(b)** 5 units
(c) 40 units **(d)** 20 units

MCQ

4 If $\triangle PQR \sim \triangle XZY$, then $\dfrac{PR}{XY}$ is equal to (1)

(a) $\dfrac{PQ}{XY}$ **(b)** $\dfrac{QR}{XZ}$ **(c)** $\dfrac{PQ}{XZ}$ **(d)** $\dfrac{PR}{ZY}$

MCQ

5 Tangents PA and PB are drawn to a circle of radius 5 cm with centre O. If $\angle APB = 90^\circ$, then $OP =$ (1)

(a) 5 cm **(b)** $5\sqrt{2}$ cm **(c)** 10 cm **(d)** $\dfrac{5}{\sqrt{2}}$ cm

ASSERTION-REASON

6 **Assertion (A):** If a chord AB of a circle subtends 80° at the centre, the tangents at A and B meet at an angle of 80°.
Reason (R): The angle between two tangents from an external point and the angle subtended at the centre by the line joining the points of contact are supplementary. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

MCQ

7 The value of $\sin^2 60^\circ + \cos^2 30^\circ$ is (1)

(a) 1 **(b)** $\dfrac{3}{2}$ **(c)** $\dfrac{1}{2}$ **(d)** $\dfrac{3}{4}$

MCQ

8 If $3\sin^2\beta - 2\cos^2\beta = 3$, then β $(0^\circ \le \beta \le 90^\circ)$ is (1)

(a) 0° **(b)** 30° **(c)** 45° **(d)** 90°

ASSERTION-REASON

9 **Assertion (A):** $\sin^2 52^\circ + \cos^2 52^\circ = 1$.
Reason (R): $\sin^2\theta + \cos^2\theta = 1$ for every angle θ with $0^\circ \le \theta \le 90^\circ$. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

MCQ

10 The height of a tower and the distance of an observer from its foot are both increased by 20%. The angle of elevation of the top (1)

(a) increases (b) decreases
(c) stays the same (d) doubles

SECTION B | SHORT ANSWERS (2 AND 3 MARKS)

VERY SHORT ANSWER

11 Find the ratio in which the x-axis divides the segment joining $A(4,-6)$ and $B(-1,9)$. Also find the point of division. (2)

VERY SHORT ANSWER

12 $A(1,-2)$, $B(3,6)$ and $C(5,10)$ are three consecutive vertices of parallelogram $ABCD$. Find D. (2)

VERY SHORT ANSWER

13 A point P is 25 cm from the centre of a circle of radius 7 cm. Find the length of the tangent from P. (2)

SHORT ANSWER

14 In $\triangle ABC$, $DE \parallel BC$ with D on AB and E on AC. If $AD = x$, $DB = x-3$, $AE = x+3$ and $EC = x-2$ (in cm), find x. (3)

SHORT ANSWER

15 If $\cos\theta = \dfrac{12}{13}$ and θ is acute, find $\sec\theta$, $\csc\theta$ and the value of $\sec\theta + \tan\theta$. (3)

SHORT ANSWER

16 A boy 1.6 m tall stands 30.4 m away from a tower. The angle of elevation of the top of the tower from his eyes is 45°. Find the height of the tower. (3)

SECTION C | LONG ANSWER AND CASE STUDY

LONG ANSWER

17 From the top of a 90 m high tower, the angles of depression of two cars on the **same side** of the tower, in a straight line with its foot, are 30° and 60°. Find the distance between the cars. (Use $\sqrt{3} = 1.732$.) (5)

CASE-BASED

18 A circular garden has centre O and radius 9 m. From a gate P outside the garden, two straight paths PA and PB touch the boundary at A and B. The gate is 15 m from O. (4)

(i) Find the length of path PA. (1)

(ii) Is PB longer than, shorter than or equal to PA? Why? (1)

(iii) Find the area of the quadrilateral $OAPB$. (2)

OR (iii) If $\angle AOB = 130^\circ$, find $\angle APB$. (2)

Day 56 Answers with worked solutions

1 **(c)**
Points of AB have $x = -2$ and $-7 \le y \le 5$. Only $(-2,-3)$ qualifies; $(-2,8)$ is on the line AB extended, not on the segment.

2 **(c)**
$PQ = \sqrt{(2+4)^2 + (-1-7)^2} = \sqrt{36+64} = 10$ units.

3 **(d)**
$DE = \sqrt{(-6)^2 + (-8)^2} = 10$.
Mid-point theorem: $DE = \dfrac{1}{2}BC \Rightarrow BC = 20$ units.

4 **(c)**
Correspondence $P \leftrightarrow X$, $Q \leftrightarrow Z$, $R \leftrightarrow Y$:
$\dfrac{PQ}{XZ} = \dfrac{QR}{ZY} = \dfrac{PR}{XY}$.

5 **(b)**
$OAPB$ has three right angles and $OA = OB$, so it is a square of side 5 cm.
OP is its diagonal $= 5\sqrt{2}$ cm.

6 **(d)**
By R, the angle between the tangents $= 180^\circ - 80^\circ = 100^\circ$, so A is false and R is true. Answer (d).

7 **(b)**
$\left(\dfrac{\sqrt{3}}{2}\right)^2 + \left(\dfrac{\sqrt{3}}{2}\right)^2 = \dfrac{3}{4} + \dfrac{3}{4} = \dfrac{3}{2}$. (Not 1: the angles differ.)

8 **(d)**
$3\sin^2\beta - 2(1-\sin^2\beta) = 3 \Rightarrow 5\sin^2\beta = 5 \Rightarrow \sin\beta = 1 \Rightarrow \beta = 90^\circ$.

9 **(a)**
A is R applied with $\theta = 52^\circ$ (the same angle in both terms). Both true, R explains A. Answer (a).

10 **(c)**
$\tan\theta = \dfrac{1.2h}{1.2d} = \dfrac{h}{d}$, unchanged, so θ stays the same.

11 **Ans:** $2:3$; $(2,0)$
Let the ratio be $k:1$. On the x-axis $y = 0$:
$\dfrac{9k-6}{k+1} = 0 \Rightarrow k = \dfrac{2}{3}$. Ratio $2:3$.
$x = \dfrac{-k+4}{k+1} = \dfrac{-2/3+4}{5/3} = 2$. Point $(2,0)$.

12 **Ans:** $D(3,2)$

Diagonals AC and BD bisect each other. Mid-point of $AC = (3,4)$.

$\left(\frac{3+x}{2}, \frac{6+y}{2}\right) = (3,4) \Rightarrow x = 3,\ y = 2$. $D(3,2)$

.

13 **Ans:** 24 cm

Radius $\perp$ tangent, so OP is the hypotenuse:

$PT^2 = OP^2 - r^2 = 625 - 49 = 576$.

$PT = 24$ cm.

14 **Ans:** $x = 4.5$

By BPT, $\frac{AD}{DB} = \frac{AE}{EC} \Rightarrow \frac{x}{x-3} = \frac{x+3}{x-2}$.

$x(x-2) = (x+3)(x-3) \Rightarrow x^2 - 2x = x^2 - 9 \Rightarrow x = 4.5$.

Check: $DB = 1.5 > 0$, $EC = 2.5 > 0$ ✓.

15 **Ans:** $\frac{13}{12}$, $\frac{13}{5}$, $\frac{3}{2}$

Adjacent $= 12k$, hypotenuse $= 13k$, opposite $= \sqrt{169 - 144}\,k = 5k$.

$\sec\theta = \frac{13}{12}$, $\csc\theta = \frac{13}{5}$, $\tan\theta = \frac{5}{12}$.

$\sec\theta + \tan\theta = \frac{13+5}{12} = \frac{3}{2}$.

16 **Ans:** 32 m

Let the top be h m above eye level. $\tan 45^\circ = \frac{h}{30.4} \Rightarrow h = 30.4$ m.

Height of tower $= 30.4 + 1.6 = 32$ m.

17 **Ans:** $60\sqrt{3} \approx 103.92$ m

Let $AB = 90$ m be the tower, cars at C (nearer, 60°) and D (farther, 30°). Elevations at the cars equal the depressions (alternate angles).

$\triangle ABC$: $\tan 60^\circ = \frac{90}{BC} \Rightarrow BC = \frac{90}{\sqrt{3}} = 30\sqrt{3}$ m.

$\triangle ABD$: $\tan 30^\circ = \frac{90}{BD} \Rightarrow BD = 90\sqrt{3}$ m.

Same side, so subtract: $CD = BD - BC = 60\sqrt{3} \approx 60 \times 1.732 = 103.92$ m.

(Had the cars been on opposite sides, we would add: $120\sqrt{3}$ m.)

18 **Ans:** 12 m

$\angle OAP = 90^\circ$: $PA = \sqrt{15^2 - 9^2} = \sqrt{144} = 12$ m.

Ans: Equal (12 m)

Tangents from an external point are equal, so $PB = PA = 12$ m.

Ans: $108\ \text{m}^2$

$OAPB$ is made of two congruent right triangles OAP and OBP.

Area $= 2 \times \frac{1}{2} \times OA \times AP = 2 \times \frac{1}{2} \times 9 \times 12 = 108\ \text{m}^2$.

OR: Ans: 50°

$\angle OAP = \angle OBP = 90^\circ$, so $\angle APB = 360^\circ - 90^\circ - 90^\circ - 130^\circ = 50^\circ$.

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Point on x-axis : put $y = 0$
2. Ratio $1 : 2$ point lies : nearer the first point
3. Parallelogram ABCD : mid-point of AC = mid-point of BD
4. BPT : $AD/DB = AE/EC$ (part to part)
5. Angle between tangents : $180^\circ -$ centre angle
6. Tangent length : $\sqrt{OP^2 - r^2}$
7. $\sin^2 30^\circ + \cos^2 60^\circ$: $\frac{1}{2}$, not 1
8. Observer's height : add it at the end

DAY 56 COMPLETE? ☐ 12 traps read ☐ 18 questions solved ☐ Answers checked ☐ My traps noted

Next: Day 57, Weak-Spot Sprint: Mensuration, Statistics and Probability Traps

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 57

WEAK-SPOT SPRINT 3 | MENSURATION, STATISTICS AND PROBABILITY

Mensuration, Statistics and Probability Traps

Weak-Spot Sprint

Aaj ka target

- ☐ 12 traps: areas, solids, statistics, probability
- ☐ 18 trap-questions, timer 50 min
- ☐ Har formula mein r, l, h, f0, f2 ka matlab bolke dekho
- ☐ Units aur table ke columns pe extra dhyan

Mensuration, statistics and probability together carry about **21 marks**, and they are the most 'scoring' part of the paper. Marks here are lost to small slips: a radius taken as the diameter, a hidden face added to a surface area, a wrong f_0 in the mode formula, or two dice treated as if (2, 3) and (3, 2) were the same outcome.

57.1 Areas Related to Circles traps

TRAP 1: PERIMETER OF A SECTOR WITHOUT THE RADII

Find the perimeter of a sector of radius 21 cm and angle 60°. ($\pi = \frac{22}{7}$)

SOLUTION

1. **WRONG:** Perimeter = arc length = $\frac{60}{360} \times 2 \times \frac{22}{7} \times 21 = 22$ cm.
2. **RIGHT:** Arc length = $\frac{\theta}{360^\circ} \times 2\pi r = 22$ cm.
3. **RIGHT:** Perimeter of the sector = arc $+ 2r = 22 + 42 = 64$ cm.

Answer: 64 cm

TRAP 2: SEGMENT = SECTOR MINUS TRIANGLE

Find the area of the minor segment of a circle of radius 10 cm cut off by a chord subtending 90° at the centre. ($\pi = 3.14$)

SOLUTION

1. **WRONG:** Area = $\frac{90}{360} \times 3.14 \times 100 = 78.5$ cm^2 (stopped at the sector).
2. **RIGHT:** Sector = 78.5 cm^2. Triangle OAB (right angle at O) = $\frac{1}{2} \times 10 \times 10 = 50$ cm^2.
3. **RIGHT:** Segment = $78.5 - 50 = 28.5$ cm^2.

Answer: 28.5 cm^2

TRAP 3: DIAMETER USED AS RADIUS

Find the area of a quadrant of a circle of diameter 28 cm.

SOLUTION

1. **WRONG:** $\frac{1}{4} \times \frac{22}{7} \times 28^2 = 616$ cm^2.
2. **RIGHT:** Underline the word **DIAMETER**: $r = 14$ cm.
3. **RIGHT:** Area = $\frac{1}{4} \times \frac{22}{7} \times 14^2 = 154$ cm^2.

Answer: 154 cm^2

57.2 Surface Areas and Volumes traps

TRAP 4: HIDDEN FACES IN A COMBINED SOLID

A solid is a cylinder (radius 7 cm, height 10 cm) with a hemisphere of the same radius on top. Find its total surface area.

SOLUTION

1. **WRONG:** TSA of cylinder + TSA of hemisphere $= 2\pi r(r+h) + 3\pi r^2 = 748 + 462 = 1210$ cm^2.
2. **RIGHT:** Only the **OUTSIDE** is counted. The top face of the cylinder and the flat face of the hemisphere are hidden at the joint.
3. **RIGHT:** TSA = CSA of cylinder + CSA of hemisphere + base $= 2\pi rh + 2\pi r^2 + \pi r^2$.
4. **RIGHT:** $= 440 + 308 + 154 = 902$ cm^2.

Answer: 902 cm^2

TRAP 5: BOWL IS CSA, NOT TSA

The inside of a hemispherical bowl of radius 21 cm is to be painted. Find the area to be painted.

SOLUTION

1. **WRONG:** $3\pi r^2 = 3 \times \frac{22}{7} \times 441 = 4158$ cm^2.
2. **RIGHT:** A bowl is open at the top, so only the curved surface: $2\pi r^2$.
3. **RIGHT:** $= 2 \times \frac{22}{7} \times 441 = 2772$ cm^2. ($3\pi r^2$ is for a **SOLID** hemisphere.)

Answer: 2772 cm^2

TRAP 6: VOLUME IN THE WRONG UNIT

A cylindrical tank has radius 0.7 m and height 2 m. How many litres of water can it hold?

SOLUTION

1. **WRONG:** $V = \frac{22}{7} \times 0.49 \times 2 = 3.08$, so 3.08 litres.
2. **RIGHT:** $V = \pi r^2 h = 3.08$ m^3.

TRAP 6: VOLUME IN THE WRONG UNIT

1. **RIGHT:** $1\ \text{m}^3 = 1000$ litres, so capacity $= 3080$ litres. (And $1000\ \text{cm}^3 = 1$ litre.)

Answer: 3080 litres

57.3 Statistics traps

TRAP 7: WRONG f_0 AND f_2 IN THE MODE

The modal class is 40-50 with frequency 12. The class before it has frequency 8, the class after it 6, and the first class of the table has frequency 3. Find the mode.

SOLUTION

1. **WRONG:** $f_0 = 3$ (first class): Mode $= 40 + \dfrac{12-3}{24-3-6} \times 10 = 46$.
2. **RIGHT:** f_0 and f_2 are the classes **JUST BEFORE AND JUST AFTER** the modal class: $f_1 = 12,\ f_0 = 8,\ f_2 = 6,\ l = 40,\ h = 10$.
3. **RIGHT:** Mode $= 40 + \dfrac{12-8}{2(12)-8-6} \times 10 = 40 + \dfrac{4}{10} \times 10 = 44$.

Answer: 44

TRAP 8: CF OF THE MEDIAN CLASS ITSELF

$N = 50$. The median class is 30-40 with $f = 10$; the cumulative frequency up to 30 is 22 and up to 40 is 32. Find the median.

SOLUTION

1. **WRONG:** Median $= 30 + \dfrac{25-32}{10} \times 10 = 23$ (used the cf of the median class; the answer is not even inside the class).
2. **RIGHT:** cf is the cumulative frequency of the class **BEFORE** the median class: $cf = 22$.
3. **RIGHT:** Median $= l + \dfrac{N/2 - cf}{f} \times h = 30 + \dfrac{25-22}{10} \times 10 = 33$.
4. **RIGHT:** For classes like 10-19, 20-29, first make them continuous (9.5-19.5, ...): l becomes 19.5, not 20.

Answer: 33

TRAP 9: SIGN OF THE ASSUMED-MEAN CORRECTION

With assumed mean $a = 25$, a table gives $\sum f_i = 50$ and $\sum f_i d_i = -40$. Find the mean.

SOLUTION

1. **WRONG:** $\overline{x} = 25 + \dfrac{40}{50} = 25.8$ (dropped the minus).
2. **RIGHT:** $d_i = x_i - a$, so $\overline{x} = a + \dfrac{\sum f_i d_i}{\sum f_i} = 25 + \dfrac{-40}{50}$.
3. **RIGHT:** $\overline{x} = 25 - 0.8 = 24.2$. A negative $\sum fd$ means the mean is **BELOW** a.

Answer: 24.2

57.4 Probability traps

TRAP 10: TWO DICE, UNORDERED OUTCOMES

Two dice are thrown. Find the probability that the sum is 5.

SOLUTION

1. **WRONG:** Sums 2 to 12 are 11 outcomes, so $P = \dfrac{1}{11}$. Or: (1,4) and (2,3) only, so $\dfrac{2}{36}$.
2. **RIGHT:** There are $6 \times 6 = 36$ equally likely ordered outcomes; (1,4) and (4,1) are different.
3. **RIGHT:** Sum 5: (1,4), (4,1), (2,3), (3,2): 4 outcomes. $P = \dfrac{4}{36} = \dfrac{1}{9}$.

Answer: $\frac{1}{9}$

TRAP 11: THE COIN HAS NO MEMORY

A fair coin shows tails 4 times in a row. Find the probability of a head on the 5th toss.

SOLUTION

1. **WRONG:** Heads is 'due', so P is more than $\dfrac{1}{2}$ (or $\dfrac{4}{5}$).
2. **RIGHT:** Each toss is independent; the outcomes are still H or T, equally likely.
3. **RIGHT:** $P(\text{head}) = \dfrac{1}{2}$.

Answer: $\frac{1}{2}$

TRAP 12: COUNTING CARDS

A card is drawn from a well-shuffled deck of 52. Find P(red face card) and P(not an ace).

SOLUTION

1. **WRONG:** Red face cards $= 12$, so $P = \dfrac{12}{52}$; $P(\text{not ace}) = \dfrac{1}{13}$.

TRAP 12: COUNTING CARDS

1. **RIGHT:** Face cards are J, Q, K: 12 in all, of which 6 are red (hearts and diamonds). $P = \frac{6}{52} = \frac{3}{26}$.
2. **RIGHT:** $P(\text{not ace}) = 1 - \frac{4}{52} = \frac{12}{13}$.

Answer: $\frac{3}{26}$ and $\frac{12}{13}$



SHIVAM BHAIYA SAYS: BOARD TIP

Combined solid ka TSA nikaalne se pehle figure pe **tick lagao** ki kaunsi surface bahar dikh rahi hai. Jo surface joint pe chhup gayi, woh nahi judegi. Har saal 3-4 mark ka sawal isi pe aata hai.



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

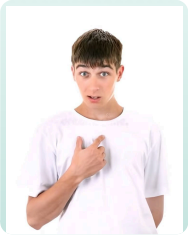
Mode mein f_0 = modal class se **theek pehle** wali class ki frequency, aur median mein cf = median class se **theek pehle** tak ki cumulative frequency. Dono jagah 'pehle wali'. Table mein arrow laga ke mark karo.



SHIVAM BHAIYA SAYS

Do dice wale sawal mein 6 × 6 ka grid bana lo (rows first die, columns second die). Sum, product, doublet sab ek nazar mein gin jaata hai, aur 36 se kam ya zyada outcome ka sawal hi nahi uthta.

MEME SAMJHO — Astra meme, maths inside



JAB TEACHER BOLE KI DO DICE MEIN (2, 3) AUR (3, 2) ALAG OUTCOMES HAIN

Main? Maine to dono ko ek hi gina tha... isliye sum 5 ki probability 2/36 aa gayi!

Asli baat: do dice ke 36 ordered outcomes hote hain. Sum 5 ke 4 outcomes hain, isliye P = 4/36 = 1/9.

Day 57 Practice

18 board-pattern questions | solve first, then check

SECTION A | MCQ AND ASSERTION-REASON (1 MARK EACH)

MCQ

1. A pizza is cut into 4 equal quadrant-shaped pieces. If its radius is 14 cm, the perimeter of one piece is ($\pi = \frac{22}{7}$) (1)

(a) 22 cm **(b)** 36 cm **(c)** 50 cm **(d)** 64 cm

MCQ

2. A square is inscribed in a circle of diameter 10 cm. The area of the square is (1)

(a) $100\ \text{cm}^2$ **(b)** $50\ \text{cm}^2$
(c) $25\ \text{cm}^2$ **(d)** $78.5\ \text{cm}^2$

ASSERTION-REASON

3. **Assertion (A):** The area of every segment of a circle is less than the area of its corresponding sector.
Reason (R): Area of a minor segment = area of the minor sector − area of the triangle formed by the two radii and the chord. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

MCQ

4. Two arcs of equal length lie on circles of radii r and $2r$. If the arc on the smaller circle subtends $60°$ at its centre, the arc on the larger circle subtends (1)

(a) $120°$ **(b)** $60°$ **(c)** $30°$ **(d)** $15°$

MCQ

5 The largest possible sphere is carved out of a solid hemisphere of radius 12 cm. The ratio of the volume of the hemisphere to that of the sphere is (1)

(a) 2 : 1 **(b)** 3 : 2 **(c)** 4 : 1 **(d)** 8 : 1

MCQ

6 The curved surface area of a cone of base radius 5 cm and height 12 cm is (1)

(a) 60π cm^2 **(b)** 65π cm^2 **(c)** 90π cm^2 **(d)** 85π cm^2

MCQ

7 In the formula Mode $= l + \dfrac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$, the symbol f_0 stands for the frequency of (1)

(a) the first class of the table
(b) the class preceding the modal class
(c) the class succeeding the modal class
(d) the modal class itself

MCQ

8 For a distribution, the mean is 26.5 and the median is 28. Using the empirical relation, the mode is (1)

(a) 29.5 **(b)** 31 **(c)** 25 **(d)** 33

MCQ

9 A fair coin is tossed and shows tails four times in a row. The probability of getting a head on the fifth toss is (1)

(a) $\frac{1}{5}$ **(b)** $\frac{4}{5}$ **(c)** $\frac{1}{2}$ **(d)** 1

MCQ

10 Which of the following **cannot** be the probability of an event? (1)

(a) $\frac{2}{3}$ **(b)** 0.7 **(c)** 15% **(d)** 1.25

ASSERTION-REASON

11 **Assertion (A):** A spinner has three regions marked 1, 2 and 3, covering $\frac{1}{4}$, $\frac{1}{4}$ and $\frac{1}{2}$ of the dial. The probability that it stops on 3 is $\frac{1}{2}$.
Reason (R): The three outcomes 1, 2 and 3 are equally likely. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B | SHORT ANSWERS (2 AND 3 MARKS)

VERY SHORT ANSWER

12 Box A has one white and one green ball; Box B has one white and one orange ball. One ball is drawn from each box. Find the probability that at least one of them is white. (2)

VERY SHORT ANSWER

13 Find the probability that a leap year chosen at random has 53 Fridays and 53 Saturdays. (2)

VERY SHORT ANSWER

14 Find the area of a quadrant of a circle of diameter 28 cm. ($\pi = \frac{22}{7}$) (2)

SHORT ANSWER

15 A toy is a cone of radius 7 cm mounted on a hemisphere of the same radius. The total height of the toy is 31 cm. Find its total surface area. ($\pi = \frac{22}{7}$) (3)

SHORT ANSWER

16 Find the mode of the data:
Class: 10-20, 20-30, 30-40, 40-50, 50-60
Frequency: 6, 9, 15, 11, 4 (3)

SECTION C | LONG ANSWER AND CASE STUDY

LONG ANSWER

17 The marks of 40 students are:
Marks: 10-19, 20-29, 30-39, 40-49, 50-59
Students: 4, 8, 12, 10, 6
Find the mean (assumed-mean method) and the median. (5)

CASE-BASED

18 In a board game, two players each throw a fair die at the same time and the two numbers are noted as an ordered pair (first die, second die). (4)

(i) How many outcomes are possible? (1)

(ii) Find the probability that the sum of the numbers is 5. (1)

(iii) Find the probability that the sum is a prime number. (2)

OR (iii) Find the probability that at least one die shows 6. (2)

Day 57 Answers with worked solutions

1 **(c)**

Arc $= \frac{1}{4} \times 2 \times \frac{22}{7} \times 14 = 22$ cm.

Perimeter $= 22 + 2(14) = 50$ cm. (22 cm forgets the two straight edges.)

2 **(b)**
The diagonal of the square = diameter = 10 cm.
Area $= \frac{d^2}{2} = \frac{100}{2} = 50$ cm^2 (not 10^2, since 10 is the diagonal, not the side).

3 **(d)**
R is true. But a **major** segment = major sector + triangle, which is more than the major sector, so A is false. Answer (d).

4 **(c)**
$\frac{60}{360} \times 2\pi r = \frac{\theta}{360} \times 2\pi(2r) \Rightarrow 60 = 2\theta \Rightarrow \theta = 30°$.

5 **(c)**
The largest sphere has diameter equal to the hemisphere's radius: sphere radius = 6 cm.
$\frac{\frac{2}{3}\pi(12)^3}{\frac{4}{3}\pi(6)^3} = \frac{1152}{288} = 4$. Ratio $4:1$.

6 **(b)**
$l = \sqrt{r^2 + h^2} = \sqrt{25 + 144} = 13$ cm.
CSA $= \pi rl = 65\pi$ cm^2. (60π uses the height instead of the slant height.)

7 **(b)**
f_1: modal class, f_0: class just before it, f_2: class just after it.

8 **(b)**
Mode $= 3\,\text{Median} - 2\,\text{Mean} = 84 - 53 = 31$.

9 **(c)**
Each toss is independent with outcomes H and T equally likely: $P = \frac{1}{2}$.

10 **(d)**
$0 \le P(E) \le 1$. $15\% = 0.15$ is allowed; $1.25 > 1$ is not.

11 **(c)**
Region 3 is half the dial, so $P(3) = \frac{1}{2}$: A is true.
The regions are unequal, so the outcomes are **not** equally likely (that would give $\frac{1}{3}$): R is false. Answer (c).

12 **Ans:** $\frac{3}{4}$
Outcomes (A, B): (W, W), (W, O), (G, W), (G, O): 4 equally likely.
At least one white: 3 outcomes. $P = \frac{3}{4}$ (or $1 - P(\text{G, O}) = 1 - \frac{1}{4}$).

13 **Ans:** $\frac{1}{7}$
A leap year has 366 days = 52 weeks + 2 extra days.
The 2 extra consecutive days can be (Sun, Mon), (Mon, Tue), ..., (Sat, Sun): 7 outcomes. Only (Fri, Sat) gives 53 Fridays and 53 Saturdays. $P = \frac{1}{7}$.

14 **Ans:** 154 cm^2
Diameter 28 cm, so $r = 14$ cm.
Area $= \frac{1}{4}\pi r^2 = \frac{1}{4} \times \frac{22}{7} \times 196 = 154$ cm^2.

15 **Ans:** 858 cm^2
Height of the cone $= 31 - 7 = 24$ cm; slant height $l = \sqrt{24^2 + 7^2} = 25$ cm.
CSA of cone $= \pi rl = \frac{22}{7} \times 7 \times 25 = 550$ cm^2.
CSA of hemisphere $= 2\pi r^2 = 2 \times \frac{22}{7} \times 49 = 308$ cm^2.
TSA $= 550 + 308 = 858$ cm^2. (No circular face is added: both bases are hidden at the joint.)

16 **Ans:** 36
Highest frequency 15, so the modal class is 30-40: $l = 30,\ h = 10,\ f_1 = 15,\ f_0 = 9,\ f_2 = 11$.
Mode $= 30 + \frac{15 - 9}{30 - 9 - 11} \times 10 = 30 + \frac{6}{10} \times 10 = 36$.

17 **Ans:** Mean = 36, Median ≈ 36.17
Class marks x_i: 14.5, 24.5, 34.5, 44.5, 54.5.
Take $a = 34.5$: $d_i = -20, -10, 0, 10, 20$.
$f_i d_i = -80, -80, 0, 100, 120$; $\sum f_i d_i = 60$, $\sum f_i = 40$.
Mean $= 34.5 + \frac{60}{40} = 36$.
For the median, make classes continuous: 9.5-19.5, 19.5-29.5, 29.5-39.5, 39.5-49.5, 49.5-59.5. cf: 4, 12, 24, 34, 40.
$\frac{N}{2} = 20$, so the median class is 29.5-39.5: $l = 29.5,\ cf = 12,\ f = 12,\ h = 10$.
Median $= 29.5 + \frac{20 - 12}{12} \times 10 = 29.5 + 6.67 \approx 36.17$.

18 **Ans:** 36
$6 \times 6 = 36$ equally likely ordered pairs.

Ans: $\frac{1}{9}$
(1,4), (4,1), (2,3), (3,2): $P = \frac{4}{36} = \frac{1}{9}$.

Ans: $\frac{5}{12}$
Prime sums 2, 3, 5, 7, 11 occur in $1 + 2 + 4 + 6 + 2 = 15$ ways.
$P = \frac{15}{36} = \frac{5}{12}$.

OR: Ans: $\frac{11}{36}$
No 6 on either die: $5 \times 5 = 25$ outcomes.
$P(\text{at least one 6}) = 1 - \frac{25}{36} = \frac{11}{36}$.

TONIGHT'S 2-MINUTE RECALL Answer from memory, then check the notes

1. Perimeter of sector : arc $+ 2r$
2. Segment : sector $-$ triangle

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Bowl paint area : $2\pi r^2$ (not $3\pi r^2$)
2. $1\ \text{m}^3$: 1000 litres
3. Mode f_0 : class just before modal class
4. Median cf : cf of class before median class
5. Two dice outcomes : 36 ordered pairs
6. Red face cards : 6

DAY 57 COMPLETE? ☐ 12 traps read ☐ 18 questions solved ☐ Answers checked ☐ My traps noted

Next: Day 58, Mock Test 4: Final Full Paper, timed

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 58

MOCK TEST 4 | FINAL PAPER | 80 MARKS | TIMED

Mock Test 4: The Final Paper, timed

Mock Test 4

Aaj ka target

- [ ] Board wala time hi rakho: same time, same table, phone door
- [ ] 15 min reading, phir 3 ghante strict timer
- [ ] Section-wise time plan follow karo, kahin atko mat
- [ ] Paper ke baad hi Day 59 kholna

Instructions: 3 hours (plus 15 minutes reading time), 80 marks, 38 questions in 5 sections: A (20 × 1), B (5 x 2), C (6 × 3), D (4 × 5), E (3 case studies x 4). All questions are compulsory; internal choice (OR) in Q25, Q26, Q29, Q32, Q34, Q35 and the last part of each case study. Use $\pi = \frac{22}{7}$ unless stated.

Section	Questions	Marks	Suggested time
A	1 to 20	20	25 min
B	21 to 25	10	20 min
C	26 to 31	18	35 min
D	32 to 35	20	50 min
E	36 to 38	12	30 min
Check	all	80	20 min

Time plan for this final paper (180 minutes)



SHIVAM BHAIYA SAYS: BOARD TIP

Ye last mock hai, isko bilkul board ki tarah do. Kisi ek sawal pe 8-10 minute se zyada atak gaye to number ghera karo aur aage badho; aakhri 20 minute wapas aao. Board mein bhi yahi rule: chhoda hua sawal zero deta hai, adha likha sawal step marks deta hai.

Day 58 Practice

Mock paper | 80 marks | 3 hours | attempt in one sitting

SECTION A | MCQ AND AR, 1 MARK EACH

MCQ

1 The smallest natural number that is divisible by each of 2, 3, 4, 5, 6 and 7 is (1)

(a) 840 (b) 210 (c) 5040 (d) 420

MCQ

2 When 3600 is written as a product of powers of primes, the sum of the exponents is (1)

(a) 6 (b) 8 (c) 7 (d) 9

MCQ

3 If α and β are the zeroes of $x^2 - 25$, then $\alpha + \beta + \alpha\beta$ equals (1)

(a) -25 (b) 25 (c) 0 (d) -5

MCQ

4 The lines $4x + 5y - 7 = 0$ and $12x + 15y - k = 0$ are parallel when (1)

(a) k = 21 (b) k = 7 (c) k ≠ 21 (d) k ≠ 7

MCQ

5 The product of the roots of $3x^2 - 7x - 12 = 0$ is (1)

(a) 4 (b) -4 (c) $\frac{7}{3}$ (d) $-\frac{7}{3}$

MCQ

6 The equation $2x^2 + 8x + k = 0$ has no real roots when (1)

(a) k < 8 (b) k = 8 (c) k > 8 (d) k > −8

MCQ

7 Which of the following lists is NOT an AP? (1)

(a) 7, 3, −1, −5, ... (b) 2.5, 5, 7.5, 10, ...
(c) −4, −4, −4, −4, ... (d) 1, 4, 9, 16, ...

MCQ

8 The distance between the points $\left(-\frac{7}{3}, -1\right)$ and $\left(\frac{5}{3}, -1\right)$ is (1)

(a) 4 units (b) $\frac{2}{3}$ unit
(c) 2 units (d) $\frac{8}{3}$ units

MCQ

9 In $\triangle ABC$ and $\triangle QRP$, $\angle A = \angle Q$, $\angle B = \angle R$ and AB = 2QR. The two triangles are (1)

(a) congruent but not similar
(b) similar but not congruent
(c) neither similar nor congruent
(d) similar and congruent

MCQ

10 The tangent at a point A of a circle with centre O makes an angle of 55° with the chord AB. Then $\angle AOB$ is (1)

(a) 55° (b) 110° (c) 125° (d) 70°

MCQ

11 If $y \sin 30^\circ \cos 60^\circ = \tan^2 60^\circ - \sec^2 45^\circ$, then y equals (1)

(a) $\frac{1}{4}$ (b) 2 (c) 4 (d) 1

MCQ

12 A tree breaks in a storm. The broken part bends so that its top touches the ground 12 m from the foot of the tree, making an angle of 30° with the ground. The original height of the tree was (1)

(a) $12\sqrt{3}$ m (b) $8\sqrt{3}$ m (c) $4\sqrt{3}$ m (d) 24 m

MCQ

13 The area of a quadrant of a circle of diameter d is (1)

(a) $\frac{\pi d^2}{4}$ (b) $\frac{\pi d^2}{8}$ (c) $\frac{\pi d^2}{16}$ (d) $\frac{\pi d^2}{2}$

MCQ

14 The area of a sector of a circle of radius 10 cm with central angle 72° is (use $\pi = 3.14$) (1)

(a) 31.4 cm² (b) 62.8 cm²
(c) 125.6 cm² (d) 78.5 cm²

MCQ

15 Two cylinders of the same height have radii in the ratio 4 : 7. The ratio of their volumes is (1)

(a) 4 : 7 (b) 64 : 343
(c) 16 : 49 (d) 8 : 14

MCQ

16 The total surface area of a solid hemisphere of radius 7 cm is (use $\pi = \frac{22}{7}$) (1)

(a) 308 cm² (b) 462 cm² (c) 154 cm² (d) 616 cm²

MCQ

17 For the distribution

Class	0-10	10-20	20-30	30-40	40-50
f	5	9	15	12	4

the sum of the lower limit of the modal class and the upper limit of the median class is (1)

(a) 40 (b) 50 (c) 60 (d) 30

MCQ

18 The probability that a non-leap year chosen at random has 53 Sundays is (1)

(a) $\frac{2}{7}$ (b) $\frac{1}{7}$ (c) $\frac{53}{365}$ (d) $\frac{1}{53}$

ASSERTION-REASON

19 **Assertion (A):** $4 + \sqrt{7}$ is an irrational number.
Reason (R): The sum of a rational number and an irrational number is irrational. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

ASSERTION-REASON

20 **Assertion (A):** The polynomial $x^2 + 4x + 5$ has two real zeroes.
Reason (R): A quadratic polynomial can have at most two zeroes. (1)

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B | 2 MARKS EACH

VERY SHORT ANSWER

21 Find the sum of all two-digit natural numbers that are multiples of 9. (2)

VERY SHORT ANSWER

22 Show that A(1, 1), B(4, 2), C(5, 5) and D(2, 4), taken in order, are the vertices of a parallelogram. (2)

VERY SHORT ANSWER

23 If $\sec\theta + \tan\theta = 4$, find $\sec\theta - \tan\theta$ and hence $\cos\theta$. (2)

VERY SHORT ANSWER

24 The sides of $\triangle ABC$ are 4 cm, 5 cm and 6 cm, and the sides of $\triangle PQR$ are 10 cm, 12.5 cm and 15 cm. Are the triangles similar? If yes, name the criterion and find the ratio of their perimeters. (2)

VERY SHORT ANSWER

25 The curved surface area of a hemisphere is 1232 cm². Find its volume. (Use $\pi = \frac{22}{7}$)
OR
Find the total surface area of a solid cone of radius 6 cm and height 8 cm. (Use $\pi = 3.14$) (2)

SECTION C | 3 MARKS EACH

SHORT ANSWER

26 Three ropes of lengths 108 cm, 162 cm and 270 cm are to be cut into pieces of equal length with nothing left over. Find the greatest possible length of each piece and the total number of pieces.
OR
Given that $\sqrt{3}$ is irrational, prove that $2 + 5\sqrt{3}$ is irrational. (3)

SHORT ANSWER

27 Find the value of k ($k \neq 0$) for which the equation $9x^2 - 3kx + k = 0$ has two equal roots. Also find the roots. (3)

SHORT ANSWER

28 Prove that the angle between the two tangents drawn from an external point to a circle is supplementary to the angle subtended at the centre by the line segment joining the points of contact. (3)

SHORT ANSWER

29 Prove that $(\cos^4 A - \sin^4 A + 1)\sec^2 A = 2$.
OR
Prove that $(\sin\theta + \cos\theta + 1)(\sin\theta + \cos\theta - 1)\sec\theta\,\text{cosec}\,\theta = 2$. (3)

SHORT ANSWER

30 Find the coordinates of the points of trisection of the line segment joining A(−4, 5) and B(8, −1). (3)

SHORT ANSWER

31 In $\triangle XYZ$, M and N are points on XY and XZ such that XM = 4 cm, MY = 6 cm, XN = 5 cm and NZ = 7.5 cm. Show that $MN \parallel YZ$. If YZ = 15 cm, find MN. (3)

SECTION D | 5 MARKS EACH

LONG ANSWER

32 Solve graphically: $x - y + 2 = 0$ and $2x + y - 8 = 0$. Find the coordinates of the vertices of the triangle formed by these lines and the x-axis, and find the area of this triangle.
OR
Ten years ago, a father was twelve times as old as his son, and ten years from now he will be twice as old as his son. Find their present ages. (5)

LONG ANSWER

33 Prove that the lengths of the tangents drawn from an external point to a circle are equal. Using this, a circle is inscribed in $\triangle ABC$, touching BC, CA and AB at D, E and F respectively. If AB = 10 cm, BC = 12 cm and CA = 8 cm, find AF, BD and CE. (5)

LONG ANSWER

34 The angle of elevation of the top of a tower from a point A on level ground is $30°$. On walking 40 m from A straight towards the tower, the angle of elevation becomes $60°$. Find the height of the tower and the distance of A from its foot.
OR
Two observers on the same side of a balloon, 120 m apart on level ground, see the balloon at angles of elevation $60°$ and $30°$. Find the height of the balloon. (5)

LONG ANSWER

35 Find the mean (by the assumed-mean method) and the mode of the following data.

Class	0-20	20-40	40-60	60-80	80-100
f	7	11	18	9	5

OR
Find the median of the following data.

Class	100-120	120-140	140-160	160-180	180-200
f	12	14	8	6	10

(5)

SECTION E | CASE-BASED, 4 MARKS EACH

CASE-BASED

36 **Case study 1.** A town is designing a rectangular park. Its length is to be 3 m more than twice its width, and its area must be 230 m². (4)
(i) Taking the width as x m, write the quadratic equation for the area. (1)
(ii) Find the discriminant of this equation. (1)
(iii) Find the length and the width of the park. (2)
OR (iii) A path 1 m wide runs along the inside of the boundary of the park. Find the area of the path. (2)

CASE-BASED

37 **Case study 2.** **Riya and Kabir play a game with two fair dice thrown together. Riya wins if the product of the two numbers is even; Kabir wins if the sum of the two numbers is 8.** (4)

(i) How many outcomes are possible? (1)

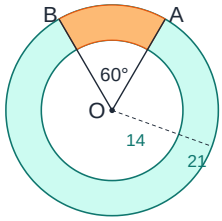
(ii) Find the probability that the sum is 8. (1)

(iii) Find the probability that Riya wins. (2)

OR (iii) Find the probability of getting a doublet whose sum is greater than 6. (2)

CASE-BASED

38 **Case study 3.** **A circular garden has a walking track between two concentric circles of radii 14 m and 21 m with centre O. Two straight paths along radii OA and OB make an angle of 60° at O (see figure). (Use $\pi = \frac{22}{7}$)** (4)



(i) Find the area of the sector of the outer circle with central angle 60°. (1)

(ii) Find the length of the arc of the outer circle between the two paths. (1)

(iii) Find the area of the part of the track that lies between the two paths (shaded). (2)

OR (iii) Find the area of the whole track. (2)

ANSWERS AND FULL WORKED SOLUTIONS FOR THIS PAPER ARE ON DAY 59. CHECK ONLY AFTER YOU FINISH THE WHOLE PAPER.

DAY 58 COMPLETE? ☐ Paper done in 3 hours ☐ Time plan followed ☐ No answers peeked ☐ Marks totalled on Day 59

Next: Day 59, Mock 4 Review: Solutions and Last Corrections

MY MISTAKES AND NOTES Write every mistake of this Day with the correct step

DAY 59

MOCK TEST 4 REVIEW | SOLUTIONS | LAST CORRECTIONS

Mock 4 Review: Solutions and Last Corrections

Mock 4 Review

Aaj ka target

- [ ] Final mock ko board ki tarah check karo
- [ ] Section-wise time bhi likho
- [ ] Last 5 traps yaad karo, kal galti nahi
- [ ] 10 fix-it sawal: aakhri practice

59.1 How to mark yourself

1. Keep your answer sheet beside the solutions below. Mark one question at a time; do not just look at the final answer.
2. MCQ and AR: 1 mark only if the letter matches. No part marks.
3. 2, 3 and 5-markers: give step marks the way the board does. Correct formula or set-up gets about a third of the marks, correct working the next part, the final answer with units the last part.
4. Proofs: full marks only if Given, To prove, Construction (if any) and the reason for each step are written.
5. Case studies: mark each part separately; for the OR part, mark only the one you attempted.
6. Total your marks section by section and write them in the table below.

Section	Max marks	My marks	Time I took
A (Q1 to Q20)	20		
B (Q21 to Q25)	10		
C (Q26 to Q31)	18		
D (Q32 to Q35)	20		
E (Q36 to Q38)	12		
Total	80		

Your score	What it means	Next step
70 to 80	Board ready	Only revise your C mistakes
55 to 69	Strong, a few gaps	Redo every wrong question tomorrow
40 to 54	Concepts shaky in 2 or 3 chapters	Re-read those chapters' notes, then the fix-it set
Below 40	Needs a reset	Formula Day (Day 42) first, then the learn days of your weakest unit

Score bands for Mock Test 4

59.2 Top traps in Mock 4



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Q4 mein parallel lines ki condition $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$ hai. k = 21 daaloge to lines coincident ho jaati hain, parallel nahi. Isliye answer "k ≠ 21" hai, "k = 21" nahi.



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Q27 mein discriminant se k = 0 aur k = 4 dono aate hain. k = 0 pe equation quadratic hi nahi rehti, to use reject karo aur ye reason likho. Board ke marking scheme mein is rejection ka alag se mark hota hai.



SHIVAM BHAIYA SAYS: GALTI MAT KARNA

Q35 mode formula mein denominator $2f_1 - f_0 - f_2$ hai, $f_1 - f_0 - f_2$ nahi. Aur f_0 modal class se pehle wali class ki frequency, f_2 baad wali. Q36 ke OR mein 1 m ka path dono taraf lagta hai, to andar ka rectangle $(23 - 2) \times (10 - 2)$ hoga.

59.3 My mistake analysis

Q no.	Topic	Marks lost	Mistake type (C / S / T)
Q4	Parallel vs coincident lines		
Q17	Modal class and median class		
Q27	Equal roots, rejecting k = 0		
Q32	Graphical solution or age problem		
Q33	Tangent lengths, incircle		
Q34	Heights and distances		
Q35	Mean and mode		
Q36 to Q38	Case studies		
Other			

C = concept not clear, S = silly or calculation slip, T = ran out of time. Fill it honestly; every C goes into your weak-spot list for Days 55 to 57.



SHIVAM BHAIYA SAYS: BOARD TIP

Q12 broken tree: height = khada hua hissa + toota hua hissa, dono alag nikaalo ($\tan$ se ek, $\cos$ se doosra). Aur Q34 jaise 5-markers mein figure banana compulsory samjho: figure ke bina examiner ko tumhara set-up samajh nahi aata aur step marks bhi kat sakte hain.

59.4 Last corrections before the board

- Write units in every final answer (cm, cm², cm³, m, km/h).
- Use the value of π the question gives; if none is given, use $\frac{22}{7}$ when the numbers are multiples of 7, else 3.14.
- Draw a neat labelled figure for every heights-and-distances, circle and triangle question.
- In OR questions, read both options fully and pick the one whose first step you can write right now.
- Keep the last 15 to 20 minutes to check Section A answers and fill any question left blank.

Finish with the 10 fix-it questions below: one last round on the topics that cost the most marks across all four mocks.

Day 59 Practice

10 board-pattern questions | solve first, then check

FIX-IT SET: LAST CORRECTIONS BEFORE THE BOARD

VERY SHORT ANSWER

1. The LCM of two consecutive even numbers is 220. Find the numbers and their HCF. (2)

VERY SHORT ANSWER

2. For what value of k will the pair $3x - ky = 7$ and $9x + 6y = 5$ have no solution? (2)

SHORT ANSWER

3. The sum of the 2nd and 6th terms of an AP is 20 and the sum of its 4th and 9th terms is 35. Find the first three terms of the AP. (3)

VERY SHORT ANSWER

4. In right triangles ABC and DEF, $\angle B = \angle E = 90^\circ$ and $\angle A = \angle D$. If AB = 5 cm, BC = 12 cm and DE = 7.5 cm, find EF and DF. (2)

VERY SHORT ANSWER

5. Tangents PA and PB to a circle with centre O are inclined to each other at 72°. Find $\angle AOB$ and $\angle OAB$. (2)

VERY SHORT ANSWER

6. Evaluate $(\operatorname{cosec} A + \cot A)(1 - \cos A)$ at $A = 30^\circ$. (2)

SHORT ANSWER

7. The angles of elevation of the top of a tower from two points 5 m and 20 m from its base, on the same straight line, are complementary. Find the height of the tower. (3)

SHORT ANSWER

8. A tent is a cylinder of height 3 m and diameter 7 m, surmounted by a cone of slant height 5 m on the same base. Find the area of canvas used. (Use $\pi = \frac{22}{7}$) (3)

SHORT ANSWER

9 Find the median of the following data.

Class	0-10	10-20	20-30	30-40	40-50
f	8	12	10	11	9

(3)

VERY SHORT ANSWER

10 A box has 80 discs numbered 1 to 80. One disc is taken out at random. Find the probability that it bears (i) a two-digit number (ii) a perfect square. (2)

Day 59 Answers with worked solutions

1 **Ans:** 20 and 22; HCF 2

Two consecutive even numbers n and n + 2 have HCF 2, so LCM $= \frac{n(n+2)}{2}$.

$n(n+2) = 440 \Rightarrow n^2 + 2n - 440 = 0 \Rightarrow (n - 20)(n + 22) = 0 \Rightarrow n = 20$.

Numbers: 20 and 22; check LCM(20, 22) = 220.

2 **Ans:** k = −2

No solution: $\frac{3}{9} = \frac{-k}{6} \neq \frac{7}{5}$.

$\frac{-k}{6} = \frac{1}{3} \Rightarrow k = -2$; and $\frac{1}{3} \neq \frac{7}{5}$, so the condition holds.

3 **Ans:** 1, 4, 7

$(a + d) + (a + 5d) = 20 \Rightarrow 2a + 6d = 20$.

$(a + 3d) + (a + 8d) = 35 \Rightarrow 2a + 11d = 35$.

Subtract: $5d = 15 \Rightarrow d = 3$; then $2a = 20 - 18 \Rightarrow a = 1$.

First three terms: 1, 4, 7.

4 **Ans:** EF = 18 cm, DF = 19.5 cm

$\angle B = \angle E$ and $\angle A = \angle D$, so $\triangle ABC \sim \triangle DEF$ (AA). Scale factor $= \frac{DE}{AB} = \frac{7.5}{5} = 1.5$.

EF $= 1.5 \times 12 = 18$ cm. AC $= \sqrt{25 + 144} = 13$ cm, so DF $= 1.5 \times 13 = 19.5$ cm.

5 **Ans:** $108°$, $36°$

$\angle AOB = 180° - 72° = 108°$ (supplementary).

OA = OB, so $\angle OAB = \frac{180° - 108°}{2} = 36°$.

6 **Ans:** $\frac{1}{2}$

$\operatorname{cosec} A + \cot A = \frac{1 + \cos A}{\sin A}$, so the product $= \frac{1 - \cos^2 A}{\sin A} = \sin A$.

At 30°: $\frac{1}{2}$. (Direct: $(2 + \sqrt{3})(1 - \frac{\sqrt{3}}{2}) = 2 - \sqrt{3} + \sqrt{3} - \frac{3}{2} = \frac{1}{2}$.)

7 **Ans:** 10 m

Let the angles be θ and $90° - \theta$, height h.

$\tan\theta = \frac{h}{20}$ and $\tan(90° - \theta) = \cot\theta = \frac{h}{5}$.

Multiply: $\tan\theta \cot\theta = 1 = \frac{h^2}{100} \Rightarrow h = 10$ m.

8 **Ans:** 121 m²

Canvas = CSA of cylinder + CSA of cone (no base): $2\pi rh + \pi rl$ with $r = 3.5$, $h = 3$, $l = 5$.

$= \pi r(2h + l) = \frac{22}{7} \times 3.5 \times 11 = 121$ m².

9 **Ans:** 25

N = 50, $\frac{N}{2} = 25$; cf: 8, 20, 30, 41, 50, so the median class is 20-30.

l = 20, cf = 20, f = 10, h = 10: median $= 20 + \frac{25 - 20}{10} \times 10 = 25$.

10 **Ans:** (i) $\frac{71}{80}$ (ii) $\frac{1}{10}$

(i) Two-digit numbers: 10 to 80, that is 71 discs: $\frac{71}{80}$.

(ii) Perfect squares: 1, 4, 9, 16, 25, 36, 49, 64, that is 8 discs: $\frac{8}{80} = \frac{1}{10}$.

Mock Test 4: worked solutions

marks as the board gives them

Mock Test 4 answers

1 **(d)**

$4 = 2^2$, $6 = 2 \times 3$; LCM $= 2^2 \times 3 \times 5 \times 7 = 420$.

2 **(b)**

$3600 = 36 \times 100 = 2^2 3^2 \times 2^2 5^2 = 2^4 \times 3^2 \times 5^2$.

Sum of exponents $= 4 + 2 + 2 = 8$.

3 **(a)**

$\alpha + \beta = -\frac{b}{a} = 0$, $\alpha\beta = \frac{c}{a} = -25$. Sum $= 0 + (-25) = -25$.

(Zeroes are 5 and −5.)

4 **(c)**

Parallel lines: $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$.

$\frac{4}{12} = \frac{5}{15} = \frac{1}{3}$, so we need $\frac{-7}{-k} \neq \frac{1}{3}$, i.e. $k \neq 21$. (k = 21 makes the lines coincident.)

5 **(b)**

Product of roots $= \frac{c}{a} = \frac{-12}{3} = -4$.

6 **(c)**

No real roots: $D < 0$: $64 - 8k < 0 \Rightarrow k > 8$.

7 **(d)**

(a) d = −4, (b) d = 2.5, (c) d = 0: all APs.

(d) differences are 3, 5, 7: not constant, so not an AP.

8 **(a)**

Same y-coordinate, so distance $= \left|\frac{5}{3} - \left(-\frac{7}{3}\right)\right| = \frac{12}{3} = 4$ units.

9 **(b)**

Two pairs of equal angles give $\triangle ABC \sim \triangle QRP$ (AA).

AB and QR are corresponding sides with AB = 2QR, so the scale factor is 2: similar but not congruent.

10 **(b)**
OA $\perp$ tangent, so $\angle OAB = 90^\circ - 55^\circ = 35^\circ$; OA = OB gives $\angle OBA = 35^\circ$.
$\angle AOB = 180^\circ - 70^\circ = 110^\circ$ (twice the tangent-chord angle).

11 **(c)**
LHS $= y \times \frac{1}{2} \times \frac{1}{2} = \frac{y}{4}$; RHS $= (\sqrt{3})^2 - (\sqrt{2})^2 = 3 - 2 = 1$.
$\frac{y}{4} = 1 \Rightarrow y = 4$.

12 **(a)**
Standing part $= 12\tan 30^\circ = \frac{12}{\sqrt{3}} = 4\sqrt{3}$ m.
Broken part $= \frac{12}{\cos 30^\circ} = \frac{24}{\sqrt{3}} = 8\sqrt{3}$ m.
Height $= 4\sqrt{3} + 8\sqrt{3} = 12\sqrt{3}$ m.

13 **(c)**
Radius $= \frac{d}{2}$; quadrant area $= \frac{1}{4}\pi\left(\frac{d}{2}\right)^2 = \frac{\pi d^2}{16}$.

14 **(b)**
Area $= \frac{72}{360} \times 3.14 \times 10^2 = \frac{1}{5} \times 314 = 62.8$ cm².

15 **(c)**
$V = \pi r^2 h$ with equal h, so $V_1 : V_2 = r_1^2 : r_2^2 = 16 : 49$.

16 **(b)**
TSA of a solid hemisphere = curved surface + flat circular face $= 2\pi r^2 + \pi r^2 = 3\pi r^2$.
$= 3 \times \frac{22}{7} \times 49 = 462$ cm².

17 **(b)**
Modal class (highest f = 15) is 20-30, lower limit 20.
N = 45, $\frac{N}{2} = 22.5$; cf = 5, 14, 29, so the median class is 20-30, upper limit 30.
Sum = 20 + 30 = 50.

18 **(b)**
365 days = 52 weeks + 1 extra day. That extra day can be any of the 7 days.
53 Sundays only if the extra day is Sunday: P $= \frac{1}{7}$.

19 **(a)**
R is a true result. Since 4 is rational and $\sqrt{7}$ is irrational, $4 + \sqrt{7}$ is irrational by R. Answer (a).

20 **(d)**
For $x^2 + 4x + 5$: $D = 16 - 20 = -4 < 0$, so no real zeroes. A is false.
R is true (degree 2). Answer (d).

21 **Ans:** 585
The numbers are 18, 27, ..., 99: an AP with a = 18, d = 9, last term 99.
$99 = 18 + (n-1)9 \Rightarrow n = 10$.
$S_{10} = \frac{10}{2}(18 + 99) = 5 \times 117 = 585$.

22 **Ans:** Diagonals bisect each other
Midpoint of AC $= \left(\frac{1+5}{2}, \frac{1+5}{2}\right) = (3, 3)$.
Midpoint of BD $= \left(\frac{4+2}{2}, \frac{2+4}{2}\right) = (3, 3)$.
The diagonals AC and BD bisect each other, so ABCD is a parallelogram.

23 **Ans:** $\sec\theta - \tan\theta = \frac{1}{4}$, $\cos\theta = \frac{8}{17}$
$(\sec\theta + \tan\theta)(\sec\theta - \tan\theta) = \sec^2\theta - \tan^2\theta = 1$, so $\sec\theta - \tan\theta = \frac{1}{4}$.
Adding: $2\sec\theta = 4 + \frac{1}{4} = \frac{17}{4} \Rightarrow \sec\theta = \frac{17}{8}$, so $\cos\theta = \frac{8}{17}$.

24 **Ans:** Yes, SSS; 2 : 5
Compare sides in increasing order: $\frac{4}{10} = \frac{2}{5}$, $\frac{5}{12.5} = \frac{2}{5}$, $\frac{6}{15} = \frac{2}{5}$.
All three ratios are equal, so the triangles are similar (SSS criterion).
Perimeters: 15 cm and 37.5 cm, ratio $15 : 37.5 = 2 : 5$ (same as the ratio of sides).

25 **Ans:** $\frac{17248}{3} \approx 5749.33$ cm³ ; OR 301.44 cm²
$2\pi r^2 = 1232 \Rightarrow r^2 = \frac{1232 \times 7}{44} = 196 \Rightarrow r = 14$ cm.
$V = \frac{2}{3}\pi r^3 = \frac{2}{3} \times \frac{22}{7} \times 2744 = \frac{17248}{3} \approx 5749.33$ cm³.
OR: Slant height $l = \sqrt{6^2 + 8^2} = 10$ cm.
TSA $= \pi r(l + r) = 3.14 \times 6 \times 16 = 301.44$ cm².

26 **Ans:** 54 cm, 10 pieces ; OR proof
Greatest length = HCF(108, 162, 270). $108 = 2^2 \times 3^3$, $162 = 2 \times 3^4$, $270 = 2 \times 3^3 \times 5$.
HCF $= 2 \times 3^3 = 54$ cm (lowest powers of common primes).
Pieces $= \frac{108}{54} + \frac{162}{54} + \frac{270}{54} = 2 + 3 + 5 = 10$.
OR: Suppose $2 + 5\sqrt{3} = r$ is rational. Then $\sqrt{3} = \frac{r - 2}{5}$.
$\frac{r-2}{5}$ is rational, so $\sqrt{3}$ would be rational, contradicting the given fact.
Hence $2 + 5\sqrt{3}$ is irrational.

27 **Ans:** k = 4; roots $\frac{2}{3}, \frac{2}{3}$
a = 9, b = −3k, c = k. Equal roots: $D = 9k^2 - 36k = 0 \Rightarrow 9k(k-4) = 0 \Rightarrow k = 0$ or 4.
k = 0 is excluded (given), so k = 4.
Equation: $9x^2 - 12x + 4 = 0 \Rightarrow (3x - 2)^2 = 0 \Rightarrow x = \frac{2}{3}, \frac{2}{3}$.

28 **Ans:** Proof
Let PA and PB be tangents from P to a circle with centre O, touching it at A and B. To prove: $\angle APB + \angle AOB = 180^\circ$.
The radius is perpendicular to the tangent at the point of contact: $\angle OAP = \angle OBP = 90^\circ$.
In quadrilateral OAPB the angles add to 360°: $\angle APB + \angle AOB + 90^\circ + 90^\circ = 360^\circ$.
So $\angle APB + \angle AOB = 180^\circ$: the two angles are supplementary.

29 **Ans:** Proof

$\cos^4 A - \sin^4 A = (\cos^2 A - \sin^2 A)(\cos^2 A + \sin^2 A) = \cos^2 A - \sin^2 A$.

So $\cos^4 A - \sin^4 A + 1 = \cos^2 A + (1 - \sin^2 A) = 2\cos^2 A$.

LHS $= 2\cos^2 A \times \sec^2 A = 2$ = RHS.

OR: $(\sin\theta + \cos\theta)^2 - 1 = 1 + 2\sin\theta\cos\theta - 1 = 2\sin\theta\cos\theta$.

LHS $= 2\sin\theta\cos\theta \times \dfrac{1}{\cos\theta} \times \dfrac{1}{\sin\theta} = 2$ = RHS.

30 **Ans:** (0, 3) and (4, 1)

P divides AB in 1 : 2: $P = \left(\dfrac{1(8)+2(-4)}{3}, \dfrac{1(-1)+2(5)}{3}\right) = (0, 3)$.

Q divides AB in 2 : 1: $Q = \left(\dfrac{2(8)+1(-4)}{3}, \dfrac{2(-1)+1(5)}{3}\right) = (4, 1)$.

(Check: Q is the midpoint of PB: $\left(\frac{0+8}{2}, \frac{3-1}{2}\right) = (4, 1)$.)

31 **Ans:** MN = 6 cm

$\dfrac{XM}{MY} = \dfrac{4}{6} = \dfrac{2}{3}$ and $\dfrac{XN}{NZ} = \dfrac{5}{7.5} = \dfrac{2}{3}$.

The sides are divided in the same ratio, so $MN \parallel YZ$ (converse of BPT).

Then $\angle XMN = \angle XYZ$ (corresponding angles) and $\angle X$ is common, so $\triangle XMN \sim \triangle XYZ$ (AA).

$\dfrac{MN}{YZ} = \dfrac{XM}{XY} = \dfrac{4}{10} \Rightarrow MN = \dfrac{2}{5} \times 15 = 6$ cm.

32 **Ans:** Solution (2, 4); vertices (2, 4), (−2, 0), (4, 0); area 12 sq units ; OR father 34 years, son 12 years

Line 1: $y = x + 2$; points (0, 2), (−2, 0), (2, 4).

Line 2: $y = 8 - 2x$; points (0, 8), (4, 0), (2, 4).

Plot both lines on the same axes: they intersect at (2, 4), so x = 2, y = 4.

Line 1 meets the x-axis at (−2, 0) and line 2 meets it at (4, 0). Vertices: (2, 4), (−2, 0), (4, 0).

Base on the x-axis $= 4 - (-2) = 6$ units; height = y-coordinate of (2, 4) = 4 units. Area $= \frac{1}{2} \times 6 \times 4 = 12$ sq units.

OR: Let the present ages be F (father) and S (son). $F - 10 = 12(S - 10) \Rightarrow F - 12S = -110$.

$F + 10 = 2(S + 10) \Rightarrow F - 2S = 10$.

Subtract: $10S = 120 \Rightarrow S = 12$, and $F = 10 + 24 = 34$.

Check: 10 years ago 24 and 2 (24 = 12 × 2); in 10 years 44 and 22. Father 34 years, son 12 years.

33 **Ans:** AF = 3 cm, BD = 7 cm, CE = 5 cm

Let PQ and PR be tangents from P to a circle with centre O. Join OP, OQ, OR.

In $\triangle OQP$ and $\triangle ORP$: OQ = OR (radii), OP common, $\angle OQP = \angle ORP = 90^\circ$ (radius $\perp$ tangent).

So $\triangle OQP \cong \triangle ORP$ (RHS) and PQ = PR (CPCT).

Application: let AF = AE = x, BF = BD = y, CD = CE = z (tangents from each vertex are equal).

$x + y = 10$, $y + z = 12$, $z + x = 8$. Adding: $2(x + y + z) = 30 \Rightarrow x + y + z = 15$.

$x = 15 - 12 = 3$, $y = 15 - 8 = 7$, $z = 15 - 10 = 5$. So AF = 3 cm, BD = 7 cm, CE = 5 cm.

34 **Ans:** Height $20\sqrt{3} \approx 34.64$ m, A is 60 m from the foot ; OR $60\sqrt{3} \approx 103.92$ m

Let the tower be PQ = h m with foot Q, and let B be the point 40 m nearer, with BQ = x m, so AQ = x + 40.

In $\triangle PBQ$: $\tan 60^\circ = \dfrac{h}{x} \Rightarrow h = \sqrt{3}x$.

In $\triangle PAQ$: $\tan 30^\circ = \dfrac{h}{x+40} \Rightarrow \sqrt{3}h = x + 40$.

Substitute: $3x = x + 40 \Rightarrow x = 20$ m, so $h = 20\sqrt{3} \approx 34.64$ m and AQ = 20 + 40 = 60 m.

OR: Let the height be h and the nearer observer be x m from the foot. $\tan 60^\circ = \frac{h}{x} \Rightarrow h = \sqrt{3}x$.

$\tan 30^\circ = \frac{h}{x+120} \Rightarrow \sqrt{3}h = x + 120 \Rightarrow 3x = x + 120 \Rightarrow x = 60$.

$h = 60\sqrt{3} \approx 103.92$ m.

35 **Ans:** Mean 47.6, mode 48.75 ; OR median ≈ 138.57

Class marks: 10, 30, 50, 70, 90. Take a = 50: d_i = −40, −20, 0, 20, 40.

$f_i d_i$ = −280, −220, 0, 180, 200; $\sum f_i d_i = -120$, $\sum f_i = 50$.

Mean $= 50 + \dfrac{-120}{50} = 50 - 2.4 = 47.6$.

Mode: modal class 40-60 ($f_1 = 18$), $f_0 = 11$, $f_2 = 9$, $l = 40$, $h = 20$.

Mode $= l + \dfrac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 40 + \dfrac{7}{36 - 20} \times 20 = 40 + 8.75 = 48.75$.

OR: N = 50, $\frac{N}{2} = 25$; cf: 12, 26, 34, 40, 50.

Median class 120-140: $l = 120$, cf = 12, f = 14, h = 20.

Median $= 120 + \dfrac{25 - 12}{14} \times 20 = 120 + \dfrac{260}{14} \approx 138.57$.

36 **Ans:** $2x^2 + 3x - 230 = 0$

Length $= 2x + 3$; $x(2x + 3) = 230 \Rightarrow 2x^2 + 3x - 230 = 0$.

Ans: 1849

$D = 3^2 - 4(2)(-230) = 9 + 1840 = 1849\ (= 43^2)$.

Ans: Width 10 m, length 23 m

$x = \dfrac{-3 \pm 43}{4} = 10$ or $-\dfrac{23}{2}$; width cannot be negative, so x = 10 m.

Length $= 2(10) + 3 = 23$ m. (Check: 10 × 23 = 230.)

OR: Ans: 62 m²

Park is 23 m by 10 m (from the equation).

Inner rectangle: $(23 - 2)$ by $(10 - 2)$ = 21 m by 8 m = 168 m².

Area of path $= 230 - 168 = 62$ m².

37 **Ans:** 36

Each die has 6 faces: $6 \times 6 = 36$ outcomes.

Ans: $\frac{5}{36}$

Sum 8: (2, 6), (3, 5), (4, 4), (5, 3), (6, 2), so 5 outcomes; P $= \frac{5}{36}$.

Ans: $\frac{3}{4}$

Product is odd only when both numbers are odd: $3 \times 3 = 9$ outcomes.

P(product even) $= 1 - \frac{9}{36} = \frac{27}{36} = \frac{3}{4}$.

OR: Ans: $\frac{1}{12}$

Doublets with sum > 6: (4, 4), (5, 5), (6, 6), so 3 outcomes.

P $= \frac{3}{36} = \frac{1}{12}$.

38 **Ans:** 231 m²

$\frac{60}{360} \times \frac{22}{7} \times 21^2 = \frac{1}{6} \times 1386 = 231$ m².

Ans: 22 m

$\frac{60}{360} \times 2 \times \frac{22}{7} \times 21 = \frac{1}{6} \times 132 = 22$ m.

Ans: $\frac{385}{3} \approx 128.33$ m²

Inner sector $= \frac{1}{6} \times \frac{22}{7} \times 196 = \frac{308}{3}$ m².

Shaded $= 231 - \frac{308}{3} = \frac{385}{3} \approx 128.33$ m².

OR: Ans: 770 m²

Area of ring $= \pi(R^2 - r^2) = \frac{22}{7}(441 - 196) = \frac{22}{7} \times 245 = 770$ m².

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Parallel lines condition : $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$
2. Mode formula : $l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$
3. Equal roots : $D = b^2 - 4ac = 0$ (reject values that make a = 0)
4. Tangents from an external point : equal in length
5. Angle between two tangents + angle at centre : $180°$
6. Complementary elevations from distances a and b : height $= \sqrt{ab}$
7. Sum of first n terms : $\frac{n}{2}[2a + (n - 1)d]$
8. Non-leap year 53 Sundays : $\frac{1}{7}$

DAY 59 COMPLETE? ☐ Mock 4 marked ☐ Time plan reviewed ☐ 10 fix-its solved ☐ Exam Eve (Day 60) next

Next: Day 60, Exam Eve: Formula Flash, Paper Strategy and Final Checklist

MY MISTAKES AND NOTES

Write every mistake of this Day with the correct step

DAY 60

EXAM EVE | LAST EVENING

Formula Flash, Paper Strategy and Final Checklist

Exam Eve

Aaj ka target

- [ ] Formula flash ek baar, bina ruke, 20 minute
- [ ] Paper strategy aur time plan dimaag mein fit karo
- [ ] Kal ka bag aaj raat hi pack
- [ ] 10:30 tak sona, 7-8 ghante ki neend

Tonight is not for new chapters. Read this flash once, from top to bottom, in one sitting. If a line feels shaky, look at Day 42 for that chapter for two minutes and move on. Then plan tomorrow, pack your bag and sleep.

60.1 Formula flash: Algebra (Ch 1 to 5)

Chapter	Must-know in one glance
1 Real Numbers	HCF = product of smallest powers, LCM = product of greatest powers of primes; HCF×LCM $= a \times b$ (two numbers only); $\sqrt{p}$ irrational for prime p (proof by contradiction)
2 Polynomials	$\alpha+\beta=-\frac{b}{a}$, $\alpha\beta=\frac{c}{a}$; polynomial: $k[x^2-(\alpha+\beta)x+\alpha\beta]$; zeroes = x-axis meeting points
3 Linear Equations	$\frac{a_1}{a_2}\neq\frac{b_1}{b_2}$: unique; $\frac{a_1}{a_2}=\frac{b_1}{b_2}=\frac{c_1}{c_2}$: infinite; $\frac{a_1}{a_2}=\frac{b_1}{b_2}\neq\frac{c_1}{c_2}$: none
4 Quadratic Equations	$x=\frac{-b\pm\sqrt{D}}{2a}$, $D=b^2-4ac$: $D>0$ two real, $D=0$ equal, $D<0$ no real roots
5 AP	$a_n=a+(n-1)d$; $S_n=\frac{n}{2}[2a+(n-1)d]=\frac{n}{2}(a+l)$; $a_n=S_n-S_{n-1}$; nth from end $=l-(n-1)d$

Algebra is the biggest unit: 20 marks.

60.2 Formula flash: Coordinate, Geometry, Trigonometry

Chapter	Must-know in one glance
7 Coordinate Geometry	$d=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$; $m:n$ point $\left(\frac{mx_2+nx_1}{m+n},\frac{my_2+ny_1}{m+n}\right)$; mid-point $\left(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2}\right)$
6 Triangles	BPT: $DE \parallel BC \Rightarrow \frac{AD}{DB}=\frac{AE}{EC}$ and its converse; similarity by AA, SSS, SAS (write vertices in matching order)
10 Circles	Radius $\perp$ tangent at the point of contact; tangents from an external point are equal; angle between tangents $= 180°-$ centre angle
8 Trigonometry	sin: $0, \frac{1}{2}, \frac{1}{\sqrt{2}}, \frac{\sqrt{3}}{2}, 1$ at $0°, 30°, 45°, 60°, 90°$ (cos in reverse); $\tan=\frac{\sin}{\cos}$; $\sin^2+\cos^2=1$, $1+\tan^2=\sec^2$, $1+\cot^2=\csc^2$
9 Heights and Distances	$\tan\theta=\frac{\text{height}}{\text{distance}}$; depression at the top = elevation at the object; add the observer's height

Coordinate 6 marks, Geometry 15 marks, Trigonometry 12 marks.

60.3 Formula flash: Mensuration, Statistics, Probability

Chapter	Must-know in one glance
11 Areas (circles)	Sector $=\frac{\theta}{360°}\pi r^2$; arc $=\frac{\theta}{360°}2\pi r$; perimeter of sector = arc $+2r$; segment = sector − triangle
12 Surface Areas, Volumes	Cylinder $2\pi rh$, πr^2h; cone πrl, $\frac{1}{3}\pi r^2h$, $l=\sqrt{r^2+h^2}$; sphere $4\pi r^2$, $\frac{4}{3}\pi r^3$; hemisphere CSA $2\pi r^2$, TSA $3\pi r^2$, $\frac{2}{3}\pi r^3$
13 Statistics	$\bar{x}=a+\frac{\sum f_id_i}{\sum f_i}$; Mode $=l+\frac{f_1-f_0}{2f_1-f_0-f_2}h$; Median $=l+\frac{N/2-cf}{f}h$; Mode = 3 Median −2 Mean
14 Probability	$P(E)=\frac{\text{favourable}}{\text{total}}$; $0\le P(E)\le 1$; $P(\text{not } E)=1-P(E)$; 2 dice: 36 outcomes; deck: 52 cards, 12 face cards

Mensuration 10 marks, Statistics + Probability 11 marks.

60.4 Paper strategy for tomorrow

Order	Section	Questions and marks	Time
1	A: MCQ + AR	20 questions x 1 = 20	30 min

Order	Section	Questions and marks	Time
2	B: Very short	5 questions x 2 = 10	15 min
3	C: Short	6 questions x 3 = 18	35 min
4	E: Case-based	3 questions x 4 = 12	30 min
5	D: Long	4 questions x 5 = 20	45 min
6	Final check	whole paper	25 min

Total 80 marks in 180 minutes. Attempt in this order if you are comfortable with it; the paper itself is printed A to E.

USING THE 15-MINUTE READING TIME

1. Read the instructions and count the questions and pages first (38 questions).
2. Read Section A fast and solve the easy MCQs in your head; you cannot write yet, so just remember them.
3. In every internal choice (OR), decide now which option you will attempt and do not switch later.
4. Read the case studies (Section E) slowly; they are long and easy to misread in a hurry.
5. Mark in your mind the 2-3 questions you will leave for the end.

PRESENTATION THAT EARNS THE MARKS

- Write the question number clearly in the margin, exactly as printed (including (i), (ii), OR).
- Write the formula first, then substitute, then the answer with its unit. Steps carry marks even if the final answer slips.
- Draw every figure for geometry, circles and heights and distances with a pencil, labelled, with the given values marked.
- Box or underline the final answer. Write 'Hence proved' at the end of a proof.
- Keep rough work on the right side of the page or at the end, headed 'Rough work', never mixed with the answer.
- Never leave a question blank: write the formula and the first step at least.



SHIVAM BHAIYA SAYS: BOARD TIP

Section A mein kisi MCQ pe 1.5 minute se zyada mat atko. Number pe circle lagao, aage badho, last ke 25 minute mein wapas aao. Ek MCQ ke liye 5 marks ka long answer adhoora chhodna sabse mehenga sauda hai.

60.5 Exam-morning checklist

- Admit card (keep a photocopy at home) and school ID.
- Two or three blue or black pens that you have already written with, a pencil, eraser, sharpener and a 15 cm scale; geometry box.
- A transparent water bottle. No calculator, phone or smart device: they are not allowed.
- Light breakfast; skim only this formula flash, nothing new.
- Leave early and reach the centre well before the reporting time on your admit card.
- Before you start writing, check the number of pages and questions in your question paper.

SHIVAM BHAIYA KI TARAF SE, AAJ RAAT KE LIYE

Pichle 59 din tumne roz thoda thoda kiya hai: notes, traps, mocks, mistakes. Woh mehnat kahin nahi jaati. Kal paper ko dushman mat samjho, woh bas wahi poochega jo tum 60 din se kar rahe ho. Pehle aasan sawal, phir mushkil; har step likho; har figure banao. Ek sawal bigde to agle pe dhyan do, poora paper ek sawal se bada hai. Aaj raat formula flash, bag pack, aur achhi neend. Kal shaant mann se jao. All the best!

MEME SAMJHO

Astra meme, maths inside



EXAM EVE, RAAT 10:30 BAJE: FORMULA FLASH HO GAYA, BAG PACK HO GAYA

Formula flash done, bag packed, alarm set. Ab bas 7-8 ghante ki neend: yeh bhi preparation ka hissa hai!

Asli baat: raat bhar jaag ke padhne se agle din calculation slips badhte hain. Neend se memory pakki hoti hai.

Day 60 Practice

15 quick questions | answer from memory

VERY SHORT ANSWER

1 **LCM of $2^3 \times 3$ and $2 \times 3^2 \times 5$.** (1)

VERY SHORT ANSWER

2 **A quadratic polynomial whose zeroes are 2 and -5.** (1)

VERY SHORT ANSWER

3. Nature of the pair $x + 2y = 3, 2x + 4y = 7$. (1)

VERY SHORT ANSWER

4. Values of k for which $x^2 + kx + 16 = 0$ has equal roots. (1)

VERY SHORT ANSWER

5. Common difference of the AP $\frac{3}{2}, 1, \frac{1}{2}, \ldots$ (1)

VERY SHORT ANSWER

6. Distance of the point $(-5, 12)$ from the origin. (1)

VERY SHORT ANSWER

7. $\triangle ABC \sim \triangle DEF$ with $AB = 3$ cm, $DE = 6$ cm, $BC = 4$ cm. Find EF. (1)

VERY SHORT ANSWER

8. Number of tangents that can be drawn to a circle from a point inside it. (1)

VERY SHORT ANSWER

9. Value of $(1 - \cos^2\theta)\csc^2\theta$. (1)

VERY SHORT ANSWER

10. $\sin 60^\circ \cos 30^\circ + \cos 60^\circ \sin 30^\circ$ (1)

VERY SHORT ANSWER

11. A pole's shadow is equal to its height. Find the sun's elevation. (1)

VERY SHORT ANSWER

12. Length of the arc of a semicircle of radius 7 cm. Use $\pi = \frac{22}{7}$. (1)

VERY SHORT ANSWER

13. Volume of a cone with $r = 3$ cm, $h = 7$ cm. Use $\pi = \frac{22}{7}$. (1)

VERY SHORT ANSWER

14. Class mark of the class 25-35. (1)

VERY SHORT ANSWER

15. A letter is chosen at random from the word MATHS. Probability that it is a vowel. (1)

Day 60 Answers with worked solutions

1. **Ans:** 360
   Greatest powers: $2^3 \times 3^2 \times 5 = 360$.

2. **Ans:** $x^2 + 3x - 10$
   Sum $= -3$, product $= -10$: $x^2 - (-3)x + (-10)$.

3. **Ans:** No solution (parallel lines)
   $\frac{1}{2} = \frac{2}{4} \neq \frac{3}{7}$.

4. **Ans:** $k = \pm 8$
   $D = k^2 - 64 = 0$.

5. **Ans:** $-\frac{1}{2}$
   $d = 1 - \frac{3}{2} = -\frac{1}{2}$.

6. **Ans:** 13 units
   $\sqrt{25 + 144} = 13$.

7. **Ans:** 8 cm
   $\frac{AB}{DE} = \frac{BC}{EF} \Rightarrow \frac{3}{6} = \frac{4}{EF}$.

8. **Ans:** 0
   Every line through an interior point cuts the circle at two points.

9. **Ans:** 1
   $\sin^2\theta \times \frac{1}{\sin^2\theta} = 1$.

10. **Ans:** 1
    $\frac{3}{4} + \frac{1}{4} = 1$.

11. **Ans:** 45°
    $\tan\theta = 1$.

12. **Ans:** 22 cm
    $\pi r = \frac{22}{7} \times 7$.

13. **Ans:** 66 cm^3
    $\frac{1}{3} \times \frac{22}{7} \times 9 \times 7 = 66$.

14. **Ans:** 30
    $\frac{25+35}{2}$.

15. **Ans:** $\frac{1}{5}$
    Only A is a vowel among 5 letters.

TONIGHT'S 2-MINUTE RECALL

Answer from memory, then check the notes

1. Biggest unit : Algebra, 20 marks
2. Order tomorrow : A, B, C, E, D, then check
3. MCQ time limit : 1.5 min each, then move on
4. Reading time : choose every OR option now
5. Every answer : formula, substitution, unit
6. Tonight : flash, pack, sleep

DAY 60 COMPLETE? ☐ Formula flash read ☐ Strategy fixed ☐ Bag packed ☐ Slept on time

All 60 Days done!

MY MISTAKES AND NOTES Write every mistake of this Day with the correct step

ASTRA SERIES | CBSE CLASS 10

MATHEMATICS IN 60 DAYS

One Day at a time: learn, solve, check, revise.

60 DAYS

A complete day-by-day plan: what to learn, which examples, how many questions, how long.

14 CHAPTERS

Board syllabus 2026-27 only. Short notes, every formula, solved examples step by step.

4 FULL MOCKS

Four 80-mark papers in board pattern, each with a full review day and worked solutions.

FORMULA DAY

Every formula of the book on a few pages, plus a last-evening formula flash.



"Roz ek Day, haath mein pen. Sixty din baad paper dekh ke darr nahi, muskaan aayegi."

Shivam Bhaiya

PUBLISHED BY
Eduspark
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